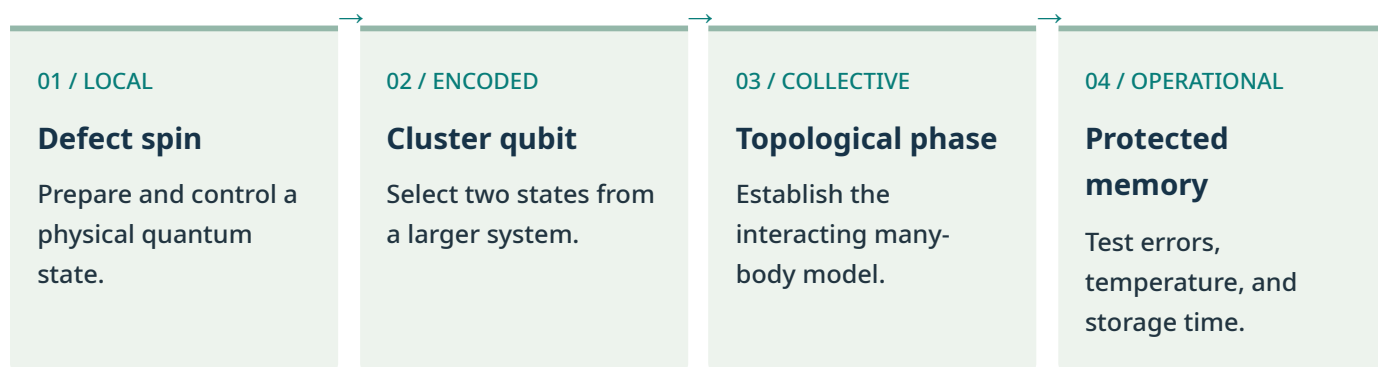


Defect-engineered topological qubits

How scientists study individual spins in crystals and investigate whether they could store quantum information reliably.

Foundations · Three assessment units · Eight mathematical appendices · Illustrated reading edition. Experimental evidence is discussed in its stated context, through August 2026.



START WITH THE QUESTION YOU HAVE

A clear route through a long subject

Start with Chapters 1–4 for the quantum mechanics used in these notes. Chapters 5–11 explain experiments with defects in crystals. Later chapters develop the proposed quantum-memory design and examine the evidence needed to establish whether it works. The links below also provide entry points for readers who already know the earlier material.

Build the foundations

Chapters 1–4 · States, measurement, entanglement, and noise.

Understand the physical hardware

Chapters 5–11 · Crystals, defects, interactions, and encoded clusters.

Connect the hardware to topology

Assess the proposed architecture

Chapters 12–36 · Anyons, target Hamiltonians, fabrication, and diagnostics.

Three connected units · Proposal and limits, decisive tests, and a worked four-cluster assessment.

In the assessment and appendices, start with the main explanation. Open **Detailed treatment** for the supporting derivation or evidence, and open the exercises when you want to check your understanding. Extra material stays with the concept it develops.

Reference numbering: Assessment I combines former Chapters 37–38; Assessment II combines 39–40; Assessment III develops Chapter 41. Original chapter and section links still work, and numbered references in the detailed notes retain their original meaning.

Keep the notation and energy scales in view

Symbol	Meaning	What to keep separate
H, H_C	Hamiltonian of the full system or one cluster	Unless divided by $\hbar$ or $\hbar$, these have energy units.
Δ_C or Δ_c	Gap from a retained cluster doublet to excluded cluster states	This energy is measured within one cluster. The collective gap is defined for the interacting system.
Δ_{topo}	Gap above a candidate many-body ground-state sector	Its size must be derived from the interacting model.
v, J, J_{eff}, K	Microscopic or effective interaction energies	The spectral gap is obtained by solving the Hamiltonian containing these coefficients.
$\nu_E = E/\hbar$	An energy expressed as an ordinary frequency	Units are Hz; $\omega_E = E/\hbar = 2\pi\nu_E$ is an angular frequency.
$\Gamma = 1/T_2$	A decay rate in an exponential-decay model	Its equivalent energy is $\hbar\Gamma$, and its equivalent ordinary frequency is $\Gamma/(2\pi)$.
P	An orthogonal projector onto retained states	It is square and obeys $P^2 = P = P^\dagger$.
W	An isometry whose columns are retained orthonormal states	It is usually rectangular: $W^\dagger W = I$ and $WW^\dagger = P$.
X, Y, Z or τ^μ	Pauli operators on a specified two-state space	State explicitly whether that space is a physical doublet or an encoded cluster.
ϵ	A small ratio of interaction strength to excitation gap	This ratio controls the approximation used in the perturbative calculation. Estimating

Symbol	Meaning	What to keep separate
		device errors also requires the noise and control models.

Symbols such as c and α are reused in the source for unrelated dimensionless coefficients or protocol costs. Each is defined locally. Chapter 41 also uses u and K_4 as ordinary frequencies; the corresponding energies are $\hbar u$ and $\hbar K_4$.

A reliable unit check is to convert every energy to hertz before comparing scales, or leave every term in energy units. For example,

$$\frac{J_{\text{eff}}}{\hbar\Gamma} = 2\pi \left(\frac{J_{\text{eff}}}{\hbar} \right) T_2.$$

The quantity $(J_{\text{eff}}/\hbar)T_2$ counts interaction-frequency cycles during the coherence time. The factor 2π converts it to the energy ratio on the left.

PART I — FOUNDATIONS OF QUANTUM MECHANICS

A beam of atoms can produce two separate spots on a detector. This part develops the quantum description needed to predict that result and understand experiments with individual atoms.

Chapter 1 — Discrete outcomes in the Stern–Gerlach experiment

In the Stern–Gerlach experiment, a beam of silver atoms travels through a magnetic field that varies with position. The field deflects the atoms according to their magnetic properties. If those properties behaved like randomly oriented classical magnetic needles, the deflections would span a continuous range. The observed beam instead separates into two branches.

Otto Stern and Walther Gerlach observed this splitting in 1922. We will idealize it as a measurement with two possible outcomes, each identified by the branch an atom follows. Repeating the experiment with the same preparation gives a fraction of atoms in each branch. The quantum model predicts the probabilities governing those fractions.

Complex amplitudes and interference

Choose the magnet's measurement axis as the z axis and label its two outputs $z+$ and $z-$. We model the internal degree of freedom responsible for this choice of output in $\mathbb{C}^2$. This leaves the atom's position and its other internal degrees of freedom outside the model.

We begin with preparations represented by a single normalized vector, called pure states. Fix an orthonormal basis associated with the two z outputs. The vector's complex coordinates in this basis are called probability amplitudes. Their relation to observed frequencies is the physical rule we need to specify.

For complex conjugation and squared magnitude, we use the notation

$$z = a + ib,$$

with real a and b , $i^2 = -1$, and conjugate $z^* = a - ib$. The squared magnitude of this number z is

$$|z|^2 = z^* z = a^2 + b^2.$$

The squared magnitude is nonnegative; normalization will make the squared magnitudes of the two coordinates sum to one. These are both requirements for assigning probabilities to two exhaustive outcomes.

Write the amplitudes for the two outputs as α and β . In Dirac notation the state vector is written as a ket:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Here the ket is displayed as a column in the fixed orthonormal basis. Its entries are the amplitudes for $z+$ and $z-$, in that order. Normalization requires

$$|\alpha|^2 + |\beta|^2 = 1.$$

The Born rule assigns probability $|\alpha|^2$ to the first output and $|\beta|^2$ to the second. The state and the chosen measurement supply the amplitudes; the Born rule converts them to probabilities. Normalization makes these probabilities sum to one, as required for the ideal measurement in which every atom gives one of the two outcomes.

A normalized column is a representative of a pure physical state. Two columns differing only by a common complex factor of magnitude one describe the same state, because this factor cancels from every Born probability.

The complex coordinates contain information beyond the two probabilities for this particular magnet. In a measurement along a different axis, contributions from the coordinates can cancel when forming the amplitude for a single output. This cancellation is an example of interference. We will see it explicitly after introducing the second measurement basis.

Basis states and superposition

The internal angular momentum tracked by this two-state model is called spin. It is an internal property, distinct from the angular momentum of the atom's motion through the apparatus. A spin component is the angular momentum measured along a chosen axis. The magnet with measurement axis z distinguishes the two possible values of that component.

The basis vectors represent preparations that give the corresponding output with certainty:

$$|z+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |z-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

In this basis the general pure-state vector is

$$|\psi\rangle = \alpha|z+\rangle + \beta|z-\rangle.$$

This expansion is a superposition in the z basis. Its coefficients give the amplitudes for a z -basis measurement, whose probabilities we have just specified. Spin is being described by a vector in a complex state space; these coefficients are not two simultaneous classical directions of the atom.

To compute amplitudes for other measurements, we need inner products. The conjugate transpose of a ket is written as a bra:

$$\langle\psi| = (\alpha^* \quad \beta^*).$$

The product of a bra with a ket is the inner product $\langle\phi|\psi\rangle$. If the first vector is a normalized basis state for a measurement outcome and the second is the prepared state, this inner product is the amplitude for that outcome. Its squared magnitude is the Born probability. In particular, normalization becomes

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1.$$

With its standard inner product, this finite-dimensional complex vector space is a Hilbert space; completeness is automatic in finite dimension. The state vectors use its linear structure, and the measurement probabilities use its inner product.

Sequential measurements along perpendicular axes

An ideal Stern–Gerlach filter separates the two outputs of a spin-component measurement and blocks one of them. Keeping only atoms from the other output is called postselection.

We assume that each retained atom has the basis state associated with its output. This is the state-update rule for an ideal projective measurement with one-dimensional output subspaces. A repeated measurement along the same axis therefore gives the same outcome with certainty. Throughout the sequence below, we also assume that the spin state does not evolve between magnets.

Now rotate the measurement axis from z to the perpendicular x axis. The new magnet measures a different spin component. Label its outputs $x+$ and $x-$. In the two-state spin model, we choose their basis vectors as

$$|x+\rangle = \frac{|z+\rangle + |z-\rangle}{\sqrt{2}}, \quad |x-\rangle = \frac{|z+\rangle - |z-\rangle}{\sqrt{2}}.$$

This relation between the two bases is part of the physical model for perpendicular spin measurements. Both vectors have unit norm, and their inner product vanishes:

$$\langle x+|x-\rangle = \frac{1}{2}(\langle z+| + \langle z-|)(|z+\rangle - |z-\rangle) = \frac{1}{2}(1 - 1) = 0.$$

The vanishing inner product has a direct measurement meaning. For an atom prepared in $|x-\rangle$, the two z -basis terms contribute equal and opposite amplitudes to the same $x+$ outcome. They cancel, so that outcome has probability zero. Such cancellation is destructive interference.

The two terms retain the relative sign fixed by the preparation, with no record distinguishing the alternatives. They form a coherent superposition. We add their contributions to a single outcome's amplitude before taking its squared magnitude. By contrast, distinct recorded outcomes are mutually exclusive, so their probabilities add. Adding the two amplitudes for the separate outputs of a z measurement would not give the probability of detecting an atom in either output.

In step 1, pass a beam of atoms through a z -oriented magnet. Block the $z-$ output and retain the $z+$ output. Under the ideal measurement assumption, each survivor is prepared in

$$|\psi_1\rangle = |z+\rangle.$$

A second z -oriented magnet would send every survivor to its $z+$ output. The inner product of the prepared state with that output state is 1, so the probability is 1.

In step 2, instead send the survivors through a perpendicular magnet and retain only its $x+$ output. The amplitude for passing this filter, often called a transition amplitude, is

$$\langle x+ | z+ \rangle = \frac{1}{\sqrt{2}} (\langle z+ | + \langle z- |) | z+ \rangle = \frac{1}{\sqrt{2}}.$$

The Born rule gives the passing probability conditional on the preparation:

$$P(x+ | z+) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

The expected surviving fraction is one half. Each retained atom, initially prepared in $z+$, now has the normalized state

$$|\psi_2\rangle = |x+\rangle = \frac{|z+\rangle + |z-\rangle}{\sqrt{2}}.$$

The probability describes the expected fraction passing the filter. The normalized state describes an atom conditional on having passed. Its norm is one even though only half the incoming atoms are expected to survive.

In step 3, measure the z component of these survivors and count both outputs. The amplitudes are

$$\langle z+ | x+ \rangle = \frac{1}{\sqrt{2}}, \quad \langle z- | x+ \rangle = \frac{1}{\sqrt{2}},$$

so

$$P(z+ | x+) = P(z- | x+) = \frac{1}{2}.$$

After selection at the second magnet, the retained state gives a certain x result but two equally probable z results. The intervening filter has changed the preparation. This rules out a description in which the middle apparatus merely reads a pre-existing value while leaving the initially certain z value intact. The sequence alone does not rule out models in which measurement disturbs pre-existing values.

Starting from 100 atoms already prepared in $z+$, the expected count at the retained $x+$ output is 50. The expected counts at the two final z outputs are 25 each. These final counts are relative to the initial 100 atoms; the final probabilities of one half are conditional on passing the middle filter. Repeated runs fluctuate around the expected counts even when the ideal probabilities remain fixed.

Overall and relative phase

The two probabilities for a fixed magnet orientation generally do not specify the whole pure state. To see what information matters, first multiply both amplitudes by the same factor $e^{i\gamma}$, with γ real. This changes their common phase. For any normalized measurement ket, the probability remains

$$|\langle a|e^{i\gamma}\psi\rangle|^2 = |e^{i\gamma}|^2 |\langle a|\psi\rangle|^2 = |\langle a|\psi\rangle|^2.$$

Thus $|\psi\rangle$ and $e^{i\gamma}|\psi\rangle$ represent the same pure state. The common factor is an *overall phase*. Mathematically, a ray consists of all nonzero complex scalar multiples of a nonzero vector. Its normalized representatives differ only by an overall phase.

Changing one amplitude's phase while holding the other fixed can change a measurement probability. With the basis vectors fixed, the phase difference between two nonzero amplitudes is their *relative phase*. The previously defined states provide an example:

$$\frac{|z+\rangle + |z-\rangle}{\sqrt{2}} = |x+\rangle, \quad \frac{|z+\rangle - |z-\rangle}{\sqrt{2}} = |x-\rangle$$

Both states give equal probabilities for the two z -basis outcomes. Their second amplitudes differ by a minus sign, so the relative phases differ by half a turn. An x -basis measurement distinguishes them with certainty: the first state gives the plus outcome, and the second gives the minus outcome. In computing each overlap, the two contributions add or cancel according to this relative phase.

The Born rule and projection

The filters can now be described for any finite-dimensional pure-state system. Consider an ideal projective measurement with a normalized ket for each outcome. These kets $\{|a_j\rangle\}$ form an orthonormal basis: each has norm one, distinct kets are orthogonal, and together they span the state space. The outcomes therefore form a complete set for this measurement. For a normalized input ket, the Born rule gives the probability of outcome j as

$$P(j) = |\langle a_j | \psi \rangle|^2.$$

Born proposed the probabilistic interpretation of the wavefunction in 1926. To describe what a filter retains, we also need the component along the selected outcome ket. The orthogonal projector onto that one-dimensional subspace is

$$\Pi_j = |a_j\rangle\langle a_j|.$$

Acting on the input ket, this projector retains its component along the selected ket and removes the orthogonal component. The squared norm of the retained component is the probability of passing the filter. Equivalently,

$$P(j) = \langle \psi | \Pi_j | \psi \rangle.$$

These orthogonal projectors are self-adjoint and idempotent. Completeness of the outcome basis gives

$$\sum_j \Pi_j = I.$$

Multiplying this identity on the left by the input bra and on the right by its normalized ket shows that the probabilities sum to one.

Now condition on the observed outcome j , assuming it has nonzero probability. The ideal filter prepares the state represented by $|a_j\rangle$. Applying the projector and normalizing gives the same state, up to overall phase:

$$|\psi_j\rangle = \frac{\Pi_j |\psi\rangle}{\sqrt{\langle \psi | \Pi_j | \psi \rangle}}.$$

The denominator is the norm of the projected ket. A zero-probability outcome has no conditional state defined by this formula.

An outcome may instead correspond to a subspace of dimension greater than one. For a complete family of mutually orthogonal subspaces, use their orthogonal projectors in the same probability formula. The normalized-projection update then specifies an ideal measurement that preserves the component within the selected subspace; this assumption is part of the measurement model.

If a measurement occurs but its outcome is not retained, subsequent predictions must average over the possible conditional states with their outcome probabilities. In general, no single ket describes that ensemble. A density operator, introduced later, will represent this statistical mixture.

Operators and observables

A filter labels an output channel. To describe a measured quantity, we must also assign a numerical reading to each channel. Define an operator that multiplies each outcome component by its assigned reading. Its action on superpositions satisfies

$$A(c_1|u\rangle + c_2|v\rangle) = c_1A|u\rangle + c_2A|v\rangle.$$

In the orthonormal basis used here, the adjoint $A^\dagger$ is represented by the conjugate transpose. Assigning real readings to mutually orthogonal outcome subspaces gives an operator satisfying

$$A = A^\dagger,$$

This is the Hermitian condition. In the ideal projective model, such an operator represents an *observable*, a quantity measured with those numerical outcomes. The spectral theorem supplies its real eigenvalues and a complete orthonormal eigenbasis. For a ket in an outcome eigenspace,

$$A|a_j\rangle = a_j|a_j\rangle.$$

The normalized ket $|a_j\rangle$ is called an eigenstate of A . Preparing this state makes the reading a_j certain. If several basis kets have the same eigenvalue, that reading corresponds to their whole eigenspace. This observable description is the ideal projective measurement model used in this chapter; more general measurement procedures require a broader description.

For the two-output magnets, first assign the dimensionless readings plus one and minus one to their respective outputs. In the fixed z basis, the resulting operators are the Pauli matrices

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Both are Hermitian. Their eigenstates are exactly the states prepared by the corresponding filters:

$$\sigma_z|z\pm\rangle = \pm|z\pm\rangle, \quad \sigma_x|x\pm\rangle = \pm|x\pm\rangle.$$

The Pauli matrices carry no units. To express the angular momentum measured in an ideal spin-1/2 model, multiply by half the reduced Planck constant, denoted by $\hbar$. This universal constant sets the quantum scale of angular momentum. The physical component operators are

$$S_z = \frac{\hbar}{2}\sigma_z, \quad S_x = \frac{\hbar}{2}\sigma_x,$$

An S_z measurement therefore returns $+\hbar/2$ or $-\hbar/2$, in joule-seconds. The labels $z+$ and $z-$ denote the eigenstates for these respective readings.

The two magnet orientations do not admit a common eigenbasis. Their Pauli operators do not commute:

$$[\sigma_z, \sigma_x] \equiv \sigma_z\sigma_x - \sigma_x\sigma_z \neq 0.$$

The commutator $[A, B]$ compares the compositions of A and B , with B applied before A in the first product. For Hermitian operators, a common orthonormal eigenbasis exists exactly when they commute. For the particular pair σ_z and σ_x , the displayed eigenstates show more: there is no common eigenvector.

Consequently, no ket predicts both a certain z outcome and a certain x outcome. The intermediate filter in the three-magnet sequence replaces one such certainty with the other. A measurement includes this conditional state change; composing observable matrices alone does not describe the filtering procedure. Neither noncommutation nor that sequence by itself rules out every model with pre-existing values. The sequence rules out a readout that leaves the original certainty undisturbed.

Repeated measurements on identically prepared atoms yield a distribution of readings. Its mean, called the expectation value of A in the normalized state $|\psi\rangle$, is

$$\langle A \rangle_\psi = \langle \psi | A | \psi \rangle = \sum_j a_j P(j).$$

A finite sample average estimates this mean. The mean need not be an allowed outcome of a single measurement.

For $|x+\rangle$, a measurement of σ_z has readings $+1$ and -1 , each with probability one half. Its expectation is 0, although an individual run never reports zero.

Hermiticity also guarantees reality directly: if $q = \langle \psi | A | \psi \rangle$ and $A = A^\dagger$, then $q^* = q$.

Unitary time evolution

A prepared atom can evolve before reaching the next filter. Suppose the tracked system is closed over that interval: interactions with untracked degrees of freedom can be neglected, while any prescribed applied field is included in its dynamics. Quantum mechanics models this evolution by a unitary operator U :

$$U^\dagger U = U U^\dagger = I.$$

If $|\psi'\rangle = U|\psi\rangle$, then

$$\langle\psi'|\psi'\rangle = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle.$$

The evolved ket remains normalized, so it still gives a complete probability distribution for any subsequent ideal measurement. Unitarity preserves inner products between any two evolving states as well.

To determine the evolution, specify the system's energy operator H , called its Hamiltonian. It is Hermitian, with energy eigenvalues measured in joules. The law connecting this operator to the changing state is Schrödinger's equation:

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle.$$

With time t in seconds and $\hbar$ in joule-seconds, the differential operator on the left has the same energy units as the Hamiltonian:

$$[\hbar d/dt] = (\text{J s})(1/\text{s}) = \text{J} = [H].$$

When the Hamiltonian H is constant in time, the initial-value solution is

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, \quad U(t) = e^{-iHt/\hbar}.$$

The exponent contains the dimensionless combination $Ht/\hbar$. Since H is Hermitian, $U(t)$ is unitary.

Consider an energy splitting between the two z -basis states, described by

$$H = \frac{\hbar\omega}{2}\sigma_z,$$

Here ω is an angular frequency, measured in radians per second. The two basis states acquire phases at opposite rates because their energies have opposite signs in this choice of energy reference. Starting with the state $|x+\rangle$ gives

$$|\psi(t)\rangle = \frac{e^{-i\omega t/2}|z+\rangle + e^{+i\omega t/2}|z-\rangle}{\sqrt{2}}.$$

Both amplitudes change with time through their phase factors. Their squared magnitudes remain one half, so the z -basis probabilities stay fixed. To detect the changing relative phase, measure in the x basis. The overlap adds the two phase-dependent contributions, giving

$$P(x+; t) = |\langle x+ | \psi(t) \rangle|^2 = \cos^2(\omega t/2).$$

Thus the probability of the plus output oscillates as the delay before measurement changes. The unchanged probabilities in the energy basis conceal a changing pure state. A measurement sensitive to the relative phase reveals that evolution.

Physical implementation of the 1922 experiment

The original apparatus sent neutral silver atoms from a heated source through a narrow beam into an inhomogeneous magnetic field. The outgoing atoms left a deposit on a collecting plate. Their positions revealed two separated branches. The sequential filters used above are an idealized extension of that experiment.

The separation arises because an atom's internal magnetic state affects the force exerted by the field. In an approximation that treats each resolved branch as a classical trajectory, the dipole interaction energy gives

$$\mathbf{F} \approx \nabla(\boldsymbol{\mu} \cdot \mathbf{B}),$$

The magnetic moment $\boldsymbol{\mu}$ describes the atom's coupling to the magnetic field and has units of joules per tesla. The field $\mathbf{B}$ is measured in tesla. Their scalar product is an energy; its spatial gradient has units of newtons. The two internal components experience different forces and can therefore reach distinguishable positions.

Stern and Gerlach published their result in 1922, three years before electron spin was proposed. The spin account is a later explanation of the observed splitting.

In the modern account, ground-state silver has electronic angular-momentum quantum number one half. The two allowed projections along the magnet axis supply the basis states of our two-dimensional model. This reduction neglects nuclear-spin structure and other atomic degrees of freedom when describing the resolved branches. The atom's position must be included as well if we want to explain how those branches form.

The magnetic field directs the two internal basis components into different spatial paths. When both components are initially present coherently, the resulting joint state generally cannot be written as an internal ket times a spatial ket: each internal component is associated with its own path. This failure to factor is called entanglement and will be developed in the next chapter. An input containing just one of the internal basis components need not become entangled in this ideal model.

The separation of paths can be described by unitary evolution of the internal and spatial degrees of freedom together. A blocker then transmits one path, or a detector registers an arrival. Conditional on a selected output, the ideal spin-state update is the projection described earlier. This is an operational model of preparation and readout; it does not derive the occurrence of an individual detector outcome from unitary evolution.

Real beams have finite width, and real devices can lose atoms or misidentify an output. Field misalignment changes which component is measured. Such effects require changes to the preparation, evolution, or detector model before its predictions are compared with counts. The Born rule remains the probability rule, but the ideal two-channel formulas alone need not describe the imperfect apparatus.

The two-component ket therefore captures a selected part of the atom's behavior. The filters already illustrate ways to prepare and read that part. To use it for quantum computation, one must also implement controlled operations and keep unwanted interactions sufficiently small over the required sequence. Those physical capabilities are additional information beyond the dimension of the state space.

Common conceptual errors

- For an outcome amplitude computed from a normalized state and a normalized measurement ket, use the squared magnitude $|c|^2 = c^*c$. Squaring the complex amplitude itself can give a negative or nonreal number.
- The column $(\alpha, \beta)^T$ gives coordinates in the fixed z basis. Choosing another coordinate basis generally changes the entries without changing the physical state. Rotating a measuring magnet changes the measurement; the state can still be expressed in the original basis.
- Equal probabilities in one basis do not establish equality of pure states. The two x-basis states have equal z-basis probabilities and are nevertheless distinguished by an x-oriented magnet. Overall phase is the freedom that leaves every measurement probability unchanged.
- An expectation value is a distribution's mean, estimated by repeated measurements. It need not be one of the values an individual measurement can report.
- In the ideal filter model, postselection conditions on an output of nonzero probability and normalizes the corresponding projection. A unitary operator on the internal state alone cannot perform this selective removal. Unitary separation of the spatial paths is an earlier, distinct part of the procedure.
- A Hermitian observable specifies the readings and outcome subspaces of an ideal projective measurement. A unitary operator specifies closed-system evolution. The Pauli matrices satisfy both algebraic conditions, so an operator's physical role must be stated along with its matrix.
- A numerical simulation can propagate the specified ket and evaluate Born probabilities for the chosen measurement. Computing these predictions does not reproduce the physical atom or its interaction with a detector.
- A two-dimensional state space does not specify how a device is operated. Preparation, control, readout, and suppression of unwanted interactions each require a physical implementation.

Self-assessment

- **What specifies a pure-state experiment?** Give a normalized initial ket, with overall phase irrelevant. Specify the Hamiltonian for each closed-system evolution interval. For each ideal projective measurement, give the outcome projectors and their numerical readings. State which outcomes, if any, are retained for subsequent steps.
- **How is an outcome probability computed?** For outcome j in an orthonormal measurement basis, the amplitude is $\langle a_j | \psi \rangle$. For the normalized input state, its squared magnitude gives $P(j) = |\langle a_j | \psi \rangle|^2$.

- **Why normalize in the z basis?** The condition $|\alpha|^2 + |\beta|^2 = 1$ makes the probabilities of the two complete z -basis outcomes sum to one. It describes the ideal two-outcome measurement, without an additional loss channel.
- **Why require a Hermitian observable?** An ideal projective observable assigns real readings to mutually orthogonal, complete outcome subspaces. An operator A constructed from those spectral data is Hermitian. A non-Hermitian operator cannot have both a real spectrum and a complete orthonormal eigenbasis, so it cannot represent an observable in this model.
- **How does evolution differ from postselection?** Closed-system unitary evolution preserves all inner products. An ideal filter instead selects an output. Conditional on that output having nonzero probability, the normalized projection describes the atoms retained for the next operation.
- **What follows the sequence $z+$, then $x+$, then z ?** Prepare the first state and retain the plus output of the perpendicular filter. Among those survivors, the final outcomes $z+$ and $z-$ each have probability one half. If the intermediate x measurement left the prepared $z+$ state unchanged, the final result would instead be certainly $z+$. The sequence demonstrates the change in preparation caused by this filter.

We can now follow a pure state through closed-system evolution and predict the outcomes of ideal filters, including the conditional state passed to the next step. The next chapter applies this framework to two atoms. Some joint pure states factor into one ket for each atom; others do not. Understanding the latter requires keeping the joint amplitudes together.

Sources

- [R037] M. Born, “Zur Quantenmechanik der Stoßvorgänge,” *Zeitschrift für Physik* 37, 863–867 (1926). DOI: [10.1007/BF01397477](https://doi.org/10.1007/BF01397477).
 - [R038] W. Gerlach and O. Stern, “Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld,” *Zeitschrift für Physik* 9, 349–352 (1922). DOI: [10.1007/BF01326983](https://doi.org/10.1007/BF01326983).
 - [R039] B. Friedrich and D. Herschbach, “Stern and Gerlach: How a Bad Cigar Helped Reorient Atomic Physics,” *Physics Today* 56(12), 53–59 (2003). DOI: [10.1063/1.1650229](https://doi.org/10.1063/1.1650229).
 - [R040] P. A. M. Dirac, *The Principles of Quantum Mechanics*, 4th ed., Oxford University Press (1958; reissued 1981). ISBN: 978-0-19-852011-5.
 - [R041] E. Schrödinger, “Quantisierung als Eigenwertproblem (Vierte Mitteilung),” *Annalen der Physik* 386, 109–139 (1926). DOI: [10.1002/andp.19263861802](https://doi.org/10.1002/andp.19263861802).
 - [R004] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information: 10th Anniversary Edition*, Cambridge University Press (2010). DOI: [10.1017/CBO9780511976667](https://doi.org/10.1017/CBO9780511976667).
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Chapter 2 — Nonfactorizable states of two two-level systems

For two classical coins, each coin can land heads or tails, giving four joint outcomes: HH, HT, TH, and TT.

For two fair coins, each joint outcome has probability $1/4$. For loaded coins, the four probabilities can have any nonnegative values whose sum is one.

The quantum analogue uses two silver atoms of the kind Chapter 1 analyzed, each with two possible readings. The pair again carries four joint labels. A quantum state assigns one complex amplitude to each label. The new feature is that the four joint amplitudes need not split into one pair belonging to the first atom and another pair belonging to the second.

No crystal enters yet. The system is two atoms, four joint labels, and four amplitudes that need not factor.

Joint basis and tensor-product dimension

Call the atoms A and B . Each gives the two answers met in Chapter 1, relabeled 0 and 1 in place of z_+ and z_- . A ket names a quantum state or a basis label. The pair has four joint basis kets:

$$|00\rangle, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle.$$

The first digit refers to A , and the second refers to B . Thus, $|01\rangle$ denotes the outcome in which A has value 0 and B has value 1.

A state of the pair lists four complex amplitudes, one per joint label, with squared magnitudes summing to one. Relative phases between these amplitudes shift the probabilities of later measurements, the same way the minus sign between z_+ and z_- decided the transverse-magnet statistics in Chapter 1.

The tensor product combines states of two systems. When A is in $\alpha|0\rangle + \beta|1\rangle$ and B is in $|0\rangle$, the joint state is

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle = \alpha|00\rangle + \beta|10\rangle.$$

The symbol $\otimes$ is the tensor product. Acting on states, it concatenates the labels of the two systems and multiplies their amplitudes; that action is the definition used here. The same symbol also combines operators.

If A has d_A independent basis states and B has d_B , then the pair has $d_A d_B$ joint basis states. Two two-state systems therefore give the four labels above.

A three-state system joined to a four-state system gives twelve joint basis states, since dimensions multiply.

An operator is a linear transformation of states. An operator that touches only A appears on the joint system as $M_A \otimes I_B$, where M_A is the original operator and the identity I_B leaves B unchanged. The joint operator $M_A \otimes I_B$ therefore acts on A while leaving B unchanged.

Each joint label names a pair of answers, one from each atom, not an additional state of a single atom.

Product states

Some joint states split into two independent single-atom states. For example,

$$|0\rangle_A \otimes |0\rangle_B = |00\rangle.$$

Here A answers 0 and B answers 0 with certainty, and specifying one atom's state says nothing new about the other.

Let both atoms instead occupy the transverse-basis state met in Chapter 1,

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Their joint state is then

$$|+\rangle_A \otimes |+\rangle_B = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}.$$

All four joint amplitudes are equal, so measuring both atoms returns each joint outcome with probability one quarter. The two readings are independent: each pair probability equals the product of the two single-atom probabilities.

A joint state of the form of one ket for A tensored with one ket for B is called a product state. The test is factorization into two local kets.

Calling a pair a product state therefore asserts that the joint amplitudes factor. Uncorrelated outcomes in one measurement basis are weaker evidence: the test is factorization of the joint amplitudes.

Entangled pure states

Consider the state

$$|\Phi^+\rangle_{AB} = \frac{|0\rangle_A |0\rangle_B + |1\rangle_A |1\rangle_B}{\sqrt{2}} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

This is a Bell state. Its two joint basis states are orthogonal, so the superposition is normalized:

$$\langle \Phi^+ | \Phi^+ \rangle = \frac{1+1}{2} = 1.$$

Measuring both atoms in the $\{|0\rangle, |1\rangle\}$ basis returns only 00 or 11, each with probability $1/2$. The two answers always agree.

Suppose, to test factorization, that the state were a product,

$$|\Phi^+\rangle = (a|0\rangle + b|1\rangle)_A \otimes (c|0\rangle + d|1\rangle)_B.$$

Expanding the product gives amplitudes ac , ad , bc , and bd for $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$. Matching the Bell state would require

$$ac = \frac{1}{\sqrt{2}}, \quad ad = 0, \quad bc = 0, \quad bd = \frac{1}{\sqrt{2}}.$$

The first equation forces a and c to be nonzero. Then $ad = 0$ forces $d = 0$, which contradicts bd equaling $1/\sqrt{2}$. No choice of a, b, c, d satisfies all four equations at once.

A pure joint state that resists factorization is called entangled. Entanglement describes the joint amplitudes: the four numbers cannot be split into independent lists for the two atoms. That structural fact calls for no signal passing between the atoms.

The Schmidt decomposition generalizes this four-equation test. Every pure state of a pair can be written as

$$|\Psi\rangle_{AB} = \sum_{k=1}^r s_k |u_k\rangle_A |v_k\rangle_B,$$

The Schmidt coefficients s_k are nonnegative numbers with $\sum_k s_k^2 = 1$, and the local states $\{|u_k\rangle_A\}$ and $\{|v_k\rangle_B\}$ form orthonormal lists: distinct entries are orthogonal and each entry has unit norm.

The count r of nonzero Schmidt coefficients is the Schmidt rank. Rank 1 marks a product state; higher rank marks an entangled one.

For the Bell pair, $r = 2$ and $s_1 = s_2 = 1/\sqrt{2}$.

Dropping the orthonormality requirement lets many states appear as sums of product terms. Those longer sums carry no factorization claim; the Schmidt rank remains the test.

Density operators for pure and mixed states

A ket describes the pair completely when the preparation is known to be exactly $|\Phi^+\rangle$. Often the preparation itself involves classical uncertainty that no single ket captures.

A fair coin may decide between preparing $|00\rangle$ or $|11\rangle$. Or the experimenter may hold only atom A while its partner sits in another laboratory. A single ket covers neither situation.

The density operator fills this gap. In a chosen basis it is a square matrix of complex numbers. For a pure state $|\psi\rangle$ it is the outer product $|\psi\rangle\langle\psi|$, the operator formed by pairing the ket with its bra. For the Bell pair,

$$\rho_{AB} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2} (|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|).$$

Once a basis is fixed the density operator is often called a density matrix. One mathematical object then covers three situations: a definite ket, a classical distribution over kets, and the description of one atom whose partner is ignored.

Three requirements single out the valid density operators ρ :

- It is Hermitian, meaning that it equals its adjoint: $\rho^\dagger = \rho$.
- It is positive semidefinite, meaning that $\langle\chi|\rho|\chi\rangle \geq 0$ for every $|\chi\rangle$.
- It has unit trace: $\text{Tr } \rho = 1$. The trace is the sum of the diagonal matrix entries in any basis.

A preparation that emits $|\psi_k\rangle$ with classical probability p_k corresponds to

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|.$$

A yes-or-no outcome m is represented by a projector P_m , an operator with $P_m^2 = P_m$. The probability of the outcome m is

$$\text{Pr}(m) = \text{Tr}(\rho P_m).$$

A density operator built from a single ket has rank 1 and satisfies $\rho^2 = \rho$, equivalently $\text{Tr}(\rho^2) = 1$. The Bell pair, taken as a complete two-atom system, obeys

$$\text{Tr}(\rho_{AB}^2) = 1.$$

The density operator is the most compact table that keeps every probability prediction when a ket falls short. Each requirement earns its place: dropping Hermiticity admits complex probabilities, dropping positivity admits negative ones, and dropping unit trace breaks normalization.

Partial trace and reduced states

Suppose only A is accessible. Every prediction for measurements on A alone follows from a reduced density operator that sums over the basis states of B :

$$\rho_A = \text{Tr}_B(\rho_{AB}) = \sum_{b=0}^1 {}_B\langle b | \rho_{AB} | b \rangle_B.$$

This operation is the partial trace over B . It describes the statistics of the first atom when the outcomes of B are discarded, without any physical action on B .

For basis operators, the partial trace satisfies

$$\text{Tr}_B(|ab\rangle\langle a'b'|) = \langle b'|b\rangle |a\rangle\langle a'| = \delta_{b'b} |a\rangle\langle a'|,$$

where the Kronecker delta $\delta_{b'b}$ equals 1 when $b' = b$ and 0 otherwise. The cross terms in the Bell-state density operator contain $|0\rangle_B\langle 1|$ or $|1\rangle_B\langle 0|$. Because these basis states are orthogonal, the cross terms vanish under the partial trace. The remaining terms give

$$\rho_A = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I_A}{2},$$

where I_A is the identity operator on A , represented by the two-by-two matrix with ones on the diagonal and zeros elsewhere.

The pair as a whole is pure, while either atom alone is maximally mixed for a two-state system:

$$\text{Tr}(\rho_A^2) = \text{Tr}\left(\frac{I_A}{4}\right) = \frac{1}{2}.$$

Global purity therefore coexists with local mixedness. The local mixture records no forgotten local ket: no ket of the subsystem alone reproduces these statistics.

Every measurement on A alone is predicted by ρ_A . An observable touching only A extends to the pair as $M_A \otimes I_B$, with statistical average

$$\langle M_A \rangle = \text{Tr}[\rho_{AB}(M_A \otimes I_B)].$$

The reduced density operator ρ_A is the unique operator that reproduces every such expectation value:

$$\text{Tr}(\rho_A M_A) = \text{Tr}[\rho_{AB}(M_A \otimes I_B)].$$

On a product operator the partial trace collapses to

$$\text{Tr}_B(X_A \otimes Y_B) = X_A \text{Tr}(Y_B).$$

Linearity extends this rule from product operators to every joint operator.

An operator fitted to one local measurement outcome generally misses the others. The partial trace is the single operator consistent with the joint state across all local measurements.

Relative phase accessible only through joint measurements

Consider the Bell state with an additional relative phase:

$$|\Phi_\theta\rangle = \frac{|00\rangle + e^{i\theta}|11\rangle}{\sqrt{2}},$$

Here θ is the relative phase between the two joint terms. The joint density operator carries cross terms $e^{-i\theta}|00\rangle\langle 11|$ and $e^{i\theta}|11\rangle\langle 00|$, and both vanish under Tr_B . The reduced state is therefore $\rho_A = I/2$ for every θ .

Joint measurements on the pair resolve this phase; measurements on A alone cannot. The phase information stays in the correlations between the atoms.

Relative phases of a pair therefore need not surface in either subsystem.

Correlation is not sufficient to establish entanglement

Consider the density operator

$$\rho_{cc} = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|.$$

Measuring both atoms in the 0/1 basis returns matching answers every time, exactly as $|\Phi^+\rangle$ does. Yet this state is a classical fifty-fifty mixture of two product states.

A mixed bipartite state is separable if it can be written as

$$\rho_{AB} = \sum_k p_k \rho_A^{(k)} \otimes \rho_B^{(k)}, \quad p_k \geq 0, \quad \sum_k p_k = 1.$$

The superscript (k) labels a possible component and is not an exponent. The state ρ_{cc} is explicitly separable. A mixed state for which no such decomposition exists is entangled.

Matching outcomes in one basis therefore cannot certify entanglement. The Bell pair and ρ_{cc} agree across every 0/1 measurement and part ways in a transverse basis.

Sharper tests rotate through several measurement settings. Bell inequalities ask whether the correlations across those settings admit an explanation in terms of shared pre-existing randomness [Ro42]; [Ro43]; [Ro44]. [Theory] Some entangled mixed states do not violate a given Bell inequality. Entanglement and Bell nonlocality are therefore distinct properties [Ro44].

Purity as a measure of mixedness

Purity, the dimensionless number $\text{Tr}(\rho^2)$, runs for a d -state system from $1/d$ for the maximally mixed state I/d up to 1 for a pure state.

For a mixed state in a finite-dimensional state space,

$$\text{Tr}(\rho^2) < 1.$$

A mixed density operator generally admits many ensemble decompositions into prepared kets. For one two-state atom,

$$\frac{I}{2} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -|,$$

where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. The density operator, rather than any preferred ensemble interpretation, contains the operational measurement predictions.

Measuring ρ leaves open which local pure states were actually prepared.

Populations and coherences

Consider one atom in the state

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad \rho_+ = |+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

The matrix uses the ordered basis $(|0\rangle, |1\rangle)$. Its diagonal entries, the populations, are the probabilities of reading 0 and 1 in this basis.

Its off-diagonal entries, the coherences, carry the relative phase that lets the two basis contributions interfere.

Contact with an uncontrolled subsystem can shrink the coherences while leaving the populations fixed. The calculation below tracks that process.

A dephasing channel

Consider noise that preserves populations and shrinks only coherences. The parameter λ , with $0 \leq \lambda \leq 1$, records the surviving fraction of each coherence. One model of that noise is

$$\mathcal{E}_\lambda(\rho) := K_0 \rho K_0^\dagger + K_1 \rho K_1^\dagger,$$

with

$$K_0 = \sqrt{\frac{1+\lambda}{2}} I, \quad K_1 = \sqrt{\frac{1-\lambda}{2}} Z, \quad 0 \leq \lambda \leq 1.$$

Here I is the 2×2 identity and

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is the phase-flip operator met in Chapter 1. The Kraus operators K_0 and K_1 satisfy

$$K_0^\dagger K_0 + K_1^\dagger K_1 = I,$$

so total probability stays normalized. A trace-preserving map built by sandwiching ρ between Kraus operators and summing the pieces is a quantum channel [Ro45].

Validity as a channel means complete positivity plus trace preservation. Complete positivity keeps the evolution legal when the system forms part of a larger entangled state; trace preservation keeps total probability at one when unobserved degrees of freedom drop out. Evaluating the channel means forming each sandwich term and adding them.

This channel describes pure dephasing: populations stay fixed while phase coherence shrinks.

For a general input state

$$\rho = \begin{pmatrix} a & c \\ c^* & b \end{pmatrix}, \quad a + b = 1,$$

with real populations a and b and complex coherence c , direct multiplication gives

$$\mathcal{E}_\lambda(\rho) = \begin{pmatrix} a & \lambda c \\ \lambda c^* & b \end{pmatrix}.$$

Populations pass through unchanged while each coherence picks up the factor λ . On $|+\rangle$ the channel returns

$$\rho'_+ = \frac{1}{2} \begin{pmatrix} 1 & \lambda \\ \lambda & 1 \end{pmatrix}.$$

The purity of the output is

$$\text{Tr}[(\rho'_+)^2] = \frac{1 + \lambda^2}{2}.$$

At $\lambda = 1$ the channel is the identity and the output stays pure. At $\lambda = 0$ the coherences vanish and $\rho'_+ = I/2$. At the intermediate value $\lambda = 1/2$,

$$\rho'_+ = \begin{pmatrix} 1/2 & 1/4 \\ 1/4 & 1/2 \end{pmatrix}, \quad \text{Tr}[(\rho'_+)^2] = \frac{5}{8}.$$

Energy relaxation acts on populations instead: it moves population between levels, usually by exchanging energy, while coherences shrink only as a consequence. A long-lived excited population therefore says nothing by itself about surviving phase coherence.

Environmental transfer of phase information

A microscopic model shows where the coherence goes. Let the environment start in $|e\rangle$, and let the joint evolution act as

$$|0\rangle|e\rangle \mapsto |0\rangle|e_0\rangle, \quad |1\rangle|e\rangle \mapsto |1\rangle|e_1\rangle.$$

An initial product $|+\rangle|e\rangle$ then becomes

$$|\Psi\rangle_{SE} = \frac{|0\rangle|e_0\rangle + |1\rangle|e_1\rangle}{\sqrt{2}},$$

where S denotes the system and E denotes the environment. Taking the partial trace over E gives

$$\rho_S = \frac{1}{2} \begin{pmatrix} 1 & \langle e_1|e_0\rangle \\ \langle e_0|e_1\rangle & 1 \end{pmatrix}.$$

The surviving system coherence equals the overlap of the two environment states. Identical environment states $|e_0\rangle$ and $|e_1\rangle$ leave the system basis states indistinguishable, so the coherence survives intact. Orthogonal environment states keep a perfectly distinguishable record of which basis state occurred, so the local coherence vanishes.

Decoherence names that transfer: uncontrolled interactions move phase information into unmonitored environmental degrees of freedom. It is a dynamical process acting on states, not a third state category alongside pure and mixed states.

Decoherence often leaves a mixed reduced state, but classically randomized preparations produce mixed states with no environment involved. A mixed state therefore does not by itself diagnose decoherence.

[Theory] Environment-induced entanglement followed by partial tracing provides the standard open-system account of decoherence [Ro46]; [Ro47]. The combined system-and-environment state may remain pure and evolve unitarily throughout the process.

This account needs no collapse postulate: system and environment evolve unitarily throughout, and the entanglement between them suppresses interference seen in the reduced system alone.

That explains the dynamics of the reduced state. Wider questions about the interpretation of measurement remain open and are not settled by this calculation.

Experimental state reconstruction and limitations

Theory writes down ρ directly. A laboratory produces detector clicks, voltages, fluorescence counts, and a documented preparation procedure.

Inferring the density operator from that raw data takes measurements in several bases plus statistical analysis. That reconstruction procedure is quantum-state tomography, and its output is an estimate conditional on detector calibration and on a model of the apparatus.

Repeatability of the preparation is likewise an experimental assumption. Slow drift lets successive trials differ.

Leakage moves population into states outside the assumed two-dimensional space and invalidates that truncation. Selection rules and detector inefficiency further bias the apparent ensemble.

The density-operator formalism still applies, but the choice of state space and noise model needs experimental justification in each setup.

[Experiment] Bell tests using separated solid-state electron spins have observed correlations incompatible with a local-realist model while closing major detection and locality loopholes under the stated experimental assumptions [Ro48]. This constitutes substantially stronger evidence than observing matching 0/1 outcomes. It does not imply that every correlated density operator is entangled.

Laboratories infer decoherence from fading interference or decaying off-diagonal coherence, summarized by fitted time constants. The parameter λ above belongs to the channel model; it is not a universal constant of nature.

Its value shifts with elapsed time, pulse sequence, temperature, noise spectrum, and correlations with previous runs. Relaxation and dephasing measurements replace this single parameter with the distinct times T_1 , T_2 , and T_2^* .

Common conceptual errors

- “Not a product state” and “mixed state” classify different things. The Bell state is a pure entangled state of a pair. The single-system state $I/2$ is mixed, and with no partner specified there is no pair on which entanglement could be defined.
- Matching outcomes in one basis leave entanglement undecided. The separable state ρ_{cc} shows perfect 0/1 correlation. The test is whether the full density operator admits a separable decomposition; one matching histogram is insufficient.
- A mixed reduced state can arise with no imprecise preparation. Either subsystem of a pure Bell pair is mixed, and ignoring the partner is enough to produce that mixture.
- Some mixed states involve no decoherence. Preparing $|0\rangle$ or $|1\rangle$ by a fair coin gives $I/2$ with no environment involved.
- Decoherence in the open-system description runs on unitary system-environment evolution whose entanglement suppresses local interference [Ro46]; [Ro47]. This account explains reduced-state dynamics but does not, by itself, resolve every interpretation of quantum measurement.
- Dephasing and relaxation act differently. Pure dephasing holds populations fixed while shrinking off-diagonal terms. Relaxation moves population, usually by exchanging energy. Real channels combine both effects.
- An ensemble decomposition of a mixed state is ambiguous. The operator $I/2$ reads equally as a 50–50 mixture of $|0\rangle$ and $|1\rangle$ or of $|+\rangle$ and $|-\rangle$, so the operator alone cannot identify which preparation occurred.
- A computer simulation of a Bell pair evaluates the mathematics of entanglement on hardware that contains no entangled atoms.

Concept checks

- **Problem:** Determine the dimension of a composite system whose subsystems have dimensions 3 and 4.

Solution: $3 \times 4 = 12$. Tensor-product dimensions multiply.

- **Problem:** Demonstrate that $|\Phi^+\rangle$ is not a product state.

Solution: Matching the four amplitudes would require $ac = 1/\sqrt{2}$, $ad = 0$, $bc = 0$, and $bd = 1/\sqrt{2}$. The first condition implies $a, c \neq 0$, so $d = 0$, after which bd cannot equal $1/\sqrt{2}$.

- **Problem:** Identify the error in interpreting matching 0/1 outcomes as sufficient evidence of entanglement.

Solution: The separable mixture ρ_{cc} produces the same matching bars. The test for entanglement is the absence of any separable decomposition of the full state; correlation in one plot cannot establish that.

- **Problem:** Explain why tracing out one subsystem of $|\Phi^+\rangle$ removes the cross terms.

Solution: Those terms contain orthogonal partner-state factors, so the partial trace produces $\langle 1|0\rangle = 0$.

- **Problem:** Show that applying $\mathcal{E}_{1/2}$ to $|+\rangle$ gives a state with purity $5/8$.

Solution: The output is $\begin{pmatrix} 1/2 & 1/4 \\ 1/4 & 1/2 \end{pmatrix}$, and $\text{Tr}[(\rho'_+)^2] = (1 + (1/2)^2)/2 = 5/8$.

- **Problem:** Identify the error in describing every mixed state as decohered.

Solution: A coin-flip preparation of $|0\rangle$ or $|1\rangle$ already produces $I/2$, with no environment involved.

The chapter leaves a working bipartite toolkit: joint states with factorizable or entangled amplitudes, the reduced state that predicts single-subsystem measurements, and the channel picture in which relative-phase information drains into unobserved degrees of freedom.

Sources

- [R042] A. Einstein, B. Podolsky, and N. Rosen, “Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?”, *Physical Review* 47, 777–780 (1935). DOI: [10.1103/PhysRev.47.777](https://doi.org/10.1103/PhysRev.47.777).
- [R043] J. S. Bell, “On the Einstein Podolsky Rosen Paradox,” *Physics Physique Fizika* 1, 195–200 (1964). DOI: [10.1103/PhysicsPhysiqueFizika.1.195](https://doi.org/10.1103/PhysicsPhysiqueFizika.1.195).
- [R044] R. F. Werner, “Quantum states with Einstein-Podolsky-Rosen correlations admitting a hidden-variable model,” *Physical Review A* 40, 4277–4281 (1989). DOI: [10.1103/PhysRevA.40.4277](https://doi.org/10.1103/PhysRevA.40.4277).
- [R045] K. Kraus, “General state changes in quantum theory,” *Annals of Physics* 64, 311–335 (1971). DOI: [10.1016/0003-4916\(71\)90108-4](https://doi.org/10.1016/0003-4916(71)90108-4).
- [R046] W. H. Zurek, “Decoherence, einselection, and the quantum origins of the classical,” *Reviews of Modern Physics* 75, 715–775 (2003). DOI: [10.1103/RevModPhys.75.715](https://doi.org/10.1103/RevModPhys.75.715); arXiv: [quant-ph/0105127](https://arxiv.org/abs/quant-ph/0105127).
- [R047] M. Schlosshauer, “Decoherence, the measurement problem, and interpretations of quantum mechanics,” *Reviews of Modern Physics* 76, 1267–1305 (2005). DOI: [10.1103/RevModPhys.76.1267](https://doi.org/10.1103/RevModPhys.76.1267); arXiv: [quant-ph/0312059](https://arxiv.org/abs/quant-ph/0312059).

- [R048] B. Hensen et al., “Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres,” Nature 526, 682–686 (2015). DOI: [10.1038/nature15759](https://doi.org/10.1038/nature15759); arXiv: [1508.05949](https://arxiv.org/abs/1508.05949).
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PART II — QUBITS AND QUANTUM INFORMATION

How can a controllable two-level system store quantum information? These chapters follow a physical qubit through an experiment and examine what limits its reliability.

Chapter 3 — Preparation, control, and measurement of a two-level quantum system

A quantum-control experiment requires three distinct operations: preparation, unitary control, and measurement.

Preparation produces a known input state. Control applies a calibrated, time-dependent Hamiltonian to that state; the resulting transformation is unitary, represented by a matrix U satisfying $U^\dagger U = I$. Measurement couples the final state to a detector and assigns a classical result to the detector output.

Controlling such a system requires spectral selectivity: the drive addresses the selected transition while leaving nearby transitions largely untouched. That selectivity requirement follows from the level structure and the drive bandwidth, developed in the pulse section below.

Every such experiment contends with two failure modes. A drive can populate a third level outside the selected pair, a process called leakage. A detector can also fire on an event unrelated to the prepared state, producing an incorrect assignment.

State preparation, phase control, and measurement

This chapter keeps the state labels of Chapter 1. The system is a spin-1/2 particle in a magnetic field pointing along the z -axis.

The spin state with Pauli- Z eigenvalue $+1$ is denoted by $|0\rangle$, and the state with eigenvalue -1 is denoted by $|1\rangle$.

Some laboratories swap the two labels. The swap is a convention that leaves all predictions unchanged.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Consider a phase-sensitive experiment consisting of five operations. First, prepare $|0\rangle$. Second, rotate the state through $+\pi/2$ about the y -axis. Third, allow a phase ϕ to accumulate; this operation is a rotation about the z -axis. Fourth, rotate the state through $-\pi/2$ about the y -axis. Finally, measure in the computational basis $\{|0\rangle, |1\rangle\}$.

A rotation through an angle ϑ about axis j , where j is x , y , or z , is represented by

$$R_j(\vartheta) = \exp\left(-\frac{i\vartheta\sigma_j}{2}\right).$$

Here $i^2 = -1$, ϑ is an angle measured in radians, and σ_j is the corresponding Pauli matrix defined in Chapter 1. The matrices $R_j(\vartheta)$ are unitary rotation operators. For the two axes required in this experiment,

$$R_y(\vartheta) = \begin{pmatrix} \cos(\vartheta/2) & -\sin(\vartheta/2) \\ \sin(\vartheta/2) & \cos(\vartheta/2) \end{pmatrix}, \quad R_z(\phi) = \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix}.$$

The half-angles in these matrices are characteristic of spin-1/2 rotations: a full turn of the drive advances the state by half that angle. Applying the first pulse gives

$$R_y(\pi/2)|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

The subsequent phase rotation gives

$$R_z(\phi)R_y(\pi/2)|0\rangle = \frac{e^{-i\phi/2}|0\rangle + e^{i\phi/2}|1\rangle}{\sqrt{2}}.$$

Measuring in the Z basis at this stage would return each outcome with probability 1/2 regardless of the accumulated phase. The final pulse translates that hidden relative phase into a population difference a Z -basis measurement can resolve:

$$|\psi_f\rangle := R_y(-\pi/2)R_z(\phi)R_y(\pi/2)|0\rangle = \cos(\phi/2)|0\rangle + i\sin(\phi/2)|1\rangle.$$

The Born rule assigns each basis outcome the squared magnitude of its amplitude, giving

$$P(0) = \cos^2(\phi/2), \quad P(1) = \sin^2(\phi/2), \quad P(0) + P(1) = 1.$$

The final pulse acts as an analyzer that makes phase visible as population. At $\phi = \pi/3$ the predicted probabilities are $P(0) = 3/4$ and $P(1) = 1/4$.

Over 100 independent repetitions about 75 zeros and 25 ones are typical, with statistical fluctuations around those averages.

The standard deviation of the number of zero outcomes is

$$\sqrt{100 \cdot (3/4) \cdot (1/4)} \approx 4.3.$$

The five-step sequence is an interferometer. The first pulse splits the state into two amplitudes, the waiting period shifts their relative phase, and the final pulse recombines them.

The final measurement records the resulting interference. Without the closing rotation the relative phase stays in the state yet leaves the Z -basis statistics at fifty-fifty.

[Experiment] Resonant control producing coherent oscillations of a single electron spin, and single-shot conversion of a spin state into a detectable charge signal, have both been demonstrated in semiconductor quantum dots [Ro52]; [Ro53]. The preceding calculation describes an ideal system. The cited experiments provide concrete physical implementations of control and readout.

Parameterization of two-level states

Every pure state in this two-dimensional Hilbert space can be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where the complex amplitudes α and β satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$. A computational-basis measurement returns 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$.

A shared global phase has no observable effect: $|\psi\rangle$ and $e^{i\chi}|\psi\rangle$ describe the same pure state for any real χ . The relative phase between α and β is the quantity the five-step interferometer converts into populations.

Normalization removes one real parameter from the four contained in two complex amplitudes, and the unobservable global phase removes a second. The two survivors serve as the polar angle θ , with $0 \leq \theta \leq \pi$, and the azimuthal angle φ , with $0 \leq \varphi < 2\pi$:

$$|\psi(\theta, \varphi)\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi} \sin(\theta/2)|1\rangle.$$

These two angles locate a point on a sphere. The link to measurable averages runs through the Pauli operators,

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These matrices repeat $\sigma_x, \sigma_y, \sigma_z$ from Chapter 1, joined by the identity I that leaves every state unchanged. For a state $|\psi\rangle$ the expectation value of A , the statistical average over repeated measurements, is $\langle A \rangle = \langle \psi | A | \psi \rangle$. The three Pauli averages form the vector

$$\mathbf{r} = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle).$$

Substitution of the two-angle state gives

$$\mathbf{r} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta).$$

Its length is one by $\sin^2 \theta + \cos^2 \theta = 1$, so every pure two-level state sits on the surface of a unit sphere.

The north pole is $|0\rangle$ and the south pole is $|1\rangle$.

The $+x$ axis carries $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and the $+y$ axis carries $(|0\rangle + i|1\rangle)/\sqrt{2}$.

This sphere is the Bloch sphere. A density operator ρ , covering both pure and mixed two-level states, decomposes as

$$\rho = \frac{1}{2} (I + \mathbf{r} \cdot \boldsymbol{\sigma}),$$

where $\boldsymbol{\sigma} = (X, Y, Z)$ and the dot product denotes $r_x X + r_y Y + r_z Z$. Pure states satisfy $|\mathbf{r}| = 1$ and therefore lie on the surface.

Statistical mixtures with no single state vector satisfy $|\mathbf{r}| < 1$ and fill the interior. The maximally mixed state $I/2$ sits at the center.

Laboratory noise shifts the direction of a Bloch vector and can also shrink its length. Once the $|\mathbf{r}| = 1$ condition is relaxed, the shortened state lies inside the sphere but remains a valid two-level state.

For any unit vector $\mathbf{n} = (n_x, n_y, n_z)$, define

$$\mathbf{n} \cdot \boldsymbol{\sigma} = n_x X + n_y Y + n_z Z.$$

The Pauli algebra gives $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$. Splitting the exponential series into even and odd powers yields

$$e^{-i\vartheta \mathbf{n} \cdot \boldsymbol{\sigma}/2} = I \cos(\vartheta/2) - i(\mathbf{n} \cdot \boldsymbol{\sigma}) \sin(\vartheta/2).$$

This unitary rotates the Bloch vector through ϑ about $\mathbf{n}$. The half-angle $\vartheta/2$ in the amplitudes reflects the double cover of spatial rotations by two-component spin states: a 2π Bloch-vector rotation multiplies the ket by -1 , and only 4π restores the ket itself.

Quantum gates as target transformations

An intended operation on a two-level system is ideally the unitary map U acting as

$$|\psi\rangle \mapsto U|\psi\rangle.$$

Up to a physically irrelevant overall phase, every such unitary represents a rotation of the Bloch vector:

$$U = \exp\left(-\frac{i\vartheta}{2}\mathbf{n} \cdot \boldsymbol{\sigma}\right).$$

Naming that target transformation and drawing it as a circuit box makes it a gate. The symbol fixes the intended map; the pulse or control procedure that implements it is specified separately.

The Pauli gates X , Y , and Z correspond, up to global phase, to π rotations about their respective axes. The Hadamard gate is

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

It sends $|0\rangle$ to $|+\rangle$, converting between the Z and X measurement bases.

A box labeled H records only that target. Laboratories synthesize H with microwaves, optical fields, voltages, exchange couplings, frame updates, or combinations of these.

Time-dependent pulse implementation

A control Hamiltonian connects the target gate to its time-dependent implementation. Suppose a resonant drive produces

$$\widehat{H}_d = \frac{\hbar\Omega}{2} (\cos \delta X + \sin \delta Y).$$

Here $\widehat{H}_d$ is an energy operator, $\hbar$ is Planck's reduced constant in joule-seconds, Ω is an angular frequency in radians per second, and δ is the drive phase in radians. Applying the drive for a duration t , measured in seconds, gives

$$U(t) = e^{-i\widehat{H}_d t/\hbar} = \exp\left[-\frac{i\Omega t}{2}(\cos \delta X + \sin \delta Y)\right].$$

The resulting rotation angle is $\vartheta = \Omega t$, with the dimensionless units an exponent requires:

$$[\Omega t] = (\text{s}^{-1})(\text{s}) = 1,$$

Setting $\delta = 0$ rotates about the x -axis; setting $\delta = \pi/2$ rotates about the y -axis.

The angular speed Ω is the Rabi frequency. For fixed Ω , the pulse duration t sets the rotation angle.

A gate names the intended transformation; a pulse applies a Hamiltonian for a set duration. The symbol $R_x(\pi/2)$ names the transformation; the waveform that realizes it is specified separately.

[Experiment] Electrically or magnetically driven spin resonance implements this control logic in solid-state spins, although the microscopic coupling and calibration differ by platform [Ro51]; [Ro52]; [Ro54].

Operational requirements for a qubit

Preparing a specified initial state with known reliability takes an explicit physical process. Waiting for relaxation toward the ground state is one route; optical pumping, reservoir-assisted loading, measurement plus a conditional pulse, and active reset are others.

An ideal computational-basis measurement is represented by the projectors

$$M_0 = |0\rangle\langle 0|, \quad M_1 = |1\rangle\langle 1|.$$

A projector selects the component of a state belonging to one measurement outcome. For a density operator ρ , outcome m occurs with probability

$$P(m) = \text{Tr}(M_m \rho), \quad m \in \{0, 1\}.$$

The trace Tr sums the diagonal elements of a matrix. Real instruments produce photons, currents, voltages, or charge-sensor traces instead of an abstract variable m , and a classifier maps that analog record to the classical value 0 or 1.

[Experiment] Energy-selective tunneling has been used to map a single electron's spin state to a charge transition detectable by a nearby sensor [Ro53]. Optically interfaced solid-state defects use spin-dependent optical dynamics to initialize and infer spin states; the relevant mechanisms and limitations vary strongly among centers [Ro54].

The complete operational loop is therefore:

- Specify $|0\rangle$ and $|1\rangle$, the associated axis, and the conditions under which the two states remain isolated.
- Prepare a known initial state and quantify the residual preparation error.
- Calibrate pulse amplitude, phase, frequency, and duration to implement target gates.
- Allow the intended single-system, two-system, or sensing energy to act.
- Rotate the desired measurement observable into the basis readable by the instrument.

- Acquire a classical record and assign an outcome.
- Reset and repeat the experiment while verifying that the calibration has not drifted.

A single result is a classical bit. Repeated results provide estimates of quantum probabilities. A measurement does not directly output the system's wavefunction.

A pair of levels together with this operational loop constitutes a qubit. The term expresses an operational claim rather than merely identifying a doublet in a spectrum. DiVincenzo organized this claim into a widely used checklist: the states must be distinguishable, one of them must be preparable, a useful set of coherent operations must be available, and the result must be readable. For computation, errors must also remain low, leakage into other levels must be limited, recalibration must be possible, and more than one qubit must be coupled [R050].

If isolation or readout is absent, the system still has two levels, but it does not satisfy these operational requirements for a qubit.

Encoded information distributed across multiple devices

In some systems, the two relevant logical outcomes are not represented by two levels of a single device. Instead, they are represented by two patterns distributed across several devices:

$$|0_L\rangle = |\text{pattern A}\rangle, \quad |1_L\rangle = |\text{pattern B}\rangle.$$

These patterns can be entangled states of many components. Quantum error-correcting codes select the pair of patterns so that specified physical errors can be detected or reversed. This capability is a property of the code and the selected error set, not a consequence of the notation L [R055].

The pair of patterns defines one encoded bit. Several devices can therefore store one encoded bit, and an individual device does not automatically constitute one encoded bit.

Counting physical hardware is consequently not equivalent to counting encoded bits. A controllable, resettable, and readable doublet is a candidate qubit.

A two-dimensional subspace distributed across several qubits is a candidate encoded bit. Programming ordinary qubits to reproduce the amplitudes of another model is a computation performed on the existing hardware.

Such programming does not change the physical identity of the hardware.

Physical degrees of freedom outside the selected subspace

A spin in a real device is not an isolated Pauli degree of freedom. The device also contains orbital states, nearby spins, phonons, electromagnetic modes, control wiring or optical components, and a detector. The states denoted by $|0\rangle$ and $|1\rangle$ are a selected pair within this larger state space.

For an ideal spin-1/2 in a static magnetic field, the Hamiltonian is often written as

$$\widehat{H}_0 = -\frac{\hbar\omega_0}{2}Z,$$

where ω_0 is the angular transition frequency in radians per second. A resonant field drives transitions between the two levels.

In an actual solid, spin–orbit coupling, hyperfine interactions, charge motion, strain, and higher-energy levels modify this idealized model. Specific defects fix these terms, with their values tabulated in the defect chapters.

[Proposal] Electron spins confined in quantum dots were proposed as physical qubits with controlled exchange interactions [Ro51]. [Experiment] Subsequent experiments demonstrated ingredients including single-spin readout and coherent single-spin rotations [Ro52]; [Ro53]. Demonstrating an individual ingredient provides evidence for that ingredient, but it does not by itself establish a fault-tolerant processor.

The same distinction applies to crystalline defects. [Experiment] Reviews of optically active solid-state spins document initialization, microwave or optical manipulation, and optical interfaces in several material platforms [Ro54].

Whether a particular defect serves as a useful qubit depends on its charge state, temperature, magnetic field, collection efficiency, nearby noise, and the exact protocol. Finding spin in a material begins that device specification; completing it requires each of those parameters.

Common conceptual and experimental errors

- **Identifying any two levels as a qubit.** A transition can be too weak to drive, too broad to address selectively, or spectrally overlapping its neighbors. Control pulses can leak into a third state. The two-level approximation holds only across stated ranges of energy, drive strength, temperature, and time.
- **Interpreting the Bloch sphere as ordinary physical space.** For a spin in a magnetic field the Bloch axes can align with physical spin components. For a superconducting circuit, a charge configuration, an orbital doublet, or a cluster encoding they are abstract coordinates fixed by the chosen basis, with no necessary direction in laboratory space.
- **Reading a probability amplitude off a single shot.** One computational-basis measurement returns 0 or 1. Estimating α , β , or a Bloch vector takes many identically prepared trials across several bases. Drift between trials can leave the reconstruction describing no state the experiment held steadily.
- **Equating relative phase with population.** The orthogonal states $(|0\rangle + |1\rangle)/\sqrt{2}$ and $(|0\rangle - |1\rangle)/\sqrt{2}$ both give 50–50 Z -basis statistics. Only an analyzer rotation such as the closing pulse of the five-step sequence makes their difference visible.
- **Equating a gate with its control waveform.** The symbol $R_x(\pi/2)$ names an ideal transformation. The laboratory applies a finite-duration pulse while unwanted Hamiltonian terms continue to act, so calibration, leakage, drift, and noise set how closely the implementation approaches the symbol.

- **Equating an encoded bit with one physical doublet.** Several qubits can hold one encoded bit with no new particle involved. Active detection of selected errors without a protective energy gap is a property of the implementation and code structure.
-

Conceptual checks

- **Operational criterion for a qubit:** Two levels form a qubit when the pair supports preparation, control, readout, and reset as a usable loop while staying isolated under stated conditions.
- **Number of parameters in a pure two-level state:** The amplitudes α and β start with four real parameters. Normalization and the unobservable global phase remove one each, leaving the Bloch-sphere angles θ and φ .
- **Location of a mixed state on the Bloch sphere:** A mixed state lies inside the sphere. Its density operator is $\rho = (I + \mathbf{r} \cdot \boldsymbol{\sigma})/2$ with $|\mathbf{r}| < 1$; the maximally mixed state lies at the center.
- **Probability produced by the five-step sequence:** The analyzer produces the state $\cos(\phi/2)|0\rangle + i \sin(\phi/2)|1\rangle$. Applying the Born rule to the first amplitude gives $P(0) = \cos^2(\phi/2)$.
- **Counting devices that store 0_L and 1_L :** The two distributed patterns 0_L and 1_L define one encoded bit. The term “encoded” identifies a selected two-dimensional subspace rather than the number of physical devices.
- **Information obtained from one computational-basis measurement:** A single shot returns 0 or 1 without fixing α and β . Estimating the amplitudes takes many identically prepared runs, usually across more than one basis.

A two-level system earns the name qubit once preparation, control, and measurement each meet specified performance. Assigning times and error rates to those operations is the task of the relaxation and dephasing analysis.

Sources

- [R049] F. Bloch, “Nuclear Induction,” *Physical Review* 70, 460–474 (1946). DOI: [10.1103/PhysRev.70.460](https://doi.org/10.1103/PhysRev.70.460).
- [R050] D. P. DiVincenzo, “The Physical Implementation of Quantum Computation,” *Fortschritte der Physik* 48, 771–783 (2000). arXiv: [quant-ph/0002077](https://arxiv.org/abs/quant-ph/0002077); DOI: [10.1002/1521-3978\(200009\)48:9/113.0.CO;2-E](https://doi.org/10.1002/1521-3978(200009)48:9<113.0.CO;2-E).
- [R051] D. Loss and D. P. DiVincenzo, “Quantum Computation with Quantum Dots,” *Physical Review A* 57, 120–126 (1998). DOI: [10.1103/PhysRevA.57.120](https://doi.org/10.1103/PhysRevA.57.120).
- [R052] F. H. L. Koppens, C. Buizert, K. J. Tielrooij, I. T. Vink, K. C. Nowack, T. Meunier, L. P. Kouwenhoven, and L. M. K. Vandersypen, “Driven Coherent Oscillations of a Single Electron Spin in a Quantum Dot,” *Nature* 442, 766–771 (2006). DOI: [10.1038/nature05065](https://doi.org/10.1038/nature05065).
- [R053] J. M. Elzerman, R. Hanson, L. H. Willems van Beveren, B. Witkamp, L. M. K. Vandersypen, and L. P. Kouwenhoven, “Single-Shot Read-Out of an Individual Electron Spin in a Quantum Dot,” *Nature* 430, 431–435 (2004). DOI: [10.1038/nature02693](https://doi.org/10.1038/nature02693).

- [R054] D. D. Awschalom, R. Hanson, J. Wrachtrup, and B. B. Zhou, “Quantum Technologies with Optically Interfaced Solid-State Spins,” *Nature Photonics* 12, 516–527 (2018). DOI: [10.1038/s41566-018-0232-2](https://doi.org/10.1038/s41566-018-0232-2).
 - [R055] E. Knill and R. Laflamme, “Theory of Quantum Error-Correcting Codes,” *Physical Review A* 55, 900–911 (1997). DOI: [10.1103/PhysRevA.55.900](https://doi.org/10.1103/PhysRevA.55.900); arXiv: [quant-ph/9604034](https://arxiv.org/abs/quant-ph/9604034).
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Chapter 4 — Relaxation, dephasing, and fidelity in qubits

A two-level quantum system loses useful information through at least two distinct mechanisms. In energy relaxation, the energy that distinguishes the excited state from the lower-energy state flows into the surroundings.

Dephasing randomizes the relative phase between the two selected states. It can proceed while the excited-state population stays high.

Energy relaxation and dephasing act on different observables, so each needs its own preparation, control, and measurement protocol. A reported lifetime is meaningful only alongside the protocol and the measured observable.

This chapter works with a single two-level system, a controlled waiting interval, and measurements that separate the two failure mechanisms.

Measurement of energy relaxation

Denote the upper level by $|1\rangle$ and the lower level by $|0\rangle$. The energy-relaxation protocol prepares $|1\rangle$, waits a time t in seconds, and measures whether the system is still in $|1\rangle$.

Repeated many times, the fraction of $|1\rangle$ outcomes estimates the excited-state population. A population is a probability, so it lies between 0 and 1. The analysis assumes a temperature low enough that the surroundings almost never re-excite the system, plus a constant decay rate: the surviving excited-state population then shrinks by the same factor in every equal time interval.

Those assumptions give exponential decay. Let ρ_{11} denote the excited-state population: the diagonal matrix element of the density operator from Chapter 2, the mathematical representation of a quantum state that accommodates statistical mixtures and quantum coherence together.

The time constant of this population decay is T_1 . Operationally, T_1 is defined by that same protocol — prepare the excited state, wait, measure its population:

$$\rho_{11}(t) = \rho_{11}(0)e^{-t/T_1}.$$

At finite temperature the population generally decays toward a nonzero thermal equilibrium population rather than toward zero.

T_1 is then the time constant of that approach to equilibrium, not a time to complete depopulation. Bloch-equation treatments of solid-state qubits identify this process as relaxation along the energy axis [\[Ro58\]](#).

A long T_1 therefore shows that the system keeps its excitation energy. Whether an initially encoded relative phase stays measurable is a separate question, answered by the coherence measurements below.

Measurement of phase coherence

A phase-coherence experiment prepares an equal superposition instead. This is the state $|x+\rangle$ from Chapter 1, sitting on the equator of the Bloch sphere from Chapter 3:

$$|x+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

For this state the decisive quantity is the off-diagonal density-matrix element ρ_{01} , called a coherence, alongside the population. This complex number encodes the relative phase and amplitude relation between the $|0\rangle$ and $|1\rangle$ components.

In the simplest memoryless model the coherence oscillates at the transition frequency while its magnitude decays. Let ω denote the angular transition frequency in radians per second.

For this model and free-evolution protocol, T_2 is the time at which the coherence magnitude has decayed to $1/e$ of its initial value. It is the decay constant of an off-diagonal density-matrix element and is distinct from T_1 :

$$\rho_{01}(t) = \rho_{01}(0)e^{-i\omega t}e^{-t/T_2}.$$

A memoryless stochastic process is called Markovian. The term records an assumption of the model about noise with negligible memory; it does not classify the atom by itself, apart from its environment and experimental conditions.

Energy relaxation feeds into the decay of ρ_{01} because a population is the squared magnitude of an amplitude. If the excited-state population decays as e^{-t/T_1} , the corresponding excited-state amplitude decays as $e^{-t/(2T_1)}$. The coherence ρ_{01} carries one factor of that amplitude, so energy relaxation alone multiplies ρ_{01} by $e^{-t/(2T_1)}$.

Independent phase noise contributes a further factor e^{-t/T_ϕ} , where T_ϕ is the time constant of dephasing beyond what energy relaxation already causes. Multiplying the independent decay factors gives

$$e^{-t/T_2} = e^{-t/(2T_1)}e^{-t/T_\phi}, \quad \boxed{\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}}.$$

Hence $T_2 \leq 2T_1$ in this model. Equality means no additional phase-only noise within the model; control pulses, readout, and computational performance still need their own tests.

A long T_1 therefore leaves the value of T_2 open: excitation energy can remain in the system after the relative phase has become unobservable.

Ramsey dephasing from shot-to-shot detuning

Chapter 3 introduced a two-pulse sequence with a fixed phase. The same sequence also probes the case where the residual frequency mismatch varies between experimental repetitions.

Work in a reference frame rotating with the control oscillator. When the two-level transition frequency differs from the oscillator frequency, the two levels accumulate opposite phases. Call the residual mismatch, or detuning, the angular frequency Δ in radians per second. In this rotating frame the residual Hamiltonian is

$$H = \frac{\hbar\Delta}{2}\sigma_z.$$

Here H is the energy operator, $\hbar$ Planck's reduced constant in joule-seconds, and σ_z the Pauli z operator, with eigenvalue $+1$ on $|0\rangle$ and -1 on $|1\rangle$. A radian is dimensionless, so Δt is a dimensionless phase.

The pulse sequence comes from Ramsey's separated-field spectroscopy [R056], is

```
|0> -- pi/2 -- free evolution for t -- -pi/2 -- measure
```

The first pulse implements

$$|0\rangle \longrightarrow |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Free evolution under the residual Hamiltonian then gives

$$|\psi(t)\rangle = \frac{e^{-i\Delta t/2}|0\rangle + e^{+i\Delta t/2}|1\rangle}{\sqrt{2}}.$$

The second pulse converts the accumulated relative phase into a measurable population difference. With this pulse convention the ideal probability of outcome 0 is

$$P_0^{\text{Ramsey}}(t) = \frac{1 + \cos(\Delta t)}{2}.$$

If Δ takes the same value in every repetition, the Ramsey fringe oscillates indefinitely at full contrast.

Now let the mismatch differ between repetitions. The additional offset is a static random variable δ with mean zero and Gaussian standard deviation σ in radians per second. “Static” means constant within one repetition, varying only between repetitions. Averaging the phase factor over repetitions gives

$$\langle e^{i\delta t} \rangle = e^{-\sigma^2 t^2 / 2}.$$

The oscillatory term then carries a Gaussian decay envelope:

$$P_0^{\text{Ramsey}}(t) = \frac{1 + e^{-\sigma^2 t^2 / 2} \cos(\Delta t)}{2}.$$

The time parameter that writes this envelope as $e^{-(t/T)^2}$ is denoted T_2^* . The asterisk marks decay that includes reversible shot-to-shot frequency variation; it is part of the symbol, not multiplication of T_2 by something else.

$$T_2^* = \frac{\sqrt{2}}{\sigma}, \quad P_0^{\text{Ramsey}}(t) = \frac{1 + e^{-(t/T_2^*)^2} \cos(\Delta t)}{2}.$$

The dimensions check out: $(\text{rad s}^{-1})^{-1} = \text{s}$ since radians carry no physical dimension. A short T_2^* can therefore accompany a long T_1 : the excitation energy remains while averaging over repetitions washes out the Ramsey-fringe contrast. This extra broadening is sometimes called inhomogeneous broadening. The defining protocol stays the same throughout: prepare the state, let it evolve freely, and measure the fringe.

Refocusing with a Hahn echo

A Hahn echo [Ro57] inserts a π pulse between two equal free-evolution intervals:

```
|0> -- pi/2 -- free tau -- pi -- free tau -- -pi/2 -- measure
```

Let the total free-evolution time be $t = 2\tau$. The π pulse flips the sign with which a static mismatch feeds the accumulated phase. The switching function $y(s)$ equals $+1$ before the pulse and -1 after it, so the unwanted phase is

$$\phi_{\text{echo}} = \delta \int_0^{2\tau} y(s) ds = \delta(\tau - \tau) = 0.$$

An ideal echo thus refocuses any mismatch that stays constant through one repetition. Energy relaxation, rapidly varying noise, and pulse errors remain unreversed: Hahn refocusing compensates the class of static phase histories described by that integral.

For a numerical example, take $\sigma/(2\pi) = 100$ kHz:

$$\sigma = 2\pi \times 10^5 \text{ s}^{-1}, \quad T_2^* = \frac{\sqrt{2}}{2\pi \times 10^5 \text{ s}^{-1}} = 2.25 \text{ } \mu\text{s}.$$

Give this hypothetical qubit also $T_1 = 1.00 \text{ ms}$ and a Markovian phase-only time $T_\phi = 100 \text{ } \mu\text{s}$. The boxed relation above then gives $T_2 = 95.2 \text{ } \mu\text{s}$.

At $t = 10 \text{ } \mu\text{s}$ the quasistatic Ramsey contrast is

$$e^{-(10/2.25)^2} \approx 2.7 \times 10^{-9},$$

whereas an idealized echo limited only by the stated exponential T_2 retains

$$e^{-10/95.2} \approx 0.900$$

of its transverse contrast. Over the same interval the excited population keeps $e^{-10/1000} \approx 0.990$ of its initial excess. One qubit and one time interval thus yield three distinct numbers — one for each observable and protocol — all mutually consistent.

In many devices the echo time exceeds T_2^* because the echo rejects frequency mismatch that stays approximately constant within one repetition. That ordering is not guaranteed: noise from the π pulse itself can make the echo result worse. A nonexponential envelope is fitted with

$$C(t) = C_0 \exp[-(t/T)^n],$$

Here C_0 is the dimensionless contrast, T the reported time in seconds, and n a dimensionless fit exponent. A complete coherence-time claim states the pulse sequence, measured observable, envelope model, fitting convention, sample conditions, and uncertainty.

Sequence-dependent filtering of frequency noise

Let $\delta\omega(s)$ denote a time-dependent fluctuation of the angular frequency. The accumulated phase is

$$\phi(t) = \int_0^t y(s) \delta\omega(s) ds.$$

For a Ramsey sequence, $y(s) = 1$. For a Hahn echo, $y(s) = +1$ during the first half of the evolution and -1 during the second half.

A zero-frequency offset feeds the Ramsey integral and cancels in the echo integral. Noise that changes appreciably between the two halves survives.

Additional pulses define further switching functions and hence further frequency-dependent noise filters. The pulses change the measurement protocol, not the material. Filter-function theory gives the formal description of this sequence dependence, computing each sequence's sensitivity to noise at each frequency [Ro59].

A longer echo time therefore says the chosen pulse sequence rejects more of the noise relevant to that measurement. Whether the material itself became less noisy needs a comparison of matched protocols under matched conditions.

Fidelity between quantum states

Saying an experiment succeeded does not say how closely its final state matches the target. State fidelity quantifies that closeness as a dimensionless number between 0 and 1.

For states with density operators ρ and σ , the fidelity is

$$F(\rho, \sigma) = \left[\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right]^2, \quad 0 \leq F \leq 1.$$

This expression follows the squared Uhlmann–Jozsa convention. Jozsa gives the mixed-state definition and its properties [Ro60]. Some authors call the unsquared square root “fidelity” instead, so every numerical report states its convention.

When the target is the pure state $|\psi\rangle$, the expression reduces to

$$F(|\psi\rangle, \rho) = \langle \psi | \rho | \psi \rangle.$$

This quantity is the probability that a projective test for $|\psi\rangle$ accepts the state ρ . A fidelity report therefore names both objects being compared. State fidelity by itself characterizes no pulse, readout apparatus, or error-correcting code.

Average fidelity of a quantum operation

A quantum gate is an operation, not a state. Let U denote the target unitary and $\mathcal{E}$ the operation the apparatus actually implements. For a d -dimensional computational space, the average gate fidelity is

$$F_{\text{avg}}(\mathcal{E}, U) = \int d\psi \langle \psi | U^\dagger \mathcal{E}(|\psi\rangle\langle\psi|) U | \psi \rangle,$$

The integral runs uniformly over pure input states with $d\psi$ a normalized dimensionless measure. Nielsen derives practical formulas for this average [Ro61]. The definition thus fixes both the assessed object — a gate — and the averaging procedure — equal weighting over all pure input states.

The complementary quantity $r = 1 - F_{\text{avg}}$ is the average gate infidelity: a dimensionless, probability-like quantity. It serves as neither a coherence time nor a worst-case error bound.

[Experiment] Randomized benchmarking estimates an average error parameter while desensitizing the result to state-preparation and measurement errors under its modeling assumptions. Interleaved benchmarking inserts a target gate to estimate that gate's contribution [Ro62]. Randomized benchmarking does not reconstruct the full noise channel. Coherent, correlated, time-dependent, and leakage errors may each need additional diagnostics.

Binary readout assignment fidelity

Readout ends in a classical decision. For binary single-shot readout, the experiment repeatedly prepares $|0\rangle$ and $|1\rangle$, records the reported labels, and averages the two correct-assignment probabilities:

$$F_{\text{RO}} = \frac{P(\hat{0} | 0) + P(\hat{1} | 1)}{2}.$$

Here $P(\hat{j} | j)$ is the conditional probability that the apparatus reports label j after state j was prepared. This balanced assignment fidelity is one common definition of readout fidelity [Ro63]. Reporting the two conditional probabilities separately keeps any asymmetry between them visible instead of averaged away.

This score absorbs any preparation errors the calibration procedure has not independently removed. It is neither Uhlmann state fidelity nor gate fidelity; it quantifies the accuracy of a classical assignment.

Leakage from the computational subspace

A bit flip maps $|0\rangle$ to $|1\rangle$ inside the selected two-level system. Excitation to $|2\rangle$ leaves that two-level system. The two processes are distinct errors.

Let P be the projector onto the computational subspace spanned by $|0\rangle$ and $|1\rangle$, and let $Q = I - P$ project onto all other levels. For a state ρ , leakage is the population outside the computational subspace:

$$L(\rho) = \text{Tr}(Q\rho) = 1 - \text{Tr}(P\rho).$$

Leakage is dimensionless and lies between 0 and 1. It separates population outside the computational subspace from errors confined within it.

A quantum channel carries leakage and seepage rates as well. Leakage transfers population from P to Q , while seepage returns population from Q to P [Ro64]. Postselecting — discarding leaked experimental outcomes — can show a high fidelity on the retained data while the unconditional operation performs poorly. The discarded fraction is therefore reported alongside the conditioned score.

Without a stated computational-subspace boundary, a high in-subspace metric can falsely suggest closed two-level dynamics.

Logical lifetimes of encoded states

The information of interest is not always held in one physical two-level system. It may be encoded across several such systems and protected by a stated protocol.

A logical lifetime is the decay time of a stated logical observable, or of a logical-state survival probability, under a stated noise model, control schedule, syndrome-extraction procedure, and decoder. Syndrome extraction measures the information used to identify errors without directly measuring the encoded logical state; a decoder uses the measured syndrome to infer a correction or logical outcome. Experimental surface-code studies commonly report the closely related logical error probability per correction cycle [Ro65]. The term “logical lifetime” applies to the complete encoded experiment rather than to any one physical subsystem.

A logical lifetime has units of seconds, or units of cycles if the cycle duration is reported separately.

As a simple example, suppose each correction cycle lasts τ_c seconds and independently causes an absorbing logical failure with probability p_L — a failure after which the state counts as failed for the rest of the survival analysis. The survival probability after N cycles is

$$S(N) = (1 - p_L)^N = \exp\left[-\frac{N\tau_c}{T_L}\right],$$

which defines

$$T_L = -\frac{\tau_c}{\ln(1 - p_L)} \approx \frac{\tau_c}{p_L} \quad \text{when } p_L \ll 1.$$

The approximation is dimensionally consistent: seconds divided by a dimensionless probability gives seconds. A logical Pauli expectation under a symmetric Pauli channel decays with a different factor, so the observable and fitting model are stated explicitly. A reported logical lifetime without them is incompletely defined.

[Experiment] Surface-code experiments report logical error per correction cycle and compare code distances. Under the conditions of one 2023 experiment, increasing the code distance improved the measured logical performance while the logical error per cycle stayed finite [Ro65].

This result evidences scaling under active error correction. It leaves open whether the superconducting hardware hosts an emergent topological phase, which needs separate many-body evidence.

A long T_L can come from an ordinary repetition code, a decoherence-free encoding, active correction, passive energy barriers, or topological order. The responsible mechanism is identified separately, since no single coherence time establishes a phase.

Experimental definitions and reporting requirements

The idealized equations omit state-preparation error, measurement error, pulse distortion, drift, multiple decay rates, and finite sampling. Laboratory measurements therefore yield protocol-dependent fitted data, not context-independent values of T_2 .

Claimed quantity | Minimal protocol | Direct observable | What it does not establish | |

T_1 | prepare excited state; vary delay; measure population | population approach to equilibrium | phase coherence, gate accuracy, or fault tolerance | |

T_2^* | Ramsey/free induction | fringe contrast versus free time | irreversible dephasing alone | |

echo T_2 | $\pi/2 - \tau - \pi - \tau$ sequence | refocused contrast versus 2τ | passive material protection or immunity to fast noise | |

gate fidelity | characterize a specified gate set and metric | tomography, randomized sequences, or another declared estimator | worst-case error, readout fidelity, or zero leakage | |

readout fidelity | prepare labels; construct assignment matrix | conditional classical outcome frequencies | premeasurement state fidelity |

leakage | resolve or infer levels outside the code space | population outside P , preferably with return dynamics | an ordinary bit/phase error rate | |

logical lifetime | prepare encoded states; run full protocol; decode at varied duration | logical survival or observable decay | topological order or universal fault tolerance by itself | |

[Experiment] Ramsey and echo data are often analyzed with more than one envelope model because different noise spectra produce Gaussian, exponential, stretched-exponential, or oscillatory decay [Ro58]; [Ro59]. Selecting the model after inspecting the data can change the reported time. Residuals, confidence intervals, and the fitting window belong to the result even when an abbreviated presentation omits them.

Meaningful comparisons also need matched experimental conditions: temperature, magnetic field, qubit transition, sample, control power, pulse sequence, number of refocusing pulses, and whether one qubit or an ensemble was measured. A coherence time from many-pulse dynamical decoupling is not reported as an “intrinsic T_2 ” without qualification. The number can be correct while the description of what was measured is wrong.

Common interpretation errors

- **Memory quality from T_1 alone.** T_1 constrains only energy relaxation. Slow frequency drift can give a short T_2^* alongside a long T_1 ; in the Markovian relation, phase-only noise can dominate $1/T_2$.
- **Echo-enhanced lifetime as passive protection.** Echo is an applied-control protocol that cancels phase from sufficiently slow fluctuations. Removing the pulse restores the Ramsey decay. Dynamical decoupling can work very well while providing no material energy barrier and no topological protection.

- **All percentages as equivalent fidelity measures.** State fidelity, average gate fidelity, postselected fidelity, readout assignment fidelity, and survival probability answer different questions. Even measurements under the same label may use squared or unsquared conventions. A complete report specifies the object, estimator, subspace, conditioning, and averaging distribution.
- **Negligible leakage from high in-subspace gate fidelity.** A reported gate metric may be conditioned on staying in the computational space, or it may be insensitive to rare excursions. F_{avg} and leakage are reported separately. Leakage can persist across cycles and produce correlated downstream errors [Ro64].
- **A logical qubit outliving one physical qubit as fault tolerance.** The physical and logical comparisons must use matched tasks. Fault tolerance means controlled scaling of logical error as code resources grow below a threshold, including the effects of operations and measurements. A single favorable lifetime can mark progress without demonstrating scalable error suppression; an encoded qubit can also be useful with entirely non-topological protection.
- **A single fitted time as a complete material characterization.** The noise spectrum and the pulse-dependent filter jointly determine coherence decay. Sample preparation, nearby spins, charge motion, temperature, and the control sequence also affect the result. The protocol is reported with the fitted time.
- **Physical-qubit times as a full account of encoded information, or a long encoded lifetime as a diagnosis of a phase.** T_1 , T_2 , T_2^* , physical-gate fidelity, physical readout fidelity, and physical leakage characterize a physical qubit and its controls. An encoded qubit instead has logical observables, logical gates, logical leakage, and a logical lifetime. A digital circuit that emulates a code can show logical behavior without an emergent many-body phase. Passive robustness from an emergent topological phase belongs to the energy operator — gap, locality, system size, temperature, and perturbations together. No single coherence time establishes it.

Conceptual checks

- In the no-memory model represented by

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi},$$

T_1 is the population-relaxation time, T_2 is the coherence time, and T_ϕ is the pure-dephasing time. Because $T_\phi > 0$, the pure-dephasing contribution $1/T_\phi$ is nonnegative. Therefore,

$$T_2 \leq 2T_1.$$

Equality requires $1/T_\phi = 0$.

- Consider a qubit with $T_1 = 5$ ms and Ramsey $T_2^* = 2$ μ s, where T_2^* is the inhomogeneous coherence time from Ramsey interferometry. The long T_1 alone does not make a good quantum memory. Population can relax slowly while quasistatic frequency noise — detuning that barely changes during one measurement but varies between measurements — rapidly suppresses phase coherence. Neither number establishes fault tolerance.

- Suppose Ramsey fringes decay in $3\ \mu\text{s}$ while Hahn-echo contrast decays in $80\ \mu\text{s}$. A Hahn echo is a pulse sequence that refocuses phase from sufficiently slow detuning fluctuations. The measurements directly support the conclusion that much of the Ramsey decay is refocusable on the echo timescale, consistent with slow detuning noise — and nothing about passive protection. Both reported times are accompanied by the pulse sequences used and the fitted decay envelopes.
- Consider a gate whose retained shots have fidelity 0.999 while 0.02 of all shots occupy $|2\rangle$. Retained-shot fidelity is conditioned on excluding specified outcomes; leakage is population that leaves the intended computational subspace. Here the escaped population is leakage $L = 0.02$, which the high retained-shot fidelity does not remove. Both quantities and the conditioning rule are reported.
- Readout fidelity differs from the fidelity of the quantum state before measurement. Balanced readout fidelity averages the two classical probabilities of correctly assigning the prepared basis states. State fidelity compares density operators, which represent quantum states. Preparation errors and measurement errors are therefore calibrated separately.
- Suppose an encoded state — quantum information stored in a larger physical Hilbert space — survives for ten times the physical T_2^* . That enhancement alone does not establish topological order, a many-body property of a topological phase. The improvement could come from echo, a decoherence-free subspace, active correction, postselection, or another encoding. Demonstrating topological order needs independent many-body evidence. Any reported logical lifetime also names the measured logical observable and the protocol used.

These times and performance measures quantify distinct physical properties, so one cannot stand in for another. The next section examines how a crystal produces two levels to which these times can be assigned.

Sources

- [R056] N. F. Ramsey, “A Molecular Beam Resonance Method with Separated Oscillating Fields,” *Physical Review* 78, 695–699 (1950). DOI: [10.1103/PhysRev.78.695](https://doi.org/10.1103/PhysRev.78.695).
- [R057] E. L. Hahn, “Spin Echoes,” *Physical Review* 80, 580–594 (1950). DOI: [10.1103/PhysRev.80.580](https://doi.org/10.1103/PhysRev.80.580).
- [R058] G. Ithier et al., “Decoherence in a superconducting quantum bit circuit,” *Physical Review B* 72, 134519 (2005). DOI: [10.1103/PhysRevB.72.134519](https://doi.org/10.1103/PhysRevB.72.134519); [arXiv:cond-mat/0508588](https://arxiv.org/abs/cond-mat/0508588).
- [R059] Ł. Cywiński, R. M. Lutchyn, C. P. Nave, and S. Das Sarma, “How to enhance dephasing time in superconducting qubits,” *Physical Review B* 77, 174509 (2008). DOI: [10.1103/PhysRevB.77.174509](https://doi.org/10.1103/PhysRevB.77.174509); [arXiv:0712.2225](https://arxiv.org/abs/0712.2225).
- [R060] R. Jozsa, “Fidelity for Mixed Quantum States,” *Journal of Modern Optics* 41, 2315–2323 (1994). DOI: [10.1080/09500349414552171](https://doi.org/10.1080/09500349414552171).
- [R061] M. A. Nielsen, “A simple formula for the average gate fidelity of a quantum dynamical operation,” *Physics Letters A* 303, 249–252 (2002). DOI: [10.1016/S0375-9601\(02\)01272-0](https://doi.org/10.1016/S0375-9601(02)01272-0); [arXiv:quant-ph/0205035](https://arxiv.org/abs/quant-ph/0205035).
- [R062] E. Magesan et al., “Efficient Measurement of Quantum Gate Error by Interleaved Randomized Benchmarking,” *Physical Review Letters* 109, 080505 (2012). DOI: [10.1103/PhysRevLett.109.080505](https://doi.org/10.1103/PhysRevLett.109.080505); [arXiv:1203.4550](https://arxiv.org/abs/1203.4550).

- [Ro63] D. Keith et al., “Benchmarking high fidelity single-shot readout of semiconductor qubits,” arXiv:1811.03630 (2018). Stable URL: <https://arxiv.org/abs/1811.03630>.
 - [Ro64] C. J. Wood and J. M. Gambetta, “Quantification and characterization of leakage errors,” Physical Review A 97, 032306 (2018). DOI: [10.1103/PhysRevA.97.032306](https://doi.org/10.1103/PhysRevA.97.032306); [arXiv:1704.03081](https://arxiv.org/abs/1704.03081).
 - [Ro65] Google Quantum AI and R. Acharya et al., “Suppressing quantum errors by scaling a surface code logical qubit,” Nature 614, 676–681 (2023). DOI: [10.1038/s41586-022-05434-1](https://doi.org/10.1038/s41586-022-05434-1); [arXiv:2207.06431](https://arxiv.org/abs/2207.06431).
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PART III — CRYSTALS AND DEFECTS

We begin with a repeating arrangement of atoms. We then change a single site and examine how this affects an electron in the crystal.

Chapter 5 — Translational symmetry and defect localization in a crystal

What does it mean for an atomic arrangement to repeat?

One-dimensional translational symmetry

Consider an ideal, infinite row of identical atoms, equally spaced by a distance $a > 0$.

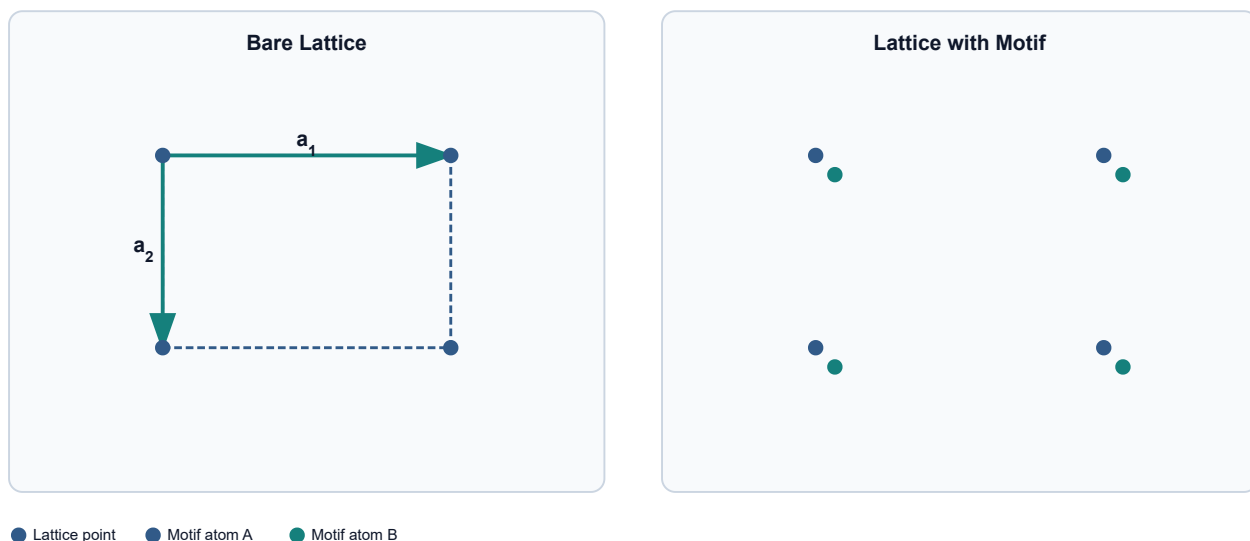
Choose one atom as the origin. Shifting the *whole row* by the spacing a carries every atom onto the site its neighbor held, so the shifted row coincides with the original. Repeating the shift in either direction gives the positions

$$\dots, -2a, -a, 0, a, 2a, \dots$$

These positions form the one-dimensional lattice $a\mathbb{Z}$. Here $\mathbb{Z}$ is the set of all integers, and a , measured in metres, is the lattice constant. The lattice names the translations reproducing the ideal pattern; the atoms and their internal structure sit on top of that translation set.

The atoms attached to one lattice point form a *motif*, the small arrangement repeated at every lattice point. A two-atom motif, for example, places two atoms in each repeated unit, so the translation spacing can differ from the nearest atom-to-atom spacing.

2D Lattice Schematic



A two-panel schematic separates lattice translation points from the repeated two-atom motif attached to each point. The motif identifies the atoms repeated at each translation point; equal spacing alone does not specify the crystal.

The figure is a two-dimensional schematic of this distinction; our present chain uses the one-dimensional set $a\mathbb{Z}$. In higher dimensions two or three independent translation vectors play the same role, and the lattice again collects the translations leaving the ideal pattern invariant.

The translation property belongs to the geometry. Later sections place a physical model on this same fixed set of sites.

Site states in a one-electron model

Now place one electron on the chain with one localized orbital $|n\rangle$ at each lattice point, where the integer n labels the site. “Localized” means the orbital concentrates around one site: an electron in $|n\rangle$ is found at n by a position measurement in this site description, instead of spreading equal weight over the whole row.

Assume these site orbitals are orthonormal:

$$\langle m | n \rangle = \delta_{mn}.$$

They form a basis, so a general one-electron state reads $|\psi\rangle = \sum_n c_n |n\rangle$. In this basis $|c_n|^2 = |\langle n | \psi \rangle|^2$ is the probability of finding the electron at site n , summing to one for a normalized state. Each localized basis vector thus supplies one position outcome.

This is a one-electron model with one orbital per site and static nuclei. It omits the further electrons and orbitals, spin, and lattice motion of a real crystal. Those omissions bound what the later calculation can claim, while the site probabilities keep their meaning within this model.

Nearest-neighbor hopping

To let the electron spread, allow it to move directly between neighboring sites. In this model that motion is the only way amplitude leaves a site, and the process is called *hopping*.

Introduce a parameter $t > 0$ to set the strength of this motion. It has units of energy, measured in joules or electronvolts.

In the Hamiltonian, choose the matrix element $\langle n+1 | H_0 | n \rangle = -t$ between neighboring sites. The reverse-direction matrix element is its complex conjugate. Its magnitude t sets how strongly the amplitude at one site affects the evolution at the other.

The Schrödinger equation then determines how the site probabilities change with time.

For an exactly solvable finite version, take N sites with periodic boundary conditions: site N is identified with site 0. The chain becomes a ring with no ends, giving every site two nearest neighbors. The Hamiltonian is

$$H_0 = -t \sum_{n=0}^{N-1} (|n\rangle\langle n+1| + |n+1\rangle\langle n|),$$

where the indices wrap around the ring. This nearest-neighbor form keeps the local geometry visible while allowing an exact calculation of the extended modes. It is the one-orbital tight-binding approximation: a model in which an electron moves weakly between localized orbitals [\[Ro67\]](#).

With $t = 0$ the coupling vanishes: every $|n\rangle$ is an energy eigenstate and nothing propagates between sites. With nonzero t , $H_0|n\rangle$ contains neighboring site states, so a single site state stops being an eigenstate of H_0 . From here on the localized basis state and the energy eigenstate are different physical objects.

Energy band of the uniform chain

Because every site couples identically, try a state whose amplitude picks up the same phase factor at each step:

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{ikna} |n\rangle.$$

Here k , measured in inverse metres, is the lattice wavevector, and the product ka is the dimensionless phase change per lattice step.

On the ring, the amplitude returns to its starting phase after N steps:

$$e^{ikNa} = 1.$$

Hence $k = 2\pi m/(Na)$ for integer m : the finite ring admits N wavevectors with orthogonal $|k\rangle$ states.

Applying the two hopping terms collects one phase factor from each neighbor:

$$H_0|k\rangle = -t(e^{ika} + e^{-ika})|k\rangle = -2t \cos(ka)|k\rangle.$$

The dispersion relation follows as

$$E(k) = -2t \cos(ka).$$

Since $E(k)$ depends on $\cos(ka)$, the distinct states with wavevectors k and $-k$ can share one energy while staying orthogonal. With the minus sign chosen in H_0 , the minimum $E = -2t$ sits at $k = 0$, where the pattern varies most slowly from site to site.

A complete set of distinct wavevectors fits in $-\pi/a < k \leq \pi/a$. On a finite ring the N energies (counting degeneracies) lie in $[-2t, 2t]$ and grow arbitrarily fine as the ring grows. The infinite chain carries a genuinely continuous wavevector and energy spectrum over this band — a limit in its own right, not shorthand for a dense finite set — with total width $4t$.

Each $|k\rangle$ eigenstate is extended: its amplitude at every site has magnitude $1/\sqrt{N}$, giving site probability

$$|1/\sqrt{N}|^2 = 1/N.$$

This is a stationary wave spread evenly around the ring: at any instant the probability sits equally everywhere, rather than a lump traveling an orbit. A prepared superposition of many $|k\rangle$ states can instead form a localized wavepacket, so a uniform eigenbasis still permits non-uniform prepared states. Once the defect below breaks translation symmetry, k stops labeling the eigenstates exactly.

A single-site defect

Change only the on-site energy at site 0 by an amount U , also measured in energy units:

$$H = H_0 + U|0\rangle\langle 0|.$$

In this model “defect” means that one changed scalar on-site energy. A real vacancy is a different object: it can also shift neighboring positions, reshape orbitals, and alter couplings. Those vacancy effects belong to Chapter 6 and later sections.

For the bound-state calculation, move from the finite ring to an infinite chain. The ring boundary disappears, so a decaying tail cannot wrap around and meet itself. Seek a reflection-symmetric state centered at the altered site:

$$\psi_n = \langle n|\psi\rangle = A\lambda^{|n|}, \quad |\lambda| < 1.$$

The condition $|\lambda| < 1$ makes the two-sided geometric tail square-summable. Normalization then fixes $|A|$, leaving no arbitrary overall scale.

At a site away from the defect, say $n \geq 1$, only the two hopping terms act, giving the eigenvalue equation

$$E A\lambda^n = -t(A\lambda^{n-1} + A\lambda^{n+1}),$$

and division by the nonzero tail amplitude leaves

$$E = -t(\lambda + \lambda^{-1}).$$

At site 0 the on-site shift contributes UA while both neighboring amplitudes equal $A\lambda$, so the equation there is

$$E = U - 2t\lambda.$$

Equating the two expressions for E gives the defect condition

$$U = t(\lambda - \lambda^{-1}).$$

For every nonzero U the resulting quadratic has exactly one root with magnitude below one, with magnitude and defect energy

$$|\lambda| = \frac{\sqrt{U^2 + 4t^2} - |U|}{2t}, \quad E_d = \text{sgn}(U)\sqrt{U^2 + 4t^2}.$$

Take first an attractive shift, $U < 0$: then $E_d < -2t$ lies below the clean-chain band, the root is positive, and the amplitudes keep the same sign on both sides of site 0.

For a repulsive shift, $U > 0$, $E_d > 2t$ lies above the band and the root is negative, so $\lambda^{|n|}$ alternates sign from site to site while decaying in magnitude.

The excluded case $U = 0$ is the clean chain again, where the defect equation gives only the non-decaying possibilities $|\lambda| = 1$ and no normalizable defect-localized state. A nonzero change at one site therefore creates one bound state outside the single clean-chain band.

The envelope can be written as $e^{-|n|a/\xi}$, where the localization length ξ is the *amplitude* e-folding length: each further distance ξ from the defect multiplies the envelope by e^{-1} .

$$\xi = -\frac{a}{\ln |\lambda|} = \frac{a}{\operatorname{arsinh}(|U|/2t)}.$$

For example, with $U = -t$:

$$E_d = -\sqrt{5}t, \quad |\lambda| = \frac{\sqrt{5}-1}{2} \approx 0.618, \quad \xi \approx 2.08a.$$

In this example normalization sums the central probability and the two geometric tails:

$$|A|^2 \left(1 + 2 \sum_{n=1}^{\infty} |\lambda|^{2n} \right) = 1, \quad |A|^2 = \frac{1 - |\lambda|^2}{1 + |\lambda|^2} = \frac{1}{\sqrt{5}}.$$

About 0.447 of the total probability then sits on the altered site, with the two tails holding the rest, summing to exactly one. [Theory] These numbers follow from this chosen one-dimensional Hamiltonian; they predict nothing about diamond, sapphire, or another real host.

The exponential above solves the infinite chain exactly; on a finite ring it is only approximate. A ring state concentrated near site 0 has a tail that eventually wraps around the periodic boundary and overlaps itself. The ring approaches the infinite-chain state only when much longer than the localization length.

Evanescient states outside the energy band

The clean-chain dispersion explains why the defect level cannot lie inside the band. With $|E| \leq 2t$ some real k satisfies $E = -2t \cos(ka)$, and the factor e^{ikna} keeps constant magnitude, so the clean-chain solution extends instead of decaying. At the band edges $k = 0$ or $k = \pi/a$ gives a constant or alternating extended pattern rather than an evanescent tail.

With $|E| > 2t$ no real wavevector satisfies the dispersion relation. Below the band, write $k = i\kappa$ with inverse length $\kappa > 0$; on the right side of the defect, for $n \geq 0$, the clean-chain factor becomes

$$e^{ikna} = e^{-\kappa na}.$$

This evanescent tail decays rightward from the defect; a matching left-hand tail decays leftward, forming the two-sided localized state used above. The defect acts as a localized energy shift supporting a level where the uniform host has no real- k extended solution, rather than as a hard wall.

Grouping two sites into each repeated unit creates the possibility of an interval between two bands.

Formation of an energy gap

What geometry creates two bands?

The model so far carries one orbital per repeated unit and one band. Keeping the nearest-site spacing a , now group two sites, A and B , into each repeated unit, giving a two-site cell of period $b = 2a$.

What makes the two sites inequivalent?

Give the two sites on-site energies $+\Delta$ and $-\Delta$, with Δ in energy units, representing different atoms or local environments. When $\Delta \neq 0$ translation by one site stops being a symmetry. Every nearest-neighbor hop keeps the same positive magnitude $t > 0$. The gap opens because the two sites in each cell sit at different energies; the hopping pattern is unchanged.

Why do we need two amplitudes per cell?

For a fixed crystal wavevector, an eigenstate assigns one amplitude to the A sublattice and one to the B sublattice. The cell translation b sets the cell-to-cell phase, so the two-component vector (A_k, B_k) represents the state in one cell.

The matrix diagonal records the two on-site energies. Two nearest-neighbor paths couple the A and B amplitudes: one path stays within the cell while the other crosses the cell boundary, contributing the phase $e^{\pm ikb}$. In the ordered basis (A, B) this gives

$$H(k) = \begin{pmatrix} \Delta & -t(1 + e^{-ikb}) \\ -t(1 + e^{ikb}) & -\Delta \end{pmatrix}.$$

Two amplitudes require a two-by-two eigenvalue problem: one equation balances energy on an A site and the other on a B site. Setting the determinant of $H(k) - EI$ to zero gives

$$E_{\pm}(k) = \pm \sqrt{\Delta^2 + 4t^2 \cos^2(kb/2)}.$$

What is the spectral gap?

As in the one-site chain, take an interval holding each distinct wavevector once. The cell period is now b , so use $-\pi/b < k \leq \pi/b$. At its boundary the cosine term vanishes, the lower branch reaches $-|\Delta|$ and the upper branch reaches $+|\Delta|$, and their minimum separation is

$$E_g = 2|\Delta|.$$

This energy gap is a spectral property: an interval with no traveling host state, not a physical void in the crystal. With $\Delta = 0$ the two branches touch, $E_g = 0$, leaving no internal interval separating a level from both bands.

When may the branches be called valence and conduction bands?

The labels add an occupation assumption to the spectral gap. In the noninteracting spinless example at zero temperature with one electron per two-site cell, the lower band is filled and the upper band empty. With E_v the upper edge of the occupied lower band and E_c the lower edge of the empty upper band, the same gap reads

$$E_g = E_c - E_v > 0.$$

Localized states within a band gap

How does a defect level enter the gap?

First turn off hopping, $t = 0$, so every site stands independent. A defect orbital with energy E_d between $-|\Delta|$ and $+|\Delta|$ then keeps its amplitude entirely on the defect: perfect localization in this decoupled limit. Restoring hopping lets the defect amplitude leak onto neighboring host sites, creating spatial tails.

Those tails become traveling waves only when the host supports propagation at that energy. While the full defect eigenvalue stays between the two host band continua, the perfect host has no propagating solution there. The recurrence relation outside the defect then selects a decaying solution over an oscillatory one. The resulting state parallels the one-band bound state: its energy lies in the spectrum's forbidden interval and its wavefunction decays away from the defect.

What do the edge separations establish?

Measure the energy separation from each band edge:

$$\delta_v = E_d - E_v, \quad \delta_c = E_c - E_d.$$

For an in-gap one-electron level both quantities are positive, which establishes *spectral isolation*: separation of the energy from the valence and conduction continua. A short spatial envelope needs a separate wavefunction check.

How is spatial localization checked?

Localization means a normalizable eigenstate keeps most of its probability in a bounded region instead of occupying a number of cells growing with sample size. A level very close to an edge can decay slowly and spread over many cells; a level farther from both edges often decays faster. Either way the wavefunction or spatial density must be calculated.

What if defect-host coupling moves the level into a band?

Stronger defect-host coupling can shift a nominal defect level into a band. Overlapping a continuum, it generically becomes a resonance mixing with extended host states rather than a normalizable bound eigenstate. A larger E_g permits larger edge separations without guaranteeing spectral isolation or localization.

What do “shallow” and “deep” mean here?

A level close to either band edge is called *shallow*: weakly separated from the continuum, its envelope may extend over many cells and respond to band carriers. A level far from both edges is called *deep*, where “deep” names its energy position inside the gap, not geometric depth below a surface. Either word still leaves charge stability, wavefunction character, and environmental coupling to be checked before calling the defect useful.

What advantage does a wide-gap host actually provide?

A wide-gap host offers a larger interval where a defect level can stay separated from both continua. [Theory] That interval is an opportunity for isolation, not a defect-generation mechanism: the gap creates no defect, spin, or qubit by itself. Candidate defects are still judged by charge-state stability, wavefunctions, lattice relaxation, optical transitions, and coupling to noise [Ro69]; [Ro70]; [Ro71].

Bloch waves in a periodic potential

What does periodicity guarantee?

For an ideal crystal, take a one-electron model with a spatially periodic potential,

$$V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r})$$

for every lattice translation $\mathbf{R}$. For the ideal periodic one-electron Hamiltonian, Bloch's theorem gives an energy eigenbasis of functions of the form

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}),$$

where $\mathbf{k}$ is the crystal wavevector, n is the band index, and the cell-periodic factor repeats in every cell:

$$u_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{n\mathbf{k}}(\mathbf{r}).$$

Such an eigenfunction is a Bloch wave, named for Felix Bloch's 1929 crystal-wave theorem; it shares only a name with a point on the two-level Bloch sphere [Ro66]. The theorem claims an eigenbasis adapted to the ideal periodic symmetry, not that every eigenfunction takes one unique Bloch form: degeneracies permit other linear combinations, and a defect breaks the translation symmetry.

How does the chain approximate this statement?

The tight-binding state $|k\rangle$ is the discrete counterpart of a Bloch wave, with site amplitudes acquiring a fixed phase from cell to cell. A nearly-free-electron description starts from the opposite limit of nearly free electrons with the periodic potential as a perturbation. The two approximations are organized around different limits. Periodicity enables the plane-wave-times-cell-periodic form; without periodicity Bloch's theorem does not apply.

How are Wannier functions different from defect-bound states?

Superpositions of Bloch waves across a band can form orbitals localized near individual cells, sometimes called Wannier functions. Whether exponentially decaying Wannier functions exist is a property of the band; the elementary chain here has no obstruction [Ro68]. A Wannier orbital represents states of a translation-invariant band: translating it gives an equivalent orbital in every cell. A defect-bound state instead needs broken translation symmetry pinning its envelope near one particular site. Its localization is a property of the physical eigenstate created by the defect, not merely a change of basis within a perfect band.

Nuclear vibrations and phonons

What was frozen in the electronic chain?

The electronic chain held nuclei at fixed equilibrium positions. A real crystal allows small displacements about those positions. Neighboring nuclei couple, so a normal mode is a collective pattern with many nuclei oscillating in a definite relative phase and polarization, rather than one atom moving independently. In the harmonic approximation each collective mode behaves like an oscillator.

How does a collective mode become a phonon?

Quantizing one mode of angular frequency ω gives total oscillator energies

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, \dots$$

The $\frac{1}{2}\hbar\omega$ term is the zero-point energy. Raising n by one adds exactly one energy quantum, $\hbar\omega$. That added quantum is one phonon: a single excitation step, distinct from both the total oscillator energy and the zero-point motion.

When are wavevector and branch labels needed?

A crystal holds many collective modes. To distinguish them, label a mode by wavevector $\mathbf{q}$ and branch index ν , where ν separates polarization/frequency families at the same $\mathbf{q}$. The mode's angular frequency is $\omega_{\mathbf{q}\nu}$, in radians per second, with total oscillator energy

$$E_{\text{phonon}} = \hbar\omega_{\mathbf{q}\nu}\left(n_{\mathbf{q}\nu} + \frac{1}{2}\right),$$

with nonnegative integer occupation $n_{\mathbf{q}\nu}$ and $\hbar$ in joule-seconds. Creating or removing one excitation changes the energy by one phonon quantum, $\hbar\omega_{\mathbf{q}\nu}$.

How can phonons affect a defect electron?

Electronic energies depend on nuclear positions, coupling a defect electron to these modes. One optical transition can appear with *sidebands*: extra absorption or emission lines displaced by phonon energies where the transition creates or removes vibrational quanta.

The same coupling shifts a defect level with nuclear fluctuations. It also dephases a superposition by tying its phase to the vibrational environment.

A *nonradiative transition* changes the electronic state with no emitted photon; multiphonon emission can carry away the required energy along one such route.

[Theory] Quantitative rates need potential-energy surfaces, vibrational modes, and electron-phonon matrix elements, which no band diagram supplies [\[Ro72\]](#).

Static nuclei are thus a model assumption rather than a property of laboratory crystals. Cooling lowers phonon occupation without fixing nuclei at mathematical points.

Additional physics in real crystals

The chain isolates one mechanism: propagation through a repeated lattice combined with an energy outside the host continuum produces an exponentially decaying state.

That isolation rests on five narrow model choices: one electron, one orbital per site, static nuclei, nearest-neighbor hopping, and one scalar U .

A real sample lies outside all five: it holds many electrons and nuclei with three-dimensional bonding, surfaces, strain, impurities, isotopes, fields, and finite temperature. The chain models one mechanism transparently; it predicts no particular material's defect properties. The calculation below adds the material details needed to study a real defect.

How is a real-material defect calculation built?

1. **Specify the structure.** Choose the nuclear species and equilibrium positions, then introduce the defect for its local electronic and geometric effects.
2. **Represent an isolated defect.** Place that structure in a periodically repeated supercell. The repetition makes the calculation tractable; because periodic images can interact, the supercell size must be checked.
3. **Approximate the electronic structure.** Use an electronic-structure method, for example density-functional theory, to calculate the electronic states for the chosen nuclei — an approximation, not a direct measurement.
4. **Relax the nuclei.** Allow the coordinates to respond to the defect until forces and local bonding reach the geometry of that charge state.
5. **Correct charged-cell artifacts.** For a charged defect, correct the artificial electrostatic interaction between repeated charged cells and the compensating background with finite-size corrections.

Only after this sequence can candidate energies and wavefunctions be interpreted. Approximate methods can misestimate the host band gap, shifting calculated defect levels relative to the band edges. The calculation therefore needs convergence checks, an appropriate charge treatment, and comparison with measurements or better methods where level placement matters. Reviews by Van de Walle and Neugebauer and by Freysoldt and collaborators give the fuller treatment of formation energies, charge corrections, chemical potentials, and transition levels [Ro69]; [Ro70].

Which “defect energy” is being discussed?

The phrase “defect energy” covers physically different quantities. Keep the following questions separate.

- **Single-particle orbital energy:** In an approximate independent-particle calculation this labels one orbital. It organizes a spectrum without automatically giving an experimentally measurable addition or removal energy.
- **Thermodynamic charge-transition level:** This comes from total-energy differences between relaxed charge states. It marks the electron chemical potential — the energy cost of exchanging an electron with a reservoir — at which two charge states share equal formation energy; moving that chemical potential changes which charge state is stable. It differs from the energy of an electron in a frozen orbital.
- **Vertical optical transition:** An optical event runs fast against nuclear rearrangement, connecting electronic states at nearly fixed nuclear geometry. Its energy can differ from a relaxed charge-transition level.
- **Zero-phonon optical transition:** This connects vibrational ground levels on the relevant electronic potential-energy surfaces, with phonon-assisted sidebands alongside whenever vibrational quanta participate.
- **Spin excitation:** This changes the spin state within one electronic configuration, possibly far below an optical transition since no charge or orbital change is required.

These quantities can differ from one another. In density-functional theory, the Kohn–Sham construction replaces the interacting electrons with an auxiliary one-electron problem reproducing their density. Most eigenvalues of that auxiliary problem are not measurable electron-addition or removal energies [Ro69]; [Ro70].

Electron-electron interactions can further make the defect's many-electron state qualitatively different from a one-electron-in-one-orbital description.

How are these quantities connected to evidence?

[Experiment] Spectroscopy detects absorption and emission lines, phonon sidebands, ionization thresholds, and spin resonances. Assigning a microscopic structure to a signal takes experiment together with electronic-structure calculation.

[Theory] A systematic defect search asks whether the host is suitable, the charge and spin stable, the relevant levels separated from the bands, the transitions controllable, and the environmental coupling tolerable. No single calculated energy answers all of these [Ro71].

What can a wide gap help with?

A wide gap provides more room to keep defect levels away from both band edges. That room helps spectral isolation without placing any level.

It can also place transitions between two defect states in the visible or near-infrared with no need to cross the full host gap. Whether such a transition is allowed and useful depends on the states and selection rules.

Finally, when the activation energy for exchanging carriers with the bands is large against $k_B T$, thermal exchange with those bands is suppressed. That suppression competes with phonon-assisted and other nonradiative pathways; level positions, selection rules, and environmental coupling decide the outcome.

Common conceptual errors

Does a line in a gap prove localization?

A line drawn inside a gap hypothesizes an energy; it gives no evidence of a localized state. Localization needs the associated wavefunction or spatial density calculated, or a model supported by experiment. A near-edge state may extend widely, and apparent localization in a finite supercell may be an artifact of repeated images or imposed boundary conditions.

Does a wide host gap trap an electron?

The defect may contribute no gap state at all: a shallow state, a resonance embedded in a band, or an energetically undesired charge state are all possible. The host gap supplies an available interval; the defect potential and charge physics determine what appears there.

Is the chain useless as an electronic-structure model?

It is a deliberately simplified electronic-structure model. Its parameters t and U , with its single basis orbital, omit three-dimensional bonding, multiple orbitals, spin, Coulomb repulsion, screening, spin-orbit coupling, and nuclear relaxation. Fitting it afterward does not make it predictive of a specific material. Its value is isolating transparently how propagation plus a forbidden energy gives exponential decay.

Is one in-gap orbital already a qubit?

A qubit needs two controllable states with initialization, gates, readout, and acceptable leakage and noise. The orbital may be empty, doubly occupied, unstable, optically dark, or strongly phonon-coupled. The operational definition from Chapter 3 still applies.

Are localization and isolation the same?

Localization describes where the wavefunction concentrates in space; isolation describes separation in energy or weakness of unwanted couplings. A localized state can still couple through electric fields, strain, phonons, photons, nuclear spins, or weak interactions with other defects — useful controls or noise sources depending on the application.

Are Bloch waves and Bloch-sphere points the same?

A Bloch wave is a crystal eigenstate labeled by crystal wavevector $\mathbf{k}$. A point on the Bloch sphere represents a state in a two-level quantum state space; the shared name does not make the concepts interchangeable.

A localized defect orbital is a microscopic electronic state. A defect spin becomes a physical qubit only after control and readout are demonstrated — a different status from an encoded qubit, an emergent quasiparticle, or topological order.

Periodic boundary conditions are a calculational convenience, not physical protection.

Verification exercises

Use these short checks to reconstruct the chapter's reasoning rather than memorize isolated labels.

Can you recover the uniform-chain dispersion?

Each hop contributes a neighboring phase factor $e^{\pm ika}$; adding the two directions gives

$$-t(e^{ika} + e^{-ika}) = -2t \cos(ka).$$

Therefore

$$H_0|k\rangle = -2t \cos(ka)|k\rangle.$$

Checking the cosine endpoints gives the band edges $-2t$ and $+2t$ in the infinite-chain limit.

Why must the defect ansatz decay?

In the localized-defect ansatz λ multiplies the amplitude from one site to the next away from the defect. The condition $|\lambda| < 1$ shrinks the tail geometrically so the infinite probability sum converges. With $|\lambda| \geq 1$ the candidate has no finite total probability on the infinite chain and is not a localized eigenstate.

Where does the one-site defect energy lie?

For nonzero U the defect energy is

$$E_d = \text{sgn}(U)\sqrt{U^2 + 4t^2},$$

and

$$\sqrt{U^2 + 4t^2} > 2t.$$

Hence $|E_d| > 2t$ whenever $U \neq 0$; the $U = 0$ case restores the uniform chain instead of producing a defect.

What happens when the two site types become equivalent?

Setting $\Delta = 0$ in the two-site-cell dispersion touches the two branches at the edge of the k interval, so $E_g = 0$ leaves no internal interval separating a level from both bands.

What is the precise implication of widening a host gap?

A wider host gap gives a larger possible energy window in which positive edge separations are easier to maintain. The gap only supplies that window. Whether a defect level exists, remains localized, keeps a stable charge or spin, supports an optical transition, or functions as a qubit requires separate wavefunction, energy, transition, and control evidence.

What cannot a band diagram tell you?

Even correct level positions provide no wavefunctions, occupations, many-electron multiplets, lattice relaxation, transition matrix elements, or environmental couplings by themselves. Those missing pieces are why a real-material assignment combines calculation with measurements.

A localized orbital alone does not say whether a defect carries a spin. Chapter 6 examines that question for concrete atomic defects.

Sources

- [R066] F. Bloch, “Über die Quantenmechanik der Elektronen in Kristallgittern,” *Zeitschrift für Physik* 52, 555–600 (1929). DOI: [10.1007/BF01339455](https://doi.org/10.1007/BF01339455).
 - [R067] J. C. Slater and G. F. Koster, “Simplified LCAO Method for the Periodic Potential Problem,” *Physical Review* 94, 1498–1524 (1954). DOI: [10.1103/PhysRev.94.1498](https://doi.org/10.1103/PhysRev.94.1498).
 - [R068] W. Kohn, “Analytic Properties of Bloch Waves and Wannier Functions,” *Physical Review* 115, 809–821 (1959). DOI: [10.1103/PhysRev.115.809](https://doi.org/10.1103/PhysRev.115.809).
 - [R069] C. G. Van de Walle and J. Neugebauer, “First-principles calculations for defects and impurities: Applications to III-nitrides,” *Journal of Applied Physics* 95, 3851–3879 (2004). DOI: [10.1063/1.1682673](https://doi.org/10.1063/1.1682673).
 - [R070] C. Freysoldt, B. Grabowski, T. Hickel, J. Neugebauer, G. Kresse, A. Janotti, and C. G. Van de Walle, “First-principles calculations for point defects in solids,” *Reviews of Modern Physics* 86, 253–305 (2014). DOI: [10.1103/RevModPhys.86.253](https://doi.org/10.1103/RevModPhys.86.253).
 - [R071] J. R. Weber, W. F. Koehl, J. B. Varley, A. Janotti, B. B. Buckley, C. G. Van de Walle, and D. D. Awschalom, “Quantum computing with defects,” *Proceedings of the National Academy of Sciences* 107, 8513–8518 (2010). DOI: [10.1073/pnas.1003052107](https://doi.org/10.1073/pnas.1003052107).
 - [R072] A. Alkauskas, Q. Yan, and C. G. Van de Walle, “First-principles theory of nonradiative carrier capture via multiphonon emission,” *Physical Review B* 90, 075202 (2014). DOI: [10.1103/PhysRevB.90.075202](https://doi.org/10.1103/PhysRevB.90.075202).
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Chapter 6 — An empty lattice site does not imply a residual spin

Placing an atom of a different element on a site where a host atom would sit in a perfect crystal creates a substitutional defect. An empty lattice site next to that foreign atom allows the two to bind into one defect complex with shared electronic structure. Putting an extra atom into a gap between regular lattice sites creates an interstitial defect.

Shifting atoms along an entire line of the crystal generates a dislocation, a separate geometric class from localized missing or added atoms. Sorting defects by this kind of geometry comes before asking which ones carry an unpaired spin or can serve as a qubit. A nitrogen atom sitting next to a missing carbon atom in diamond illustrates the next step, where charge, spin, strain, and surrounding nuclei become separately identifiable.

Classification of structural defects

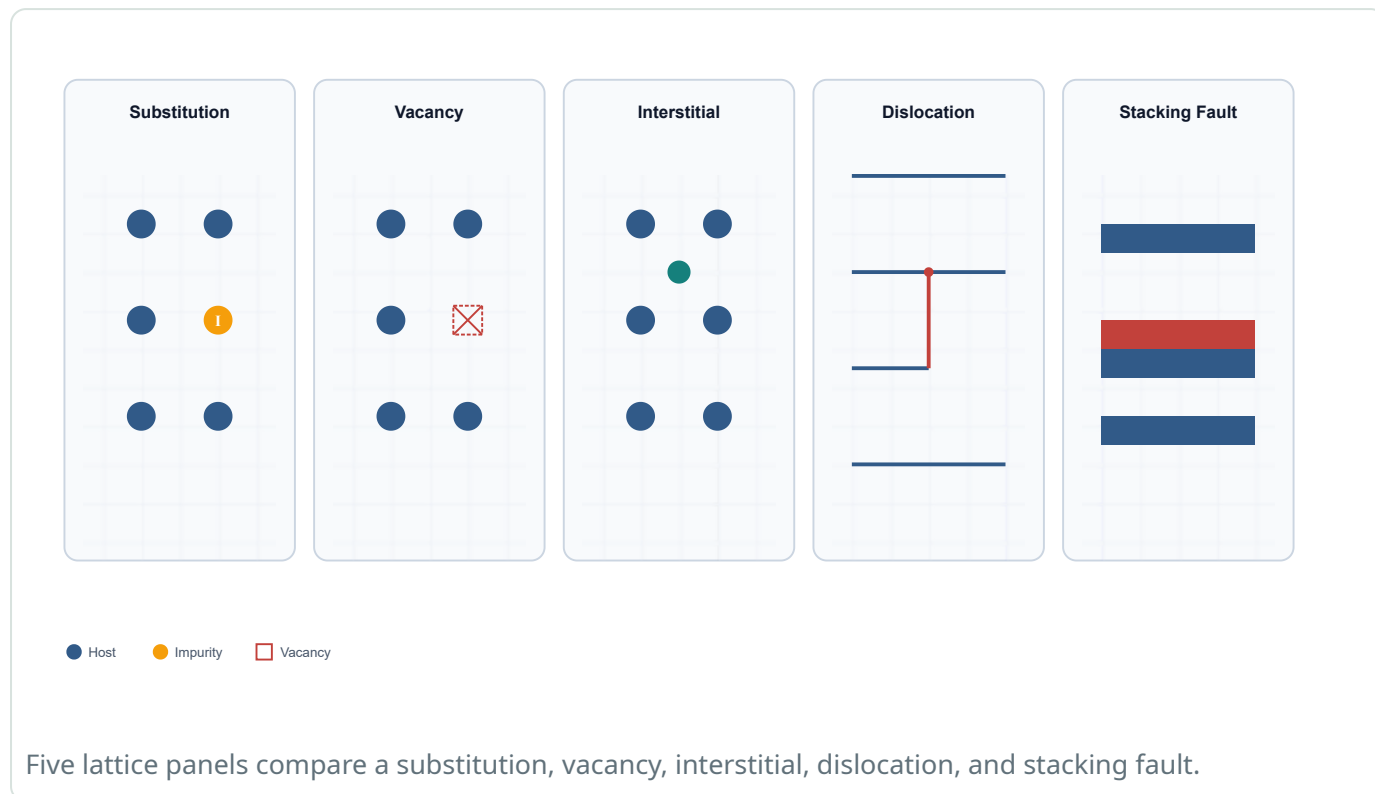
A perfect crystal repeats the same atomic arrangement in every unit cell. Any local departure from that repetition counts as a structural defect, a phrase that names the altered atomic positions while leaving open the optical response, the spin state, and any use as a computational bit.

Counting how many dimensions the disrupted region spans gives a practical first sorting of defects.

Kind	What happened	What the electrons often notice	What not to assume
Wrong atom on a lattice site	A host atom is replaced by a different element	A different valence, size, electronegativity, and local potential	That it leaves an unpaired spin. It may just donate or accept charge.
Empty site	A lattice site has no atom	Neighboring leftover bonds may sit in the gap and the neighbors may lean in	That reconstruction has not paired every electron. Charge matters.
Extra atom stuffed in	An atom sits off the regular sites	Strong local squeeze and new bonds; the extra atom may wander	That there is only one geometry or one charge.
Two accidents bound together	Nearby defects lock into one structural unit	New symmetry and molecular-like combinations of the leftover bonds	That the pair is the sum of the two isolated level diagrams.
A slipped line	A one-dimensional line defect, marked by a Burgers vector — how far the lattice slipped	A messy core plus a long-range elastic field; sometimes a band of core states	That the whole line is one localized emitter.
A mis-stacked plane	The usual stacking of planes is interrupted	A local change of stacking, a shift of the bands, and an extended squeeze	That a sheet of electronic states is a point-like two-level system.

Occupation of individual sites defines the localized types. A host atom replaced by a different atom at its regular site is a substitution, an empty regular site is a vacancy, and an atom in a position outside the regular lattice is an interstitial. Two defects close enough that their structures and electron distributions must be calculated as one system form a bound defect pair.

Some disruptions extend beyond a single site. A line along which the bonding pattern shifts continuously forms a dislocation. An interruption in the repeating sequence of atomic planes forms a stacking fault.



A foreign atom on one lattice site and an empty lattice site are each localized to roughly one site, which is the defining character of a point defect. A dislocation perturbs atomic positions along an extended curve through the crystal, giving a line defect. A stacking fault changes the registry of atomic layers across an extended interface, giving a planar defect.

Describing structural disorder therefore starts from the energies and spatial forms of the quantum states supported by the host crystal together with its defects.

A point defect sometimes confines an electron to an orbital near the defect site with an energy level deep inside a wide forbidden energy range. Under other charge or chemical conditions the same kind of defect supports a weakly bound state whose wavefunction spreads into a nearby conduction or valence band. The defect electrons can also pair completely to give total spin zero. Another outcome is capture of an electron and a hole followed by emission of lattice vibrations, so the stored electronic energy leaves as heat rather than light.

An extended defect alters nearby electronic behavior by localizing a moving electron or hole, by distorting bond lengths around a neighboring point defect, and by adding fluctuations to electrical or optical signals. Assigning the defect a dimension from its geometry selects the model used at the outset, while calculation or measurement remains necessary to establish how the defect behaves electronically [R070]; [R073].

A uniform distortion of bond lengths belongs with elastic deformation rather than with the catalog of structural defects, and a quantized lattice vibration belongs with excitations rather than with defect species. Both shift defect energy levels and alter transition rates between those levels.

Charge and spin states of a vacancy

A missing atom leaves three or four neighbors with orbitals that formerly formed bonds to that atom. These residual orbitals are called dangling-bond orbitals in the literature. A separate calculation of electron occupation and pairing is required to decide whether those orbitals hold an unpaired electron or carry nonzero spin.

Neighboring atoms shift to new equilibrium positions and form new bonding arrangements around the missing atom. Electrons move into or out of the defect region whenever that transfer lowers the total energy. The lowest-energy configuration reached in this way often has all electron spins paired into a closed shell.

The charge state counts electrons in the defective region relative to the electron count of the neutral atoms. A vacancy holding one extra electron and the same vacancy missing one electron are two different physical states. Each charge state carries its own equilibrium structure, optical transitions, and total spin.

Specifying the vacancy geometry leaves the spin open until the electron occupancy is also specified. The charge label identifies which occupancy is present. Including that label keeps physically distinct systems separate under their shared geometric picture.

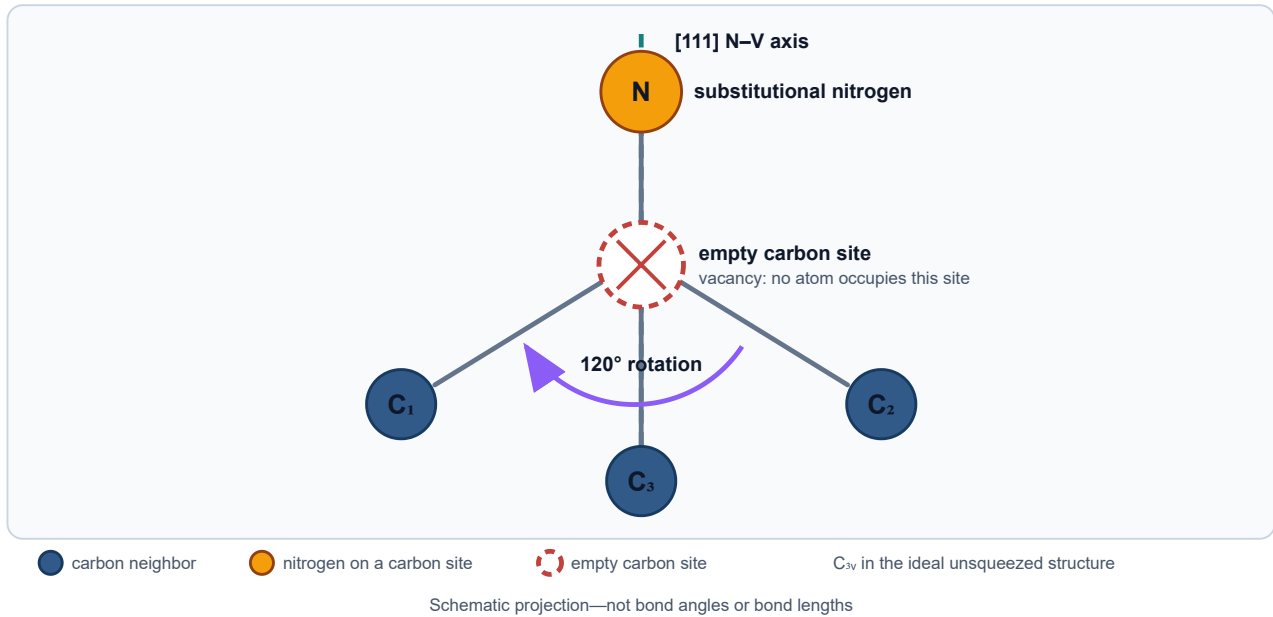
Nitrogen adjacent to a carbon vacancy

A concrete system for working through these ideas contains one nitrogen atom substituted for carbon in diamond next to a missing carbon atom. This nitrogen-vacancy pair allows direct analysis of the remaining symmetry. Results obtained for this pair carry over to another impurity-vacancy complex only after separate checking.

The axis running from the nitrogen atom through the vacant site selects one preferred direction, usually one of the $[111]$ directions of diamond. With no applied strain, the three carbon atoms around the vacancy remain equivalent to one another. A 120° rotation about the defect axis and reflection through any of three mirror planes each map the structure onto itself.

Rotations and reflections that map the local structure onto itself make up its local point group. The nitrogen-vacancy pair has the local point group C_{3v} [R074]; [R075].

NV center: one substitution beside one empty carbon site



A schematic diamond projection shows substitutional nitrogen beside an empty carbon site, three neighboring carbons, and the nitrogen-vacancy axis.

An ideal NV center places a substitutional nitrogen atom next to a missing carbon atom. Three carbon neighbors point their unsatisfied bonds into that empty site around the line joining nitrogen to the vacancy. The projection records this neighborhood arrangement without representing bond lengths.

Terms in the Hamiltonian respect the symmetries left intact by the defect, and an external or local perturbation that removes one of those symmetries allows a spectroscopic line to split. The remaining geometric pattern fixes which degeneracies can occur, while the size of an energy separation requires a separate calculation of matrix elements. When the three carbon atoms become inequivalent, the local pattern departs from C_{3v} , and the degeneracies associated with that point group can split.

Symmetry-adapted combinations of the carbon orbitals

Let $|c_1\rangle$, $|c_2\rangle$, and $|c_3\rangle$ denote the three carbon dangling-bond orbitals directed toward the vacancy. Weighting these directed orbitals by coefficients chosen from the local threefold geometry gives symmetry-adapted linear combinations, each labeled by how it changes under the point-group operations. For the threefold axis, they are

$$|a_C\rangle = \frac{|c_1\rangle + |c_2\rangle + |c_3\rangle}{\sqrt{3}},$$

$$|e_x\rangle = \frac{2|c_1\rangle - |c_2\rangle - |c_3\rangle}{\sqrt{6}}, \quad |e_y\rangle = \frac{|c_2\rangle - |c_3\rangle}{\sqrt{2}}.$$

The first combination adds the three carbon orbitals with equal weight, so a 120° rotation leaves it unchanged. This rotationally invariant behavior identifies it as belonging to an a representation.

The other two combinations reshuffle into mixtures of each other under that rotation, so the threefold equivalence of the carbon sites gives them a shared energy. This paired behavior identifies them as forming an e representation.

The nitrogen dangling-bond orbital is also unchanged by rotation about the defect axis, so it belongs to an a representation. That shared rotational behavior allows it to mix with $|a_C\rangle$, producing two a_1 combinations at different energies. In this restricted basis, the carbon e pair finds no nitrogen orbital with the corresponding paired transformation behavior, so it stays doubly degenerate as long as ideal C_{3v} symmetry holds.

These combinations give the qualitative pattern of energy levels: two totally symmetric levels and one doublet.

```
N substitution + neighboring vacancy
|
v
C3v symmetry
|
+-----+-----+
| |
two a1 levels one e doublet
```

The intact symmetry fixes the allowed degeneracies and the allowed mixing among the four dangling-bond orbitals; the spread of the full defect orbitals into the lattice puts their energies in the domain of electronic-structure calculation.

The overall energy scale comes from electronic-structure calculations [Ro70]; [Ro75]. Loss of the threefold rotation axis removes the requirement that the two states in the e pair share one energy.

Electron occupancy and the spin-triplet ground multiplet

The positions of the atoms leave open how many electrons occupy these orbitals. The negatively charged nitrogen-vacancy center, denoted NV^- , holds six electrons in the configuration

$$a_1(1)^2 a_1(2)^2 e^2.$$

Each occupied a_1 orbital holds two electrons with paired spins and forms a closed shell. The remaining two electrons sit in the e pair, where Coulomb repulsion and exchange favor parallel spins and give a ground-state manifold of total electron spin $S = 1$, conventionally labeled 3A_2 .

That spin-triplet assignment rests on the charge state, the ordering of orbital energies, and the interacting many-electron problem [Ro74]; [Ro75]. [Theory]

A multiplet collects the many-electron states that possess a given total spin and a given remaining spatial symmetry. The collection forms a family of energy levels with those shared quantum numbers.

The spin belongs to the occupied many-electron state. For fixed atomic geometry, adding or removing an electron or interchanging two orbital energies selects a different multiplet.

Ground-state zero-field splitting

The three lowest spin arrangements of the ground-state triplet evolve under an effective Hamiltonian expressed in frequency units:

$$\frac{H_{\text{gs}}}{h} = D \left[S_z^2 - \frac{S(S+1)}{3} \right] + E (S_x^2 - S_y^2) + \frac{\mu_B}{h} \mathbf{B} \cdot \mathbf{g} \cdot \mathbf{S}.$$

H_{gs} carries units of energy and division by h , Planck's constant, converts each contribution to frequency. S_x , S_y , and S_z are the dimensionless operators for a spin-1 system.

D and E give zero-field-splitting frequencies in hertz. The magnetic contribution combines μ_B , the Bohr magneton in joules per tesla, with $\mathbf{B}$, the magnetic field in tesla, and $\mathbf{g}$, the dimensionless electron g -tensor that encodes how strongly the electron spin responds along each direction.

The first term places the $m_s = 0$ level apart from the $m_s = \pm 1$ levels when no magnetic field is applied. That spin-level separation defines the zero-field splitting, while the separate orbital splitting comes from a nonspherical neighboring environment.

Transverse strain or another perturbation that makes the x and y directions inequivalent activates the second term. The third term describes the electron Zeeman interaction, the coupling of the applied magnetic field to the electron spin.

For NV^- near room temperature, the axial splitting is approximately $D = 2.87 \text{ GHz}$ [Ro74]; [Ro76]. [Experiment] The units agree because multiplying $2.87 \times 10^9 \text{ s}^{-1}$ by h produces an energy.

The ideal C_{3v} model has $E = 0$. A real center commonly carries a small transverse term.

The split triplet spectrum and its measured frequencies describe this Hamiltonian. The Hamiltonian alone does not establish qubit operation: if $D = 0$ and $E = 0$, all three spin projections share one energy until an applied magnetic field separates them.

Requirements for a two-level qubit

A magnetic field along the defect axis shifts the $m_s = +1$ level upward and the $m_s = -1$ level downward. That separation supports a two-level working subspace with $|0\rangle = |m_s = 0\rangle$ and $|1\rangle = |m_s = -1\rangle$. A resonant microwave magnetic field drives transitions within this subspace.

NV^- also has a sharp optical transition between its triplet ground-state and excited-state families near 637 nm, corresponding to approximately 1.945 eV [Ro74]. [Experiment] The conversion follows from $hc \approx 1240 \text{ eV nm}$:

$$\frac{hc}{\lambda} \approx \frac{1240 \text{ eV nm}}{637 \text{ nm}} \approx 1.95 \text{ eV}.$$

Electrons excited by light can relax through intermediate singlet states with rates that depend on the starting spin level, so repeated optical cycling builds population preferentially in one spin level and makes the emitted fluorescence brightness report which spin level is occupied. This spin-selective pumping and readout, combined with preparation of the negative charge state, microwave rotation of the spin, and efficient collection of emitted photons, makes the measured optical and spin response usable as a laboratory qubit [Ro74]. [Experiment]

The label for a nitrogen atom beside a missing carbon atom records the atomic arrangement. “NV[−]” adds the extra electron, and “³A₂” labels the collective triplet ground configuration. The $m_s = 0, -1$ pair then isolates two magnetic orientations that serve as a qubit when they can be initialized, rotated coherently, read out optically, and protected from the environment through the operation. These descriptions are related but are not synonymous. Two spectral levels therefore supply only a candidate computational subspace: without initialization, control, readout, and charge stability, the defect does not constitute an operational qubit.

Defect formation energy

Evaluating the thermodynamic cost of a defect starts from two calculated total energies: the energy of a simulation cell that contains the defect and the energy of the matching cell of perfect crystal.

Adding the energy accounting for atoms transferred in or out of the cell and for electrons exchanged with an external reservoir to that energy difference gives the defect formation energy.

The expression below applies to defect D in charge state q .

$$E_f(D^q) = E_{\text{tot}}(D^q) - E_{\text{tot}}(\text{bulk}) - \sum_i n_i \mu_i + q(E_F + E_{\text{VBM}}) + E_{\text{corr}}.$$

Each quantity entering the expression is an energy, commonly expressed in electronvolts, with $E_{\text{tot}}(D^q)$ giving the total energy of the simulation cell containing the defect and $E_{\text{tot}}(\text{bulk})$ giving the total energy of the perfect simulation cell.

Atomic exchange enters through the integer n_i for the number of atoms of species i added to the cell and through μ_i for the energy cost of supplying one such atom, while electronic exchange enters through E_F for the electron energy measured upward from the valence-band maximum and through E_{VBM} for the absolute energy of that maximum in the same reference as the total energies.

Finite size of the periodically repeated simulation cell introduces spurious electrostatic and elastic interactions, and E_{corr} adds the estimated removal of those artifacts.

Publications write n_i with different sign choices for atoms added or removed, so a calculation reports its choice directly [Ro70].

The formation energy gives thermodynamic favorability for fixed chemical potentials and electron chemical potential. Implantation yield, kinetic trapping, optical brightness, and coherence require separate kinetic or excited-state calculations.

A charge-transition level marks the electron chemical potential where two charge states share the same formation energy, so the lower-energy charge state changes there. A compensating impurity, a surface, a gate, or illumination can shift the electron chemical potential across that value and convert one atomic geometry between a spin-active state and a spinless state.

The $q(E_F + E_{\text{VBM}})$ term adds the energy exchanged with the surrounding electron reservoir when electrons enter or leave. Including this term puts formation energies of different charge states on a common energy scale at specified atomic positions.

Crystal-field splitting

A free atom has equal energy in every orientation. Neighboring atoms introduce particular bond directions that break that rotational symmetry.

Surrounding atoms at the corners of a tetrahedron or an octahedron create an electrostatic and covalent environment that separates the five equal-energy d -orbital states of a free atom into a twofold group and a threefold group. Tetrahedral or octahedral coordination together with the character of the bonding sets which group lies higher in energy.

Neighboring atoms generate electrostatic attraction and repulsion together with overlap of electron clouds, and this surrounding arrangement lacks spherical symmetry. That nonspherical environment is called the crystal field. Its energy contribution is written with the Hamiltonian term H_{CF} .

H_{CF} selects particular combinations of orbitals adapted to the symmetry left over by the neighbors. Those combinations carry labels from symmetry classification, for example A , E , or T . A further distortion that lowers the symmetry can separate the members of an E or T family in energy.

Crystal-field splitting sets the energy separation among orbital states. The spin-triplet zero-field splitting D sets a separate energy scale, and both energy scales remain present when no external magnetic field is applied.

Zero-field splitting adjusts spin levels that belong to one chosen orbital and spin multiplet.

With neighbors arranged into a perfectly spherical shell, the five d -orbital states would share one common energy.

Spin-orbit coupling

An electron circling a nucleus creates a magnetic moment, and the electron also carries a magnetic moment from its intrinsic spin, so the two moments exert torques on each other. Treating a fixed set of levels as isolated gives this interaction as

$$H_{\text{SO}} = \lambda \mathbf{L} \cdot \mathbf{S},$$

where $\mathbf{L}$ and $\mathbf{S}$ quantify orbital circulation and intrinsic electron magnetism in dimensionless units, respectively, and λ sets the energy scale of their coupling. A realistic defect may require a tensorial coupling or a representation expressed in terms of the residual point-group symmetry.

This interaction carries the name spin–orbit term. Evaluation of useful qubit properties requires criteria beyond the size of this interaction.

A crystal field that leaves several orbital states at the same lowest energy allows spin–orbit coupling to shift those levels in direct proportion to the spin-orbit coupling strength. A crystal field that isolates one orbital ground state quenches the orbital angular momentum.

Mixing of the ground orbital with higher orbitals in intermediate steps of perturbation theory allows spin–orbit coupling to act even when the orbital moment is quenched. Such mixing can make the g -factors direction dependent, contribute to the zero-field splitting, and produce spin-selective optical selection rules [Ro73]; [Ro75].

[Theory]

The spin–orbit interaction links optical excitation to spin orientation and links spin energy to lattice motion. The strength of control enabled by the spin-orbit coupling strength grows together with the strength of this lattice-induced noise. If $\lambda = 0$, both the first-order coupling and these virtual-mixing effects vanish, eliminating both mechanisms.

Interactions between unpaired electron spins

Each unpaired electron carries a magnetic moment, and the moments of two or more such electrons exert forces on one another. After projection into a total-spin multiplet, their dipolar and spin–orbit-mediated effects are often represented by

$$H_{\text{SS}} = \mathbf{S} \cdot \mathbf{D} \cdot \mathbf{S},$$

where $\mathbf{D}$ is a zero-field-splitting tensor with units of energy. The tensor encodes the energy cost of orienting the spin along different spatial axes at zero magnetic field. Under axial symmetry, orientation dependence collapses to a single axial parameter, giving, up to an additive constant, $\hbar D[S_z^2 - S(S+1)/3]$. This axial parameter is the same D introduced for NV^- .

Thus, the zero-field splitting of an integer-spin multiplet arises at zero applied magnetic field from interactions among the electrons. A half-integer spin supports a twofold degeneracy at zero magnetic field when time-reversal symmetry remains intact.

An integer-spin multiplet admits complete lifting of degeneracy into singlets. Long coherence calls for additional isolation beyond either degeneracy pattern.

A symmetry-protected doublet shifts in energy under particular lattice and electromagnetic perturbations.

Electron Zeeman and hyperfine interactions

An applied magnetic field shifts electron spin energies in proportion to field strength and spin projection along the field, a contribution called the electron Zeeman term

$$H_Z = \mu_B \mathbf{B} \cdot \mathbf{g} \cdot \mathbf{S}.$$

A localized electron spin couples magnetically to nearby nuclei carrying nonzero nuclear magnetic moments.

If nucleus k has nuclear spin $\mathbf{I}_k$, the magnetic moment of the electron senses the magnetic moment of that nucleus through the electron–nuclear coupling

$$H_{\text{hf}} = \sum_k \mathbf{S} \cdot \mathbf{A}_k \cdot \mathbf{I}_k,$$

where $\mathbf{A}_k$ is the hyperfine tensor for that nucleus and has units of energy. The name hyperfine coupling denotes this magnetic interaction between an electron spin and a nuclear spin.

The hyperfine tensor encodes how electron spin density overlapping the nuclear volume produces an approximately direction-independent contact shift. It also encodes how the through-space magnetic field of the electron spin produces a direction-dependent dipolar shift. Nuclei with spin quantum number $I_k \geq 1$ may also carry a nonspherical charge distribution that couples to the local electric field gradient through $\mathbf{I}_k \cdot \mathbf{Q}_k \cdot \mathbf{I}_k$, where $\mathbf{Q}_k$ is an energy tensor.

Nuclear spins add a magnetic degree of freedom beside the electron spin. That added degree of freedom lets hyperfine coupling resolve different nuclear configurations by their electron transition frequencies, hold a quantum state in the long-lived nuclear spin, and condition electron evolution on the nuclear state for a conditional gate.

Host nuclei left uncontrolled shift the electron transition frequency from shot to shot and shorten phase coherence. Identified and driven nuclei give calibrated splittings and gate operations governed by the same hyperfine term [\[Ro73\]](#).

If every nearby nucleus has $I = 0$, those nuclei carry no magnetic moment for the electron to sense, so the hyperfine interaction vanishes.

Static strain coupling

$\mathbf{u}(\mathbf{r})$ gives the vector by which the lattice at position $\mathbf{r}$ moves from its ideal site, measured in metres. For small distortions, the dimensionless strain tensor is

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial r_j} + \frac{\partial u_j}{\partial r_i} \right).$$

Static strain is a time-independent field describing local stretching and rotation of the lattice. A defect generates such a field around itself, changes its energy levels under an externally imposed field, and exchanges energy with lattice vibrations.

Projecting onto the low-energy defect states leaves a strain coupling built from operators O_α compatible with the symmetry remaining at the defect site:

$$H_\varepsilon = \sum_{\alpha} g_{\alpha} \varepsilon_{\alpha} O_{\alpha}.$$

ε_{α} combines Cartesian strain components into forms matched to that site symmetry, O_{α} acts as a dimensionless orbital or spin operator within the projected manifold, and the strain susceptibility g_{α} carries units of energy. Multiplying dimensionless strain by this energy scale yields a Hamiltonian term with units of energy.

Strain along the principal defect axis moves the overall optical or spin transition energy while preserving a doublet degeneracy. Strain applied perpendicular to that axis lowers the remaining site symmetry and thereby splits orbital levels or mixes distinct spin projections.

The point group fixes which combinations of strain and defect operators can appear together. Numerical values of those allowed couplings come from measurement or from microscopic electronic-structure calculation [Ro77].

[Theory]

Static strain contributes a fixed deformation energy to the model, evaluated separately from lattice vibrations. Setting $\varepsilon = 0$ makes that deformation energy zero, so degeneracies that strain would split stay unsplit.

Dynamic strain and phonon coupling

Chapter 5 introduces phonons, which make the local deformation oscillate in time. A phonon is one quantum of energy in a collective sinusoidal motion of the atoms. For normal mode k , ω_k gives the oscillation rate in radians per second, while $b_k^\dagger$ adds one quantum to that motion and b_k removes one quantum from it. Then

$$H_{\text{ph}} = \sum_k \hbar \omega_k \left(b_k^\dagger b_k + \frac{1}{2} \right), \quad H_{\text{def-ph}} = \sum_{k,\alpha} \kappa_{k\alpha} (b_k + b_k^\dagger) O_{\alpha},$$

$\kappa_{k\alpha}$ sets the energy scale for coupling of that vibrational motion to the defect. A phonon drives a defect transition when the phonon energy matches the required energy change and the coupled distortion preserves the symmetry required by that transition.

Spin relaxation and dephasing proceed through one-phonon emission or absorption, through two-phonon Raman scattering, and through thermal excitation to higher-energy states followed by decay. Which pathway carries most of the rate is set by the defect energy levels, the number of phonon modes available at each energy, and the temperature [Ro78].

[Theory]

Cooling lowers the number of thermally occupied phonons. Static strain persists at low temperature, and spontaneous phonon emission remains possible.

Static strain describes a frozen displacement of atoms around the defect. Phonons describe quantized oscillations of atoms around their equilibrium positions. Both interact with the defect through matching residual-symmetry channels. Deleting the operators b_k from the model keeps the frozen-displacement energy and removes all transitions driven by the oscillations.

Combined effective Hamiltonian

The contributions relevant to a defect-state family collect into the working Hamiltonian below, the operator that determines the modeled energies and dynamics.

$$H_{\text{system}} = H_{\text{CF}} + H_{\text{SO}} + H_{\text{SS}} + H_Z + H_{\text{hf}} + H_Q + H_{\epsilon} + H_{\text{ph}} + H_{\text{def-ph}}.$$

Residual symmetry decides which of these contributions act within a given defect-state family. The existence of this sum does not establish that the defect is a functional device.

Not every term contributes in every defect-state family: residual symmetry can force a matrix element to vanish, and projection into a restricted state manifold can reduce an orbital operator to zero. An isotope may have $I = 0$, eliminating its nuclear-spin terms.

Conversely, an interaction omitted from a simplified level diagram can dominate the experimentally observed linewidth.

The analysis therefore begins with defect geometry and charge state. The orbital localized at the defect is solved first, and the interacting states of many electrons are built from those orbitals afterward.

Only after those steps is the model projected into a selected spin or orbital manifold and restricted to the terms that act within it. Starting from a spin Hamiltonian and working backward treats that projection as given, while the preceding geometric, electronic, and many-body analysis is what justifies it.

Selection rules imposed by residual symmetry

Consider an initial physical configuration $|i\rangle$ and a final physical configuration $|f\rangle$ that change under the remaining rotations and reflections of the defect according to the transformation rules Γ_i and Γ_f , with a perturbing physical quantity O changing according to Γ_O . A representation is the tabulated rule for such changes. The transition amplitude produced by O , namely the matrix element $\langle f|O|i\rangle$, can differ from zero only when the tensor product

$$\Gamma_f^* \otimes \Gamma_O \otimes \Gamma_i$$

includes the fully invariant transformation rule. In that condition, Γ_f^* is the conjugate representation, $\otimes$ denotes a tensor product, and the totally symmetric representation stays unchanged under every remaining rotation and reflection.

The selection rule sets the transition amplitude exactly to zero whenever the ideal remaining symmetry forbids the transition. When the product of transformation rules permits the transition, symmetry establishes the possibility of a nonzero matrix element. The spatial overlap of wavefunctions, the coupling strength, and the lifetime require separate calculation. A deformation of the crystal lattice, called strain, can mix configurations belonging to different transformation rules and open a matrix element that was forbidden.

Remaining symmetry fixes the locations of exact zeros and leaves numerical values undetermined. A deformation of the lattice, a neighboring defect, or a surface can remove the remaining symmetry, and a matrix element that vanished under that symmetry can then acquire a nonzero value.

Experimental properties of fabricated defects

A fabricated sample holds many defect sites with slightly different neighboring atoms, strain fields, and charge surroundings, so measurements average over that spread of local defect environments.

[Experiment] Electron-paramagnetic resonance detects transitions between electron-spin states at magnetic fields set by their energy separation, and records spin splittings together with the further hyperfine splitting produced by coupling between electronic and nuclear degrees of freedom. Photoluminescence records the allowed photon energies emitted after optical excitation.

Optically detected magnetic resonance correlates the two: fluorescence intensity changes when a microwave drive hits spin resonance. Stress spectroscopy and electric-field spectroscopy measure how applied mechanical deformation and applied voltage shift spectral lines in patterns fixed by the remaining point symmetry of the site.

A level model gains support when spin splittings, hyperfine structure, optical transition energies, symmetry-dependent shifts, and many-electron calculations point to the same electronic structure, and that combined agreement carries more weight than any single spectral line [Ro73]; [Ro74]; [Ro75].

Atomic arrangement and strain change from one defect site to another. Replacement of an atom by another isotope with different nuclear spin and magnetic moment changes the strength of the electron-nuclear coupling.

Extra chemical species left from growth or processing move the Fermi-level interval where a given charge state has the lowest formation energy. Termination of the crystal and contact to another material bend the electronic bands with position.

Ion bombardment displaces atoms from lattice sites, leaving vacancies and interstitials together with larger damaged regions alongside the intended defect center. Annealing makes defects mobile so they migrate and meet: that motion sometimes assembles the intended pair and sometimes forms a recombination center without the intended optical and spin behavior.

Lines along which one part of a crystal has slipped relative to another and planes stacked in the wrong sequence create lattice distortion that changes with position and introduce localized distributions of charge that support their own energy levels.

Ideal residual symmetry keeps only the undisturbed spatial symmetry operations as a reference for the energy levels. A weak distortion that lowers that symmetry splits previously degenerate spectral lines and rotates the stationary charge patterns (eigenstates) into new superpositions.

Strain shifts the photon energies of two emitters toward a common value (resonance), and hyperfine coupling gives access to a long-lived nuclear spin for storage. Reliable device fabrication still requires measuring the local distortion and fields and adding their measured strengths directly to the energy model.

The nitrogen-vacancy (NV) center, a nitrogen atom next to an empty lattice site, has unusually complete measured spin, optical, and charge behavior. Those values describe this center specifically; each other vacancy complex needs its own measurement.

Reviews of candidate spin defects identify requirements that hold together: charge that remains in one stable state during operation, spin levels and photon-emitting transitions at usable energies, a procedure that prepares the spin and converts its state into a measurable photon signal, phase stability that persists long enough for control, and processing that fits with device fabrication and optical confinement [Ro73]. [Theory/Experiment synthesis] A host material can meet several of those requirements durably and still fail persistently on one remaining requirement.

Common conceptual errors

- Establishing a spin at a vacancy with a dangling electron requires determining the charge state and many-electron occupancy, because atomic reconstruction and charge transfer can form a closed electronic shell.
- Use as a qubit requires more than an electronic energy level within the host material's band gap: a single in-gap level supplies only one state, while a pair of levels functions as a qubit when it supports initialization, coherent control, and readout with acceptably small leakage and noise.
- A color center absorbs or emits visible light and carries a total spin that is independent of that optical activity, including the possibility of total spin zero. A spin-active center carries nonzero spin and may operate without a useful optical cycle.
- Crystal-field splitting acts primarily on orbital states. Zero-field splitting separates spin sublevels in the absence of an applied field and arises from electron spin-spin interactions and spin-orbit-mediated effects.
- Static strain enters the relevant model as a time-independent deformation, while phonons enter as quantized dynamical modes of lattice motion. Both couple through related residual-symmetry channels and produce different experimental signatures.
- A slipped line is an extended defect with a line core and a long-range elastic field.

A plane where the stacking order shifts extends across two dimensions and acts as a planar defect. The electronic states tied to such an extended defect spread along the plane, with a spatial structure set by that extended geometry.

- Collections of many-electron states linked by their spin and orbital structure are called multiplets, and determining their energies defines the multiplet spectrum. One-electron orbital diagrams from density-functional calculations give starting energies and symmetry labels for that problem, while exchange, correlation,

excited-state relaxation, and spin–orbit coupling set the spacings and orderings of the many-electron states. Quantitative assignment of optical and spin transitions therefore calls for a many-body treatment checked against experiment [Ro70]; [Ro75].

- The rotations and reflections that leave a local atomic arrangement unchanged make up its point group, and the subgroup that survives around a defect is its residual local symmetry. That residual symmetry forbids selected matrix elements through selection rules that hold for as long as the atomic positions preserve the symmetry.

A system with emergent fractionalized excitations and degenerate sectors distinguished by global measurements realizes a many-body topological phase. Protection supplied by local symmetry concerns matrix elements at one site, while protection in such a phase concerns the global degeneracy and braiding of those emergent excitations.

In the classical theory of ordered media, a vacancy, a slipped line, or a mis-stacked plane carries a winding number or a Burgers vector, and that conserved distortion field is the reason for the name topological defect. Quantum topological order with anyons and a protected logical qubit belongs to a separate many-body phenomenon that calls for its own evidence.

Placing many structural defects in a regular array creates a lattice of microscopic degrees of freedom. Showing that those degrees of freedom enter an emergent topological phase calls for writing down a specific many-body Hamiltonian and mapping its phase diagram.

Conceptual checks

- Cyclic permutation of the three carbon orbitals acts on $|a_C\rangle$, $|e_x\rangle$, and $|e_y\rangle$ as a direct symmetry check. Under

$$c_1 \rightarrow c_2 \rightarrow c_3 \rightarrow c_1,$$

this permutation, the equal-weight sum $|a_C\rangle$ stays unchanged, while the two e combinations $|e_x\rangle$ and $|e_y\rangle$ transform into linear combinations of one another.

- Electronic occupancy sets total spin, so reconstruction and charge transfer can produce a closed electronic shell at an empty site.
- The photon-energy relation gives approximately 1.95 eV for a line at 637 nm:

$$hc/\lambda \approx 1240/637 \approx 1.95 \text{ eV}.$$

- For NV^- , the residual C_{3v} symmetry organizes the residual-bond combinations into a_1 singlets and an e doublet, constrains their allowed mixing, and specifies the transverse deformation channel that lifts the doublet degeneracy, while leaving numerical energies undetermined.
- Static strain denotes a time-independent deformation, and phonons denote quantized dynamical modes, with coupling to the defect proceeding through related residual-symmetry channels.

- A spin doublet functions as a qubit after experimental work establishes preparation in a known spin state, rotation of that spin with preserved phase, measurement that distinguishes the two spin projections, stabilization of the surrounding charge configuration during those steps, and confinement of evolution to the two chosen levels with phase memory extending across gate durations.

Characterizing a lattice vacancy together with its neighbors gives the relaxed atomic positions and the wavefunctions trapped at the defect, and counting the electrons in those wavefunctions fixes the total spin while the size of crystal-field, spin–orbit, spin–spin, hyperfine, strain, and phonon couplings sets the splitting and mixing of the resulting levels. Deciding whether a center in diamond meets the operational requirements for a qubit calls for testing those level structures against initialization, control, readout, and stability.

Sources

- [R070] C. Freysoldt, B. Grabowski, T. Hickel, J. Neugebauer, G. Kresse, A. Janotti, and C. G. Van de Walle, “First-principles calculations for point defects in solids,” *Reviews of Modern Physics* 86, 253–305 (2014). DOI: [10.1103/RevModPhys.86.253](https://doi.org/10.1103/RevModPhys.86.253).
 - [R073] G. Wolfowicz et al., “Quantum guidelines for solid-state spin defects,” *Nature Reviews Materials* 6, 906–925 (2021). DOI: [10.1038/s41578-021-00306-y](https://doi.org/10.1038/s41578-021-00306-y).
 - [R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* 528, 1–45 (2013). DOI: [10.1016/j.physrep.2013.02.001](https://doi.org/10.1016/j.physrep.2013.02.001); arXiv: [1302.3288](https://arxiv.org/abs/1302.3288).
 - [R075] Á. Gali, “Ab initio theory of the nitrogen-vacancy center in diamond,” *Reviews of Modern Physics* 91, 015004 (2019). DOI: [10.1103/RevModPhys.91.015004](https://doi.org/10.1103/RevModPhys.91.015004); arXiv: [1906.00047](https://arxiv.org/abs/1906.00047).
 - [R076] M. W. Doherty, N. B. Manson, P. Delaney, and L. C. L. Hollenberg, “The negatively charged nitrogen-vacancy centre in diamond: the electronic solution,” *New Journal of Physics* 13, 025019 (2011). DOI: [10.1088/1367-2630/13/2/025019](https://doi.org/10.1088/1367-2630/13/2/025019); arXiv: [1008.5224](https://arxiv.org/abs/1008.5224).
 - [R077] P. Udvarhelyi, V. O. Shkolnikov, A. Gali, G. Burkard, and A. Pályi, “Spin-strain interaction in nitrogen-vacancy centers in diamond,” *Physical Review B* 98, 075201 (2018). DOI: [10.1103/PhysRevB.98.075201](https://doi.org/10.1103/PhysRevB.98.075201); arXiv: [1712.02684](https://arxiv.org/abs/1712.02684).
 - [R078] A. Norambuena, E. Muñoz, H. T. Dinani, A. Jarmola, P. Maletinsky, D. Budker, and J. R. Maze, “Spin-lattice relaxation of individual solid-state spins,” *Physical Review B* 97, 094304 (2018). DOI: [10.1103/PhysRevB.97.094304](https://doi.org/10.1103/PhysRevB.97.094304); arXiv: [1711.10280](https://arxiv.org/abs/1711.10280).
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PART IV — DEFECT-SPIN PLATFORMS

Several crystals contain defects whose electron spins can be prepared and measured. These chapters examine what experiments establish about their usefulness as qubits.

Chapter 7 — Nitrogen-vacancy centers in diamond

A diamond illuminated with a green laser can shine red from a single atomic-sized spot, with the rest of the crystal staying dark.

Diamond forms a repeating three-dimensional pattern of carbon atoms, and at the glowing spot that pattern has two changes side by side: one carbon position stands empty, forming a vacancy, while an adjacent carbon position holds a nitrogen atom.

One extra electron stays bound at that nitrogen plus vacancy combination, and transitions of that electron give off the observed light.

This irregularity in the crystal pattern is the object under study.

Structure of the nitrogen-vacancy center

Taking out one carbon atom and putting nitrogen in place of a neighboring carbon atom creates a nitrogen–vacancy defect. The structure lies inside the crystal at the atomic scale. Its line connecting nitrogen to vacancy sets a distinguished direction, and no mirror operation interchanges the two ends of that direction.

Green light at 532-nm shone on this defect makes it glow red.

Red photons from the sample spread across a wide range of wavelengths, with a sharp emission peak concentrated near 637 nm.

The sharp emission peak near 637 nm signals a nitrogen atom adjacent to a missing carbon atom carrying a specific net charge.

A localized crystal defect absorbs photons of one wavelength and re-emits photons of another wavelength, and this behavior defines a *center* in the usage here. The re-emitted wavelength gives the observable laboratory color, and the word *center* refers to the fixed lattice site where the defect sits.

The charge configuration used in most applications carries one extra electron, denoted NV^- . Its lowest-energy electronic manifold contains three spin sublevels with total electronic spin $S = 1$, which is called a spin triplet.

NV^- retains its spin-dependent optical response at room temperature. Its optical transition frequency shifts with the surrounding electric field, and the sharp emission peak at 637 nm carries only a small share of the total emitted photons.

Identifying NV^- by its absorption and emission colors does not by itself establish a usable qubit. Qubit operation additionally requires physical processes that set the spin state, drive transitions among the three ground-state sublevels, and read out the resulting spin population optically.

Optical zero-phonon line

Two separated defects can emit photons with matching energy only within the narrow part of the red emission. Most decays transfer part of the energy to shaking of the surrounding crystal, which spreads emitted photon energies over a broad band.

Emission that leaves the crystal shaking unchanged forms a sharp spectral feature. A phonon is a quantized lattice vibration. Physicists call this sharp feature the zero-phonon line, or ZPL, where *zero-phonon* specifies exchange of zero phonons during emission of the photon. Emission in the ZPL carries the full electronic energy difference, so it supplies photons capable of matching the transition of another center.

For NV^- , the ZPL is at 637 nm. Removal of the additional electron leaves neutral NV^0 , with ZPL at about 575 nm [Ro74].

The measured emission wavelength reports the charge occupying a fixed nitrogen-vacancy structure. The nitrogen atom and empty lattice site stay in place while the electron count shifts the optical wavelength.

Room-temperature green illumination drives NV^- through a repeating excitation and emission sequence:

```
532-nm light
|
v
3E excited triplet ---- red fluorescence ----> 3A2 ground triplet
| ^
+-- spin-selective leak into singlets -----+
(stronger for  $m_s = \pm 1$ )
```

NV^- contains a triplet ground-state manifold 3A_2 , an optically excited triplet manifold 3E , and intermediate singlet states [Ro74]. A manifold is a set of related quantum states, and a singlet has total electronic spin $S = 0$. Green illumination lifts population from 3A_2 to 3E . Return to the ground triplet proceeds by direct red fluorescence or by an indirect transition through the singlet states.

The rate for decay from the excited triplet through the singlet manifold depends on spin projection along the NV axis, with larger transfer for $m_s = \pm 1$ than for $m_s = 0$. Under repeated green excitation, population accumulates in $m_s = 0$ because return from the singlets favors that projection. During the initial fluorescence interval, an NV starting in $m_s = 0$ emits more red photons than one starting in $m_s = \pm 1$.

A single green laser drives optical spin preparation by accumulating population in $m_s = 0$ and enables statistical spin readout by mapping spin projection onto fluorescence brightness. At room temperature, the photon counts from one excitation pulse leave overlap between the distributions for different spin projections, so repeated cycles are used to assign the spin state.

Readout fidelity increases with resonant optical excitation at low temperature or with mapping of the electron spin onto a nearby nuclear spin used as an auxiliary memory. These approaches operate under separate experimental conditions from room-temperature fluorescence counting.

Without the spin-selective singlet pathway, green excitation continues to generate red fluorescence with equal emission probability for each spin projection. The steady-state population retains its initial distribution across $m_s = 0$ and $m_s = \pm 1$, and the detected brightness carries information about optical cycling rather than m_s .

Repeated optical excitation changes the defect charge state, converting NV^- into NV^0 .

Illumination at a shorter wavelength returns the extra electron to NV^- with a probability below unity per pulse. Loss of the red signal through conversion to NV^0 reflects electron transfer to the lattice, while loss through spin relaxation reflects transitions among magnetic sublevels at fixed charge.

Electron count and orbital occupancy

The structural designation “NV” identifies a nitrogen atom adjacent to a vacant lattice site, and that geometric label leaves open the count of electrons localized on the defect. Each allowed value of that count defines a distinct electronic system.

Chapter 6 introduced the integer electron-count classification called the charge state. The NV^- charge state carries one additional electron, and that electron configuration gives the $S = 1$ ground state employed in most sensing and qubit applications.

NV^0 possesses its own electronic levels and optical transitions, with a spin structure distinct from the triplet qubit. NV^+ remains generally optically dark under the usual experimental conditions.

The stable charge state is set by nearby donors, which supply electrons, and nearby acceptors, which capture electrons, together with the surface, the optical power, and applied electrodes [Ro74].

Reference to the qubit uses NV^- because the geometric label “NV” leaves the electron number unspecified. The charge state belongs in the physical specification.

The same specificity applies to the group-IV defects discussed below. The established spin–photon charge states are SiV^- , GeV^- , and SnV^- , alongside PbV^- , which is at a less mature stage of study.

Neutral SiV^0 forms a separate spin-triplet system with $S = 1$ and emits on an optical line near 946 nm. Single-center qubit operation with neutral GeV and neutral SnV has a smaller published evidence base.

Charge-state stabilization sets a fabrication and operating condition for devices using these centers [Ro82].

Neutral nitrogen-vacancy centers

A related family places one group-IV impurity atom M —silicon, germanium, tin, or lead—halfway between two missing carbon atoms on neighboring lattice sites:

```
carbon lattice ... vacancy - M - vacancy ... carbon lattice
^ inversion center
```

The impurity with an empty site on either side along the axis gives the split-vacancy structure. Its midpoint acts as an inversion center, so reflecting every atom through that point reproduces the same atomic arrangement.

Rotations about the defect axis, combined with that inversion, leave the split-vacancy geometry approximately unchanged, and this set of operations carries the label D_{3d} . A point group names the collection of such localized symmetry operations. The NV defect has one distinguished axis without such a midpoint mapping and carries the label C_{3v} .

Inversion symmetry makes the optical transition energy insensitive to a uniform electric field at linear order. Separate split-vacancy defects therefore show a narrower spread of optical frequencies, and that suppressed linear Stark shift follows directly from the geometry.

The useful negative charge state, MV^- , has effective electronic spin $S = 1/2$ inside ground and excited manifolds in which several orbital states share the same energy before spin–orbit and related interactions act. The coupling between electron spin and orbital motion splits each manifold, and heavier impurity atoms generally give a larger splitting. At a given cryogenic temperature, a larger splitting weakens some transitions that need phonons to carry away the energy difference. The same large splitting shifts direct microwave spin transitions to frequencies or selection rules that require mixing by strain or by the orientation of the magnetic field [Ro82].

Inversion symmetry accounts for the stability of the optical line against first-order shifts from electric fields, not for straightforward spin control. Without inversion symmetry, first-order electric-field shifts return. Even with inversion symmetry retained, strain, higher-order field dependence, nearby charges, and fabrication damage still move the levels; a Stark shift is a change in an energy level or transition frequency caused by an electric field.

Microwave transitions between spin-triplet sublevels

Quantitative modeling of the NV^- ground-state spin specifies how microwave pulses drive controlled transitions between selected spin levels and thereby define an operational two-level system.

Let the NV symmetry axis define the z direction. Let S_x , S_y , and S_z be the dimensionless spin-1 matrices, with $S_z|m_s\rangle = m_s|m_s\rangle$ and $m_s \in \{-1, 0, +1\}$. The Hamiltonian is expressed in frequency units by dividing the energy operator H by Planck’s constant h . A useful ground-state model is

$$\frac{H}{h} = DS_z^2 + \gamma_e \mathbf{B} \cdot \mathbf{S} + E(S_x^2 - S_y^2) + \mathbf{S} \cdot \mathbf{A} \cdot \mathbf{I} - \gamma_n \mathbf{B} \cdot \mathbf{I} + PI_z^2.$$

This Hamiltonian separates the principal interactions affecting the electronic and nuclear spins. Here D is the axial zero-field splitting in hertz; γ_e is the electron gyromagnetic ratio in hertz per tesla; $\mathbf{B}$ is the magnetic field in tesla; E is a transverse strain- or electric-field-induced splitting in hertz; $\mathbf{I}$ is the nitrogen nuclear-spin operator; $\mathbf{A}$ is the hyperfine tensor in hertz, describing coupling between the electronic and nuclear spins; γ_n is the nuclear gyromagnetic ratio in hertz per tesla; and P is the quadrupole coefficient, present when the nucleus has $I \geq 1$. The term $-\gamma_n \mathbf{B} \cdot \mathbf{I}$ is the nuclear Zeeman interaction, while PI_z^2 represents the nuclear quadrupole interaction.

The parameter D is the zero-field splitting introduced in Chapter 6. The commonly quoted value 2.87 GHz comes from the crystal environment itself. It is the intrinsic crystal-field splitting that separates $m_s = 0$ from the degenerate $m_s = \pm 1$ levels before an external field is applied.

The abundant nitrogen isotope in the NV^- center is ^{14}N , whose nucleus carries spin $I = 1$. The rarer isotope in $^{15}\text{NV}^-$ carries spin $I = 1/2$, and spin-1/2 nuclei have no electric quadrupole moment, so that interaction term vanishes. At room temperature the energy separation between the $m_s = 0$ level and the $m_s = \pm 1$ levels corresponds to $D = 2.87$ GHz, and the electron gyromagnetic ratio is $\gamma_e \approx 28.0$ GHz/T [Ro74].

A compact estimate of the microwave transition frequencies follows from setting crystal distortion and coupling to the nitrogen nucleus to zero and placing the magnetic field along z . Under these assumptions,

$$\frac{E_{m_s}}{h} = Dm_s^2 + \gamma_e B_z m_s.$$

The resulting expression gives the energy of each electron-spin orientation, divided by Planck's constant so that it reads directly in frequency units. At $B_z = 10$ mT = 0.010 T, the shift produced by coupling of the electron magnetic moment to the applied field has magnitude

$$(28.0 \text{ GHz/T})(0.010 \text{ T}) = 0.280 \text{ GHz}.$$

Adding that shift to the zero-field separation gives the two microwave frequencies connecting $|0\rangle$ to the split excited spin orientations as

$$f_{0 \rightarrow +1} = 3.150 \text{ GHz}, \quad f_{0 \rightarrow -1} = 2.590 \text{ GHz}.$$

A microwave pulse tuned to one of those two frequencies drives only that spin-flip transition. With $|0\rangle$ and $|-1\rangle$ serving as the computational states, $|+1\rangle$ persists as a physical level outside that chosen pair, called a leakage state. The system functions as a qubit because it supports preparation by optical pumping, rotation by resonant microwaves, and spin-dependent fluorescence readout, not merely because two energy levels can be identified.

With the magnetic field tilted away from the NV axis, field components perpendicular to that axis together with the E term blend the nominal spin orientations. A two-level model then misses microwave-driven transfer of population into $|+1\rangle$, and that missing population leaves the calculated gate error incomplete.

Hyperfine interaction with the adjacent nitrogen nucleus

Diamond carbon consists mainly of ^{12}C , whose nucleus carries $I = 0$. About 1.1% of natural diamond is ^{13}C , whose nucleus carries $I = 1/2$.

Growth enriched in ^{12}C contains fewer surrounding magnetic nuclei, so the ensemble of uncontrolled spins that perturbs the central electronic spin becomes more dilute. A chosen neighboring ^{13}C can be separately addressed and operated as a long-lived quantum memory [Ro74].

The implanted atom contributes its own magnetic nucleus coupled to the defect electron:

nucleus | nuclear spin I | practical consequence | |

^{14}N / ^{15}N | 1 / $1/2$ | intrinsic NV hyperfine register; ^{14}N also has quadrupole structure | |

^{29}Si | $1/2$ | optional intrinsic SiV nuclear memory; spin-zero Si isotopes remove it | |

^{73}Ge | $9/2$ | supports a larger intrinsic register with a denser hyperfine spectrum | |

^{117}Sn , ^{119}Sn | $1/2$ | supports an isotope-selectable SnV hyperfine degree of freedom | |

^{207}Pb | $1/2$ | supports a possible intrinsic PbV nuclear degree of freedom | |

The group-IV review summarizes these group-IV isotope spins together with their spin-zero alternatives [\[Ro82\]](#).

[Experiment] A nearby nuclear spin provides longer quantum-state storage and repetitive readout through added spectral transitions that require calibration. Experimental control determines the description of that coupling. Unresolved and uncontrolled couplings form a spin bath. Resolved and controllable couplings form a register, a set of addressable quantum degrees of freedom.

A particular ^{13}C nucleus with unresolved and uncontrolled coupling adds to the bath, so isotopic purification improves coherence for that case.

A nucleus with a distinct resonance frequency under active microwave or radio-frequency driving functions as a storage register. Isotopic purification eliminates that storage element along with the surrounding bath spins.

Four crystallographic orientations and their experimental conditions

For NV^- , green laser light excites the defect away from resonance, microwaves rotate the electron spin, and the intensity of the emitted red light reports the spin state. Radio-frequency fields rotate ^{14}N , ^{15}N , or an individually resolved ^{13}C nucleus through its lower-frequency transition.

NV operation proceeds at room temperature in ordinary laboratory conditions. Cooling to cryogenic temperature narrows the optical emission line to the range needed for interference of photons in a network link.

Defects formed by a group-IV atom centered between two vacancies possess inversion symmetry, and that symmetry sets distinct requirements for optical excitation and spin control.

SiV^- responds to laser light tuned to its optical resonance, which pumps population into a chosen spin state and produces fluorescence for readout. The excited state decays back to the starting spin state across many cycles, with only a small probability of leakage per cycle.

The SiV spin rotates under a microwave magnetic field or under a pair of laser fields detuned from an excited state that together drive the spin transition. The optical line stays protected from first-order shifts by the inversion symmetry, so separate emitters radiate photons at closely matched frequencies.

Near 4 K the crystal lattice carries enough vibrational energy to move electrons between the two low-energy orbital branches that lie only tens of gigahertz apart, and each such hop randomizes the spin phase. Cooling the diamond to about 100 mK freezes out those resonant lattice vibrations, which allowed one bulk-device experiment to reach $T_2 = 13$ ms, $T_1 > 1$ s, and 89% single-shot spin readout [\[Ro83\]](#).

These three performance values apply specifically to operation at about 100 mK.

GeV⁻. This defect emits its sharp optical line near 602 nm, and the energy spacing between its two ground-state orbital branches exceeds the corresponding spacing in SiV⁻ [Ro82].

Microwave rotations of the spin together with optical detection of the spin state have been achieved. [Experiment] In a millikelvin device the free precession signal decayed in $T_2^* \approx 1.43 \mu\text{s}$, a single refocusing pulse extended the decay to roughly 0.44 ms, and a train of refocusing pulses in the Carr–Purcell–Meiboom–Gill, or CPMG, sequence extended it past 20 ms [Ro84]. Ramsey T_2^* records dephasing during undisturbed precession, while CPMG inserts repeated refocusing pulses that cancel selected noise, so the value above 20 ms describes coherence under active refocusing.

SnV⁻. This defect emits its sharp optical line near 619 nm, and its two ground-state orbital branches are separated by about 0.85 THz. Because that spacing is larger than in SiV⁻, lattice vibrations at helium-like temperatures excite electrons across the gap less often [Ro82].

Static distortion of the crystal lattice mixes the spin-orbit states enough to give oscillator strength to magnetic transitions that are inactive in the undistorted defect. [Experiment] A cryogenic strained device coupled to a superconducting microwave waveguide gave about 0.43 ms with one refocusing pulse in a Hahn-echo sequence, and T_2 of 10(1) ms with a 64-pulse CPMG sequence; the reported single-pulse fidelity was about 99.1% [Ro85]. The notation 10(1) ms reports the stated uncertainty in the final digit.

These measurements characterize a single center operated with the stated pulse sequences, and array-level median performance calls for separate characterization across many centers.

PbV⁻ optical emission has been assigned to a configuration in which a lead atom sits between two missing carbon sites, a geometry called the lead-vacancy split-vacancy family, and this configuration shows large spin–orbit splittings [Ro82].

Coherent spin initialization, gate operations, readout, and pairwise coupling for the lead-vacancy system remain at an earlier stage of demonstration than the corresponding operations for NV⁻, SiV⁻, GeV⁻, and SnV⁻, and its larger spin–orbit splitting can strengthen thermal isolation while requiring higher-frequency microwave access.

[Proposal] PbV⁻ remains a candidate for experimental evaluation, with directly deployable qubit operation still to be demonstrated.

Optical contrast, phonon isolation, and microwave accessibility combine differently in each group-IV center, so a quoted T_2 value is tied to its temperature and pulse sequence because altered conditions constitute a distinct measurement.

Tabulated properties and associated experimental conditions

The following rows collect published benchmarks obtained under particular temperatures, magnetic fields, samples, pulse sequences, and measurement definitions, and the final column tests those results against a deliberately strict criterion for scalable interacting arrays.

center | useful electronic/optical facts | representative spin result and conditions | readout/control demonstrated | demonstrated coupling | evidence for a scalable interacting array | |

NV^- | $S = 1$; $D \approx 2.87$ GHz; 637-nm ZPL; weak ZPL fraction | $T_2 = 1.8$ ms in an isotopically engineered bulk sample at room temperature under Hahn echo [Ro79] | room-temperature optical preparation/fluorescence and microwave control; cryogenic resonant single-shot methods | two NV electron spins at about 25 nm were entangled at room temperature with dipolar coupling 4.93(5) kHz [Ro80]; remote NV registers later supported a teleported CNOT [Ro81] | No. Pair and network-node demonstrations do not establish a uniform dense lattice | |

SiV^- | $S = 1/2$ Kramers branches; about 737-nm ZPL; strong ZPL emission and inversion symmetry [Ro82] | $T_2 = 13$ ms and $T_1 > 1$ s at about 100 mK in bulk diamond [Ro83] | resonant optical initialization/readout, Raman and microwave control; 89% single-shot readout in that experiment [Ro83] | two SiV optical emitters coupled through one nanocavity showed superradiant/subradiant collective modes [Ro86] | No. The cavity experiment was a selected emitter pair, not a fabricated spin lattice | |

GeV^- | $S = 1/2$; about 602-nm ZPL; larger orbital splitting than SiV^- [Ro82] | $T_2^* \approx 1.43$ μs , echo ~ 0.44 ms, CPMG > 20 ms at millikelvin temperature [Ro84] | optical spin readout and coherent microwave control | local hyperfine coupling is usable; no multi-GeV spin entangling result is used as evidence here | No. Long coherence of one center says nothing about placement yield or coupling uniformity | |

SnV^- | $S = 1/2$; about 619-nm ZPL; ground orbital splitting ~ 0.85 THz [Ro82] | echo ~ 0.43 ms; CPMG-64 10(1) ms in a cryogenic strained, superconducting-waveguide device [Ro85] | optical preparation/readout and strain-enabled microwave control; about 99.1% single-pulse fidelity in that device [Ro85] | coupling to a nearby ^{13}C was observed [Ro85]; photonic-interface work does not yet equal a two-SnV gate | No. Site-controlled creation exists, but a coherent interacting array has not been shown | |

$\text{PbV}^- / \text{SiV}^0$ | PbV^- : less mature group-IV optical center; SiV^0 : distinct neutral $S = 1$ center near 946 nm | no directly comparable benchmark adopted | partial optical/spin functionality, platform dependent [Ro82] | no array-level coupling evidence adopted | No; exploratory. Do not average these distinct charge states into the rows above | |

[Experiment] The 2026 NV network experiment implemented an unconditional teleported controlled-NOT, or CNOT, operation between remote diamond registers. The ^{13}C nuclei served as data qubits, while the NV electron spins provided local logic, readout, and photonic entanglement [Ro81]. This result demonstrates substantial system-level integration. However, the interaction was modular and operated through measurement and feed-forward, meaning that measurement outcomes were used to determine subsequent operations. It was not a static nearest-neighbor interaction energy in a dense crystal array.

Similarly, the two-SiV nanocavity experiment observed coherent photon-mediated collective optical states [Ro86]. In that experiment, light confined in a shared cavity mode linked the optical transitions of selected emitters.

Establishing a deterministic two-spin gate, a uniform graph of couplings, or passive many-body order requires evidence beyond such coupling. A claim of an interaction specifies both the coupled physical quantities and the protocol under which the coupling operates.

An NV electron spin provides a physical two-level system used as a qubit. A collection of controlled electronic and nuclear spins functions as a processor register under a specified encoding of logical information into those spins.

Two remote registers connected through heralded photons form a quantum network. Heralding means that a measurement signal indicates successful generation of the desired photonic link. Establishing an emergent anyon, a topologically ordered medium, or a topologically protected logical qubit requires evidence beyond these observations.

Three characteristic spatial scales

Color centers form through several fabrication routes. Impurity atoms enter the lattice during chemical-vapor-deposition, or CVD, growth. Focused ion implantation followed by annealing places impurities and then heals the lattice. Vacancy creation near pre-existing dopants allows complexes to assemble. Femtosecond-laser processing writes vacancies locally. Annealing is controlled heating that promotes vacancy motion and defect formation. Nanopillars, waveguides, and photonic-crystal cavities subsequently improve photon collection or coupling to an optical cavity. Each fabrication step can introduce strain, surface charge noise, paramagnetic damage, or spectral diffusion, which is a time-dependent fluctuation of the optical transition frequency [Ro82].

Three spatial metrics describe different stages of the fabrication process.

- The spot where an ion beam or laser deposits energy has a spatial precision called delivery resolution.
- Implanted ions scatter along random paths in the crystal, a spread called ion straggle, and vacancies also move and bind into complexes afterward, so the resulting spatial pattern of defects is called final defect distribution.
- Processed sites vary in charge state, orientation, optical transition frequency, and coherence class, and the fraction that meets all required values is called usable-qubit yield.

[Experiment] A 2025 SnV^- study combined focused implantation, local laser annealing, and in-situ fluorescence feedback. It reported sub-50-nm site positioning, implantation doses as low as a Poisson mean of one implanted ion per site, and site-selective formation of individual SnV^- emitters [Ro87]. A Poisson mean of one describes a stochastic implantation process whose expected ion count is one per site.

The same study identified high-yield activation as an unresolved scaling problem. Sub-50-nm site registration does not demonstrate that every site contains exactly one mutually resonant and coherent SnV^- center at a selected three-dimensional coordinate.

CVD growth generally provides high crystalline quality but statistical defect positions. Implantation provides mask-defined coordinates but introduces lattice damage and incurs penalties from conversion yield and ion straggle.

Laser feedback improves control over activation but does not establish an interacting array. To realize a specified defect Hamiltonian, fabrication must simultaneously satisfy requirements on position, orientation, charge state, optical frequency, and coherence.

If five independent fabrication criteria each have a yield below unity, their combined yield is the product of those five probabilities and can therefore be small. This limitation follows from elementary probability rather than from topological physics.

Residual spin-decoherence mechanisms

For NV^- , decoherence and control-error channels with concrete physical origins include precession in the field of the ^{13}C nuclear-spin bath, coupling to substitutional-nitrogen electron spins, coupling to surface spins for shallow defects, magnetic-field drift, temperature- and strain-induced shifts of D , microwave-amplitude errors, optical spectral diffusion, and charge conversion [Ro74]. Isotopic purification removes much of the ^{13}C bath while leaving surface effects and noise from the control electronics in place.

For negatively charged group-IV centers, inversion symmetry lowers the first-order electric-field sensitivity of optical transitions to a small value. Sensitivity to strain, higher-order Stark shifts, local charges, and fabrication damage remains finite.

A principal low-temperature limitation of these defects is scattering of orbital states by lattice vibrations, a process that transfers population or coherence between orbital states. Cooling lowers thermal occupation, and increasing the spin–orbit splitting also lowers the scattering rate.

Mechanical deformation applied for tuning shifts energy levels and mixes orbitals, and that mixing changes which optical and microwave transitions carry oscillator strength and where they appear in frequency [Ro82]; [Ro83]; [Ro84]; [Ro85]. Selection rules specify which transitions are allowed or suppressed by the symmetries and matrix elements of the system.

Packing centers close together activates additional physical processes beyond those seen for a single isolated center. Accumulated lattice damage from implantation degrades the crystal environment, overlapping optical lines complicate frequency-resolved addressing, spatial variation of the driving field produces position-dependent rotation angles, magnetic coupling between neighboring spins provides both a controllable interaction and fluctuating background fields, and confinement in a thin photonic membrane increases coupling to surface charge and surface spins.

[Speculation] A defect that performs best in a pristine bulk sample may perform less well after the fabrication required for coupling and individual addressing. Measurements on fabricated ensembles and arrays are required to determine the actual performance ordering.

Common analytical errors

- A reported value of T_2 acquires meaning only together with the pulse sequence that produced it. T_2^* records free-induction decay under inhomogeneous broadening, Hahn-echo T_2 records decay after one refocusing pulse, and CPMG-protected coherence records decay under a train of refocusing pulses, so each probes a distinct combination of noise spectrum and control. A many-body gate requires compatibility between the decoupling sequence and the interaction, which sets whether a CPMG-protected memory time applies during that gate.
- An optical interface requires specification beyond vacuum wavelength. The relevant physical quantities include the fraction of emission in the sharp zero-phonon line, the homogeneous linewidth, slow wandering of the line center, the probability that excitation yields a photon, collection loss along the detection path, and stability of the charge configuration. Cavity cooperativity quantifies the relative strength of coherent emitter–cavity coupling compared with dissipative losses. An optical line that is narrow in bulk diamond can broaden after nanofabrication.
- Every reported defect requires its charge and isotope designation because those designations fix distinct level structures. $\text{NV}^0 \neq \text{NV}^-$, and $\text{SiV}^0 \neq \text{SiV}^-$. A ^{73}GeV center has a different hyperfine state space from a GeV center containing a spin-zero germanium isotope.
- Scaling from one center to N centers requires characterizing the full array. A single-center coherence time gives one sample, while array operation depends on the distributions of coherence, transition frequency, charge stability, and coupling strength across all sites. Pairwise entanglement validates the control protocol for that measured pair, and lattice-scale operation requires additional characterization of uniformity and crosstalk across the array.

- Remote links and local lattice couplings draw on different physical mechanisms. Heralded photonic links, cavity-mediated optical modes, direct magnetic dipole interactions, and hyperfine interactions each impose distinct requirements for range, timing, and auxiliary control. A teleported gate implements a quantum coupling operation through photonic measurement and feedforward, with operational overhead specific to that photonic process.
- The NV and group-IV results in this chapter establish physical qubits, memories, and network primitives in small registers. Emergent topological order in an array of diamond color centers would require measurements of long-range entanglement or topological ground-state degeneracy, and those measurements are absent from the experiments cited here.

Conceptual checks

- **Charge-state notation for NV centers.** “NV” denotes a nitrogen atom occupying a carbon lattice site next to an empty lattice site. The charge superscript specifies the electron count around that vacancy complex, and the electron count sets the spin and optical level structure used for qubit operation. NV^- carries the extra electron that yields the standard $S = 1$ qubit, while NV^0 and NV^+ hold different electron counts and therefore exhibit different spin and optical spectra.
- **Microwave transition frequencies at $B_z = 10$ mT.** The Zeeman interaction is the energy shift produced by a magnetic field. For the axial field $B_z = 10$ mT, its frequency contribution is

$$(28.0 \text{ GHz/T})(0.010 \text{ T}) = 0.280 \text{ GHz}.$$

Therefore, the two microwave transition frequencies from $|0\rangle$ are given by $D \pm \gamma_e B_z$, where D is the zero-field splitting and γ_e is the electron gyromagnetic ratio. The resulting frequencies are

$$2.87 + 0.28 = 3.15 \text{ GHz}$$

and

$$2.87 - 0.28 = 2.59 \text{ GHz}.$$

Thus, the two frequencies are 3.150 GHz and 2.590 GHz.

- **Temperature dependence of SiV^- operation.** SiV^- denotes the silicon atom at an interstitial position between two vacant sites with one additional trapped electron. Near 4 K, thermal lattice vibrations drive population transfer between the nearby orbital branches, and that transfer shortens spin coherence. Cooling to about 100 mK suppresses that phonon process, and the reported $T_2 = 13$ ms, $T_1 > 1$ s, and 89% single-shot readout were measured under that colder condition. Here, T_2 is the spin-coherence time, T_1 is the longitudinal spin-relaxation time, and single-shot readout is the determination of the spin state from one measurement attempt.

- **Interpretation of a 10 ms CPMG result.** CPMG denotes the Carr–Purcell–Meiboom–Gill dynamical-decoupling sequence, in which a train of refocusing pulses applied to the spin suppresses specified dephasing noise. The 10 ms value gives the coherence time measured while those refocusing pulses were applied, so it quantifies storage under active decoupling. A many-body gate acquires that full 10 ms budget only when its pulse timing remains compatible with the CPMG refocusing pulses.
- **Scope of direct two-NV dipolar entanglement.** A direct dipolar interaction is the magnetic coupling between two spin dipole moments. Experiments using this interaction have established coherent coupling and entanglement for a selected pair of NV centers at room temperature. Controlled yield and uniformity for a large array remain unestablished in those experiments.
- **Limitations of sub-50-nm site registration.** Site registration specifies the measured location of a defect-delivery site. A scalable interacting array requires occupancy, charge state, resonance frequency, coherence, and designed coupling as independent requirements alongside sub-50-nm registration. The cited SnV study, concerning tin-vacancy centers, identified high-yield activation as an unresolved problem.

NV[−] provides mature room-temperature spin control but has a comparatively difficult optical interface, meaning the mechanism that couples the spin qubit to optical photons. Inversion-symmetric group-IV negative centers, which are negatively charged vacancy defects containing group-IV impurity atoms and possessing inversion symmetry, provide stronger coherent optical emission and generally require cryogenic operation that accounts for phonon-induced processes.

The charge state and isotope are both part of the device specification. Direct dipolar coupling, hyperfine coupling between electron and nuclear spins, cavity-mediated coupling through a confined optical mode, and remote photonic coupling have all been demonstrated in limited settings.

No cited result involving a single center, a pair of centers, a cavity, a network, or site placement establishes a scalable interacting diamond array.

Sources

- [Ro74] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* 528, 1–45 (2013). DOI: [10.1016/j.physrep.2013.02.001](https://doi.org/10.1016/j.physrep.2013.02.001).
- [Ro79] G. Balasubramanian et al., “Ultralong spin coherence time in isotopically engineered diamond,” *Nature Materials* 8, 383–387 (2009). DOI: [10.1038/nmat2420](https://doi.org/10.1038/nmat2420).
- [Ro80] F. Dolde et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* 9, 139–143 (2013). DOI: [10.1038/nphys2545](https://doi.org/10.1038/nphys2545).
- [Ro81] M. Iuliano et al., “Unconditionally teleported quantum gates between remote solid-state qubit registers,” *Nature Communications* 17 (2026). DOI: [10.1038/s41467-026-72818-6](https://doi.org/10.1038/s41467-026-72818-6).
- [Ro82] C. Bradac, W. Gao, J. Forneris, M. E. Trusheim, and I. Aharonovich, “Quantum nanophotonics with group IV defects in diamond,” *Nature Communications* 10, 5625 (2019). DOI: [10.1038/s41467-019-13332-w](https://doi.org/10.1038/s41467-019-13332-w).
- [Ro83] D. D. Sukachev et al., “Silicon-vacancy spin qubit in diamond: a quantum memory exceeding 10 ms with single-shot state readout,” *Physical Review Letters* 119, 223602 (2017). DOI: [10.1103/PhysRevLett.119.223602](https://doi.org/10.1103/PhysRevLett.119.223602).

- [Ro84] K. Senkalla, G. Genov, M. H. Metsch, P. Siyushev, and F. Jelezko, “Germanium vacancy in diamond quantum memory exceeding 20 ms,” *Physical Review Letters* 132, 026901 (2024). DOI: [10.1103/PhysRevLett.132.026901](https://doi.org/10.1103/PhysRevLett.132.026901); arXiv: [2308.09666](https://arxiv.org/abs/2308.09666).
 - [Ro85] I. Karpatzakis et al., “Microwave control of the tin-vacancy spin qubit in diamond with a superconducting waveguide,” *Physical Review X* 14, 031036 (2024). DOI: [10.1103/PhysRevX.14.031036](https://doi.org/10.1103/PhysRevX.14.031036).
 - [Ro86] R. E. Evans et al., “Photon-mediated interactions between quantum emitters in a diamond nanocavity,” *Science* 362, 662–665 (2018). DOI: [10.1126/science.aau4691](https://doi.org/10.1126/science.aau4691); arXiv: [1807.04265](https://arxiv.org/abs/1807.04265).
 - [Ro87] X. Cheng et al., “Laser activation of single group-IV colour centres in diamond,” *Nature Communications* 16, 5124 (2025). DOI: [10.1038/s41467-025-60373-5](https://doi.org/10.1038/s41467-025-60373-5).
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Chapter 8 — Chromium and other defects in corundum

Ruby is a hard, transparent, red crystal of aluminum and oxygen in which a small fraction of aluminum atoms are replaced by chromium. Light absorption by those chromium atoms gives the crystal its red color.

Sapphire is the same aluminum-oxygen crystal, called corundum, with different atoms replacing aluminum. The replacement species and its charge determine the absorption spectrum, so corundum appears colorless, blue, yellow, or another color.

Gemological grading sorts these corundum crystals mainly by color. The technical question is whether the established impurities in corundum support initialization, coherent rotation, and readout of individual quantum systems, operations already demonstrated with a defect in diamond consisting of a nitrogen atom on a carbon site next to an empty carbon site, called a nitrogen-vacancy center.

The first operating laser used ruby, and early devices that amplify microwaves into a coherent beam through stimulated emission, called masers, also used ruby.

Corundum in which titanium replaces aluminum, called titanium-doped sapphire, continues to serve in many laboratories as a light-amplifying crystal whose output wavelength is adjustable over a broad range. Impurities in corundum have supplied useful optical and microwave behavior for approximately sixty years.

The central issue is why defects in diamond, rather than established impurities in corundum, are routinely initialized, coherently rotated, and read out as individual quantum systems.

Corundum, ruby, and sapphire

Corundum holds oxygen ions in an approximately close-packed stack, with aluminum ions filling two thirds of the regions between host ions that can accommodate another ion, called interstitial sites.

Corundum names the aluminum oxide host crystal, in which small changes in impurity composition produce the gemstones called *ruby* and *sapphire*.

Replacement of some aluminum ions by chromium ions makes the crystal ruby, while the same aluminum oxide lattice serves as a window, watch crystal, or microwave resonator under the name sapphire.

Chromium substitutes readily for aluminum because both elements commonly carry a +3 charge, so chromium can occupy an aluminum site without necessarily requiring an additional charge-compensating defect.

Six oxygen ions surround each chromium ion in a slightly distorted octahedron, and this neighboring oxygen geometry sets the electronic energy levels of the chromium impurity.

The distortion at the chromium site favors particular orientations of the spins of the three chromium electrons at zero applied magnetic field, and absorption of light lifts these electrons into higher-energy electronic states.

Microwave fields drive transitions among the four lowest spin states of the chromium impurity. Optical pumping through its optical transitions supplied the population inversion for the first laser, and microwave-driven transitions between its electron-spin energy levels support electron spin resonance spectroscopy, abbreviated ESR.

Ruby and sapphire share the same corundum host lattice and differ in the impurity species they contain. A crystal with many equivalent local coordination environments provides repeated copies of the same defect environment, and establishment of a register of individually identified and addressable qubits requires demonstration of single-site identification and control beyond that repetition.

At chromium concentrations used in a ruby laser, the optical field in a laser spot and the microwave field of a resonator address a collection of nominally similar emitters or spins measured collectively, called an ensemble, containing many ions. At dilution sufficient for spatial isolation of individual ions, emitted photons or microwave absorption from one ion produce a signal that can fall below the experimental noise floor.

Sapphire supports a microwave resonator with exceptionally low loss, meaning stored microwave energy decays slowly. Verification of a many-body phase calls for thermodynamic and correlation measurements, and verification of an individually controllable defect system calls for single-site initialization, manipulation, and readout, so assessment of corundum defects calls for the same operational criteria already satisfied by diamond defects.

Ground-state spin structure of Cr^{3+}

A chromium ion that has lost three electrons keeps three electrons in its outer $3d$ shell. Exchange energy favors parallel alignment of the three electron spins, giving total spin $3/2$ with four allowed spin projections:

$$m = -3/2, -1/2, +1/2, +3/2.$$

An applied magnetic field shifts the energies of the four spin levels through the Zeeman interaction.

The six oxygen ions around chromium form a cage that is stretched along one direction, and that uneven electric environment shifts the four spin orientations into two distinct energy pairs even when laboratory magnets apply no field. Physicists call that remaining separation zero-field splitting, defined in Chapter 6, and spectrometers measure it directly as the energy difference between the spin states at zero applied magnetic field.

A chromium atom that has given up three electrons to bonding becomes an ion written Cr^{3+} , and the three electrons it keeps reside in its d shell, written $3d^3$.

In the cage of oxygen ions, the lowest available d orbitals each hold one of the three electrons, and the electrons point their spins in the same direction to reduce mutual repulsion. That arrangement has total spin $S = 3/2$ and a single orbital ground configuration, written as the term symbol 4A_2 , where a single orbital ground configuration means the orbital part has no remaining degeneracy. The oxygen cage is stretched slightly along the crystal c -axis, and the electric field from that stretched arrangement creates the zero-field splitting [Ro88].

Spin direction is tracked with the dimensionless operators S_x, S_y, S_z , where S_z takes the four m values listed above as its eigenvalues. The energy scale of the crystal distortion is D , the conversion between electron magnetic moment and energy is the Bohr magneton μ_B in joules per tesla, the laboratory field components in tesla are B_i , and the

proportionality between spin and magnetic moment along and across the crystal axis is given by the dimensionless numbers $g_{\parallel}, g_{\perp}$ parallel and perpendicular to the c -axis. With those quantities, the effective ground-state Hamiltonian is

$$H = g_{\perp}\mu_B(B_x S_x + B_y S_y) + g_{\parallel}\mu_B B_z S_z + D \left[S_z^2 - \frac{S(S+1)}{3} \right].$$

The first two terms give the energy cost of tilting the electron spin relative to the applied magnetic field, which is the Zeeman interaction. The final term gives the energy cost imposed by the stretched oxygen cage, which is the zero-field splitting. This Hamiltonian describes processes whose energies stay within the four orientations of the $S = 3/2$ ground-state manifold.

With no applied field, the combination $S(S+1)/3 = 5/4$ enters the Hamiltonian as a constant shift, giving eigenenergies

$$E_{\pm 3/2} = D, \quad E_{\pm 1/2} = -D.$$

For ruby, $D < 0$. The $|\pm 3/2\rangle$ pair is therefore lower in energy, and the frequency separation between the two doublets is

$$\nu_0 = \frac{-2D}{h} \simeq 11.493 \text{ GHz},$$

The splitting is expressed with h as Planck's constant [Ro88]; [Ro89]. At zero magnetic field, reversal of time leaves the energy of a spin projection $+m$ equal to the energy of $-m$. The four states therefore form two degenerate doublets rather than four separate spectral lines.

The energy of the zero-field splitting corresponds to a thermal scale through the following evaluation

$$\frac{h\nu_0}{k_B} = \frac{(6.626 \times 10^{-34} \text{ J s})(11.493 \times 10^9 \text{ s}^{-1})}{1.381 \times 10^{-23} \text{ J K}^{-1}} \approx 0.552 \text{ K}.$$

Multiplying joule-seconds by inverse seconds gives an energy in joules. Dividing this energy by joules per kelvin gives the temperature in kelvin whose thermal energy matches the zero-field transition frequency.

Cooling to temperatures well below approximately 0.5 K makes the thermal population of the upper doublet small relative to the lower doublet. Operation of a ruby laser at room temperature establishes the existence of an optical excitation and emission cycle, but it does not establish that the ground-state spin is predominantly initialized in its lowest-energy doublet.

A magnetic field directed along the crystal c -axis adds Zeeman energy proportional to the spin projection, so the Hamiltonian remains diagonal in the $|m\rangle$ basis:

$$E_m = D \left(m^2 - \frac{5}{4} \right) + g_{\parallel} \mu_B B_z m.$$

A microwave magnetic field drives spin flips between states of definite magnetic projection $|m\rangle$, with appreciable strength only when the projection changes by one unit, $\Delta m = \pm 1$. Tilting the applied magnetic field away from the c -axis mixes these basis states, so each energy eigenstate acquires components with several projections and the flip strengths redistribute. At particular tilt angles and field magnitudes the transition frequency passes through an extremum versus field, giving operating points called clock transitions, where the frequency is insensitive to magnetic-field fluctuations to first order.

A clock transition suppresses magnetic-field-induced phase spreading, which can lengthen the persistence of a superposition. That protection is distinct from a determination of the persistence time: the existence of such a transition does not itself constitute a measurement of the coherence time T_2 . Here, T_2 is the timescale over which a coherent superposition retains its relative phase.

Breaking time-reversal symmetry supplies an energy difference within each zero-field doublet and separates all four levels. When time-reversal symmetry is retained, the four levels remain grouped as two degenerate pairs. Selecting two of the four levels for computation leaves the remaining pair physically accessible, so describing the device as an ideal two-level atom introduces possible leakage into states outside the selected qubit subspace.

Optical and microwave energy scales

The chromium ion absorbs red light in addition to microwaves. At cryogenic temperature, the absorption spectrum shows two sharp optical lines near 693.6 nm and 692.2 nm, corresponding to approximately 1.788 eV and 1.792 eV [Ro88].

Absorption of these red photons promotes the ion from any of the four ground-state spin levels to a higher-energy pair of electronic states denoted 2E in spectroscopic notation.

The numerical transition energies identify the optical energy scale, while the 2E label specifies the excited-state term required for comparison with the spectroscopy literature.

The optical transition is formally spin-forbidden, meaning that it violates the leading-order spin selection rule. Coupling to lattice vibrations nevertheless gives the transition a nonzero amplitude.

Resonant optical excitation can distinguish the ground-state sublevels, and phonon-assisted fluorescence can be used to infer their populations. These processes enable ensemble optical initialization and ensemble detection of magnetic resonance, but they do not automatically provide single-shot measurement of an individual chromium spin.

The optical and microwave energy scales differ because the oxygen coordination environment first separates orbital terms on the electron-volt scale. The smaller local distortion, together with spin–orbit coupling between the electrons’ spin and orbital degrees of freedom, then produces a residual gigahertz-scale splitting within the ground term. Optical transitions involve an energy of approximately 1.8 eV, whereas microwave transitions involve energies of tens of microelectron-volts.

The optical and microwave transitions therefore do not constitute equivalent descriptions of a single two-level atom. A complete model must include their distinct selection rules and the leakage channels associated with the four-level ground-state structure.

Microwave losses and the nuclear-spin environment

Sapphire is transparent over a broad optical range, mechanically hard, chemically stable, and an exceptional microwave dielectric. In a high-purity sapphire disk, a microwave field can circulate by repeated internal propagation near the boundary of the dielectric, forming a resonant electromagnetic mode called a whispering-gallery mode.

Stored energy divided by the energy lost per radian of oscillation defines the resonator quality factor Q . A large Q describes low loss in the host sapphire resonator, and it is not a measurement of the coherence time of an impurity spin.

[Experiment] At about 100 mK, energy stored in a high-purity sapphire whispering-gallery resonator leaks slowly enough to give microwave Q values in the 10^8 – 10^9 range. For an 11 GHz mode with $Q = 2 \times 10^9$, that slow leakage corresponds to a photon energy-decay timescale of $Q/(2\pi f) \approx 29$ ms [Ro90].

Aluminum nuclei in corundum form a magnetic environment that fluctuates more strongly than the nuclear environment of isotopically enriched diamond. Natural aluminum consists essentially entirely of ^{27}Al , which has nuclear spin $I = 5/2$. Every substitutional chromium ion therefore sits inside a dense bath of host nuclear spins, meaning an ensemble of surrounding nuclei whose magnetic fluctuations can perturb the electron spin, called a nuclear-spin bath.

The chromium nucleus itself can be chosen to carry no spin by selecting the dominant isotope. Natural ^{52}Cr , the dominant chromium isotope, has $I = 0$, whereas ^{53}Cr has $I = 3/2$ [Ro88]; [Ro90]. Isotopic selection can therefore remove the chromium ion's own nuclear spin.

Aluminum has no spin-zero isotope available, so the aluminum nuclear-spin bath cannot be eliminated by isotopic enrichment.

Coupling between the chromium electron spin and surrounding aluminum nuclear spins, plus time-dependent fluctuation of the transition frequency from changes in that surroundings, remain present even in a perfectly grown crystal. The electron-nuclear coupling produces splitting called hyperfine structure. The time-dependent frequency fluctuation is called spectral diffusion. A model that omits the aluminum nuclear spins does not describe actual corundum.

Experimental measurements on ruby

A laser spot focused on ruby illuminates a volume holding many chromium ions. The optical signal averages over that volume, so the resulting measurement is an ensemble measurement.

An ensemble measurement collects light emitted by many chromium ions at once. Such a measurement must not be interpreted as control or readout of one isolated spin even when the averaged numbers look favorable.

[Experiment] Sewani and co-workers studied a 0.005% ruby crystal in a dilution refrigerator. Under 15 nW continuous resonant excitation, they inferred a ruby temperature of 143 ± 7 mK, measured a maximum timescale over which spin-state populations return toward thermal equilibrium, called the spin-lattice relaxation time $T_1 = 3.67 \pm 0.35$ s, and observed transitions detected as changes in an optical signal, called optically detected magnetic resonance, or ODMR, [Ro88]. The spin-lattice relaxation time T_1 is the timescale over which spin-state populations return toward thermal equilibrium. ODMR detects spin transitions through changes in an optical signal.

Their resonant confocal measurement collected emission from an area of approximately $1 \mu\text{m}^2$, which contains many ions rather than a single ion. The paper identified coherence-time measurement as future work.

It did not report Ramsey fringes, Hahn-echo T_2 , single-qubit gates, single-shot readout, or two-ion entanglement. Ramsey and Hahn-echo measurements are pulse protocols used to characterize phase coherence and refocus selected sources of dephasing, respectively.

The distinction between T_1 and T_2 prevents the incorrect substitution of a population-relaxation time for a phase-coherence time. A seconds-long T_1 at millikelvin temperature is favorable, but phase coherence may decay much more rapidly because of aluminum nuclear spins, chromium–chromium dipolar interactions, magnetic-field noise, and inhomogeneous strain.

[Experiment] A 2025 experiment placed a 0.05% ruby sample on a 5.5 GHz high- T_c superconducting resonator and measured ensemble spin–photon coupling from 30 mK to 16 K [Ro89]. A high- T_c superconductor is a material with a comparatively high superconducting transition temperature.

At 65 mK, one transition in ruby exchanged energy with the cavity field at a collective rate of $\bar{\Omega}/2\pi \approx 21$ MHz while spins lost phase coherence at $\gamma_s/2\pi \approx 44$ MHz. The ratio comparing coherent exchange with dissipative losses, called cooperativity, was approximately 15, and the coupling did not satisfy the strict strong-coupling condition $\bar{\Omega} > \gamma_s$ [Ro89]. The sample contained an estimated 2.6×10^{15} spins.

The authors divided the measured collective interaction by the participating spin population to obtain the interaction strength belonging to one spin averaged over the crystal [Ro89]. That averaging procedure gave approximately 0.41 Hz over all spins, or 1.25 Hz when only the subset in the high-field-volume region was counted.

These results establish collective exchange of energy between the spin ensemble and the cavity field, and the measurements do not provide strong coupling between individually selected neighboring spins.

The measured 21 MHz rate includes growth by the square root of the number of participating spins, $\sqrt{N}$, where N is the number of participating spins. It is not the coupling of one defect. In that experiment, spin damping removed coherence too rapidly for a photon to be coherently re-emitted.

Attributing the full 21 MHz to one spin would contradict the same paper's model, which places the single-spin coupling in the hertz range because collective enhancement from the large spin population accounts for the observed megahertz rate.

Other transition-metal impurities

Sapphire hosts transition-metal ions that replace aluminum in the lattice, where the ions settle into several charge states depending on growth and charge compensation. Each valence and compensation arrangement sets its own electronic levels and optical transitions, so different impurity histories produce substantially different electronic and

optical systems.

Species in sapphire	Established behavior	Relevance to defect-qubit operation
$\text{Cr}^{3+}, S = 3/2$	Ruby R-line fluorescence, laser/maser gain, EPR/ODMR, seconds-scale T_1 at mK, ensemble cavity coupling [R088]; [R089]	Best-characterized candidate, but the complete set of modern single-center coherent-control capabilities is absent
$\text{Fe}^{3+}, S = 5/2$	EPR and microwave transitions in nominally pure sapphire; useful or parasitic in frequency standards [R090]	Demonstrated ensemble spectroscopy, not optical single-spin initialization/readout
$\text{V}^{2+}, S = 3/2, {}^{51}\text{V}$ with $I = 7/2$	Eight-line hyperfine structure resolved by whispering-gallery ESR [R090]	Multiple spin and hyperfine levels, but substantial nuclear-spin complexity and no demonstrated qubit lifecycle
$\text{Ti}^{3+}, 3d^1$	Broad vibronic gain underlying the tunable Ti:sapphire laser, approximately 650–1100 nm [R091]	Excellent classical gain medium; broad electron–phonon optical transitions are not evidence of a narrow spin–photon qubit interface
Co, Ni, Mn and other ions	Optical absorption/EPR reported under charge-state- and growth-dependent conditions	A spectroscopic line identifies a candidate species but does not establish a scalable defect center

Vibronic gain describes optical amplification in which an electronic transition changes together with vibrational excitation of the lattice. A qubit lifecycle describes initialization, coherent control, and readout acting on the same physical degree of freedom.

[Experiment] Farr and co-workers detected native Fe^{3+} , Cr^{3+} , and V^{2+} impurities at parts-per-billion to parts-per-million concentrations in high-purity sapphire near 115 mK using 8–19 GHz whispering-gallery spectroscopy [R090]. This sensitivity carries two physical consequences. Dilute spins couple measurably to microwave modes, and nominally pure sapphire already contains uncontrolled spin species.

A catalog of impurity species is not a catalog of operational qubits. A spectroscopic line identifies a candidate species present in the crystal, but it does not demonstrate initialization, coherent rotation, or readout of an individual spin.

Oxygen vacancies

Removing an O^{2-} ion leaves an unoccupied oxygen lattice site denoted V_{O} . Charge-state terminology for this site counts how many electrons remain trapped at the vacancy.

A nominally neutral oxygen vacancy that traps two electrons is called an F center. In the literature, this term denotes a specific defect species with that two-electron occupancy.

Removing one electron from the same vacancy produces an F^+ center, also denoted V_{O}^+ , with one remaining electron. Aggregates of vacancies are labelled F_2 , F_2^+ , and so forth, although historical assignments can depend on sample-processing conditions.

[Experiment] Aluminum oxide crystals that have been exposed to radiation or heated in an oxygen-poor atmosphere contain missing oxygen atoms that trap electrons. The vacancy trapping two electrons absorbs light near 6.1 eV and gives off light near 3.0 eV in the defect known as the F center. The vacancy trapping one electron absorbs near 4.8

and 5.4 eV and gives off light near 3.8 eV in the defect known as the F^+ center [R092]; [R095].

The F -center emission lifetime was measured as 36 ± 4 ms, consistent with a relaxed transition in which the electron spin must flip [R092]. [Theory] Calculations using a modern electronic-structure approximation reproduce the principal absorption and emission assignments and distinguish transitions in which the electron stays localized at the vacancy from transitions involving band edges [R093]. A hybrid density functional is an electronic-structure approximation that combines components of local or semilocal density-functional theory with nonlocal exact exchange.

Matching calculated optical energies to measured absorption and emission energies checks the spectroscopic assignment, and agreement between measured and calculated optical energies does not establish qubit operation.

The vacancy with one unpaired trapped electron carries a magnetic moment and can therefore be observed by EPR in suitable samples, which is the property described by the term paramagnetic for the F^+ center. The two trapped electrons of the neutral F center can form singlet and triplet configurations involved in its luminescence. A singlet has total spin zero, whereas a triplet has total spin one.

For the literature survey used here through 24 August 2026, no peer-reviewed report was located that demonstrated an isolated oxygen-vacancy center in bulk sapphire with all of the following capabilities: photon antibunching tied to that center, spin-selective optical initialization, coherent single-spin rotations, a measured single-center T_2 , and single-shot readout. Photon antibunching is a suppression of simultaneous photon detections and is commonly used as evidence of emission from an individual quantum emitter.

The available literature establishes measurements averaged over many defects and calculations of defect properties for color-center spectroscopy and theoretical modeling [R092]; [R093]; [R095]. The complete operational sequence required for a single-defect qubit has not been demonstrated.

Oxygen vacancies form when irradiation knocks oxygen atoms out, when growth uses little oxygen, or when annealing drives oxygen out. Illumination shifts their charge state and lets them cluster together. Each of these handles adjusts how many vacancies sit in the crystal.

Those vacancy-creation steps knock host atoms out of place and leave different atomic surroundings from site to site. Excitation with ultraviolet light at approximately 5–6 eV couples with higher loss in nanophotonic structures and with lower compatibility with biological samples and fiber optics, while visible or near-infrared excitation couples with lower loss and broader compatibility in those settings.

Calling an oxygen vacancy a deep defect in a wide-band-gap material assigns its electronic level far from the band edges. That electronic assignment does not define a device architecture.

Chromium implantation and spatial control

Bulk ruby growth spreads chromium atoms through the crystal at a uniform average concentration with random positions. Focused-ion-beam or masked implantation drives chromium atoms into a chosen surface area with a chosen depth range.

[Experiment] Chromium implanted into sapphire at 150 keV, followed by annealing at 1450 °C, produced substitutional ruby-like R-line luminescence in a thin implanted layer [R094]. The demonstrated fluences were ensemble-scale, 6×10^{14} to 4×10^{15} cm⁻², and the reported internal optical efficiency was approximately 50% under those processing conditions [R094].

That experiment shows implanted chromium turning optically active after annealing. Building a nanometre-scale interaction graph from this route requires managing the high annealing temperature, the statistical spread in implanted-ion positions from scattering in the target, called collision straggle, the leftover lattice damage, the selection of charge state, and the motion of chromium by diffusion.

No sapphire result located for this chapter combined deterministic registration of a single implanted ion, verified optical and spin behavior of one chromium center, and controllable coupling between two chromium centers. Diamond implantation also exhibits straggle and surface noise, but single-defect verification has been demonstrated many times in diamond [R074].

Chromium implantation places chromium atoms into a sapphire crystal as a materials-processing result, and the sharp red R-line emission from a thin implanted layer shows occupation of aluminum sites by chromium ions. That layer-averaged optical signal does not determine the exact occupied sites and does not demonstrate coherent interaction between two selected ions, so the work is not a demonstration of deterministic defect placement.

Comparison with diamond defects

A direct comparison rests on capabilities demonstrated together under stated experimental conditions, rather than assembling hypothetical optimum features from separate experiments.

Criterion	Sapphire/corundum evidence	Diamond reference point	Consequence for a selected-neighbor qubit register
Single-defect identification	Ruby and vacancy work reviewed here is predominantly ensemble spectroscopy; no intrinsic sapphire antibunching-plus-spin-control demonstration was located [R088]; [R092]; [R095]	Single NV centers support room-temperature confocal fluorescence, ODMR, initialization and coherent control; mature review in [R074]	Diamond decisively ahead
Coherence	Cr ruby: $T_1 = 3.67 \pm 0.35$ s at mK; that study did not measure T_2 [R088]	Single NV Hahn-echo $T_2 = 1.8$ ms at room temperature in isotopically purified bulk diamond [R079]	Values are not directly comparable; sapphire's needed single-spin T_2 datum is absent
Optical interface	Narrow ruby R lines and ensemble resonant readout near 693 nm at cryogenic temperature [R088]	Single-NV spin-photon entanglement and remote-node components have been demonstrated [R074]	Sapphire has spectroscopy, not a demonstrated single-spin network node
Microwave interface	Very high- Q sapphire resonators and collective coupling; 2025 ruby device had 21 MHz ensemble coupling versus 44 MHz spin decay [R089]; [R090]	Diamond spin ensembles and single NVs couple to microwave structures; single-defect control is routine [R074]	Sapphire excels as a resonator, not yet as a register of individually coupled defects
Placement	Random bulk doping; ensemble Cr implantation and activation after 1450 °C anneal [R094]	Implantation, delta doping, nanofabrication, and registration all demonstrated, with nonzero yield/straggle limitations [R074]	Neither is perfect; sapphire lacks the single-center end-to-end evidence
Density and selected interactions	Cr-Cr effects and collective $\sqrt{N}$ enhancement exist, but chosen pairwise coherent gates were not located [R089]	Dipolar-coupled NV pairs and multi-spin registers have been coherently controlled [R074]	Ensemble interaction does not define an individually addressable interaction graph

Criterion	Sapphire/corundum evidence	Diamond reference point	Consequence for a selected-neighbor qubit register
Temperature	Optical ruby operation is possible at room temperature, but the strongest quantum-spin characterization here used 65–186 mK [R088]; [R089]	NV spin initialization, control and readout operate at room temperature; high-quality optical networking usually uses cryogenics [R074]	The consequence depends on the application, but sapphire does not show NV-like room-temperature qubit operation
Host spin bath	Unavoidable dense ^{27}Al , $I = 5/2$	^{12}C , $I = 0$, can be isotopically enriched [R074]; [R079]	Diamond has a fundamental materials advantage

Claim	Status	Platform	Evidence	Reference
Optical population readout, ODMR and seconds-long T_1	Experimentally demonstrated	Cr:ruby ensemble, dilution refrigerator	Resonant PLE/ODMR and all-optical relaxation	[R088]
Coherent strong coupling of one Cr spin to one cavity photon	Not demonstrated	Cr:ruby	Inferred average single-spin coupling in hertz; collective mode failed $\bar{\Omega} > \gamma_s$	[R089]
Trace transition-metal spins couple to sapphire microwave modes	Experimentally demonstrated	Fe, Cr, V ensembles	Avoided crossings and ESR/hyperfine spectra	[R090]
Oxygen vacancies are optically active deep defects	Experimentally demonstrated and theoretically modeled	F , F^+ in alpha-alumina	Ensemble absorption/luminescence plus hybrid-DFT assignments	[R092]; [R093]
Oxygen vacancy is an optically readable single spin qubit	No supporting demonstration located	Sapphire V_O	Required antibunching/control/read out sequence absent from located literature	[R092]; [R093]
Implanted chromium can be optically activated	Experimentally demonstrated, ensemble	Cr-implanted sapphire	R-line luminescence after high-temperature anneal	[R094]
Deterministic interacting array of sapphire defects	No supporting demonstration located	Any sapphire center	No selected pair gates, array calibration, or many-body Hamiltonian validation located	[R088]; [R089]; [R090]; [R091]; [R092]; [R093]; [R094]
Sapphire defects realize topological order	Speculation	Sapphire defect lattice	No microscopic implementation or phase diagnostic	—

A chromium ion in ruby carries an electronic spin with addressable magnetic sublevels, which is a physical spin. Calibrated coherent interaction between two such selected spins, where the quantum state of one controls evolution of the other, would constitute a coupled-qubit primitive.

A macroscopic spin ensemble shares one microwave cavity photon through collective magnetic coupling, forming a distinct collective degree of freedom with one hybrid resonance. An individual physical spin, a coupled pair, and a collective ensemble mode become an encoded qubit, an emergent anyon, or a realization of topological order only with the additional encoding, braiding, or ground-state degeneracy specified for those phases, so existence alone does not establish them.

The cavity anticrossing reported in 2025 shows hybridized spin–photon modes from collective magnetic coupling, which is evidence for a collective hybrid spin–photon system and not for a topological material.

Common analytical errors

A long longitudinal relaxation time T_1 , which characterizes the decay of spin-state populations toward thermal equilibrium, must not be interpreted as a long transverse coherence time T_2 , which characterizes the loss of relative phase coherence. Populations persist in an excited state for seconds in systems where relative phase washes out much faster, so an array proposal needs direct phase measurements on the proposed defect species at the proposed density, temperature, and device geometry: Ramsey measurement of free-induction decay, spin-echo measurement that refocuses slow frequency wander, and error measurement of driven gates.

When N similar spins couple to one electromagnetic mode, the symmetric in-phase superposition forms a bright collective mode with coupling larger by approximately $\sqrt{N}$, while the remaining orthogonal superpositions form dark modes with no direct coupling to that mode. Consequently, the measured collective coupling of 21 MHz cannot be assigned to a single spin. The ruby model placed average single-spin coupling in the hertz range [Ro89], and selection of one addressed ion remains unavailable alongside a spread of spin-transition frequencies across the ensemble, called inhomogeneous broadening.

Optical color does not by itself establish spin readout. An absorption band shows electronic transitions that take up light, and luminescence shows that part of the excitation energy comes back out as light. Confirmation of a readout-capable emitter calls for separate tests: single-photon antibunching that identifies one individual emitter, spin-dependent change in optical signal, stable repeated optical cycling, photon indistinguishability, and stable charge state.

Sapphire carries a large microwave quality factor Q , defined as the ratio of stored resonator energy to the energy dissipated per cycle up to the conventional angular-frequency factor, and that low loss makes it an excellent resonator substrate. The same sensitivity makes unwanted paramagnetic impurities at parts-per-billion concentrations experimentally detectable [Ro90]. Resonator performance and qubit performance must therefore be assessed independently because low microwave loss in the host says nothing about coherence, control, or readout of a defect qubit placed there.

A random increase in defect density is not equivalent to controlled fabrication. Raising chromium concentration shortens the average distance between chromium ions, and it simultaneously creates Cr pairs, energy transfer, dipolar broadening, and local strain. It leaves unspecified the interaction graph, defined by the set of interacting defects and their pairwise couplings. A designed Hamiltonian calls for calibrated coupling signs, anisotropies, and connectivities, not only many nearby spins.

A theoretically calculated vacancy spectrum does not by itself establish a qubit platform. Match between first-principles calculations and measured optical bands supports assignment of a spectral feature to a particular defect [Ro93]. Construction of a qubit platform additionally requires measured fabrication yield, measured distribution of coherence times, measured readout fidelity, and measured two-qubit gate performance.

Evidence-supported assessment

[Experiment] Sapphire hosts impurity ions carrying nonzero magnetic moments that serve as paramagnetic dopants; chromium ions emitting spectrally narrow optical lines; spin populations that return to equilibrium over long times at low temperature, giving long spin-relaxation times; microwave cavity modes that lose energy exceptionally slowly; readout of large groups of spins by optical and microwave fields, giving optical and microwave ensemble readout; vacancy defects created by radiation exposure that absorb and emit light as color centers; and light emission from chromium introduced by ion bombardment, giving activation of implanted chromium [Ro88]; [Ro89]; [Ro90]; [Ro91]; [Ro92]; [Ro93]; [Ro94]. These capabilities are substantial components of masers, sensors, frequency standards, ensemble memories, and hybrid-resonator experiments.

For the specific objective of a dense and controllable architecture of interacting defects, the evidence available as of 24 August 2026 does not make sapphire competitive with diamond in demonstrated end-to-end qubit capability. The missing demonstrations required for that capability are spin-selective optical readout from a single center, modern measurements of single-center T_2 and gate fidelities, deterministic and verified defect placement, coherent gates between selected sapphire defects, and scalable nanophotonic integration around those defects. Aluminum nuclei are present in every lattice cell, so the unavoidable ^{27}Al nuclear-spin bath provides a source of magnetic noise and dephasing, and this magnetic noise is an additional materials disadvantage.

This assessment does not prove that such an architecture is impossible. [Proposal] A credible program for establishing sapphire as a defect-qubit platform would first demonstrate one identified Cr^{3+} or F^+ center. It would then measure the spread of precession frequencies across centers, called the inhomogeneous dephasing time T_2^* , the refocused coherence lifetime called the Hahn-echo coherence time T_2 , repeated photon scattering that preserves the spin state, called optical-cycling behavior, and stability of the defect charge state, called charge-state stability. The next requirement would be the creation and spectral resolution of a registered pair, meaning two defects with verified placement or identity, whose coherent coupling exceeds both transition linewidths.

An array-level Hamiltonian proposal would become evidence-based only after the single-center and registered-pair milestones described above had been achieved. [Speculation] Extrapolating directly from ruby ensemble measurements to Fibonacci-like order omits all of these necessary intermediate demonstrations.

Quantitative and conceptual checks

- Explain the four ground-state spin levels of Cr^{3+} in ruby.

Three electrons occupy the localized d-shell, giving the $3d^3$ electronic configuration with total spin $S = 3/2$, so the allowed spin projections are $m = \pm 3/2, \pm 1/2$. The distorted arrangement of neighboring oxygen ions produces a local crystal-field environment that splits these four states into two pairs. Reversal of all spins and momenta leaves the energy unchanged, so this time-reversal symmetry keeps the members of each pair degenerate.

- An 11.493 GHz zero-field splitting corresponds to an equivalent temperature of about 0.55 K, obtained by equating the photon energy to the thermal energy.

The zero-field splitting is the transition-frequency separation present without an applied magnetic field. Its thermal-energy equivalent is

$$h\nu_0/k_B = (6.626 \times 10^{-34} \text{ J s})(11.493 \times 10^9 \text{ s}^{-1})/(1.381 \times 10^{-23} \text{ J K}^{-1}) \approx 0.552 \text{ K}.$$

Thus, the 11.493 GHz splitting corresponds to only about half a kelvin. Strong passive thermal polarization therefore favors sub-kelvin operation.

- Treating the 3.67 s T_1 as a coherence result confuses two distinct decay processes: the cited time describes how long excess population in the excited spin level takes to relax toward thermal equilibrium, while coherence describes how long a superposition phase survives.

T_1 measures population relaxation. Magnetic-field fluctuations and other phase noise can destroy the superposition phase well before the population relaxes, which makes T_2 much shorter than T_1 possible, and the cited millikelvin ruby study did not measure T_2 .

- The 21 MHz ruby-cavity result is not a single-spin coupling. The cavity photon interacted simultaneously with a macroscopic spin ensemble, so the observed splitting reports their joint interaction strength.

Approximately 2.6×10^{15} spins shared the electromagnetic mode. The paper inferred an average coupling of 0.41 Hz per spin, or 1.25 Hz for the subset within the high-field mode volume. The measured 21 MHz value is the collective rate enhanced by $\sqrt{N}$. Even this collective rate did not satisfy $\bar{\Omega} > \gamma_s$, where $\bar{\Omega}$ is the average collective coupling rate and γ_s is the spin linewidth.

- Classifying an F^+ center as a qubit from paramagnetism plus luminescence alone leaves the qubit evidence missing: a magnetic resonance signal shows an unpaired electron and emitted light shows an optical transition, while qubit operation requires demonstrated preparation, microwave-driven rotation, and readout of the spin state.

Isolation with coherent control, spin-dependent readout, stability, and measured error rates still lacks reported measurements in this material. The red optical absorption of a large collection of chromium ions describes ensemble color, and ensemble color leaves that set of single-qubit operations undemonstrated.

- Name the physical property where sapphire has the most direct measured support for defect arrays, and name the missing measurement that most limits its use for defect arrays.

Microwave fields decay unusually slowly in sapphire because its dielectric loss is extraordinarily low. Demonstrations of an isolated single defect operated as a qubit, and of a controlled coupling between two chosen defects, remain sparse beside the corresponding demonstrations in diamond.

The measurements support a chromium ion in ruby with four spin levels in the ground state, a splitting near 11.5 GHz, and a seconds-long T_1 that does not establish T_2 . The operational comparison therefore continues to favor diamond. The next analysis checks the same operational criteria in silicon carbide, since a band gap alone leaves qubit performance untested.

Sources

- [Ro88] V. K. Sewani, R. J. Stöhr, R. Kolesov, H. H. Vallabhapurapu, T. Simmet, A. Morello, and A. Laucht, “Spin thermometry and spin relaxation of optically detected Cr³⁺ ions in Al₂O₃ (ruby),” *Physical Review B* 102, 104114 (2020). DOI: [10.1103/PhysRevB.102.104114](https://doi.org/10.1103/PhysRevB.102.104114).

- [R089] Z. Velluire-Pellat, E. Maréchal, C. Feuillet-Palma, and N. Bergeal, “Spin-photon interaction between a ruby crystal and a high-critical-temperature superconducting microwave cavity,” *Communications Physics* 8, 236 (2025). DOI: [10.1038/s42005-025-02159-1](https://doi.org/10.1038/s42005-025-02159-1).
 - [R090] W. G. Farr, D. L. Creedon, M. Goryachev, K. Benmessai, and M. E. Tobar, “Ultrasensitive microwave spectroscopy of paramagnetic impurities of sapphire crystals at millikelvin temperatures,” *Physical Review B* 88, 224426 (2013). DOI: [10.1103/PhysRevB.88.224426](https://doi.org/10.1103/PhysRevB.88.224426).
 - [R091] P. F. Moulton, “Spectroscopic and laser characteristics of Ti:Al₂O₃,” *Journal of the Optical Society of America B* 3, 125–133 (1986). DOI: [10.1364/JOSAB.3.000125](https://doi.org/10.1364/JOSAB.3.000125).
 - [R092] K. H. Lee and J. H. Crawford Jr., “Luminescence of the F center in sapphire,” *Physical Review B* 19, 3217–3221 (1979). DOI: [10.1103/PhysRevB.19.3217](https://doi.org/10.1103/PhysRevB.19.3217).
 - [R093] C. Wilhelmer, M. E. Turiansky, D. Waldhör, L. Cvitkovich, C. G. Van de Walle, and T. Grasser, “Optical properties of vacancies in aluminum oxide (alpha-Al₂O₃) from first principles,” *Physical Review Materials* 9, 096202 (2025). DOI: [10.1103/5kvl-hktd](https://doi.org/10.1103/5kvl-hktd).
 - [R094] T. M. Hensen, M. J. A. de Dood, and A. Polman, “Luminescence quantum efficiency and local optical density of states in thin film ruby made by ion implantation,” *Journal of Applied Physics* 88, 5142–5147 (2000). DOI: [10.1063/1.1314322](https://doi.org/10.1063/1.1314322).
 - [R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* 528, 1–45 (2013). DOI: [10.1016/j.physrep.2013.02.001](https://doi.org/10.1016/j.physrep.2013.02.001).
 - [R079] G. Balasubramanian et al., “Ultralong spin coherence time in isotopically engineered diamond,” *Nature Materials* 8, 383–387 (2009). DOI: [10.1038/nmat2420](https://doi.org/10.1038/nmat2420).
 - [R095] B. D. Evans and M. Stapelbroek, “Optical properties of the F^+ center in crystalline Al₂O₃,” *Physical Review B* 18, 7089–7098 (1978). DOI: [10.1103/PhysRevB.18.7089](https://doi.org/10.1103/PhysRevB.18.7089).
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Chapter 9 — Silicon carbide and related defect-spin platforms

Silicon carbide is a semiconductor manufactured at industrial scale for devices such as electric-vehicle inverters, and it is available as large wafers. Removing a neighboring silicon atom and carbon atom from the crystal leaves localized electrons whose spin can be addressed in the laboratory.

This adjacent pair of missing atoms belongs to the same family as the vacancy defects in diamond. Its setting is a crystal grown and processed by the semiconductor industry.

The silicon-carbide platform already supports wafer processing, ion implantation, etching, electronic integration, and optical waveguides. This chapter examines how effectively those technologies support defect-spin systems and compares them with other material platforms.

Each host material combines physical properties with fabrication capabilities in its own way, so the choice of host depends on which combination a device needs. The chapter works through silicon carbide in detail and then measures the alternatives against the same array requirements: hexagonal boron nitride with its atomically thin geometry, rare-earth-doped crystals with their high-performance quantum memories, silicon with its mature semiconductor platform, and the broad class of oxide candidates. Diamond, treated two chapters earlier, serves as the benchmark.

The comparison used here asks which host lets many spins be placed, distinguished, coupled, tuned, and measured together under one set of operating conditions. The longest reported memory time alone does not answer that question.

Single-defect quality and array performance

A single defect may support reliable initialization, stable quantum evolution, and measurable signals. Initialization prepares a known quantum state; coherent control applies a deliberate unitary change to that state; coherence preserves the quantum phase information during the evolution; and readout infers the state from an experimental signal. These four properties characterize one spin.

An array asks for more. Defects of the required type must form at selected sites, individual spins must respond to control without adjacent spins responding too, each spin must give a measurable signal, and fabrication must leave the properties of neighboring defects intact.

A long-lived memory in an unsuitable device geometry stays a memory result, and a patterned grid of optical emitters whose identities and spin properties are unknown stays a fabrication result. Either achievement matters, and neither substitutes for the other.

The chapter therefore tracks two performance categories. Single-defect quality covers the initialization, coherent control, coherence, and readout of one spin. Array performance covers placing, distinguishing, coupling, tuning, and measuring many spins under one compatible set of conditions.

Ten high-quality but isolated spins illustrate the distinction: as a collection of single spins they can be excellent, while as an array they are missing every collective requirement.

The two performance categories can initially be represented as qualitative labels rather than numerical figures of merit:

- Q_{single} is the quality of initialization, coherent control, coherence, and readout for one defect;
- Q_{array} is placement yield, interaction reproducibility, addressability, routing, and compatibility with fabrication.

Array performance is governed by a bottleneck condition:

Q_{array} is limited by its weakest required operation.

A long transverse coherence time T_2 , which measures the persistence of spin-phase coherence, cannot replace a missing two-spin interaction. Atomic-scale placement cannot replace an unavailable readout mechanism. Similarly, combining record values obtained from different samples, temperatures, and pulse sequences does not describe a realizable device operating under one set of conditions.

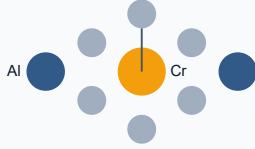
A ranking by longest published memory time would put the six-hour nuclear coherence at the top. The bottleneck criterion gives its actual standing: without a designed nearest-neighbor interaction, that coherence result does not establish an array.

Adjacent vacancy defects in silicon carbide

The two silicon-carbide defects considered here are a neighboring missing silicon atom and carbon atom, commonly written $V_{\text{Si}}V_{\text{C}}^0$, and a negatively charged missing silicon atom, V_{Si}^- . A vacancy is an unoccupied atomic site in an otherwise ordered crystal lattice, and the superscripts specify the defect charge state.

The adjacent vacancy pair is called a divacancy.

Substitution: Cr³⁺ in Corundum
Ruby as dilute chromium in corundum



Divacancy: SiC in Silicon Carbide
Adjacent missing silicon and carbon sites



Two panels compare chromium substituting for aluminum in corundum with adjacent silicon and carbon vacancies in silicon carbide.

Ruby contains chromium substituted on an aluminum site within an oxygen coordination cage. The silicon-carbide divacancy consists of adjacent unoccupied sites, one on each sublattice.

The neutral divacancy has electronic spin $S = 1$, where S is the total spin quantum number. The silicon vacancy commonly used in 4H-SiC has $S = 3/2$.

Silicon carbide grows in several polytypes: crystal structures with the same chemical composition but different periodic stacking sequences. The common wafer polytype considered here is 4H-SiC, and different stacking sequences and inequivalent lattice sites give different optical and spin spectra.

That structural variety widens the choice of operating points, and it multiplies the number of defect configurations that must be identified and controlled.

Stating that a device holds “a spin in SiC” therefore leaves the charge state, crystal stacking, and lattice site unspecified, and each of these shifts the spectrum that a control pulse must address.

Single-spin control

Take one spin-1 divacancy in 4H-SiC. With dimensionless spin operators S_x, S_y, S_z , the ground-state Hamiltonian in frequency units reads

$$\frac{H}{\hbar} = D \left(S_z^2 - \frac{S(S+1)}{3} \right) + E(S_x^2 - S_y^2) + \gamma_e \mathbf{B} \cdot \mathbf{S}.$$

In this expression H is energy in joules, h is Planck's constant in joule-seconds, D and E are axial and transverse zero-field-splitting frequencies in hertz, $\mathbf{B}$ is the magnetic field in tesla, and $\gamma_e = g\mu_B/h$ is the electron-spin gyromagnetic ratio in hertz per tesla. Zero-field splitting is the energy splitting between spin sublevels in the absence of an applied magnetic field. The electron g -factor is near 2, so $\gamma_e \approx 28 \text{ GHz T}^{-1}$.

Suppose the magnetic field has a component B_z along the defect axis and that E is small compared with the other Hamiltonian terms. The microwave transition frequencies from $m_s = 0$ to $m_s = \pm 1$, where m_s is the spin projection quantum number, are approximately

$$\nu_{\pm} = D \pm \gamma_e B_z.$$

The axial field moves the two transitions in opposite directions. A 2024 SiC-on-insulator experiment measured a PL6 center with $D = 1340.4 \pm 1.3 \text{ MHz}$ and a field slope of $2.82 \pm 0.02 \text{ MHz G}^{-1}$, while coherently controlling the implanted single spin at room temperature [R099]. For $B_z = 5.0 \text{ mT} = 50 \text{ G}$, the Zeeman shift is

$$\gamma_e B_z = (2.82 \text{ MHz G}^{-1})(50 \text{ G}) = 141 \text{ MHz},$$

and the two transition frequencies are therefore

$$\nu_- \approx 1.199 \text{ GHz}, \quad \nu_+ \approx 1.481 \text{ GHz}.$$

The units check out because $\text{MHz/G} \times \text{G} = \text{MHz}$. An optical pulse initializes the spin. A microwave tone near one transition then drives Rabi oscillations, which are coherent oscillations of the spin-state population under resonant driving. Spin-dependent fluorescence, in which the emitted light depends on the spin state, or spin-to-charge conversion, in which the spin state is mapped onto a measurable charge state, provides readout. [Experiment] Each of these operations has been demonstrated for SiC divacancies, although the record values come from different devices rather than one room-temperature dense array [R096]; [R097]; [R098]; [R099].

An implanted PL6 spin can therefore be spectrally selected and coherently controlled. That result raises Q_{single} , while leaving the identity and interaction of neighboring sites open.

Site-selective control and crosstalk

Addressability means controlling one selected site without appreciably affecting another. Take, as a design condition rather than a reported array result, local strain or a magnetic-field gradient separating the transition frequencies of neighboring sites by $\Delta = 20 \text{ MHz}$. Let one site be driven with cyclic-frequency Rabi rate $\Omega = 2 \text{ MHz}$. A resonant π pulse, which transfers the population between the two driven levels, has duration

$$t_{\pi} = \frac{1}{2\Omega} = 250 \text{ ns}.$$

The neighboring spin then sits off resonance. Where Ω is small compared with Δ , the leading residual population transferred at the unintended site scales approximately as

$$p_{\text{xtalk}} \sim \left(\frac{\Omega}{\Delta} \right)^2 = 10^{-2}.$$

This residual excitation is crosstalk: an intended operation on one spin also perturbs another spin. It results from incomplete spectral selectivity.

Raising Ω shortens the gate time but raises crosstalk at fixed Δ . Increasing Δ improves site selection, but reproducible gradients or local tuning become necessary, and nominally identical couplings can turn unequal.

Controlling sites one by one still leaves the interaction to be established. A two-spin coupling rate J must also exceed the relevant spectral linewidth and decoherence rates. The next chapter measures this trade-off instead of leaving scalability as an unspecified property.

The implanted object described above is a physical defect-spin qubit: a defect electron coupled to a nearby ^{13}C nuclear spin forms a small quantum register, meaning a set of coupled quantum degrees of freedom that can store and process information. Patterning 64 sites changes how many fabricated sites exist, without changing what kind of qubit each site holds.

Without detuning, the same pulse drives both sites. With very large detuning, the interaction strengths J_{ij} can stop being uniform. Addressability and coupling pull the design in opposite directions.

Interaction topology in a defect array

Labeling the defects by i gives a minimal many-spin Hamiltonian expressed in frequency units:

$$\frac{H_{\text{array}}}{h} = \sum_i \frac{H_i}{h} + \sum_{i < j} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j + \frac{H_{\text{drive}}}{h}.$$

Here H_i is the single-defect Hamiltonian defined above, J_{ij} is a coupling frequency in hertz between sites i and j , and H_{drive} contains the optical, microwave, electric, and strain control terms. The isotropic scalar product $\mathbf{S}_i \cdot \mathbf{S}_j$ is a simplified stand-in for the real interaction. Magnetic dipolar coupling varies with the orientation of the spins and their displacement. Exchange coupling falls off rapidly as electronic wavefunction overlap decreases, while cavity-mediated and phonon-mediated interactions follow the spatial shape of the relevant electromagnetic or vibrational mode.

The interaction graph lists the coupled site pairs, including couplings the design did not intend. Physical geometry alone does not fix this graph: the positions of two optical spots in an image do not determine J_{ij} .

A complete array proposal therefore specifies five mappings:

- site map: intended position $\rightarrow$ actual active defect and charge state;

- spectrum map: actual defect $\rightarrow$ optical and microwave transition frequencies;
- control map: control line or beam $\rightarrow$ addressed subset and crosstalk;
- interaction map: geometry $\rightarrow J_{ij}$, including unwanted edges;
- measurement map: physical signal $\rightarrow$ inferred local or collective observable.

Leaving out any one of these maps leaves a major part of device operation unspecified.

Experimentally demonstrated silicon-carbide capabilities

[Experiment] Individual neutral divacancies in high-purity 4H-SiC have been optically isolated and coherently controlled; ensemble Hahn-echo coherence exceeded 1 ms at low temperature [Ro96]. A Hahn echo is a pulse sequence that refocuses reversible dephasing in a spin ensemble. In isotopically purified 4H-SiC at 5 K, a single divacancy reached $T_2 = 5.3 \pm 1.3$ s using up to 16,384 dynamical-decoupling pulses, with end-to-end spin-to-charge readout fidelity $80.8 \pm 0.6\%$ [Ro97]. Dynamical decoupling uses repeated control pulses to suppress selected environmental noise, and readout fidelity is the probability of correctly inferring the prepared state. The operating conditions qualify every part of that sentence: the five-second figure used up to 16,384 decoupling pulses at 5 K on an isolated spin, rather than a bare Hahn echo at room temperature in a dense implanted array.

Fabrication adds a separate test, which SiC passes unusually well. [Experiment] Implanted V_{Si}^- centers retained nearly lifetime-limited optical emission and high spin coherence in etched nanophotonic waveguides, with control of nearby nuclear spins [Ro98]. Lifetime-limited emission has an optical linewidth close to the minimum set by the excited-state lifetime.

[Experiment] In 2024, selective carbon implantation through an 8×8 mask of 100-nm-diameter holes created a designed PL6 array in SiC-on-insulator; the same study integrated an electron–nuclear register into a waveguide and reported ambient-condition entangled-state fidelities of 0.89 before and 0.88 after integration [Ro99]. Together these results cover materials processing, spin control, and photonic integration in unusual breadth.

[Experiment] The electronic readout route also advanced in 2025: room-temperature photoelectrical magnetic-resonance readout of a single V_{Si}^- spin produced a signal-to-noise ratio 1.7–2 times that of optical detection in the same study [R104]. Photoelectrical magnetic resonance detects spin-dependent electrical signals under optical excitation. This result makes integrated charge collection look feasible, while parallel high-fidelity readout of a coupled array remains to be shown.

None of these experiments produced the required interacting electron-spin lattice. The mask pitch and defect-activation statistics do not fix nanometer-accurate final defect coordinates.

One optically bright location may contain zero, one, or several relevant defects. The 2024 entanglement was between one defect electron and a nearby nuclear spin, not between selected electron defects across the 8×8 pattern [Ro99].

SiC leads the industrially processable optical defect hosts because several necessary technologies coexist in one material system. A controlled interaction graph has yet to be demonstrated in any of them.

SiC therefore has substantial Q_{single} , and several components of Q_{array} have been demonstrated. Controlled electron–electron interaction edges remain the outstanding component.

Atomically thin hexagonal boron nitride

A monolayer or few-layer sheet of hexagonal boron nitride, abbreviated hBN, places every defect close to a surface, electrostatic gate, resonator, and potential neighboring defect. Planar interaction graphs and photonic integration benefit directly from this closeness, and noise from adsorbed atoms and molecules enters through the same short distance.

[Experiment] Single carbon-related hBN defects showed room-temperature optically detected magnetic resonance: the fluorescence changed when a microwave tone hit the spin. Optically detected magnetic resonance, or ODMR, identifies spin transitions through microwave-induced changes in fluorescence. In the reported material only 27 of more than 400 investigated isolated defects showed a measurable signal — about 5% — and saturated linewidths were about 35 MHz [R100]. The microscopic structures and spin multiplicities were not uniquely established.

Selected individual sites therefore behave promisingly, while defect yield and microscopic identity stay poorly controlled. The short distance to an engineered interface and the added exposure to surface contamination and surface-induced noise are two sides of the same geometry.

Chemical identity, uniformity, lifetime, and yield each need independent demonstration before the thin host can carry an array.

Rare-earth-ion-doped crystals

Some crystals are doped with rare-earth ions such as Eu^{3+} , Er^{3+} , and Yb^{3+} . Doping is the intentional incorporation of impurity ions into a host crystal. The partly filled $4f$ electron shell of a rare-earth ion is shielded by closed outer shells, so its optical transitions can be extremely narrow and its spin coherence times can be extremely long.

[Experiment] $^{151}\text{Eu}^{3+} : \text{Y}_2\text{SiO}_5$ reached 370 ± 60 minutes of hyperfine coherence at 2 K with dynamical decoupling [R101]. Hyperfine coherence refers to coherence between states split by interactions involving electronic and nuclear magnetic moments. [Experiment] A single $^{171}\text{Yb}^{3+}$ ion in a YVO_4 nanophotonic cavity showed an optical linewidth below 1 MHz, spin coherence beyond 30 ms, and conditional single-shot readout above 95% [R102]. Conditional single-shot readout is a state measurement performed in one experimental trial, with the quoted fidelity conditioned on the specified experimental acceptance criteria. These measurements establish strong performance as quantum-memory and network nodes.

Against a dense local interaction Hamiltonian, dilute dopants are commonly distributed randomly in space, direct interactions are weak or inhomogeneous, and operation is generally cryogenic. Spectral multiplexing, which distinguishes ions by transition frequency within one optical mode, can identify many ions without producing a designed nearest-neighbor lattice.

A long memory then increases Q_{single} without improving Q_{array} . For rare-earth systems, the principal bottleneck is the interaction map rather than the coherence time.

Silicon spin-qubit platforms

Silicon is the strongest alternative to an exclusively optical-defect approach. Isotopically enriched ^{28}Si , in which magnetic isotopes are strongly reduced, provides a low-noise nuclear environment. Silicon also has unmatched industrial electronic-processing infrastructure. In addition, scanning-tunneling-microscope hydrogen lithography can place phosphorus donors on selected lattice sites.

An electron in silicon can occupy more than one equivalent minimum, or valley, in the electronic dispersion relation in momentum space. Interference between these valley components causes the exchange energy of two nearby donors to depend strongly on the direction of their displacement as well as on their separation.

A valley is a local minimum of the electronic energy surface in momentum space, and interference between valley components is called a valley effect. [Experiment/Theory] Atomic-scale imaging and modeling show that aligning donors along favorable crystallographic directions reduces, without removing, exchange variability [R103].

Gate-defined silicon spin qubits already support electrically controlled two-qubit logic in devices that are not color centers. A color center is an optically active point defect whose electronic transitions can provide spin initialization or readout; a cycling transition is one that repeatedly produces photons while approximately preserving the measured state. Relative to an optical defect lattice, silicon platforms require millikelvin operation plus elaborate gate stacks and charge reservoirs, and standard phosphorus-donor nodes lack a naturally bright cycling optical transition. Where the goal is a programmable electrical spin Hamiltonian rather than a spin–photon network, silicon is the preferred material.

Drop the optical-interface requirement and silicon has the strongest Q_{array} case in this chapter. If that requirement is retained, the absence of a cycling optical transition is the bottleneck.

Wide-gap oxide hosts

“Oxide” denotes a broad material class covering chemically and structurally distinct hosts such as MgO , ZnO , TiO_2 , Y_2O_3 , and many others.

Candidate spin systems include transition-metal defects, oxygen vacancies, and rare-earth dopants. Rare-earth oxides therefore overlap with the rare-earth-ion systems discussed above.

[Theory/Experiment] Reviews identify low nuclear-spin abundance, large band gaps, and mature thin-film processing as useful search criteria, while also emphasizing charge-state stability, optical cycling, and reproducible defect formation as independent requirements [R073]. As of the evidence surveyed here, deterministic single-spin placement, a high-fidelity optical lifecycle, preserved coherence after nanofabrication, and a demonstrated controlled interaction graph have not yet coincided in one wide-gap oxide. Oxides therefore constitute a broad discovery space awaiting a first validated interacting-array platform.

Comparative assessment

The following matrix evaluates the hosts specifically as platforms for interacting arrays. “High coherence” reports only what the cited experiment measured under its stated conditions. The ratings state comparative judgments, with no hidden numerical data behind them.

Host and representative spin | Native/engineered coupling route | Placement and yield | Coherence evidence |
Photonics and readout | Maturity and typical temperature | Platform assessment | |

4H-SiC: neutral divacancy, V_{Si}^- | Nearby nuclear hyperfine demonstrated; electron dipolar/exchange at short range; cavity, strain, and electrical mediation plausible | Implantation, masks, annealing, commercial wafers; designed 8×8 PL6 pattern demonstrated, but not an atomically registered interacting electron array [R099] | > 1 ms ensemble Hahn echo at low temperature [R096]; single-spin 5.3 ± 1.3 s at 5 K with isotopic purification and 16,384-pulse decoupling [R097] | Near-infrared optical transitions, spin-to-charge readout, waveguides and SiC-on-insulator integration [R097]; [R098]; [R099] | Highest combined materials/device maturity here; room-temperature control exists, best resonant optical/readout records often cryogenic | Most complete combination of optical and wafer-processing capabilities considered here; activation statistics, final position, spectral disorder, and controlled electron–electron edges still open | |

hBN: boron vacancy and carbon-related spins | Short-range exchange/dipolar interactions are geometrically accessible in 2D; resonator coupling possible | Transfer, growth, irradiation and local writing are available; microscopic identity and reproducible spin-active yield remain weak | Room-temperature single-defect ODMR; about 5% ODMR-active yield in one >400 -defect survey and roughly 35 MHz saturated linewidth [R100] | Bright visible emitters and easy planar integration; spectral and photodynamic variability substantial [R100] | Rapidly developing, room-temperature demonstrations | High-potential planar platform, but not yet a reproducible many-spin component library | |

Rare-earth crystals: Eu:YSO, Yb:YVO₄, Er:YSO | Weak direct ion–ion coupling; cavity-mediated and spectral-multiplexed links are natural | Dopants are usually dilute and spatially random; individual ions selected spectrally or by cavity overlap | Eu nuclear hyperfine 370 ± 60 min at 2 K with decoupling [R101]; single Yb spin >30 ms in a cavity [R102] | Exceptional narrow optical lines; single-ion cavity readout $>95\%$ conditional fidelity [R102] | Mature memories and emerging nanophotonics; usually kelvin-scale | Strong capabilities for memory and network applications; less natural for a dense prescribed local interaction graph | |

Silicon: P donors or gate-defined electron spins | Electrically tunable exchange is strong; capacitive and resonator links available | CMOS gates plus STM donor placement can approach lattice-site precision; exchange remains sensitive to valley phase and interfaces [R103] | Long coherence in enriched ²⁸Si is established, but values depend strongly on electron versus nucleus, ensemble versus single device, and pulse sequence [R073] | Excellent electrical readout/control; standard donor architecture lacks a bright cycling optical interface | Most mature semiconductor control stack; generally dilution-refrigerator temperatures | Strongest electrical programmable-array platform; less suited to an optical color-center route | |

Wide-gap oxides: oxygen vacancies, transition-metal or rare-earth dopants | Dipolar/exchange possible in principle; oxide electronics and strain offer mediators | Thin-film processing is mature for some hosts, but quantum-active defect identity, charge state, activation and placement are host-specific [R073] | Isolated or ensemble coherence exists in selected systems; no transferable “oxide value” is meaningful [R073] | Some excellent emitters or microwave ensembles, rarely a complete single-spin lifecycle | Fragmented; temperatures range from ambient sensing to cryogenic spectroscopy | Discovery portfolio, not presently a validated interacting-array platform | |

Diamond baseline: NV and group-IV vacancies | Dipolar/exchange, nuclear registers, and strong nanophotonic interfaces | Implantation and delta doping mature but deterministic nanometer-scale coupled arrays remain difficult | Platform-leading single-node demonstrations, with defect- and temperature-dependent trade-offs [R073] | Most mature color-center networking evidence; nanofabrication and surfaces can degrade performance | Mature, spanning room-temperature NV control to cryogenic group-IV optics | Benchmark rather than automatic winner; SiC offers stronger wafer/electronics integration, not yet stronger end-to-end coupled-array evidence | |

The matrix keeps two claims separate that are often run together:

- [Experiment] single-defect quality: SiC, rare-earth ions, silicon, diamond, and now hBN all contain individually controllable quantum systems, although the depth of evidence and operating conditions differ substantially.
- [Proposal] array suitability: none of these hosts has demonstrated a dense, designed, coherently interacting defect lattice.

The right material follows from the device architecture. For an optically addressable interacting-defect program, SiC and diamond are the two primary platforms. SiC suits programs where monolithic wafer electronics and photonics dominate the requirements. hBN is a high-risk planar alternative. Rare-earth crystals suit cavity-mediated or spectral architectures better than short-range lattices, and selected oxides remain discovery targets.

For an entirely electrical exchange-coupled array, silicon leads. This ordering stays open to revision: a convincing two-dimensional hBN placement-and-yield result or a deterministic rare-earth implantation method could change the comparative assessment.

Common analytical errors

- Combining independent record values does not characterize a single device. A five-second spin-coherence time T_2 , where T_2 is the transverse decoherence time, was measured in SiC at 5 K using 16,384 control pulses [Ro97]. A room-temperature implanted array was reported separately [Ro99], as was a nanophotonic linewidth measurement [Ro98]. These results were obtained from different devices and cannot be combined into the specification of one device.
- A fabricated pattern does not establish the final defect positions. A lithographic aperture constrains the nominal ion trajectory during implantation. The final location and identity of an active center are also determined by collision straggle, meaning the statistical spread of implanted-ion trajectories; vacancy diffusion; annealing; charge-state conversion; and the possible occupation of one site by multiple defects. Consequently, observing an optical spot at every mask location does not determine the atomic coordinates of the associated defects.
- Addressability does not establish coupling. Spectral addressability is the ability to distinguish and control sites through differences in their transition frequencies. Resolving two sites spectrally demonstrates that the controls can distinguish them, but it does not determine the pairwise coupling J_{ij} , demonstrate coherent exchange of quantum states, or establish entanglement.
- A high defect density does not necessarily produce a useful interaction graph. An interaction graph represents defects as vertices and selected couplings as edges. Increasing the defect density simultaneously increases the intended coupling strengths, unintended couplings, implantation damage, spectral crowding, and charge noise. The physical interactions in the crystal are therefore not restricted to the edges specified by the intended design.
- A long memory time does not imply fast processing. Rare-earth nuclear spins can exhibit extraordinary coherence [R101], while their direct local interactions remain weak. An operational array requires a favorable ratio of the gate or coupling rate to the error rate; a large T_2 alone is insufficient.
- A two-dimensional host does not by itself provide a complete qubit platform. Hexagonal boron nitride (hBN) provides geometric access to defects in an atomically thin material, but it does not automatically provide controlled chemical identity, uniformity, lifetime, or fabrication yield. The reported optically detected magnetic

resonance (ODMR) evidence for single carbon-related defects also documents substantial variability [R100]. ODMR is the optical measurement of spin-dependent changes produced by resonant microwave excitation.

- Semiconductor compatibility does not imply that a fabrication process is qualified. Silicon carbide (SiC) can support electronics and photonics, but every implantation, etching, oxidation, metallization, and annealing step must be evaluated for its effects on charge stability and spin-optical coherence. Establishing compatibility therefore requires systematic process development and testing.
- Coupled qubits do not automatically realize a new phase of matter. Even a perfect SiC or silicon spin array would initially constitute an ordinary interacting quantum system. A many-body phase is a collective state characterized by properties not reducible to isolated constituents. The host material alone does not imply the existence of a gapped many-body phase, meaning a phase separated from excitations by a finite energy gap, or of nonlocal observables, which depend on spatially extended degrees of freedom.

Required operation rates

Let J denote the frequency associated with a desired coupling, and let $\Gamma_2 = 1/T_2$ denote the effective decoherence rate during an operation. Let σ_J be the standard deviation of coupling disorder, which quantifies the variation of coupling frequencies across the array. Let Ω be the control rate, and let Δ_{addr} be the spectral detuning between the addressed transition and the nearest unwanted transition. A useful operating regime requires, schematically,

$$J \gg \Gamma_2, \quad J \gg \sigma_J, \quad \Gamma_2 \ll \Omega \ll \Delta_{\text{addr}}.$$

The first inequality requires the desired interaction to act much faster than decoherence. The second requires the intended coupling to exceed its disorder-induced variation. The final pair of inequalities requires control to be faster than decoherence but slower than the frequency separation needed for selective addressing. All symbols are frequencies in hertz, so these comparisons are dimensionally consistent. Across a large, deliberately designed defect array, every host in the comparison matrix still fails at least one of these inequalities. The next several chapters quantify these conditions.

Conceptual checks

- The host with the longest reported T_2 is not necessarily the best array platform. Placement, coupling, addressability, readout, yield, and mutually compatible operating conditions remain independent bottlenecks. The array-level performance metric Q_{array} is limited by the weakest required operation.
- SiC's principal array-level advantage is the coexistence of optically addressable spins with commercial wafers, ion implantation, electronics, mechanics, and monolithic photonics.
- For the cited PL6 center, a 5.0 mT magnetic field produces microwave transition frequencies of approximately 1.199 GHz and 1.481 GHz. Using the electron gyromagnetic ratio γ_e and an axial magnetic field B_z ,

$$\gamma_e B_z = (2.82 \text{ MHz G}^{-1})(50 \text{ G}) = 141 \text{ MHz},$$

$$\nu_{\pm} = 1340.4 \pm 141 \text{ MHz.}$$

- Treating the 8×8 optical-spot pattern as an interaction graph is not justified. The pattern does not determine nanometer-registered electron coordinates, and the entanglement reported in 2024 was electron–nuclear entanglement rather than entanglement along selected electron–electron edges.
- Silicon can outrank SiC when the intended platform is an all-electrical, exchange-coupled, programmable spin array and a direct optical interface is unnecessary.
- Under the stated design assumptions, the estimated crosstalk probability is 10^{-2} . With a control rate $\Omega = 2 \text{ MHz}$ and detuning $\Delta = 20 \text{ MHz}$,

$$p_{\text{xtalk}} \sim (\Omega/\Delta)^2 = (1/10)^2 = 10^{-2}.$$

This value is not a measured array error. It is the crosstalk predicted directly from the specified drive rate and detuning.

Sources

- [R096] D. J. Christle, A. L. Falk, P. Andrich, et al., “Isolated electron spins in silicon carbide with millisecond coherence times,” *Nature Materials* 14, 160–163 (2015). DOI: [10.1038/nmat4144](https://doi.org/10.1038/nmat4144).
- [R097] C. P. Anderson, E. O. Glen, C. Zeledon, et al., “Five-second coherence of a single spin with single-shot readout in silicon carbide,” *Science Advances* 8, eabm5912 (2022). DOI: [10.1126/sciadv.abm5912](https://doi.org/10.1126/sciadv.abm5912); arXiv: [2110.01590](https://arxiv.org/abs/2110.01590).
- [R098] C. Babin, R. Stöhr, N. Morioka, et al., “Fabrication and nanophotonic waveguide integration of silicon carbide colour centres with preserved spin-optical coherence,” *Nature Materials* 21, 67–73 (2022). DOI: [10.1038/s41563-021-01148-3](https://doi.org/10.1038/s41563-021-01148-3).
- [R099] H. Hu, Y. Zhou, A. Yi, et al., “Room-temperature waveguide integrated quantum register in a semiconductor photonic platform,” *Nature Communications* 15, 10256 (2024). DOI: [10.1038/s41467-024-54606-2](https://doi.org/10.1038/s41467-024-54606-2).
- [R100] H. L. Stern, Q. Gu, J. Jarman, et al., “Room-temperature optically detected magnetic resonance of single defects in hexagonal boron nitride,” *Nature Communications* 13, 618 (2022). DOI: [10.1038/s41467-022-28169-z](https://doi.org/10.1038/s41467-022-28169-z).
- [R101] M. Zhong, M. P. Hedges, R. L. Ahlefeldt, et al., “Optically addressable nuclear spins in a solid with a six-hour coherence time,” *Nature* 517, 177–180 (2015). DOI: [10.1038/nature14025](https://doi.org/10.1038/nature14025).
- [R102] J. M. Kindem, A. Ruskuc, J. G. Bartholomew, et al., “Control and single-shot readout of an ion embedded in a nanophotonic cavity,” *Nature* 580, 201–204 (2020). DOI: [10.1038/s41586-020-2160-9](https://doi.org/10.1038/s41586-020-2160-9); arXiv: [1907.12161](https://arxiv.org/abs/1907.12161).

- [R103] B. Voisin, J. Bocquel, A. Tankasala, et al., “Valley interference and spin exchange at the atomic scale in silicon,” Nature Communications 11, 6124 (2020). DOI: [10.1038/s41467-020-19835-1](https://doi.org/10.1038/s41467-020-19835-1).
 - [R073] G. Wolfowicz, F. J. Heremans, C. P. Anderson, et al., “Quantum guidelines for solid-state spin defects,” Nature Reviews Materials 6, 906–925 (2021). DOI: [10.1038/s41578-021-00306-y](https://doi.org/10.1038/s41578-021-00306-y).
 - [R104] T. Nishikawa, N. Morioka, H. Abe, et al., “Coherent photoelectrical readout of single spins in silicon carbide at room temperature,” Nature Communications 16, 3405 (2025). DOI: [10.1038/s41467-025-58629-1](https://doi.org/10.1038/s41467-025-58629-1).
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PART V — INTERACTING DEFECT SYSTEMS

What happens when nearby defect spins interact? We first study a pair, then ask how three spins can store quantum information with reduced sensitivity to some kinds of noise.

Chapter 10 — Interactions between two defect spins

Two magnetic moments held at separate positions interact through their magnetic fields, with no particles exchanged. The displacement vector joining them sets the geometry: both the distance and the orientation of the moments against that vector fix the strength and sign of the interaction, and for some orientations the interaction vanishes.

Dipole–dipole coupling here means the energy of one magnetic moment sitting in the magnetic field generated by the other. This chapter computes that energy for two electron spins in a crystal and tests whether the resulting coupling can drive coherent evolution. A coupling frequency in megahertz settles nothing on its own; feasibility comes from comparing it with the relevant linewidths and decoherence rates.

Magnetic field of a localized moment

A magnetic moment is a vector recording the strength and orientation of a magnetic source; the classical picture is a small current loop. Far from the source, every localized current distribution with the same moment produces the same leading-order field.

Call the first magnetic moment μ_1 , and let the second moment be displaced from it by $\mathbf{r}$. Define r as the magnitude of $\mathbf{r}$, measured in metres, and define $\hat{\mathbf{r}} = \mathbf{r}/r$ as the unit vector directed from the first moment to the second.

The magnetic field produced by the first moment at the position of the second is

$$\mathbf{B}_1(\mathbf{r}) = \frac{\mu_0}{4\pi r^3} [3(\mu_1 \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mu_1],$$

where μ_0 is the vacuum permeability, measured in N A^{-2} . The energy of a second moment μ_2 in this field is $-\mu_2 \cdot \mathbf{B}_1$. Substitution gives the magnetic dipole–dipole Hamiltonian

$$H_{dd} = \frac{\mu_0}{4\pi r^3} [\mu_1 \cdot \mu_2 - 3(\mu_1 \cdot \hat{\mathbf{r}})(\mu_2 \cdot \hat{\mathbf{r}})].$$

Here a Hamiltonian is an energy operator, and H_{dd} denotes the dipole–dipole contribution. The field of a localized magnetic moment decreases as $1/r^3$. Because the interaction energy is the product of one moment and the field produced by the other, the coupling has the same $1/r^3$ dependence.

This result assumes that the moments are localized and sufficiently separated for the far-field approximation to apply. If the electronic wavefunctions overlap, the $1/r^3$ law is not a general theorem. Overlapping electronic orbitals produce a distinct interaction discussed below, and that interaction is not represented by the point-dipole formula.

Magnetic coupling between two electron spins

Electron spin is a quantum-mechanical angular momentum carrying a magnetic moment. Let $\mathbf{S}$ denote the spin angular-momentum operator, measured in joule-seconds, and let $\hbar = h/(2\pi)$ denote the reduced Planck constant.

Define the dimensionless spin operator as $\mathbf{s} = \mathbf{S}/\hbar$. The associated magnetic moment is $\boldsymbol{\mu} = -g\mu_B\mathbf{s}$, where g is the dimensionless electron g -factor and μ_B is the Bohr magneton, measured in J T^{-1} .

Substituting these two magnetic moments into the dipole–dipole energy cancels the two negative signs. Dividing the resulting energy by Planck’s constant h converts joules to hertz:

$$H_{dd} = h\,d(r)\left[\mathbf{s}_1\cdot\mathbf{s}_2 - 3(\mathbf{s}_1\cdot\hat{\mathbf{r}})(\mathbf{s}_2\cdot\hat{\mathbf{r}})\right], \qquad d(r) = \frac{\mu_0(g\mu_B)^2}{4\pi\hbar r^3}.$$

The quantity $d(r)$ is the characteristic dipolar coupling coefficient in hertz, fixing both the angular tensor and the $1/r^3$ distance dependence.

Using $g = 2.0023$, $\mu_B = 9.2740101 \times 10^{-24} \text{ J T}^{-1}$, and $h = 6.62607015 \times 10^{-34} \text{ J s}$, the CODATA constants give [\[R105\]](#)

$$d(r) = \frac{52.04 \text{ MHz nm}^3}{r^3}.$$

Dimensional analysis checks this expression. The quantity $\mu_0\mu_B^2/r^3$ has units of joules. Division by h , measured in joule-seconds, gives $\text{s}^{-1} = \text{Hz}$, with no residual factor of nanometres.

The following table lists the characteristic coefficient for two pointlike electron spins with $g \approx 2$. Its entries leave out the angular factor and the matrix elements belonging to any particular spin transition, where a matrix element is the amplitude with which an operator connects two selected quantum states.

Separation r	$d(r)$	$1/d$	Noisy benchmark: $d/\Delta\nu^*$	Quiet benchmark: $d/\Delta\nu^*$
1 nm	52.04 MHz	19.2 ns	164	16,400
2 nm	6.505 MHz	154 ns	20.4	2,044
5 nm	416.3 kHz	2.40 μs	1.31	131
10 nm	52.04 kHz	19.2 μs	0.164	16.4
20 nm	6.505 kHz	154 μs	0.0204	2.04
50 nm	416.3 Hz	2.40 ms	0.00131	0.131

The two rightmost columns rest on assumed coherence times, not measured performance records for a specific platform. The noisy benchmark assumes a free-induction coherence time $T_2^* = 1 \text{ }\mu\text{s}$, whereas the quiet benchmark assumes $T_2^* = 100 \text{ }\mu\text{s}$. The free-induction coherence time T_2^* characterizes phase loss without refocusing pulses. For

exponential phase decay, the corresponding Lorentzian full width at half maximum is

$$\Delta\nu^* = \frac{1}{\pi T_2^*},$$

which gives linewidths of 318 kHz and 3.18 kHz, respectively. A 52 kHz coupling at 10 nm clears the quiet linewidth yet sits inside the noisy linewidth. A quoted coupling of 52 kHz therefore leaves gate feasibility open until it is compared with a linewidth measured under the same conditions.

At separations of 1–2 nm, the point-dipole calculation stays a useful reference, but the electronic wavefunctions may overlap. The exchange interaction treated below can then grow significantly, so a model with point-dipole coupling alone is incomplete.

At 50 nm the magnetic interaction is still nonzero; whether it is usable turns on the available coherence and control resources.

From here on, every Hamiltonian in this chapter uses frequency units, so $H/\hbar$ is measured in hertz. When two defects i and j have a coherent matrix element J_{ij} , J_{ij} denotes the coupling that actually connects the selected states. Different spin components or transition directions can carry different values.

Orientation relative to the quantization axis

An applied magnetic field defines a quantization axis: the axis that spin projections are measured against. Call this axis z , and let θ be the angle between $\hat{r}$ and z .

Take the Zeeman splitting or zero-field splitting to be much larger than d . The Zeeman splitting is the field-induced separation between spin-energy levels, and zero-field splitting names the separation persisting without an applied field. Rapidly rotating terms then average to zero and drop out. The raising and lowering operators are $s^\pm = s^x \pm is^y$.

The remaining energy-conserving contribution is

$$\frac{H_{dd}^{\text{sec}}}{\hbar} = d(r)(1 - 3 \cos^2 \theta) \left[s_1^z s_2^z - \frac{1}{4}(s_1^+ s_2^- + s_1^- s_2^+) \right].$$

That surviving part is the secular interaction: the piece remaining observable in the high-field or large-splitting limit since the other terms average away.

The first term shifts one spin's transition frequency conditional on the state of the other. The second term moves one spin excitation across when the two transitions are resonant.

The angular factor is -2 when the displacement vector lies along z , $+1$ when it lies in the transverse plane, and zero at the magic angle

$$\theta = \cos^{-1}(1/\sqrt{3}) \approx 54.7^\circ.$$

The magic angle is the orientation at which the secular angular factor vanishes, so defect pairs at equal separations on a lattice can carry coupling edges of different magnitudes and signs.

Model the relevant two-state subspace by

$$H/h = j(|10\rangle\langle 01| + |01\rangle\langle 10|),$$

where j is measured in hertz. This Hamiltonian moves population coherently between the states $|10\rangle$ and $|01\rangle$. A complete excitation swap then requires

$$t_{\text{swap}} = \frac{1}{4|j|}.$$

A rough gate condition follows: $4|j|T_2 \gg 1$, where T_2 is the coherence time measured with refocusing. Resolving the interaction spectrally generally needs $|j|$ to exceed the relevant linewidth.

Pulse sequences can hold back slowly varying single-spin noise while keeping a selected two-spin interaction. For a carefully designed gate, T_2 rather than T_2^* may therefore determine the available coherent evolution time.

Near the magic angle, the dropped nonsecular terms and small orientation errors matter. If the relevant level splitting is not much larger than d , the secular approximation stops being justified. Pulse sequences also leave relaxation, pulse errors, and some fluctuating couplings unremoved.

Direct exchange from overlapping orbitals

When two localized electronic wavefunctions overlap, the indistinguishability of electrons contributes a further interaction. Electrons are fermions: the two-electron wavefunction changes sign when the particle labels are exchanged, so different spin configurations can carry different Coulomb energies.

Let $\phi_i(\mathbf{x})$ and $\phi_j(\mathbf{x})$ denote localized orbitals. The exchange energy shift is associated with the Coulomb matrix element in which the orbital labels are crossed:

$$J_{ij}^{\text{ex}} \sim \iint d^3x d^3x' \phi_i^*(\mathbf{x}) \phi_j(\mathbf{x}) \frac{e^2}{4\pi\epsilon|\mathbf{x} - \mathbf{x}'|} \phi_j^*(\mathbf{x}') \phi_i(\mathbf{x}'),$$

where e is the elementary charge and ϵ is the permittivity of the host material. This energy is direct exchange, named for the direct overlap of the orbitals. The virtual mechanisms treated later need no overlap of the defect orbitals themselves.

When an orbital envelope decays over a localization length a , its overlap is approximately proportional to $e^{-r/a}$. Direct exchange then commonly scales as a polynomial multiplying $e^{-2r/a}$. The prefactor, sign, orbital-orientation dependence, and precise exponent are material-specific.

[Numerics] Atomistic calculations for several diamond spin-center pairs found exchange larger than magnetic dipolar coupling below roughly 3 nm for NV pairs, with strong crystallographic and defect-orientation dependence [R107]. This is a calculated trend rather than a demonstrated, reproducible NV–NV gate: stronger interaction comes with stronger sensitivity to placement errors.

Neglecting orbital overlap removes the direct-exchange term while the dipolar term remains. Direct exchange and magnetic dipole coupling are distinct physical mechanisms.

Superexchange through virtual charge hopping

Consider two half-filled sites. Let t , measured in joules, be the amplitude for an electron to hop between them. Let U , also measured in joules, be the energy cost of placing two electrons on the same site, and assume U is large.

Real hopping is blocked within the low-energy subspace, but a doubly occupied state can still enter as a virtual intermediate state: an intermediate configuration that contributes perturbatively without becoming a populated long-lived state.

Second-order perturbation theory then lowers the singlet energy relative to the triplet energy:

$$H_{\text{SE}} = J_{\text{SE}} \mathbf{s}_1 \cdot \mathbf{s}_2, \quad J_{\text{SE}} \approx \frac{4t^2}{U}.$$

Here J_{SE} is an energy, with corresponding frequency J_{SE}/h . Hopping that is forbidden as a real low-energy process can thus still generate an effective spin–spin interaction, with magnitude set by the energy cost of the virtual doubly occupied state.

This virtual-hopping mechanism is superexchange, distinct from direct exchange.

Because the microscopic mechanism is virtual charge hopping, if $t \propto e^{-r/a}$, then the superexchange scale decreases approximately as $e^{-2r/a}$.

With an explicit bridge orbital participating, the effective interaction contains products of the hopping amplitudes from each defect to the bridge, and bond angles and bridge chemistry can outweigh the direct defect-to-defect distance.

Systems with multiple orbitals, Hund coupling, or different electronic filling can change the numerical factor and sign, as Anderson’s original analysis shows [R108]. If U is not large compared with t , the low-energy spin-only description is no longer the appropriate starting point.

Photon-mediated interactions

Two optical transition dipoles interact through the electromagnetic field, where an optical transition dipole is the electric-dipole matrix element connecting two emitter states. In empty three-dimensional space, the coherent near-field interaction scales as $1/r^3$.

Once the separation becomes comparable to the optical wavelength, retardation adds oscillatory terms proportional to $1/r^2$ and $1/r$. The electromagnetic field also generates collective radiative decay. A coherent level shift must therefore be compared with the optical homogeneous linewidth: the linewidth from dynamical dephasing and decay, not static ensemble variation.

A resonator replaces free-space distance dependence with dependence on a cavity-mode profile. Let a annihilate one cavity photon, let σ_i^- lower emitter i , let g_i be the emitter–cavity coupling frequency, and let Δ_i be the detuning between emitter i and the cavity. Detuning is the difference between the relevant transition and mode frequencies. Where $|\Delta_i| \gg |g_i|$, the cavity stays only virtually occupied. Eliminating the cavity mode then gives the effective interaction scale

$$J_{12}^{\text{cav}} \approx \frac{g_1 g_2}{2} \left(\frac{1}{\Delta_1} + \frac{1}{\Delta_2} \right).$$

There is no universal $1/r^n$ law in this case since g_i follows the cavity electric-field amplitude at each defect position. With cavity energy-decay rate κ , cavity elimination also contributes loss of order $(g_i/\Delta_i)^2 \kappa$. Coherent operation therefore needs $|J_{12}^{\text{cav}}|$ to exceed both emitter dephasing and cavity-induced decay; a nonzero value alone is not enough.

The cavity mode profile, not Euclidean distance alone, sets the spatial range of this cavity-mediated coupling.

[Experiment] Two SiV centers coupled to a common diamond nanocavity exhibited spectrally resolved superradiant and subradiant optical states [Ro86]. Superradiant and subradiant states are collective emitter states with enhanced and suppressed radiative decay, respectively. This observation demonstrates coherent photon-mediated interaction between the emitters, which is a weaker claim than a high-fidelity gate between long-lived ground-state spin qubits.

Where the large-detuning condition fails, the cavity becomes a real populated mediator rather than a virtual one. A result reported only as $g_1 g_2 / \Delta$, without quoting the cavity decay rate κ , temperature, and emitter linewidth, leaves the performance estimate incomplete.

Phonon-mediated interactions

Strain is the spatial derivative of mechanical displacement, and a defect whose orbital or spin energy changes with strain couples locally to the strain field as an elastic dipole.

In an isotropic three-dimensional bulk solid, the displacement from a point force falls off approximately as $1/r$. Derivatives at both the source and receiver turn that displacement law into a static elastic-dipole interaction typically scaling as $1/r^3$.

[Theory] A 2024 lattice treatment derived this $1/r^3$ phonon-mediated law for NV centers and identified a quadrupole–quadrupole spin structure [R111]. Crystal anisotropy, surfaces, and nanobeams can alter both the interaction tensor and the distance power law.

A quantized mechanical mode is a vibration shared by both defects. Let b annihilate one quantum of that vibration, let g_i^{ph} be the defect–mode coupling frequency, and let Δ_i^{ph} be the corresponding detuning. In the dispersive regime the mode stays sufficiently detuned to remain virtually occupied, and eliminating it gives

$$J_{12}^{\text{ph}} \sim \frac{g_1^{\text{ph}} g_2^{\text{ph}}}{2} \left(\frac{1}{\Delta_1^{\text{ph}}} + \frac{1}{\Delta_2^{\text{ph}}} \right).$$

A quantum of lattice vibration is a phonon. In a waveguide or structured mechanical system, the waveguide Green function—the response at one position to a localized source at another—sets the interaction range, not Euclidean distance alone.

A single ideal mechanical mode can connect distant sites, but its decay rate κ_m and thermal occupation

$$\bar{n} = [\exp(\hbar f_m / k_B T) - 1]^{-1}$$

introduce noise. Here f_m is the mechanical-mode frequency, k_B is Boltzmann’s constant, and T is temperature.

[Proposal] A quasi-one-dimensional diamond waveguide linking SiV centers through propagating phonons has been quantitatively analysed, including state emission and reabsorption [R109]. This result proposes a network architecture; it is not evidence that a many-defect phonon bus already outperforms dipolar links.

Where the solid departs from isotropy, the bulk $1/r^3$ law need not apply. Unless the thermal occupation is small, the same mode that mediates J_{12}^{ph} also adds noise of order $\kappa_m(2\bar{n} + 1)$.

Nuclear-spin-mediated interactions

An electron defect can couple to a nearby nuclear spin through a hyperfine tensor $\mathbf{A}$, measured in hertz, connecting the electron spin $\mathbf{s}$ to the nuclear spin $\mathbf{I}$:

$$H_{\text{hf}}/\hbar = \mathbf{s} \cdot \mathbf{A} \cdot \mathbf{I}.$$

The hyperfine interaction couples electronic and nuclear magnetic degrees of freedom, so a nucleus can serve as a mediator as well as a quantum memory.

Two electrons coupled off-resonantly to the same nucleus exchange virtual nuclear-spin flips, generating an electron–electron interaction scale

$$J_N \sim A_{1\perp} A_{2\perp} / \delta_N,$$

where $A_{i\perp}$ is a transverse hyperfine component and δ_N is the detuning of the virtual nuclear transition. The exact numerical factors depend on the selected energy levels.

Dipolar hyperfine terms fall off as $1/r_i^3$, where r_i is the distance between electron i and the nucleus. At fixed detuning, their product can therefore scale as $1/(r_1^3 r_2^3)$. Contact hyperfine coupling instead tracks the electron density at the nucleus and stays short-ranged, so no single scaling law in terms of the electron–electron separation alone covers both cases.

The reverse architecture—dipolar-coupled NV electrons mediating an interaction between their nuclear memories—is theoretically developed [R110]. [Proposal] This proposal illustrates a hierarchy of mediators without providing experimental evidence for a nuclear bus connecting arbitrary defects.

Nuclear spins hold memory long but generally couple slowly, so mediator initialization, spectral crowding, and electron back-action belong in the linewidth and control budget.

Dropping the off-resonance assumption turns the nucleus into a real dynamical participant rather than a virtual mediator, and the product-of-hyperfine-couplings estimate stops being the right starting point.

Electric dipole–dipole interactions

Beyond magnetic moments, some defect states carry a state-dependent electric dipole moment $\mathbf{d}$, measured in coulomb-metres, whose electrostatic interaction energy in an isotropic dielectric is

$$H_{ee} = \frac{1}{4\pi\epsilon r^3} [\mathbf{d}_1 \cdot \mathbf{d}_2 - 3(\mathbf{d}_1 \cdot \hat{\mathbf{r}})(\mathbf{d}_2 \cdot \hat{\mathbf{r}})].$$

That interaction shares the angular tensor and $1/r^3$ near-field dependence of magnetic dipole coupling, with different material constants and moments: the electric interaction contains the permittivity ϵ rather than μ_0 , and electric dipole moments rather than magnetic moments.

For an illustrative upper scale, two 1-debye dipoles in a dielectric of relative permittivity $\epsilon_r = 5.7$ have the coefficient

$$H_{ee}/h \approx 26.5 \text{ GHz nm}^3/r^3.$$

Here $1 \text{ D} = 3.33564 \times 10^{-30} \text{ C m}$.

At 10 nm that estimate gives 26.5 MHz. Where the two qubit states of each defect differ in electric dipole moment by only 0.01 D, the state-dependent product drops by 10^{-4} , giving 2.65 kHz at 10 nm.

These dipole moments are assumed values for illustration, not measured NV values.

Ground-state spin levels couple to electric fields only through admixture with spin–orbit, orbital, or strain degrees of freedom, while optical orbital transitions couple much more strongly at the price of faster decay.

The NV electric and strain response, including its symmetry, is reviewed in [Ro74]. Useful electric coupling must exceed the Stark linewidth from charge-noise-induced frequency shifts while staying inside the computational subspace.

A large electric susceptibility strengthens electrical control and simultaneously sensitizes the qubit to unwanted electric fields. A state-dependent moment below one debye scales the megahertz-scale estimate down by the product of the reductions in the two moments.

Experimental benchmark for an NV pair

Dipolar-coupled NV centers provide a direct experimental benchmark. [Experiment] A pair separated by 25 ± 2 nm showed a measured effective dipolar coupling of 4.93 ± 0.05 kHz, and refocused optimal-control sequences produced electron-spin entanglement with reported fidelity above 0.82 at room temperature [R106].

The point-dipole coefficient at 25 nm is

$$52.04 \text{ MHz}/25^3 = 3.33 \text{ kHz}.$$

An orientation or transition-matrix-element factor of order unity brings the measured value into agreement with magnetic dipolar coupling. Spin echo and control engineering were therefore central to the experiment: free-induction noise would otherwise have obscured the interaction, which uncontrolled evolution under a 4.93 kHz coupling alone could not reveal.

[Experiment] Millisecond-scale Hahn-echo coherence—up to 1.8 ms at room temperature in a specially isotopically engineered, low-spin-bath diamond sample—has been reported for single NV centers [Ro79]. A Hahn echo is a refocusing sequence suppressing slowly varying phase noise. That coherence time stays with its specially prepared sample rather than transferring to dense implanted arrays, where implantation damage, surfaces, unwanted paramagnetic defects, and inter-defect frequency disorder affect both T_2 and fabrication yield.

Mechanism	Range or placement law	Evidence level relevant here	Coherent-use comparison
Magnetic dipole	$1/r^3$, angular tensor	Two-NV gates and entanglement demonstrated [R106]	j versus spin linewidth, $1/T_2$, pulse error
Direct exchange	roughly exponential; atomistic orientation dependence	Diamond-pair dominance below about 3 nm is numerical [R107]	$J/\hbar$ versus dephasing, leakage, placement disorder
Superexchange	$4t^2/U$; hopping often exponential	General mechanism established; defect implementation is material-specific [R108]	$J/\hbar$ versus charge/orbital noise and validity gap $U/\hbar$
Free-space optical	near $1/r^3$, retarded oscillatory tail	Collective optical effects are platform-specific	coherent shift versus optical linewidth and collective decay
Cavity photon	cavity mode profile, not a simple power	Two-SiV coherent optical interaction demonstrated [R086]	J_{cav} versus emitter dephasing and induced κ loss

Mechanism	Range or placement law	Evidence level relevant here	Coherent-use comparison
Static strain	often $1/r^3$ for bulk elastic dipoles	Single-defect response established [R074]; NV pair interaction derived [R111]	shift versus strain disorder and spectral diffusion
Phonon bus	mechanical Green function/mode profile	Bulk NV theory [R111]; SiV waveguide proposal [R109]	J_{ph} versus $\kappa_m(2\bar{n} + 1)$ and spin dephasing
Nuclear bus	product of hyperfine couplings	Closely related electron-mediated nuclear scheme proposed [R110]	J_N versus nuclear/electron decoherence and control time
Electric dipole	near-field $1/(\epsilon r^3)$	Electric susceptibility established; pair gate is defect-specific [R074]	state-dependent shift versus Stark linewidth and leakage

The mechanisms in this table resist ranking by coupling magnitude alone. For quantum logic, a 10 MHz orbital interaction against a 100 MHz optical linewidth is weaker than a 10 kHz spin interaction against a 100 Hz refocused linewidth.

Two coherently coupled defect spins are two interacting physical qubits. Projecting several such spins onto a two-level subspace or doublet instead forms one encoded qubit.

Either construction still falls short of topological order. Emergent anyons need a many-body Hamiltonian, a gapped phase, and nonlocal observables, and an interaction graph sharing the geometry of a theoretical model does not supply them.

Common technical errors

- Do not report an angular frequency Ω , measured in rad s^{-1} , as though it were a cyclic frequency f , measured in hertz. They satisfy $\Omega = 2\pi f$. This chapter consistently uses $H/\hbar$ and hertz.
- Do not interpret 52.04 MHz nm^3 as the coupling for every transition. The observable secular coupling also contains angular factors and transition matrix elements. At the magic angle, the secular term vanishes even though $d(r)$ remains finite.
- Do not classify every indirect interaction as superexchange. Superexchange specifically arises from virtual charge hopping. Virtual cavity photons produce photon-mediated coupling, and virtual phonons produce phonon-mediated coupling. Matching second-order mathematical forms still leaves the microscopic noise mechanisms distinct.
- Do not eliminate a cavity or mechanical mode and report only $g_1 g_2 / \Delta$. The same elimination produces inherited loss. Quoting the coherent term without κ , temperature, and emitter linewidth leaves the performance budget incomplete.
- Do not select the largest available J without considering uniformity and control. Very short-range exchange can exceed dipolar coupling while varying strongly among nominally identical pairs. A target Hamiltonian can therefore favor a weaker but more predictable interaction edge.
- Do not combine the best coherence measured in one sample with the smallest defect spacing achieved in another and treat the result as an integrated-device demonstration. A valid feasibility analysis must evaluate coupling, linewidth, fabrication yield, addressability, and temperature under the same fabrication conditions. Combining

results from different papers can bound the possibilities without proving the performance of an integrated device.

Conceptual checks

- **Magnetic dipolar coupling as a function of separation.** A localized magnetic dipole confines its moment to a finite region of space. At separation r , its magnetic field scales as the dipole moment divided by r^3 . The second dipole contributes an interaction energy proportional to the scalar product of its moment with that field, giving a magnetic dipolar coupling scaling as $1/r^3$.
- **Effect of doubling the electron-spin separation.** Let d denote the magnetic dipolar coupling strength between two electron spins, with distance dependence

$$d(r) \propto 1/r^3,$$

so doubling the separation gives $d(2r) = d(r)/8$. For an electron-spin g -factor $g \approx 2$, the corresponding coupling is 52.04 MHz at 1 nm and 6.505 MHz at 2 nm.

- **Why a bare coupling number misleads: $d = 50$ kHz as a coherence criterion.** Coherent dynamics and gate viability turn on more than the coupling strength. The relevant matrix element—the Hamiltonian matrix element connecting the selected states—competes with the homogeneous linewidth of intrinsic spectral broadening, the refocused T_2 , measured under a refocusing sequence, relaxation, detuning disorder, and control errors. For example, a 52 kHz coupling at 10 nm sits inside a noisy linewidth of 318 kHz yet clears a quiet linewidth of 3.18 kHz.
- **Zero of the secular angular factor.** The secular approximation keeps interaction terms conserving energy to leading order and drops rapidly oscillating nonsecular terms. Its angular factor satisfies

$$1 - 3 \cos^2 \theta = 0$$

when $\cos^2 \theta = 1/3$, so that

$$\theta = \cos^{-1}(1/\sqrt{3}) \approx 54.7^\circ.$$

Near this angle the retained secular contribution vanishes, so the terms dropped by the secular truncation, together with any small angular tilt, can dominate the residual coupling.

- **Distinction between direct exchange and superexchange.** Direct exchange needs spatially overlapping orbitals: fermionic antisymmetry, the sign change of a fermionic many-body wavefunction under exchange of two identical fermions, splits the Coulomb energy by spin configuration. Superexchange instead proceeds by virtual hopping through energetically forbidden charge configurations. In the simplest model its characteristic scale is $4t^2/U$, where t is the hopping amplitude and U is the energy cost of the intermediate charge configuration. Treating the two as equivalent gets the interaction-range law, the noise assumptions, and the placement tolerances wrong.

- **Cavity loss belongs in any J_{12}^{cav} report.** J_{12}^{cav} is a cavity-mediated coherent interaction between emitters 1 and 2. The emitter positions enter through the local cavity-mode coupling amplitudes g_i , so the interaction follows no universal $1/r^n$ distance law. The cavity loss rate κ also contributes induced decay

$$(g_i/\Delta_i)^2\kappa,$$

where Δ_i is the detuning of emitter i from the cavity mode. A gate needs the coherent shift to exceed the induced decay and emitter dephasing rates; a nonzero shift alone does not qualify.

These numerical values and interaction tensors equip a small-cluster Hamiltonian—the Hamiltonian restricted to a finite set of interacting degrees of freedom—and test whether its spectrum holds an isolated low-energy doublet: two low-energy states separated from the rest by an energy gap.

Sources

- [R105] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor, “CODATA recommended values of the fundamental physical constants,” NIST/CODATA database (2022 adjustment, accessed 2026), <https://physics.nist.gov/cuu/Constants/>.
- [R106] F. Dolde et al., “High-fidelity spin entanglement using optimal control,” Nature Communications 5, 3371 (2014). DOI: [10.1038/ncomms4371](https://doi.org/10.1038/ncomms4371). arXiv:1309.4430.
- [R107] V. R. Kortan, C. Şahin, and M. E. Flatté, “Nanometer-scale exchange interactions between spin centers in diamond,” Physical Review B 93, 220402(R) (2016). DOI: [10.1103/PhysRevB.93.220402](https://doi.org/10.1103/PhysRevB.93.220402). arXiv:1603.03485.
- [R108] P. W. Anderson, “Antiferromagnetism. Theory of superexchange interaction,” Physical Review 79, 350–356 (1950). DOI: [10.1103/PhysRev.79.350](https://doi.org/10.1103/PhysRev.79.350).
- [Ro86] R. E. Evans et al., “Photon-mediated interactions between quantum emitters in a diamond nanocavity,” Science 362, 662–665 (2018). DOI: [10.1126/science.aau4691](https://doi.org/10.1126/science.aau4691). arXiv:1807.04265.
- [R109] M.-A. Lemonde et al., “Phonon networks with silicon-vacancy centers in diamond waveguides,” Physical Review Letters 120, 213603 (2018). DOI: [10.1103/PhysRevLett.120.213603](https://doi.org/10.1103/PhysRevLett.120.213603). arXiv:1801.01904.
- [R110] A. Bermudez, F. Jelezko, M. B. Plenio, and A. Retzker, “Electron-mediated nuclear-spin interactions between distant nitrogen-vacancy centers,” Physical Review Letters 107, 150503 (2011). DOI: [10.1103/PhysRevLett.107.150503](https://doi.org/10.1103/PhysRevLett.107.150503). arXiv:1107.2617.
- [Ro74] M. W. Doherty et al., “The nitrogen-vacancy colour centre in diamond,” Physics Reports 528, 1–45 (2013). DOI: [10.1016/j.physrep.2013.02.001](https://doi.org/10.1016/j.physrep.2013.02.001). arXiv:1302.3288.
- [Ro79] G. Balasubramanian et al., “Ultralong spin coherence time in isotopically engineered diamond,” Nature Materials 8, 383–387 (2009). DOI: [10.1038/nmat2420](https://doi.org/10.1038/nmat2420).
- [R111] J. J. Nakane, K. Tahara, K. Kutsuki, and A. Yamakage, “Phonon-mediated spin-spin interaction: A general theory and application to diamond nitrogen vacancy centers,” Physical Review B 110, 064428 (2024). DOI: [10.1103/PhysRevB.110.064428](https://doi.org/10.1103/PhysRevB.110.064428).

Chapter 11 — Encoding a qubit in three spins subject to noise

Three spin degrees of freedom can fluctuate independently. Environmental perturbations such as magnetic-field variations and temperature changes can flip one spin and destroy information stored in that physical spin.

Strong interactions among the three spins reshape the spectrum. When the interaction energy makes most collective configurations costly, only two states remain at low energy; moving elsewhere requires excitation to a higher-energy state.

The construction leaves the intrinsic noise of each physical spin unchanged. It defines an energy for the three-spin cluster and works within its two-dimensional low-energy subspace. The sections below identify that subspace, calculate the action of physical operators inside it, and assess the resulting protection and limits.

Spectrum and low-energy states of three spins

Take three physical spin-1/2 objects labeled 1, 2, 3. Along the applied-field direction each has two eigenstates, conventionally shown by up and down arrows, so their product basis contains eight states.

The construction selects two of those states as the low-energy pair rather than assigning a separate logical state to every basis state.

Call the interacting three-spin system a cluster. Several clusters can later be coupled weakly, with each cluster represented by an effective two-level system.

Let $\mathbf{S}_i = (S_i^x, S_i^y, S_i^z)$ be the dimensionless spin operator at site i , with S_i^z eigenvalues $\pm 1/2$. The cluster Hamiltonian assigns energy through antiferromagnetic Heisenberg exchange and a uniform longitudinal field:

$$H_C = J \sum_{i < j} \mathbf{S}_i \cdot \mathbf{S}_j - b S_{\text{tot}}^z, \quad S_{\text{tot}}^z = \sum_{i=1}^3 S_i^z.$$

Heisenberg exchange is the rotationally invariant pair interaction $\mathbf{S}_i \cdot \mathbf{S}_j$. Antiferromagnetic exchange has $J > 0$ and favors lower total spin; the field-coupling coefficient also has $b > 0$. Both J and b are energies, so every term in H_C has energy units. Experiments often quote the corresponding frequencies J/h and b/h , with h Planck's constant.

The Hilbert space of three spin-1/2 objects decomposes into one total-spin $S_{\text{tot}} = 3/2$ quartet and two copies of a total-spin $S_{\text{tot}} = 1/2$ doublet. First restrict the system to total magnetization $m = S_{\text{tot}}^z = +1/2$. Two states in this sector can be written in the product basis as

$$|\tilde{0}\rangle = \frac{|\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle}{\sqrt{2}},$$

$$|\tilde{1}\rangle = \frac{|\uparrow\downarrow\uparrow\rangle + |\downarrow\uparrow\uparrow\rangle - 2|\uparrow\uparrow\downarrow\rangle}{\sqrt{6}}.$$

In $|\tilde{0}\rangle$, the first two spins form a singlet, which is their total-spin-zero state. In $|\tilde{1}\rangle$, the first two spins occupy the triplet channel, meaning their total spin is one, and combine with spin 3 to produce total spin 1/2. Direct evaluation of the inner products gives unit norm for each state and $\langle\tilde{0}|\tilde{1}\rangle = 0$.

The exchange energy can be evaluated using the total-spin identity

$$\sum_{i<j} \mathbf{S}_i \cdot \mathbf{S}_j = \frac{1}{2} \left(S_{\text{tot}}^2 - \sum_i S_i^2 \right),$$

together with $S_i^2 = 3/4$. Both states consequently have the same energy,

$$E_L = -\frac{3J}{4} - \frac{b}{2}.$$

The two states therefore span a two-dimensional eigenspace of H_C . A useful qubit also requires controls that act within this pair and measurements that distinguish its states.

If $J < 0$, the exchange is ferromagnetic, the quartet lies lower in energy, and this pair is not the ground-state sector. The assumption $J > 0$ is therefore essential.

Definition of the encoded qubit

Two low-energy eigenstates become a computational system when controls can act on them, measurements can read them, excluded states remain unoccupied, and physical operations on the three spins induce the intended pair operations.

An encoded qubit is a chosen two-dimensional subspace of a larger Hilbert space into which quantum information is deliberately mapped. “Encoded” specifies where the information resides; protection is a separate property. A repetition-code state, this exchange-controlled pair, and an arbitrary pair of molecular energy levels all qualify when information is intentionally assigned to the selected subspace.

The retained basis states are denoted by $|\tilde{0}\rangle$ and $|\tilde{1}\rangle$. Every other state in the eight-dimensional Hilbert space represents departure from the encoding.

The encoding becomes protective only after its response to a specified noise model has been established.

Leakage gap and energetic isolation

Transitions from the selected pair to excluded states require additional energy. Excluded states stay weakly occupied when the perturbations driving the transitions are small compared with that energy.

Define the projectors

$$P = |\tilde{0}\rangle\langle\tilde{0}| + |\tilde{1}\rangle\langle\tilde{1}|, \quad Q = I - P,$$

where I is the identity operator on the full eight-dimensional three-spin Hilbert space. The projector P selects the encoded subspace, while the complementary projector Q selects all states outside it.

For an exactly degenerate pair, the minimum excitation energy into the excluded subspace is

$$\Delta_{\text{leak}} = \min_{|a\rangle \in Q} (E_a - E_L),$$

When every excluded eigenstate lies above E_L , this quantity is the leakage gap: the energetic separation from states outside the encoded subspace. It says nothing about noise acting within the pair.

With a small internal splitting, measure the leakage gap from the higher state of the pair to the nearest excluded level. This keeps the qubit's own splitting out of the leakage gap.

The other $S_{\text{tot}} = 1/2$ states have $m = -1/2$ and lie an energy b above the chosen pair. The lowest state in the $S_{\text{tot}} = 3/2$ quartet has $m = +3/2$ and lies an energy $3J/2 - b$ above it. Therefore,

$$\boxed{\Delta_{\text{leak}} = \min \left(b, \frac{3J}{2} - b \right)}, \quad 0 < b < \frac{3J}{2}.$$

At $b = 0$, the opposite-magnetization partners are degenerate with the chosen pair, leaving the fixed- m subspace without energetic isolation. Increasing b raises those partners relative to the encoded pair; eventually the polarized quartet becomes the nearest excluded sector. At $b = 3J/4$, both excitation energies equal $3J/4$.

If a perturbation has a characteristic matrix element v , also expressed in energy units, then v/Δ_{leak} is dimensionless. A valid low-energy description requires this ratio to be small. Similarly, a resonant control field with angular Rabi rate Ω requires $\hbar\Omega/\Delta_{\text{leak}} \ll 1$, unless a selection rule makes the relevant leakage matrix element vanish exactly.

A spectral gap suppresses transitions only for perturbations that can produce them. Noise within the encoded pair remains possible: even a perfectly isolated two-level subspace can dephase.

Projection of physical operators into the encoded subspace

Projecting a physical operation on the three spins produces its effective two-by-two matrix on the encoded pair.

The encoded Pauli operators are

$$\begin{aligned}\bar{Z} &= |\tilde{0}\rangle\langle\tilde{0}| - |\tilde{1}\rangle\langle\tilde{1}|, \\ \bar{X} &= |\tilde{0}\rangle\langle\tilde{1}| + |\tilde{1}\rangle\langle\tilde{0}|, \\ \bar{Y} &= -i|\tilde{0}\rangle\langle\tilde{1}| + i|\tilde{1}\rangle\langle\tilde{0}|.\end{aligned}$$

For any physical operator O , projection gives its first-order action within the encoded subspace:

$$O_{\text{eff}} = POP = c_0P + c_x\bar{X} + c_y\bar{Y} + c_z\bar{Z}.$$

The four coefficients come from the four matrix elements of O in the ordered basis $\{|\tilde{0}\rangle, |\tilde{1}\rangle\}$. Thus POP restricts the physical operator to the encoded subspace and returns the corresponding two-by-two matrix.

For the three-spin system, direct projection gives

$$\begin{aligned}P(\mathbf{S}_1 \cdot \mathbf{S}_2)P &= -\frac{1}{4}P - \frac{1}{2}\bar{Z}, \\ P(\mathbf{S}_2 \cdot \mathbf{S}_3)P &= -\frac{1}{4}P + \frac{1}{4}\bar{Z} - \frac{\sqrt{3}}{4}\bar{X}, \\ P(\mathbf{S}_1 \cdot \mathbf{S}_3)P &= -\frac{1}{4}P + \frac{1}{4}\bar{Z} + \frac{\sqrt{3}}{4}\bar{X}.\end{aligned}$$

Suppose the three pairwise exchange coefficients J_{12}, J_{23}, J_{13} can be tuned independently. Their projection into the encoded pair is

$$PH_C P = E_{\text{off}}P + h_x\bar{X} + h_z\bar{Z},$$

where

$$E_{\text{off}} = -\frac{J_{12} + J_{23} + J_{13}}{4} - \frac{b}{2}, \quad h_x = \frac{\sqrt{3}}{4}(J_{13} - J_{23}), \quad h_z = \frac{J_{13} + J_{23} - 2J_{12}}{4}.$$

All three coefficients have energy units. The term $E_{\text{off}}P$ shifts both logical states equally, so it contributes no relative logical phase. The coefficients h_x and h_z rotate the encoded state about two nonparallel Bloch-sphere axes, where the Bloch sphere represents a two-level quantum state.

Combining noncommuting rotations in pulse sequences generates rotations about the third axis. This is the central exchange-only control principle proposed for three-spin qubits [R114] and later demonstrated in semiconductor quantum dots [R115].

[Theory/Experiment] Isotropic exchange commutes with both S_{tot}^2 and S_{tot}^z , so ideal exchange pulses leave this encoded pair uncoupled from the quartet. Real systems can open leakage channels through anisotropy, field gradients, timing errors, and coupling to higher orbital or defect levels.

For several nearby clusters, the full Hamiltonian is

$$H = \sum_C H_C + \lambda \sum_{\langle C,D \rangle} V_{CD}.$$

Here, C and D label clusters, H_C is the strong interaction within cluster C , V_{CD} couples neighboring clusters, and the dimensionless parameter λ tracks the weak coupling between clusters. If every H_C has a well-isolated pair, the large microscopic Hilbert space may admit an effective description containing one two-level system per cluster.

This reduction works when the selected states are controllable and readable, excluded states remain weakly populated, projected interactions have the required operator form, and fabrication reproducibly generates the assumed spectrum.

For the projection of multiple clusters, let P_C project cluster C onto its selected pair and define $P_{\text{all}} = \prod_C P_C$.

To first order in the weak intercluster coupling,

$$H_{\text{eff}} = P_{\text{all}} H P_{\text{all}} = \sum_C P_C H_C P_C + \lambda \sum_{\langle C,D \rangle} (P_C P_D) V_{CD} (P_C P_D).$$

Each projected pair interaction can be expanded in products $\bar{\sigma}_C^\mu \bar{\sigma}_D^\nu$, where $\mu, \nu \in \{0, x, y, z\}$ and $\bar{\sigma}^0 = P$.

Virtual transitions into the subspace selected by Q , meaning temporary occupation of states outside the encoded subspace, generate corrections of characteristic scale $\lambda^2 \|V\|^2 / \Delta_{\text{leak}}$. Chapter 22 derives this result.

If individual clusters are not energetically isolated, the product projector does not define a valid low-energy description. A nonzero QVP produces leakage and virtual corrections. The relevant small parameter is a coupling matrix element divided by Δ_{leak} .

Decoherence-free behavior under collective longitudinal noise

Encoding alone does not imply protection. Protection must be demonstrated for a specific interaction between the system and its environment.

Suppose the three-spin cluster couples to a bath through

$$H_{SB} = S_{\text{tot}}^z \otimes B_z,$$

where B_z is an unspecified bath operator and $\otimes$ denotes the tensor product between system and bath operators. Within the fixed- $m = +1/2$ encoded pair,

$$PS_{\text{tot}}^z P = \frac{1}{2}P.$$

The bath coupling is therefore identical for $|\tilde{0}\rangle$ and $|\tilde{1}\rangle$. It cannot distinguish the two encoded states and adds the same phase to both.

A subspace on which every relevant noise operator acts as a scalar is a decoherence-free subspace for that noise model [R112]. [Theory] The qualification by noise model is essential: the term does not imply immunity to all sources of decoherence. The relevant noise operators must be identified, and each must be shown to act as a scalar on the code.

The pair is not a decoherence-free subspace for independent noise of the form $\sum_i S_i^z \otimes B_i$. Distinct bath operators B_i can distinguish the different spatial spin patterns. An environment that couples approximately identically to all three spins does not replace this calculation.

Noiseless subsystem under fully collective noise

For fully collective noise, the system–bath interaction can contain all three total-spin components:

$$H_{SB} = \sum_{\alpha=x,y,z} S_{\text{tot}}^\alpha \otimes B_\alpha.$$

A collective transverse rotation changes m , so this fixed- m pair does not form a decoherence-free subspace for the full interaction. The entire total-spin-1/2 sector factorizes as

$$\mathcal{H}_{1/2} \cong \mathbb{C}_{\text{logical}}^2 \otimes \mathbb{C}_{\text{gauge}}^2.$$

This factorization separates a logical two-dimensional space from a gauge two-dimensional space. Collective spin operators act on the gauge factor and leave the logical factor unchanged. Information stored in the first factor is therefore a noiseless subsystem [R113]. [Theory] The second factor is called a gauge spin; its state can change without erasing the logical information.

This construction handles collective longitudinal noise and the transverse components in the full interaction. It remains conditional on the coupling being genuinely collective.

Three equal-exchange spins have a four-dimensional $S = 1/2$ sector. A two-state projector within that sector requires fixed magnetization, another splitting mechanism, or the noiseless-subsystem interpretation. The additional two states carry the gauge degeneracy and must remain in the description.

Exchange-only control

An exchange-only qubit uses controllable pairwise exchange interactions in place of separately addressed single-spin rotations [R114]; [R115]. This describes the available control operations; reduced noise requires a separate argument.

Because exchange interactions commute with collective rotations, they naturally act on the logical factor of the three-spin noiseless subsystem without depending on the gauge orientation.

Magnetic dipolar coupling is anisotropic, so a spin–spin interaction need not supply isotropic exchange. Project every physical interaction using its actual operator form and compare it with the interaction required by the encoding.

Symmetry-based selection rules

Symmetry can make a leakage matrix element exactly zero. An energy gap instead reduces transitions when the perturbation is weak.

If a symmetry generator G commutes with H_C , the eigenstates carry symmetry-sector labels. A perturbation that also commutes with G has zero matrix elements between inequivalent sectors. This selection rule is symmetry protection: the relevant transition matrix element vanishes.

In the model considered here, isotropic exchange preserves total spin, while the longitudinal field preserves total magnetization. These symmetries can forbid some leakage transitions.

These symmetries leave the logical splitting unprotected. Both encoded states have $S_{\text{tot}} = 1/2$ and $m = +1/2$, and unequal exchanges that respect these symmetries generate $\bar{X}$ and $\bar{Z}$ terms at first order. A protection claim must therefore identify the symmetry, the symmetry representation carried by the code, and the transformation properties of the actual perturbation.

If the perturbation does not commute with G , the corresponding matrix element need not vanish. Anisotropy, field gradients, and orbital mixing commonly break the relevant symmetry assumptions.

Local distinguishability and absence of topological order

The two encoded states occupy three nearby spins, and a local pair observable distinguishes them. In particular, $S_1 \cdot S_2$ has different projected values in $|\tilde{0}\rangle$ and $|\tilde{1}\rangle$.

Unequal local exchange consequently splits the states at first order. A closed path among the three sites also fails to acquire a logical phase determined by the geometric shape of that path.

The logical operators therefore have support on three nearby spins, and the degeneracy is not set by the topology of a macroscopic surface. This construction is an encoding, not topological order.

Topological order, when present, is a property of an entire many-body phase. Small local perturbations cannot read the encoded information or split the relevant degeneracy at first order, and the excitations carrying the information can be transported over a large lattice. Such excitations are later referred to as anyons.

The present cluster is finite: its three physical spins are ordinary qubits, the pair selected by P is one encoded qubit, and controlled evolution that reproduces a target model constitutes digital or analog emulation. Such a finite construction has no emergent many-body phase.

Relation to color-center experiments and proposals

The exact Heisenberg triangle is an instructional model. Color-center hardware includes spin-1 NV electron states, spin-1/2 or higher-spin nuclear memories, anisotropic dipolar and hyperfine couplings, optical reset, strain, and imperfect spatial placement. Selecting two levels of a spin-1 center can define an effective physical qubit, while magnetic dipolar coupling remains distinct from isotropic exchange.

A literature search through August 2026 is most informative with explicit inclusion criteria. A direct realization of the model derived here would require: (i) at least three deliberately used spins associated with color centers; (ii) strong static interactions within each cluster; (iii) two logical levels isolated in the spectrum; and (iv) pair interactions projected into the encoded space. The related literature falls into four categories.

Encoded center–nuclear registers have been demonstrated. [Experiment] Reiserer et al. encoded a diamond–network memory in two nearby ^{13}C spins whose antiparallel states suppress phase shifts shared by both spins during repeated NV optical operations [R116].

Cramer et al. encoded a qubit across three ^{13}C spins and repeatedly extracted and corrected error syndromes using an NV ancilla [R117].

These systems are encoded clusters associated with color centers. Their protection arises respectively from a decoherence-free subspace and active error correction. The three-center exchange model instead relies on a ground-state doublet.

Spin-pair encodings have produced high-performance experimental memories. [Experiment] Bartling et al. used pairs of ^{13}C nuclei near NV nodes as antiparallel spin-pair qubits, reported intrinsic dephasing times exceeding one minute, and entangled two such encoded qubits [R118].

The statement “exceeding one minute” applies to those measured nuclear-spin pairs and experimental protocols, not to arbitrary color-center clusters. These results provide strong evidence for noise-selective encoding; they do not establish passive many-body topological protection.

Large local registers and coupled color centers have been demonstrated. These systems meet only subsets of the four criteria. [Experiment] A ten-qubit diamond register combined one NV electron, its nitrogen nucleus, and eight ^{13}C nuclei under gate-based control [R119].

Separately, two NV electron spins have been entangled through their direct dipolar interaction [R106]. The first system is a programmable register, while the second is an interacting pair.

Together, these papers report no low-energy encoded qubit formed from three color centers.

Architectures using multiple NV centers in decoherence-free subspaces have also been proposed. [Proposal] Yun et al. proposed geometric gates in a decoherence-free subspace using NV centers coupled through cavity-mediated interactions [R120].

This proposal uses a color-center decoherence-free subspace. Its logical space and gate operations are engineered with drives and a mediator; it does not demonstrate that a fabricated, strongly coupled NV triangle naturally realizes the spectrum and projected exchange interactions derived above.

The resulting literature assessment is limited but specific. Existing work includes encoded qubits constructed from spins near color centers, coupled color centers, and proposals for multi-center decoherence-free-subspace gates. Among the sources identified here, no experiment or proposal simultaneously establishes the sequence “three deliberately positioned color-center electronic spins $\rightarrow$ static exchange-dominated isolated doublet $\rightarrow$ projected exchange-only qubit.” This conclusion is a search result, not a proof that no such paper exists. It identifies the missing element that a defect-cluster proposal must establish independently; semiconductor quantum-dot results cannot establish it.

Common conceptual and modeling errors

Two low-energy levels form a protected qubit only after the noise model and leakage channels have been analyzed. The leakage gap controls transitions out of the code. Noise projecting to $\bar{Z}$ causes dephasing within it.

The additional two states in the $S = 1/2$ sector must remain in the model. Three equal-exchange spins have a four-dimensional $S = 1/2$ sector. A two-state projector requires fixed magnetization, another splitting mechanism, or the noiseless-subsystem interpretation.

Projecting onto P while permanently neglecting Q omits both leakage and virtual processes. A nonzero QVP produces leakage and virtual corrections of scale $\lambda^2 \|V\|^2 / \Delta_{\text{leak}}$.

A spin interaction is not necessarily exchange. Magnetic dipolar coupling is anisotropic, hyperfine coupling connects unlike spins, and optical and strain-mediated interactions introduce driven dynamics and loss. Project each interaction from its actual physical operator.

A decoherence-free subspace cancels specified correlated noise. Independent fields, coupling mismatch, relaxation, optical back-action, and control errors remain possible and require separate analysis.

Active error correction and a static energy gap are distinct mechanisms. Repeated syndrome extraction can increase a logical lifetime, and that protection ends if the correction cycle stops.

A leakage gap is a static property of the energy spectrum. Active correction and spectral protection can coexist; they remain distinct mechanisms.

Local pair correlations distinguish the two states, so the cluster is a finite encoded object rather than an anyon or a topological encoding. Local exchange splits the states at first order, unlike a deconfined excitation of a many-body phase.

Verification exercises

The orthonormality of $|\tilde{0}\rangle$ and $|\tilde{1}\rangle$ follows directly from their product-basis coefficients.

Expanding in the product basis,

$$\langle \tilde{0} | \tilde{1} \rangle = \frac{1}{\sqrt{12}}(1 - 1 + 0) = 0.$$

For each state, the squared magnitudes of the coefficients sum to one. Equivalently, the singlet channel of the first two spins is orthogonal to their triplet channel.

The leakage gap satisfies $\Delta_{\text{leak}} = \min(b, 3J/2 - b)$ for $0 < b < 3J/2$.

The $m = -1/2$ partners require energy b , while the lowest quartet member requires energy $3J/2 - b$. The leakage gap is the smaller of these two excitation energies.

At $b = 0$, the fixed- m encoding ceases to be energetically isolated.

The opposite-magnetization partners then have the same energy as the chosen pair. The leakage-gap formula assumes $0 < b < 3J/2$.

The projected total magnetization is $PS_{\text{tot}}^z P = \frac{1}{2}P$.

Both $|\tilde{0}\rangle$ and $|\tilde{1}\rangle$ have $m = +1/2$, so S_{tot}^z acts as the scalar $1/2$ throughout the range of P . This is why collective longitudinal noise cannot distinguish the encoded states.

For noise of the form $\sum_i S_i^z \otimes B_i$, with independent bath operators in place of the common $S_{\text{tot}}^z \otimes B_z$, the decoherence-free-subspace condition generally fails.

Different bath operators B_i can distinguish the spatial spin patterns. The encoded pair therefore fails the decoherence-free condition for this noise; assuming a common bath cannot replace the projection calculation.

The encoded qubit is supported on three local spins; its protection is therefore not topological.

Local pair operators distinguish $|\tilde{0}\rangle$ from $|\tilde{1}\rangle$, and unequal local exchange splits them at first order. The logical operators have support on three nearby spins, making this a finite encoding rather than a topologically ordered phase.

Sources

- [R112] P. Zanardi and M. Rasetti, “Noiseless quantum codes,” Physical Review Letters 79, 3306–3309 (1997). DOI: [10.1103/PhysRevLett.79.3306](https://doi.org/10.1103/PhysRevLett.79.3306).
- [R113] E. Knill, R. Laflamme, and L. Viola, “Theory of quantum error correction for general noise,” Physical Review Letters 84, 2525–2528 (2000). DOI: [10.1103/PhysRevLett.84.2525](https://doi.org/10.1103/PhysRevLett.84.2525). arXiv:quant-ph/9908066.

- [R114] D. P. DiVincenzo, D. Bacon, J. Kempe, G. Burkard, and K. B. Whaley, “Universal quantum computation with the exchange interaction,” *Nature* 408, 339–342 (2000). DOI: [10.1038/35042541](https://doi.org/10.1038/35042541). arXiv:quant-ph/0005116.
- [R115] E. A. Laird, J. M. Taylor, D. P. DiVincenzo, C. M. Marcus, M. P. Hanson, and A. C. Gossard, “Coherent spin manipulation in an exchange-only qubit,” *Physical Review B* 82, 075403 (2010). DOI: [10.1103/PhysRevB.82.075403](https://doi.org/10.1103/PhysRevB.82.075403). arXiv:1005.0273.
- [R116] A. Reiserer, N. Kalb, M. S. Blok, K. J. M. van Bemmelen, D. J. Twitchen, M. Markham, T. H. Taminiau, and R. Hanson, “Robust quantum-network memory using decoherence-protected subspaces of nuclear spins,” *Physical Review X* 6, 021040 (2016). DOI: [10.1103/PhysRevX.6.021040](https://doi.org/10.1103/PhysRevX.6.021040). arXiv:1603.01602.
- [R117] J. Cramer, N. Kalb, M. A. Rol, B. Hensen, M. S. Blok, M. Markham, D. J. Twitchen, R. Hanson, and T. H. Taminiau, “Repeated quantum error correction on a continuously encoded qubit by real-time feedback,” *Nature Communications* 7, 11526 (2016). DOI: [10.1038/ncomms11526](https://doi.org/10.1038/ncomms11526). arXiv:1508.01388.
- [R118] H. P. Bartling, M. H. Abobeih, B. Pingault, M. J. Degen, S. J. H. Loenen, C. E. Bradley, J. Randall, M. Markham, D. J. Twitchen, and T. H. Taminiau, “Entanglement of spin-pair qubits with intrinsic dephasing times exceeding a minute,” *Physical Review X* 12, 011048 (2022). DOI: [10.1103/PhysRevX.12.011048](https://doi.org/10.1103/PhysRevX.12.011048). arXiv:2103.07961.
- [R119] C. E. Bradley, J. Randall, M. H. Abobeih, R. C. Berrevoets, M. J. Degen, M. A. Bakker, M. Markham, D. J. Twitchen, and T. H. Taminiau, “A ten-qubit solid-state spin register with quantum memory up to one minute,” *Physical Review X* 9, 031045 (2019). DOI: [10.1103/PhysRevX.9.031045](https://doi.org/10.1103/PhysRevX.9.031045). arXiv:1905.02094.
- [R106] F. Dolde et al., “High-fidelity spin entanglement using optimal control,” *Nature Communications* 5, 3371 (2014). DOI: [10.1038/ncomms4371](https://doi.org/10.1038/ncomms4371). arXiv:1309.4430.
- [R120] M.-R. Yun, F.-Q. Guo, L.-L. Yan, E. Liang, Y. Zhang, S.-L. Su, C. X. Shan, and Y. Jia, “Parallel-path implementation of nonadiabatic geometric quantum gates in a decoherence-free subspace with nitrogen-vacancy centers,” *Physical Review A* 105, 012611 (2022). DOI: [10.1103/PhysRevA.105.012611](https://doi.org/10.1103/PhysRevA.105.012611).

PART VI — TOPOLOGY AND ANYONS

Some properties survive continuous changes to a system. We begin with simple examples of this idea, then examine what changes when identical particles exchange positions in two dimensions.

Chapter 12 — Winding, homotopy, and topological protection

Remove one point from the plane and watch what happens to closed loops drawn in what remains. We call this space the punctured plane.

$$X = \mathbb{R}^2 \setminus \{\mathbf{p}\},$$

Here $\mathbf{p}$ is the removed point. A closed loop in X is a continuous path that starts and ends at the same place without leaving the punctured plane — formally a continuous map from a circle into X .

Such a loop can be pushed around and reshaped freely — into a circle, a polygon, an irregular curve, or a curve winding several times around the missing point. Reshaping never changes its winding number, the integer counting its net turns around the puncture. Only three violent events change that integer: the curve runs into the puncture, opens up, or breaks.

Shape changes while the loop's class stays fixed. Because a bounded segment cannot determine how the rest of the curve closes around the puncture, winding is a global label; the quantum analogue below uses a nonlocal label of the whole system to encode a bit.

Continuous deformations and permitted operations

Start with a closed loop drawn in the punctured plane. Its shape is unrestricted: circle, triangle, or an irregular curve all qualify.

The loop may be moved and reshaped, provided three rules hold throughout:

- it must remain in the punctured plane;
- it must remain a closed loop;
- it may not intersect the missing point or be cut and rejoined.

A loop that misses the puncture contracts down to a tiny loop around some ordinary point. A loop encircling the puncture once admits no such contraction: shrinking it to a point would drag some intermediate curve across the missing point, which the rules forbid.

A loop winding twice around the puncture forms a third class of its own. Reshaping changes its geometry freely and leaves its winding number fixed at two.

A **deformation** is a continuous one-parameter family of curves, each obeying the three rules. The definition constrains the family, not any particular shape within it.

The three rules carry the whole classification. Allowing cuts would let every loop open up and slip off the puncture; allowing passage through the missing point would let a loop step across it and change its winding. Either relaxation collapses the distinct classes into one.

Survival under deformation is therefore always relative: it depends on the space and on which transformations count as permitted.

The missing point plays no other role: it is simply the one point of the plane the curve must avoid, and that exclusion is what the word puncture means.

Schematically:

$w = 0$ closed contractible loop not enclosing • • = missing point	$w = 1$ closed noncontractible loop enclosing •
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The $w = 0$ loop contracts clear of the missing point; the $w = 1$ loop has no contraction that avoids it.

Definition and calculation of winding number

Put the missing point at Cartesian coordinates

$$\mathbf{p} = (a, b),$$

Here a and b are lengths along two perpendicular axes. Trace the loop with the parameterized curve

$$\mathbf{r}(t) = (x(t), y(t)), \quad 0 \leq t \leq 1.$$

The parameter t marks position along the loop and carries no units. Closure means

$$\mathbf{r}(0) = \mathbf{r}(1).$$

Avoiding the puncture means

$$\mathbf{r}(t) \neq \mathbf{p}$$

for every t .

Track the direction from the puncture to the moving point $\mathbf{r}(t)$ with a continuously **unwrapped angle** $\theta(t)$ in radians: instead of snapping back from 2π to 0 after each counterclockwise turn, it keeps accumulating, reaching four pi after two turns.

The net revolution count is then

$$w = \frac{\theta(1) - \theta(0)}{2\pi}.$$

Angles are dimensionless, so w is dimensionless too. The loop ends where it begins, so its final direction from the puncture matches its initial direction; the accumulated angle differs between the endpoints by a whole number of turns:

$$\theta(1) - \theta(0) = 2\pi w, \quad w \in \mathbb{Z},$$

Here $\mathbb{Z}$ is the set of integers. Counterclockwise winding counts positive and clockwise counts negative.

That integer w is the loop's **winding number** about the puncture: its net revolution count. Length and curvature play no part in it.

For a concrete circle of radius $R > 0$, where R is a length:

$$x(t) = a + R \cos(2\pi t), \quad y(t) = b + R \sin(2\pi t).$$

Its direction angle grows linearly,

$$\theta(t) = 2\pi t.$$

so

$$w = \frac{2\pi - 0}{2\pi} = 1.$$

Running the circle backward, replacing t with $-t$ modulo one, gives $w = -1$. Traversing it twice gives $\theta(t) = 4\pi t$ and $w = 2$. A small circle lying entirely to one side of the puncture swings its direction one way and then back by the same amount, for net zero change and $w = 0$.

Where the loop is **differentiable** — its coordinate derivatives exist — the winding number follows directly from the coordinates:

$$w = \frac{1}{2\pi} \int_0^1 \frac{[x(t) - a] \dot{y}(t) - [y(t) - b] \dot{x}(t)}{[x(t) - a]^2 + [y(t) - b]^2} dt.$$

A dot is differentiation with respect to t . Numerator and denominator are each length squared, keeping the integrand and w dimensionless.

For the circle above the derivatives are

$$\dot{x}(t) = -2\pi R \sin(2\pi t), \quad \dot{y}(t) = 2\pi R \cos(2\pi t).$$

Substituting gives integrand 2π throughout, from numerator $2\pi R^2$ over denominator R^2 :

$$w = \frac{1}{2\pi} \int_0^1 2\pi dt = 1.$$

Deforming the circle into an ellipse changes its circumference and curvature while the winding number stays one, provided the ellipse never touches the puncture. The integer keeps the global fact — the loop still goes around once — and drops the geometric detail.

A deformation that stays clear of the puncture moves the angle continuously, so the winding number would have to move continuously too. An integer has no continuous path from 1 to 0 — every route passes through nonintegers — so the only way the number changes is for its formula to break down, which happens exactly when the loop hits the puncture.

Integer invariants hold steady until a singular event resets them.

Both hypotheses do work. An open arc sweeps any real angle, so without closure w need not be integral. At the puncture the direction itself is undefined, so the formula breaks at the crossing instant.

Local and global geometric information

Examine only one bounded segment of a loop with nonzero winding. Local measurements there give the segment's curvature — how fast its tangent turns — and its length. They give no verdict on whether the rest of the curve closes around the puncture with nonzero winding.

A property is **local** when a bounded neighborhood decides it. Curvature at a point is local, and so is a sharp nondifferentiable bend: smoothing the bend touches only a short segment.

A **global** property depends on the whole object or on how it sits in the full space. Winding is global: no bounded segment carries the completed loop's winding number.

In table form:

Question	Kind of information	Changed by
What is the loop's length?	geometric, global	ordinary stretching
What is its curvature here?	geometric, local	smoothing nearby
Does it pass through this point?	geometric / incidence	moving one segment
What is its winding about the hole?	global, survives stretching	crossing the hole, cutting, or changing closure

Deformation changes most geometric features and preserves selected discrete information. **Topology** studies the distinctions that survive a specified class of allowed transformations — so the permitted operations, not the visual look of a lattice or device, decide what counts.

A honeycomb, a ring, or a drilled device looks geometrically nontrivial, yet coordinates and connectivity alone leave open which quantities survive deformation. Only the permitted operations settle that.

Nonlocal encoding and protection mechanisms

Two homotopy classes can serve as a classical bit: assign one logical value to $w = 0$ loops and the other to $w = 1$ loops.

A local deformation touches only a bounded segment, and the rules forbid exactly the crossings that would change the winding number.

Flipping the bit takes a singular event: the curve hits the puncture, breaks open, or the space itself loses the puncture. Each acts across a nonlocal region — spanning the distance from loop to puncture or around the loop — so larger loops and more distant punctures cost more effort to flip.

Knots show the same pattern in three dimensions. A knot is a loop embedded in space — formally an embedding of S^1 — and its knot type survives any reshaping that never cuts the string (ambient isotopy). A deformation confined to a bounded section cannot turn a knotted loop into the unknot, just as a bounded reshaping cannot change winding number. Both invariants belong to the whole embedding, not to any segment.

That local-versus-global split supplies one ingredient of a protected quantum bit. A classical curve and a quantum state remain different mathematical objects, so the analogy stops there until quantum labels replace winding numbers.

A classical loop carries no entanglement, and its winding number computes nothing by itself. The chapter accordingly retires the loop model here and carries forward only its lesson — global labels resist local changes — recast in quantum operators below.

A correspondence counts as a protection mechanism when every disturbance confined to a sufficiently small region leaves the global label fixed.

Defeating that protection takes an extended process: local errors lining up into a chain, an excitation traveling a noncontractible path — one that cannot shrink to a point without crossing a hole — or a perturbation strong enough to close the gap and destroy the phase. Each costs effort that grows with the relevant distance — the chain length, the path length, the gap size.

Harder does not mean impossible. Local errors accumulate over time; boundaries shorten the path an error must cross; thermal excitations wander; finite devices keep residual splitting between labels; correlated noise strikes non-locally; initialization and measurement fail on their own. Each mechanism below gets its own estimate — difficulty is a scaling argument, not a guarantee.

A stable global label therefore shifts failure from a single local move to processes whose rates depend on path length, gap, boundaries, temperature, and control.

classical loop	quantum many-body system
-----	-----
local deformation	local operator or local noise
winding number	nonlocal quantum label
puncture intersection or loss	extended error or gap-closing event
of continuity or closure	
winding-number measurement	loop measurement or charge label

The rest of the chapter gives these correspondences precise names. Names alone protect nothing; protection comes from the physics — a local probe can neither read nor change the global label — developed in the sections below.

Homotopy between closed loops

Take an initial loop $\mathbf{r}_0(t)$ and a final loop $\mathbf{r}_1(t)$. A continuous deformation between them is a family of intermediate closed loops, each missing the puncture, varying without jumps.

Such a family is called a **homotopy** and written

$$\mathbf{H}(t,s), \qquad 0 \leq t \leq 1, \quad 0 \leq s \leq 1.$$

Here t runs along each loop and s runs across the deformation from start to finish. The endpoints match:

$$\mathbf{H}(t,0) = \mathbf{r}_0(t), \qquad \mathbf{H}(t,1) = \mathbf{r}_1(t).$$

Each intermediate curve stays closed,

$$\mathbf{H}(0,s) = \mathbf{H}(1,s).$$

each avoids the missing point,

$$\mathbf{H}(t,s) \neq \mathbf{p}$$

for every t and s : no intermediate curve touches $\mathbf{p}$. The removed point $\mathbf{p}$ is excluded from every intermediate curve. That excluded point is the **puncture**, and the plane without it the **punctured plane**.

Two loops joined by a homotopy are homotopic in the punctured plane, and homotopic loops share their winding number w .

Changing w takes a singular event — an intermediate loop crossing the puncture, opening up, or breaking — each violating the rules above.

Winding number therefore labels the homotopy classes of the punctured plane, one integer per class.

For loops with a chosen common starting point, that single integer classifies all closed loops in the punctured plane up to homotopy. Richer spaces need more: one invariant found does not finish the classification.

Global sectors and nonlocal loop operators

Each integer w owns one homotopy class: all $w = 1$ loops in one class, all $w = 0$ loops in another, and so on. Permitted local deformations stay inside a class; crossing between classes takes a singular event.

A **sector** is a family of configurations or quantum states sharing one global label that permitted local operations leave fixed. Sectors sort states by label, ignoring geometric shape.

For a quantum system, the **Hilbert space** $\mathcal{H}$ — the vector space of possible states — splits schematically under a global label q as

$$\mathcal{H} = \bigoplus_q \mathcal{H}_q,$$

Here $\mathcal{H}_q$ holds the states of sector q , and $\oplus$ is the direct sum of distinct subspaces.

In lattice models a **loop operator** $W(C)$ acts along a closed path C , and its measured value labels the sector.

A short **contractible** loop — one that shrinks to a point — is typically built from local constraints. A loop winding around a hole or around a periodic direction of a torus admits no such shrinking, so its measured value reports on the whole system rather than any patch.

Kitaev's toric-code model realizes this correspondence exactly: its noncontractible loop operators distinguish degenerate ground states [\[Ro30\]](#).

That correspondence mirrors winding-number classification in the punctured plane. Whether a given loop operator yields a working topological phase still needs its own demonstration.

Sectors also arise from ordinary symmetry or conservation laws — total particle-number parity, for example. Such a label is useful without being topological. Telling the two cases apart takes three further checks: what enforces the sectors, whether any local measurement distinguishes them, and whether they survive generic local perturbations.

Where local operations change the label, sectors classify on paper only. Where generic local perturbations close the gap or break the enforcing symmetry, the separation disappears with them.

Quantum topological order in many-body systems

Consider a many-particle system with several lowest-energy states that no measurement confined to a bounded region can tell apart. Every local observable returns the same statistics in each version.

Telling the versions apart takes an operator reaching nonlocally across the sample. Weak local noise mixes them only by closing the gap above the low-energy manifold or by assembling an extended error chain.

That many-body property is **quantum topological order**: order in the quantum phase itself, not a visible hole in the device.

In the setting relevant here, quantum topological order comes with **long-range entanglement**: entanglement organized globally, beyond what any local order parameter captures. Foundational work tied that organization to topology-dependent ground-state degeneracy and robustness against local disorder [R122]. Exactly solvable models demonstrate how nonlocal loop operators and unusual pointlike excitations can realize it [R030]. Reviews [R015]; [R124] describe the broader formalism and its qualifications.

For a two-dimensional gapped phase intended to store quantum information, use these practical diagnostics:

- a local many-body Hamiltonian;
- an energy gap above the lowest-energy states;
- ground states that are locally indistinguishable in a large system;
- nonlocal operators that distinguish or transform global sectors;
- excitations with a conserved type and, in appropriate phases, unusual exchange and fusion behavior;
- stability of the phase under sufficiently weak local perturbations that do not close the gap.

Some phases host pointlike excitations whose exchange does more than attach a plus or minus sign; those excitations are **anyons**. An anyon is not an electron on a fancier path but an emergent excitation whose exchange and fusion properties the many-body state determines.

[Theory] In an ideal gapped topologically ordered system, matrix elements of an operator O_R supported within a bounded region R approximately satisfy

$$\langle \psi_a | O_R | \psi_b \rangle = c_O \delta_{ab} + \text{finite-size corrections}.$$

Here $|\psi_a\rangle$ and $|\psi_b\rangle$ are different ground states, c_O varies with the operator while staying sector-independent, and δ_{ab} is one for $a = b$ and zero otherwise. In many gapped models the corrections fall exponentially in L/ξ , where L is a system length and ξ the **correlation length** setting how far local correlations reach [R123]; [R015]. Both L and ξ are lengths, keeping L/ξ dimensionless as an exponential argument must be.

A genuinely local probe thus neither identifies the global ground state nor converts one into another — the two operations a stored bit must resist — so the logical bit lives nonlocally.

Test any claim of topological protection against four separate criteria:

- the geometry, including the actual lengths, angles, coordinates, and shape;
- the classical invariant that remains unchanged under the permitted continuous deformations;
- the global quantum sectors, meaning the quantum labels that no sufficiently small local operation can change;
- the presence of quantum topological order rather than only nontrivial geometry, a global conserved quantity, or a programmed simulation.

A loop with winding $w = 1$ carries a classical invariant; a quantum state picked out by a nonlocal loop measurement occupies a global quantum sector.

Where robust local indistinguishability joins long-range entanglement, the phase is a candidate for quantum topological order, and a logical qubit in its sectors is a **topological encoding**.

A circuit on ordinary physical qubits that prepares the model state is a **digital emulation**. It graduates beyond emulation only where the hardware's own equilibrium Hamiltonian produces the phase.

Classical invariant, quantum sector, topological order, topological encoding, and digital emulation mark successive claims of increasing strength; passing one leaves the next to be demonstrated.

Limitations of the classical-loop analogy

Classical winding and quantum topological order both separate local reshaping from changes of a global label. A classical loop is one geometric configuration traced by one definite curve.

A topologically ordered ground state is a coherent superposition of many microscopic configurations, entangled across the system. Its order lives in the wavefunction together with the Hamiltonian that stabilizes it — no lattice diagram exhibits it by itself.

A physical hole earns its place in a topological code where logical operators run on noncontractible paths around it. Drilling that hole into an ordinary magnet creates geometry without creating topological order.

Arranging defect centers around an empty site likewise establishes a geometry. The required quantum order comes only from the interactions among those centers and the many-body state they produce.

Experimental evidence and finite-system limitations

A laboratory system is a finite sample or processor: interactions of finite range, boundaries, disorder, imperfect controls and detectors, nonzero temperature. Exact mathematical phase labels belong to idealized infinite systems; the laboratory reaches them through finite-size diagnostics.

Evidence therefore accumulates in stages, each checking one layer of the idealization:

[Experiment] Superconducting-qubit processors have prepared a toric-code ground state using a circuit, measured topological entanglement entropy, and simulated anyon interferometry [R125]. These results demonstrate controlled preparation and probing of a topological model state. A circuit-prepared state shows what the processor can enact; whether the processor material hosts the phase at equilibrium is a separate measurement the circuit result leaves open.

[Experiment] Programmable Rydberg-atom arrays have implemented frustrated dynamics and measured nonlocal string observables consistent with a toric-code-type spin-liquid regime [R126]. This is analog quantum-simulation evidence for the engineered model: the dynamics follows the target Hamiltonian. Long-lived passive protection is the further claim, testable by lifetime measurements under perturbation.

[Theory] For an ideal local Hamiltonian, stability results keep the topological ground-state structure intact over an appropriate range of weak local perturbations, short of a phase transition [R123]. The theorem assumes weak local perturbations of an ideal Hamiltonian. Temperature, loss, drive errors, long-range couplings, and gap-closing perturbations fall outside those hypotheses, so finite devices face them separately.

For defect-engineered matter the evidence sequence runs:

- fabricate and characterize the defects;
- establish the intended microscopic couplings;
- derive or measure the effective many-body Hamiltonian;
- show a gapped regime and nonlocal sector structure;
- demonstrate local indistinguishability or equivalent diagnostics;
- create, move, and identify the predicted excitations;
- test robustness as size, disorder, temperature, and evolution time are varied.

A regular defect pattern completes the first step and part of the second. Calling the pattern topological jumps ahead: steps three through seven still need their own measurements.

Common conceptual errors

- **Equating a visually suggestive graph with a topological invariant.** A honeycomb, kagome lattice, ring, or punctured device can have nontrivial geometry. The same graph supports a trivial product state under one Hamiltonian and a topologically ordered state under another; the interactions decide.
- **Equating one robust number with quantum topological order.** Winding is robust under the classical rules, while entanglement, excitation statistics, an energy gap, and quantum coherence belong to the quantum system. The classical invariant and quantum topological order answer different questions.
- **Equating a conserved global quantity with topological protection.** A symmetry can divide a Hilbert space into sectors. When a generic local symmetry-breaking perturbation removes the protection, the mechanism is symmetry protection rather than intrinsic topological order. Symmetry protection remains useful for engineering, with a mechanism of its own.

- **Equating an encoding with an emergent phase.** Several physical spins can encode one logical qubit, and a surface-code patch can use nonlocal logical operators. Those constructions concern the code space; an emergent topologically ordered ground state concerns the hardware Hamiltonian. Report active error correction, digital state preparation, and passive Hamiltonian protection as separate mechanisms.
- **Equating a simulation with the simulated material.** A programmable device can reproduce the amplitudes, string observables, or braiding protocol of a topological model. These are genuine experimental achievements [R125]; [R126], establishing properties of the prepared state or simulated dynamics under the stated controls. The underlying superconducting circuits or atoms still require evidence of intrinsic anyonic matter.
- **Equating topology with error-free operation.** Topology suppresses particular local transitions between sectors. Leakage outside the modeled Hilbert space, readout errors, thermal population, correlated disturbances, fabrication failures, and control faults remain separate channels that the architecture must address. A noncontractible logical error can still be assembled from many ordinary local errors.
- **Conflating different meanings of “defect.”** An atomic vacancy in diamond is a microscopic crystal defect; a vortex is a classical defect of an order parameter; an anyon is an emergent quantum excitation; a puncture or boundary in an error-correcting code is also called a defect. The shared word hides different physical objects that may interact within one architecture.

Verification exercises

- **Effect of a smooth deformation on a loop.** Smoothly deforming a circle into an ellipse without crossing the hole changes its length and curvature while preserving its winding number w , the integer counting net oriented turns around the hole.
- **Invariance of the winding number.** A deformation that avoids the hole varies continuously, so w would vary continuously as well. Since w is integer-valued, continuity makes it constant: every loop in the deformation has the initial loop's winding number.
- **Consequence of crossing the hole.** At the crossing the winding formula is undefined; afterward w can jump. The hole then ceases to separate loops into distinct deformation families.
- **Limits of a global quantum sector as evidence for topological order.** A global quantum sector is a subspace labeled by a quantum number that local inspection cannot assign. Such a sector can arise from an ordinary symmetry or an imposed code constraint; evidence for topological order additionally needs the relevant many-body phase, locally indistinguishable ground states, nonlocal operators, a suitable gap, and stability.
- **Failure of local indistinguishability.** Let O_R be a bounded operator with finite norm, supported in local region R . If its leading-order matrix elements distinguish two ground states, a local probe reads or mixes their global labels and the logical bit loses its nonlocal storage.
- **Distinction between digital emulation and emergence.** Digital emulation uses controlled quantum gates to prepare a model state or its evolution. Emergence occurs when the hardware's physical many-body Hamiltonian produces the low-energy phase and its excitations.

The next chapter moves from one loop around one hole to worldlines of indistinguishable particles in two dimensions. Their winding and exchanges form braids — topological classes of intertwined worldlines — that in some systems implement noncommuting operations, whose results depend on order, on a degenerate quantum space.

Sources

- [R121] N. D. Mermin, “The topological theory of defects in ordered media,” *Reviews of Modern Physics* 51, 591–648 (1979). DOI: [10.1103/RevModPhys.51.591](https://doi.org/10.1103/RevModPhys.51.591).
 - [R122] X.-G. Wen and Q. Niu, “Ground-state degeneracy of the fractional quantum Hall states in the presence of a random potential and on high-genus Riemann surfaces,” *Physical Review B* 41, 9377–9396 (1990). DOI: [10.1103/RevModPhys.41.9377](https://doi.org/10.1103/RevModPhys.41.9377).
 - [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* 303, 2–30 (2003). DOI: [10.1016/S0003-4916\(02\)00018-0](https://doi.org/10.1016/S0003-4916(02)00018-0); arXiv: [quant-ph/9707021](https://arxiv.org/abs/quant-ph/9707021).
 - [R123] M. B. Hastings and X.-G. Wen, “Quasi-adiabatic continuation of quantum states: The stability of topological ground-state degeneracy and emergent gauge invariance,” *Physical Review B* 72, 045141 (2005). DOI: [10.1103/PhysRevB.72.045141](https://doi.org/10.1103/PhysRevB.72.045141); arXiv: [cond-mat/0503554](https://arxiv.org/abs/cond-mat/0503554).
 - [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083); arXiv: [0707.1889](https://arxiv.org/abs/0707.1889).
 - [R124] X.-G. Wen, “Colloquium: Zoo of quantum-topological phases of matter,” *Reviews of Modern Physics* 89, 041004 (2017). DOI: [10.1103/RevModPhys.89.041004](https://doi.org/10.1103/RevModPhys.89.041004); arXiv: [1610.03911](https://arxiv.org/abs/1610.03911).
 - [R125] K. J. Satzinger et al., “Realizing topologically ordered states on a quantum processor,” *Science* 374, 1237–1241 (2021). DOI: [10.1126/science.abi8378](https://doi.org/10.1126/science.abi8378); arXiv: [2104.01180](https://arxiv.org/abs/2104.01180).
 - [R126] G. Semeghini et al., “Probing topological spin liquids on a programmable quantum simulator,” *Science* 374, 1242–1247 (2021). DOI: [10.1126/science.abi8794](https://doi.org/10.1126/science.abi8794); arXiv: [2104.04119](https://arxiv.org/abs/2104.04119).
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Chapter 13 — Exchange of two identical particles

Consider two identical particles initially located at distinct positions on a two-dimensional surface. After they are moved so that each occupies the other's original position, the initial and final configurations are physically indistinguishable because the particles have no observable labels.

The trajectory connecting these configurations carries information absent from the endpoints themselves. One particle may pass around the other clockwise or counterclockwise, and it may wind around the other more than once. Each choice leaves the endpoints unchanged while recording a different history between them.

Confined to a two-dimensional surface, no trajectory can use a third spatial direction to lift one path over another and unwind that history continuously. The winding number survives as a physical record of the motion.

For quantum particles forbidden to occupy the same point, the exchange history acts on the quantum state in one of four ways: multiplication by $+1$, multiplication by -1 , multiplication by a general complex phase, or transformation by a matrix. The chapter develops each possibility in turn, from the familiar exchange signs through the matrix-valued case.

Worldline representation

An endpoint configuration leaves the trajectory between its endpoints unspecified. Drawing one spatial direction horizontally and time vertically makes that trajectory visible: each particle traces a curve through spacetime, and an exchange appears as a crossing of the two curves.

```
time
^  \ /
| X exchange
| / \
+-----> plane
```

The crossing records that one particle moved around the other while both remained in the plane. The two particles never touched: with collisions forbidden, one worldline cannot pass continuously through the other.

A **worldline** is the trajectory of an object through spacetime. The physical object of interest here is therefore the history of the particle configuration, tracked through time, alongside any single-time snapshot of the positions.

Two histories count as equivalent when one diagram deforms continuously into the other with the worldlines kept apart, the endpoints held fixed, and the direction of time preserved. Each permitted deformation leaves the identity of the history unchanged.

Allowing collisions erases this information, because one worldline then passes through another and the apparent crossing loses its invariant meaning. Keeping collisions forbidden preserves the record of how the particles moved.

Comparison with three spatial dimensions

The exchange looks different when the particles move in three spatial dimensions. The extra direction lets one path lift over the other, so a double exchange unwinds continuously. Under the usual assumptions for identical point particles, the surviving record is the final permutation of the particles: winding counts wash out once the third direction is available.

Confined to a surface, the lift is impossible, so clockwise and counterclockwise exchanges stay distinct. Repeating the same neighboring exchange twice winds the two worldlines once around each other, and that winding cannot deform into the trivial history while collisions stay forbidden.

Leinaas and Myrheim traced this enlarged set of possibilities to the topology of the space of allowed particle configurations. [Theory] [R127] Their analysis shows that identical endpoints on a surface admit several inequivalent connecting trajectories, each carrying its own exchange history.

Bosonic exchange statistics

Identical endpoint configurations still leave room for the quantum state to change. In the simplest case, a counterclockwise exchange leaves every amplitude unchanged, multiplying the whole state by $+1$.

This exchange law defines a **boson**. Photons are an example: two photons can occupy the same quantum state, consistent with a symmetric exchange factor.

Ordinary label permutations admit exactly two exchange factors, $+1$ and -1 , discussed next. Two dimensions admit further possibilities, developed below, while retaining $+1$ as an allowed exchange law.

Fermionic exchange statistics

A second exchange law multiplies the complete quantum state by a global minus sign. After the same counterclockwise exchange, every amplitude changes sign, so the state acquires the factor -1 .

This exchange law defines a **fermion**. Electrons are an example, and the exchange minus sign underlies the Pauli exclusion principle: two electrons cannot share the same single-particle quantum state.

Bosonic and fermionic exchange laws both remain available to particles confined to two dimensions. A two-dimensional particle can therefore be an ordinary boson or fermion, and $+1$ and -1 exhaust the usual exchange possibilities for identical point particles in three or more spatial dimensions. Removing the planar restriction lets the winding of a double exchange unwind; keeping the restriction turns $+1$ and -1 into special cases of a broader family of exchange laws.

Abelian anyonic exchange phase

On a surface, an exchange can multiply the state by a phase with an intermediate angle,

$$e^{i\theta},$$

where the real angle θ is measured in radians and is defined modulo 2π . A counterclockwise exchange contributes this phase and a clockwise exchange contributes its complex conjugate, $e^{-i\theta}$. The bosonic value $+1$ corresponds to $\theta = 0$, and the fermionic value -1 corresponds to $\theta = \pi$.

Wilczek named this general case the **anyon**, since the exchange phase can take any angular value. [Theory] [R128] The name records the physical claim that exchange produces a phase with values beyond ± 1 .

Phase-only exchange operations commute with one another, so reordering them leaves every internal basis state in place. Such a phase shows up experimentally when two paths are compared, most directly through interference. [Theory] [R128]; [R129]

Counting exchanges keeps the histories distinct. One neighboring exchange contributes $e^{i\theta}$, while carrying one particle completely around the other uses two exchanges and contributes $e^{i2\theta}$. A full winding therefore differs physically from a single exchange.

Abelian anyonic statistics names this one-dimensional exchange representation. Bosons and fermions sit inside it as the two familiar phase values.

The braid group

Two strands offer only one neighboring pair, so no ordering of distinct crossings can be tested. Three particles, stationed left, middle, and right, supply the smallest informative case: exchange the left pair counterclockwise and then the right pair counterclockwise, and compare with the same two exchanges in reverse order. The resulting worldline histories visibly differ, before any matrices enter.

left then right right then left

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\ / | | \ /
x | | x
/ \ | | / \
| \ / \ / |
| x x |
| / \ / \ |

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A **braid** is a worldline tangle considered up to continuous deformations that keep the strands apart. Stacking one braid above another composes them. Equipped with this stacking operation, the set of all braids on N strands forms the **braid group**.

The braid group on N strands is denoted by B_N . Its generators are the counterclockwise exchanges of neighboring strands.

Denote these generators by $\sigma_1, \dots, \sigma_{N-1}$. The generator σ_i exchanges the objects currently occupying neighboring positions i and $i + 1$.

Its inverse, σ_i^{-1} , performs the corresponding clockwise exchange, and therefore

$$\sigma_i \sigma_i^{-1} = 1.$$

No relation $\sigma_i^2 = 1$ is imposed. A braid therefore preserves winding information alongside the final permutation. Imposing that extra relation would collapse the braid group onto the ordinary permutation group used for identical-particle exchanges in three dimensions.

Crossings involving well-separated pairs do not affect one another and therefore commute:

$$\sigma_i \sigma_j = \sigma_j \sigma_i \quad \text{when } |i - j| \geq 2.$$

Adjacent generators obey an additional deformation rule. Sliding one crossing through the junction formed by two others changes the drawing but not the braid class:

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}.$$

Together these form the complete presentation used here: well-separated generators commute, neighboring generators satisfy the three-strand braid relation, and generator squares remain nontrivial. Neighboring exchanges can therefore fail to commute, each still exchanging its own neighboring pair.

That missing $\sigma_i^2 = 1$ relation carries the two-dimensional physics. Keeping the square nontrivial lets a double exchange record a winding, and lets clockwise and counterclockwise exchanges remain distinct, rather than reducing everything to a permutation of final positions.

Configuration space of identical particles

Braids record the collision-free histories of identical particles: each braid class stands for one family of motions that deform into one another.

For N point-like objects in the plane, an ordered list of positions is

$$(\mathbf{r}_1, \dots, \mathbf{r}_N) \in (\mathbb{R}^2)^N,$$

where $\mathbf{r}_j = (x_j, y_j)$ has units of length. The set of configurations containing at least one collision is

$$\Delta = \{(\mathbf{r}_1, \dots, \mathbf{r}_N) : \mathbf{r}_j = \mathbf{r}_k \text{ for at least one } j \neq k\}.$$

Exchange paths keep every object at a distinct position, so the collision set Δ stays excluded. For identical objects, permuting temporary particle labels leaves the physical configuration unchanged. Dividing out those permutations gives the **configuration space**, the space of physically distinct allowed arrangements,

$$\mathcal{C}_N(\mathbb{R}^2) = \frac{(\mathbb{R}^2)^N \setminus \Delta}{S_N},$$

where S_N is the group of all permutations of N labels.

A motion that starts and ends at the same physical configuration traces a loop in $\mathcal{C}_N$. Two loops share a class when one deforms continuously into the other without crossing Δ . The deformation classes of loops based at a fixed configuration form the **fundamental group**. For particles in the plane, that group is the braid group:

$$\pi_1(\mathcal{C}_N(\mathbb{R}^2)) = B_N.$$

[Theory] [R127]; [R129] This equation establishes the relation between collision-free exchange paths and braids. In three or more spatial dimensions, the analogous ordinary-exchange classes reduce to the permutation group.

A wavefunction can return with a changed complex value after transport around such a loop. Equivalently, paths lift to a simply connected covering space, and a unitary transformation applies whenever the projected path closes in the original configuration space.

Consistency requires this transformation to track only the loop class and to respect composition by stacking loops. A map meeting both requirements is a **unitary representation** of B_N : a group homomorphism from braid classes to unitary linear operators.

[Theory] [R127]; [R129]

Every braid matrix later in the book therefore implements the linear action belonging to one loop class in configuration space. Keeping the label identification and the exclusion of Δ matters: dropping the identification restores unphysical particle labels, while allowing collisions collapses the fundamental group.

A two-dimensional matrix representation of three-strand braids

A **representation** assigns a matrix to each braid so that stacking braids corresponds to multiplying matrices. Writing $\rho(b)$ for the matrix assigned to a braid b gives the composition law

$$\rho(b_1 b_2) = \rho(b_1) \rho(b_2).$$

Operator products act on a ket from right to left. Here ρ names the representation, and it carries no relation to a density matrix. An overall phase of a closed, isolated system can pass unobserved, while a relative phase between distinct paths shows up in interference.

Two strands supply only one neighboring exchange, so no ordering of generators can be tested there. Three strands give the smallest test of noncommutativity. Take a two-dimensional state space with orthonormal basis kets $|0\rangle$ and $|1\rangle$ and Pauli matrices

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

and assign

$$A \equiv \rho(\sigma_1) = e^{-i\pi Z/4} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1-i & 0 \\ 0 & 1+i \end{pmatrix},$$

$$B \equiv \rho(\sigma_2) = e^{-i\pi X/4} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}.$$

The imaginary unit satisfies $i^2 = -1$. Each matrix is unitary: its inverse equals its conjugate transpose, and direct multiplication confirms $A^\dagger A = B^\dagger B = I$, with $\dagger$ for conjugate transpose and I for the identity matrix.

Multiplying in the two possible orders gives

$$AB = \frac{1}{2} \begin{pmatrix} 1-i & -1-i \\ 1-i & 1+i \end{pmatrix}, \quad BA = \frac{1}{2} \begin{pmatrix} 1-i & 1-i \\ -1-i & 1+i \end{pmatrix}.$$

The matrices differ. Acting on $|0\rangle$ and dropping an overall phase gives

$$AB|0\rangle \sim \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad BA|0\rangle \sim \frac{|0\rangle - i|1\rangle}{\sqrt{2}}.$$

An X -basis measurement returns the outcome $(|0\rangle + |1\rangle)/\sqrt{2}$ with certainty for the first state and with probability $1/2$ for the second. The two operation orders are therefore experimentally distinguishable.

Noncommutativity alone does not qualify the pair as a braid representation; the braid-group relation must hold as well. Direct calculation gives

$$ABA = BAB = -\frac{i}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

This identity is the three-strand braid relation. Up to basis and overall-phase conventions, the pair forms an Ising-type two-dimensional braid representation.

[Theory] [R015]; [R134] The matrices serve here as an algebraic model: loading them into a pulse sequencer reproduces their algebra without creating Ising anyons.

The pair therefore satisfies $AB \neq BA$ together with $ABA = BAB$. The first relation records noncommutativity and the second records the braid constraint; a valid example needs both.

Four classes of exchange action

A one-dimensional representation assigns each neighboring exchange a single complex number of unit magnitude. For a boson that number is

$$\rho(\sigma_i) = +1.$$

For a fermion,

$$\rho(\sigma_i) = -1.$$

For an anyon whose exchange action is only a phase,

$$\rho(\sigma_i) = e^{i\theta}.$$

Bosons and fermions sit at the special values $\theta = 0$ and $\theta = \pi$. Every pair of complex phase factors commutes, so reordering phase-only exchanges keeps each internal basis state in place.

The three-strand example works differently. The same particle positions support a two-dimensional state space, and the two neighboring exchanges act as matrices satisfying

$$AB \neq BA.$$

When exchange acts by matrices on a state space of dimension greater than one, with at least one noncommuting pair, the exchange statistics are called **non-Abelian**. The adjective classifies the representation; electric charge is a separate property, determined by coupling to electromagnetic fields, and the particle paths still avoid collision.

Demonstrating this property takes at least two operations whose order can be compared, hence at least three strands as in the preceding calculation; a single exchange matrix on its own carries no such test. Candidate matrices must also satisfy the braid relations and be unitary, and the matrices A and B meet both requirements.

Non-Abelian statistics therefore describes a representation of B_N , with the matrices as its values, rather than a phase belonging to one isolated exchange.

Degenerate fusion spaces at fixed particle positions

The two-dimensional state space in the preceding example persists with the particle positions held fixed. The basis states $|0\rangle$ and $|1\rangle$ are distinct quantum states belonging to the same spatial configuration.

One source of such multiplicity is the collective appearance of a pair viewed from far away. Two anyons of the same type can jointly look like no additional excitation or like one residual excitation, and a prepared pair can sit in a superposition of the two outcomes until a measurement of the pair selects one.

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a a
 \ /
 \ /
 ?
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The question mark stands for the residual label. With the particle positions and the total residual label both fixed, the allowed residual outcomes span a **fusion space**.

Combining particles already acts as a quantum operation: the system gains multiple states without any particle motion.

This chapter needs only the existence of those multiple basis states, on which the matrices A and B act. The next chapter introduces the residual labels in full, with the tree diagrams that organize them and the F -matrices that convert between different tree groupings and support explicit braid representations.

Removing the degeneracy collapses that stage: each braid then contributes at most a phase. Keeping degeneracy while dropping the braid relations leaves a degenerate subspace without exchange statistics.

Adiabatic transport and holonomy

Take a Hamiltonian $H(\lambda)$ whose control parameters λ record the positions of well-separated quasiparticles. The **Hamiltonian** sets the system energies and time evolution. Varying the parameters slowly around a closed path, while keeping the system inside a degenerate low-energy subspace, lets the state acquire a matrix-valued geometric transformation.

Subtracting the ordinary dynamical phases, which accumulate with elapsed time, leaves the geometric part, called the **holonomy**. It depends on the path taken through parameter space; in an ideal topological phase it depends only on the braid class of that path. [Theory] [R015]

Slow motion and large separation translate into physical inequalities. The motion stays slow enough to avoid exciting states across the many-body energy gap, yet finishes before decoherence or stray quasiparticle motion damages the state.

The quasiparticles stay separated by distances much larger than the phase correlation length, the characteristic distance over which local correlations decay. That spacing suppresses local couplings enough to keep the intended degeneracy intact. Later chapters convert these requirements into energy, length, and timing budgets.

A braid operator and a statistical angle are dimensionless, while laboratory time runs in seconds and energy gaps in joules or electronvolts. The inequalities governing a given device therefore depend on its specific Hamiltonian; braid topology alone supplies no numbers.

Fast motion populates excited states and moves the transformation out of the intended-subspace holonomy. Insufficient separation splits the degeneracy through local couplings and reduces the motion to ordinary dynamical evolution of a unique ground state.

Emergent quasiparticles and microscopic defects

An **emergent quasiparticle** is a localized excitation of an interacting many-body Hamiltonian that behaves as an effective particle at low energy. It has no counterpart among the microscopic constituents taken individually: a phonon, a quantized lattice vibration, is a familiar example. An **emergent anyon** is the narrower case in which exchanges of such an excitation realize a nontrivial braid representation in an effectively two-dimensional topological phase.

A diamond defect is a microscopic imperfection with electronic and nuclear degrees of freedom. Point-like confinement and qubit encodings qualify it as a controllable microscopic object; anyon status additionally requires exchange behavior with the braid statistics described above.

Diamond defects could still enter as microscopic building blocks of a two-dimensional Hamiltonian whose collective excitations are anyons. Tracing the path from those ingredients to an emergent topological phase is most of the engineering problem.

Fractional quantum Hall devices supply evidence for Abelian anyonic statistics. [Experiment] At filling factor $\nu = 1/3$, collider measurements reported correlations consistent with fractional exchange statistics, while an electronic Fabry–Pérot interferometer reported a statistical phase associated with quasiparticle braiding.

[R130]; [R131] Both experiments probe collective excitations of an interacting electron system, not software labels attached to ordinary particles. Extracting a statistical angle from an interference or correlation signal still depends on models and device details, so “evidence consistent with” describes the outcome more accurately than a claim of direct braid-matrix observation.

A physical anyon is therefore an excitation of a many-body phase, not a renamed microscopic defect.

Emulated braids and hardware realizations

A programmable processor can prepare an encoded many-qubit state and apply gates whose logical action realizes a braid representation. An **emulated braid** is a braid implemented through a designed sequence of gates, measurements, or code deformations rather than by adiabatic transport of persistent quasiparticle eigenstates of the hardware Hamiltonian.

[Experiment] In 2023, a superconducting processor implemented a generalized stabilizer code and a unitary protocol for projective Ising-type graph defects, testing fusion and non-Abelian exchange effects. [R132] The paper reports that gates deformed the stabilizer graph to move its defects; the mobile objects did not persist as quasiparticle eigenstates throughout Hamiltonian evolution. The procedure substantively emulates braid physics while leaving open whether the processor material occupies a non-Abelian topological phase.

In a defect-engineered platform, realizing a braid matrix with programmed microwave pulses demonstrates control of the encoded state space. Confirming an emergent excitation spectrum takes a separate measurement of the many-body excitations themselves.

A stronger claim starts from the material or effective analog Hamiltonian supporting the relevant phase and its localized excitations, with exchanges implementing the braid representation inside the low-energy subspace.

[Proposal] Engineered defect arrays could be designed toward this objective.

[Speculation] Defect positions on a planar lattice, or drawn crossing paths between them, acquire anyonic meaning only together with such a Hamiltonian and its measured excitation spectrum.

A programmed braid and an emergent anyon thus name distinct physical achievements, even when the same matrix describes both.

Distinct physical and computational concepts

A **physical defect qubit** is a controlled microscopic degree of freedom, such as a defect spin.

An **encoded qubit** is a selected subspace distributed across multiple physical degrees of freedom.

An **emulated braid** is a protocol whose logical gates reproduce a braid representation.

An **emergent quasiparticle** is an excitation of the many-body Hamiltonian. It is an anyon only if its physical exchanges exhibit the required braid statistics.

Topological order is a property of a many-body phase rather than of an individual gate sequence.

One experiment can involve every description on this list. Each one still needs its own evidence: establishing one leaves the next to be shown separately.

Common conceptual errors

- **Incorrect claim: Only bosons and fermions exist.** This conclusion assumes the permutation topology applicable in three or more spatial dimensions. Permutation topology classifies exchanges only by the resulting permutation of particle positions. In two spatial dimensions, however, the collision-free configuration space—the space of all allowed particle positions with coincident positions excluded—has braid-group topology. The braid group retains information about how particle trajectories wind around one another. [\[R127\]](#); [\[R129\]](#)
- **Incorrect claim: Every two-dimensional excitation is an anyon.** Ordinary bosons and fermions can also exist in two dimensions. The particle statistics are determined by the exchange representation, which assigns a phase or operator to each allowed exchange.
- **Incorrect claim: Every unusual exchange phase is non-Abelian.** A scalar exchange factor $e^{i\theta}$ defines Abelian statistics at every value of θ , since scalars commute. Non-Abelian statistics additionally requires a multidimensional state space with braid operators that fail to commute.

- **Incorrect claim: If AB and BA are different matrices, then A and B automatically define valid braids.** Noncommuting matrices qualify only with the braid relations, which encode consistent deformations of braid trajectories, plus unitarity, which preserves quantum-state norms and probabilities. The matrices used in the worked example satisfied both requirements.
- **Incorrect claim: A full winding is equivalent to one exchange.** One exchange of neighboring objects is represented by the braid-group generator σ_i . Moving one object completely around the other corresponds to two successive exchanges and is represented by σ_i^2 . For an Abelian anyon, whose exchange operation is a scalar phase, the corresponding factors are $e^{i\theta}$ and $e^{i2\theta}$, respectively.
- **Incorrect claim: A quasiparticle is a small constituent particle.** A quasiparticle is a collective low-energy excitation of a many-body system. It can serve as an effective particle while arising from the coordinated motion of many constituents. Its identity depends on the system's Hamiltonian, which specifies its dynamics, and on the many-body phase realized by that Hamiltonian.
- **Incorrect claim: Executing the required unitary operators proves that the associated particles are emergent.** A conventional processor can compile the operators A and B exactly. Such an implementation demonstrates an emulated braid representation. Emergent anyons additionally require that the representation arise physically from exchanging quasiparticle excitations supported by the system's many-body phase.
- **Incorrect claim: Topological protection is perfect.** Topological dependence is an ideal low-energy property in which an operation depends on the topological class of a trajectory, fixed by which winding and ordering of exchanges occurred, instead of its detailed geometric shape. Finite separation, finite temperature, disorder, nonadiabatic motion, leakage, measurement error, and stray quasiparticles can all damage a braid.

[Theory] [R015]; [R030] Topological protection suppresses the specified local errors under the specified conditions; the remaining error mechanisms need separate treatment.

Concept checks

- **Explain why exchange in two spatial dimensions can be richer than exchange in three spatial dimensions.**

In two dimensions, collision-free loops in the configuration space of identical particles form the braid group. Consequently, a double exchange need not be continuously deformable to the trivial path. The exchange history can therefore retain winding information that is absent from the corresponding permutation description in three or more dimensions.

- **Show that imposing $\sigma_i^2 = 1$ reduces the braid group to the permutation group.**

The additional relation $\sigma_i^2 = 1$ makes a double exchange trivial. It therefore identifies clockwise and counterclockwise exchanges and removes winding information. Only the permutation of the final particle positions remains.

- **State the relation among a boson, a fermion, and an Abelian anyon.**

All three have one-dimensional exchange representations. Their exchange factors are, respectively, $+1$, -1 , and the general phase $e^{i\theta}$. Bosons and fermions are the special cases $\theta = 0$ and $\theta = \pi$.

- **Determine what fails when $AB \neq BA$ but $ABA \neq BAB$.**

The matrices do not represent the braid group. The condition $AB \neq BA$ establishes noncommutativity, but noncommutativity alone does not define particle statistics. The matrices must also satisfy the three-strand braid relation $ABA = BAB$, which expresses the equivalence of the corresponding continuous rearrangements of three strands.

- **State the defining feature of a non-Abelian anyon.**

Braids act as unitary matrices on a multidimensional degenerate state space, meaning a space of distinct quantum states with the same energy, and at least some braid operations do not commute.

- **Assess whether a gate sequence implementing A and B establishes the existence of emergent anyons.**

It establishes an emulated braid representation, which leaves emergence open. Emergence additionally requires exchangeable quasiparticle excitations produced by the many-body Hamiltonian.

This chapter has assembled worldlines, the trajectories of particles in spacetime; the allowed exchange laws; the braid group that classifies ordered exchanges; a worked pair of noncommuting matrices; and the fusion space on which those matrices act, namely the state space of possible collective fusion outcomes of anyons. The next chapter constructs that space from residual labels and tree representations.

Sources

- [R127] J. M. Leinaas and J. Myrheim, “On the theory of identical particles,” *Il Nuovo Cimento B* 37, 1–23 (1977). DOI: [10.1007/BF02727953](https://doi.org/10.1007/BF02727953).
- [R128] F. Wilczek, “Quantum mechanics of fractional-spin particles,” *Physical Review Letters* 49, 957–959 (1982). DOI: [10.1103/PhysRevLett.49.957](https://doi.org/10.1103/PhysRevLett.49.957).
- [R129] Y.-S. Wu, “General theory for quantum statistics in two dimensions,” *Physical Review Letters* 52, 2103–2106 (1984). DOI: [10.1103/PhysRevLett.52.2103](https://doi.org/10.1103/PhysRevLett.52.2103).
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083); arXiv: [0707.1889](https://arxiv.org/abs/0707.1889).
- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* 303, 2–30 (2003). DOI: [10.1016/S0003-4916\(02\)00018-0](https://doi.org/10.1016/S0003-4916(02)00018-0); arXiv: [quant-ph/9707021](https://arxiv.org/abs/quant-ph/9707021).
- [R130] H. Bartolomei et al., “Fractional statistics in anyon collisions,” *Science* 368, 173–177 (2020). DOI: [10.1126/science.aaz5601](https://doi.org/10.1126/science.aaz5601); arXiv: [2006.13157](https://arxiv.org/abs/2006.13157).
- [R131] J. Nakamura, S. Liang, G. C. Gardner, and J. M. Manfra, “Direct observation of anyonic braiding statistics,” *Nature Physics* 16, 931–936 (2020). DOI: [10.1038/s41567-020-1019-1](https://doi.org/10.1038/s41567-020-1019-1).

- [R132] M. B. Andersen et al. (Google Quantum AI and Collaborators), “Non-Abelian braiding of graph vertices in a superconducting processor,” *Nature* 618, 264–269 (2023). DOI: [10.1038/s41586-023-05954-4](https://doi.org/10.1038/s41586-023-05954-4); arXiv: [2210.10255](https://arxiv.org/abs/2210.10255).
 - [R133] A. Stern, “Anyons and the quantum Hall effect—a pedagogical review,” *Annals of Physics* 323, 204–249 (2008). DOI: [10.1016/j.aop.2007.10.008](https://doi.org/10.1016/j.aop.2007.10.008); arXiv: [0711.4697](https://arxiv.org/abs/0711.4697).
 - [R134] D. A. Ivanov, “Non-Abelian statistics of half-quantum vortices in p-wave superconductors,” *Physical Review Letters* 86, 268–271 (2001). DOI: [10.1103/PhysRevLett.86.268](https://doi.org/10.1103/PhysRevLett.86.268); arXiv: [cond-mat/0005069](https://arxiv.org/abs/cond-mat/0005069).
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Chapter 14 — Fusion and topological charge

Consider two particles enclosed within a region whose interior an outside observer cannot resolve. Each external measurement then reports one coarse property: the type of source the region presents as a whole.

Local rearrangements of the particles inside leave that external report unchanged. The region carries a label fixed by all such local rearrangements.

The previous chapter established that exchanging two such particles can rotate an internal state. This chapter identifies the structure of that state.

The discussion begins with a multiplication table describing how particle labels combine. It then introduces two diagrammatic operations: an F -move, which changes the grouping of particles, and an R -move, which exchanges two particles.

Topological charge as a locally invariant label

Consider a loop enclosing several quasiparticles, where a quasiparticle is an emergent particle-like excitation of a many-body system. An observer outside the loop can characterize the type of source contained inside it. Operations performed locally within the loop cannot change that external characterization.

This characterization is called a charge. More specifically, it is often called a topological charge because a perturbation confined to one location cannot change it. Throughout this chapter, charge denotes this collective label of the enclosed region.

Possible charges carry symbols such as a , b , and c . Every allowed local operation preserves the total charge of an isolated system: a region of total charge a keeps that label under all local operations, and never acquires a different total charge b through them.

A region containing no residual excitation is assigned the distinguished label 1, called the vacuum charge. The symbol 1 is a charge label rather than an ordinary numerical value.

An operation reaching in from outside can change the total charge by acting on the region nonlocally. Restricting to local operations recovers the sector structure of the previous two chapters: each charge label names one sector, and allowed local operations never connect states of different total charge.

Fusion of two charge labels

Two regions can be combined and treated as a single larger region. The total charge of the combined region need not be unique. Charges a and b can admit several possible total charges.

The fusion product records these allowed totals:

$$a \times b = \sum_c N_{ab}^c c.$$

The quantity N_{ab}^c is a nonnegative integer called a fusion coefficient. The value zero means that total charge c is forbidden. The value one means that there is one independent way for a and b to have total charge c . The value two means that there are two independent ways to obtain the same externally visible total charge c .

This combination process is called fusion. Fusion names the total charge seen when the two particles are treated as one composite region; the quasiparticles persist, and no collision debris is implied.

The plus sign in the fusion product lists alternative allowed totals. It records the set of possible measurement outcomes, not a statistical mixture already prepared in the system, and not a simultaneous realization of every listed result. A quantum state becomes a superposition of these alternatives only when the full experimental configuration supports coherence between them.

A fusion product therefore lists allowed total charges while leaving the prepared superposition and its probability distribution to be specified separately.

Fusion rules of the Ising model

The Ising model contains three charge labels: 1 , ψ , and σ . Its fusion products are

$$1 \times a = a, \quad \psi \times \psi = 1, \quad \psi \times \sigma = \sigma, \quad \sigma \times \sigma = 1 + \psi.$$

In the first relation, a denotes any of the three charge labels. Fusion with the vacuum charge leaves a charge unchanged.

The relation $\psi \times \psi = 1$ states that two ψ charges have vacuum total charge. The relation $\psi \times \sigma = \sigma$ states that combining ψ with σ gives total charge σ .

The final relation gives two σ charges a choice of total charge: 1 or ψ .

The same fusion rules can be represented as a table in which each entry lists the allowed total charges:

$\times \mid 1 \mid \psi \mid \sigma \mid \mid$

$1 \mid 1 \mid \psi \mid \sigma \mid \mid$

$\psi \mid \psi \mid 1 \mid \sigma \mid \mid$

$\sigma \mid \sigma \mid \sigma \mid 1 + \psi \mid \mid$

Every nonempty entry in this table contains one total charge except the final entry. Each allowed total occurs with coefficient one.

The integer multiplying a total charge, conventionally omitted when it equals one, is the fusion coefficient. It counts independent fusion states and is not a probability.

All nonzero fusion coefficients in the Ising model equal one. The model is therefore multiplicity-free, meaning that each allowed total has exactly one fusion channel and each forbidden total has none.

The vector space of ways in which charges a and b can have total charge c is denoted by

$$V_{ab}^c, \quad \dim V_{ab}^c = N_{ab}^c.$$

This space is called a fusion space, and its dimension equals the corresponding fusion coefficient. For two σ charges, $V_{\sigma\sigma}^1$ and $V_{\sigma\sigma}^\psi$ are each one-dimensional, whereas $V_{\sigma\sigma}^\sigma$ is zero-dimensional and therefore represents a forbidden channel.

If one were instead to propose $N_{\sigma\sigma}^\sigma = 1$, two σ charges would also be allowed to have total charge σ . The resulting fusion table would not be the Ising table. The absence of this channel is part of the defining fusion data.

Diagrammatic representation of a fusion vertex

A basis vector in V_{ab}^c can be represented by a trivalent vertex, meaning a vertex with three incident edges. Diagrams in this chapter are read from top to bottom: the incoming charges enter at the top, and their total charge exits at the bottom.

```
a b
 \ /
 \ /
  c
```

When $N_{ab}^c > 1$, a multiplicity label $\mu = 1, \dots, N_{ab}^c$ must be placed beside the vertex. The fusion channel is then specified by the complete tuple $(a, b; c, \mu)$, rather than by c alone.

Three or more charges must be represented using a sequence of pairwise fusion operations. Such a diagram is called a fusion tree.

The internal edges of a fusion tree carry intermediate charges. Selecting the labels on these edges, together with any required multiplicity labels, selects a basis state.

A fusion tree organizes a many-particle state by groupings, recording which pair fuses first in the bookkeeping. That ordering carries no claim that the system underwent a literal temporal sequence of collisions.

If multiplicity labels are omitted when $N_{ab}^c > 1$, multiple independent states with the same total charge are incorrectly represented as a single vector. The statement that an internal-edge label specifies the state is valid only when every relevant fusion coefficient is zero or one.

Three-particle fusion bases

Consider three Ising σ charges constrained to have total charge σ . In the left-associated basis, the left pair is fused first. Its intermediate charge x can be 1 or ψ , and either intermediate result can then fuse with the third σ to produce total charge σ :

Left pairing: $((\sigma \sigma)_x \sigma)_\sigma$

$\sigma \sigma \sigma$

$\backslash / |$

$\backslash / |$

$x |$

$\backslash /$

$\backslash /$

σ

$x = 1 \text{ or } \psi$

Alternatively, the right pair can be fused first. Its intermediate charge y can likewise be 1 or ψ :

Right pairing: $(\sigma (\sigma \sigma)_y)_\sigma$

$\sigma \sigma \sigma$

$| \backslash /$

$| \backslash /$

$| y$

$\backslash /$

$\backslash /$

σ

$y = 1 \text{ or } \psi$

These two trees share external charges and total charge. They define two bases of one vector space: the three-particle fusion space, which is the state space on which the braid operations of the previous chapter act.

The dimension follows directly from the fusion table. In the left pairing, the first pair can have total charge 1 or ψ , and either intermediate charge can fuse with the remaining σ to give total charge σ .

The left pairing therefore contains two states, and the right pairing describes the same two-dimensional space in a different basis. Three σ charges constrained to total charge σ thus support a two-dimensional fusion space; changing the total-charge constraint generally changes the space.

The F -move as a change of fusion basis

The following diagram represents a change of pairing rather than motion of the particles:

```

σ σ σ σ σ σ
\ / | | \ /
x | --> | y
\ / \ /
σ σ

```

This change of pairing is called an F -move. A distinct operation introduced later, the R -move, exchanges particle positions.

Let $|x\rangle_L$ denote orthonormal states in the left-pairing basis, and let $|y\rangle_R$ denote orthonormal states in the right-pairing basis. The F -move is the basis transformation

$$|x\rangle_L = \sum_{y \in \{1, \psi\}} [F_{\sigma}^{\sigma\sigma\sigma}]_{xy} |y\rangle_R.$$

In a standard gauge for Ising anyons,

$$F_{\sigma}^{\sigma\sigma\sigma} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

where the rows correspond to $x = (1, \psi)$ and the columns correspond to $y = (1, \psi)$ [Ro15]; [Ro23]. Here a gauge is a choice of phases for the basis vectors at fusion vertices. The explicit basis transformation is

$$|1\rangle_L = \frac{|1\rangle_R + |\psi\rangle_R}{\sqrt{2}}, \quad |\psi\rangle_L = \frac{|1\rangle_R - |\psi\rangle_R}{\sqrt{2}}.$$

Suppose the first two charges are prepared with definite fusion channel $x = 1$. If the last two charges are measured instead, the amplitudes are $1/\sqrt{2}$ for $y = 1$ and $1/\sqrt{2}$ for $y = \psi$.

By the Born rule, which assigns a measurement probability equal to the squared magnitude of the corresponding amplitude, each result has probability $1/2$. These probabilities are determined by the F -matrix, while the fusion table lists only the available channels.

The amplitudes satisfy the normalization condition

$$\left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2 = 1.$$

In addition, $F^{\dagger}F = I$, where $F^{\dagger}$ is the conjugate transpose and I is the 2×2 identity matrix. Thus, changing the pairing preserves inner products and measurement probabilities.

An F -move is therefore a unitary change of basis within a fixed fusion space, leaving particle positions in place.

If F were not unitary, inner products and therefore Born probabilities would depend on which pairing served to represent the state, and the two fusion trees could not both count as bases of one physical state space.

Antiparticles and the vacuum channel

Each simple charge a , where a simple charge is an irreducible charge type, has a partner $\bar{a}$ such that their fusion contains the vacuum channel:

$$N_{a\bar{a}}^1 \geq 1.$$

The charge $\bar{a}$ is called the antiparticle of a . The pair can fuse to vacuum, analogous to the disappearance of a particle-antiparticle pair into the vacuum sector.

In the common simple setting, the vacuum channel occurs exactly once. A charge can be its own antiparticle. In the Ising fusion table, $\bar{\sigma} = \sigma$ and $\bar{\psi} = \psi$.

A physical fusion experiment can give probabilistic outcomes while every fusion coefficient N_{ab}^c is zero or one, because a coefficient counts basis states and the prepared state supplies the amplitudes. The value $N_{ab}^c = 2$ instead means two orthogonal states sharing the external labels a, b, c , so a multiplicity label must join the description before amplitudes enter.

Every charge needs its antiparticle partner: with no $\bar{a}$, charge a would have no fusion path to vacuum, leaving the vacuum sector unreachable from that charge and the theory without duals in the sense used later.

Associativity and dimension counting

For fixed total charge d , the left-associated fusion space of three charges decomposes as

$$V_{abc}^d \cong \bigoplus_e V_{ab}^e \otimes V_{ec}^d.$$

Here e ranges over allowed intermediate charges, $\oplus$ denotes a direct sum over alternative intermediate sectors, and $\otimes$ combines the two vertex spaces along a selected fusion path. The same fusion space has the right-associated decomposition

$$V_{abc}^d \cong \bigoplus_f V_{bc}^f \otimes V_{af}^d.$$

Because these are two decompositions of the same vector space, their dimensions must agree. This requirement gives an integer consistency condition on the fusion table before any phases are assigned:

$$\sum_e N_{ab}^e N_{ec}^d = \sum_f N_{bc}^f N_{af}^d.$$

Coefficients violating this equation admit no phase assignment that renders the fusion rules associative.

For three σ charges with total charge σ , the left-associated dimension is

$$N_{\sigma\sigma}^1 N_{1\sigma}^\sigma + N_{\sigma\sigma}^\psi N_{\psi\sigma}^\sigma + N_{\sigma\sigma}^\sigma N_{\sigma\sigma}^\sigma = (1)(1) + (1)(1) + (0)(0) = 2.$$

The right-associated calculation is the same sum with the first pair replaced by the last pair, and it also equals 2. The Ising fusion table therefore passes this consistency test.

For n charges, the construction is repeated. External edges record the n individual charges, internal edges record intermediate charges, and the root records the total charge. The number of admissible edge labelings, including vertex multiplicities, equals the dimension of the fusion space with fixed total charge.

This multiplicity of internal labelings explains how non-Abelian charges can store information. Even when the total charge is fixed, several distinct internal fusion states can remain.

A local measurement enclosing only one separated particle cannot determine the complete internal fusion pattern. [Theory] In an ideal topological phase, operations supported far from one another act on this nonlocal fusion space according to the theory's data, up to corrections caused by finite separation and unwanted excitations [R015]; [R017].

This identity belongs to the effective theory; a fabricated device needs separate experimental confirmation. Associativity of fusion therefore imposes a numerical identity on the fusion coefficients at the level of dimensions, and a failing identity leaves no consistent three-charge fusion space.

Quantum dimensions and fusion-space growth

The quantum dimension d_a measures the asymptotic contribution of charge a to the growth of large fusion spaces. For a unitary theory, the quantum dimensions are chosen as positive numbers satisfying

$$d_a d_b = \sum_c N_{ab}^c d_c, \quad d_1 = 1.$$

Equivalently, define the fusion matrix $(N_a)^c_b = N_{ab}^c$. Then d_a is its largest positive eigenvalue in the relevant sector. This follows from the Perron–Frobenius theorem and is not a fitted parameter [R017]; [R016].

Applying these relations to the Ising fusion table, $\psi \times \psi = 1$ gives $d_\psi^2 = 1$. Positivity therefore requires $d_\psi = 1$. Similarly, $\sigma \times \sigma = 1 + \psi$ gives

$$d_\sigma^2 = d_1 + d_\psi = 2, \quad d_\sigma = \sqrt{2}.$$

The value $\sqrt{2}$ cannot count states in an ordinary local vector space, whose dimensions are integers.

Instead, d_σ characterizes asymptotic growth. For $2n$ Ising σ charges constrained to have total charge 1, the fusion-space dimension is 2^{n-1} : two charges give one state, four give two, six give four, and the pattern continues.

The characteristic multiplicative contribution associated with each additional σ approaches $\sqrt{2}$.

In a unitary fusion theory, a charge with $d_a = 1$ is Abelian. A charge with $d_a > 1$ supports non-Abelian growth of its fusion spaces. The total quantum dimension is

$$\mathcal{D} = \sqrt{\sum_a d_a^2},$$

where the sum extends over all simple charges. For the Ising theory, $\mathcal{D} = \sqrt{1 + 1 + 2} = 2$.

Thus $d_\sigma = \sqrt{2}$ reads as an asymptotic growth rate, not as a count of local on-site levels: local Hilbert-space dimensions take integer values.

General F -moves with fusion multiplicities

Let a, b, c be three incoming charges with total charge d . In the left-associated tree, let e denote the intermediate charge, let μ label a basis of V_{ab}^e , and let ν label a basis of V_{ec}^d . In the right-associated tree, let f, α , and β denote the corresponding intermediate charge and multiplicity labels. The general basis transformation is

$$|((ab)_e c)_d; \mu, \nu\rangle = \sum_{f, \alpha, \beta} [F_d^{abc}]_{(e, \mu, \nu)(f, \alpha, \beta)} |(a(bc)_f)_d; \alpha, \beta\rangle.$$

This equation defines the F -matrix convention used in this chapter. Other sources may use the inverse transformation, take a complex conjugate, or arrange the labels differently. Such alternatives are equivalent provided that the convention is stated and applied consistently.

An F -move remains a change of basis and does not exchange particle positions. Its individual matrix entries depend on the phase choices assigned to the vertex basis states.

These phase choices constitute a gauge. Observable probabilities and complete braid amplitudes are gauge-independent.

Rephasing a vertex basis changes the individual F -symbols. It does not change closed-process probabilities, consistency relations, or physically defined interference signals. An individual gauge-dependent matrix entry is therefore not itself an observable.

The R -move as particle exchange

An R -move exchanges two neighboring particles. For a counterclockwise exchange of charges a and b in total channel c , it is the unitary map

$$R_c^{ab} : V_{ab}^c \longrightarrow V_{ba}^c.$$

Diagrammatically, an R -move contains a crossing rather than a change of pairing:

```
a b b a
 \ / --> \ /
 c c
```

The R -move represents an actual braid and is distinct from an F -move.

In a multiplicity-free theory, each allowed R_c^{ab} is a phase. In the same standard Ising gauge used above,

$$R_1^{\sigma\sigma} = e^{-i\pi/8}, \quad R_\psi^{\sigma\sigma} = e^{3i\pi/8}$$

for a chosen counterclockwise convention [Ro15]; [Ro23]. Reversing the direction of exchange applies the inverse phases.

In the left-pairing basis of three σ charges, exchanging the first two particles is represented by the diagonal matrix

$$B_{12} = \begin{pmatrix} R_1^{\sigma\sigma} & 0 \\ 0 & R_\psi^{\sigma\sigma} \end{pmatrix}.$$

To exchange the last two particles, the state is first transformed into the basis in which those particles fuse directly. Their R -move is then applied, followed by the inverse basis transformation:

$$B_{23} = F B_{12} F^{-1}$$

for the matrix convention used in the worked example. Because F mixes the two fusion channels and the two R phases are different, B_{12} and B_{23} do not commute. This provides the explicit mechanism for a non-Abelian braid action.

A braid involving particles that are not adjacent fusion partners generally requires both operations: F regroups the fusion tree to bring the pair together, and R then performs the exchange. Regrouping alone performs no exchange.

Equal R phases would collapse both B_{12} and B_{23} to the same global phase, so the two operations would commute and the non-Abelian action on this two-dimensional fusion space would disappear with the fusion table unchanged.

Coherence conditions for fusion and braiding data

Four charges can be regrouped through different sequences of F -moves. When two sequences begin and end with the same fusion trees, they must define the same linear map.

This consistency requirement is the pentagon equation. Similarly, two equivalent sequences of regrouping and exchanging three charges lead to the hexagon equations, which constrain F and R jointly.

These coherence equations ensure that a complicated diagram has a unique value independent of the intermediate bookkeeping choices [Ro22].

The associated algebraic structure is a fusion category: a finite semisimple linear category whose simple objects represent charge types, whose tensor product represents fusion, whose unit object is 1, and whose associator is represented by the F -matrices. Semisimplicity means that objects decompose into finite direct sums of simple objects.

Dual objects represent antiparticles. A braided fusion category additionally contains a braiding isomorphism represented by R .

Braiding is optional at the fusion-category level: a fusion category carries no R -moves until a braiding is added. A unitary modular tensor category further imposes unitarity and a nondegeneracy condition on braiding, providing the standard algebraic structure for a complete two-dimensional anyon theory [Ro17]; [Ro16].

The fusion table, F -moves, and R -moves are the explicit computational data. The categorical terminology organizes these data into a unified mathematical structure.

The same fusion coefficients N_{ab}^c can admit inequivalent but internally consistent choices of F - and R -data. Fusion rules are necessary input but do not uniquely characterize the theory [Ro16]. If the pentagon or hexagon equations fail, a larger diagram does not have a well-defined value because different evaluation routes give different results.

Operational evidence in laboratory systems

The preceding symbols define an effective theory. A laboratory system instead consists of a finite sample, control mechanisms, detectors, unwanted excitations, boundaries, and a microscopic Hamiltonian. Fusion-category data are therefore inferred from measurements rather than read directly from a dedicated instrument.

[Experiment] Evidence for fusion data must be based on operationally distinguishable outcomes. Relevant procedures include creating or identifying excitations, combining spatial regions, measuring total charge, and showing that the observed channels agree with a consistent fusion rule. Interferometry provides one possible method for measuring charge because a probe can respond to the monodromy of the enclosed charge, where monodromy is the effect produced by winding the probe around that charge [Ro23]. A single spectral peak or degeneracy is insufficient to reconstruct N , and still less sufficient to determine F and R .

[Experiment] Evidence for an F -matrix requires comparison of incompatible fusion-tree measurements while maintaining enough phase coherence to recover amplitudes rather than only the set of allowed outcomes. Evidence for R requires controlled exchanges, or an operationally equivalent braid, together with measurements of channel-dependent phases. Multiple operation sequences should satisfy the braid and coherence relations within the measured uncertainty.

[Theory] Even exact agreement with a limited set of matrices establishes the effective operations tested, and those alone. A claim of emergent anyons additionally requires excitations arising from a many-body phase, deconfinement over the relevant length scale, and data stable under allowed local perturbations.

A claim of topological protection must also bound errors from non-topological processes, thermal excitation, finite-size splitting, and control faults. These are physical requirements beyond the fusion and braiding algebra [Ro15]; [Ro17].

Digital emulation, defects, and emergent anyons

A digital processor can encode the two Ising fusion-tree basis states in ordinary qubits and implement the Hadamard matrix displayed above. Such an implementation is a digital emulation of an F -move.

A collection of defect spins can similarly encode a two-dimensional subspace. Both constructions operate on ordinary local degrees of freedom; emergent σ anyons and topological order need the many-body phase evidence described above.

Extrinsic defects can carry nontrivial projective fusion and braiding properties, but they must be distinguished from intrinsic quasiparticles that propagate freely within the host phase [Ro21]. The same algebra can occur in several physical settings, and experimental evidence from the hardware determines which setting is realized.

Assigning anyonic labels to defect-spin energy levels leaves the physical claim unestablished until the proposed microscopic Hamiltonian is shown to generate emergent excitations whose low-energy states obey consistent fusion and braiding data; later chapters examine whether this more demanding physical requirement is plausible.

A programmed braid and an emergent anyon therefore remain distinct claims when the relevant algebra concerns fusion rather than exchange.

Common conceptual errors

- Interpreting “+” as a prepared quantum superposition is incorrect. In a fusion rule, “+” denotes the allowed total-charge sectors. A quantum state, including its amplitudes in those sectors, can be specified only with additional preparation information.
- Interpreting N_{ab}^c as a probability is incorrect. The fusion multiplicity N_{ab}^c is the integer dimension of the space of fusion channels in which charges a and b combine to total charge c . Probabilities instead follow from normalized state amplitudes together with a specified measurement protocol.
- Interpreting quantum dimension as the number of states at a single site is incorrect. Quantum dimension characterizes the asymptotic growth of fusion-space dimension as the number of charges increases. Thus, $d_\sigma = \sqrt{2}$ cannot be the dimension of a finite local vector space.

- Not every basis transformation is a braid. An F -move changes the grouping, or parenthesization, of a fusion tree without exchanging the spatial positions of charges. An R -move exchanges two charges. Braiding nonadjacent fusion partners generally requires both F - and R -moves.
- The total-charge constraint cannot be omitted. Three σ charges constrained to have total charge σ form the two-dimensional example discussed above. A different total-charge constraint generally produces a different fusion space.
- Multiplicity labels cannot be omitted in a general theory. The statement that an internal-edge label completely specifies a fusion-tree basis state is valid only when all relevant fusion multiplicities satisfy $N_{ab}^c \leq 1$. If a fusion vertex has multiplicity greater than one, an additional vertex label is required to distinguish the independent fusion channels.
- Gauge-dependent matrix entries are not themselves observables. Here, a gauge choice is a choice of basis phases at fusion vertices. Rephasing such a vertex basis changes individual F - and R -symbols, which are the matrix elements associated with reassociation and exchange operations. By contrast, probabilities for closed processes, consistency relations, and physically defined interference signals are gauge invariant, meaning that they do not depend on this basis choice.
- Fusion rules do not uniquely determine a topological phase. The same fusion coefficients N_{ab}^c can admit inequivalent sets of consistent F - and R -data. Fusion rules are therefore necessary input, but they are not a complete specification of the phase [Ro16].
- Labels used in a simulator must not be identified automatically with material excitations. A quantum circuit can reproduce fusion-space algebra accurately even when every underlying carrier is an ordinary physical qubit, meaning a physical two-level quantum system. Such a construction is a simulation rather than, by itself, evidence for emergent material excitations.
- Algebraic data do not establish the existence of a microscopic energy gap. A consistent set of F - and R -tables does not prove that a proposed defect Hamiltonian realizes those data. Such a realization additionally requires a gapped many-body phase, appropriate excitations, and stability under realistic perturbations.

Verification exercises

- The equation $N_{ab}^c = 2$ states that there are two linearly independent fusion channels in which charges a and b combine to total charge c . A multiplicity label is therefore required to distinguish them.
- Both pairings of three Ising σ charges with total charge σ have dimension two. For the left pairing,

$$N_{\sigma\sigma}^1 N_{1\sigma}^\sigma + N_{\sigma\sigma}^\psi N_{\psi\sigma}^\sigma = 1 + 1 = 2.$$

The right pairing gives the same sum when it is applied to the last pair. The two fusion trees therefore provide two different bases for the same two-dimensional fusion space.

- A fusion tree records the organization of successive fusion operations. Its external edges specify the initial charges, its internal edges specify intermediate charges, its vertices may carry multiplicity labels, and its root specifies the total charge.

- If F were not unitary, inner products would depend on the pairing used to represent the state. A unitary transformation preserves inner products, so failure of unitarity would cause Born probabilities—the probabilities obtained from squared state amplitudes—for the same physical question to disagree between pairings.
- An F -move changes the pairing of charges without exchanging their positions. An R -move exchanges neighboring charges within a specified fusion channel.
- Reproducing an F -matrix with defect-spin qubits demonstrates an encoded operation. Emergent anyons need independent evidence for both a many-body topological phase and its emergent excitations.

Fusion rules list the allowed outcomes, an F -move regroups three charges, and an R -move exchanges two charges. The next analysis applies this machinery to a single nontrivial charge τ , whose fusion-space dimensions generate the Fibonacci numbers.

Sources

- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian Anyons and Topological Quantum Computation,” *Reviews of Modern Physics* 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083); [arXiv:0707.1889](https://arxiv.org/abs/0707.1889).
 - [R017] A. Kitaev, “Anyons in an Exactly Solved Model and Beyond,” *Annals of Physics* 321, 2–111 (2006). DOI: [10.1016/j.aop.2005.10.005](https://doi.org/10.1016/j.aop.2005.10.005); [arXiv:cond-mat/0506438](https://arxiv.org/abs/cond-mat/0506438).
 - [R023] P. Bonderson, K. Shtengel, and J. K. Slingerland, “Interferometry of Non-Abelian Anyons,” *Annals of Physics* 323, 2709–2755 (2008). DOI: [10.1016/j.aop.2008.01.012](https://doi.org/10.1016/j.aop.2008.01.012); [arXiv:0707.4206](https://arxiv.org/abs/0707.4206).
 - [R016] E. C. Rowell and Z. Wang, “Mathematics of Topological Quantum Computing,” *Bulletin of the American Mathematical Society* 55, 183–238 (2018). DOI: [10.1090/bull/1605](https://doi.org/10.1090/bull/1605); [arXiv:1705.06206](https://arxiv.org/abs/1705.06206).
 - [R022] G. Moore and N. Seiberg, “Classical and Quantum Conformal Field Theory,” *Communications in Mathematical Physics* 123, 177–254 (1989). DOI: [10.1007/BF01238857](https://doi.org/10.1007/BF01238857).
 - [R021] M. Barkeshli, P. Bonderson, M. Cheng, and Z. Wang, “Symmetry Fractionalization, Defects, and Gauging of Topological Phases,” *Physical Review B* 100, 115147 (2019). DOI: [10.1103/PhysRevB.100.115147](https://doi.org/10.1103/PhysRevB.100.115147); [arXiv:1410.4540](https://arxiv.org/abs/1410.4540).
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PART VII — FIBONACCI ANYONS

The preceding chapter introduced fusion rules, which specify the possible outcomes when topological charges combine. Here we study the Fibonacci fusion rule in detail.

Chapter 15 — Two fusion outcomes for a pair of Fibonacci anyons

Consider a topological charge denoted by τ . When two such charges are combined, their total charge can be either the vacuum or another τ :

$$\tau \times \tau = 1 + \tau.$$

This equation defines the Fibonacci fusion rule. The symbol 1 denotes the vacuum charge, so the vacuum outcome means that the two τ charges can annihilate.

The plus sign lists alternative fusion outcomes; unlike ordinary addition, it records that a measurement of the pair's total charge returns 1 or τ , with one outcome realized per measurement.

Self-duality and fusion channels

This theory contains only two topological-charge labels: the vacuum 1 and the nontrivial charge τ .

The vacuum is the identity element of fusion. For either charge a ,

$$1 \times a = a \times 1 = a.$$

A single τ therefore retains charge τ . The only fusion process with multiple possible outcomes is the fusion of two τ charges.

The charge τ is its own antiparticle: fusing it with another τ can produce vacuum, which is the defining property called self-duality. Thus τ is self-dual, and self-duality places the vacuum outcome in $\tau \times \tau$, as the rule shows.

The two terms on the right-hand side identify two orthogonal fusion outcomes, called fusion channels — the terminology introduced for Ising anyons in the previous chapter — rather than pieces of divided particles. In the Fibonacci theory, the channels of $\tau \times \tau$ are exactly 1 and τ .

Counting fusion histories

For an ordered collection of τ charges, sequential fusion defines an intermediate total charge that can be either 1 or τ . Adding one more τ produces the following transitions. A previous total charge 1 must become τ , because $1 \times \tau = \tau$. A previous total charge τ can become either 1 or τ , because $\tau \times \tau = 1 + \tau$.

```
old total add τ new total
1 -----> τ
τ -----> 1 or τ
```

A valid sequence of intermediate charge labels, called a fusion history, specifies one basis state of the many-particle fusion space. Each added τ charge opens further histories, so their number grows with the collection size.

Let A_n denote the number of fusion histories for n copies of τ with total charge 1. Let B_n denote the number with total charge τ . Adding another τ gives

$$A_{n+1} = B_n, \qquad B_{n+1} = A_n + B_n.$$

The first equation counts the single route to vacuum: only a previous total charge τ combines with the new τ to produce 1. The second equation counts both routes to τ : from a previous 1 or a previous τ . The fusion rule and the initial conditions therefore fix both sequences completely.

For one τ , there are no histories with total charge 1 and one history with total charge τ . Therefore,

$$(A_1, B_1) = (0, 1).$$

For two τ charges,

$$A_2 = B_1 = 1, \qquad B_2 = A_1 + B_1 = 1.$$

For three,

$$A_3 = B_2 = 1, \qquad B_3 = A_2 + B_2 = 2.$$

Continuing through seven anyons gives:

Number n of τ anyons	total 1: A_n	total τ : B_n	sum of sectors
2	1	1	2
3	1	2	3
4	2	3	5
5	3	5	8
6	5	8	13
7	8	13	21

The final column sums the dimensions of two distinct total-charge sectors, recording a direct sum rather than a single coherent computational space.

A superselection rule forbids ordinary coherent superpositions between total charge 1 and total charge τ states. Computation therefore fixes the total charge and works within one sector; the direct sum across sectors is not generally usable as a qubit.

The two columns contain the sequences 1, 1, 2, 3, 5, 8, ... and 1, 2, 3, 5, 8, 13, In each sequence, every entry is the sum of the preceding two entries.

These are the Fibonacci numbers. Define them by $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 1$. Then

$$A_n = F_{n-1}, \quad B_n = F_n.$$

Thus, the table follows solely from the fusion rule [R135]. The theory is named after this Fibonacci growth of its fusion spaces.

Matrix recurrence, Fibonacci growth, and quantum dimension

The recurrence relation can be represented by a 2×2 matrix. In the ordered charge basis $(1, \tau)$, define the fusion matrix

$$N_\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

The matrix N_τ updates the pair of fusion-path counts according to the recurrence above. Its powers therefore generate Fibonacci numbers.

Let $V_c^{a_1 \cdots a_n}$ denote the complex vector space of fusion states in which the charges $a_1, \dots, a_n$ have fixed total charge c . The dimensions of the two fixed-charge sectors are

$$\dim V_1^{\tau^n} = F_{n-1}, \quad \dim V_\tau^{\tau^n} = F_n.$$

Here τ^n denotes n copies of τ , not an ordinary numerical power. Every fusion multiplicity in this theory is either zero or one. Therefore, each valid labeling of a fusion tree corresponds to one basis vector.

These dimensions grow exponentially at large n . The ratio of neighboring Fibonacci numbers approaches the largest eigenvalue of N_τ . Its characteristic equation is $\lambda^2 - \lambda - 1 = 0$, whose positive root is

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618.$$

The number φ is the golden ratio. In this context, it appears as the asymptotic growth rate of the fusion-space dimension.

The quantum dimension d_a of a charge a is the positive number compatible with the fusion coefficients N_{ab}^c , where N_{ab}^c is the multiplicity of charge c in the fusion of a and b :

$$d_a d_b = \sum_c N_{ab}^c d_c.$$

Set $d_1 = 1$. Applying this relation to $\tau \times \tau = 1 + \tau$ gives

$$d_\tau^2 = 1 + d_\tau.$$

Solving $d_\tau^2 - d_\tau - 1 = 0$ yields $(1 \pm \sqrt{5})/2$. A unitary fusion theory selects the positive Perron–Frobenius solution, where the Perron–Frobenius solution is the positive eigenvalue associated with a nonnegative fusion matrix:

$$d_\tau = \varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618.$$

Therefore, φ has three equivalent interpretations: it is the largest eigenvalue of N_τ , the limit of F_{n+1}/F_n , and the quantum dimension of τ . The number of fusion states consequently scales as a constant times φ^n .

Quantum dimension is dimensionless and need not be an integer. It characterizes the asymptotic growth of a nonlocal fusion space rather than the number of local energy levels [R135]; [R015]. In particular, a single τ does not possess “1.618 states.”

Channel-dependent phases under exchange

Fusion histories label a degenerate state space nonlocally, through the collective fusion outcomes, rather than through local degrees of freedom attached independently to each particle. Exchanging neighboring τ charges acts as a unitary transformation on this fusion space.

An exchanged pair in a definite fusion channel contributes the phase belonging to that channel. Exchanging a different pair calls for a change of basis first: express the state with that pair fusing first, apply the exchange phases, and transform back to the original basis. In the notation of the previous chapter, this procedure is $F^{-1} R F$.

A basis transformation followed by unequal channel-dependent phases generally fails to commute with the same steps in a different order, so exchanges of different neighboring pairs generate different rotations of the encoded state.

Repeated exchanges build up a large set of rotations. For the Fibonacci theory, this set approximates the gates required for quantum computation; the density statement below makes the claim precise and states its limitations.

An engineered Hamiltonian earns identification with the Fibonacci theory by reproducing both the fusion spaces and the braiding data: the value 1.618 appearing in one observable is not enough on its own. A claim of passive protection further requires these data to arise as the low-energy physics of a gapped two-dimensional many-body phase.

Three-anyon qubit encoding

Consider three τ anyons with total charge fixed to τ . Since $B_3 = 2$, this sector is two-dimensional and can encode one logical qubit:

$$\begin{aligned} |0_L\rangle &= |((\tau\tau)_1\tau)_\tau\rangle, \\ |1_L\rangle &= |((\tau\tau)_\tau\tau)_\tau\rangle. \end{aligned}$$

The subscript on a parenthesized group denotes that group's total charge. The logical qubit is encoded in whether the first pair fuses to 1 or to τ , while the total charge of all three anyons remains fixed at τ .

Each individual particle keeps local topological charge τ , so the logical information resides in the pair outcome rather than on any single anyon.

Four τ anyons with total charge 1 also provide a two-dimensional fusion space because $A_4 = 2$. This encoding is often preferred because each encoded block has vacuum total charge. The three-anyon encoding with total charge τ is the smallest example, so it is used for the following calculation.

Dropping the total-charge constraint leaves two states in different superselection sectors, which do not constitute an ordinary qubit: coherence between such sectors is generally unavailable as a computational degree of freedom.

Braid matrices for the first and second pairs

Adopt the convention of Ref. [R135]: fusion trees point downward, basis channels are ordered as $(1, \tau)$, and the positive braid generator is the review's right-handed exchange with time directed upward. Define

$$r_1 = e^{4\pi i/5}, \quad r_\tau = e^{-3\pi i/5}.$$

Exchanging anyons 1 and 2 is diagonal in the logical basis:

$$\rho(\sigma_1) = R = \begin{pmatrix} r_1 & 0 \\ 0 & r_\tau \end{pmatrix}.$$

Here σ_i denotes the positive exchange of anyons i and $i + 1$, and ρ is the matrix representation of that exchange on the fusion space. To exchange anyons 2 and 3, the basis must first be changed so that those two anyons fuse first. The required transformation is

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad \varphi = \frac{1 + \sqrt{5}}{2}.$$

In this gauge, F is real and symmetric, and $F^{-1} = F$. Therefore,

$$\rho(\sigma_2) = F R F = \begin{pmatrix} \varphi^{-2} r_1 + \varphi^{-1} r_\tau & \varphi^{-3/2} (r_1 - r_\tau) \\ \varphi^{-3/2} (r_1 - r_\tau) & \varphi^{-1} r_1 + \varphi^{-2} r_\tau \end{pmatrix}.$$

Applying this braid once to $|0_L\rangle$ produces a state determined by the first column of $\rho(\sigma_2)$. Both entries are nonzero, so the braid creates a coherent superposition of the two fusion channels. If the first pair is then fused and its charge measured, this prepared state gives

$$P(1) = \varphi^{-2}, \quad P(\tau) = \varphi^{-1}.$$

These probabilities sum to one because $\varphi^{-2} + \varphi^{-1} = 1$. This identity provides their normalization.

The values apply only to this initial state, this braid, and this measurement. They are not universal probabilities for arbitrary encounters between two τ anyons.

Because R is diagonal whereas $F R F$ has nonzero off-diagonal entries,

$$\rho(\sigma_1)\rho(\sigma_2) \neq \rho(\sigma_2)\rho(\sigma_1).$$

Thus, neighboring exchanges do not commute on this two-dimensional fusion space. This is the smallest explicit demonstration of non-Abelian statistics in the Fibonacci theory.

If the two channel phases were instead equal, so that $r_1 = r_\tau$, then R would be proportional to the identity and would represent only a global phase. The matrix $F R F$ would be the same global phase, and the neighboring braid generators would commute.

The off-diagonal entries therefore require both ingredients: the recoupling move mixes fusion channels, and unequal exchange phases convert that mixing into a nontrivial transformation.

Consistency and gauge dependence of the F - and R -symbols

For three τ anyons with total charge τ , the two fusion bases are

$$|((\tau\tau)_x\tau)_\tau\rangle, \quad |(\tau(\tau\tau)_y)_\tau\rangle,$$

where $x, y \in \{1, \tau\}$. The matrix $F_{\tau}^{\tau\tau\tau}$ transforms between these two parenthesizations.

In the gauge adopted above, $F_{\tau}^{\tau\tau\tau}$ is the matrix F already used. All other allowed F -moves in this theory are scalars and can be chosen to equal one.

The pentagon equation is the consistency condition requiring every sequence of F -moves between two parenthesizations of four charges to give the same result. It expresses associativity of fusion at the level of fusion-space basis transformations.

The scalar $R_c^{\tau\tau}$ is the phase associated with the chosen handed exchange of two τ charges whose combined charge is c . In the present convention,

$$R_1^{\tau\tau} = e^{4\pi i/5}, \quad R_{\tau}^{\tau\tau} = e^{-3\pi i/5}.$$

The hexagon equations impose compatibility between recoupling transformations and braiding. Together, the fusion rules, unitary F -symbols, and R -symbols define the braid representation used here [R135]; [R015].

The individual matrix entries displayed above are not convention-independent observables. Rephasing the one-dimensional fusion and splitting vertices changes individual F -symbols and can also change the representatives of the R -symbols, while leaving complete diagrams and physical probabilities invariant. This freedom is a gauge freedom in the sense introduced in the previous chapter. A matrix representation must therefore be accompanied by its basis order, fusion-tree orientation, gauge choice, and braid orientation.

Reversing braid handedness replaces R by $R^{-1} = R^{\dagger}$. Passing to the mirror theory complex-conjugates the braiding data.

Different references may therefore display complex-conjugate phases while describing equivalent physical content. Comparisons should use gauge-invariant braid words, fusion probabilities, and link amplitudes rather than isolated signs or matrix entries.

Density and braid universality

For n anyons, the braid generators obey the two relations introduced in Chapter 13:

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, \quad \sigma_i \sigma_j = \sigma_j \sigma_i \quad (|i - j| \geq 2).$$

The F - and R -data map these abstract generators to unitary matrices $\rho(\sigma_i)$ acting on a fixed-total-charge fusion space.

[Theory] For the Fibonacci theory's data, the resulting braid representations are dense, up to a physically irrelevant overall phase, in the unitary transformations on the computational fusion spaces. In particular, the two three-anyon matrices above generate a dense set of single-qubit rotations.

Suitable braids involving multiple anyons can approximate entangling operations. This property is the mathematical basis of braid universality [R015]; [R136].

Numerical compilation algorithms explicitly construct braid words that approximate specified target gates [R137].

More precisely, density means that for any target unitary U and any tolerance $\varepsilon > 0$, there is a finite braid word w such that

$$\min_{\alpha} \|\rho(w) - e^{i\alpha} U\| < \varepsilon,$$

where $\|\cdot\|$ is an operator norm and $e^{i\alpha}$ is a global phase. This definition promises approximation by some finite braid word, leaving open both single-exchange implementation and knowledge of the shortest word.

This density property is conventionally called universality. The term must be interpreted specifically as the ability to approximate the relevant unitary transformations. It leaves two further questions separate: which braids serve as valid computational gates, and whether a laboratory realization constitutes a complete quantum computer.

The fusion rule alone establishes Fibonacci growth and, in a unitary theory, fixes d_τ ; braid universality needs more. The universality result depends on the particular consistent and unitary F/R data of the braided category.

A fusion ring specifying fusion algebra defines no braid representation on its own.

Leakage outside the computational subspace

For encodings containing several logical qubits, the full fixed-charge fusion space is generally larger than the selected tensor-product computational subspace. A braid that does not preserve the chosen subspace can populate these additional states.

This transition out of the computational subspace is called leakage. Universality does not imply that every braid is leakage-free [R015]; [R137]. In the ideal theory, universal compilation can suppress leakage to arbitrary accuracy. A complete computer also requires initialization and fusion measurement.

If the additional fusion states are omitted from the analysis, a braid can appear to act correctly on the intended logical states even though it transfers amplitude outside the computational subspace. An error estimate that ignores this leakage is therefore incomplete.

Relation to laboratory systems

The preceding equations define an ideal topological theory rather than a specific material realization.

[Theory] The $k = 3$ Read–Rezayi fractional quantum Hall state was proposed to support a non-Abelian sector related to these anyons [R139]. Observation of a Hall plateau at a compatible filling would not, by itself, determine the complete fusion and braiding data. In the literature surveyed through August 2026, intrinsic Fibonacci exchange and fusion have not been established in a defect crystal.

[Experiment] In 2024, Xu and collaborators used 27 superconducting transmon qubits to prepare and manipulate a digital representation of a Fibonacci Levin–Wen string-net state. They applied circuits representing creation, fusion, and braiding and measured signatures consistent with the target model [R138].

This result constitutes a substantial laboratory demonstration of compiled string-net operations. Its microscopic excitations were actively driven defects of the gate sequence rather than passively emergent, mobile quasiparticles of a static transmon Hamiltonian.

The implemented string-net target was doubled Fibonacci order, and its operations were compiled gates acting on ordinary physical qubits [R138].

[Experiment] A 2026 trapped-ion experiment prepared a 54-qubit state of the S_3 quantum double and demonstrated a universal gate set by combining braiding with fusion [R140]. This experiment provides a relevant control case because its anyons were digitally encoded and its topological order was S_3 , not Fibonacci. The phrase “universal anyon computation” does not uniquely identify the underlying anyon theory.

A processor executing an F gate simulates an algebraic basis transformation. Establishing an equilibrium Fibonacci phase or passive topological memory requires evidence of emergent excitations in a gapped phase together with measured fusion and braiding data.

Distinct meanings of Fibonacci implementations

- **Encoded Fibonacci qubit:** Information is stored in a fixed-charge fusion space of τ anyons.
- **Emergent Fibonacci anyons:** These are quasiparticles of a gapped two-dimensional many-body phase whose adiabatic exchanges realize the Fibonacci braid representation.
- **Doubled Fibonacci:** This is the nonchiral Drinfeld-center theory produced by a Levin–Wen model using this chapter’s input data. Its complete charge set and topological order are larger than those of the two-object chiral theory [R018].
- **Fibonacci-like digital simulation:** Ordinary hardware qubits encode fusion labels, while programmed gates implement F , R , or string operators. Such a simulation can test the algebra without providing intrinsic quasiparticles or passive protection [R138].

Doubled Fibonacci is a genuine topological order distinct from the chiral Fibonacci theory, containing computationally useful Fibonacci-type sectors. A digital simulation is scientifically useful for testing circuits and measurement protocols. The distinction among these implementations is which physical system carries the topological order and what mechanism, if any, supplies protection.

Common conceptual errors

- **Inferring universality from $\tau \times \tau = 1 + \tau$ alone.**

The fusion rule determines state counting but does not determine braid phases. Pentagon- and hexagon-consistent unitary F/R data are essential. Without the braid data, Fibonacci growth remains, but no density theorem follows.

- **Interpreting $d_\tau = 1.618 \dots$ as a local degeneracy.**

A single anyon does not have “1.618 states.” Quantum dimension describes the asymptotic growth of a nonlocal fusion space. The appearance of φ in a spectrum does not by itself demonstrate the required fusion spaces.

- **Combining distinct total-charge sectors as one encoding space.**

The sum $A_n + B_n$ does not automatically define one coherent Hilbert space available for encoding. The boundary charge must be fixed before logical states are counted. Otherwise, the count refers to a direct sum of superselection sectors rather than a qubit.

- **Quoting matrices without their conventions.**

Changes in basis phases, basis order, braid handedness, or mirror chirality alter the displayed matrices. Gauge-invariant braid words, fusion probabilities, and link amplitudes should be compared instead of isolated signs.

- **Equating braid universality with a fault-tolerant device.**

[Theory] Ideal adiabatic braids are insensitive to small path deformations when the anyons remain separated in a gapped phase [R015].

Real systems can nevertheless experience thermal anyon creation, quasiparticle poisoning, finite-separation splitting, diabatic transitions, control errors, and faulty readout. Universality specifies which gates can be approximated. Protection specifies how physical errors scale.

- **Ignoring leakage in multi-qubit encodings.**

The physical fusion space can contain states outside the computational tensor product. A compilation procedure must control leakage as well as gate error [R137].

- **Treating a circuit implementation as evidence of emergence.**

A processor that executes an F gate has simulated an algebraic transformation. It has not thereby acquired an equilibrium Fibonacci phase or passive topological memory.

- **Classifying every non-Abelian platform as Fibonacci.**

Ising anyons, finite-group quantum doubles, and Fibonacci anyons have different fusion and braid data. “Non-Abelian” describes a broad class rather than a specific model.

Exercises and answers

- **Derive the recurrence relations $A_{n+1} = B_n$ and $B_{n+1} = A_n + B_n$ from the fusion rule.**

Only a previous total charge τ can fuse with a new τ to produce 1, so $A_{n+1} = B_n$. A new total charge τ can arise from either a previous 1 or a previous τ , so $B_{n+1} = A_n + B_n$.

- **Find the positive solution of $d_\tau^2 = 1 + d_\tau$.**

Solving the quadratic gives $(1 \pm \sqrt{5})/2$. Unitarity selects the positive root,

$$\varphi = \frac{1 + \sqrt{5}}{2}.$$

The same quadratic is the characteristic equation of N_τ .

- **Identify where the logical information is stored in the three-anyon encoding.**

It is stored in whether the first pair has fusion channel 1 or τ , while the total charge of all three anyons remains fixed at τ .

- **Determine the consequence of equal channel phases in R .**

If the channel phases are equal, R is a global phase on the two-dimensional space. Then FRF is the same global phase, neighboring braids commute, and the non-Abelian action disappears.

- **State what cannot be concluded from the fusion rule alone.**

The fusion rule establishes Fibonacci growth and, in a unitary theory, $d_\tau = \varphi$. It does not provide the braid phases or establish the density theorem. A fusion ring is not a braid representation.

- **Assess whether the 27-transmon experiment created intrinsic chiral Fibonacci matter.**

The experiment digitally implemented a doubled-Fibonacci string-net state and associated operations on ordinary qubits. That implementation does not establish intrinsic chiral-Fibonacci matter [R138].

The next chapter turns to a simpler non-universal phase, the toric code, and shows how local Hamiltonian terms produce anyons and nonlocal logical sectors.

Sources

- [R135] S. Trebst, M. Troyer, Z. Wang, and A. W. W. Ludwig, “A short introduction to Fibonacci anyon models,” Progress of Theoretical Physics Supplement 176, 384–407 (2008). DOI: [10.1143/PTPS.176.384](https://doi.org/10.1143/PTPS.176.384). arXiv: [0902.3275](https://arxiv.org/abs/0902.3275).
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” Reviews of Modern Physics 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083). arXiv: [0707.1889](https://arxiv.org/abs/0707.1889).
- [R136] M. H. Freedman, M. Larsen, and Z. Wang, “A modular functor which is universal for quantum computation,” Communications in Mathematical Physics 227, 605–622 (2002). DOI: [10.1007/s002200200645](https://doi.org/10.1007/s002200200645). arXiv: [quant-ph/0001108](https://arxiv.org/abs/quant-ph/0001108).
- [R137] L. Hormozi, G. Zikos, N. E. Bonesteel, and S. H. Simon, “Topological quantum compiling,” Physical Review B 75, 165310 (2007). DOI: [10.1103/PhysRevB.75.165310](https://doi.org/10.1103/PhysRevB.75.165310). arXiv: [quant-ph/0610111](https://arxiv.org/abs/quant-ph/0610111).

- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” Physical Review B 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110). arXiv: [cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617).
 - [R138] S. Xu et al., “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” Nature Physics 20, 1469–1475 (2024). DOI: [10.1038/s41567-024-02529-6](https://doi.org/10.1038/s41567-024-02529-6). Stable article: nature.com/articles/s41567-024-02529-6.
 - [R139] N. Read and E. Rezayi, “Beyond paired quantum Hall states: Parafermions and incompressible states in the first excited Landau level,” Physical Review B 59, 8084–8092 (1999). DOI: [10.1103/PhysRevB.59.8084](https://doi.org/10.1103/PhysRevB.59.8084). arXiv: [cond-mat/9809384](https://arxiv.org/abs/cond-mat/9809384).
 - [R140] C. F. B. Lo et al., “Universal gates from braiding and fusing anyons on quantum hardware,” Nature 655, 591–597 (2026). DOI: [10.1038/s41586-026-10709-y](https://doi.org/10.1038/s41586-026-10709-y). Stable article: nature.com/articles/s41586-026-10709-y.
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PART VIII — TOPOLOGICAL LATTICE MODELS

This part examines how local spin interactions and lattice paths produce global topological behavior. It then distinguishes a calculation or controlled simulation from a physical material that realizes the same phase.

Chapter 16 — Local stabilizer checks and encoded information

Take a square lattice whose opposite sides are identified. The resulting surface is a torus, and each edge carries one spin-1/2 that serves as a qubit.

Each vertex and each face has its own binary-valued operator. We call these operators checks because their eigenvalues report whether the associated local constraint is satisfied.

A check reports a local violation. The collection of check outcomes identifies where defects occur, while a separate global question asks whether the flipped edges form a loop winding around a noncontractible cycle of the torus.

Here “toric code” refers to this lattice model: qubits occupy edges, vertex and face checks are defined below, and the same checks specify either a Hamiltonian or a quantum code.

A color-center spin is one physical qubit in a crystal. Assigning one defect to each drawn edge does not by itself produce the required checks or an energy gap.

Eight-qubit periodic lattice

Consider a 2×2 square grid with opposite sides identified. The coordinates $x, y \in \{0, 1\}$ are evaluated modulo 2. Let $h_{x,y}$ denote the horizontal edge from (x, y) to $(x + 1, y)$, and let $v_{x,y}$ denote the vertical edge from (x, y) to $(x, y + 1)$.

The lattice contains four horizontal edges and four vertical edges. Because each edge carries one qubit, the Hilbert space has dimension $2^8 = 256$.

A planar representation of this periodic patch is

```
h00 h10
(0,0) -----> (1,0) -----> (0,0)
| | |
v00| v10 | | periodic
v v v
(0,1) -----> (1,1) -----> (0,1)
h01 h11
| |
v01| v11 |
v v
(0,0) (1,0) (bottom = top)
```

Repeated vertices in the drawing mark periodic identifications, not extra sites. A 1×1 periodic patch cannot display four distinct incident edges at each vertex. The 2×2 patch is the smallest one that preserves this local incidence pattern and still contains two independent noncontractible cycles.

This construction is therefore already a lattice on a torus rather than a crystal. The eight spins label lattice edges. They are not eight nitrogen-vacancy centers that acquire topology merely by being placed near one another.

If periodicity is removed, the same local checks can still be defined on a planar patch with a boundary. The resulting count of unconstrained states is different because the global topology and boundary conditions have changed.

Vertex stabilizer operators

Each vertex is incident on four edges: the horizontal edge directed to the right, the horizontal edge arriving from the left, the vertical edge directed upward, and the vertical edge arriving from below.

The vertex operator is the product of Pauli X operators on these four edges. For the vertex at (x, y) , it is

$$A_{x,y} = X_{h_{x,y}} X_{h_{x-1,y}} X_{v_{x,y}} X_{v_{x,y-1}},$$

where X_e is the Pauli X operator acting on edge e , and every index is evaluated modulo 2.

This four-edge product is the star operator. A star eigenvalue $+1$ satisfies the vertex check; -1 marks a violated vertex constraint.

Two stars that share an edge commute because both act with Pauli X on that edge and contain only X operators. Consequently, all star operators can be assigned simultaneous definite eigenvalues.

The choice of Pauli type is essential. If a face operator acted with X , rather than Z , on only one edge shared with a star, that star and face operator would anticommute. The division between X -type vertex checks and Z -type face checks is therefore part of the model's defining operator algebra.

Face stabilizer operators

Each square face is bounded by four edges: its bottom, right, top, and left edges.

The face operator is the product of Pauli Z operators on these four edges. For the square with lower-left corner at (x, y) , it is

$$B_{x,y} = Z_{h_{x,y}} Z_{v_{x+1,y}} Z_{h_{x,y+1}} Z_{v_{x,y}},$$

where Z_e is the Pauli Z operator acting on edge e . The indices are again evaluated modulo 2.

This four-edge product is the plaquette operator. A plaquette eigenvalue $+1$ satisfies the face check; -1 marks a violated face constraint.

Stars are products of X , whereas plaquettes are products of Z . This distinction determines the local algebra of the model.

If the four- Z plaquette product were replaced by four X operators on the same face, its algebra with the stars would change. A star and that modified face operator would act with matching Pauli types on their shared edges, and the two canceling minus signs associated with the original X - Z overlaps would no longer be guaranteed.

Pair creation and transport around a plaquette

Let $|\psi_0\rangle$ be a state for which every star and every plaquette has eigenvalue $+1$. Applying Z to the single edge $h_{0,0}$ creates two violated star checks.

Each star contains Pauli X operators, and X and Z anticommute when they act on the same edge. The operator $Z_{h_{0,0}}$ therefore anticommutes with the two stars incident on $h_{0,0}$, namely the stars at $(0, 0)$ and $(1, 0)$.

The eigenvalues of these two stars change from $+1$ to -1 . The resulting violated vertices are called e anyons, where an anyon is a localized excitation characterized by its fusion and exchange properties in two spatial dimensions.

The label e denotes a violated star; violated plaquettes receive a different label below.

Applying $Z_{v_{1,0}}$ next flips the shared endpoint $(1, 0)$ a second time and restores its star eigenvalue to $+1$.

A new violated endpoint appears at $(1, 1)$. Thus the second operation transports one member of the pair.

Applying $Z_{h_{0,1}}$ moves that endpoint to $(0, 1)$. Applying $Z_{v_{0,0}}$ then removes both remaining violated endpoints.

The product of the four operations is one plaquette operator:

$$Z_{h_{0,0}} Z_{v_{1,0}} Z_{h_{0,1}} Z_{v_{0,0}} = B_{0,0}.$$

Since $B_{0,0}|\psi_0\rangle = |\psi_0\rangle$, the completed square leaves every check unchanged. The intermediate states contain an excitation pair; the full closed path is itself a stabilizer.

A different result is obtained if the sequence continues around a periodic direction rather than closing around one square.

When the endpoints meet, they annihilate, and every local check again has eigenvalue $+1$. The remaining closed loop is noncontractible, meaning that it cannot be continuously reduced to a point on the torus.

Such a loop is not a product of local plaquette operators. The local checks therefore detect no excitation even though the winding sector has changed.

If the sequence stops after three edges, one e remains. The state still carries excitation energy and has not returned to the codespace, defined as the simultaneous $+1$ eigenspace of all stabilizer checks.

Direct- and dual-lattice string operators

A path γ along the grid's ordinary (direct-lattice) edges defines a Z -string operator:

$$W_e(\gamma) = \prod_{e \in \gamma} Z_e.$$

At each interior vertex, the path uses two of the four incident edges. The string therefore anticommutes twice with that star, producing no net sign change. At each endpoint, the path uses one incident edge and flips the corresponding star eigenvalue. These endpoint excitations are the e anyons described above.

The dual lattice is obtained by placing a dual vertex at the center of every face and drawing dual edges across the original edges. A path γ^* on this dual lattice crosses a set of ordinary edges. Its associated X -string operator is

$$W_m(\gamma^*) = \prod_{e \perp \gamma^*} X_e.$$

This operator flips the plaquette checks at the two endpoint faces of the dual path. The resulting violated faces are called m anyons.

Star checks detect e excitations, and plaquette checks detect m excitations. An open Z string creates e excitations at its endpoints, while an open X string creates m excitations at its endpoints.

If both types of check are violated at the same location, the composite excitation is $\varepsilon = e \times m$. It carries one electric violation and one magnetic violation.

A Pauli Y error on one edge does not define a third independent excitation species. Since $Y = iXZ$, it contributes to both neighboring star and plaquette syndromes.

Commutation of the stabilizer checks

Each star A_v and each plaquette B_p is Hermitian and squares to the identity. Its only possible eigenvalues are therefore $+1$ and -1 . Here v labels a vertex and p labels a face.

Any two stars commute because they contain only Pauli X operators. Any two plaquettes commute because they contain only Pauli Z operators.

A star and a plaquette share either no edge or two edges. On one shared edge, the Pauli operators satisfy $XZ = -ZX$. When two edges are shared, the two minus signs cancel:

$$A_v B_p = (-1)^2 B_p A_v = B_p A_v.$$

Every check therefore commutes with every other check, so all check outcomes can be specified simultaneously.

These checks are the stabilizers of the model. A stabilizer is an operator that acts as the identity on the valid code states and has eigenvalue $+1$ on those states.

If a star and a plaquette shared only one edge, they would anticommute. They could not then have simultaneously definite eigenvalues, and a Hamiltonian containing both operators would not be a sum of commuting penalties.

Energetic penalties for violated checks

When the checks are implemented as persistent energy penalties, the lattice Hamiltonian is

$$H_{\text{TC}} = -J_e \sum_v A_v - J_m \sum_p B_p,$$

where $J_e > 0$ and $J_m > 0$ have units of energy. Pauli products are dimensionless, so each coefficient J , and hence H_{TC} , must carry units of energy.

A check operator is not itself a projector because its eigenvalues are ± 1 . The projectors onto its two eigenspaces are

$$P_v^\pm = \frac{I \pm A_v}{2}, \quad P_p^\pm = \frac{I \pm B_p}{2}.$$

Let N_v and N_p denote the numbers of vertices and faces. The Hamiltonian can then be written as

$$H_{\text{TC}} = E_0 + 2J_e \sum_v P_v^- + 2J_m \sum_p P_p^-, \quad E_0 = -J_e N_v - J_m N_p.$$

A violated star raises the energy by $2J_e$, and a violated plaquette raises it by $2J_m$. On a closed torus, either species must occur in pairs, so the minimum pair-creation energy is $4J_e$ for e excitations or $4J_m$ for m excitations.

The model is therefore a commuting-projector Hamiltonian: it is a sum of local projector penalties whose terms all commute. The exact solution and the interpretation of violated checks as anyons are due to Kitaev [\[Ro30\]](#).

A diamond crystal does not intrinsically realize H_{TC} . The four-body products must be engineered directly, approximated through effective interactions, or compiled into digital operations. Without those interactions, a system of eight or eight thousand spins remains a collection of ordinary spins rather than this energy model.

Four-dimensional ground space

A ground state of H_{TC} satisfies

$$A_v|\psi\rangle = |\psi\rangle, \quad B_p|\psi\rangle = |\psi\rangle$$

for every vertex and every face. The 2×2 torus has eight written checks, but these checks are not all independent.

Every edge is incident on two vertices and borders two faces. Consequently,

$$\prod_v A_v = I, \quad \prod_p B_p = I.$$

Thus three stars and three plaquettes are independent. Six independent binary constraints acting on eight qubits leave a ground-space dimension

$$\dim \mathcal{H}_0 = 2^{8-6} = 4.$$

Equivalently, the patch encodes $k = 8 - 6 = 2$ logical qubits. For the ideal Hamiltonian on a torus, the fourfold degeneracy is exact. More generally, the toric code has 4^g ground states on a closed orientable surface of genus g ; a torus has $g = 1$ [Ro30].

One of the four states can be constructed explicitly. The state $|0\rangle^{\otimes 8}$ already satisfies $B_p = +1$ for every face. Averaging it over products of stars gives

$$|\psi_{00}\rangle \propto \prod_{v \neq v_0} (I + A_v) |0\rangle^{\otimes 8}.$$

One star v_0 is omitted because the product of all stars is already I . The resulting state is an equal superposition of contractible loop configurations. Applying either of two independent noncontractible wrapping operators generates the other three sectors.

The ground space consequently contains four states that no local operator can distinguish.

Incorrectly treating all eight checks as independent would imply that no state remains unconstrained. That conclusion fails because the two global products of checks equal the identity.

Noncontractible logical loop operators

On the eight-edge patch, two Z loops that wind around the torus and cannot be contracted to a point are

$$\bar{Z}_x = Z_{h_{0,0}} Z_{h_{1,0}}, \quad \bar{Z}_y = Z_{v_{0,0}} Z_{v_{0,1}}.$$

Two X loops that cross them once can be chosen as

$$\bar{X}_x = X_{h_{0,0}} X_{h_{0,1}}, \quad \bar{X}_y = X_{v_{0,0}} X_{v_{1,0}}.$$

Each closed loop commutes with every star and every plaquette. Matching pairs share one edge and therefore anticommute; for example,

$$\bar{X}_x \bar{Z}_x = -\bar{Z}_x \bar{X}_x.$$

Mismatched pairs share either zero edges or two edges, so they commute. These relations are exactly the Pauli algebra of two encoded qubits.

The wrapping products are logical operators: they change the encoded ground-state sector without producing a locally detectable violated check.

A local operator cannot measure the eigenvalue of a wrapping operator. It also cannot transform one locally indistinguishable ground-state sector into another.

On an $L \times L$ lattice, completing a noncontractible loop requires at least L single-edge Pauli operations. The minimum weight of such a wrapping operator is the code distance, $d = L$. It is the minimum number of local errors required to change a winding sector while leaving all checks satisfied.

For sufficiently weak local perturbations that preserve a many-body gap, topological-order stability theorems imply that local indistinguishability persists and that the splitting among the four sectors is exponentially small in system size, subject to the locality assumptions of those theorems [Theory] [R142].

If the many-body gap closes, these theorems no longer apply. Local perturbations can then mix the sectors at a scale that need not vanish as L increases.

Abelian fusion and braiding phases

The fusion rules of the four charges are

$$e \times e = 1, \quad m \times m = 1, \quad e \times m = \varepsilon,$$

$$e \times \varepsilon = m, \quad m \times \varepsilon = e, \quad \varepsilon \times \varepsilon = 1,$$

where 1 denotes the vacuum, meaning that no excitation remains. Every pair of charges has exactly one fusion outcome. Consequently, every fixed-charge fusion space is one-dimensional, so there is no multidimensional basis on which braiding could act by mixing states.

The string operators nevertheless have nontrivial algebra. Transporting an e around an m produces a Z string and an X string that cross once, so

$$W_e W_m = -W_m W_e.$$

The wavefunction therefore acquires a phase of -1 . Thus e and m have mutual semionic statistics. Individually, e and m have bosonic exchange statistics, while their composite ε is a fermion [Ro30].

These anyons are Abelian because each braid multiplies the state by a scalar phase. Non-Abelian anyons act with noncommuting matrices on a fusion space of dimension greater than one.

The toric code exhibits topological order, but braiding its anyons does not provide universal quantum computation.

The term “Abelian” does not imply that braiding has no observable effect. The mutual phase -1 is nontrivial and measurable. It remains a single scalar and therefore cannot implement the matrix-valued braids associated with a non-Abelian fusion space.

Topological order does not imply non-Abelian anyons.

Syndrome endpoints and decoding

Suppose unwanted Z errors occur on a set of edges E . The corresponding error operator is

$$Z(E) = \prod_{e \in E} Z_e.$$

A star check returns -1 exactly at the boundary ∂E , defined here as the set of vertices incident on an odd number of errored edges.

These violated vertices form the syndrome supplied to a decoder. The decoder sees the endpoints rather than the full error path and must infer a recovery operation.

A decoder selects a recovery chain R with the same boundary as E . After recovery, the net chain is the symmetric difference $E \oplus R$, because applying Z twice gives the identity. Since $\partial(E \oplus R) = 0$, the residual chain is closed. There are two possible topological outcomes:

- If $E \oplus R$ is contractible, it is a product of plaquettes and does not alter the encoded sector.
- If $E \oplus R$ winds around the torus, it implements a logical $\bar{Z}$ and changes an encoded bit without leaving a nonzero syndrome.

The analogous process for X errors uses plaquette syndromes and recovery chains on the dual lattice, and it can produce logical $\bar{X}$ failures. A general Pauli error $Y = iXZ$ contributes to both syndromes. Dennis and collaborators formulated this error-chain description and related it to statistical-mechanical decoding [R141].

A vanishing syndrome is therefore not sufficient evidence of successful correction. Both a harmless contractible loop and a damaging noncontractible loop have no endpoints.

A decoder that merely pairs and cancels observed defects without inferring the homology class can apply a recovery that leaves a wrapping loop and changes the logical sector. Homology class here distinguishes closed loops according to whether they can be continuously contracted or instead wind around the torus.

Passive Hamiltonian protection and active syndrome extraction

The same stabilizer checks support two physically different architectures.

In passive Hamiltonian protection, the terms $-J_e A_v$ and $-J_m B_p$ remain continuously present in the device Hamiltonian. Leaving the ground space then requires energy.

At low temperature, a thermal environment must supply this energy to create an anyon pair. A many-body gap also supports stability against sufficiently weak local perturbations [Theory] [R142].

This constitutes genuine Hamiltonian protection only if the physical system actually realizes the required interactions and occupies their gapped phase.

The two-dimensional toric code is not a self-correcting quantum memory. After a pair has been created, either pointlike anyon can move across the lattice without increasing the number of excitations.

The maximum energy encountered along a logical-string process therefore remains of order J_e or J_m . It does not increase with L .

At nonzero temperature, diffusion can complete a noncontractible loop. Consequently, increasing the lattice size without bound does not increase the passive lifetime without bound [Theory] [R143]; [R144]. The gap suppresses pair creation by making pairs energetically costly.

The gap does not localize pairs after they have been created.

In active syndrome extraction, hardware repeatedly measures the star and plaquette operators, usually using additional ancilla qubits and a gate circuit. Classical software compares measurement rounds, infers space-time error chains, and updates either a physical correction or a Pauli frame.

In this architecture, the checks need not appear as energy terms in the data-qubit Hamiltonian. Measurement errors add time as an additional decoding direction, so a single round with an apparently consistent syndrome is insufficient [R141]; [R145].

A static H_{TC} with thermally occupied energy levels is an analog many-body model. A circuit may prepare a toric-code wavefunction without making H_{TC} the equilibrium Hamiltonian of the hardware.

Repeated stabilizer measurement is active quantum error correction even when no four-body energy penalty exists. Evidence for a robust emergent $\mathbb{Z}_2$ topological phase in a material is stronger than evidence from preparing a state or implementing a short digital braid.

Eight defect spins assigned to the eight edges of a diagram remain eight defect spins unless the star interactions, plaquette interactions, and a many-body gap are shown to exist.

Passive and active strategies can be combined, but they have different error models: passive energy penalties modify transition energies, while active correction obtains syndrome information at the cost of control, measurement, and decoding resources. Both architectures may be described as toric-code implementations only if the relevant physical realization is specified.

Physical requirements for laboratory implementations

The ideal square-lattice Hamiltonian uses weight-four Pauli products, where weight counts the qubits acted on nontrivially. Candidate hardware usually supplies one- or two-body interactions. Implementing the commuting-projector model therefore needs direct multiqubit terms, perturbative effective terms, extra constraints, or digital sequences, each with its own energy scales and control errors.

For a defect array, assigning one defect spin to each edge is only the initial hardware specification. The star and plaquette constraints and a many-body gap must still be demonstrated [Proposal]. A diamond device with well-controlled single-spin coherence is physically distinct from this lattice model.

[Experiment] Satzinger and collaborators prepared and characterized a toric-code ground state using 31 superconducting qubits, measured a topological entanglement signature, and performed anyon operations [R125]. These results provide strong evidence that a programmable processor can synthesize and probe the state. They do not show that the processor's native equilibrium material realizes a passive toric-code Hamiltonian.

[Experiment] Repeated stabilizer extraction has been demonstrated in planar surface-code devices. Surface codes use boundaries rather than a periodic torus but retain the same local check algebra. Krinner and collaborators operated a distance-three, 17-qubit surface code through repeated correction cycles [R145]. This result demonstrates active syndrome extraction and decoding rather than thermally passive storage.

[Experiment] A trapped-ion processor has also prepared toric-code topological order using measurement and feed-forward under periodic connectivity [R146]. In this case, the preparation protocol supplied the required resource. Turning off the controls does not leave an autonomous topological material.

For defect-engineered crystals, the toric code is therefore a benchmark rather than an automatic consequence of arranging defects. A convincing realization would require spectroscopy or dynamical measurements establishing the intended effective terms, a separated low-energy manifold, nonlocal ground-state sectors, and controlled string excitations. Long coherence times for individual defects, entanglement among a few neighboring defects, or a software-defined stabilizer graph address different physical questions.

Common conceptual errors

- A stabilizer should not be identified with its projector. The operators A_v and B_p have eigenvalues ± 1 . The projectors onto their $+1$ eigenspaces are $(I + A_v)/2$ and $(I + B_p)/2$. This distinction affects the interpretation of Hamiltonian coefficients.
- The global stabilizer relations must be included in the constraint count. Treating all eight checks on eight qubits as independent would predict no remaining state. Because two products of checks equal the identity, only six constraints are independent, leaving two logical qubits.

- A vanishing syndrome does not prove that correction succeeded. Both a contractible, harmless loop and a noncontractible, damaging loop have no endpoints. Decoding must infer the homology class rather than merely cancel observed defects.
- An energy gap does not imply a self-correcting memory. The two-dimensional toric code has a finite pair-creation cost but no energy barrier that increases with system size [R143]; [R144]. Passive suppression is useful, but it does not provide indefinite thermal protection.
- Abelian braiding is not physically trivial. The mutual phase -1 is topological and measurable. Because it is a scalar rather than a matrix, however, it cannot implement the matrix-valued braids of a non-Abelian fusion space.
- Digital state preparation is not equivalent to emergent topological order. A gate sequence can produce the exact wavefunction and reproduce the anyon algebra. Emergent topological order additionally requires a local physical Hamiltonian for which that sector is robust low-energy physics.
- A torus demonstration should not be interpreted as a direct device layout. The torus provides mathematically simple boundary conditions, whereas laboratory devices usually use planar patches with boundaries. Boundaries alter the ground-state degeneracy and the geometry of logical operators while preserving the local star and plaquette algebra.
- A diamond defect array is not automatically a toric-code realization. Assigning one defect to each edge specifies only a hardware arrangement. Without four-body checks and a many-body gap, the crystal remains a collection of ordinary qubits.

Verification exercises

- **Commutation of star and plaquette checks.** A star operator A_v is a stabilizer check formed from Pauli X operators on the edges incident on vertex v , whereas a plaquette operator B_p is formed from Pauli Z operators around plaquette p . Checks of the same type commute because they use the same Pauli type. A star and a plaquette share either zero edges or two edges. Because X and Z anticommute on each shared edge, each overlap contributes an XZ minus sign. The number of such signs is even, so their product is $+1$, and the star and plaquette commute.
- **Number of logical qubits on the eight-edge torus.** The eight edges support $n = 8$ physical qubits. There are six independent stabilizer generators: three independent stars and three independent plaquettes. The remaining checks are constrained by the relations $\prod_v A_v = I$ and $\prod_p B_p = I$, where I is the identity operator. Thus the stabilizer rank is $r = 6$. For a stabilizer code, the number of encoded logical qubits is $k = n - r$, so $k = n - r = 8 - 6 = 2$. The ground-state subspace therefore contains four states.
- **Creation of e and m anyon pairs.** An anyon is a localized topological excitation identified by a violated stabilizer check. An open direct-lattice Z string, defined as a product of Pauli Z operators along a path on the original lattice, flips the star checks at the two endpoints and creates a pair of e anyons. An open dual-lattice X string, defined as a product of Pauli X operators along a path on the dual lattice, flips the endpoint plaquette checks and creates a pair of m anyons.

- **Failure of decoding based only on a vanished syndrome.** A syndrome is the set of violated stabilizer-check outcomes. After decoding, both a shrinkable residual string and a wrapping residual string can have an empty boundary and therefore produce no syndrome. A shrinkable string can be continuously contracted and corresponds to a stabilizer. A wrapping string follows a noncontractible cycle of the torus and is a logical operator. Consequently, a vanished syndrome does not by itself imply successful recovery.
- **Abelian character of the anyons.** Fusion specifies the total topological charge obtained by combining anyons. The fusion outcomes are unique, so each fusion space—the state space associated with a fixed set of anyons and fusion outcomes—is one-dimensional. Braiding therefore multiplies the state by a phase, such as -1 , rather than mixing a vector of fusion amplitudes. The anyons are Abelian even though the braiding phase is not $+1$.
- **Insufficiency of the many-body gap for self-correction.** A many-body gap is the energy separation between the ground-state subspace and the lowest excited states. A self-correcting memory would additionally require an error-process energy barrier that increases with the linear system size L . Once an anyon pair exists, either anyon can move without increasing the number of excitations. The energy barrier along a path that wraps around the system therefore remains of order J_e or J_m , where J_e and J_m set the excitation-energy scales, and does not grow with L .

Local commuting checks define a gapped subspace whose remaining labels are nonlocal. Open string operators create anyons at their endpoints.

Closed shrinkable strings are stabilizers, whereas closed wrapping strings are logical operators.

Crossings between string operators produce the braiding phase. A many-body gap alone does not provide a scalable finite-temperature memory.

Active extraction, in which stabilizer syndromes are repeatedly measured and decoded, is distinct from passive protection by a Hamiltonian. The one-dimensional fusion spaces establish that these anyons are Abelian.

The subsequent lattice model obtains related order from two-body honeycomb bonds rather than four-spin checks.

Sources

- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* 303, 2–30 (2003). DOI: [10.1016/S0003-4916\(02\)00018-0](https://doi.org/10.1016/S0003-4916(02)00018-0). arXiv: [quant-ph/9707021](https://arxiv.org/abs/quant-ph/9707021).
- [R141] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, “Topological quantum memory,” *Journal of Mathematical Physics* 43, 4452–4505 (2002). DOI: [10.1063/1.1499754](https://doi.org/10.1063/1.1499754). arXiv: [quant-ph/0110143](https://arxiv.org/abs/quant-ph/0110143).
- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* 51, 093512 (2010). DOI: [10.1063/1.3490195](https://doi.org/10.1063/1.3490195). arXiv: [1001.0344](https://arxiv.org/abs/1001.0344).
- [R143] R. Alicki, M. Fannes, and M. Horodecki, “On thermalization in Kitaev’s 2D model,” *Journal of Physics A: Mathematical and Theoretical* 42, 065303 (2009). DOI: [10.1088/1751-8113/42/6/065303](https://doi.org/10.1088/1751-8113/42/6/065303). arXiv: [0810.4584](https://arxiv.org/abs/0810.4584).
- [R144] B. M. Terhal, “Quantum error correction for quantum memories,” *Reviews of Modern Physics* 87, 307–346 (2015). DOI: [10.1103/RevModPhys.87.307](https://doi.org/10.1103/RevModPhys.87.307). arXiv: [1302.3428](https://arxiv.org/abs/1302.3428).

- [R145] S. Krinner et al., “Realizing repeated quantum error correction in a distance-three surface code,” *Nature* 605, 669–674 (2022). DOI: [10.1038/s41586-022-04566-8](https://doi.org/10.1038/s41586-022-04566-8). arXiv: [2112.03708](https://arxiv.org/abs/2112.03708).
 - [R125] K. J. Satzinger et al., “Realizing topologically ordered states on a quantum processor,” *Science* 374, 1237–1241 (2021). DOI: [10.1126/science.abi8378](https://doi.org/10.1126/science.abi8378). arXiv: [2104.01180](https://arxiv.org/abs/2104.01180).
 - [R146] M. Iqbal et al., “Topological order from measurements and feed-forward on a trapped ion quantum computer,” *Communications Physics* 7, 205 (2024). DOI: [10.1038/s42005-024-01698-3](https://doi.org/10.1038/s42005-024-01698-3). arXiv: [2302.01917](https://arxiv.org/abs/2302.01917).
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Chapter 17 — Bond-directional interactions on the honeycomb lattice

Place a spin-1/2 at every vertex of a honeycomb lattice. Each site touches three nearest-neighbor bonds, pointing along three distinct lattice directions. Call these bond directions x , y , and z .

Each bond couples a single spin component, selected by the bond direction. The two spins on an x -type bond interact through their x components only. A y -type bond couples only the y components, and a z -type bond only the z components. Every spin therefore takes part in three interactions at once, each involving a different component of that spin.

No single spin orientation satisfies all three bonds at once: fixing one component to please one bond constrains the others. For a classical vector the three demands compete directly. For a quantum spin the competition is sharper, because the x , y , and z components do not commute and cannot take definite values simultaneously at the same site.

An interaction whose selected spin component follows the bond direction is called a compass interaction. On each nearest-neighbor bond of type $\alpha \in \{x, y, z\}$, the energy contains $\sigma_i^\alpha \sigma_j^\alpha$ and no other spin components. Here, σ_i^α denotes the Pauli operator for component α at site i . This interaction differs from the Heisenberg interaction, which couples every spin component on a bond.

Conserved flux on a single hexagonal plaquette

The three bond demands generally cannot be minimized independently; this competition is called frustration. Frustration here has a precise consequence: around each hexagonal plaquette, a product of six suitably chosen spin operators forms a loop operator that the Hamiltonian leaves unchanged.

Label the six sites of one hexagon clockwise by 1, 2, ..., 6. Let the bond types around its perimeter be

```
1
y / \ z
6 2
x | | x
5 3
z \ / y
4
```

```
perimeter sequence: (1,2)=z, (2,3)=x, (3,4)=y,
(4,5)=z, (5,6)=x, (6,1)=y
```

Each vertex also has one bond extending away from the hexagon. The type of this outward bond is the missing member of $\{x, y, z\}$ at that vertex. The outward bond types at sites 1 through 6 are therefore x, y, z, x, y, z .

Place one spin-1/2 at each vertex. Write σ_i^α for the dimensionless Pauli operator of component $\alpha \in \{x, y, z\}$ at site i . The corresponding plaquette loop operator is

$$W_p = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z.$$

Pauli operators on different sites commute, and each Pauli operator squares to the identity. Hence

$$W_p^2 = 1,$$

A measurement of W_p therefore returns one of the eigenvalues $w_p = +1$ or $w_p = -1$. This binary eigenvalue is called the flux through the hexagon. It describes a collective pattern of the six spins; no applied magnetic field threads the plaquette.

Flux conservation follows from how the bond terms move past the loop operator. Consider the z -bond term on sites 1 and 2,

$$K_{12} = \sigma_1^z \sigma_2^z.$$

At site 1, σ^z anticommutes with the σ^x factor in W_p , contributing one minus sign. At site 2, it anticommutes with the σ^y factor, contributing a second. Moving K_{12} through W_p therefore produces two minus signs, which cancel:

$$K_{12} W_p = (-1)^2 W_p K_{12} = W_p K_{12}.$$

Every perimeter bond gives the same pairwise cancellation. An outward bond carries the same spin component as the W_p factor at its shared endpoint, so it commutes with that factor and touches none of the other five. Every bond term therefore commutes with W_p , and so does their sum:

$$[H, W_p] = 0.$$

Every plaquette flux is conserved in the ideal model [Ro17]. The bond coloring is part of the Hamiltonian rather than a graphical convention. If an x coupling is replaced by an isotropic coupling, the cancellation of two anticommutation signs may no longer apply.

A single spin operator flips the fluxes of the two plaquettes sharing the corresponding bond. For example, σ_i^α anticommutes with the two plaquette operators adjacent to the α bond leaving site i , and commutes with all the others.

Acting with σ_i^α on a flux eigenstate therefore reverses the signs of exactly those two w_p eigenvalues. On a closed periodic lattice, where every bond borders two plaquettes, flux excitations always appear in pairs. A short string operator in the toric code creates its two anyons the same way, at the two ends of the string.

Bond-dependent Kitaev Hamiltonian

Let $\langle ij \rangle_\alpha$ denote a nearest-neighbor bond of type α . The Kitaev honeycomb Hamiltonian is

$$H_K = -J_x \sum_{\langle ij \rangle_x} \sigma_i^x \sigma_j^x - J_y \sum_{\langle ij \rangle_y} \sigma_i^y \sigma_j^y - J_z \sum_{\langle ij \rangle_z} \sigma_i^z \sigma_j^z.$$

The couplings J_x, J_y, J_z are real numbers with units of energy. The Pauli products are dimensionless, so each term in the Hamiltonian, and H_K itself, carries units of energy.

The standard phase diagram takes $J_\alpha \geq 0$. Spin rotations or gauge choices map many other sign patterns onto this case. Open boundaries and finite samples need separate analysis.

This is the compass interaction written out: each bond selects its own component, unlike the isotropic Heisenberg form $J \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j$, which couples all three components on every bond.

The Hamiltonian contains only two-body bond terms, with no direct six-spin plaquette coupling. The loop operator is conserved through the algebra of those bond terms.

All the W_p commute with H_K and with one another, so each energy eigenstate carries a definite set of eigenvalues $\{w_p\}$. One complete assignment of these eigenvalues is called a flux sector.

On a torus, two further labels record the gauge flux around the two noncontractible directions; a Wilson loop is a loop operator wrapping such a direction. The plaquette signs then obey global constraints, which exact state counting must respect. For the local physics, what matters is that $\{w_p\}$ is conserved.

For the translationally invariant model, the ground state has $w_p = +1$ on every plaquette, the sector conventionally called flux-free [R017]; [R148]. [Theory] The term “flux-free” specifies the plaquette eigenvalues; it does not imply a trivial excitation spectrum. The remaining spectrum can be gapless or topological.

Chapter 16 built topology from explicit four-spin stabilizers. A defect array is more likely to offer pairwise exchange, dipolar, or mediated couplings. The honeycomb model shows that strictly local two-body spin interactions can still be enough: their collective effect produces flux sectors, fractionalized quasiparticles, and topological order. [Theory] This holds exactly for this special Hamiltonian. It does not imply that an arbitrary honeycomb-shaped array is topological, so such an array needs its own evidence before any topological claim follows [R017]; [R148].

Majorana-fermion representation and gauge constraint

Fix the flux sector, and the interacting spin problem becomes free fermions hopping through a fixed pattern of signs. The auxiliary operators that make this possible are Hermitian, square to one, and each carries half the degrees of freedom of one ordinary complex fermion.

At each site, introduce four operators,

$$b_i^x, \quad b_i^y, \quad b_i^z, \quad c_i,$$

which anticommute when distinct and square to one. Represent the physical spin operators as

$$\sigma_i^\alpha = ib_i^\alpha c_i.$$

An operator γ with

$$\gamma^\dagger = \gamma, \quad \gamma^2 = 1$$

is called a Majorana operator. Here these operators are calculational variables that solve the many-spin problem; they are not electrons split into physical halves.

Four Majorana operators describe a larger Hilbert space than one spin-1/2. The constraint

$$D_i = b_i^x b_i^y b_i^z c_i = +1.$$

selects the physical subspace. Changing the signs of all four Majorana operators at one site leaves every physical spin operator unchanged. This redundancy is a local $\mathbb{Z}_2$ gauge freedom, meaning that distinct auxiliary-variable descriptions correspond to the same physical spin state. It is not an additional experimentally controllable degree of freedom.

Orient every α bond from site i toward site j and define

$$u_{ij} = ib_i^\alpha b_j^\alpha, \quad u_{ij} = \pm 1.$$

One bond term then reads

$$-J_\alpha \sigma_i^\alpha \sigma_j^\alpha = iJ_\alpha u_{ij} c_i c_j,$$

with the sign set by the chosen bond orientation. Summing over bonds,

$$H_K = i \sum_{\langle ij \rangle_\alpha} J_\alpha u_{ij} c_i c_j.$$

Every u_{ij} commutes with this ideal Hamiltonian. With all the u_{ij} signs fixed, H_K is quadratic in the c Majorana operators: a free-fermion hopping problem that diagonalizes directly.

Multiplying the u_{ij} variables around a plaquette reproduces W_p , up to the fixed bond-orientation convention. Each u_{ij} sign separately depends on the gauge choice; the product around the loop does not, and that product is the physical flux.

Physical spins live on the lattice sites. The b and c Majorana operators act in an enlarged Hilbert space, and only states satisfying $D_i = +1$ at every site correspond to spin states. A fixed- u solution computed before this projection still contains unphysical states. A flux eigenvalue is likewise a collective label of the many-body state.

This machinery gives delocalized matter bands in an enlarged representation. These properties do not by themselves demonstrate a localized, independently controllable zero mode in a material, and they do not by themselves define an encoded qubit: a localized zero mode belongs to the gapped phase with vortices, after projection onto the physical subspace described below.

The representation separates two kinds of excitation:

- Fluxes, which change the loop eigenvalues.
- Matter Majoranas, whose allowed energies depend on the flux background.

Physical spin excitations combine the two sectors under the constraint. Either sector on its own, with the constraint ignored, describes auxiliary variables outside the spin Hilbert space, not a physical excitation of the spins.

Gapless and gapped regions of the phase diagram

Work with a two-site unit cell, with sublattices A and B . Let $\mathbf{a}_1$ and $\mathbf{a}_2$ be the lattice translations from a chosen z bond to its neighboring x and y bonds. In the flux-free gauge, the matter-Majorana excitation energy is

$$E(\mathbf{k}) = 2|f(\mathbf{k})|, \quad f(\mathbf{k}) = J_z + J_x e^{i\mathbf{k} \cdot \mathbf{a}_1} + J_y e^{i\mathbf{k} \cdot \mathbf{a}_2},$$

Here $\mathbf{k}$ is the crystal momentum, the quantum number labeling eigenstates of lattice translations. The overall factor of 2 follows from the Hamiltonian normalization; whether the energy can reach zero does not.

The function f adds three complex numbers of lengths J_x, J_y, J_z . They can cancel to zero exactly when those lengths form a triangle. The gapless B phase is therefore the region

$$J_x \leq J_y + J_z, \quad J_y \leq J_z + J_x, \quad J_z \leq J_x + J_y.$$

B is Kitaev's name for this triangle region. The isotropic point $J_x = J_y = J_z$ lies inside it; its spectrum has two Majorana Dirac cones, points where the bands touch with linear dispersion.

Outside the triangle lie three gapped regions:

$$\begin{aligned}
A_x : J_x &> J_y + J_z, \\
A_y : J_y &> J_z + J_x, \\
A_z : J_z &> J_x + J_y.
\end{aligned}$$

[Theory] At zero field, the A phases possess Abelian topological order. In a strongly anisotropic limit, their low-energy theory becomes toric-code-like [Ro17]; [R148]. The unperturbed B phase is gapless and therefore is not yet the gapped non-Abelian phase often associated with the model.

Violating any one triangle inequality keeps $f(\mathbf{k})$ nonzero everywhere, so the excitation spectrum stays gapped and the Dirac cones are gone.

Perturbative generation of an effective loop interaction

Take the strongly anisotropic regime $J_z \gg J_x, J_y > 0$. One z bond contributes $-J_z \sigma_i^z \sigma_j^z$, with aligned states at energy $-J_z$ and anti-aligned states at $+J_z$. The two aligned states sit $2J_z$ below the excited pair, so each aligned z dimer behaves as one effective spin.

One x - or y -bond interaction kicks dimers out of this low-energy subspace. The first process that disturbs dimers and returns all of them, while tracing a closed loop, appears at fourth order. Degenerate perturbation theory then gives, in the standard convention,

$$H_{\text{eff}} = \text{constant} - J_{\text{eff}} \sum_p \widetilde{W}_p + \dots, \quad J_{\text{eff}} = \frac{J_x^2 J_y^2}{16 J_z^3},$$

where $\widetilde{W}_p$ is a four-effective-spin plaquette operator on the contracted lattice [Ro17]. The dimensions are consistent because

$$[J_{\text{eff}}] = \frac{(\text{energy})^4}{(\text{energy})^3} = \text{energy}.$$

Thus no microscopic four-body force is required: repeated virtual excursions into higher-energy dimer states generate the four-spin interaction from a starting Hamiltonian with only two-body terms. But when $J_x, J_y \ll J_z$, this induced topological scale sits far below the microscopic coupling.

At only moderate anisotropy, higher-order terms and processes that leave dimers broken stop being small, and the toric-code description loses its controlled perturbative footing.

Time-reversal breaking and Ising topological order

A weak Zeeman term breaks time-reversal symmetry:

$$H_h = - \sum_i (h_x \sigma_i^x + h_y \sigma_i^y + h_z \sigma_i^z),$$

Each h_α carries units of energy; for an applied magnetic field it absorbs the g factor and Bohr magneton. When the product $h_x h_y h_z$ is nonzero, third-order perturbation theory produces an effective three-spin interaction with scale

$$\kappa \sim \frac{h_x h_y h_z}{J^2},$$

Here J stands for a representative Kitaev exchange, and $[\kappa] = (\text{energy})^3/(\text{energy})^2 = \text{energy}$. This interaction gaps the Dirac cones.

[Theory] In the weak-field effective model, the gapped Majorana band carries Chern number $\nu = +1$ or -1 , an integer invariant of a two-dimensional band structure. Vortices in this phase bind Majorana zero modes, giving non-Abelian Ising topological order [R017]; [R148]. A generic microscopic magnetic field also destroys exact flux conservation. Consequently, the full finite-field problem is not the zero-field free-Majorana solution with a larger gap: a generic field mixes flux sectors, so the gapped phase must be analyzed with its own flux-mixing dynamics.

“Ising” names three different things, kept separate here:

- Conventional Ising magnetic order breaks a $\mathbb{Z}_2$ symmetry and shows up in a local order parameter such as magnetization. The gapped honeycomb phase has no such order.
- Ising topological order has topological charges 1 (vacuum), σ (vortex), and ψ (fermion), with

$$\sigma \times \sigma = 1 + \psi, \quad d_\sigma = \sqrt{2}.$$

Here, the fusion rule states that two σ anyons can combine into either 1 or ψ , and d_σ is the quantum dimension of σ . The associated degeneracy and braiding act in a nonlocal fusion space. No local magnetization is required.

- Fibonacci topological order instead has charges 1 and τ , with

$$\tau \times \tau = 1 + \tau, \quad d_\tau = \varphi = \frac{1 + \sqrt{5}}{2}.$$

The two fusion rules are not equivalent under relabeling: no renaming of charges turns one into the other. Braiding Ising anyons alone does not generate a dense set of single-qubit unitaries.

In standard encodings, Ising braiding provides Clifford operations; universal computation needs one further non-topological resource. Fibonacci braiding, developed in Chapter 15, is universal on its own.

The field-gapped honeycomb model thus traces a rigorous path from two-body interactions to Ising-type non-Abelian anyons. Its fusion rules, quantum dimensions, and braid representations differ from the Fibonacci data, so it belongs to a different topological phase.

Experimental realizations and limitations

The leading microscopic proposal starts from magnetic ions whose spin and orbital motion are locked together by strong spin-orbit coupling, sitting in edge-sharing octahedra. Electron hopping along the two exchange paths interferes destructively for ordinary Heisenberg exchange, leaving the bond-directional Kitaev coupling dominant.

[Proposal] Jackeli and Khaliullin derived this interference mechanism for certain Mott insulators [R147]. A Mott insulator is a material in which electron-electron interactions prevent conduction despite a band structure that would otherwise permit it. The mechanism is a materials-design principle, not a proof that every edge-sharing honeycomb realizes H_K .

Real candidates, including honeycomb iridates and α - RuCl_3 , add further ingredients on top of the Kitaev coupling: Heisenberg exchange, symmetric off-diagonal Γ terms that mix spin components, longer-range couplings, lattice distortions, phonons, stacking faults, and interlayer coupling [R149]; [R150]. Here, Γ terms are symmetric exchange interactions that couple different spin components. α - RuCl_3 develops zigzag magnetic order near 7 K at zero field, with details depending on sample and stacking history [R151]. [Experiment] This local symmetry-breaking order directly demonstrates that the clean zero-field sample is not the ideal Kitaev spin liquid.

Neutron and Raman measurements do observe broad continua and field-dependent spectra consistent with sizable Kitaev interactions and fractionalized descriptions [R149]; [R150]; [R151]. [Experiment] Compatibility does not constitute unique identification, because magnons, disorder, and multiparticle continua can occupy overlapping frequency ranges.

A sharper predicted signature is heat carried by a single chiral Majorana edge mode, a boundary excitation that travels in one direction only. The ideal two-dimensional theory predicts the transverse thermal conductance

$$\frac{\kappa_{xy}}{T} = \frac{1}{2} \frac{\pi^2 k_B^2}{3h_P} = \frac{\pi k_B^2}{12\hbar},$$

where κ_{xy} is the transverse thermal conductance, T is temperature, k_B is Boltzmann's constant, h_P is Planck's constant, and $\hbar = h_P/(2\pi)$. The units are W K^{-2} .

Kasahara and collaborators reported a plateau-like signal near half quantization in field-driven α - RuCl_3 , interpreting it as the Majorana edge contribution [R152]. [Experiment] Subsequent measurements found strong sample dependence and substantial phonon or bosonic contributions [R150]; [R153].

A 2026 acoustic study measured phonon Hall viscosity directly and assigned a substantial part of the intrinsic thermal Hall response to phonons [R154]. [Experiment] As of 2026, the existence of a field-induced non-Abelian Kitaev phase in α - RuCl_3 remains actively disputed rather than experimentally established [R150]; [R154].

No cited experiment has tuned engineered color-center defects to this compass Hamiltonian and demonstrated intrinsic honeycomb-model Ising order there; none has obtained Fibonacci order from such a system either.

For defect engineering, the model therefore sets a target Hamiltonian and a checklist. Geometry, pairwise coupling, and a suggestive continuum are insufficient by themselves: a convincing realization must pin down the interaction tensor, the many-body gap, the flux structure, and the topological response.

Common conceptual and experimental errors

- **Reading the Kitaev Hamiltonian off honeycomb geometry.** A graph records which sites connect. The Kitaev Hamiltonian further assigns one spin component to every edge and fixes the competing coupling strengths. An isotropic Heisenberg magnet drawn on the same hexagonal graph obeys a different Hamiltonian.
- **Mistaking the auxiliary Majorana operators for physical zero modes.** The c_i operators form a delocalized matter band inside an enlarged representation with gauge redundancy; a parton is such an auxiliary degree of freedom used to represent a physical spin. Only in the gapped topological regime, at a vortex, and after projecting onto the physical-state constraint does a localized zero mode appear. A parton band structure computed without that projection is not yet a physical excitation spectrum.
- **Expecting control fields to respect flux conservation.** Only H_K and selected integrable extensions conserve every flux exactly. Generic Zeeman, Heisenberg, Γ , disorder, and drive terms mix flux sectors. A gapped topological phase can survive small perturbations, yet the exact labeling of every eigenstate by fixed w_p values no longer holds.
- **Calling every non-Abelian phase Fibonacci.** Ising and Fibonacci anyons both have multidimensional fusion spaces, but they differ in fusion rules, quantum dimensions, braid representations, and computational power. The non-Abelian phase of the honeycomb model is Ising-type; labeling it Fibonacci discards the data that define it.
- **Confusing a programmed simulation with intrinsic emergence.** A gate array can encode the Majorana Hamiltonian, prepare a flux sector, and reproduce the expected spectrum. When the processor's own equilibrium Hamiltonian lacks the phase and its protection, that procedure remains a digital emulation carried out by ordinary physical qubits. It tests the model without showing that intrinsic anyons inhabited the processor between gate pulses.
- **Treating an exactly solvable point as a complete architecture.** Solvability pins down the phase precisely. A working architecture additionally needs sector initialization, cooling below the many-body gap, controlled creation and motion of excitations, fusion readout, disorder tolerance, and scalable boundaries and defects. Exact solvability does not supply these engineering functions; each of them remains to be built on top of solvability.

Verification exercises

- **Plaquette-flux conservation.** Move each bond term past W_p : a perimeter bond anticommutes with two of its factors, and the two minus signs cancel. An outward bond shares its spin component with the plaquette factor at their common site, so it commutes. Every bond term therefore commutes with W_p .
- **Gauge-dependent and physical quantities.** Each u_{ij} separately depends on the gauge choice. The product around a plaquette, which is the flux W_p , does not, and that product is the physical quantity.
- **Condition for a gapless zero-field matter spectrum.** Since $E(\mathbf{k}) = 2|f(\mathbf{k})|$, the spectrum reaches zero exactly when the three complex terms in f can cancel. That cancellation is possible exactly when J_x, J_y, J_z satisfy the triangle inequalities, the B region containing the isotropic point.

- **Why the ungapped B phase cannot itself host protected non-Abelian anyons.** The bulk of the unperturbed B phase is gapless. A weak time-reversal-breaking perturbation gaps its Dirac Majoranas and yields Ising topological order, but the ungapped bulk lacks the required protection.
- **Honeycomb non-Abelian order against Fibonacci order.** Honeycomb vortices fuse as $\sigma \times \sigma = 1 + \psi$ with $d_\sigma = \sqrt{2}$; Fibonacci charges fuse as $\tau \times \tau = 1 + \tau$ with $d_\tau = \varphi$. Braiding the two sets supports different computational operations.
- **Small induced scale in the anisotropic limit.** Virtual transitions through excited dimer states generate the loop interaction, but its scale $J_x^2 J_y^2 / (16 J_z^3)$ lies far below the microscopic exchange when $J_x, J_y \ll J_z$.

The chapter leaves a two-body compass model with conserved fluxes, free fermions within each flux sector, and a field-induced gapped phase with Ising anyons rather than Fibonacci anyons.

Sources

- [R017] A. Kitaev, “Anyons in an exactly solved model and beyond,” *Annals of Physics* 321, 2–111 (2006). DOI: [10.1016/j.aop.2005.10.005](https://doi.org/10.1016/j.aop.2005.10.005). arXiv: [cond-mat/0506438](https://arxiv.org/abs/cond-mat/0506438).
 - [R147] G. Jackeli and G. Khaliullin, “Mott Insulators in the Strong Spin-Orbit Coupling Limit: From Heisenberg to a Quantum Compass and Kitaev Models,” *Physical Review Letters* 102, 017205 (2009). DOI: [10.1103/PhysRevLett.102.017205](https://doi.org/10.1103/PhysRevLett.102.017205). arXiv: [0809.4658](https://arxiv.org/abs/0809.4658).
 - [R148] M. Hermanns, I. Kimchi, and J. Knolle, “Physics of the Kitaev Model: Fractionalization, Dynamic Correlations, and Material Connections,” *Annual Review of Condensed Matter Physics* 9, 17–33 (2018). DOI: [10.1146/annurev-conmatphys-033117-053934](https://doi.org/10.1146/annurev-conmatphys-033117-053934). arXiv: [1705.01740](https://arxiv.org/abs/1705.01740).
 - [R149] H. Takagi, T. Takayama, G. Jackeli, G. Khaliullin, and S. E. Nagler, “Concept and realization of Kitaev quantum spin liquids,” *Nature Reviews Physics* 1, 264–280 (2019). DOI: [10.1038/s42254-019-0038-2](https://doi.org/10.1038/s42254-019-0038-2). arXiv: [1903.08081](https://arxiv.org/abs/1903.08081).
 - [R150] Y. Matsuda, T. Shibauchi, and H.-Y. Kee, “Kitaev quantum spin liquids,” *Reviews of Modern Physics* 97, 045003 (2025). DOI: [10.1103/3m4m-3v59](https://doi.org/10.1103/3m4m-3v59). Stable URL: [APS](https://arxiv.org/abs/APS).
 - [R151] A. Banerjee et al., “Neutron scattering in the proximate quantum spin liquid α -RuCl₃,” *Science* 356, 1055–1059 (2017). DOI: [10.1126/science.aah6015](https://doi.org/10.1126/science.aah6015). arXiv: [1702.01688](https://arxiv.org/abs/1702.01688).
 - [R152] Y. Kasahara et al., “Majorana quantization and half-integer thermal quantum Hall effect in a Kitaev spin liquid,” *Nature* 559, 227–231 (2018). DOI: [10.1038/s41586-018-0274-0](https://doi.org/10.1038/s41586-018-0274-0). arXiv: [1805.05022](https://arxiv.org/abs/1805.05022).
 - [R153] É. Lefrançois et al., “Evidence of a Phonon Hall Effect in the Kitaev Spin Liquid Candidate α -RuCl₃,” *Physical Review X* 12, 021025 (2022). DOI: [10.1103/PhysRevX.12.021025](https://doi.org/10.1103/PhysRevX.12.021025). arXiv: [2111.05493](https://arxiv.org/abs/2111.05493).
 - [R154] A. Shragai, E. Horsley, S. Kim, Y.-J. Kim, and B. J. Ramshaw, “Phonon Hall viscosity and the intrinsic thermal Hall effect of α -RuCl₃,” *Nature* 652, 1166–1172 (2026). DOI: [10.1038/s41586-026-10420-y](https://doi.org/10.1038/s41586-026-10420-y). Stable URL: [Nature](https://arxiv.org/abs/Nature).
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Chapter 18 — Fusion constraints in string-net models

Picture a honeycomb lattice with every edge either blank or carrying one string type. Three edges meet at each vertex, so the lattice is trivalent. A local branching rule lists which triples of edge labels may meet at a vertex.

In the rules used below, an allowed triple may mix two edges of one type with one of another, or use three edges of the same type. A vertex joining one string to two blank edges is forbidden.

No single edge-label pattern is the physical state. The physical state is a quantum superposition of many legal patterns, each entering with its own amplitude. Local moves create a small closed loop, carry it across a vertex by recoupling the neighboring strings, and absorb it into the surrounding network.

A Levin–Wen model is a lattice Hamiltonian built from three ingredients: a finite set of edge labels on a trivalent lattice, a branching rule for the triples meeting at each vertex, and a second local rule that inserts contractible loops and recouples them into the network. A contractible loop shrinks continuously to a point on the surface. The branching rule alone cannot produce a topological phase; the loop rule is also required.

Allowed trivalent-vertex label combinations

The smallest label set used here comes from Fibonacci fusion. The blank edge is the vacuum label 1 ; the string label is τ . Both labels are self-dual: reversing a string's orientation leaves its label unchanged. Two labels combine according to

$$1 \times a = a, \quad \tau \times \tau = 1 + \tau,$$

with a equal to 1 or τ . Fusion lists the possible outcomes of combining two labels. The plus sign separates two distinct allowed channels; it does not place two particles side by side.

On a trivalent lattice, the allowed unordered label triples are

$$(1, 1, 1), \quad (1, \tau, \tau), \quad (\tau, \tau, \tau),$$

with all permutations included. A vertex with exactly one τ label is forbidden. The triple (τ, τ, τ) lets two incoming τ strings continue as τ in the third leg.

This list is the branching rule: a local constraint on the three labels at each vertex. The constraint alone does not make a topological phase.

Dropping the triple (τ, τ, τ) leaves an Abelian rule: two τ strings may meet only through the vacuum channel. The loop algebra of that truncated rule cannot reproduce Fibonacci fusion.

Superpositions of admissible string-net configurations

The branching rule throws out configurations with forbidden vertices. It leaves open which superposition of the remaining legal configurations forms the ground state.

The ground state is a coherent superposition of many legal configurations: the relative amplitudes and phases between configurations carry physical meaning.

Local moves create a small closed loop, carry it across a vertex, and absorb it into the network. Repeated moves build closed, branching configurations on every length scale.

A string-net is a fluctuating superposition of this kind. Levin and Wen showed that such a wavefunction arises as the exact ground state of a Hamiltonian built from commuting local projectors [Ro18]. [Theory]

The toric code is the simplest member of this family, with Abelian string rules and recoupling amplitudes of ± 1 . Here strings branch, and the recoupling amplitudes take more general values.

Each edge label is a microscopic basis state of the lattice Hilbert space, and each closed labeled network contributes one component to the many-body wavefunction.

An endpoint or ribbon excitation counts as an emergent quasiparticle when no local equivalence removes it. Drawing an endpoint or ribbon on paper does not by itself create a defect-center spin or a digitally emulated anyon.

Equivalent representations on a single hexagonal plaquette

Isolate one hexagonal plaquette and set all six outward legs to vacuum 1. The branching rule then permits two boundary configurations:

```
|0> : six boundary edges are 1
|R> : six boundary edges are  $\tau$ , forming a ring
```

Growing a τ loop inside $|0\rangle$ and expanding it onto the plaquette boundary gives $|R\rangle$, because $\tau \times 1 = \tau$. Call this loop-insertion operator B^τ . Fusing a second τ loop into the result follows $\tau \times \tau = 1 + \tau$, so

$$B^\tau|0\rangle = |R\rangle, \quad B^\tau|R\rangle = |0\rangle + |R\rangle.$$

In the ordered basis $(|0\rangle, |R\rangle)$ this operator reads

$$B^\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

With the golden ratio

$$\varphi = \frac{1 + \sqrt{5}}{2}.$$

the label weights, called quantum dimensions, are $d_1 = 1$ and $d_\tau = \varphi$. Each weight is a positive algebraic number attached to its label type. Their squared total is

$$\mathcal{D}^2 = d_1^2 + d_\tau^2 = 1 + \varphi^2 = \varphi + 2.$$

This operation inserts and fuses a loop according to the fusion algebra: the second tau loop splits into the vacuum and tau channels, so the empty and ring configurations mix. It is a correlated loop insertion followed by fusion, which a flip of six edge bits with fitted coefficients misses.

Once any outward leg carries τ , further intermediate fusion channels open and the two-dimensional matrix no longer suffices. Moving between those channels uses an F -move, a change of basis between different orders of fusion.

Loop-weight factors

The empty network alone does not satisfy the plaquette rule. The plaquette operation averages over loop types, weighted by their quantum dimensions:

$$B_p = \frac{1}{\mathcal{D}^2} \sum_{s \in \{1, \tau\}} d_s B_p^s = \frac{I + \varphi B_p^\tau}{\varphi + 2},$$

Here B_p^s inserts a loop of type s into plaquette p , with $B_p^1 = I$, where I is the identity. The weighted average is the plaquette projector: it selects the superposition whose amplitudes satisfy the input fusion data, with the vacuum loop entering with unit weight and the tau loop entering with the golden-ratio weight.

In the two-state sector,

$$B_p = \frac{1}{\varphi + 2} \begin{pmatrix} 1 & \varphi \\ \varphi & \varphi^2 \end{pmatrix} = |g\rangle\langle g|,$$

with

$$|g\rangle = \frac{|0\rangle + \varphi|R\rangle}{\sqrt{\varphi + 2}}.$$

This calculation fixes three facts: $B_p^2 = B_p$, so B_p projects; the selected ground-state component combines the empty network with the ring; and the relative amplitude φ comes from the input labels. Loop insertion is always followed by fusion and coherent recoupling.

These two states omit the extra channels of that case. Carrying an inserted loop through such a vertex repartitions three successive fusions, and that change of fusion basis is an F -move.

Equal weights in place of d_s generally destroy idempotence: in the two-state example the empty network and the ring would no longer span the image of a rank-one projector.

Edge-label assignments

The single-hexagon construction extends to an oriented trivalent lattice, commonly a honeycomb. Orientation puts an arrow on each edge; in a general theory the arrow distinguishes a label from its dual.

Each oriented edge e carries a label i_e from a finite set $I = \{0, 1, \dots\}$. The label 0, also written 1, is the vacuum.

Reversing an edge arrow replaces its label i by the dual i^* , the same string type read in the opposite direction. Fibonacci labels are self-dual, so the arrow reversal leaves them unchanged. For a general label set the i against i^* distinction is needed.

A basis vector assigns a label to every edge:

$$|\{i_e\}\rangle = \bigotimes_e |i_e\rangle.$$

The fusion multiplicity N_{ij}^k counts the independent channels in which i and j combine into k . The models used here are multiplicity-free, so each N_{ij}^k is zero or one.

With labels i, j, k suitably oriented, a trivalent vertex is allowed when $N_{ij}^{k*} > 0$. Multiplicities above one would need a further vertex-channel index to tell the independent channels apart; omitting that index uses the multiplicity-free assumption.

The quantum dimension $d_i > 0$ is the positive number satisfying

$$d_i d_j = \sum_k N_{ij}^k d_k.$$

For Fibonacci fusion this gives $d_\tau^2 = 1 + d_\tau$, solved by φ : the same weights that entered the hexagon loop average.

The honeycomb edge labels are the microscopic basis of the many-body Hilbert space, rather than drawings overlaid on separate physical degrees of freedom.

Vertex projectors

Let Q_v act diagonally in the edge-label basis:

$$Q_v|i, j, k\rangle = \begin{cases} |i, j, k\rangle, & N_{ij}^{k*} > 0, \\ 0, & N_{ij}^{k*} = 0. \end{cases}$$

Its eigenvalues are zero or one, so $Q_v^2 = Q_v$: it preserves allowed triples and annihilates forbidden ones. A state annihilated by Q_v violates the local fusion rule at vertex v . Used as a Hamiltonian term, Q_v is the vertex projector enforcing the branching rule. In the Hamiltonian below, changing one satisfied vertex from eigenvalue one to zero raises the energy by J_v , counted before any accompanying plaquette violations.

Writing the vertex energy as the penalty $J_v(1 - Q_v)$ instead of $-J_vQ_v$ shifts all energies by a constant and leaves dynamics and eigenstates unchanged.

Without Q_v , illegal triples such as a vertex with a single τ stay in the low-energy space, and the loop operators that follow no longer act within a fusion-consistent subspace.

Fusion-order transformations

Three labels admit two orders of pairwise fusion:

$$((a \times b) \rightarrow x) \times c \rightarrow d \quad \longleftrightarrow \quad a \times ((b \times c) \rightarrow y) \rightarrow d.$$

The unitary basis change between the two orders is

$$|((ab)xc)d\rangle = \sum_y [F_d^{abc}]_{xy} |a(bc)y; d\rangle.$$

Here a, b, c, d are outer labels and x and y are the allowed intermediate channels. The coefficients $[F_d^{abc}]_{xy}$ are the matrix elements of an F -matrix. The F -matrices obey the pentagon equation.

The pentagon equation demands that two different recoupling sequences around a five-leg fusion tree give the same transformation. This consistency is what lets neighboring plaquette operations commute [\[Ro18\]](#); [\[R155\]](#).

[Theory]

In the usual basis convention, where the F -matrices are unitary, the only nontrivial two-dimensional Fibonacci matrix is

$$F_{\tau}^{\tau\tau\tau} = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix},$$

with rows and columns ordered by the intermediate channels $(1, \tau)$. The matrix is real, symmetric, and unitary. Gauge choices move its signs and phases around without changing measurable topological data.

Carrying a τ loop through a vertex whose outward legs already carry τ uses exactly these recoupling amplitudes. The isolated two-state hexagon calculation extends to sectors with extra fusion channels only through this matrix.

Without the pentagon equation, neighboring B_p operators need not commute, and the model leaves the commuting-projector family.

Action of a loop operator on a string-net configuration

For each string type s , the operator B_p^s runs this graphical procedure:

- Insert a closed loop of type s just inside plaquette p .
- Fuse that loop into the boundary edges.
- Apply an F -move at each boundary vertex to recouple the inserted loop through the existing network.
- Sum coherently over all allowed intermediate fusion channels.

On a hexagonal plaquette the procedure applies six local recouplings. The result is no simple product of six Pauli matrices: its matrix elements depend on the labels of the legs extending outward from the plaquette.

The loop operators multiply like the fusion algebra:

$$B_p^r B_p^s = \sum_t N_{rs}^t B_p^t.$$

The weighted sum

$$B_p = \frac{1}{\mathcal{D}^2} \sum_s d_s B_p^s, \quad \mathcal{D} = \sqrt{\sum_s d_s^2},$$

is a projector: the quantum-dimension identities give $B_p^2 = B_p$, the pentagon equation gives $[B_p, B_{p'}] = 0$ even when p and p' are adjacent plaquettes, and the branching plus recoupling data give $[Q_v, B_p] = 0$ [Ro18]; [R155].

The weights come from a short algebraic calculation. Define the formal weighted loop

$$\Omega = \sum_s d_s s.$$

Fusing it with a label r gives

$$r \times \Omega = \sum_{s,t} d_s N_{rs}^t t = d_r \Omega.$$

Consequently, fusing two weighted loops gives $\Omega \times \Omega = \mathcal{D}^2 \Omega$. Dividing loop insertion by $\mathcal{D}^2$ cancels the extra total-dimension-squared factor and gives an idempotent operator. The plaquette projector is a normalized projector onto vacuum flux.

Hamiltonian enforcement of vertex and plaquette constraints

Index vertices by v and plaquettes by p . The dimensionless projector Q_v accepts allowed label triples at vertex v ; the dimensionless projector B_p enforces the plaquette recoupling rule at p . The positive energy scales J_v and J_p set the cost of violating each rule. The many-body Hamiltonian is

$$H = -J_v \sum_v Q_v - J_p \sum_p B_p.$$

Every term in H carries units of energy. The common choice $J_v = J_p = 1$ fixes those units.

The vertex terms exclude illegal local fusion configurations, while the plaquette terms bind locally equivalent legal networks into one coherent quantum state.

The construction can be organized as follows:

```
microscopic edge degree of freedom
↓
allowed fusion at every vertex Q_v = 1
↓
coherent loop fluctuations B_p = 1
↓
nonlocal ground sectors and emergent anyons
```

This Hamiltonian is an exact theoretical construction: it proves the string-net wavefunction is the ground state of local commuting projectors. Real materials realize only approximate couplings, so the construction serves as the reference point that later sections compare against approximate realizations.

Common eigenspace of all projectors

All Hamiltonian terms commute, so the model is frustration-free: one ground state minimizes every local term at once. Each ground state $|\Psi_0\rangle$ satisfies

$$Q_v|\Psi_0\rangle = |\Psi_0\rangle, \quad B_p|\Psi_0\rangle = |\Psi_0\rangle$$

for every v and p . [Theory] The first equation restricts the wavefunction to legal networks. The second makes it invariant under weighted insertion and recoupling of a contractible loop [Ro18]. Tensor-network constructions provide explicit representations of this coherent superposition [R159].

“Condensed” here describes closed networks fluctuating over arbitrarily large scales in the ideal ground state, with local reconnections governed by the fusion and recoupling data. It says nothing about high occupation probability on any single edge.

This proliferation of closed, branching networks across scales is called string-net condensation [Ro18]. The term denotes a property of the ground-state wavefunction. It is distinct from anyon condensation, in which an already-emergent bosonic quasiparticle is identified with the vacuum.

On a surface with noncontractible cycles, closed paths that never shrink to a point, local moves cannot always erase global winding information. The surviving winding sectors form the topological ground-state space.

Enforcing $Q_v = 1$ alone leaves a constrained space of legal configurations. Without the coherent dynamics of B_p , that space stays a classical collection of networks instead of becoming a topological quantum phase.

Open-string endpoints

A local operator can create violations of the projectors. A violated vertex constraint behaves as an electric charge.

A violated plaquette constraint behaves as a flux. In a non-Abelian string-net model the freely mobile quasiparticles generally combine charge with flux.

Ribbon operators run along extended paths. Each ribbon creates a quasiparticle pair, transports the endpoints, and records their fusion and braiding in the algebra of intersecting ribbons.

The edge label τ is a microscopic basis label, not an emergent quasiparticle. An emergent anyon is an excitation of the full Hamiltonian that stays distinguishable under every local equivalence.

The bulk superselection sectors, excitations that no local operator converts into one another, are described by the Drinfeld center $Z(\mathcal{C})$ of the input fusion category $\mathcal{C}$ [R156]; [R157]. [Theory] The Drinfeld center equips each object with consistent data for passing every string type.

The half-braiding data carried by the Drinfeld center, which specify how each object passes every string type, supply the emergent braiding. The lattice Hamiltonian needs no separate microscopic braid operator on each edge.

Treating an edge label as an anyon therefore counts the input data rather than the excitations of H .

Intrinsic doubled topological order

For modular Fibonacci input,

$$Z(\text{Fib}) \simeq \text{Fib} \boxtimes \overline{\text{Fib}},$$

where the overline is the mirror theory with opposite braiding chirality and $\boxtimes$ is the product of the two sectors. The four bulk charges are

$$(1, \bar{1}), \quad (\tau, \bar{1}), \quad (1, \bar{\tau}), \quad (\tau, \bar{\tau}),$$

with quantum dimensions $1, \varphi, \varphi, \varphi^2$, respectively. On a torus, the ideal doubled-Fibonacci model consequently has four topological ground sectors [R156]; [R158]. [Theory]

This is doubled-Fibonacci order: a Fibonacci sector paired with its time-reversed partner, whose opposite chiralities cancel.

The edge label τ is input data rather than an emergent anyon, and the bulk phase is doubled rather than a single chiral Fibonacci theory. Chapter 19 traces the computational consequences of that doubling.

Dropping the mirror sector defines a different phase. With Fibonacci input, the ordinary two-dimensional Levin–Wen construction yields the doubled liquid, not an isolated chiral one.

Limitations of physical crystal lattices

The Levin–Wen Hamiltonian is an exact existence proof and a precise theoretical benchmark. Its couplings do not occur naturally in diamond, sapphire, or silicon carbide.

A defect-based implementation would need an isolated local Hilbert space for each effective edge, with $|1\rangle$ and $|\tau\rangle$ encoded in two reliably controlled states. It would need a large energy penalty enforcing the three-edge projector Q_v . Most demandingly, it would need plaquette matrix elements built from a product of F -moves with the exact ratio $d_1 : d_\tau = 1 : \varphi$.

Leading microscopic defect interactions are commonly pairwise. The plaquette operator B_p is instead a correlated, label-dependent operation on a whole plaquette plus its attached legs, distinct from any uniform ring-exchange interaction.

[Proposal] Ancilla-mediated perturbative gadgets, periodic drives, or digital circuits could synthesize such an operator. Each route must separately establish its effective coefficients, unwanted terms, many-body gap, and regime of validity. The code and Turaev–Viro literature provides mathematical and computational constructions [R155]; [R158], but those results do not establish a passive defect-material phase.

[Speculation] A defect-cluster architecture might encode one edge label per cluster and use mediators to generate Q_v and B_p . Placing spins on a honeycomb lattice is the easy part. The architecture must reproduce the category-consistent multi-edge recoupling amplitudes accurately enough that residual two-body terms, disorder, leakage, and temperature leave the topological gap open. No cited source demonstrates that hierarchy in a crystalline defect array.

Topological order of commuting-projector type survives sufficiently weak local perturbations while the many-body gap stays open, so exact commutation is needed only at the ideal point, not everywhere in the phase [R142]. [Theory] However, the meaning of “sufficiently weak” must be derived or measured for the proposed effective Hamiltonian. It does not justify omitting an inconvenient interaction term without analysis.

Common conceptual errors

- **Identifying an edge label with an anyon.** The microscopic τ label is input data. Emergent anyons are excitations of the collective Hamiltonian and belong to $Z(\text{Fib})$.
- **Equating branching constraints with topological order.** Enforcing $Q_v = 1$ constrains the Hilbert space to legal networks. Coherent B_p dynamics must still bind those networks into a topological liquid; without it they may stay a classical collection.
- **Replacing recoupling with a bit flip.** The Fibonacci operator B_p acts through fusion channels and F -symbols. A six-edge flip with fitted coefficients generally writes down a different Hamiltonian.
- **Calling every superposition a condensate.** String-net condensation means scale-spanning fluctuations of closed networks with the associated topological equivalences. A superposition of two configurations shows no such structure.
- **Confusing string-net condensation with anyon condensation.** String-net condensation builds a ground-state liquid of fluctuating networks. Anyon condensation starts from emergent quasiparticles and changes the topological order by condensing a bosonic sector.
- **Omitting the mirror sector.** Fibonacci input to the ordinary two-dimensional Levin–Wen construction yields doubled Fibonacci, not an isolated chiral phase.
- **Treating digital enforcement as passive emergence.** Measuring projectors and applying corrections emulates or stabilizes a code space by active intervention. Passive topological order means the material Hamiltonian selects the phase energetically, with no intervention.
- **Assuming exact solvability is the protection mechanism.** Commuting projectors simplify the analysis. Physical protection comes from locality, a persistent many-body gap, system size, temperature, and control of perturbations. The exact algebra characterizes the ideal model without establishing that robustness.

Verification exercises

- Show that Q_v is a projector. It is diagonal in the edge-label basis, with eigenvalue 1 on allowed triples and 0 on forbidden triples. Therefore, $Q_v^2 = Q_v$.

- What does B_p^s do? It inserts an s -type loop into plaquette p , fuses that loop into the boundary, and recouples the fusion trees with F -moves.
- Show that the weights $d_s/\mathcal{D}^2$ make B_p a projector. The formal loop $\Omega = \sum_s d_s s$ satisfies $\Omega \times \Omega = \mathcal{D}^2 \Omega$. Dividing the insertion operator by $\mathcal{D}^2$ makes it idempotent, so $B_p^2 = B_p$.
- What is missing if $Q_v = 1$ holds without coherent B_p dynamics? A constrained space of legal networks, which loop recoupling has not yet bound into a topological liquid.
- What topological order follows from Fibonacci input? The bulk theory is $Z(\text{Fib})$, which is equivalent to Fibonacci times mirror Fibonacci and therefore has doubled-Fibonacci order.
- What fails if the edge label τ is treated as an emergent anyon? The label τ is microscopic input data. The bulk charges belong to $Z(\text{Fib})$ and include the mirror sector.

Sources

- [R018] Michael A. Levin and Xiao-Gang Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110); arXiv: [cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617).
 - [R155] Lukasz Fidkowski, Michael Freedman, Chetan Nayak, Kevin Walker, and Zhenghan Wang, “From string nets to nonabelions,” *Communications in Mathematical Physics* 287, 805–827 (2009). DOI: [10.1007/s00220-009-0757-9](https://doi.org/10.1007/s00220-009-0757-9); arXiv: [cond-mat/0610583](https://arxiv.org/abs/cond-mat/0610583).
 - [R156] Alexei Kitaev and Liang Kong, “Models for gapped boundaries and domain walls,” *Communications in Mathematical Physics* 313, 351–373 (2012). DOI: [10.1007/s00220-012-1500-5](https://doi.org/10.1007/s00220-012-1500-5); arXiv: [1104.5047](https://arxiv.org/abs/1104.5047).
 - [R157] Alexander Kirillov Jr., “String-net model of Turaev–Viro invariants,” arXiv:1106.6033 (2011). DOI: [10.48550/arXiv.1106.6033](https://doi.org/10.48550/arXiv.1106.6033); stable URL: [arXiv:1106.6033](https://arxiv.org/abs/1106.6033).
 - [R158] Robert König, Greg Kuperberg, and Ben W. Reichardt, “Quantum computation with Turaev–Viro codes,” *Annals of Physics* 325, 2707–2749 (2010). DOI: [10.1016/j.aop.2010.08.001](https://doi.org/10.1016/j.aop.2010.08.001); arXiv: [1002.2816](https://arxiv.org/abs/1002.2816).
 - [R159] Oliver Buerschaper, Miguel Aguado, and Guifré Vidal, “Explicit tensor network representation for the ground states of string-net models,” *Physical Review B* 79, 085119 (2009). DOI: [10.1103/PhysRevB.79.085119](https://doi.org/10.1103/PhysRevB.79.085119); arXiv: [0809.2393](https://arxiv.org/abs/0809.2393).
 - [R142] Sergey Bravyi, Matthew B. Hastings, and Spyridon Michalakis, “Topological quantum order: Stability under local perturbations,” *Journal of Mathematical Physics* 51, 093512 (2010). DOI: [10.1063/1.3490195](https://doi.org/10.1063/1.3490195); arXiv: [1001.0344](https://arxiv.org/abs/1001.0344).
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Chapter 19 — Chirality and reversed chirality

A quasiparticle braid runs in a definite spatial orientation: an orientation-preserving motion keeps clockwise exchanges clockwise, while spatial reflection turns clockwise exchanges into counterclockwise ones. A braid and its mirror image are the two orientations of one exchange sequence, and keeping both gives two oppositely oriented sectors.

Chapter 15 introduced one fusion rule, and Chapter 18 built a lattice Hamiltonian from it. The fusion rule alone leaves open whether the phase is chiral, reverse-chiral, or doubled.

Orientation dependence of braiding

A braid is the spacetime trajectory traced out by exchanging quasiparticles. For non-Abelian quasiparticles, whose exchanges act noncommutatively on a degenerate fusion space, running the braid in one orientation transforms the fusion state by a unitary matrix U , a transformation that preserves inner products and probabilities.

Reflection turns clockwise exchanges into counterclockwise ones. With consistent basis choices, the reflected braid acts as the complex-conjugate matrix U^* .

A topological medium is chiral when it realizes only one orientation of the braiding data, so that spatial reflection changes the braiding. Chirality is a physical property of the phase, not a naming convention.

The chiral Fibonacci theory $\mathcal{F}$ takes one such orientation. Its nontrivial charge τ labels a superselection sector of quasiparticle excitations.

“Fibonacci” alone names a family of related theories, so it cannot by itself pick out one phase or one quasiparticle.

The fusion rule alone leaves the theory incomplete:

$$\tau \times \tau = 1 + \tau$$

holds in both the chiral theory and its mirror, whose braid matrices differ.

Reverse-braided Fibonacci theory

Reflection produces a second theory, the mirror chiral Fibonacci theory $\overline{\mathcal{F}}$, with nontrivial charge $\bar{\tau}$.

The bar marks reverse braiding, not an antiparticle. Both τ and $\bar{\tau}$ are self-dual: each is isomorphic to its dual charge and fuses with itself into the vacuum channel.

$\mathcal{F}$ and $\overline{\mathcal{F}}$ are therefore distinct phases with opposite braiding orientations. A Hamiltonian realizing one need not realize the other.

Reading the bar as “antiparticle” wrongly demands a distinct charge fusing with τ into the vacuum channel. Because τ is self-dual, $\tau \times \tau$ already contains the vacuum channel, so no distinct partner charge is needed.

Paired charge labels and doubling

The two theories enter as independent categorical factors, so an excitation of the combined theory carries the pair of labels

$$(\text{charge in } \mathcal{F}, \text{ charge in } \overline{\mathcal{F}}).$$

Paired labels are a convenient way to write the doubling. They are not its formal definition, and they ask for no device with two literal layers.

Formally, the construction pairs each object of a fusion theory with every consistent half-braiding, each coherent prescription for braiding that object past all others. For a modular input theory, a braided fusion theory with nondegenerate braiding, this construction factors into the original theory and its reverse-braided mirror [R16o].

The resulting doubled-Fibonacci theory is denoted by DFib :

$$\text{DFib} = \mathcal{F} \boxtimes \overline{\mathcal{F}}.$$

Here $\boxtimes$ is the product of the two independent theories. “Doubled” means combined with its reverse: the two oppositely oriented sectors survive as distinct factors, rather than duplicating one theory or trivializing it.

[Theory] A Levin–Wen model whose input is the Fibonacci fusion category has bulk anyons described by the Drinfeld center of that category. Because the chiral Fibonacci category is modular, its center is equivalent to a product of the chiral theory and its reverse-braided mirror [R018]; [R16o]. The standard commuting-projector construction therefore realizes DFib , not $\mathcal{F}$ alone.

This distinction fixes the bulk-charge list, the torus ground-state degeneracy, the edge structure, the thermal Hall response, and the operations a computation must control. A defect lattice aimed at one Fibonacci-family member but landing on another realizes different topological data, however well the engineering otherwise succeeds.

Simple bulk charges

The chiral theory $\mathcal{F}$ has charges 1 and τ ; its mirror $\overline{\mathcal{F}}$ has $\bar{1}$ and $\bar{\tau}$. The doubled theory DFib therefore has four simple bulk charges, with “simple” meaning an irreducible superselection sector.

Doubled-Fibonacci charge	Pair label	Quantum dimension	Topological spin
vacuum $\mathbf{1}$	$(1, \bar{1})$	1	1
left-handed charge x	$(\tau, \bar{1})$	φ	$e^{4\pi i/5}$
right-handed charge y	$(1, \bar{\tau})$	φ	$e^{-4\pi i/5}$
diagonal charge b	$(\tau, \bar{\tau})$	φ^2	1

Here

$$\varphi = (1 + \sqrt{5})/2$$

is the golden ratio. Quantum dimension measures how fast the fusion-state space grows with more charges; topological spin is the phase from a 2π rotation of a charge. The table uses $\theta_\tau = e^{4\pi i/5}$, which complex-conjugates when every braid reverses. A paired charge has the product of its components’ quantum dimensions [\[R160\]](#); [\[Ro15\]](#).

Doubled Fibonacci is therefore more than the two-object theory $\{1, \tau\}$ renamed: it holds the simple charges x and y , each generating one chiral Fibonacci fusion subtheory, plus the diagonal charge b .

The label b carries no ordinary-boson implication for this non-Abelian charge. Although b has trivial topological spin, its quantum dimension $\varphi^2 > 1$ separates it from the vacuum.

Componentwise fusion

Paired charges fuse independently in the two factors:

$$(a, \bar{c}) \times (a', \bar{c}') = (a \times a', \bar{c} \times \bar{c}').$$

This is the componentwise fusion rule of the product theory. For example,

$$x \times y = (\tau, \bar{1}) \times (1, \bar{\tau}) = (\tau, \bar{\tau}) = b.$$

This fusion has a single outcome. Fusing the resulting b with itself gives

$$\begin{aligned} b \times b &= (\tau \times \tau, \bar{\tau} \times \bar{\tau}) \\ &= (1 + \tau, \bar{1} + \bar{\tau}) \\ &= \mathbf{1} + x + y + b. \end{aligned}$$

Collecting the nontrivial cases,

$$\begin{aligned}
x \times x &= \mathbf{1} + x, & y \times y &= \mathbf{1} + y, \\
x \times y &= b, & x \times b &= y + b, \\
y \times b &= x + b, & b \times b &= \mathbf{1} + x + y + b.
\end{aligned}$$

The charge x inherits the self-fusion rule of τ from Chapter 15, and y inherits the mirror version. The diagonal charge b is a further object beyond those two, not another copy of τ .

Keeping only the first components reproduces the chiral fusion table at the cost of collapsing y and b into one bulk charge, which the doubled theory keeps distinct.

Total quantum dimension

Every anyon theory has a total quantum dimension $\mathcal{D}$ defined by

$$\mathcal{D}^2 = \sum_a d_a^2,$$

with the sum over all simple charges, each contributing its quantum dimension d_a for charge a . It measures the combined size of the simple superselection sectors. For doubled Fibonacci,

$$\begin{aligned}
\mathcal{D}_{\text{DFib}}^2 &= 1 + \varphi^2 + \varphi^2 + \varphi^4 \\
&= (1 + \varphi^2)^2, \\
\mathcal{D}_{\text{DFib}} &= 1 + \varphi^2 = \varphi + 2.
\end{aligned}$$

The last step uses $\varphi^2 = \varphi + 1$. In categorical dimension, then, $\mathcal{D}_{\text{DFib}}$ is the product of the single-sector values from the two independent factors [\[R160\]](#)

On a torus the ideal doubled-Fibonacci theory has four ground-state sectors, one per simple bulk charge; the chiral theory has two. A reported count of two rules out doubled-Fibonacci order.

Braiding in individual and diagonal sectors

Braiding only x charges leaves the mirror component in vacuum, so their fusion spaces, F -moves (changes of fusion-ordering basis), and braid matrices match those of $\mathcal{F}$. Braiding only y charges leaves the chiral component in vacuum and exercises only the mirror factor, giving the complex-conjugate matrices; the F -moves there are the mirror recoupling data.

Diagonal charges engage both components. A braid word acting as U in $\mathcal{F}$ acts on the factorized doubled fusion space schematically as

$$U \otimes U^*.$$

Here $\otimes$ combines the actions on the two factors. In general $U \otimes U^*$ differs from the identity: opposite orientations cancel the net chiral anomaly, but each factor still transforms its own fusion space nontrivially.

Constraining the two factors to travel as an inseparable pair makes the computational operation $U \otimes U^*$ rather than U , so the single-sector universality argument does not transfer directly.

The b fusion rule alone cannot rescue that argument either. A computational specification must name the charges the hardware can transport and measure.

Chiral central charge and thermal Hall response

A topological phase is chiral when its low-energy data keep a nonzero net orientation dependence: reflection replaces its braiding with reverse braiding.

The chiral central charge c_- , defined modulo the appropriate invertible phase, quantifies that residue. At a vacuum interface it fixes the universal low-temperature thermal Hall coefficient [R015]:

$$\frac{\kappa_{xy}}{T} = c_- \frac{\pi^2 k_B^2}{3h},$$

Here κ_{xy} is the thermal Hall conductance in watts per kelvin, T the temperature in kelvin, k_B Boltzmann's constant, and h Planck's constant. The combination k_B^2/h carries units W K^{-2} , matching κ_{xy}/T .

The chiral Fibonacci theory carries nonzero c_- ; its mirror carries the opposite value.

In doubled Fibonacci, the two contributions sum to zero [R160]; [R015]. Correspondingly, the topological spins of x and y are complex conjugates, and the diagonal charge has topological spin one.

A vanishing c_- leaves topological order possibly nontrivial and says nothing about microscopic time-reversal invariance. It establishes that the doubled data carry no net chiral anomaly.

Half-braiding and the Drinfeld center

A Levin–Wen edge carries only the input labels 1 and τ , while the bulk theory holds four charges. The counts differ because a bulk object pairs an input object with a coherent prescription for braiding it past every other input object.

That prescription is called a half-braiding: for an object X , a consistent family of isomorphisms from $X \otimes Y$ to $Y \otimes X$ for every Y , obeying naturality and coherence conditions. Those conditions promote local fusion and recoupling data into genuine two-dimensional bulk braiding.

Pairing each object of an input fusion category $\mathcal{C}$ with every consistent half-braiding produces its Drinfeld center $Z(\mathcal{C})$.

[Theory] Levin–Wen edge labels are input labels, while deconfined bulk quasiparticles are simple objects of $Z(\mathcal{C})$ [R018]; [R161]. For $\mathcal{C} = \mathcal{F}$, modularity gives

$$Z(\mathcal{F}) \simeq \mathcal{F} \boxtimes \overline{\mathcal{F}} = \text{DFib}.$$

Edge labels alone therefore list only allowed fusion and recoupling data. The full bulk-charge classification needs the half-braiding as well.

Doubled order from commuting projectors

The ideal string-net Hamiltonian is

$$H = - \sum_v Q_v - \sum_p B_p,$$

with local vertex and plaquette projectors Q_v and B_p , all commuting. The ground state comes from local recouplings that select no net direction of propagation.

The Drinfeld center bundles every consistent braiding prescription with the reverse-braided data a nonchiral bulk needs. That bundling is why the standard Levin–Wen commuting-projector construction yields doubled topological orders [R018]; [R161].

This conclusion covers that construction only. Noncommuting local Hamiltonians can realize chiral topological order on a lattice; a standard Levin–Wen commuting-projector Hamiltonian always yields the doubled kind.

Boundary chirality and gapped edges

Where a standalone chiral Fibonacci phase meets the trivial vacuum, its nonzero net chirality forbids a fully gapped edge [R156]; [R015]. Low-energy edge modes carry the bulk anomaly: disorder can reshape the edge dynamics but cannot stop the net chiral energy flow short of changing the bulk phase.

The doubled theory behaves differently: couplings between its oppositely chiral edge sectors can gap the edge. [Theory] It admits gapped boundaries for suitable boundary data; one categorical description condenses the diagonal combination $1 \oplus b$ [R156].

An allowed gapped boundary need not describe any particular physical termination, whose couplings or symmetries may keep it gapless. A geometric outline of a boundary is not yet a complete boundary condition.

A gapped doubled-Fibonacci boundary also decides which charges stay distinguishable there and which confine. Proposals using holes, punctures, or boundary quasiparticles must name the boundary type they assume.

Computational access to chiral charge sectors

Consider four charges of type $x = (\tau, \bar{1})$, constrained to have total charge 1. Their fusion space has two possible fusion paths. If a denotes the intermediate charge of the first pair, define

$$|0_L\rangle : a = 1, \quad |1_L\rangle : a = x.$$

These two states span a two-dimensional fusion space encoding a logical qubit. Braiding the x charges acts through the same dense braid-group representation as braiding charges in the chiral Fibonacci theory $\mathcal{F}$, a dense representation whose braid operations approximate arbitrary target operations in the relevant unitary group.

[Theory] Those representations can approximate arbitrary logical unitaries, up to overall phase and with standard encoding and compilation qualifications [R136]; [R015]. The y sector provides the complex-conjugate construction.

An ideal doubled-Fibonacci phase supports universal topological computation once the architecture creates, transports, braids, fuses, and measures one chosen chiral sector, x or y , while holding leakage and cross-sector mixing down. Gates confined to the chosen sector act as the identity on the mirror factor, whose components sit in vacuum, so they implement the chiral-sector unitary.

Three limitations qualify this conclusion.

- The fusion rule alone leaves universality open. Universality needs the F - and R -data, the R -data giving the action of elementary exchanges, plus access to the correct charge sector.
- Hardware with access only to the diagonal charge b , or with mirror charges locked into identical trajectories as inseparable pairs, falls outside the standard chiral-sector universality argument.
- Universal braid representations supply none of initialization, topological-charge measurement, low thermal anyon density, or an adequate many-body gap. Those are further physical requirements beyond the categorical data.

[Theory] Turaev–Viro codes make this computational statement constructive: with suitable access to encoded surfaces and topological operations, doubled theories built from universal modular data can implement quantum computation [R158]. This supports the ideal-model conclusion without showing that any proposed defect Hamiltonian realizes the required operations.

Four descriptive levels need separating. (1) The microscopic qubit: a physical spin or transmon.

(2) A fusion space of several x charges can encode a logical qubit. (3) A digital circuit storing that space's amplitudes emulates the doubled-Fibonacci model.

(4) An emergent x charge needs a many-body phase whose low-energy bulk sectors are those of DFib. Only there does braiding happen through adiabatic quasiparticle transport inside an intrinsic topological medium.

Digital simulation with 27 qubits

The chiral and doubled Fibonacci theories differ mathematically in bulk-charge list, torus degeneracy, and thermal Hall response; which of them a device realizes is decided by those measured data.

[Experiment] In 2024, Xu and collaborators used 27 superconducting transmon qubits to prepare a three-plaquette state of the Fibonacci-input Levin–Wen model by digital circuit, then ran open-string, recoupling, and braid circuits [R138]. They reported average vertex- and plaquette-projector expectation values of 0.94 and 0.58 for their optimized prepared state, under the device and measurement conditions given in that paper. The target string-net is a doubled-type topological model even though the paper uses a shorter anyon label.

The paper states explicitly that its braids involved no Hamiltonian dynamics of quasiparticle excitations and hence drew no protection from a many-body gap [R138]. The transmons instead executed gates that emulated the intended fusion-space transformations.

This result gives no evidence for an intrinsic chiral Fibonacci material phase, nor for an equilibrium doubled-Fibonacci phase stabilized in defect matter.

[Proposal] A static defect-spin architecture with a Levin–Wen target must therefore name DFib in its requirements and derive a low-energy Hamiltonian with the correct Drinfeld center, many-body gap, ribbon operators, and controllable sector-resolved excitations. No result cited here shows that diamond, sapphire, silicon carbide, or any other defect host meets these conditions today.

A chiral Fibonacci target needs a different engineering program with a time-reversal-breaking mechanism producing nonzero net chirality. Striking the word “doubled” from a commuting-projector string-net proposal leaves its topological order doubled.

Common conceptual errors

Labeling every nontrivial object τ collapses real distinctions. In the doubled-Fibonacci bulk, x , y , and b are inequivalent, and assigning the same label τ to all three discards their topological-spin and fusion data.

Input string labels and emergent anyons sit on different descriptive levels. The two Levin–Wen edge labels 1 , τ define local Hilbert-space states, while the four charges 1 , x , y , b classify emergent bulk excitations.

“Doubled” never meant trivial: the net thermal Hall response cancels while non-Abelian fusion spaces survive. The doubled-Fibonacci theory keeps four bulk sectors, total quantum dimension $\varphi + 2$, and nontrivial braiding.

Opposite chirality cancels the net chiral central charge, not every computational gate. A braid confined to the x sector runs no accompanying braid in y , so its computational action stays that of the chiral Fibonacci theory.

A gappable edge is not necessarily a gapped one: a consistent gapped boundary fixes neither the boundary Hamiltonian nor its gap size.

A prepared wavefunction is not an equilibrium phase. A circuit can prepare a small state with ideal-model correlations and apply an ideal braid matrix; an analog topological phase further demands a local Hamiltonian, a robust many-body gap, a scalable bulk, and adiabatic quasiparticle motion. Digital simulation tests the model

without turning the processor's microscopic excitations into doubled-Fibonacci anyons.

Conceptual checks

- **Topological order produced by the standard Levin–Wen model with Fibonacci fusion-category input.**

The model produces doubled-Fibonacci order,

$$Z(\mathcal{F}) \simeq \mathcal{F} \boxtimes \overline{\mathcal{F}},$$

rather than a single chiral Fibonacci order.

- **Derivation of $x \times y = b$ and $b \times b = 1 + x + y + b$.**

The pair labels fuse independently:

$$(\tau, \bar{1}) \times (1, \bar{\tau}) = (\tau, \bar{\tau}) = b,$$

and

$$(\tau \times \tau, \bar{\tau} \times \bar{\tau}) = (1 + \tau, \bar{1} + \bar{\tau}) = 1 + x + y + b.$$

- **Derivation of $\mathcal{D}_{\text{DFib}} = \varphi + 2$.**

$$\mathcal{D}^2 = 1 + \varphi^2 + \varphi^2 + \varphi^4 = (1 + \varphi^2)^2,$$

so $\mathcal{D} = 1 + \varphi^2$. Using $\varphi^2 = \varphi + 1$ gives

$$1 + \varphi^2 = \varphi + 2.$$

- **Limitation when hardware provides access only to the diagonal charge b .**

The standard chiral-sector universality argument does not apply. The fusion rule of b differs from that of τ , and braiding inseparable pairs acts as $U \otimes U^*$, not as U .

- **Limitation of a digital braid as evidence for an intrinsic doubled-Fibonacci phase.**

A digital braid demonstrates controlled emulation of the model's state and operations. It does not establish a protecting many-body Hamiltonian or emergent quasiparticles.

- **Reason doubled Fibonacci is nonchiral but nontrivial.**

The two sectors have opposite chiral central charges, so their net chiral central charge vanishes. Their non-Abelian fusion and braiding data remain nontrivial.

A Levin–Wen model with the Fibonacci fusion category as input realizes DFib. Its four bulk charges are $1, x, y, b$.

The charges x and y have opposite chiral Fibonacci braid data. The doubled edge can be gapped even though a single chiral edge cannot.

Ideal universal computation needs sector-resolved control with the full exchange data, which the fusion rule alone does not supply. Chapter 20 applies these distinctions to what programmable hardware has prepared, fused, braided, and protected.

Sources

- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110); arXiv: [cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617).
 - [R160] M. Müger, “From subfactors to categories and topology II: The quantum double of tensor categories and subfactors,” *Journal of Pure and Applied Algebra* 180, 159–219 (2003). DOI: [10.1016/S0022-4049\(02\)00248-7](https://doi.org/10.1016/S0022-4049(02)00248-7); arXiv: [math/0111205](https://arxiv.org/abs/math/0111205).
 - [R161] C.-H. Lin, M. Levin, and F. J. Burnell, “Generalized string-net models: A thorough exposition,” *Physical Review B* 103, 195155 (2021). DOI: [10.1103/PhysRevB.103.195155](https://doi.org/10.1103/PhysRevB.103.195155); arXiv: [2012.14424](https://arxiv.org/abs/2012.14424).
 - [R156] A. Kitaev and L. Kong, “Models for gapped boundaries and domain walls,” *Communications in Mathematical Physics* 313, 351–373 (2012). DOI: [10.1007/s00220-012-1500-5](https://doi.org/10.1007/s00220-012-1500-5); arXiv: [1104.5047](https://arxiv.org/abs/1104.5047).
 - [R136] M. H. Freedman, M. J. Larsen, and Z. Wang, “A modular functor which is universal for quantum computation,” *Communications in Mathematical Physics* 227, 605–622 (2002). DOI: [10.1007/s002200200645](https://doi.org/10.1007/s002200200645); arXiv: [quant-ph/0001108](https://arxiv.org/abs/quant-ph/0001108).
 - [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083); arXiv: [0707.1889](https://arxiv.org/abs/0707.1889).
 - [R158] R. König, G. Kuperberg, and B. W. Reichardt, “Quantum computation with Turaev–Viro codes,” *Annals of Physics* 325, 2707–2749 (2010). DOI: [10.1016/j.aop.2010.08.001](https://doi.org/10.1016/j.aop.2010.08.001); arXiv: [1002.2816](https://arxiv.org/abs/1002.2816).
 - [R138] S. Xu et al., “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* 20, 1469–1475 (2024). DOI: [10.1038/s41567-024-02529-6](https://doi.org/10.1038/s41567-024-02529-6); stable article: [nature.com/articles/s41567-024-02529-6](https://www.nature.com/articles/s41567-024-02529-6).
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Chapter 20 — Digital emulation and intrinsic anyons

A programmable quantum processor can encode labels such as vacuum and τ , apply a braid matrix, and reproduce the fusion histogram predicted by an anyon theory. An anyon is a two-dimensional quasiparticle whose exchange statistics differ from those of bosons and fermions.

That procedure tests an encoded quantum model. The processor becomes a material host for intrinsic anyons only when its own physical Hamiltonian produces and stabilizes those excitations.

Four distinct physical and computational objects

“Fibonacci anyons were braided” can describe four physically distinct objects. Fibonacci anyons are non-Abelian anyons whose charges include vacuum and τ , with the fusion rules given below.

First is the apparatus's physical two-level system: a superconducting circuit bit, trapped ion, or nuclear spin in chloroform. Such a system is a physical qubit. During these experiments it stays at a fixed location while control pulses act on it; the hardware qubits do not trace exchange paths around one another.

Second is a pattern spread across several physical qubits and interpreted as an anyonic charge in a target model. The pattern belongs to the encoding; the hardware Hamiltonian does not generate it as an excitation. The apparatus may use transmons, ions, or nuclei. A transmon is a superconducting qubit designed to reduce charge-noise sensitivity.

Third is the complete register state at one time. It can be entangled, reproduce loop observables, and match the model's fusion table. A circuit prepares that state, after which hardware dynamics and noise begin to degrade it.

Fourth is an excitation generated by the material after cooling. Here the physical Hamiltonian determines the wavefunction and excitation, without a control circuit continually constructing it.

These objects therefore have separate names. The elementary hardware system is a physical qubit. A multi-qubit pattern interpreted through a target model is an encoded anyonic degree of freedom. A register state with the model's observables is a digitally prepared topological wavefunction. An excitation of the laboratory Hamiltonian is an intrinsic anyon.

“Encoded” marks a theory represented on a processor; its physical Hamiltonian may realize a different theory.

[Experiment] Superconducting and trapped-ion processors have prepared wavefunctions with Fibonacci, finite-group quantum-double, and twist-defect structure. A quantum-double model is a topological lattice model associated with a group. A twist defect is a defect that implements a symmetry or permutation of the model's anyon types. These processors have also executed circuit operations representing fusion and braiding [R162]; [R132]; [R163]; [R164]; [R138]; [R165]; [R140]; [R166]. These results do not show that the processor's physical Hamiltonian has a Fibonacci phase, a protecting many-body gap, or thermally stable quasiparticles. A many-body gap is a nonzero energy separation between the relevant low-energy sector and higher-energy excitations.

A future defect array can be a useful digital processor without its physical Hamiltonian realizing a Fibonacci phase. Digital simulation supports computation through encoded operations; intrinsic material anyons would be excitations stabilized by the material Hamiltonian.

Fusion probabilities after a braid

Let $\mathbf{1}$ denote vacuum and τ a Fibonacci charge, with

$$\tau \times \tau = \mathbf{1} + \tau.$$

This fusion rule states that two τ charges can have total charge either $\mathbf{1}$ or τ . Four τ anyons with total charge vacuum encode a two-dimensional logical space. Define $|\bar{0}\rangle$ as the state in which the first pair and the last pair each fuse to $\mathbf{1}$. Define $|\bar{1}\rangle$ as the state in which those pairs each fuse to τ , while all four still have total charge $\mathbf{1}$.

Creating the two pairs from vacuum prepares the initial state $|\bar{0}\rangle$. Exchanging the middle two anyons once applies a braid generator, which is a unitary operator representing an elementary exchange. In a common phase convention, the braid generator σ_2 gives

$$\sigma_2|\bar{0}\rangle = \phi^{-1}e^{4\pi i/5}|\bar{0}\rangle + \phi^{-1/2}e^{-3\pi i/5}|\bar{1}\rangle,$$

where

$$\phi = \frac{1 + \sqrt{5}}{2}$$

is the golden ratio. The complex phases depend on chirality, meaning the orientation convention for braiding, and on the chosen basis convention. The probabilities in this experiment are independent of those phase choices. Fusing the original pairs and measuring their fusion channels gives

$$P(\mathbf{1}) = |\phi^{-1}|^2 = \phi^{-2} \approx 0.382, \quad P(\tau) = |\phi^{-1/2}|^2 = \phi^{-1} \approx 0.618.$$

The identity $\phi^2 = \phi + 1$ normalizes the probabilities, since $\phi^{-2} + \phi^{-1} = 1$. Their ratio is

$$\frac{P(\tau)}{P(\mathbf{1})} = \phi.$$

In the ideal model, braiding the middle pair once and fusing the original pairs gives this golden-ratio probability ratio. Recent superconducting experiments use it as a fusion signature [R138]; [R165]. The test contains more information than the application of a precomputed 2×2 braid matrix alone because creation, transport within an

encoded string net, and fusion readout are all represented on a spatial many-qubit register. A string net is a graph-based representation of topological degrees of freedom in which edge labels satisfy local fusion constraints.

An implementation that keeps only the final 2×2 matrix can still reproduce the histogram, even without a spatial register. That histogram alone cannot test whether intermediate encoded states corresponded to spatially separated quasiparticles, because the final matrix contains no spatial support information.

The digital implementation is ultimately a circuit acting on physical qubits:

```
initialize physical qubits
|
prepare target vacuum wavefunction U_prep
|
create two encoded anyon pairs U_create
|
transport/exchange middle pair U_braid
|
change to a fusion basis U_fuse
|
measure physical qubits -----> fusion histogram
```

The logical model assigns worldlines—spacetime trajectories—to encoded excitations, and those worldlines describe the gates' logical action. The transmons themselves remain in place rather than following spatial trajectories around one another.

Circuit preparation and equilibrium cooling

Let N count the physical qubits. A digital experiment typically starts in the product state $|0\rangle^{\otimes N}$, with all N qubits independently initialized in the computational-basis state $|0\rangle$, and applies a compiled circuit:

$$|\psi_{\text{prep}}\rangle = U_{\text{prep}}|0\rangle^{\otimes N}.$$

Here U_{prep} is assembled from the hardware's available gates. The target may be the ground state $|G\rangle$, the lowest-energy eigenstate, of an ideal Hamiltonian

$$H_{\text{target}} = -\sum_v Q_v - \sum_p B_p,$$

Here Q_v enforces an allowed fusion rule at vertex v , while B_p enforces the desired flux around plaquette p , an elementary lattice face. In a Fibonacci Levin–Wen model, a string-net Hamiltonian with doubled topological order, Q_v can involve three edge qubits and B_p many qubits around a plaquette [R138].

The circuit may be exact, variationally synthesized, measurement-assisted, or adaptive. Variational synthesis optimizes circuit parameters; measurement-assisted circuits use intermediate outcomes during the experiment, while adaptive circuits use those outcomes to choose later operations. The experiment can then measure $\langle Q_v \rangle$, $\langle B_p \rangle$, Wilson loops, or entanglement combinations. A Wilson loop is a nonlocal operator evaluated around a closed path to diagnose charge or flux.

Agreement with model predictions verifies properties of the prepared state, but a circuit can produce those properties even when the laboratory Hamiltonian differs from H_{target} .

Equilibrium preparation requires engineering $H_{\text{physical}} \approx H_{\text{target}}$, cooling below its gap, and letting the Hamiltonian select and stabilize the phase.

Even an exact circuit with $|\psi_{\text{prep}}\rangle = |G\rangle$ establishes that equality only at the end of preparation. The chip's idle Hamiltonian can differ from H_{target} , so the prepared ground-state wavefunction alone does not constitute a physical phase.

Gate-based transport of encoded charges

An open string operator $W_a(\gamma)$ of charge type a acts along path γ . Acting on that connected path, it ideally creates charges at the endpoints while leaving the intervening region in the code space, the subspace satisfying the model's local constraints:

$$|a, \bar{a}; \gamma\rangle = W_a(\gamma)|G\rangle.$$

In a digital string net, a local graph rewrite can change fusion parenthesization—the order in which several charges are fused—or deform a string's support path while preserving its topological content. This rewrite is an F -move. A compiled braid consists mainly of these rewrites rather than spatial motion of transmons.

An R -move supplies the exchange phase or transformation when strings cross in the graphical representation. Hardware implements both F - and R -moves as multi-qubit gates, ultimately decomposed into native one- and two-qubit operations.

A compiled braid is therefore a product

$$U_{\text{braid}} = U_m U_{m-1} \cdots U_2 U_1,$$

Each U_j is a local graph deformation, string extension, or crossing operation. Their order matters because the factors need not commute; different final observables from two such ordered products provide an operational signature of non-Abelian behavior.

Two compiled sequences differing only by a topologically trivial local deformation should give the same histogram. A difference means the hardware implements a different circuit while assigning it the same anyonic label.

Fusion-channel measurement

The expression $a \times b = \sum_c N_{ab}^c c$ lists the allowed total charges c produced by fusing charges a and b . The nonnegative integer N_{ab}^c is the fusion multiplicity of channel c . A fusion experiment must determine which channel occurred. Let $\Pi_c^{(ab)}$ be the projector onto channel c for the selected pair. For a density matrix ρ , which represents a possibly mixed quantum state,

$$P(c) = \text{Tr}(\rho \Pi_c^{(ab)}) .$$

This equation gives the probability of observing fusion channel c .

Hardware rarely measures $\Pi_c^{(ab)}$ directly. Instead, a circuit maps the fusion basis to computational-basis bit strings, after which physical-qubit readout estimates $P(c)$.

Other experiments measure stabilizers or a nonlocal string operator whose eigenvalue identifies the charge. A stabilizer is an operator whose specified eigenvalue defines a constrained code subspace. Error mitigation, meaning post-processing or control procedures that reduce bias in estimated observables without implementing full error correction, can improve the estimate.

Mitigation changes the estimate but not the underlying physical protection; the two describe different physical properties.

A fusion histogram therefore characterizes a measurement circuit and its readout channel. It does not automatically demonstrate an anyon produced intrinsically by the material.

Four distinct protection mechanisms

Protection can arise through four mechanisms.

First, the material's energy gap can suppress departures from the code space. Under passive protection, local perturbations split logical states only weakly as system size grows, and the Hamiltonian supplies stability without continuous corrective control.

Second, active error correction repeatedly measures syndromes, decodes them, and applies corrections. A syndrome is a set of outcomes that diagnoses errors without directly measuring encoded logical information. This mechanism corrects errors in the hardware but requires continual intervention.

Third, error mitigation processes noisy data to estimate a less-noisy result. It includes dynamical decoupling (pulse sequences average unwanted evolution), readout correction, zero-noise extrapolation (measurements at several noise levels estimate the zero-noise value), twirling (randomizing errors into a simpler channel), and postselection (discarding runs that fail checks).

These procedures improve an estimated observable; they do not create a protected logical memory.

Fourth, in an ideal encoded model, a selected class of simulated errors may act only as a global phase. This demonstrates an algebraic property of the model; uncontrolled physical noise still affects the hardware.

The 2024 Fibonacci experiment states this distinction explicitly: its braids used no Hamiltonian dynamics of quasiparticle excitations and therefore lacked the topological protection supplied by an energy gap [R138]. The 2025 IBM experiment likewise required deep circuits and composite error mitigation [R165].

A correct final braid transformation is not sufficient to establish spatial anyon dynamics. Suppose calibration produces a direct unitary V equal to the desired logical braid on the encoded two-dimensional subspace.

Measurement of the expected fusion probabilities then shows that V was implemented on that subspace. It does not show that the intermediate states represented spatially separated quasiparticles or that local path deformations leave the operation unchanged.

Stronger evidence combines model constraints measured before and after the operation, localized endpoint excitations, dependence on path or operation order, fusion-channel readout, noncommuting braid sequences, and negative controls that omit the relevant operation. These tests probe encoded locality and exchange behavior, but they still establish only a digital realization unless the experiment also demonstrates a physical Hamiltonian that realizes the target phase and supplies physical protection.

Experimental implementations and their interpretation

As of 24 August 2026, the table includes direct Fibonacci demonstrations and the closest non-Abelian comparison experiments because “non-Abelian,” “universal,” and “Fibonacci” are not interchangeable. Qubit counts distinguish the active experimental register from the total chip where the paper makes that distinction.

Date and primary source | Hardware and qubits | Target model and preparation | Fusion measurement | Braid operation and evidence | Protection mode | Intrinsic-versus-emulated verdict | |

May 2023, Xu et al. [R162] | Programmable square-lattice superconducting processor; up to 68 transmons | Toric/surface-code states with twist defects; finite-depth gate circuits prepare stabilizer ground states | Stabilizers and logical/string observables distinguish $\sigma \times \sigma = \mathbf{1} + \varepsilon$ channels | Pauli-string operations and graph deformations implement Ising-type logical gates and noncommuting/projective braids | No passive gap; circuit preparation and readout mitigation | Emulated projective Ising anyons, not Fibonacci and not excitations of the transmon hardware Hamiltonian | |

May 2023, Andersen et al. [R132] | Google superconducting processor; 25 transmons in a 5×5 register | Generalized surface-code stabilizer state; four CZ layers prepare the ground state | Destructive stabilizer and nonlocal Pauli-string measurements reveal vacuum or fermion after fusing twist vertices | Local two-qubit Clifford graph-deformation gates transport and braid degree-3 vertices in the encoded model; fusion after crossing reveals changed fermion parity | No repeated error correction and no passive gap; paper identifies future error correction as necessary | Encoded twist defects with projective Ising statistics; spatial digital braid, but not intrinsic or Fibonacci | |

July 2023, Fan et al.

[R163] | Liquid-state NMR, ^{13}C -labelled chloroform; 2 nuclear-spin qubits at 295 K | Minimal disk-code subspace representing three boundary Fibonacci anyons; shaped RF pulses prepare logical states | No pair-fusion experiment; logical-state tomography/readout in the two-spin basis | Fifteen compiled braid blocks approximate a logical

Hadamard; randomized-benchmarking fidelity reported as 97.18% [R163] | Selected simulated local disturbances act as a global phase in the ideal code; physical dephasing remains, so no passive protection | Direct Hilbert-space emulation of three Fibonacci anyons; no spatial quasiparticles or material realization of a Fibonacci phase | |

February 2024, Iqbal et al. [R164] | Quantinuum H2 trapped-ion processor; 27 qubits on an encoded kagome lattice | Adaptive measurement-and-feed-forward circuit prepares the D_4 quantum-double ground-state wavefunction | Anyon pairs are created and fused; charge/interferometric measurements diagnose outcomes | Ribbon operations transport charges through a Borromean-ring spacetime braid; interferometry distinguishes the intrinsically non-Abelian process within the model | Adaptive preparation and verification, but no passive D_4 Hamiltonian gap during storage/operations | Digitally emergent D_4 anyons in a prepared wavefunction; non-Abelian but not Fibonacci, and not intrinsic ion excitations | |

July 2024, Xu et al.

[R138] | Flip-chip, tunable-transmon square lattice; 27 neighboring qubits mapped to a three-plaquette honeycomb string net | Doubled-Fibonacci Levin–Wen state; product-state loops, U_S , CNOT and variationally synthesized F -move circuits; final preparation depth 53 | Two anyon pairs are recombined; tail/fusion qubits are measured in the computational basis to recover channel probabilities and monodromy | Open-string creation plus five-qubit F - and three-qubit R -moves braid the middle pair; measured $d_\tau \approx 1.60$ [R138] | Echo and readout mitigation only; authors explicitly say there is no Hamiltonian topological protection | Direct 2024 digital Fibonacci-string-net demonstration in which transmons emulate doubled-Fibonacci anyons rather than realize them as excitations of the physical Hamiltonian | |

July 2025, Mineev et al.

[R165] | IBM superconducting hardware: 7-qubit charge register and 9-qubit braid register selected on the 133-qubit Heron processor `ibm_torino`; a 3-qubit vacuum also ran on `ibm_peekskill` | Doubled-Fibonacci string-net condensate via dynamical string-net preparation: modular- S , exact F -moves, tails, and graph sewing, meaning the circuit operation that joins encoded graph components consistently with the string-net constraints | Graph deformation maps charge/fusion information to single-qubit/root-edge readout; reported charge-measurement accuracy 94% [R165] | Four five-qubit F -moves exchange the middle τ_1 anyons; an R - then F -move fuses a pair; measured probability ratio 1.65 ± 0.14 , consistent with ϕ [R165] | Dynamical decoupling, twirling, zero-noise extrapolation, readout mitigation; no passive gap or demonstrated logical lifetime | Emulated doubled-Fibonacci anyons in a digitally prepared string net; exact local moves improve the correspondence with the target model but do not make the anyons intrinsic excitations | |

July 2026, Lo et al. [R140] | Quantinuum H2 trapped ions; 54 qubits | Quantum double $D(S_3)$; adaptive circuit gauges charge-conjugation symmetry and prepares the non-Abelian ground-state wavefunction | Charge-pair braiding and fusion implement logical $\mathcal{X}$ - and $\mathcal{Z}$ -basis measurements on fusion-space qutrits | Ribbon/pull-through braids, implemented as encoded operations that deform and compose ribbon operators according to the quantum-double model, entangle topological qutrits; braiding plus fusion prepares a magic state and supplies a universal gate set. A magic state is a non-stabilizer resource state that enables universal quantum computation when combined with the available operations | Measurement/feed-forward in preparation and fusion-based operations; no native passive S_3 Hamiltonian protection demonstrated | Emulated S_3 quantum-double anyons; universal through braid plus fusion, not Fibonacci braid universality | |

August 2026, Hayata, Hidaka & Kikuchi [R166] | Quantinuum H2 trapped ions; 18 model qubits plus 2 ancillas for compiled multi-control gates | $(2 + 1)$ -D q -deformed $SU(2)_3$ Yang–Mills model restricted to its integer-spin/Fibonacci sector; Trotterized real-time evolution | No anyon-pair fusion readout; computational-basis/Wilson-loop observables test fusion-constrained gauge dynamics | No exchange braid; circuits explicitly execute up to 47

sequential F -moves while evolving the gauge model [R166] | Dynamical decoupling and parallelization mitigate dominant idling error; no passive topological protection | Fibonacci fusion algebra is digitally encoded, but the experiment is gauge dynamics, not a Fibonacci braid demonstration | |

Hardware moved from direct matrix emulation with two nuclear spins to spatial many-qubit string nets and adaptively prepared non-Abelian wavefunctions. Larger registers and sharper topological observables improve tests of the encoded model, but they do not alter the laboratory's physical Hamiltonian; these experiments therefore remain digital emulations.

The two superconducting Fibonacci experiments address complementary parts of the digital program. Xu et al. used 27 transmons for a three-plaquette fixed-point string-net state and measured entanglement plus braid/fusion data [R138].

Mineev et al. reduced local graph resources, used exact graph moves, and simplified fusion readout, while relying on very deep circuits and extensive mitigation [R165]. Together, these experiments show increasingly detailed digital string-net operations, but neither realizes a topological computer with passive protection.

Lo et al. obtained universality from S_3 anyons through braid operations combined with fusion measurements [R140]; this differs from the Fibonacci claim that braiding alone densely generates logical gates.

Hayata, Hidaka, and Kikuchi encoded the Fibonacci fusion algebra in a non-Abelian lattice-gauge simulation and executed long sequences of F -moves, without braiding or fusing localized anyon pairs [R166]. Experiments implementing Fibonacci mathematics therefore outnumber those demonstrating Fibonacci fusion and braiding.

Common interpretive errors

One error is to classify every non-Abelian object as Fibonacci. Ising twist defects obey $\sigma \times \sigma = \mathbf{1} + \varepsilon$, whereas Fibonacci charges obey $\tau \times \tau = \mathbf{1} + \tau$. The D_4 and S_3 quantum doubles have larger, more complex charge sets.

These systems provide useful comparison cases with different braid representations and computational capabilities. A universal protocol based on S_3 braiding and fusion does not establish Fibonacci braid universality.

A second error is to treat a state prepared at one time as a stable phase. A finite-depth or adaptive circuit can produce target stabilizers and long-range entanglement. After the circuit ends, the hardware evolves under its own control and noise Hamiltonian; passive stabilization requires a sustained target Hamiltonian or repeated error correction.

A third error is to equate a more accurate observable estimate with a better quantum memory. Zero-noise extrapolation estimates a hypothetical low-noise observable from several noisy runs; readout mitigation inverts a calibrated measurement channel; postselection discards detected faulty runs. These procedures improve scientific inference, while stored quantum information still lacks a logical-memory guarantee within an individual run.

A protection claim is meaningful only relative to an error model. The NMR experiment demonstrated robustness against a programmed class of local disturbances within its encoded model [R163]. This tests invariance of the encoded logical operation under that programmed disturbance.

It does not imply that the liquid-state nuclear spins were protected against RF miscalibration, relaxation, or dephasing; the measured gate error was in fact dominated by ordinary hardware decoherence [R163]. A protection claim must therefore identify the physical errors involved, the protection duration, and the comparison with an unencoded control.

A further error is to infer a material realization from a compiled result: a direct logical unitary can reproduce the final fusion histogram of a spatial braid. Stronger demonstrations localize endpoints, transport them with local operations, compare braid orders, and perform fusion or interferometric readout. In a digital implementation, locality refers to the programmed encoding and gate connectivity, not to spatial locality in the material. Intrinsic emergence additionally requires the physical Hamiltonian to generate and stabilize the quasiparticle sector.

Finally, small systems limit the physical conclusions. Three plaquettes can demonstrate exact algebraic relations, but their size prevents tests of asymptotic degeneracy splitting, a correlation length much smaller than system size, stability under generic perturbations, or a logical-error rate that improves with code distance. They therefore provide weak evidence for thermodynamic protection even when they measure topological data effectively.

Verification exercises

- **Operational sequence for a digital braid experiment.** A digital braid experiment represents anyons—quasiparticle excitations characterized by their fusion and exchange operations—within an encoded qubit Hilbert space. The experiment first prepares a target vacuum wavefunction, meaning the encoded state with no anyonic excitations. It then creates encoded anyon pairs, transports or exchanges them using compiled sequences of local quantum gates, transforms the resulting state into a fusion basis, and measures the fusion channel. The fusion basis labels states by the possible total topological charges obtained when anyons are combined, and the measured fusion channel identifies the resulting charge.
- **Fusion probabilities after braiding the middle pair of four Fibonacci anyons.** Let $P(\mathbf{1})$ and $P(\tau)$ denote the probabilities of measuring the trivial fusion channel $\mathbf{1}$ and the nontrivial Fibonacci charge τ , respectively. After the middle pair is braided, the corresponding amplitudes have squared magnitudes ϕ^{-2} and ϕ^{-1} . Thus,

$$P(\mathbf{1}) = \phi^{-2}, \quad P(\tau) = \phi^{-1},$$

and therefore

$$P(\tau)/P(\mathbf{1}) = \phi.$$

Here ϕ is the positive solution of $\phi^2 = \phi + 1$. This identity also gives

$$\phi^{-2} + \phi^{-1} = 1,$$

so the two fusion probabilities are normalized.

- **Physical status of Fibonacci anyons in the superconducting experiments.** The experiments did not contain physically intrinsic Fibonacci anyons. Instead, ordinary transmon qubits—superconducting circuit qubits—digitally represented doubled-Fibonacci string-net states and their associated operations. A string-net state is a lattice state whose encoded connectivity and local constraints represent a topologically ordered phase. These experiments did not realize a native Fibonacci Hamiltonian that supplied passive protection.

- **Distinction between adaptive control and passive protection.** Adaptive measurement and feed-forward use measurement outcomes to select subsequent operations. Such adaptive circuits can prepare a target state or actively maintain it. Passive protection instead results from the physical Hamiltonian and its energy gap, without continual corrective intervention. Describing adaptive measurement and feed-forward as passive protection therefore misidentifies the physical mechanism responsible for maintaining the state.
- **Distinction among non-Abelian digital simulations.** Non-Abelian anyon models denoted by Ising, D_4 , and S_3 have different fusion rules, braid representations, and computational power. Fusion rules specify the charges that can result when anyons are combined, while braid representations specify the transformations produced by exchanging them. Consequently, a universal S_3 protocol that uses braiding and fusion is not evidence of Fibonacci braid universality.
- **Methods transferable to a defect-spin array.** A defect-spin array can adopt state-preparation circuits, encoded fusion spaces, compiled F/R moves, stabilizer or Wilson-loop measurements, adaptive feedback, and error-mitigation methods without becoming topological matter. An F move changes the fusion basis, whereas an R move represents the exchange of two anyons. Stabilizer and Wilson-loop measurements probe encoded constraints or charge information. Adaptive feedback applies operations conditioned on measurement outcomes, and error mitigation reduces the effects of experimental imperfections without establishing passive topological protection.

A braid experiment is classified by which of the four objects it constructs. A physical realization requires replacing the compiled state preparation $U_{\text{prep}}|0\rangle^{\otimes N}$, in which a preparation circuit acts on N qubits initialized in $|0\rangle$, with the requirement that the laboratory Hamiltonian itself approximate the target Hamiltonian.

Sources

- [R162] Shibo Xu et al., “Digital Simulation of Projective Non-Abelian Anyons with 68 Superconducting Qubits,” Chinese Physics Letters 40, 060301 (2023). DOI: [10.1088/0256-307X/40/6/060301](https://doi.org/10.1088/0256-307X/40/6/060301).
- [R132] Trond I. Andersen et al., “Non-Abelian braiding of graph vertices in a superconducting processor,” Nature 618, 264–269 (2023). DOI: [10.1038/s41586-023-05954-4](https://doi.org/10.1038/s41586-023-05954-4).
- [R163] Yu-ang Fan et al., “Experimental quantum simulation of a topologically protected Hadamard gate via braiding Fibonacci anyons,” The Innovation 4, 100480 (2023). DOI: [10.1016/j.xinn.2023.100480](https://doi.org/10.1016/j.xinn.2023.100480); [arXiv:2210.12145](https://arxiv.org/abs/2210.12145).
- [R164] Mohsin Iqbal et al., “Non-Abelian topological order and anyons on a trapped-ion processor,” Nature 626, 505–511 (2024). DOI: [10.1038/s41586-023-06934-4](https://doi.org/10.1038/s41586-023-06934-4); [arXiv:2305.03766](https://arxiv.org/abs/2305.03766).
- [R138] Shibo Xu et al., “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” Nature Physics 20, 1469–1475 (2024). DOI: [10.1038/s41567-024-02529-6](https://doi.org/10.1038/s41567-024-02529-6); [arXiv:2404.00091](https://arxiv.org/abs/2404.00091).
- [R165] Zlatko K. Mineev et al., “Realizing string-net condensation: Fibonacci anyon braiding for universal gates and sampling chromatic polynomials,” Nature Communications 16, 6225 (2025). DOI: [10.1038/s41467-025-61493-8](https://doi.org/10.1038/s41467-025-61493-8).
- [R140] Chiu Fan Bowen Lo et al., “Universal gates from braiding and fusing anyons on quantum hardware,” Nature 655, 591–597 (2026). DOI: [10.1038/s41586-026-10709-y](https://doi.org/10.1038/s41586-026-10709-y); [arXiv:2601.20956](https://arxiv.org/abs/2601.20956).

- [R166] Tomoya Hayata, Yoshimasa Hidaka & Yuta Kikuchi, “Digital quantum simulation of q -deformed SU(2) Yang–Mills theory on a trapped-ion quantum computer,” Physical Review Research 8, 033137 (2026). DOI: [10.1103/vlpv-n8dy](https://doi.org/10.1103/vlpv-n8dy); [arXiv:2601.13530](https://arxiv.org/abs/2601.13530).
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Chapter 21 — Digital preparation and Hamiltonian phases

A digital protocol is a timed sequence of gate operations. Executing the sequence prepares a chosen physical state. Once the sequence stops, the gates stop acting, so nothing keeps the state in place.

A Hamiltonian describes energies that act continuously. The system's own interactions favor states near an energy minimum, and that preference persists with no further commands from a controller.

A toric-code state can be reached either way: by a circuit that assembles the wavefunction, or by a material whose lowest-energy sector contains it. The two routes can arrive at the same wavefunction. They differ after preparation ends. The material keeps penalizing departures from the state, while an idle processor applies no such penalty.

Digital state preparation and analog realization

The preceding chapter described digital simulation. A processor applies gates at specified times, prepares a state, and measures it. The hardware runs instructions that stand for the model. It does not host the model's interactions as its own physical couplings.

An analog realization claims something more demanding. The material's local interactions stay on permanently. Its lowest-energy sector carries long-range entanglement, an entanglement structure spread across the whole sample rather than built from independent local correlations. Excited states sit above this sector, separated by a fixed energy cost.

Two things can spoil this picture. Fabrication errors can rearrange the energy landscape until the protected sector disappears. The environment can also create anyons, the emergent localized excitations of the topological model, and drag them across the sample. A logical operator is any operation that changes the encoded quantum information, and a wandering anyon can effect just such a change when its path wraps the system.

A single defect spin supplies a physical qubit: one two-level quantum degree of freedom. A group of defects can share one logical bit or qubit, with the information spread redundantly across the larger joint space of the group. A processor can even prepare a state with Fibonacci-like fusion structure. Each of these is a statement about a prepared state. A topologically ordered phase is a property of the Hamiltonian and its low-energy sector, so prepared-state demonstrations leave that claim open.

For a closed quantum system, the full energy landscape is encoded in one Hermitian operator H , called the Hamiltonian. Every subsequent inequality in this chapter concerns properties of H , rather than the properties of a single prepared wavefunction.

The analog target is a family of materials described by

$$H_{\text{physical}} = H_{\text{topological}} + V,$$

where $H_{\text{topological}}$ is the target Hamiltonian and V is an unwanted perturbation. The perturbation must stay local and weak enough to leave the phase intact. A logical qubit can be written into a prepared state or carried by the low-energy sector of an analog Hamiltonian. Anyons that appear as equilibrium excitations of the material, and topological order as an equilibrium property, require the Hamiltonian itself to have the right structure.

In the Hamiltonian setting, protection comes from energy penalties combined with locality of the interactions. In a digitally corrected processor, it comes from a controller that repeatedly measures error syndromes and feeds back corrections. The two mechanisms belong to different parts of the experiment.

Toric-code Hamiltonian and circuit synthesis

Consider a square lattice with one qubit on each edge. Let X_e and Z_e denote the Pauli X and Z operators acting on edge e . For every vertex v and plaquette p , where a plaquette is an elementary square face of the lattice, define the star and plaquette operators

$$A_v = \prod_{e \ni v} X_e, \quad B_p = \prod_{e \in \partial p} Z_e.$$

Here $e \ni v$ denotes the edges incident on vertex v , and $e \in \partial p$ denotes the edges on the boundary of plaquette p . Each product contains four edges on the square lattice. A star and a plaquette share either zero or two edges. On one shared qubit, Pauli X and Z anticommute, but produce two cancelling minus signs on two shared qubits, all A_v and B_p commute.

The static toric-code Hamiltonian is

$$H_{\text{TC}} = -J_s \sum_v A_v - J_p \sum_p B_p, \quad (21.1)$$

where $J_s > 0$ and $J_p > 0$ are interaction energies measured in joules (J) or electronvolts (eV). A ground state, defined as a state of minimum energy, satisfies $A_v = B_p = +1$ for every term. On a torus, which is a periodic surface with two independent noncontractible cycles, the ideal model has four ground states. These states are distinguished by noncontractible loop operators, whose paths cannot be continuously contracted to a point [\[Ro30\]](#). [Theory]

To compare this Hamiltonian with a digital implementation, isolate one plaquette, label its edges 1, 2, 3, 4, and define $B = Z_1 Z_2 Z_3 Z_4$. During a digital time step of duration δt , the target evolution generated by the plaquette term $-J_p B$ is

$$U_p(\delta t) = \exp\left(+i \frac{J_p \delta t}{\hbar} B\right), \quad (21.2)$$

where $\hbar$, the reduced Planck constant, has units J·s. The exponent must be dimensionless. This condition is satisfied because $J_p \delta t / \hbar$ has units (J)(s)/(J s) = 1.

A gate processor can synthesize this four-body evolution by computing the parity of the four qubits. Define

$$W = \text{CNOT}_{1 \rightarrow 4} \text{CNOT}_{2 \rightarrow 4} \text{CNOT}_{3 \rightarrow 4}.$$

Here $\text{CNOT}_{i \rightarrow j}$ is a controlled-NOT gate with control qubit i and target qubit j . Conjugation by this circuit gives $W^\dagger Z_4 W = Z_1 Z_2 Z_3 Z_4 = B$. With the single-qubit rotation $R_z(\phi) = \exp(-i\phi Z/2)$, choose $\phi = -2J_p \delta t / \hbar$. Then

$$W^\dagger R_z(\phi)_4 W = \exp\left(+i \frac{J_p \delta t}{\hbar} B\right). \quad (21.3)$$

Repeating the corresponding gate blocks for all stars and plaquettes implements evolution under Eq. (21.1).

At this commuting point every ideal evolution factor commutes with every other, so splitting the evolution into star and plaquette blocks introduces no Trotter error. Trotter error is the approximation error from splitting an evolution into separately applied pieces that fail to commute. The digital route still carries control errors and decoherence on real gates, and the processor's idle Hamiltonian is generally unrelated to Eq. (21.1).

A material with a permanent term $-J_p B$ assigns it energy rather than time evolution. If $b = \pm 1$ is the measured eigenvalue of B , the corresponding energy contribution is $-J_p b$. Changing b from $+1$ to -1 therefore costs $2J_p$.

A local X_e anticommutes with the two plaquette operators adjacent to edge e . In a periodic bulk, it therefore creates two magnetic anyons and costs $4J_p$. Similarly, a local Z_e creates two electric anyons and costs $4J_s$. The periodic model's lowest bulk pair-creation gap is therefore

$$\Delta_{\text{pair}} = 4 \min(J_s, J_p), \quad (21.4)$$

while the energy assigned to one well-separated anyon is $\epsilon_a = 2J_s$ or $2J_p$. Boundaries can permit a single excitation, so any quoted gap must specify both the gap convention and the geometry [Ro30]; [R141]. [Theory]

The circuit and the static Hamiltonian can produce the same unitary evolution or the same state. The static Hamiltonian alone keeps the energy penalty switched on after preparation. If a circuit prepares $|\psi_{\text{TC}}\rangle$ and the hardware subsequently idles under a trivial H_{idle} , then the target state has been prepared, but it has not become an equilibrium phase of the hardware.

The many-body topological gap

List the eigenenergies of the full interacting Hamiltonian in increasing order as $E_0 \leq E_1 \leq \dots$. If the topological ground-state manifold—the collection of nearly degenerate ground states associated with distinct logical sectors—occupies the lowest q levels, define the finite-size bulk many-body gap and its thermodynamic limit by

$$\Delta(L) = E_q(L) - E_{q-1}(L), \quad \Delta_{\text{topo}} = \liminf_{L \rightarrow \infty} \Delta(L), \quad (21.5)$$

where L is the sample's linear size in metres and Δ is an energy. The lower limit, $\liminf$, accounts for possible nonmonotonic size dependence. Material topological order requires $\Delta_{\text{topo}} > 0$, rather than merely a nonzero level spacing in a small cluster.

The name many-body gap records what Δ_{topo} belongs to: the full interacting spectrum in the large-sample limit. A defect's optical gap lives on one site, and a cluster's leakage gap lives on a few sites. The collective topological gap lives on the whole sample. The three energies answer different questions and cannot stand in for one another.

The gap defines a microscopic response time

$$\tau_{\Delta} = \hbar / \Delta_{\text{topo}}.$$

Drive the system slowly and weakly compared with this response time, and the bulk stays unexcited. That protection covers bulk excitations only. The same perturbation can still split the ground-state manifold or carry existing anyons across the sample, and either effect corrupts the logical state.

A splitting measured on two sites describes that cluster alone, so it cannot be identified with Δ_{topo} . Such a measurement characterizes a cluster, not the thermodynamic gap above the topological manifold.

Local perturbations and their strength

A lattice Hamiltonian is a sum of terms that each touch only a few nearby sites:

$$H = \sum_X h_X,$$

Here X labels one bounded cluster of nearby sites and h_X acts only there. The Hamiltonian counts as local when each cluster has finite range, or when the strength of larger clusters falls off fast enough with the diameter of X . Locality says nothing about the size of the global operator norm $\|H\|$, defined as the largest magnitude by which the operator can act on a normalized state, is small. That norm generally increases with the number of sites.

For an unwanted perturbation $V = \sum_X V_X$, a useful measure of local strength is

$$g_{\text{loc}} = \max_i \sum_{X \ni i} \|V_X\|, \quad (21.6)$$

which has units of energy. The relevant engineering ratio is $g_{\text{loc}} / \Delta_{\text{topo}}$, not $\|V\| / \Delta_{\text{topo}}$.

The sum in Eq. (21.6) gathers every perturbation term touching one site, and the maximum selects the worst site. A perturbation spread across the sample grows in total strength with system size while staying weak at each site. A single strong defect does the reverse: the spatial average looks small while one site violates the stability condition.

Stability theorems cover broad classes of locally perturbed commuting-projector Hamiltonians. Below a nonzero local threshold, and given suitable topological-order conditions, the spectrum keeps separated bands and the ground-state band stays exponentially narrow [R142]. [Theory] No theorem supplies a universal number, such as “ten percent,” valid for every material. Each model carries its own threshold.

Defect arrays need particular care here. Dipolar interactions decay as $1/r^3$ rather than terminating at a selected neighbor. In spatial dimension D , the contribution from distance shells scales as

$$\int dr r^{D-1} / r^\alpha,$$

so the long-distance sum converges only when $\alpha > D$. A $1/r^3$ tail is summable in two dimensions. Summability answers whether distant shells add up to a finite total. Finite-range stability theorems assume more, so summability alone does not bring the interaction under them.

An analysis must therefore keep the long-range tail, the control crosstalk, and the disorder, either as explicit terms or as quantitative bounds. Dropping them inside a nearest-neighbor approximation assumes the answer.

Correlation length and system size

For local observables O_i and O_j separated by distance r , define their connected ground-state correlation by

$$C(r) = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle.$$

Subtracting the product of the separate averages leaves the part the averages cannot explain. A gapped short-range system generally keeps this remainder exponentially small:

$$|C(r)| \leq C_0 e^{-r/\xi}, \tag{21.7}$$

where C_0 has the units of the observable product and ξ is the correlation length in metres, the distance over which ordinary local measurements stay correlated [R167]. [Theory] The correlation length sets the range of ordinary local correlations. The ground state can still carry entanglement stretched across the whole sample.

Its size follows from the gap and the speed at which disturbances travel:

$$\xi \sim \hbar v_{\text{LR}} / \Delta,$$

where v_{LR} is a characteristic information-propagation velocity in m/s. The units are $(\text{J s})(\text{m/s})/\text{J} = \text{m}$.

Rapid decay of local correlators leaves topological order intact. The logical information sits in loop operators that wrap the sample, which local measurements cannot see.

At the exact toric-code fixed point, many connected local correlations vanish beyond zero range. A generic perturbation spreads them over a finite ξ without necessarily destroying the phase.

The practical finite-size requirement is

$$L \gg \xi, \quad (21.8)$$

Non-Abelian anyons, whose exchanges combine like noncommuting operations rather than simple phases, additionally need a separation $R \gg \xi$. If these inequalities are not satisfied, the topological sectors overlap appreciably, and the physical encoding is not effectively nonlocal.

Local indistinguishability and finite-size splitting

In the thermodynamic limit, no measurement confined to a bounded region can tell which logical sector the system occupies. A weak local perturbation then slightly deforms each ground state, a deformation perturbation theory calls dressing, rather than collapsing the manifold onto one sector [R142]. This result governs how the zero-temperature phase responds to static deformations. It says nothing about how long stored information survives.

A finite torus admits a tunneling process: a virtual anyon pair appears, one partner winds once around a noncontractible cycle, and the pair reannihilates. The winding records which sector the system was in, so the process couples different logical sectors. For a gapped local phase, the resulting ground-state splitting takes the form

$$\delta E_0(L) \sim C \Delta_{\text{topo}} e^{-L/\xi}, \quad (21.9)$$

up to model-dependent powers, paths, and coefficients. The splitting between fusion states of anyons separated by R is similarly exponentially small in R/ξ [Ro15]. [Theory] Here C is dimensionless, so δE_0 has units of energy.

During a storage time t_{store} , this splitting produces a relative phase of order $\delta E_0 t_{\text{store}}/\hbar$. A necessary condition for this phase to remain small is

$$\frac{\delta E_0 t_{\text{store}}}{\hbar} \ll 1. \quad (21.10)$$

Controlled adiabatic motion therefore faces two opposing demands, giving an ideal operating-time window

$$\hbar/\Delta_{\text{topo}} \ll t_{\text{op}} \ll \hbar/\delta E_0.$$

The left inequality keeps the motion slow enough to avoid creating bulk excitations; the right inequality keeps it fast enough to avoid resolving the unwanted splitting [R015]. [Theory] Real control pulses add diabatic errors, meaning unwanted excitations left behind when the pulse moves too fast for the state to follow, plus noise and timing limits on top of this window.

At finite L , replacing the splitting by $\delta E_0 = 0$ misses the physics: the splitting is generally nonzero, and the relative phase it generates keeps accumulating. Over sufficiently long t_{store} , the logical sectors become resolvable, so exact degeneracy cannot generally be inferred from a finite-size system.

Thermal creation of anyons

Heat changes the accounting. Coupling to the environment lets the system exchange energy, so excitations appear with thermal probabilities.

At temperature T in kelvin, a weakly coupled system in thermal equilibrium has density matrix

$$\rho_T = \frac{e^{-H/(k_B T)}}{\text{Tr}(e^{-H/(k_B T)})},$$

where Boltzmann's constant k_B has units J/K. For a dilute anyon species with excitation energy ϵ_a , the two-dimensional equilibrium occupation is proportional to

$$n_a \sim \frac{g_a}{a^2} e^{-\epsilon_a/(k_B T)} \quad (21.11)$$

where a is the lattice spacing and g_a is a dimensionless degeneracy factor. This is only a dilute, noninteracting estimate.

The expected number multiplies this density by the sample area. Larger samples offer more places for a pair to appear. Hence, $k_B T \ll \epsilon_a$ is necessary for suppressing excitations, but it does not guarantee that a macroscopic sample contains no anyons [R141]; [R169]; [R143].

[Theory]

To fix the size of these factors, take numbers as an arithmetic exercise, unrelated to any particular material. Suppose that $J_s = J_p = h \times 1 \text{ GHz}$. Since $h \times 1 \text{ GHz}/k_B \approx 48 \text{ mK}$, one anyon costs $2J \approx k_B \times 96 \text{ mK}$, and a bulk pair costs $4J \approx k_B \times 192 \text{ mK}$.

At $T = 20 \text{ mK}$, the pair Boltzmann factor is $e^{-192/20} \approx 6.8 \times 10^{-5}$, whereas the single-anyon density factor is $e^{-96/20} \approx 8.2 \times 10^{-3}$. Thermal suppression keeps each factor small, but the logical error rate needs more: the sample area, the frequency spectrum of the bath, and how far anyons travel once created.

In a two-dimensional toric-code-like memory, creating a pair costs a fixed energy, and dragging one partner across the sample costs no additional energy. A bigger sample then adds more creation sites without raising the barrier, so passive memory performance need not improve with size.

Anyon transport and localization

A logical fault needs two stages: creation of an anyon, then travel far enough to wrap the code. Each local hop lengthens the error string attached to the anyon. When the motion is diffusive in two dimensions with diffusion constant D in m^2/s , its root-mean-square displacement after time t is

$$\ell_{\text{diff}}(t) = \sqrt{4Dt}, \quad (21.12)$$

which has units $\sqrt{(\text{m}^2/\text{s})(\text{s})} = \text{m}$. For a code with distance d and lattice spacing a , a rough transport requirement is

$$\ell_{\text{move}}(t_{\text{store}}) \ll da,$$

together with a sufficiently low excitation-creation rate. The code distance d is the minimum number of local errors required to implement a nontrivial logical operation.

Static disorder can Anderson-localize the defects of an ideal toric code. Interference between scattering paths suppresses coherent propagation, leaving wavefunctions that decay over a localization length ξ_{loc} [R170]. [Theory] [Numerics] This effect can slow coherent motion.

Localization leaves the creation gap unchanged. A thermal bath supplies the energy for inelastic hops between localized states, and at finite anyon density the anyons interact, so a one-particle localization calculation stops applying.

Localization can still help by slowing motion, but only when demonstrated for the actual noise model. It differs from topological protection in what it does: it slows travel rather than penalizing creation. Deliberate pinning, meaning locally trapping anyons with engineered potential wells, carries a further cost, since gates that braid anyons need those same anyons to move on command.

Laboratory Hamiltonians and evidence for a phase

A fabricated defect array always contains more than the target model:

$$H_{\text{lab}} = H_{\text{target}} + V_{\text{static}} + H_{\text{drive}}(t) + H_{\text{bath}} + H_{\text{system-bath}}. \quad (21.13)$$

Here V_{static} represents fabrication disorder and unwanted couplings, $H_{\text{drive}}(t)$ represents time-dependent control, H_{bath} describes environmental modes, and $H_{\text{system-bath}}$ describes coupling between the system and those modes. An analog realization must remain valid in the presence of V_{static} and must quantitatively characterize H_{bath} and $H_{\text{system-bath}}$.

Periodic driving can shape the dynamics observed once per drive period, a technique called Floquet engineering. The stroboscopic motion then follows a drive-induced effective Hamiltonian. Such drives can help, but a protection claim built on one must account for heating, for the micromotion inside each period, for calibration errors, and for what happens when the drive fails.

A persuasive demonstration goes past preparing a target state. It maps the bulk gap and tracks how it changes with system size. It checks that local observables agree across candidate ground sectors up to finite-size corrections. It measures loop operators or braiding data directly. It verifies that weak local perturbations leave the gap open and the topological data unchanged. And it measures thermal creation and thermal transport as separate rates.

Each of these probes a different requirement, so one signature from one finite device cannot settle whether a thermodynamic phase exists.

For defect-cluster architectures, the proposed correspondence $H_{\text{physical}} \approx H_{\text{string-net}}$ remains a [Proposal] until the microscopic terms, corrections, gap, and diagnostics have been derived or measured. When a programmed circuit reproduces target fusion data, that success demonstrates control and simulation. Whether the undriven material hosts emergent anyons remains a separate question.

Self-correction and passive protection

A self-correcting quantum memory is a memory whose storage lifetime increases without bound with system size at fixed nonzero temperature and with no active syndrome measurement or feedback [R169]. This requirement is stronger than demonstrating a long lifetime in one finite sample.

In the ordinary two-dimensional toric code the energy barrier stays constant with system size. Creating an anyon pair costs a fixed energy, moving one endpoint across the lattice adds no further energy, and reannihilating the pair leaves behind a string that can act as a logical operator.

For geometrically local two-dimensional stabilizer Hamiltonians, the Bravyi–Terhal no-go theorem proves an $O(1)$ energy barrier. It therefore rules out conventional self-correction within that class [R168].

[Theory] Models that are non-Abelian, long-range, driven, or held out of equilibrium sit outside those assumptions, so the theorem leaves them unclassified.

A gapped 2D topological phase still helps: it suppresses local error matrix elements, shrinks finite-size splitting exponentially, and lowers the thermal anyon density. Arbitrarily long storage may still need active error correction. Analyses of the toric code coupled to thermal reservoirs draw the same line between stability against static perturbations and stability against heat [R143]. [Theory] A passive-protection claim should therefore quote measured improvement at stated values of time, temperature, size, and bath properties. Unlimited lifetime is a separate claim that needs its own scaling evidence.

A useful set of necessary conditions is

$$\begin{array}{l}
\Delta_{\text{topo}} > 0, \quad L/\xi \gg 1, \\
g_{\text{loc}}/\Delta_{\text{topo}} < c_{\text{stab}}, \\
k_B T/\epsilon_a \ll 1, \\
\delta E_0 t_{\text{store}}/\hbar \ll 1, \\
\ell_{\text{move}}(t_{\text{store}}) \ll da.
\end{array} \tag{21.14}$$

Here c_{stab} is a dimensionless stability threshold that varies from model to model. Passing every inequality in Eq. (21.14) is required, yet the set remains incomplete. A complete assessment additionally counts the available creation sites, tracks leakage out of the encoded clusters, verifies the target topological data, and models the bath.

Common conceptual errors

A phase describes a family of Hamiltonians whose low-energy sectors share the same stable structure. One prepared wavefunction shows that a pulse sequence reached a state; it says nothing about whether neighboring Hamiltonians keep that structure.

The topological gap is the many-body gap above the topological manifold after all effective corrections have been included. A single-site or few-site energy difference cannot stand in for Δ_{topo} , which belongs to the full interacting spectrum in the large-sample limit.

Local stability is judged per site by g_{loc} . An extensive perturbation can have a large $\|V\|$ while remaining weak per site. Conversely, one strong local defect can be dangerous despite a small spatial average.

A finite correlation length governs ordinary connected correlations, which decay over that length. Long-range entanglement across the sample is a separate property that survives this decay.

An exponentially small splitting still generates a relative phase that grows with storage time. At finite L , δE_0 is generally nonzero, so long experiments resolve sectors that short ones cannot distinguish.

Localization slows motion without raising the creation gap. Thermal creation continues, and bath-assisted or interacting dynamics can defeat the slowdown.

Perturbative stability asks whether weak static changes near zero temperature keep the phase. Thermal self-correction asks whether storage lifetime grows with system size in open-system dynamics at nonzero temperature. The two questions involve different perturbations and different limits.

Active digital error correction protects information through measured syndromes and feedback. Passive material protection works through energy penalties and locality with the controller idle. Both help, through different mechanisms.

Conceptual checks

- **Question:** A circuit exactly implements $e^{-iH_{\text{TCT}}t/\hbar}$. Does this establish an analog toric-code phase?

Answer: No. The idle physical Hamiltonian must itself lie in the phase. A timed gate sequence implements dynamics but need not provide the equilibrium energy landscape.

- **Question:** Show that a local X_e on the periodic toric code costs $4J_p$.

Answer: X_e anticommutes with the two adjacent plaquette operators. Each flipped plaquette changes its energy by $2J_p$, so creating the pair costs $4J_p$.

- **Question:** What error results from using $\|V\|$ instead of g_{loc} ?

Answer: $\|V\|$ usually increases with array size even when every site remains weakly perturbed. The resulting engineering ratio can incorrectly suggest instability while the local theory remains in the same phase. Conversely, a spatial average can conceal one strong and dangerous defect.

- **Question:** Show that if $J_s = J_p = h \times 1 \text{ GHz}$, the pair Boltzmann factor at $T = 20 \text{ mK}$ is about 6.8×10^{-5} .

Answer: $h \times 1 \text{ GHz}/k_B \approx 48 \text{ mK}$, so a pair costs $4J \approx k_B \times 192 \text{ mK}$, and $e^{-192/20} \approx 6.8 \times 10^{-5}$.

- **Question:** Why does $k_B T \ll \Delta_{\text{pair}}$ not establish passive fault tolerance?

Answer: Logical errors can still depend on the number of creation sites, bath properties, anyon mobility, code geometry, and the constant energy barrier. A dilute anyon density is not zero, and increasing the sample size does not automatically improve protection.

- **Question:** What does the two-dimensional stabilizer no-go theorem rule out?

Answer: Within its assumptions of geometrically local two-dimensional stabilizer Hamiltonians, it rules out a growing energy barrier and therefore conventional self-correction. It does not classify every possible topological material.

This chapter compared a compiled circuit with a Hamiltonian in a topological phase and collected the necessary inequalities Eq. (21.14). The next chapter asks whether a microscopic defect Hamiltonian actually generates the required low-energy terms, and how large the leftover corrections are.

Sources

- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* 303, 2–30 (2003). DOI: [10.1016/S0003-4916\(02\)00018-0](https://doi.org/10.1016/S0003-4916(02)00018-0); [arXiv:quant-ph/9707021](https://arxiv.org/abs/quant-ph/9707021).
- [R167] M. B. Hastings and T. Koma, “Spectral gap and exponential decay of correlations,” *Communications in Mathematical Physics* 265, 781–804 (2006). DOI: [10.1007/s00220-006-0030-4](https://doi.org/10.1007/s00220-006-0030-4); [arXiv:math-ph/0507008](https://arxiv.org/abs/math-ph/0507008).
- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* 51, 093512 (2010). DOI: [10.1063/1.3490195](https://doi.org/10.1063/1.3490195); [arXiv:1001.0344](https://arxiv.org/abs/1001.0344).
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083); [arXiv:0707.1889](https://arxiv.org/abs/0707.1889).

- [R141] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, “Topological quantum memory,” *Journal of Mathematical Physics* 43, 4452–4505 (2002). DOI: [10.1063/1.1499754](https://doi.org/10.1063/1.1499754); [arXiv:quant-ph/0110143](https://arxiv.org/abs/quant-ph/0110143).
 - [R168] S. Bravyi and B. Terhal, “A no-go theorem for a two-dimensional self-correcting quantum memory based on stabilizer codes,” *New Journal of Physics* 11, 043029 (2009). DOI: [10.1088/1367-2630/11/4/043029](https://doi.org/10.1088/1367-2630/11/4/043029); [arXiv:0810.1983](https://arxiv.org/abs/0810.1983).
 - [R169] B. J. Brown, D. Loss, J. K. Pachos, C. N. Self, and J. R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* 88, 045005 (2016). DOI: [10.1103/RevModPhys.88.045005](https://doi.org/10.1103/RevModPhys.88.045005); [arXiv:1411.6643](https://arxiv.org/abs/1411.6643).
 - [R170] C. Stark, L. Pollet, A. Imamoğlu, and R. Renner, “Localization of toric code defects,” *Physical Review Letters* 107, 030504 (2011). DOI: [10.1103/PhysRevLett.107.030504](https://doi.org/10.1103/PhysRevLett.107.030504); [arXiv:1101.6028](https://arxiv.org/abs/1101.6028).
 - [R143] R. Alicki, M. Fannes, and M. Horodecki, “On thermalization in Kitaev’s 2D model,” *Journal of Physics A: Mathematical and Theoretical* 42, 065303 (2009). DOI: [10.1088/1751-8113/42/6/065303](https://doi.org/10.1088/1751-8113/42/6/065303); [arXiv:0810.4584](https://arxiv.org/abs/0810.4584).
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PART IX — ENGINEERING TOPOLOGY FROM DEFECT CLUSTERS

The models studied so far require interactions involving several spins at once. This part investigates how interactions between pairs could generate those effects in groups of defect spins.

Chapter 22 — Effective interactions induced by high-energy states

A cluster holds more states than the low-energy model keeps. A coupling between two clusters can leave the retained states, visit a higher-energy level briefly, and come back. Viewed from inside the retained subspace, that excursion looks like a new operator: the system starts low, borrows a costly state, and ends low.

Put the retained states at energy zero and the eliminated states at least an energy Δ above them, and the mixing has characteristic energy scale g , with $|g|/\Delta \ll 1$. Any induced operator that begins and ends in the retained subspace must then contain at least two applications of the mixing. The leading contribution therefore has scale g^2/Δ .

This induced operator is the effective Hamiltonian of this chapter. Simply deleting the high-energy states drops the round-trip contribution, since projection keeps only processes that never leave the retained space. Keeping every microscopic state avoids that error but carries many degrees of freedom the low-energy question never uses.

Second-order transitions through a high-energy subspace

Divide the Hilbert space into two parts. The low-energy part holds the states kept in the effective model. The high-energy part holds states whose energies exceed those of the retained states by at least Δ . A weak coupling with characteristic strength g connects the two subspaces.

Both g and Δ have units of energy, and the perturbative assumption is $|g|/\Delta \ll 1$.

One application of the coupling moves the state upstairs, so a process that starts and ends downstairs needs two applications:

$$\text{cheap } a \xrightarrow{g} \text{expensive } m \xrightarrow{g} \text{cheap } b.$$

The resulting amplitude has the characteristic scale

$$\frac{g g}{E_{\text{low}} - E_m} \approx -\frac{g^2}{\Delta}.$$

Three things matter in this expression. Two matrix elements of the coupling produce the factor g^2 . Second, a large excitation gap suppresses the process as $1/\Delta$. Third, when the intermediate state lies above the retained states, the energy denominator is negative.

The high-energy state appears only midway through the amplitude; no energy measurement finds the system there at the end. An intermediate contribution of this kind is called virtual.

The leading induced operator within the low-energy subspace therefore has energy scale g^2/Δ , rather than g .

The virtual description applies only when $|g|/\Delta$ is small. If the coupling is sufficiently strong to produce substantial real occupation of the high-energy state, elimination of that state is not perturbatively justified.

The sign of one spin-coupling coefficient depends on the phases of its matrix elements and on the operator convention of the effective Hamiltonian, so no blanket sign rule covers every case. One statement does hold generally: when the retained states couple only upward in energy, the second-order energy-correction operator is negative-semidefinite, with all expectation values nonpositive.

Subtracting a constant energy redefines the zero without changing splittings, and the exchange coefficient left behind can be positive. The two-site Hubbard example below works through exactly that case.

The round trip has the shape:

```

low-energy subspace
|
| coupling v
v
high-energy subspace, energy separation Delta
|
| return to the low-energy subspace
v
effective coupling ~ v^2 / Delta

```

Three-state model with one high-energy state

Take three orthonormal states: two low-energy states $|L\rangle$ and $|R\rangle$, and one high-energy state $|e\rangle$. Let the unperturbed Hamiltonian be

$$H_0 = \Delta |e\rangle\langle e|,$$

where $\Delta > 0$ has units of energy. The low-energy states have energy zero. Introduce a coupling of strength g , also with units of energy, between each low-energy state and the high-energy state:

$$V = g(|e\rangle\langle L| + |e\rangle\langle R| + \text{h.c.}).$$

Here “h.c.” denotes the Hermitian conjugate, which includes the reverse transitions from $|e\rangle$ to $|L\rangle$ and $|R\rangle$. The perturbative assumption is $|g|/\Delta \ll 1$.

The operator that selects the retained pair is

$$P = |L\rangle\langle L| + |R\rangle\langle R|.$$

The operator that selects the eliminated state is

$$Q = |e\rangle\langle e|.$$

Together they span the full Hilbert space, so $P + Q = I$. Each operator is idempotent, and the two subspaces are orthogonal:

$$P^2 = P, \quad Q^2 = Q, \quad PQ = QP = 0.$$

An operator with $P^2 = P$ is a projector: it selects a subspace and leaves vectors inside it unchanged. Here the retained projector keeps the working low-energy subspace; it is a bookkeeping selection, not a measurement. The complementary projector Q collects the high-energy states left out of that working model.

The coupling always moves the system between the two subspaces, so $PVP = 0$: no first-order matrix element connects two retained states directly. The first nonzero contribution is second order:

$$H_{\text{eff}}^{(2)} = -PVQ \frac{1}{\Delta} QVP = -\frac{g^2}{\Delta} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}_{\{|L\rangle, |R\rangle\}}.$$

The microscopic coupling never directly connects $|L\rangle$ and $|R\rangle$; it generates the matrix element:

$$\langle L | H_{\text{eff}} | R \rangle = -\frac{g^2}{\Delta}.$$

Thus, a virtual path through a single high-energy state produces an off-diagonal low-energy coupling that is absent from the original operator V .

Adding the two paths with opposite relative phase gives the antisymmetric combination $(|L\rangle - |R\rangle)/\sqrt{2}$ as a dark state, meaning that its matrix element with $|e\rangle$ vanishes. It therefore remains at zero energy. The symmetric, or bright, combination couples to $|e\rangle$ and receives the energy shift $-2g^2/\Delta$.

The three-state model can be diagonalized exactly, which exposes what the truncated series keeps and drops. The lower bright-state energy is

$$E_- = \frac{\Delta - \sqrt{\Delta^2 + 8g^2}}{2} = -\frac{2g^2}{\Delta} + \frac{4g^4}{\Delta^3} + O\left(\frac{g^6}{\Delta^5}\right).$$

Expanding the square root in powers of the small ratio shows perturbative control directly. The leading retained energy shift is of order g^2/Δ .

In this model, symmetry eliminates the cubic term, so the first omitted energy correction is fourth order. Without that symmetry, truncation at second order generally leaves third-order corrections.

Decomposition into retained and eliminated subspaces

Separate the solvable part from the coupling that mixes the subspaces:

$$H = H_0 + \lambda V.$$

Here H_0 is the exactly solvable part and V carries units of energy. The dimensionless number λ only counts powers in the expansion; once the physically small ratio is identified, set 1 to recover the real Hamiltonian.

The projector P selects the retained subspace, while $Q = I - P$ selects all eliminated states. Choose P to be spanned by eigenstates of H_0 , so that $PH_0Q = 0$. The Hamiltonian then has the block form

$$H = \begin{pmatrix} PHP & PHQ \\ QHP & QHQ \end{pmatrix}.$$

The diagonal blocks act inside one subspace each. The off-diagonal blocks carry the mixing between them. Keeping only the retained diagonal block replaces H by PHP and therefore removes the off-diagonal blocks. A higher-order elimination instead retains the effects of transitions from P into Q and back.

Let E_a denote an unperturbed energy in P , and let E_m denote an unperturbed energy in Q . Define the minimum spectral separation by

$$\Delta = \min_{a \in P, m \in Q} |E_m - E_a|.$$

When P is a genuine low-energy subspace, the relevant differences $E_m - E_a$ are positive. The perturbation is controlled not by the statement that V is small in isolation, but by the dimensionless ratio

$$\epsilon = \frac{|\lambda|v}{\Delta} \ll 1,$$

where v is the characteristic off-diagonal coupling scale, such as a suitable operator norm of $PVQ + QVP$. Both v and Δ have units of energy, so ϵ is dimensionless.

What controls the expansion is therefore a ratio of energy scales, not the formal split $H = H_0 + V$.

When a level in Q drifts toward a level in P , the gap Δ shrinks and ϵ grows. The nearly resonant state belongs inside the retained subspace, so enlarge P to include it. A small denominator signals that the chosen low-energy model has broken down; it never offers an arbitrarily large usable coupling.

Exact energy-dependent effective Hamiltonian

Split an exact eigenstate into its retained and eliminated parts $|\psi\rangle = |p\rangle + |q\rangle$, where $|p\rangle = P|\psi\rangle$ and $|q\rangle = Q|\psi\rangle$. Projecting the eigenvalue equation $H|\psi\rangle = E|\psi\rangle$ into the two subspaces gives

$$PHP|p\rangle + PHQ|q\rangle = E|p\rangle,$$

$$QHP|p\rangle + QHQ|q\rangle = E|q\rangle.$$

When $E - QHQ$ can be inverted, the second row gives the eliminated part in terms of the retained part:

$$|q\rangle = (E - QHQ)^{-1}QHP|p\rangle.$$

Insert that expression in the first row. The retained part then obeys a closed equation with an energy-dependent effective Hamiltonian:

$$H_{\text{eff}}(E) = PHP + PHQ(E - QHQ)^{-1}QHP.$$

This projection procedure is associated with Feshbach [\[R171\]](#). The inverse operator provides the energy denominator. Replacing E and QHQ by their unperturbed values generates the perturbation series.

When the retained manifold is degenerate with unperturbed energy E_0 , the effective Hamiltonian through second order is

$$H_{\text{eff}} = E_0 P + \lambda PVP + \lambda^2 PVQ \frac{1}{E_0 - QH_0Q} QVP + O\left(\frac{|\lambda|^3 v^3}{\Delta^2}\right)$$

for a bounded finite system whose relevant perturbation scales are represented by v . If every eliminated state lies above E_0 , the second-order term can equivalently be written as

$$H_{\text{eff}}^{(2)} = -PVQ \frac{1}{QH_0Q - E_0} QVP.$$

These expressions are equivalent. The second form makes the sign explicit because $QH_0Q - E_0$ is positive.

Through second order, the effective low-energy Hamiltonian therefore combines the projected first-order block PHP , a sum over virtual paths through Q , and a remainder of order v^3/Δ^2 .

For basis states $|a\rangle, |b\rangle \in P$ and $|m\rangle \in Q$,

$$\langle a|H_{\text{eff}}^{(2)}|b\rangle = \sum_{m \in Q} \frac{\langle a|V|m\rangle \langle m|V|b\rangle}{E_0 - E_m}.$$

Every allowed virtual path contributes one term to this sum, and different paths interfere with their own relative phases. A small denominator enlarges one term while shrinking the expansion parameter that justifies keeping only a few orders.

Block diagonalization by a unitary transformation

Elimination can instead proceed as a perturbative change of basis. Pick an anti-Hermitian generator S , defined by $S^\dagger = -S$, and take it to be off-diagonal with respect to the P and Q subspaces. Define the transformed Hamiltonian

$$\widetilde{H} = e^S H e^{-S}.$$

Because S is anti-Hermitian, e^S is unitary. Therefore, H and $\widetilde{H}$ have the same exact spectrum. Expand the generator as $S = \lambda S_1 + \lambda^2 S_2 + \dots$, and use the nested-commutator expansion

$$e^S H e^{-S} = H + [S, H] + \frac{1}{2}[S, [S, H]] + \dots.$$

Split $V = V_d + V_{\text{od}}$ into a block-diagonal part V_d acting inside P or Q and a mixing part V_{od} connecting them. Removing the first-order mixing imposes

$$[S_1, H_0] = -V_{\text{od}}.$$

It follows that

$$\langle a|S_1|m\rangle = \frac{\langle a|V|m\rangle}{E_a - E_m}.$$

Projecting the transformed Hamiltonian into P gives

$$H_{\text{eff}} = PH_0P + \lambda PVP + \frac{\lambda^2}{2}P[S_1, V_{\text{od}}]P + O(\lambda^3).$$

If the retained states are not exactly degenerate, the second-order matrix element contains the Hermitian average of the two energy denominators:

$$\frac{1}{2} \sum_m V_{am} V_{mb} \left(\frac{1}{E_a - E_m} + \frac{1}{E_b - E_m} \right).$$

The second-order interaction therefore comes from a change of basis that removes retained-eliminated mixing one order at a time.

Different bases inside the retained subspace redistribute the effective coefficients while leaving all low-energy predictions unitarily equivalent [R025]. Any comparison of coefficients must therefore specify the basis convention.

This change of basis is the Schrieffer–Wolff transformation. Schrieffer and Wolff introduced it for the Anderson-to-Kondo reduction, and it now serves clusters and many-body systems with a low-energy block coupled to high-energy states [R024]; [R025]. It is a formulation of degenerate perturbation theory as a change of basis.

Two-site Hubbard model and antiferromagnetic exchange

The Hubbard model follows electrons that hop between sites, with each site charging an on-site energy cost when two electrons share it [R172]. Consider two sites, labeled 1 and 2, occupied by exactly two electrons. Let $c_{i\sigma}^\dagger$ create an electron with spin $\sigma \in \{\uparrow, \downarrow\}$ on site i , and define the number operator $n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$. The Hamiltonian components are

$$H_0 = U \sum_{i=1}^2 n_{i\uparrow} n_{i\downarrow},$$

$$V = -t \sum_{\sigma} (c_{1\sigma}^\dagger c_{2\sigma} + c_{2\sigma}^\dagger c_{1\sigma}).$$

Here $U > 0$ is the energy cost of double occupancy, and t is the hopping matrix element. Both quantities have units of energy. Set $\lambda = 1$ and assume $|t|/U \ll 1$.

The retained sector P contains states with one electron on each site:

$$|\uparrow, \uparrow\rangle, \quad |\uparrow, \downarrow\rangle, \quad |\downarrow, \uparrow\rangle, \quad |\downarrow, \downarrow\rangle.$$

All four states have unperturbed energy $E_0 = 0$. The eliminated sector Q contains the two doublon-hole states, where a doublon is a doubly occupied site and the other site is empty:

$$|D_1\rangle = |\uparrow\downarrow, 0\rangle, \quad |D_2\rangle = |0, \uparrow\downarrow\rangle,$$

both with energy U . For this two-site problem, the gap is $\Delta = U$, and $PVP = 0$ because a single hop always produces one doublon and one hole.

Reorganize the retained states by total spin:

$$|S\rangle = \frac{|\uparrow, \downarrow\rangle - |\downarrow, \uparrow\rangle}{\sqrt{2}},$$

$$|T_0\rangle = \frac{|\uparrow, \downarrow\rangle + |\downarrow, \uparrow\rangle}{\sqrt{2}}, \quad |T_+\rangle = |\uparrow, \uparrow\rangle, \quad |T_-\rangle = |\downarrow, \downarrow\rangle.$$

Two electrons sharing one orbital must form a spin singlet, since the triplet spatial wavefunction vanishes there. The polarized triplets therefore have no hopping amplitude out, and the two hopping paths from $|T_0\rangle$ cancel. With a consistent convention for fermionic phases,

$$V|S\rangle = -\sqrt{2}t(|D_1\rangle + |D_2\rangle), \quad V|T_\mu\rangle = 0.$$

The total squared matrix element connecting the singlet to Q is $4t^2$. The energy denominator is $0 - U = -U$, giving

$$\delta E_S^{(2)} = -\frac{4t^2}{U}, \quad \delta E_T^{(2)} = 0.$$

The singlet drops below the triplets. That splitting defines antiferromagnetic exchange: a positive exchange coefficient J favors antiparallel spins. Within the singly occupied sector, the effective operator is

$$\boxed{H_{\text{eff}}^{(2)} = J \left(\mathbf{S}_1 \cdot \mathbf{S}_2 - \frac{1}{4} n_1 n_2 \right), \quad J = \frac{4t^2}{U} > 0,}$$

where $\mathbf{S}_i$ is the spin-1/2 operator at site i , and $n_i = n_{i\uparrow} + n_{i\downarrow}$. At half filling within P , $n_1 n_2 = 1$. Because $\mathbf{S}_1 \cdot \mathbf{S}_2 = -3/4$ in the singlet and $+1/4$ in a triplet, this expression gives energies $-J$ and 0 , respectively.

Two virtual hops through the doubly occupied intermediates therefore build an antiferromagnetic Heisenberg coupling with coefficient $4t^2/U$.

Many texts drop the constant $-J/4$, giving the shorter convention $J\mathbf{S}_1 \cdot \mathbf{S}_2$. This convention shifts every retained energy by $+J/4$ but leaves the singlet-triplet splitting equal to J . Comparing the two conventions without accounting for this constant produces an apparent discrepancy equal to that constant.

Units check out because $[t] = [U] = \text{energy}$:

$$[J] = \frac{[t]^2}{[U]} = \text{energy}.$$

For a concrete arithmetic illustration, unrelated to any particular defect, take $t = 1 \text{ meV}$ and $U = 10 \text{ meV}$. Then $t/U = 0.1$, and

$$J^{(2)} = 4 \frac{(1 \text{ meV})^2}{10 \text{ meV}} = 0.4 \text{ meV}.$$

These numbers fix the size of the effect; they describe no particular defect.

The two-site problem also permits direct evaluation of the truncation error. The exact singlet-triplet splitting is

$$J_{\text{exact}} = \frac{\sqrt{U^2 + 16t^2} - U}{2} = \frac{4t^2}{U} - \frac{16t^4}{U^3} + O\left(\frac{t^6}{U^5}\right).$$

At the illustrative ratio $t/U = 0.1$, $J_{\text{exact}} \approx 0.385 \text{ meV}$, approximately 3.9% below the second-order result. Odd perturbative orders vanish in this specific two-site, half-filled model.

That cancellation comes from the symmetry and path structure of this minimal model. Larger lattices or fillings away from half filling admit odd-order contributions.

Carrying the expansion further in t/U for the Hubbard model produces further-neighbor and multi-spin interactions at higher order [Ro26]. The superexchange mechanism and the origin of its antiferromagnetic sign also have a broader history beyond this minimal model [R173].

Quantitative conditions for perturbative control

For one finite cluster with typical mixing matrix element v , the required hierarchy is

$$\underbrace{|\lambda|v}_{\text{mixing}} \ll \underbrace{\Delta}_{\text{leakage cost}}.$$

Under this hierarchy, a low-energy eigenstate carries an eliminated-sector amplitude of order $\epsilon = |\lambda|v/\Delta$. A dressed state means an exact or perturbatively corrected state mixing components from both P and Q . A generated second-order energy has scale

$$\epsilon^2 \Delta = \frac{|\lambda|^2 v^2}{\Delta},$$

and a generic third-order remainder has scale

$$\epsilon^3 \Delta = \frac{|\lambda|^3 v^3}{\Delta^2}.$$

These relations estimate scales; they are not error bounds with unit coefficients. Multiple bonds, near-degeneracies, and many virtual paths each bring combinatorial factors.

For an array of clusters where each cluster touches z , defined as the number of locally coupled neighboring clusters, a conservative local condition is often $zv/\Delta \ll 1$. This estimate should be supplemented by an explicit bound or convergence study for the actual local Hamiltonian. A global operator norm generally grows with system size and can therefore impose an unnecessarily restrictive criterion.

Local Schrieffer–Wolff methods and linked-cluster expansions handle extensive systems systematically [Ro25].

A usable perturbative derivation therefore names four quantities: the retained space P , the smallest relevant gap Δ , the dimensionless expansion parameter, and the leading omitted operators together with their characteristic scale.

A derivation that quotes only the desired generated term leaves perturbative control undemonstrated: without the gap, the expansion parameter, and the omitted scale, no reader can check the series.

Experimental identification and validation

The Schrieffer–Wolff transformation is [Theory]. A laboratory cluster is characterized by measured energy levels, transition matrix elements, applied drives, disorder, dissipation, and temperature. Claiming that such a cluster realizes a Hamiltonian derived by this method requires matching those laboratory observables to H_0 , V , P , and Q .

In a defect cluster, P might be an isolated spin doublet: two nearly degenerate spin states split from the rest. The gap Δ could represent a many-spin excitation gap, an orbital splitting, or a charge-transfer energy.

Each denominator belongs to a physically distinct excitation — a many-spin rearrangement, an orbital promotion, or a charge transfer — and each responds differently to noise. The coupling V could arise from magnetic dipole coupling, direct exchange, strain coupling, or a driven interaction.

A magnetic dipole interaction already acts inside the doublet, so it enters at first order through PVP ; calling it superexchange mislabels a direct process. Superexchange means the system leaves P for a charge-transfer or mediator state and returns, which needs at least two applications of the coupling.

When a ligand or an extra defect orbital mediates the exchange, the single denominator U is replaced by several intermediate-state energies and hopping amplitudes. The corresponding contributions can compete.

Orbital occupancy and Hund coupling — the intra-site exchange favoring parallel spins — can outweigh the kinetic mechanism and flip the sign of the effective interaction. The textbook result $4t^2/U$ from the two-site model is therefore a derivation template rather than a universal materials formula [R173].

Validation is spectroscopic. Measure or calculate the retained manifold and the leakage levels, estimate the matrix elements that connect them, diagonalize at least the smallest relevant microscopic cluster, and compare its low-energy spectrum and projected observables with the truncated effective model. [Numerics] Exact diagonalization tests both coefficients and truncation errors without assuming the perturbation series converges. Agreement at a single tuned point counts less than agreement over a range of v/Δ .

[Proposal] Later cluster constructions will use this expansion to test whether available two-body defect couplings generate specified encoded interactions. The transformation by itself guarantees none of the three things such a proposal needs: a large desired coefficient, cancellation of unwanted terms, or a topological low-energy phase.

Elimination is a change of description, not a change of device. The crystal keeps all its microscopic defect and mediator states. The operator H_{eff} describes an encoded low-energy sector of the same physical system.

A generated spin interaction records a two-cluster energy shift. Emergent anyons and topological order require the full many-body phase with its excitation gap and nonlocal observables, so they need separate evidence. A digitally programmed gate sequence can emulate the same operator while lacking the static microscopic hierarchy derived here; simulation and material realization remain distinct categories.

Common errors in effective-Hamiltonian derivations

- **Using the wrong denominator sign.** Write the denominator from the retained energy E_0 and a higher intermediate energy E_m , the denominator is $E_0 - E_m < 0$. Equivalently, the term can be written using an explicit minus sign and the positive energy cost $E_m - E_0$. These two sign conventions must not be applied simultaneously.
- **Eliminating high-energy states before including return processes.** Include the round trip: replacing H by PHP omits $PVQ(E_0 - QH_0Q)^{-1}QVP$. First-order projection and second-order elimination compute different approximations.
- **Applying the expansion without spectral separation.** Check the gap first: if a Q level approaches P , the corresponding denominator becomes small and the mixing becomes large. The retained space must then be enlarged. The divergence indicates failure of the selected low-energy description rather than an available source of arbitrarily strong coupling.
- **Comparing basis-dependent coefficients term by term.** Fix the basis convention first: effective Hamiltonians related by a unitary rotation within P can distribute coefficients differently while predicting the same spectrum. Comparisons should use invariant predictions or consistently transformed operators [R025].
- **Neglecting dressed observables.** Transform the observable too: physical low-energy states contain a Q -sector admixture of order v/Δ . When this correction is relevant, an observable O must be transformed as $Pe^SOe^{-S}P$, rather than approximated only by POP .

- **Ignoring lower-order unwanted terms.** List all symmetry-allowed terms through the working order before celebrating a fourth-order interaction: it may sit far below an allowed first- or second-order contribution. Lower-order terms need a symmetry reason to vanish, an explicit cancellation, or tuned parameters. Renaming operators leaves their size unchanged.
 - **Assuming that the existence of a small coefficient implies its accuracy.** Compare a nearly canceled coefficient against the remainder, not against zero: when virtual paths nearly cancel, the leftover can lie below the nominal perturbative remainder. Good control of the full Hamiltonian still leaves a large relative error on that specially suppressed term.
 - **Inferring coherence from a closed-system Hamiltonian.** Carry the environment through the transformation: block diagonalization leaves phonons, fluctuating fields, thermal activation, and control noise in place. Coupling to an environment produces real leakage, so effective noise operators need their own projection.
 - **Identifying generated interactions with a topological phase.** Treat a generated multi-body operator as one entry in a checklist: the full operator content, coefficient hierarchy, system size, temperature, and perturbative stability each need separate evidence.
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Verification exercises

- **Definitions of P and Q .** A projector selects a specified subspace and leaves vectors inside it unchanged. P projects onto the low-energy states retained in the effective model. $Q = I - P$, where I is the identity operator, projects onto the complementary high-energy states excluded from that model.
- **Scale of a second-order virtual process.** A virtual process passes through an intermediate state that appears in no initial or final state. Let v denote the characteristic coupling between the retained and excluded subspaces, and let Δ denote the excitation energy of the intermediate state. A process that leaves the P subspace and returns to it requires two applications of the coupling. The intermediate resolvent, which is the inverse energy-difference operator, contributes one inverse excitation energy. The resulting second-order scale is therefore v^2/Δ , because the contribution has the form $v \cdot (1/\Delta) \cdot v$.
- **Breakdown near an intermediate-state resonance.** Let E_0 be the energy of a retained low-energy state and E_m the energy of an intermediate excluded state. If E_m approaches E_0 , the perturbative denominator $E_0 - E_m$ becomes small. Consequently, the dimensionless expansion parameter $\epsilon = v/\Delta$ is no longer small, and the retained low-energy subspace must be enlarged to include the nearly degenerate state. The effective-model formula does not imply an arbitrarily large coupling in this regime; instead, its assumptions have failed.
- **Sign of Hubbard exchange.** In the Hubbard model, t is the hopping amplitude and U is the on-site interaction energy. At second order, only the spin singlet receives the energy correction $-4t^2/U$, while the triplet states remain at zero. Expressing this singlet-triplet splitting as $J(\mathbf{S}_1 \cdot \mathbf{S}_2 - 1/4)$, where $\mathbf{S}_1$ and $\mathbf{S}_2$ are the two spin operators, gives $J = 4t^2/U > 0$. A positive J corresponds to antiferromagnetic exchange, meaning that the interaction energetically favors the singlet over the triplets. Thus, the negative singlet energy correction is consistent with antiferromagnetic exchange.
- **Required information for a second-order effective Hamiltonian.** A second-order effective Hamiltonian must be accompanied by the definition of P , the leakage gap Δ , a dimensionless small parameter such as v/Δ , and the scale and operator form of the leading omitted corrections. The leakage gap is the energy separation between the retained subspace and the excluded states into which the system can couple.

- **Limitations of a generated plaquette term.** A Schrieffer–Wolff transformation perturbatively decouples retained low-energy states from excluded high-energy states. A plaquette term from this transformation lives on the plaquette degrees of freedom and stays one term among others in the effective Hamiltonian. Topological order needs the many-body phase, its energy gap and excitations, its robustness, and its finite-size behavior as separate evidence.
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Sources

- [R024] J. R. Schrieffer and P. A. Wolff, “Relation between the Anderson and Kondo Hamiltonians,” *Physical Review* 149, 491–492 (1966). DOI: [10.1103/PhysRev.149.491](https://doi.org/10.1103/PhysRev.149.491).
 - [R025] S. Bravyi, D. P. DiVincenzo, and D. Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” *Annals of Physics* 326, 2793–2826 (2011). DOI: [10.1016/j.aop.2011.06.004](https://doi.org/10.1016/j.aop.2011.06.004); arXiv: [1105.0675](https://arxiv.org/abs/1105.0675).
 - [R171] H. Feshbach, “Unified theory of nuclear reactions,” *Annals of Physics* 5, 357–390 (1958). DOI: [10.1016/0003-4916\(58\)90007-1](https://doi.org/10.1016/0003-4916(58)90007-1).
 - [R172] J. Hubbard, “Electron correlations in narrow energy bands,” *Proceedings of the Royal Society A* 276, 238–257 (1963). DOI: [10.1098/rspa.1963.0204](https://doi.org/10.1098/rspa.1963.0204).
 - [R026] A. H. MacDonald, S. M. Girvin, and D. Yoshioka, “ t/U expansion for the Hubbard model,” *Physical Review B* 37, 9753–9756 (1988). DOI: [10.1103/PhysRevB.37.9753](https://doi.org/10.1103/PhysRevB.37.9753).
 - [R173] P. W. Anderson, “New Approach to the Theory of Superexchange Interactions,” *Physical Review* 115, 2–13 (1959). DOI: [10.1103/PhysRev.115.2](https://doi.org/10.1103/PhysRev.115.2).
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Chapter 23 — Perturbative gadgets

Consider four spins around a square. The target topological Hamiltonian assigns an energy set by the product of all four spin operators. A crystal offers a narrower starting point: pairwise interactions between neighboring spins, with no direct four-spin coupling built in.

A perturbative gadget bridges that gap. It adds an auxiliary spin, called an ancilla or mediator, with a large excitation cost, and couples each of the four target spins weakly to it through two-body links. A virtual process then visits the excited mediator once per coupling and returns it to its initial state. Seen from the low-energy subspace, the round trip acts as an effective four-body interaction built only from available few-body pieces.

The construction charges two prices. It needs extra hardware: the mediator and its links. The generated four-body energy also sits far below the two-body couplings it comes from. Ordinary pair interactions appear at lower perturbative order, so they outweigh the desired term until additional counterterm interactions cancel them.

Single-mediator construction for four target spins

Four target spins surround one central mediator:

```
target 1
|
target 4 -- ancilla -- target 2
|
target 3

strong cost to excite ancilla:  $\Delta$ 
weak two-body links:  $g_i$ 
```

Give the mediator a large excitation energy Δ . A single weak interaction can transfer it from the low-energy subspace to the excited subspace. Any process contributing to an effective operator within the low-energy subspace must eventually return the mediator to its initial state. If all four target spins participate before this return, the resulting amplitude can depend on the product of four target-spin operators.

The mediator thus ferries the influence of the targets without ending excited. Each intermediate visit is virtual: it enters the transition amplitude without leaving a stable excited population for an energy measurement to find. Every visit contributes one weak coupling to the numerator and one energy denominator, so a process with four weak interactions carries the characteristic scale

$$K \sim \frac{g^4}{\Delta^3},$$

where g , Δ , and the effective coupling K all have units of energy. Dimensional consistency follows from $E^4/E^3 = E$, as required for a Hamiltonian coefficient.

A four-body interaction built from four two-body links therefore has scale g^4/Δ^3 , rather than scale g . This scaling estimate omits combinatorial coefficients and lower-order terms, both of which must be included in a complete analysis.

Lower-order pair interactions

Let the four targets be spin-1/2 particles numbered $i = 1, 2, 3, 4$. For each target, the Pauli operator Z_i has eigenvalues $z_i = \pm 1$. Let the mediator have ground state $|0\rangle_a$, excited state $|1\rangle_a$, and excitation energy $\Delta > 0$. Define

$$H_0 = \Delta |1\rangle_a \langle 1|, \quad V = X_a S, \quad S = \sum_{i=1}^4 g_i Z_i.$$

Here H_0 is the unperturbed mediator Hamiltonian, V is the perturbation, X_a is the Pauli operator that flips the mediator, and g_i is the energy coefficient of the two-body coupling $X_a Z_i$. An interaction is two-local if it acts nontrivially on at most two subsystems; every term in V is therefore two-local. The projector onto the low-energy mediator state is

$$P = |0\rangle_a \langle 0| \otimes I_{1\dots 4}.$$

All target operators Z_i commute, so the problem can be evaluated separately in each simultaneous target-spin eigenstate. For fixed target eigenvalues, S is a scalar, and the mediator Hamiltonian has the matrix representation

$$\begin{pmatrix} 0 & S \\ S & \Delta \end{pmatrix}.$$

Its lower eigenvalue can then be interpreted as an operator on the target spins:

$$E_-(S) = \frac{\Delta - \sqrt{\Delta^2 + 4S^2}}{2} = -\frac{S^2}{\Delta} + \frac{S^4}{\Delta^3} - \frac{2S^6}{\Delta^5} + O\left(\frac{S^8}{\Delta^7}\right).$$

This expansion assumes that S/Δ is sufficiently small for perturbation theory to apply. The first nonconstant contribution occurs at second order rather than fourth order. Expanding it gives

$$-\frac{S^2}{\Delta} = -\frac{\sum_i g_i^2}{\Delta} I - \sum_{i < j} \frac{2g_i g_j}{\Delta} Z_i Z_j.$$

The term proportional to the identity I shifts all energies by the same amount and does not change energy differences. The remaining six terms are pairwise target interactions. They are unwanted and are parametrically larger than the desired four-body interaction. At fourth order,

$$\frac{S^4}{\Delta^3} \supset \frac{24g_1g_2g_3g_4}{\Delta^3} Z_1Z_2Z_3Z_4.$$

The factor $24 = 4!$ counts the possible orders in which the four distinct mediator–target links can act. Changing the sign of one g_i changes the sign of the resulting four-body coefficient. Deliberately added two-body counterterms, with coefficients tuned to negate the unwanted pair contributions, can cancel the pair interactions through fourth order. For completeness, the fourth-order coefficient multiplying a particular Z_iZ_j is

$$\frac{4g_i g_j (g_i^2 + g_j^2) + 12g_i g_j \sum_{k \neq i,j} g_k^2}{\Delta^3}.$$

A calibrated counterterm must negate this fourth-order coefficient together with the corresponding second-order coefficient. After this cancellation, the effective low-energy Hamiltonian is

$$H_{\text{eff}} = K Z_1 Z_2 Z_3 Z_4 + O(g^6/\Delta^5), \quad K = \frac{24g_1g_2g_3g_4}{\Delta^3}.$$

This conversion produces a many-body interaction from two-body couplings, with auxiliary hardware and a reduced energy scale as costs. That combination is what makes the construction a perturbative gadget.

The four-body term is present but subleading. The leading nonconstant terms are pair interactions, larger by two powers of g/Δ . They must be cancelled before the four-body term becomes the dominant nonconstant operator in the low-energy subspace.

Count the hardware: one mediator, four mediator–target links, and six nominal target–target counterterm links. The algebra shows the interaction structure exists; whether a specific crystal can supply all ten links is a separate materials question.

The mediator and the four targets may be bare physical spins or logical spins encoded into clusters. In either case, $Z_1Z_2Z_3Z_4$ is an effective interaction. Neither the use of encoded spins nor the generation of this operator alone produces an emergent anyon, meaning a quasiparticle with topological exchange and fusion properties. Toric-code quasiparticles arise only in an extended, gapped lattice Hamiltonian within the appropriate topological phase.

A digital four-qubit gate can reproduce the same time evolution. That gate sequence simulates the operator in time; it does not install it as a passive static energy term.

Magnitude and accuracy of the four-body interaction

For equal coupling magnitudes $|g_i| = g$, define the dimensionless perturbative expansion parameter $\eta = g/\Delta$. Then

$$\frac{|K|}{\Delta} = 24\eta^4.$$

The sixth-order contribution to the same four-body operator is $-960g^6/\Delta^5$, so its magnitude relative to the leading coefficient K is $40\eta^2$.

At $\eta = 0.05$, the leading four-body scale is only $1.5 \times 10^{-4}\Delta$, the sixth-order correction is 10% of it, and the leading unwanted pair scale $2g^2/\Delta = 5 \times 10^{-3}\Delta$ is about 33 times larger.

Keeping the residual pair interaction below $0.1K$ requires approximately 0.3% cancellation in this illustrative cell. At $\eta = 0.025$, the sixth-order fraction decreases to 2.5%, but K/Δ decreases to 9.375×10^{-6} .

Smaller η tightens the perturbative expansion and simultaneously shrinks the desired effective coupling K . [Theory] These numbers follow directly from the expansion above; they are not measured device performance.

As an illustrative conversion of units, suppose—not claim—that $\Delta/h = 1$ GHz, where h is Planck’s constant. At $\eta = 0.05$, $K/h = 150$ kHz, and the dynamical time is

$$\hbar/K = 1/[2\pi(K/h)] \approx 1.1 \mu\text{s}.$$

A gigahertz microscopic scale therefore produces a kilohertz-scale effective coupling here. Large bare energies do not guarantee a large topological gap.

A toric-code plaquette takes the four edge Z_i operators around one square as its targets. A star interaction uses the same geometry with each Z_i replaced by X_i .

On a periodic square lattice with E target edges, the number of vertices V and plaquettes P satisfies $V + P = E$. This direct construction therefore uses one mediator for each stabilizer: E mediators for E target spins, four mediator-link incidences per stabilizer, and six counterterm incidences per stabilizer. Each target participates in four gadgets.

This resource count excludes routing devices and any additional encoding required to represent one target spin using a defect cluster.

Extension from one gadget to a lattice

A general gadget follows the same pattern. Write its Hamiltonian as

$$H = H_0 + V,$$

where H_0 has a low-energy subspace with projector P , the complementary projector is $Q = I - P$, and the smallest excitation energy connecting the P and Q subspaces is Δ . A perturbative sequence contributing at order n has the schematic form

$$PVQ \frac{1}{E_0 - QH_0Q} QVQ \cdots QVP,$$

with n copies of V and $n - 1$ resolvents. A resolvent is the inverse energy-denominator operator left behind after eliminating one intermediate excited state. If every weak matrix element has scale g , the resulting contribution has scale g^n / Δ^{n-1} .

Jordan and Farhi provide systematic gadget constructions at arbitrary perturbative order. Bravyi, DiVincenzo, Loss, and Terhal show that bounded-strength local target Hamiltonians can be simulated with an extensive error $O(\epsilon N)$, where epsilon denotes the per-target simulation error, without requiring simulator couplings to increase with the number N of target spins [R174]; [R175]. [Theory] These results concern Hamiltonian simulation.

Bounded-strength simulation still needs the lattice connectivity and the coherence budget from real hardware.

Three energies enter the design, and confusing them overstates the protection:

- **Penalty gap Δ** : the energy required to leave a gadget's encoded subspace.
- **Effective stabilizer coefficient K** : the coefficient of the generated low-energy stabilizer term, typically suppressed as g^n / Δ^{n-1} .
- **Topological many-body gap Δ_{top}** : the lowest allowed excitation energy of the extended effective model.

For the ideal toric-code Hamiltonian

$$H_{\text{TC}} = -K_A \sum_v A_v - K_B \sum_p B_p,$$

A_v and B_p are the star and plaquette stabilizers, respectively. An individual violated stabilizer costs $2K_A$ or $2K_B$. On a closed surface, local operations create violations in pairs, so the smallest such pair costs $4 \min(K_A, K_B)$. The useful low-energy gap of the simulator is therefore of order K , not of order Δ . A basic hardware criterion is

$$\max(k_B T, \hbar/T_2, |\delta J|, |\delta K|) \ll \Delta_{\text{top}} \sim K \ll g \ll \Delta.$$

Here $k_B T$ is the thermal energy, T_2 is the relevant coherence time, δJ is a residual unwanted two-body energy, and δK denotes disorder in the stabilizer coefficients. Every quantity in this inequality has units of energy.

Protection therefore tracks the suppressed effective coefficient K , rather than the mediator penalty Δ .

This inequality is a design criterion rather than a universal phase boundary. The analysis of one gadget cell does not prove the behavior of an extensive lattice.

Neighboring gadgets share target spins, so a virtual path can visit several mediators before returning. The number of such cross-gadget paths grows with the coordination number, and a bound built from the norm of the full perturbation V grows with system size along with it.

Controlling local errors in a lattice of gadgets needs tools built for parallel composition: parallel-gadget constructions, locality-based Schrieffer–Wolff analyses, or exact subsystem symmetries [R175]; [R177]. A Schrieffer–Wolff analysis is a perturbative block-diagonalization that derives the effective low-energy Hamiltonian by eliminating couplings to high-energy states. Finally, proving that an approximate effective Hamiltonian has the intended ground space is weaker than proving a stable thermodynamic gap and the intended excited anyons.

Established perturbative and nonperturbative constructions

Target	Microscopic construction	First target order / resource cost	What is established	Important limit
Toric-code phase from the honeycomb model	Nearest-neighbor XX , YY , or ZZ bonds; strong J_z dimers encode effective spins	Fourth order; two physical spins per effective dimer	In the anisotropic limit, $K = J_x^2 J_y^2 / (16 J_z ^3)$ multiplies an effective four-dimer plaquette operator [R017]; [R176]. [Theory]	Natural and exactly solvable, but its non-Abelian phase is Ising type, not Fibonacci anyons (the distinct non-Abelian fusion theory).
Toric code by code gadgets	Each target edge qubit is encoded into four physical qubits; inter-gadget couplings are two-body	Fourth order on a square lattice; four physical qubits per target qubit	Encoded star and plaquette terms arise; odd orders vanish and second order is a constant because the gadget detects single-qubit errors [R177]. [Theory]	The low-energy model is approximate, frustrated, and has a suppressed gap.
Finite-group quantum doubles $D(G)$	Two-body code gadgets using $ G $ -level qudits	Four physical qudits per target edge in the square-lattice construction [R177]	Includes non-Abelian anyonic models when G is non-Abelian. [Theory]	A non-Abelian quantum double is not automatically Fibonacci; local dimension and operator control grow with group structure.
Simplified quantum doubles	A hopping/clock gadget generates an ordered operator product	Order follows the clock path; König reduces quantum-double interactions to simpler local terms [R178]. [Theory]	Ordered noncommuting products demand more structure than the commuting parity toy above.	
String-net PEPS	Strong two-body terms define an encoded PEPS subspace; weak bonds generate a parent Hamiltonian	Finite order set by the local interaction region; additional virtual degrees of freedom per PEPS tensor	A two-body parent construction is proved for a broad PEPS class, with an explicit double-semion string-net example [R018]; [R179]. [Theory]	Double semion is Abelian. The paper is not an experimental doubled-Fibonacci defect design.

Target	Microscopic construction	First target order / resource cost	What is established	Important limit
Exact two-body alternative	Enlarged “composite particles” and two-body interactions	Extra local Hilbert space rather than a small perturbation parameter	Ocko and Yoshida construct nonperturbative two-body models for toric-code and quantum-double topological codes [R180]. [Theory]	Avoids perturbative gap suppression but retains substantial local-state and coupling complexity.

The honeycomb model tests the same numerator-denominator structure on an exactly solvable lattice. In the strong- J_z limit, each z -bond dimer (a paired two-spin unit) has a two-dimensional low-energy doublet.

A closed sequence containing two J_x bond operations and two J_y bond operations returns every dimer to this low-energy subspace. Lower perturbative orders only shift energies, while fourth order generates the plaquette term with coefficient $J_x^2 J_y^2 / (16 |J_z|^3)$ [R017]; [R176].

The coefficient again puts coupling factors upstairs and excitation energies downstairs, as in the mediator model. In the honeycomb lattice, symmetry and exact solvability pin down the unwanted terms instead of leaving them for counterterms.

Non-Abelian excitations are not automatically Fibonacci anyons. König’s constructions and those of Brell et al. produce quantum doubles of finite groups, including some theories with non-Abelian excitations [R177]; [R178].

Levin–Wen models belong to the broader class of categorical string-net models [R018]. The PEPS gadget framework treats the double-semion model explicitly and argues for a broad class of suitable PEPS [R179]. A PEPS, or projected entangled-pair state, is a tensor-network representation of a many-body state. These results do not constitute a worked, defect-mapped two-body Hamiltonian for doubled Fibonacci order.

[Proposal] Turning this machinery into a doubled-Fibonacci defect device needs two further steps: picking a specific doubled-Fibonacci tensor and mapping its interaction graph onto fabricable defect couplings. General string-net existence theorems complete neither step.

Physical requirements for a defect implementation

A defect device starts from electronic and nuclear energy levels, selection rules (which transitions are allowed), dipolar interaction tensors (direction-dependent couplings), exchange pathways, strain responses, optical excited states, and disorder. Abstract Pauli operators come later, as a description of the encoded subspace. Mapping a perturbative gadget onto that hardware means answering six concrete questions.

First, the penalty needs a physical origin: Δ . Possible sources include a crystal-field splitting between local levels, a driven dressed-state gap created by control fields, or an intra-cluster penalty. Regardless of its origin, the control fields and strain that establish Δ must preserve a well-isolated low-energy doublet. Additional leakage levels replace the simple two-level energy denominator with several denominators, potentially having opposite signs.

Second, every operator the gadget uses must exist in hardware. The parity gadget requires $Z_i X_a$, while a complete toric-code Hamiltonian also requires the basis-rotated analogue used for star operators.

A fixed magnetic dipole interaction gives a geometry-dependent tensor, not independently switchable pure Pauli products. Echo sequences or dressing fields can average away selected terms. Periodic driving used this way produces its effective Hamiltonian by Floquet engineering, meaning that the drive shapes the time-averaged interactions. The resulting driven realization must budget heating and micromotion, the residual within-period motion.

Third, one mediator must couple to its four targets selectively. Dipolar interactions decay with distance but never switch off past a strict range.

Exchange couplings fall off faster but swing sharply with atomic placement. The six counterterms include diagonal target pairs, which sit farther apart and couple less controllably than the four radial mediator–target links.

Published code-gadget constructions establish locality on abstract lattices rather than solving defect implantation and routing [R177]; [R179].

Fourth, the signs and magnitudes need calibration. At $\eta = 0.05$, the toy construction requires sub-percent cancellation merely to reduce residual pair terms to one tenth of K .

Disorder shifts both the couplings g_i and the energy denominators. A small fractional error relative to g can consequently produce a large error relative to K .

Fifth, the array must start cold inside the low-energy subspace P . The penalty suppresses real mediator excitation only when the thermal energy, drive-induced transition rates, and linewidths are small compared with Δ . Topological defects are controlled by the much smaller scale Δ_{top} , so the condition $k_B T \ll \Delta$ is insufficient.

Sixth, coherence must last across the effective dynamical timescale. That time is $\hbar/K$, not $\hbar/g$.

Mediator dephasing leaks into the low-energy dynamics through the nonzero virtual admixture of excited mediator states. Projecting the Hamiltonian alone is not enough; the noise operators need the same projection into the effective subspace.

[Experiment] None of the gadget papers reviewed in this chapter reports a crystal-defect array that realizes its complete two-body simulator Hamiltonian and diagnoses the resulting topological phase [R175]; [R177]; [R178]; [R179]; [R180]. Their results are theoretical constructions. Defect hardware may eventually supply the required components; assembling connectivity, anisotropy, scale hierarchy, cooling, and calibration into one working array remains a proposal-level mapping problem.

Common conceptual and implementation errors

- **Identifying the penalty gap with the protection scale.** Compare each threat to the right gap: the mediator excitation may cost Δ , while anyonic excitations cost only order K . Thermal and coherent protection must therefore be evaluated relative to K .
- **Neglecting lower perturbative orders.** In the single-mediator example, count orders from the bottom: the desired term first appears at fourth order, while larger pair interactions already appear at second order. The order carrying the desired operator is rarely the first nonzero order.
- **Treating counterterms as purely formal corrections.** Budget each counterterm as hardware: an additional physical interaction with its own noise, range, and calibration error.

- **Extrapolating directly from one cell to an extended lattice.** Shared spins open virtual paths that visit neighboring gadgets and write cross-gadget operators. Extensivity and a thermodynamic gap need their own analysis past the single cell.
- **Identifying an encoded target spin with a topological degree of freedom.** The four-qudit code gadget of Brell et al. detects local errors without creating extended order. Topological order belongs to the extended low-energy phase [R177].
- **Identifying every non-Abelian model with Fibonacci order.** Check fusion and braiding data explicitly: non-Abelian quantum doubles and the non-Abelian Ising phase of the honeycomb model carry data different from Fibonacci anyons.
- **Inferring the excitation structure from an effective ground-state Hamiltonian alone.** Verify excitations separately: the intended quasiparticle dispersion, localization, and braiding can all shift under higher-order corrections even when ground-space order is correct.
- **Taking the ratio arbitrarily small.** Decreasing g/Δ tightens the perturbative expansion but suppresses K , slows the effective dynamics, and imposes more stringent temperature and coherence requirements.

Verification exercises

- Derive the usual scaling g^n/Δ^{n-1} for an order- n virtual process. Such a process contains n weak matrix elements and $n - 1$ intermediate energy denominators.
- Account for the resources in the single-mediator example: one ancilla, four mediator–target links, six nominal pair counterterms, and fourth-order gap suppression for one four-spin interaction.
- Explain why Δ cannot be identified with the toric-code gap. The scale Δ penalizes departure from the gadget subspace, whereas stabilizer violations occur within the low-energy effective theory and cost order $K \ll \Delta$. Thermal and coherent protection must therefore be evaluated relative to K .
- Verify that reducing g/Δ improves the relative sixth-order error while suppressing K more rapidly. For equal $|g_i|$, the relative sixth-order contribution is $40\eta^2$, while $|K|/\Delta = 24\eta^4$. Halving η reduces the relative error by a factor of four and multiplies K by $1/16$.
- Assess whether existing non-Abelian quantum-double gadgets establish doubled-Fibonacci defect hardware. They do not. They establish Hamiltonian reductions for a different family of theories; the category, local Hilbert space, physical couplings, and phase diagnostics still require an explicit mapping.
- State the additional results required after deriving one plaquette interaction: controlled parallel composition, local errors small compared with K , a stable thermodynamic gap, and the intended extended ground-state and excitation structure.

A two-body perturbative gadget can generate a four-body interaction at fourth order and above, with its coefficient scaling as a high power of g/Δ .

Mediators, counterterms, and virtual paths shared between cells are physical resource requirements. The useful topological gap is of order K , not the penalty energy Δ .

Perturbative reductions for the toric code and finite-group quantum doubles are theoretically established results. A defect-specific doubled-Fibonacci construction remains a proposal: the target tensor, the defect mapping, and the phase diagnostics are all still open.

Sources

- [R174] S. P. Jordan and E. Farhi, “Perturbative gadgets at arbitrary orders,” *Physical Review A* 77, 062329 (2008). DOI: [10.1103/PhysRevA.77.062329](https://doi.org/10.1103/PhysRevA.77.062329); [arXiv:0802.1874](https://arxiv.org/abs/0802.1874).
 - [R175] S. Bravyi, D. P. DiVincenzo, D. Loss, and B. M. Terhal, “Quantum simulation of many-body Hamiltonians using perturbation theory with bounded-strength interactions,” *Physical Review Letters* 101, 070503 (2008). DOI: [10.1103/PhysRevLett.101.070503](https://doi.org/10.1103/PhysRevLett.101.070503); [arXiv:0803.2686](https://arxiv.org/abs/0803.2686).
 - [R017] A. Kitaev, “Anyons in an exactly solved model and beyond,” *Annals of Physics* 321, 2–111 (2006). DOI: [10.1016/j.aop.2005.10.005](https://doi.org/10.1016/j.aop.2005.10.005); [arXiv:cond-mat/0506438](https://arxiv.org/abs/cond-mat/0506438).
 - [R176] J. Vidal, K. P. Schmidt, and S. Dusuel, “Perturbative approach to an exactly solved problem: Kitaev honeycomb model,” *Physical Review B* 78, 245121 (2008). DOI: [10.1103/PhysRevB.78.245121](https://doi.org/10.1103/PhysRevB.78.245121); [arXiv:0809.1553](https://arxiv.org/abs/0809.1553).
 - [R177] C. G. Brell, S. T. Flammia, S. D. Bartlett, and A. C. Doherty, “Toric codes and quantum doubles from two-body Hamiltonians,” *New Journal of Physics* 13, 053039 (2011). DOI: [10.1088/1367-2630/13/5/053039](https://doi.org/10.1088/1367-2630/13/5/053039); [arXiv:1011.1942](https://arxiv.org/abs/1011.1942).
 - [R178] R. König, “Simplifying quantum double Hamiltonians using perturbative gadgets,” *Quantum Information and Computation* 10, 292–324 (2010). DOI: [10.26421/QIC10.3-4-9](https://doi.org/10.26421/QIC10.3-4-9); [arXiv:0901.1333](https://arxiv.org/abs/0901.1333).
 - [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110); [arXiv:cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617).
 - [R179] C. G. Brell, S. D. Bartlett, and A. C. Doherty, “Perturbative 2-body parent Hamiltonians for projected entangled pair states,” *New Journal of Physics* 16, 123056 (2014). DOI: [10.1088/1367-2630/16/12/123056](https://doi.org/10.1088/1367-2630/16/12/123056); [arXiv:1407.4829](https://arxiv.org/abs/1407.4829).
 - [R180] S. A. Ocko and B. Yoshida, “Nonperturbative gadget for topological quantum codes,” *Physical Review Letters* 107, 250502 (2011). DOI: [10.1103/PhysRevLett.107.250502](https://doi.org/10.1103/PhysRevLett.107.250502); [arXiv:1107.2697](https://arxiv.org/abs/1107.2697).
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Chapter 24 — Assessment of an eight-stage architecture

The proposed architecture chains together the following sequence of transformations.

```
host crystal
-> addressable defect spins
-> strongly coupled cluster
-> isolated two-state pseudospin
-> edge label {1, tau}
-> vertex and plaquette operators
-> doubled-Fibonacci phase
-> emergent anyons and a logical qubit
```

A host crystal is the bulk solid containing localized defects. An addressable defect spin is an electron or nuclear spin associated with such a defect that can be individually initialized, controlled, and measured. Each arrow in the sequence denotes a separate physical or computational hypothesis rather than an established transformation.

Each arrow needs its own test: name the objects on both sides and state the measurement that could reject the link. The first worked calculation uses three exchange-coupled spins. Every reduction after that adds experimental and theoretical requirements.

A defect electron or nucleus supplies a physical spin. A cluster doublet packs several physical spins into one encoded two-state subspace, separated from higher-energy cluster states when the splitting holds.

Calling the two cluster states 1 and τ only matches one basis to another; it does not create an anyon. An anyon is an emergent quasiparticle in two spatial dimensions whose exchange behavior can depart from boson or fermion statistics, and it appears only when the lattice Hamiltonian realizes the Levin–Wen construction.

A processor that prepares the same state amplitudes digitally simulates the target state. Intrinsic topological order instead requires the many-body phase to emerge from the system Hamiltonian itself.

The diagram therefore includes four distinct types of object: physical defect spins, encoded cluster states, abstract string-net labels, and emergent many-body excitations. These objects must not be treated as equivalent.

Minimal three-defect cluster

A proposed cluster size of five to twenty defects is not established by the available evidence.

The evidence discussed below does not establish a designed, mutually coupled array of five to twenty optically addressable centers forming a protected low-energy edge state. The analysis therefore starts with the smallest viable example: three effective spin-1/2 objects. This is the shortest open antiferromagnetic chain with a spinful ground doublet; here antiferromagnetic means the interaction favors lower total spin.

The chapter uses this size because the algebra is tractable, not because an optimum cluster size has been found.

Let $\mathbf{S}_i = (S_i^x, S_i^y, S_i^z)$ denote the dimensionless spin-1/2 operator at site i , with $\mathbf{S}_i^2 = 3/4$. The coupling J sets the energy scale and is measured here in joules. Spectroscopic measurements may instead report J/h in hertz, where h is Planck's constant. Consider the cluster Hamiltonian

$$H_C = J(\mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_2 \cdot \mathbf{S}_3), \quad J > 0.$$

This Hamiltonian is an ideal isotropic exchange model. Isotropic exchange means that the interaction has the same strength for the x , y , and z spin components. [Theory] The model is a worked target. It does not imply that three nearby nitrogen-vacancy (NV) centers, silicon-vacancy (SiV) centers, or defects in silicon carbide (SiC) naturally realize equal antiferromagnetic Heisenberg exchange.

Exchange-only encodings, in which logical operations are constructed from exchange interactions among three spins, are theoretically established. Controlled three-electron implementations have also been pursued in quantum dots [R114]. These results do not establish the same Hamiltonian for color centers, which are optically active point defects in a crystal.

Define the combined spin of the two end sites by $\mathbf{S}_{13} = \mathbf{S}_1 + \mathbf{S}_3$, and define the total cluster spin by $\mathbf{S} = \mathbf{S}_{13} + \mathbf{S}_2$. The Hamiltonian can then be written as

$$H_C = \frac{J}{2} (\mathbf{S}^2 - \mathbf{S}_{13}^2 - \mathbf{S}_2^2).$$

This form expresses the exchange energy in terms of angular-momentum quantum numbers and therefore determines the spectrum without direct diagonalization of the full 8×8 matrix:

End spin S_{13}	Total spin S	Energy	Degeneracy
1	1/2	$-J$	2
0	1/2	0	2
1	3/2	$J/2$	4

The ground manifold is a doublet, meaning that it contains two degenerate states. The smallest excitation energy out of this manifold is

$$\Delta_C = 0 - (-J) = J.$$

Here Δ_C is the cluster excitation gap. A convenient ground-doublet basis, with spin states ordered by sites 1, 2, 3, is

$$|\tilde{\uparrow}\rangle = \sqrt{\frac{2}{3}} |\uparrow\downarrow\uparrow\rangle - \sqrt{\frac{1}{6}} (|\uparrow\uparrow\downarrow\rangle + |\downarrow\uparrow\uparrow\rangle),$$

with $|\tilde{\downarrow}\rangle$ obtained by reversing every spin. Define the projector onto this two-state subspace by

$$P_C = |\tilde{\uparrow}\rangle\langle\tilde{\uparrow}| + |\tilde{\downarrow}\rangle\langle\tilde{\downarrow}|.$$

The effective spin operator within the projected subspace is

$$\tilde{\mathbf{S}} = P_C \mathbf{S} P_C.$$

The two projected states define a pseudospin: an effective spin-1/2 degree of freedom encoded in three physical spins. The pseudospin is not an additional particle in the crystal.

Angular-momentum addition gives the projected local spin operators:

$$P_C \mathbf{S}_1 P_C = P_C \mathbf{S}_3 P_C = \frac{2}{3} \tilde{\mathbf{S}}, \quad P_C \mathbf{S}_2 P_C = -\frac{1}{3} \tilde{\mathbf{S}}.$$

Consequently, if the three-spin chain is accurately described by this Hamiltonian, it has a unique ground doublet and its three local magnetic responses occur in the ratio $2 : -1 : 2$.

The relative signs and magnitudes are experimentally testable predictions. Failure to observe the predicted spectrum and matrix elements would falsify this cluster model, making construction of a lattice from such clusters unjustified.

Effective coupling between two cluster pseudospins

Consider two clusters, A and B , coupled through their endpoints by

$$V_{AB} = j \mathbf{S}_{3,A} \cdot \mathbf{S}_{1,B},$$

where j is an energy. Let $P = P_A P_B$ project both clusters onto their respective ground doublets. First-order projection gives

$$P V_{AB} P = \frac{4j}{9} \tilde{\mathbf{S}}_A \cdot \tilde{\mathbf{S}}_B.$$

This result follows because each endpoint spin projects to 2/3 of its cluster pseudospin.

Virtual leakage, meaning temporary perturbative occupation of cluster states outside the ground doublet, produces corrections of order j^2/J . The perturbative regime is therefore controlled when $|j|/J \ll 1$. The dimensional consistency is explicit: j^2/J has units of energy.

[Theory] This derivation establishes only an effective two-body exchange interaction between the pseudospins. It does not generate a string-net vertex projector, a plaquette recoupling operator, or Fibonacci exchange statistics.

The leading intercluster term is therefore Heisenberg exchange between the two pseudospins. Its strength is $4/9$ of the microscopic endpoint coupling, accompanied by leakage corrections of order j^2/J .

Each projection eliminates high-energy internal dynamics and retains a lower-energy effective two-state system. A later perturbative gadget—a construction that uses auxiliary mediator degrees of freedom to generate an effective interaction—would similarly eliminate mediator dynamics and retain a still lower-energy vertex or plaquette term.

A reduction is useful only when the input Hamiltonian is characterized accurately enough and the unwanted residual terms stay below the desired interaction. Those unwanted contributions include leakage, disorder, residual lower-order terms, and decoherence.

Each reduction lowers the relevant energy scale. The sequence can simplify the Hilbert space while pushing the desired interaction below experimentally resolvable energies.

The pseudospin is not automatically protected from magnetic noise. Uniform magnetic noise couples to $\tilde{\mathbf{S}}$ at first order because

$$P_C(\mathbf{S}_1 + \mathbf{S}_2 + \mathbf{S}_3)P_C = \tilde{\mathbf{S}}.$$

The cluster suppresses leakage when the noise spectrum has negligible weight near the excitation energy J . It does not suppress ordinary logical dephasing, which is loss of phase coherence within the encoded doublet. An encoded subspace is not necessarily a decoherence-free subspace.

Mapping a two-state basis to Fibonacci labels

A Fibonacci string-net edge has two possible labels: the vacuum label 1 and the nontrivial label τ . A string-net is a lattice description in which labeled edges obey local fusion constraints and are transformed by specified recoupling rules. The Fibonacci fusion rule is

$$\tau \times \tau = 1 + \tau.$$

This equation states that two τ labels may fuse either to the vacuum channel 1 or to the τ channel.

Because a cluster doublet has the required local Hilbert-space dimension, one may define

$$|1\rangle_e \equiv |\uparrow\rangle_e, \quad |\tau\rangle_e \equiv |\downarrow\rangle_e$$

for edge e . [Proposal] This definition is a basis correspondence. Renaming the physical spin states does not give their operators Fibonacci fusion data.

On a trivalent graph, in which three edges meet at each vertex, the allowed unordered vertex triples are

$$(1, 1, 1), \quad (1, \tau, \tau) \text{ and permutations,} \quad (\tau, \tau, \tau).$$

Let Q_v project the three edges meeting at vertex v onto these allowed states. Thus, Q_v imposes a three-edge fusion constraint.

A two-dimensional local Hilbert space does not by itself enforce the vertex rule. Without the constraint, the labels 1 and τ are only alternative names for a spin basis.

Fibonacci plaquette operator

Let $\varphi = (1 + \sqrt{5})/2$ denote the golden ratio and the quantum dimension of τ . A quantum dimension characterizes the asymptotic growth of the fusion-state space associated with repeated anyons. The total squared quantum dimension is

$$\mathcal{D}^2 = 1 + \varphi^2.$$

For plaquette p , let B_p^s denote the operation that inserts a loop carrying label s and recouples it into the surrounding edges using the Fibonacci F -symbols. The F -symbols are the amplitudes that relate different orders of fusion for the same collection of labels. The doubled-Fibonacci plaquette projector is [\[Ro18\]](#)

$$B_p = \frac{B_p^1 + \varphi B_p^\tau}{1 + \varphi^2}.$$

The ideal Levin–Wen Hamiltonian is

$$H_{\text{LW}} = -K_v \sum_v Q_v - K_p \sum_p B_p, \quad K_v, K_p > 0,$$

where K_v and K_p are energies. [Theory] In the ideal construction, the projectors commute and realize doubled rather than single chiral Fibonacci order [\[Ro18\]](#). “Doubled” means that the theory includes both a chiral sector and its time-reversed counterpart.

The plaquette operator transforms superpositions of edge labels with specified irrational relative amplitudes. An arbitrary six-spin interaction does not approximate this operator merely because it acts on six spins surrounding a hexagon.

Generic perturbative constructions can reduce interaction locality for broad classes of encoded projected entangled-pair-state (PEPS) parent Hamiltonians. A PEPS is a tensor-network representation of a many-body state, and a parent Hamiltonian is a Hamiltonian for which that state is a ground state. An explicit double-semion string-net

example also exists [R179]. [Theory] That result establishes existence in an abstract two-body model; it does not provide a defect-specific doubled-Fibonacci construction.

For a perturbative gadget whose desired interaction first appears at order n , dimensional analysis gives an effective scale of the form

$$K_{\text{eff}} \sim c j \left(\frac{j}{\Delta_m} \right)^{n-1},$$

where Δ_m is a mediator excitation energy and c is a dimensionless coefficient that depends on the construction. When $|j|/\Delta_m \ll 1$, each additional perturbative order reduces the output energy scale. Lower-order unwanted terms must either be canceled or demonstrated to be harmless.

Producing one desired term perturbatively does not reproduce the complete Hamiltonian H_{LW} . Terms omitted from a truncated derivation remain physically present unless they are explicitly suppressed or canceled.

Necessary energy-scale inequalities

Define δ_C as the energy norm of disorder within a cluster, Γ as a logical decoherence rate in s^{-1} , σ_{eff} as the energy scale of spatial variation in effective couplings, and H_{err} as the sum of all unwanted effective terms. Let k_B denote Boltzmann's constant, T the temperature in kelvin, $\hbar = h/(2\pi)$, and $\|\cdot\|$ the operator norm. Let Δ_{topo} denote the actual many-body energy gap above the intended ground sector, rather than merely the coefficient multiplying one projector.

A minimally viable passive hierarchy must satisfy

$$J \gg \max(|j|, \delta_C, k_B T),$$

and

$$\Delta_{\text{topo}} \gg \max(k_B T, \hbar \Gamma, \sigma_{\text{eff}}, \|H_{\text{err}}\|).$$

Every term in these inequalities has units of energy. The conditions are necessary but not sufficient.

The architecture also requires local interactions, a system larger than the correlation length, an initialization procedure, and a method for identifying topological sectors. The correlation length is the characteristic distance over which local correlations decay. Stability theorems protect suitable topologically ordered commuting-projector models against sufficiently weak local perturbations [R142].

These theorems do not establish that a poorly approximated microscopic Hamiltonian belongs to the target phase.

A large cluster coupling J is therefore insufficient to protect a logical qubit. The relevant protection scale is the many-body gap remaining after all projections and perturbative-gadget suppressions. That gap must exceed the energy scales associated with temperature, decoherence, disorder, and residual interactions.

Evidence and tests for each architectural link

Link	Status	What is actually supported	Falsifiable next test
Crystal $\rightarrow$ addressable defect spin	[Experiment], host-dependent	Individual solid-state defect spins can be initialized, controlled, and read out; this is mature for selected centers, not for arbitrary defects.	Demonstrate the chosen charge state, coherence, and readout under the density and temperature required by the array.
Separate defects $\rightarrow$ coherent cluster	[Experiment] for small diamond systems; [Speculation] at 5–20 designed centers	Coupled diamond color centers separated by 98 ± 3 Å were coherently controlled [R181], and two single defect spins were entangled at room temperature [R080]. A sensed three-electron-spin cluster has shown coherent dynamics [R182]. None is the proposed large regular cluster.	Fabricate repeated three-center units and reproduce one coupling graph and spectrum across the chip.
Three spins $\rightarrow$ isolated pseudospin	[Theory] for the model above; [Proposal] for defects	The exact projection works if nearly isotropic antiferromagnetic J is realized. Exchange-only logic has a broader theoretical basis [R114].	Resolve the $-J, 0, J/2$ multiplets and the projected $2 : -1 : 2$ local response.
Pseudospin $\rightarrow$ edge label $\{1, \tau\}$	[Proposal]	A two-state Hilbert space has the correct dimension. No fusion constraint follows from dimension alone.	Tomographically implement one Q_v , and verify all allowed and forbidden triples.
Two-body defect coupling $\rightarrow Q_v, B_p$	[Theory] in generic gadget frameworks; [Speculation] in defect hardware	Locality reduction is possible in abstract models [R179]. No cited defect experiment implements the Fibonacci F -symbol amplitudes as a static effective plaquette operator.	Measure the full effective operator, including phases and all lower-order residual terms, on one vertex and one plaquette.
Approximate projectors $\rightarrow$ doubled-Fibonacci phase	[Theory] for the ideal model; [Numerics] for selected perturbations [R183]; [Speculation] for this architecture	The ideal phase and some phase boundaries are known. The basin of attraction cannot be inferred from term names alone.	On increasing patches, show a stable gap, correlation-length saturation, and the expected ground-sector/topological data under measured disorder.
Phase $\rightarrow$ emergent Fibonacci-type excitations	[Theory] conditional on the phase	Excitations of the doubled theory carry the corresponding non-Abelian data. A finite circuit demonstrating fusion does not prove a material phase.	Create separated excitations, perform path-deformation tests, and recover fusion/braid matrices insensitive to local path details.
Emergent excitations $\rightarrow$ logical qubit	[Proposal] for defect hardware	Topological encodings and braiding are known theoretically. Initialization, motion, fusion measurement, and error budgets are unspecified here.	Demonstrate a logical operation whose error decreases with separation or code distance under fixed local noise.

Here, tomography means experimental reconstruction of an operator or state from measurement data. The table states the architecture together with the evidence and falsification criterion for each link; the arrow diagram only lists those links.

Current experimental capabilities

[Experiment] Small interacting systems have been demonstrated in diamond. Neumann and colleagues coherently controlled a pair of color centers at a measured separation of $98 \pm 3 \text{ \AA}$ [R181].

Dolde and colleagues generated entanglement between individual defect electron spins [Ro80]. Rosenfeld and colleagues observed coherent dynamics involving an NV center and two dark electronic spins, where “dark” denotes spins not directly observed through the same optical channel [R182].

These experiments establish coherent coupling among only a few defects or nearby spins. They do not establish uniform antiferromagnetic Heisenberg bonds, repeatable cluster spectra, or a two-dimensional array.

[Experiment] Cluster-sized registers also exist as actively controlled collections of electron and nuclear spins. Bradley and colleagues controlled a ten-qubit diamond register consisting of one NV electron spin and nine nuclear spins, at 3.7 K, and entangled as many as seven spins [R119].

This result provides evidence for multispin control. It does not demonstrate ten engineered defect centers, a passive exchange-coupled cluster, or an analog string-net Hamiltonian.

The electron-mediated pulse-controlled register is therefore an actively controlled system rather than a passive analog material.

[Experiment] Programmable processors have dynamically prepared Fibonacci string-net states and implemented fusion and braiding protocols [R165]. This demonstrates control of the encoded mathematical structure, not spontaneous emergence from a static defect Hamiltonian.

Such processors provide useful measurement procedures and target data. They do not validate the proposed hierarchy of material energy scales and effective interactions.

For sapphire, corundum, SiC, or diamond, host selection must be based on measured interaction tensors, fabrication distributions, charge stability, and readout performance rather than on qualitative properties of the crystal alone.

[Speculation] No source cited here supports treating a five-to-twenty-defect cluster in any of these hosts as an available component. The proposed count is a resource hypothesis that can be optimized only after a microscopic Hamiltonian has been established.

Eight falsifiable requirements

The proposal remains viable only if all of the following requirements can be tested and satisfied.

- **A1 — Repeatable local object [Proposal].** The selected defect and charge state must behave as the same effective spin throughout a dense array. Spectroscopy must place quantitative bounds on site-to-site variation in the local Hamiltonian.

- **A2 — Cluster window [Proposal].** A reproducible cluster must have one doublet separated by Δ_C , with Δ_C exceeding $k_B T$, disorder, the energy scale of leakage-driving controls, and the intercluster coupling by specified margins.
- **A3 — Useful projection [Theory → experiment].** Experimentally determined microscopic operators must project to the predicted effective tensors. For the worked model, endpoint exchange must approach $4j/9$, and the corrections must scale as j^2/J .
- **A4 — No cluster-count assumption [Speculation].** Three, five, or twenty spins are acceptable only if they improve a measured objective such as the gap, addressability, noise susceptibility, or gadget order. Otherwise, additional spins merely introduce additional fabrication variables and disorder channels.
- **A5 — Operator completeness [Proposal].** The available interactions and mediators must generate both Q_v and B_p with the required Fibonacci matrix elements. Hamiltonian tomography must include unwanted terms; fitting only the desired coefficients does not test this requirement.
- **A6 — Surviving energy scale [Proposal].** The measured Δ_{topo} , after every projection and perturbative-gadget suppression, must exceed $k_B T$, $\hbar\Gamma$, coupling disorder, and residual interactions.
- **A7 — Phase evidence [Numerics → experiment].** Finite-size calculations using measured parameters must show convergence toward doubled-Fibonacci topological data, and the result must persist across the measured disorder distribution. A single carefully tuned small patch is insufficient.
- **A8 — Operational topology [Proposal].** Excitations must be created, moved, and fused without closing the gap. In addition, a nonlocal observable must become less sensitive to local perturbations as the system size increases.

The experimental order follows directly from the links: measure one cluster spectrum, verify the two-cluster projection, implement a three-edge vertex, reconstruct a plaquette operator, and build several connected plaquettes only after those checks pass. Development stops at the first failed inequality.

A small patch can falsify a proposed microscopic mapping. By itself, it cannot establish a thermodynamic phase, which is a phase defined in the limit of increasing system size.

End-to-end verdict — [Assessment]. The hierarchy is highly speculative but physically coherent.

No link violates a known physical principle, and the first projection is explicit. Current experimental support extends only to small coherent spin systems and digitally controlled registers.

The defect-specific cluster Hamiltonian, Fibonacci vertex and plaquette operators, surviving topological gap, and protected logical operation remain unestablished. Neither a five-to-twenty-defect component nor the feasibility of defect-based Fibonacci order can be assumed as an input.

Common conceptual errors

The local-dimension error. Dimension two can encode 1 and τ , the two outcomes of a coin toss, or a binary record of whether lunch occurred. Dimension names the count of basis states. Fusion additionally needs local constraints and recoupling amplitudes.

The five-to-twenty error. No scaling law presented in this chapter selects a cluster size between five and twenty.

Increasing the cluster size may produce a cleaner code space, but it can also increase spectral crowding, placement requirements, control cross-talk, and the number of disorder parameters. Until calculation and experiment identify an optimum, the proposed interval has status [Speculation].

The gadget-equals-phase error. A perturbative construction that yields one desired term has not produced the full target Hamiltonian.

Lower-order terms may dominate, higher-order terms may be smaller than experimental linewidths, and the perturbative parameter may simultaneously be too large for a reliable expansion and too small for observation of the desired term. All perturbative orders remain part of the physical effective Hamiltonian unless they are canceled or shown to be negligible.

The digital-equals-emergent error. Pulse sequences can prepare a string-net wavefunction and compile braid operations [R165].

This is [Experiment: digital emulation]. Passive topological order instead requires the material Hamiltonian, its many-body gap, and its perturbative stability to preserve the phase between control pulses.

The gap-label error. The coupling J is the cluster leakage gap, not the topological gap, and the coefficient K_p is a Hamiltonian parameter rather than the many-body gap. Only the spectrum of the complete disordered effective model determines Δ_{topo} .

The protection error. The worked cluster is spinful, so magnetic noise acts within its logical doublet.

Even a genuine two-dimensional topological phase at nonzero temperature contains thermally activated anyons. Topological order does not eliminate thermal excitation processes.

The chirality error. The Levin–Wen target considered here is doubled Fibonacci. Referring to it only as “Fibonacci” omits the time-reversed sector and can lead to incorrect claims about edge physics and control requirements.

Conceptual checks

- **Justification for using three defects rather than assuming five to twenty.** Three is the smallest exactly tractable cluster with a spinful ground doublet, whereas the cited evidence provides no demonstrated defect-cluster basis for the larger count.
- **Projection of endpoint exchange.** Endpoint exchange between two such clusters projects to $(4j/9)\tilde{\mathbf{S}}_A \cdot \tilde{\mathbf{S}}_B$. Each endpoint spin satisfies $P_C \mathbf{S}_{\text{end}} P_C = (2/3)\tilde{\mathbf{S}}$, so multiplying the two projection factors gives $4/9$. Corrections of order j^2/J are outside this first-order result.
- **Limitation of the basis identification.** Mapping $|\tilde{\uparrow}\rangle$ to $|1\rangle$ only assigns a name to a local basis state. Vertex constraints, plaquette recoupling, a gapped phase, and emergent excitations are still required. A two-dimensional local Hilbert space does not determine fusion rules.
- **Decisive energy scale.** The relevant scale is the measured or reliably calculated Δ_{topo} of the full model after projection and gadget suppression, not the bare cluster coupling J .

- **Consequence of failure at the cluster level.** If the first cluster does not exhibit the predicted doublet and the $2 : -1 : 2$ local response, the cluster model is falsified. Constructing a lattice from that cluster would then lack a valid microscopic basis.
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Sources

- [R114] D. P. DiVincenzo, D. Bacon, J. Kempe, G. Burkard, and K. B. Whaley, “Universal quantum computation with the exchange interaction,” *Nature* 408, 339–342 (2000). DOI: [10.1038/35042541](https://doi.org/10.1038/35042541).
 - [R181] P. Neumann, R. Kolesov, B. Naydenov, et al., “Quantum register based on coupled electron spins in a room-temperature solid,” *Nature Physics* 6, 249–253 (2010). DOI: [10.1038/nphys1536](https://doi.org/10.1038/nphys1536).
 - [Ro80] F. Dolde, I. Jakobi, B. Naydenov, et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* 9, 139–143 (2013). DOI: [10.1038/nphys2545](https://doi.org/10.1038/nphys2545).
 - [R182] E. L. Rosenfeld, L. M. Pham, M. D. Lukin, and R. L. Walsworth, “Sensing coherent dynamics of electronic spin clusters in solids,” *Physical Review Letters* 120, 243604 (2018). DOI: [10.1103/PhysRevLett.120.243604](https://doi.org/10.1103/PhysRevLett.120.243604).
 - [R119] C. E. Bradley, J. Randall, M. H. Abobeih, et al., “A ten-qubit solid-state spin register with quantum memory up to one minute,” *Physical Review X* 9, 031045 (2019). DOI: [10.1103/PhysRevX.9.031045](https://doi.org/10.1103/PhysRevX.9.031045).
 - [Ro18] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110); [arXiv:cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617).
 - [R179] C. G. Brell, S. D. Bartlett, and A. C. Doherty, “Perturbative 2-body parent Hamiltonians for projected entangled pair states,” *New Journal of Physics* 16, 123056 (2014). DOI: [10.1088/1367-2630/16/12/123056](https://doi.org/10.1088/1367-2630/16/12/123056); [arXiv:1407.4829](https://arxiv.org/abs/1407.4829).
 - [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* 51, 093512 (2010). DOI: [10.1063/1.3490195](https://doi.org/10.1063/1.3490195); [arXiv:1001.0344](https://arxiv.org/abs/1001.0344).
 - [R183] M. D. Schulz, S. Dusuel, K. P. Schmidt, and J. Vidal, “Topological phase transitions in the golden string-net model,” *Physical Review Letters* 110, 147203 (2013). DOI: [10.1103/PhysRevLett.110.147203](https://doi.org/10.1103/PhysRevLett.110.147203); [arXiv:1212.4109](https://arxiv.org/abs/1212.4109).
 - [R165] Z. K. Mineev, K. Najafi, S. Majumder, et al., “Realizing string-net condensation: Fibonacci anyon braiding for universal gates and sampling chromatic polynomials,” *Nature Communications* 16, 6225 (2025). DOI: [10.1038/s41467-025-61493-8](https://doi.org/10.1038/s41467-025-61493-8); [arXiv:2406.12820](https://arxiv.org/abs/2406.12820).
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Chapter 25 — Target, placement, and interaction graphs

A string-net model assigns a label to every edge of a honeycomb graph. At each vertex, a vertex term tests the three labels meeting there against the fusion rules, which state which combinations of labels may meet. Around each hexagonal face, a plaquette term recouples the six labels on its boundary, changing between equivalent ways of combining them. This labeled honeycomb is the first of three graphs in this chapter.

Each edge label can be built as a physical object: one encoded cluster placed at the midpoint of that edge. Joining two midpoints whenever their original edges share a vertex yields a second graph. Its repeating pattern holds triangles and hexagons rather than honeycomb cells.

A third graph records the couplings the laboratory Hamiltonian actually generates. Every pair of midpoints joined by a physical interaction becomes an edge of this graph, including the long-range $1/r^3$ tail and any auxiliary component sitting at a vertex.

A tiling drawing by itself fixes no Hamiltonian. The interactions realized in the crystal must reproduce the incidence relations the string-net model assumes: which labels meet at each vertex and which surround each face. When they differ, operators taken from the Levin–Wen construction act on a different graph than the one the physical couplings connect.

The target graph of the operators

A string-net Hamiltonian needs two adjacency relations: which edge labels meet at each vertex, and which edge labels surround each plaquette. The graph recording both relations is the target graph.

Write the target graph as $G = (V, E)$, with V the set of vertices and E the set of edges.

In the primary construction every vertex of G joins exactly three edges. The vertex term at that vertex tests the labels carried by those three edges. Trivalent here means three edges incident on each vertex.

The string-net operators are therefore defined on G . How the fabricated chip looks leaves those operators unchanged.

The host crystal does not realize G on its own. It holds defects at positions allowed by fabrication, with couplings set by distance, orientation, crystal axis, and residual fabrication damage. Two further graphs therefore sit between G and the physical couplings, and they need not agree with each other.

The design sequence is

$$\begin{aligned} \text{topological data} &\longrightarrow \text{target graph} \longrightarrow \text{cluster positions} \\ &\longrightarrow \text{physical couplings} \longrightarrow \text{effective operators.} \end{aligned}$$

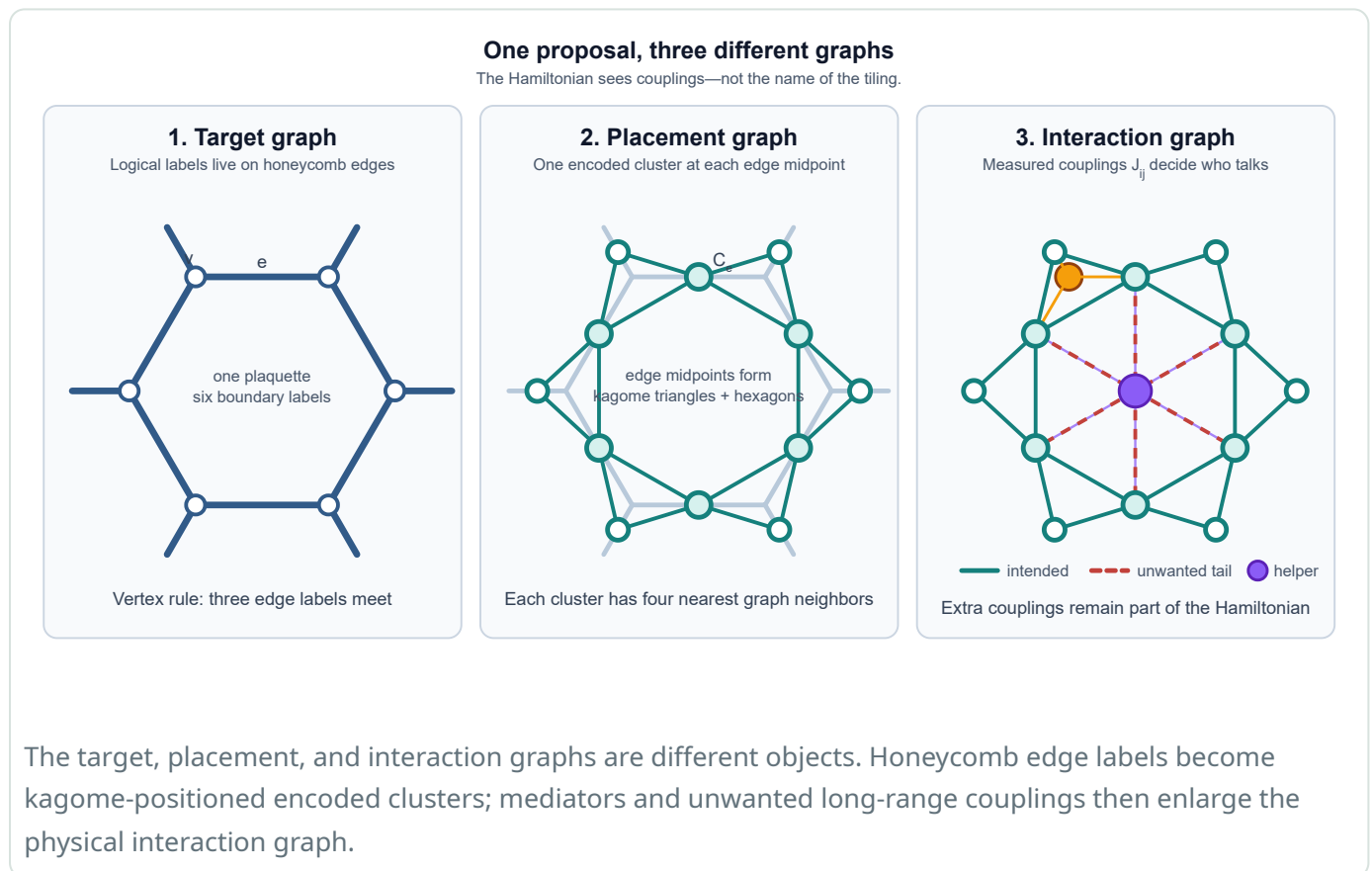
Each step in this chain needs a separate derivation. Skipping a step assumes physical connections that have not been established.

Edge-midpoint clusters and the placement graph

Chapter 24 grouped several physical defects into one low-energy cluster and used that cluster as a single string-net edge label. This chapter places one such encoded cluster C_e at the midpoint of every target edge e . An encoded cluster is a group of microscopic degrees of freedom whose chosen low-energy subspace stands for one effective local degree of freedom. The cluster stores the edge label, 1 or τ in the Fibonacci input theory, rather than the state of one microscopic defect spin.

The placement graph joins two such midpoints whenever their target edges share a vertex.

The three graphs separate as follows. The first panel draws the logical honeycomb with a label on each edge. The second panel puts one encoded cluster at each edge midpoint; joining clusters whose honeycomb edges share a vertex gives a kagome lattice of triangles and hexagons. The third panel adds the helpers and extra couplings that can appear in the laboratory Hamiltonian.



Each panel therefore answers a different design question: the first fixes the intended vertex and plaquette incidence, the second fixes the cluster coordinates and constraint neighbors, and the third lists the couplings the Hamiltonian actually generates, including mediator and long-range terms.

The six central cluster sites show only part of the local structure. Each C_i also sits in a triangle of three edge centers near each endpoint of its original honeycomb edge. Repeating the construction over the whole lattice puts the clusters on a kagome lattice.

Two clusters touch in the placement graph exactly when their target edges share an endpoint. This is the line graph $L(G)$ of G . A line graph is defined by replacing each edge of G with a vertex of $L(G)$, with two vertices in $L(G)$ adjacent when the corresponding edges in G share an endpoint.

An edge of a degree-three graph has two endpoints and meets two further edges at each endpoint, so every edge cluster has four nearest constraint-neighbors in this counting.

$$z_{L(G)} = 2(3 - 1) = 4$$

with $z_{L(G)}$ the degree of a vertex in the line graph. A honeycomb target and a kagome device therefore describe one architecture without conflict: the honeycomb is the target graph and the kagome is the placement graph.

Joining every nearby pair of clusters instead of only those sharing a vertex leaves $L(G)$ behind. The result belongs to the interaction graph fixed by the Hamiltonian.

The physical interaction graph

If clusters coupled only along placement-graph edges, the placement graph would double as the pair-interaction graph of the Hamiltonian. Dipolar tails, residual exchange, cavities, drives, and perturbative gadgets can add couplings beyond the drawn placement edges, so the interaction graph records the couplings actually present.

The ideal Levin–Wen string-net energy contains a vertex term Q_v and a plaquette term B_p ,

$$H_{\text{SN}} = - \sum_{v \in V} J_v Q_v - \sum_p J_p B_p,$$

where J_v and J_p are energies. This equation defines the target string-net Hamiltonian H_{SN} as a weighted sum of local vertex and plaquette operators.

The operator Q_v checks whether the three labels incident on vertex v satisfy the allowed fusion rule. Fusion is the rule specifying which topological charge labels can consistently combine at a vertex.

The operator B_p inserts and recouples a loop around plaquette p . Recoupling is a change of fusion basis among equivalent ways of combining labels. On a honeycomb plaquette, B_p acts on six boundary edge labels and also depends on nearby labels through the recoupling.

The exact commuting-projector construction is [Theory], not an interaction already known among color centers [Ro18]. A commuting-projector Hamiltonian sums local projection operators that commute with each other.

Drawing G does not implement H_{SN} . The physical Hamiltonian acts only through the couplings recorded in the interaction graph.

The defects inside C_e are physical spins. The chosen doublet of C_e , a two-dimensional low-energy subspace, is an encoded local degree of freedom.

The labels on G are string-net variables. Excitations count as emergent anyons, quasiparticles in two spatial dimensions whose exchange or braiding statistics can differ from bosonic and fermionic statistics, only when the many-body Hamiltonian enters the corresponding topological phase.

Programming a circuit or measuring a limited set of fusion-like outcomes does not, by itself, establish intrinsic anyons.

Geometric spacing and the 0.19 coupling ratio

Give each logical honeycomb edge length a in metres. The distance between the midpoints of two edges meeting at a vertex is then fixed by the honeycomb geometry:

$$d_1 = \frac{\sqrt{3}}{2}a.$$

The numerical coefficient carries no units, so $[d_1] = [a] = \text{m}$. In a regular embedding, all four nearest placement-graph bonds at a cluster share this length. That uniformity is the main geometric reason for choosing the honeycomb layout.

To gauge the long-range tail, suppose pair couplings fall off as $J(r) = C/r^3$, with C in units of energy times cubic metres. The next kagome distance is $d_2 = \sqrt{3} d_1$. Hence the displayed ratio follows:

$$\frac{J(d_2)}{J(d_1)} = \left(\frac{d_1}{d_2}\right)^3 = \frac{1}{3\sqrt{3}} \approx 0.19.$$

The next kagome shell therefore couples at roughly one fifth of the nearest-neighbor scale. Chapter 26 must keep these off-target terms or exhibit a working refocusing mechanism.

This number follows from the r^{-3} assumption alone; it is not a universal law of defect couplings. Magnetic dipole interactions also carry angular factors.

Exchange couplings need not obey a power law, and driven interactions can have their own spatial range.

0.19 is large enough that dropping it needs a justification. Without the r^{-3} hypothesis the number changes, but the extra interaction edges remain part of the model.

A vertex helper M_v , an auxiliary physical degree of freedom that mediates an effective interaction, can sit near each original honeycomb vertex and couple to the three surrounding clusters C_e . A plaquette helper M_p can sit at each hexagon center and couple radially to the six boundary clusters.

The radial bonds meet only at their shared helper, so this planar layout avoids geometric crossings. Still, a helper with six neighbors does not by itself act as a six-edge operator.

Building the six-edge operator still calls for a controlled perturbative gadget or projection.

The geometry must meet five requirements:

- The logical vertex degree must fit the branching rule.
- Cluster separations must be usable and sufficiently uniform.
- Unwanted interaction edges must be avoided or made correctable.
- The layout must leave physical room for vertex and plaquette helpers without wire crossings.
- The architecture must absorb placement error, missing defects, annealing, surfaces, and crystal-axis constraints.

When any one requirement fails, the remaining four cannot establish the intended Hamiltonian on their own.

Trivalent branching constraints

Write z for the vertex degree, the number of target edges meeting at a target vertex. The standard Levin–Wen construction works directly on a trivalent graph, $z = 3$, where each local branching constraint tests three labels [Ro18].

A vertex of higher degree can be broken into a tree of trivalent vertices by adding short edges and fixing a fusion basis. That repair adds degrees of freedom and operators at a cost.

Placing edge clusters on the edges of a regular degree- z graph gives a line graph of degree set by the displayed formula:

$$z_{\text{edge clusters}} = 2(z - 1).$$

A triangular target has $z = 6$ and gives degree 10 among edge clusters; a square or kagome target has $z = 4$ and gives degree 6.

A trivalent target gives degree 4. Smaller target degree generally means fewer intended pair channels, fewer spectral collisions, and fewer paths by which one defective site disturbs its surroundings.

Dropping the trivalent assumption alters the local algebra tested by the vertex term, and not only the drawing.

Propagation of placement error into coupling disorder

With a distance-dependent interaction, fabrication error spreads into the couplings. Take $J(r) = Cr^{-3}$, and let a bond length shift by a small δr . Differentiation gives the displayed sensitivity:

$$\frac{\delta J}{J} \approx -3 \frac{\delta r}{r}.$$

δr and r need the same length unit, so their ratio is dimensionless. A 5% radial error then gives about a 15% coupling error, before angular dependence and defect-to-defect variation enter. This follows from the assumed power law geometrically; it predicts no specific wafer.

Equal drawn lengths still leave spin couplings potentially unequal. For magnetic dipoles, the coupling depends on the bond direction relative to the quantization axes, the axes fixing the spin projection directions.

Defects grown in crystallographically inequivalent orientations can therefore turn a single-length lattice into a Hamiltonian with several coupling strengths. Shaped pulse sequences might average this directional variation away.

Until that averaging is derived for the chosen defects, it stays at the [Proposal] stage.

Without the power-law assumption the factor of three changes. With it retained, even a millimetre-scale drawing error moves J by a non-negligible amount.

Quantifying target–interaction graph mismatch

Write a target adjacency matrix A_{ij} , with $A_{ij} = 1$ when clusters i and j should couple at the leading pairwise level and $A_{ij} = 0$ otherwise. Let J_{ij} be the realized pair coupling. A workable geometry puts the large values of $|J_{ij}|$ on pairs with $A_{ij} = 1$, while keeping the displayed mismatch ratio:

$$\epsilon_{\text{graph}} = \frac{\max_{A_{ij}=0} |J_{ij}|}{\min_{A_{ij}=1} |J_{ij}|}$$

This dimensionless ratio compares the strongest unwanted pair coupling against the weakest wanted one. It flags graph mismatch before a full many-body calculation starts; it establishes no topological stability.

For long-range dipolar interactions, $A_{ij} = 0$ seldom gives $J_{ij} = 0$. Hence ϵ_{graph} serves as a diagnostic and never justifies deleting the interaction tail. Counting drawn edges answers a different question than mapping the graph seen by the Hamiltonian.

Helper-mediated multi-body interactions

Every tiling named below is planar: its nearest-neighbor target edges draw without crossings. Whether the operator-incidence graph stays planar after helpers are added is a separate question. That graph records which helpers or physical components take part in each intended local operator.

One helper per vertex fits naturally into the drawing. One helper per face also fits, but its coordination must match the face size: six for honeycomb, three and twelve for decorated honeycomb, four or eight for square–octagon.

A central helper offers one shared site, cavity, or mode to several bonds. Pairwise microscopic couplings to that helper still need a derivation before they yield the required product operator.

[Theory] Perturbative gadgets generate multi-body terms only at the price of energy denominators, unwanted corrections, and a reduced effective scale. Chapter 23 derived that price.

That price applies to every lattice geometry considered here.

A six-leg star of pair couplings is therefore distinct from B_p . Without a derivation of the required gadget, the device holds six ordinary bonds and no plaquette recoupling operator.

Fabrication coordinates and realized defects

A written coordinate does not place a working defect there. Fabrication contributes pattern registration, aperture or beam size, ion straggle, vacancy diffusion during annealing, conversion yield, depth distribution, and damage-induced noise. Ion straggle is the statistical spread of implanted-ion trajectories around their nominal path.

[Experiment] Toyli and colleagues combined electron-beam-defined apertures with nitrogen implantation to pattern single-spin and spin-array sites in diamond; for 20 keV ^{15}N , their modeling gave about 9 nm lateral straggle under those implantation conditions [R189].

[Experiment] In a separate 2017 experiment, focused Si implantation made SiV centers with about 32 nm lateral standard deviation and 48 ± 21 nm accuracy relative to nanocavities; the measured conversion yield reached about 2.5% at 100 keV before an added electron-irradiation step [R190]. These numbers measure placement precision under different definitions.

The two experiments use different ions, implantation energies, devices, and reported metrics, so their numbers cannot be interchanged.

A workable geometry must allow repeated masks or direct-write coordinates, post-fabrication characterization, and spare sites or repair strategies. No cited experiment demonstrates a dense, uniformly coupled lattice of encoded clusters realizing a string-net Hamiltonian.

Reading beam width as the endpoint distribution after annealing omits the straggle, diffusion during annealing, registration error, and conversion variation that enter J_{ij} .

Candidate target and placement graphs

The table keeps each target graph separate from its edge-cluster graph. “Mediator burden” means the machinery needed for local vertex and face operators, rather than the nearest-neighbor lines visible in a drawing.

Candidate target	Target degree	Edge-cluster degree	Spacing/graph issue	Crossing or mediator issue	Fabrication verdict
Honeycomb	3	4; positions are kagome	One intended midpoint spacing in regular embedding; dipolar tails remain	No planar crossings; 3-leg vertex and 6-edge face	Primary: simplest trivalent incidence and a single repeated coordinate motif
Triangular	6	10	Dense, strongly overconnected for edge encoding	No geometric crossings, but a six-label vertex must be decomposed	Reject as direct string-net graph; useful only for a different derived model
Kagome	4	6 if its edges carry labels	Triangles and hexagons create two	4-label vertices require resolution; 3-	Keep as the placement graph of

Candidate target	Target degree	Edge-cluster degree	Spacing/graph issue	Crossing or mediator issue	Fabrication verdict
			face scales; degree exceeds trivalent target	and 6-face mediators	honeycomb edge clusters, not the primary logical graph
Decorated honeycomb / star (3, 12 ²)	3	4	Intra-triangle and inter-triangle bonds naturally form at least two classes	3-edge and 12-edge faces; large face operator or mediator tree	Interesting for Kitaev-type physics, but too many scales for the first defect prototype
Ruby	4	6	Crowded local triangles and colored bond classes	Higher local degree and mediator congestion	Strong theoretical precedent for two-body color-code physics, not a direct Fibonacci edge-label map
Square-octagon (4.8.8)	3	4	Incident midpoint distances are unequal in regular embedding	3-leg vertices; separate 4- and 8-edge face machinery	Fallback: orthogonal registration is attractive, but two bond/face scales must be calibrated
Square	4	6	One site spacing but wrong degree for direct trivalent branching	4-label vertex resolution and 4-edge faces	Natural for toric-code stabilizers, not the cleanest doubled-Fibonacci map
Brick-wall honeycomb	3	4	Graph-equivalent to honeycomb but geometrically anisotropic	Same incidence as honeycomb	A mask-friendly deformation of the primary, not a distinct topological candidate

The verdict column recommends a graph; it reports no demonstrated material implementation. Each section below supplies the reasoning behind one verdict.

Honeycomb target graph

The honeycomb graph earns selection through its incidence structure: trivalent, bipartite, planar, with a single face size. A bipartite graph splits its vertices into two disjoint sets with every edge joining vertices from different sets.

The Levin–Wen construction works directly with trivalent string-net branching [R018]. Kitaev’s exactly solved honeycomb model shows separately that a simple trivalent interaction graph can host emergent gauge structure and non-Abelian Ising-type excitations under suitable bond-dependent interactions, a phase distinct from Fibonacci order [R017].

[Theory] This precedent shows how strongly graph structure shapes the physics, and it leaves the proposed defect Hamiltonian still to be validated.

The honeycomb stays primary because its local incidence matches the required algebra; the hexagonal appearance is incidental.

The cost is the one computed above: encoding labels on edges moves the clusters onto a degree-four kagome lattice.

Plaquette terms still span six clusters, and long-range tails add further unwanted edges. The anyons of Kitaev's model live in a different phase, so they cannot stand in for the excitations of the proposed Fibonacci construction.

Triangular target graph

A triangular lattice surrounds each vertex with six nearest neighbors. That coordination serves frustrated-magnet and quantum-dimer models; a gapped resonating-valence-bond phase, a quantum phase built from superpositions of short-range singlet-pair coverings, was established theoretically in the triangular-lattice quantum dimer model [R187].

[Theory] That result does not show that triangular geometry implements the Fibonacci string net needed here.

Taken as a direct target, the triangular lattice turns every trivalent branching check into a six-way junction, whose resolution calls for internal fusion-tree edges.

With edge clusters the local pair graph reaches degree ten. Dense coordination adds connectivity together with more unintended dipolar paths and stricter demands on local addressability.

The triangular lattice is therefore set aside for the first architecture. It could underlie a different derived model, but its dimer variables become τ labels only through a derivation not yet given.

Kagome as a placement graph

The kagome graph joins four edges at each vertex and mixes triangular with hexagonal faces. It is studied for geometric frustration, and theory has found fractionalized phases, excitations whose quantum numbers or statistics differ from those of the microscopic constituents, in specific kagome Hamiltonians [R188]. Finding such a phase in one model leaves the required string-net phase for the available couplings still to be shown.

Kagome keeps a definite role in this architecture: it appears on its own as the physical midpoint lattice of honeycomb edge variables. In this picture each kagome triangle stands for one logical honeycomb vertex, and each kagome hexagon for one logical honeycomb plaquette.

That incidence correspondence is more useful than promoting kagome to the target graph. Promoting it sacrifices trivalent branching and introduces two face scales.

Decorated-honeycomb target graph

Replacing every honeycomb vertex with a triangle gives the decorated honeycomb, or star, lattice with face sequence (3, 12, 12). The face sequence lists the polygons met around each vertex. The lattice stays trivalent.

[Theory] Yao and Kivelson solved a Kitaev-type model on this lattice and found a chiral spin liquid with non-Abelian anyons in part of its phase diagram [R184]. A chiral spin liquid is a topologically ordered phase that breaks time-reversal symmetry and carries chiral boundary behavior. Its excitations come from Kitaev/Majorana physics, rather than doubled Fibonacci string-net order.

For fabrication, the decoration brings internal triangles that could serve as strongly coupled mini-clusters. It also creates two geometric bond classes, within triangles and between triangles, and faces of size three and twelve.

That structure helps when the microscopic Hamiltonian naturally provides two interaction scales. It hinders the design when even one uniform scale is hard to achieve.

A plaquette operation around a twelve-edge boundary needs a helper tree or several perturbative stages.

The decorated honeycomb stays as a research variant rather than the primary layout. Without a justified two-scale microscopic interaction, the decoration adds cost for no demonstrated physical gain.

Ruby target graph

The ruby lattice meets four edges at each vertex, and the best-known topological construction on it uses colored bond types. [Theory] Kargarian, Bombin, and Martin-Delgado wrote a two-body spin Hamiltonian on the ruby lattice whose low-energy sector realizes topological color-code order [R185]. Their result gives concrete evidence that microscopic two-body interactions can encode more elaborate low-energy stabilizers.

Their result leaves a ruby-lattice Fibonacci phase in defect clusters still to be shown. The degree-four junctions, small triangles, and multiple bond classes of the ruby lattice raise control and placement demands. Its value for this architecture is methodological: it motivates asking whether a two-body parent Hamiltonian can generate the required low-energy constraints.

The anyons of the color-code result belong to the Hamiltonian that produces them; they cannot be relabeled as τ excitations.

Square–octagon target graph

The Archimedean square–octagon tiling has face sequence (4, 8, 8) and degree three, so it keeps trivalent branching. An Archimedean tiling paves the plane with regular polygons so that every vertex looks the same. Kitaev-type models on this lattice support a rich set of Abelian and non-Abelian phases in theory [R186], which leaves Fibonacci order and feasibility in a defect system still to be shown.

The straight, orthogonal square motif suits lithographic registration. The regular embedding is still geometrically nonuniform at the edge midpoints: at each vertex the incident edges meet at 90° , 135° , 135° . For equal target-edge length a , the distances between adjacent edge midpoints follow in the displayed formulas:

$$d_{90} = a \sin 45^\circ = \frac{a}{\sqrt{2}}, \quad d_{135} = a \sin 67.5^\circ \approx 0.924a.$$

Under a pure r^{-3} interaction, the intended couplings at these two distances differ by the displayed factor:

$$\left(\frac{d_{135}}{d_{90}} \right)^3 \approx 2.23.$$

The square and octagonal plaquettes need four-edge and eight-edge operations. Calibration, geometric distortion, or driven equalization is therefore mandatory; one of these corrections must be carried out.

The square–octagon lattice stays a viable fallback through its logical degree and planarity, while its regular embedding yields nonuniform intended couplings. Relaxing equal edges might shrink the factor of 2.23, but only a derivation of the distorting geometry makes that claim stick.

Square and brick-wall lattice embeddings

A square lattice meets four edges at each vertex. It carries many representations of the toric code, but its vertex degree does not match the three-edged vertices of a trivalent Fibonacci fusion tree, the diagram recording successive fusion operations.

A brick-wall embedding deforms honeycomb edges so rows and columns look rectangular while vertex connectivity stays fixed. Two graphs related this way are graph-isomorphic: they share the same adjacency relations after relabeling vertices. Idealized topological lattice models keep their form under such graph-preserving deformations.

Physical coupling strengths generally change under these deformations. Brick-wall coordinates may simplify the fabrication pattern, but the resulting bond lengths and orientations must enter the Hamiltonian, the operator fixing the system's energies and dynamics.

The brick-wall embedding therefore counts as a fabrication variant of the primary honeycomb graph, rather than the fallback phase graph. Isomorphism of the abstract target graph G leaves the interaction graph, which records the couplings actually realized between physical degrees of freedom, still to be matched.

Current fabrication capabilities and requirements

[Experiment] Patterned arrays and individual nanostructure targets have been implanted [R189]; [R190]. In particular experiments, these methods support coordinate lists with length scales of tens to hundreds of nanometres. An interaction matrix, the table of couplings between each pair of physical degrees of freedom, still lies beyond what they provide.

The proposed string-net prototype calls for substantially more than implanted coordinate lists:

- Several defects must combine into each reproducible cluster doublet, a controlled pair of cluster states serving as one encoded degree of freedom.
- Couplings within one cluster, between its own defects, must dominate couplings between different clusters.
- Every intended edge needs the required coupling sign, tensor structure, and strength.
- Unwanted long-range couplings must enter the model explicitly or be physically suppressed.
- Missing defects and defects in the wrong charge state need detection and repair.
- Vertex and plaquette terms must emerge with an energy gap above the scales of disorder and decoherence. A vertex term touches the degrees of freedom at a vertex, while a plaquette term runs around the boundary of a face. Decoherence is the loss of quantum phase information into uncontrolled degrees of freedom.

No cited experiment meets all of these conditions. The geometric design therefore stays at the [Proposal] stage: use a shallow, planar patterned layer in a host where the chosen defect species can be created and read out, and start by testing isolated triangles and hexagons.

Diamond has direct experimental evidence for patterned implantation [R189]; [R190]. Silicon carbide may offer wafer-scale advantages, but this chapter does not assume equal placement yield, coherence, or coupling properties for diamond and silicon carbide without evidence tied to the specific host and defect species.

A graph-first experimental sequence starts with one logical vertex, continues to two vertices sharing an edge cluster, and then tests one hexagonal plaquette with its neighboring legs. A logical vertex belongs to the encoded target graph; it need not be a single physical site. Each projected constraint, the constraint after projection into the encoded low-energy subspace, needs measurement before the coordinate cell is repeated into a larger lattice.

Sources of modeling and design error

Keep the target and placement lattices distinct. Calling the device “kagome” without saying that kagome sites encode honeycomb edges drops the map that defines the branching structure. Calling it only “honeycomb” hides the degree-four physical neighbor graph.

Treat drawn edges as a subset of the physical couplings. Dipolar interactions reach beyond the nearest-neighbor edges of a geometric diagram.

Evaluation must cover the pair coupling J_{ij} , its angular factors, and ϵ_{graph} , the graph-error measure. Here, A_{ij} is the adjacency-matrix element telling whether an intended graph edge joins sites i and j . Setting $J_{ij} = 0$ wherever $A_{ij} = 0$ deletes the long-range tail that Chapter 26 still has to include.

Equal separation leaves interaction strength partly undetermined. Crystal orientation, strain, charge state, and quantization axis also feed into J_{ij} , so geometry fixes only part of the pair interaction.

A six-leg star of pairwise couplings is not the plaquette operator B_p . The effective operator from the mediator, with all its correction terms, is part of what has to be derived.

An anyon type belongs to the Hamiltonian that produces it; moving it to a different Hamiltonian calls for a derivation. An anyon is a two-dimensional quasiparticle with exchange statistics beyond bosonic and fermionic cases. Decorated-honeycomb and square–octagon Kitaev models supply non-Abelian precedents [R184]; [R186], and ruby models supply color-code order [R185]. These precedents leave Fibonacci anyons in the proposed defect system still to be established.

The nominal beam size measures the beam, rather than the final defect-position uncertainty. Implantation straggle, annealing, registration error, and stochastic defect conversion stay relevant after the beam specification is fixed [R189]; [R190].

Optimization must cover more than the nearest-neighbor spacing. Reducing a , the lattice-spacing parameter, strengthens wanted and unwanted interactions together, shrinks the room for clusters and auxiliary mediators, and can let damage volumes overlap.

Increasing a eases addressability and weakens local interactions. Chapter 28 must therefore optimize the full error and coupling budget.

Absence of geometric edge crossings is a useful design condition, but it does not establish that the required Hamiltonian can be implemented.

Selected primary and fallback graphs

Primary — honeycomb logical graph with kagome cluster positions. The honeycomb graph has degree three, so it matches trivalent branching with no auxiliary fusion-tree vertices. In its regular embedding, all nearest edge-midpoint separations are equal. Its line graph, which assigns one vertex to each original edge and joins two such vertices when the corresponding edges share an endpoint, has degree four. Vertex helpers and hexagonal plaquette helpers fit without planar crossings, and the pattern repeats from a single coordinate motif. Fabrication stays challenging because the required clusters, yields, long-range tails, and six-edge effective terms are undemonstrated.

Fallback — square–octagon logical graph. This graph keeps degree three and planarity, and its orthogonal square motif may help registration, routing, or device boundaries.

The fallback graph gives two intended distances and two face sizes in its regular edge-midpoint layout, so it needs calibration or engineered couplings. It earns selection only when fabrication tests show that orthogonal patterning and access outweigh the factor-of-about-2.23 geometric spread in a nominal r^{-3} bond scale.

Both graph choices are proposals; neither reports an observed topological material. The selected target graph is the trivalent honeycomb graph.

One encoded cluster sits at the midpoint of each honeycomb edge. The resulting physical nearest-constraint graph is the degree-four kagome lattice.

A vertex term touches three clusters, while a hexagonal plaquette term touches at least the six clusters on the plaquette boundary. The regular nearest midpoint spacing is $\sqrt{3}a/2$.

Long-range pair terms stay in the Hamiltonian. The square–octagon graph stays as fallback, with two intended midpoint distances and faces of four and eight edges.

The next stage selects no further tiling. It builds the zero-field, Zeeman, dipolar, exchange, strain, phonon, and drive terms that can generate the selected graph and its operators. A Zeeman term couples spins to a magnetic field; dipolar and exchange terms give two distinct mechanisms of interaction between localized degrees of freedom; strain and phonon terms couple to static deformation and lattice vibrations; drive terms are externally applied time-dependent controls.

Consistency checks

- **Derive the degree-four kagome placement graph from honeycomb edge clusters.**

Two cluster sites touch when their honeycomb edges share a vertex. This line-graph construction on the honeycomb lattice produces the kagome lattice. Because the honeycomb graph has degree 3, the line-graph degree is

$$z_{L(G)} = 2(3 - 1) = 4.$$

Here, $z_{L(G)}$ is the coordination number, or vertex degree, of the line graph $L(G)$.

- **Distinguish the three graphs used in the design.**

The three graphs are the string-net target graph, the physical placement graph, and the realized interaction graph with its unwanted and mediated couplings. The target graph fixes the intended logical connectivity; the placement graph fixes the physical coordinates and local neighbor relations; the realized interaction graph lists every coupling in the physical Hamiltonian.

- **Calculate the relative strength of a pure r^{-3} interaction tail on the next kagome shell.**

The next-neighbor distance is $d_2 = \sqrt{3} d_1$, with d_1 the nearest-shell distance. For an interaction with $J(r) \propto r^{-3}$,

$$J(d_2)/J(d_1) = (d_1/d_2)^3 = 1/(3\sqrt{3}) \approx 0.19.$$

Thus, the next-shell interaction is about 0.19 of the nearest-shell interaction in this idealized distance-only model.

- **Identify the limitation of using planarity as evidence that mediators are straightforward to implement.**

A six-leg star of pairwise interactions is not the plaquette operator B_p .

- **State the primary and fallback graph decision.**

The primary design uses a honeycomb target graph with kagome edge-cluster positions. The square–octagon target graph serves as fallback, accepted only when its two spacing classes and two face classes can be compensated, including the factor-of-about-2.23 spread in a nominal r^{-3} bond scale.

- **Distinguish beam width from placement accuracy.**

Beam width, lateral straggle, the final defect distribution, and registration relative to a device are four different quantities. Each cited numerical value needs its associated definition and experimental conditions.

A geometric tiling alone defines no Hamiltonian. The honeycomb lattice is the target graph because it is trivalent and has one face size.

The clusters sit on kagome sites. The realized coupling pattern is a third, distinct graph, and the 0.19 dipolar tail is large enough that its additional edges stay in the Hamiltonian.

Sources

- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110); [arXiv:cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617).
 - [R017] A. Kitaev, “Anyons in an exactly solved model and beyond,” *Annals of Physics* 321, 2–111 (2006). DOI: [10.1016/j.aop.2005.10.005](https://doi.org/10.1016/j.aop.2005.10.005); [arXiv:cond-mat/0506438](https://arxiv.org/abs/cond-mat/0506438).
 - [R184] H. Yao and S. A. Kivelson, “An exact chiral spin liquid with non-Abelian anyons,” *Physical Review Letters* 99, 247203 (2007). DOI: [10.1103/PhysRevLett.99.247203](https://doi.org/10.1103/PhysRevLett.99.247203); [arXiv:0708.0040](https://arxiv.org/abs/0708.0040).
 - [R185] M. Kargarian, H. Bombin, and M. A. Martin-Delgado, “Topological color codes and two-body quantum lattice Hamiltonians,” *New Journal of Physics* 12, 025018 (2010). DOI: [10.1088/1367-2630/12/2/025018](https://doi.org/10.1088/1367-2630/12/2/025018); [arXiv:0906.4127](https://arxiv.org/abs/0906.4127).
 - [R186] G. Kells, J. Kailasvuori, J. K. Slingerland, and J. Vala, “Kaleidoscope of topological phases with multiple Majorana species,” *New Journal of Physics* 13, 095014 (2011). DOI: [10.1088/1367-2630/13/9/095014](https://doi.org/10.1088/1367-2630/13/9/095014); [arXiv:1012.5276](https://arxiv.org/abs/1012.5276).
 - [R187] R. Moessner and S. L. Sondhi, “Resonating valence bond phase in the triangular lattice quantum dimer model,” *Physical Review Letters* 86, 1881–1884 (2001). DOI: [10.1103/PhysRevLett.86.1881](https://doi.org/10.1103/PhysRevLett.86.1881); [arXiv:cond-mat/0007378](https://arxiv.org/abs/cond-mat/0007378).
 - [R188] L. Balents, M. P. A. Fisher, and S. M. Girvin, “Fractionalization in an easy-axis kagome antiferromagnet,” *Physical Review B* 65, 224412 (2002). DOI: [10.1103/PhysRevB.65.224412](https://doi.org/10.1103/PhysRevB.65.224412); [arXiv:cond-mat/0110005](https://arxiv.org/abs/cond-mat/0110005).
 - [R189] D. M. Toyli, C. D. Weis, G. D. Fuchs, T. Schenkel, and D. D. Awschalom, “Chip-scale nanofabrication of single spins and spin arrays in diamond,” *Nano Letters* 10, 3168–3172 (2010). DOI: [10.1021/nl102066q](https://doi.org/10.1021/nl102066q).
 - [R190] T. Schröder et al., “Scalable focused ion beam creation of nearly lifetime-limited single quantum emitters in diamond nanostructures,” *Nature Communications* 8, 15376 (2017). DOI: [10.1038/ncomms15376](https://doi.org/10.1038/ncomms15376); [stable full text](#).
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Chapter 26 — Microscopic operators supported by diamond

Start with three negatively charged nitrogen-vacancy centers, NV^- , a few nanometres apart in diamond. Each defect pairs a missing carbon atom with a neighboring substitutional nitrogen atom and traps an extra electron. Its electronic state reports through optical fluorescence.

The NV^- ground state splits by 2.87 GHz even at zero field. Microwave control isolates the $m_s = 0$ and $m_s = -1$ electron-spin states as a two-level system. The remaining $m_s = +1$ state is a leakage state: it sits outside the chosen qubit subspace and does not serve as a third qubit value.

The electron spins couple through their magnetic dipole moments, whose strength falls with separation r as $1/r^3$. At five nanometres the characteristic frequency is hundreds of kilohertz, far below gigahertz scales. Each defect also carries a crystallographically fixed axis, a nitrogen nuclear spin, strain and phonon couplings, and externally applied microwave interactions.

A host-supported microscopic Hamiltonian, as used here, keeps every term either because diamond supplies the corresponding interaction or explicitly, so that its coefficient can be set to zero under a stated approximation.

The microscopic state space narrows in two stages. First, each spin-1 electron truncates to a driven two-level qubit. Second, each triangular cluster of three qubits projects onto two selected cluster states. This chain turns the proposed physical ingredients into an operator with a tractable low-energy description.

Geometry and spectrum of a three-vacancy cluster

Take three NV^- centers sharing one crystallographic axis, the local z direction. Ideally they sit at the vertices of an equilateral triangle in the plane perpendicular to z .

A diamond (111) plane has the required symmetry. Still, fabricating many identical nanometre-scale triangles remains a proposal; no demonstrated device does this yet.

Each ground-state NV electron carries spin $S = 1$. Call $|0\rangle$ the state with $m_s = 0$ and $|1\rangle$ the state with $m_s = -1$. Define the Pauli operators τ^α , with $\alpha = x, y, z$, inside this two-state subspace by the displayed action:

$$\tau^z|0\rangle = |0\rangle, \quad \tau^z|1\rangle = -|1\rangle.$$

The omitted $m_s = +1$ state is a leakage state rather than a third qubit value.

Apply a static magnetic field B_0 along z plus a microwave field near the $|0\rangle \leftrightarrow |1\rangle$ transition. Describe the dynamics in a frame rotating at the microwave frequency, a time-dependent representation with the rapid phase evolution at the drive frequency removed.

Terms still oscillating rapidly in this frame can be dropped when their frequencies lie far above all relevant coupling strengths. This step is the rotating-wave approximation. The secular part of the dipole interaction is the slowly varying component left after the gigahertz-frequency oscillations are removed.

For two identical, resonant NV qubits, the secular pair energy takes the displayed form:

$$\frac{H_{ij}^{\text{sec}}}{h} = \frac{c_{ij}}{4}(1 - 3 \cos^2 \theta_{ij}) (\tau_i^z \tau_j^z - \tau_i^x \tau_j^x - \tau_i^y \tau_j^y) + \text{one-body shifts}.$$

Here h is Planck's constant, $c_{ij} > 0$ is a positive frequency defined below, and θ_{ij} is the angle between the pair displacement and the z axis. For the planar triangle, $\theta_{ij} = \pi/2$. When all three sides share one length, $c_{ij} = c$, and the cluster Hamiltonian becomes the displayed operator:

$$\frac{H_C}{h} = \frac{c}{4} \sum_{i < j \in C} (\tau_i^z \tau_j^z - \tau_i^x \tau_j^x - \tau_i^y \tau_j^y). \quad (26.1)$$

This three-qubit Hamiltonian solves exactly. In the one-excitation basis $\{|100\rangle, |010\rangle, |001\rangle\}$, every diagonal matrix element equals $-c/4$ and every off-diagonal hopping element equals $-c/2$. The symmetric state defined below is therefore an eigenstate:

$$|W_1\rangle = \frac{|100\rangle + |010\rangle + |001\rangle}{\sqrt{3}}$$

has quasienergy $-5hc/4$. The two states with relative phases $1, e^{\pm 2\pi i/3}, e^{\mp 2\pi i/3}$ have quasienergy $+hc/4$. A quasienergy is an eigenvalue of a periodically driven (Floquet) Hamiltonian. These rotating-frame quasienergies exist only while the drive and its phase reference persist, and only modulo the drive frequency; they are not laboratory-frame energies. Switching the drive off removes their connection to measured energy splittings.

Flipping every qubit gives the symmetric two-excitation state defined below,

$$|W_2\rangle = \frac{|011\rangle + |101\rangle + |110\rangle}{\sqrt{3}},$$

which also has quasienergy $-5hc/4$. The fully polarized states have quasienergy $+3hc/4$. In the ideal resonant rotating frame, $|W_1\rangle$ and $|W_2\rangle$ thus form a doublet set apart from the nearest cluster states by the displayed gap:

$$\Delta_C/h = \frac{3c}{2}. \quad (26.2)$$

The ideal triangular cluster therefore opens a gap of $3c/2$ in frequency units between the symmetric W -state doublet and the chiral states, the eigenstates carrying phases $1, e^{\pm 2\pi i/3}, e^{\mp 2\pi i/3}$.

Define the cluster Pauli operators X_C, Y_C, Z_C by $Z_C|W_1\rangle = |W_1\rangle$ and $Z_C|W_2\rangle = -|W_2\rangle$. Let P_C project onto this doublet. Direct evaluation gives, for any site i in the symmetric triangle, the displayed projection relations:

$$P_C \tau_i^z P_C = \frac{1}{3} Z_C, \quad P_C \tau_i^x P_C = \frac{2}{3} X_C, \quad P_C \tau_i^y P_C = \frac{2}{3} Y_C. \quad (26.3)$$

Equation (26.3) maps microscopic qubit operators onto encoded cluster operators. It also marks a limit: a physical pair interaction projects to an encoded pair interaction at first order. Projection rescales coefficients and operator components, yet it does not generate a plaquette operator, a multibody operator around a lattice plaquette, on its own.

If the sides or axes become unequal, the doublet in (26.2) splits, and the mapping in (26.3) gives way to a different projected matrix representation.

Complete host-supported microscopic Hamiltonian

The Hamiltonian below is divided by H/h , so every coefficient reads in hertz. The electron spin operators S_i^α are dimensionless with eigenvalues $m_s = 0, \pm 1$.

Each NV center carries a local orthonormal frame (x_i, y_i, z_i) , with z_i along the nitrogen–vacancy axis. The array design uses only one of diamond’s four NV orientations, because mixing orientations would split levels differently and change the interaction tensors.

A host-supported microscopic Hamiltonian takes the displayed form:

$$\begin{aligned} \frac{H}{h} = & \sum_i \left[D_i \left((S_i^{z_i})^2 - \frac{2}{3} \right) + E_{x,i} \left((S_i^{x_i})^2 - (S_i^{y_i})^2 \right) + E_{y,i} \{ S_i^{x_i}, S_i^{y_i} \} \right] \\ & + \sum_i \gamma_e \mathbf{B}_i \cdot \mathbf{S}_i + \sum_i (\mathbf{S}_i \cdot \mathbf{A}_i \cdot \mathbf{I}_i + P_i (I_i^{z_i})^2 - \gamma_{n,i} \mathbf{B}_i \cdot \mathbf{I}_i) \\ & + \sum_{i < j} c_{ij} [\mathbf{S}_i \cdot \mathbf{S}_j - 3(\mathbf{S}_i \cdot \hat{\mathbf{r}}_{ij})(\mathbf{S}_j \cdot \hat{\mathbf{r}}_{ij})] \\ & + \sum_{i < j} J_{ij}^{\text{ex}} \mathbf{S}_i \cdot \mathbf{S}_j + \frac{H_{\text{strain}}(t) + H_{\text{ph}} + H_{\text{drive}}(t)}{h}. \end{aligned} \quad (26.4)$$

The anticommutator is $\{A, B\} = AB + BA$. Every term in (26.4) either comes from the diamond host or is kept explicitly so that setting it to zero names a physical approximation.

The first line holds the zero-field splitting and static-strain terms. The coefficient D_i is the axial zero-field splitting; for the NV^- ground state near room temperature, $D \approx 2.87$ GHz [R074]. Subtracting $2/3$ drops an irrelevant trace, shifting all energies by one constant. The frequencies $E_{x,i}$ and $E_{y,i}$ describe transverse symmetry breaking from static strain and electric fields.

The NV ground-state spin Hamiltonian and its electric response are experimentally established [Experiment] [R074]; [R191].

Controllable or fluctuating strain separates from the static terms in the displayed time-dependent contribution:

$$\frac{H_{\text{strain}}(t)}{h} = \sum_i [\delta D_i(t)(S_i^{z_i})^2 + \epsilon_{x,i}(t)((S_i^{x_i})^2 - (S_i^{y_i})^2) + \epsilon_{y,i}(t)\{S_i^{x_i}, S_i^{y_i}\}] . \quad (26.5)$$

The coefficients in (26.5) fold in the spin–strain susceptibilities and read in hertz. Mechanical control of NV spin transitions has been demonstrated [Experiment] [R192]; [R193]. Equation (26.5) does not claim that strain couples spatially separated defects to each other.

The Zeeman term couples the spins to a magnetic field. The electron gyromagnetic ratio in frequency units is $\gamma_e = g_e \mu_B / h \approx 28 \text{ GHz/T}$, with g_e the electron g -factor and μ_B the Bohr magneton. The field $\mathbf{B}_i$ reads in tesla and $\mathbf{S}_i$ is dimensionless, so $\gamma_e B$ reads in s^{-1} , as required.

The hyperfine term couples each electron spin to its nuclear spin. The operator $\mathbf{I}_i$ is the dimensionless ^{14}N or ^{15}N nuclear spin, and $\mathbf{A}_i$ is its hyperfine tensor in hertz. The coefficient P_i is the ^{14}N nuclear quadrupole coefficient, and $\gamma_{n,i}$ is the nuclear gyromagnetic ratio. Nearby ^{13}C nuclei contribute terms of the same form.

The hyperfine interaction needs an explicit justification before it can be dropped: its megahertz-scale structure can outweigh the dipolar coupling in the numerical example below [Ro74]. The proposal therefore requires isotopic ^{12}C enrichment and preparation of every nitrogen nuclear spin in a specified state. Without these steps the NV centers differ from one another.

The magnetic dipole–dipole term is the direct interaction of the electron magnetic moments. The vector $\mathbf{r}_{ij}$ runs from defect i to defect j , $r_{ij} = |\mathbf{r}_{ij}|$, and $\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/r_{ij}$. Its coefficient is the displayed frequency:

$$c_{ij} = \frac{\mu_0 (g_e \mu_B)^2}{4\pi \hbar r_{ij}^3} . \quad (26.6)$$

Checking units gives $(\text{T m/A})(\text{J/T})^2/(\text{J s m}^3) = \text{s}^{-1}$. Numerically, the coefficient follows the displayed scaling:

$$c_{ij} \approx 52 \text{ MHz}(1 \text{ nm}/r_{ij})^3 .$$

Coherent coupling and entanglement between separate NV electron spins have been demonstrated [Experiment] [Ro80]. That experiment supports keeping the dipolar term, while it does not establish the fabrication yield a full array would need.

The coefficient J_{ij}^{ex} is an isotropic exchange frequency from overlap of the electronic wave functions. Diamond offers no verified, lithographically tunable, long-range NV–NV exchange bus of this kind.

Electronic wave-function overlap falls rapidly with separation, so the 5–10 nm model array below sets $J_{ij}^{\text{ex}} \approx 0$. The symbol stays in (26.4) to name the discarded interaction explicitly. The antiferromagnetic open-chain Hamiltonian sometimes used for an abstract three-spin encoded qubit does not follow from this NV geometry.

Phonons are the quantized normal modes of lattice vibration. Let b_q annihilate a phonon in normal mode q of frequency ν_q . A compact host-supported form for the phonon Hamiltonian and spin–phonon coupling is the displayed operator:

$$\frac{H_{\text{ph}}}{\hbar} = \sum_q \nu_q b_q^\dagger b_q + \sum_{i,q} (b_q + b_q^\dagger) \left[g_{iq}^\parallel (S_i^{z_i})^2 + g_{iq}^x Q_i^x + g_{iq}^y Q_i^y \right], \quad (26.7)$$

$Q_i^x = (S_i^{x_i})^2 - (S_i^{y_i})^2$, $Q_i^y = \{S_i^{x_i}, S_i^{y_i}\}$, with every coefficient g a frequency: phonons can coherently drive a selected mechanical mode [R192]; [R193], while an uncontrolled phonon bath causes relaxation and dephasing. Relaxation changes state populations; dephasing destroys phase coherence without necessarily changing populations.

Measured NV longitudinal relaxation depends strongly on temperature [Experiment] [R194]. The uncontrolled bath must therefore stay in the model with its noise included.

A microwave magnetic field $\mathbf{B}_{1,i}(t)$ generates the displayed drive term:

$$\frac{H_{\text{drive}}(t)}{\hbar} = \sum_i \gamma_e \mathbf{B}_{1,i}(t) \cdot \mathbf{S}_i. \quad (26.8)$$

Optical fields initialize and read out the NV center through excited electronic states [Ro74], which (26.4) leaves out. Adiabatic elimination removes rapidly evolving excited states perturbatively and would produce light shifts, Raman couplings, scattering, and optical decay together. Keeping only the useful coherent interaction while dropping the associated decay would make the elimination inconsistent.

Every symbol in (26.4) is thus measured, estimated, or deliberately set to zero. Dropping the list of zeroed terms changes the assumed host Hamiltonian.

Projection from spin 1 to a driven qubit

The single-NV projector is the displayed operator:

$$P_i = |0\rangle\langle 0| + |1\rangle\langle 1|.$$

Inside this two-state subspace, the projected spin-1 operators take the displayed form:

$$P_i S_i^z P_i = \frac{\tau_i^z - \mathbb{1}}{2}, \quad P_i S_i^x P_i = \frac{\tau_i^x}{\sqrt{2}}, \quad P_i S_i^y P_i = \frac{\tau_i^y}{\sqrt{2}}. \quad (26.9)$$

With the magnetic field along the NV axis, the two transition frequencies from $m_s = 0$ are approximately $\omega_- = D - \gamma_e B_0$ and $\omega_+ = D + \gamma_e B_0$. A drive near ω_- supports rotating-wave and leakage expansions when ω_- , ω_+ , and their separation all lie far above the coupling strengths and Rabi frequencies, the coherent transition rates from the applied drive. Projecting the interaction of (26.6) and dropping terms oscillating near gigahertz frequencies then gives the secular pair Hamiltonian used for the triangle.

This derivation is [Theory]. Its result is a Floquet quasienergy Hamiltonian that lasts only while an external phase reference runs.

Switching the resonant drive off brings back the large laboratory-frame energy difference between $|W_1\rangle$ and $|W_2\rangle$. The encoded doublet is therefore a driven construct rather than a passive ground-space doublet: the first projection truncates to two driven levels, while the three microscopic NV spins stay physical spins and the two retained states at each NV stay physical qubit levels. Equation (26.9) does not generate an anyon, a quasiparticle excitation with nontrivial exchange or braiding statistics.

When the residual oscillation frequencies are comparable to the couplings, the rotating-wave approximation fails. Gigahertz-frequency terms then feed the dynamics, and (26.1) no longer gives the pair energy.

Projection from a triangle to a cluster doublet

Let $P = \prod_C P_C$ project every triangle onto the doublet of (26.2). Let V collect intercluster dipolar bonds, detuning disorder, and weak drives. Schrieffer–Wolff perturbation theory, the perturbative separation of kept and eliminated energy sectors from Chapter 22, gives the displayed effective Hamiltonian:

$$\frac{H_{\text{eff}}}{h} = P \frac{H_C + V}{h} P - P \frac{V}{h} Q \left(Q \frac{H_C - E_0}{h} Q \right)^{-1} Q \frac{V}{h} P + O\left(\frac{v^3}{(\Delta_C/h)^2}\right), \quad (26.10)$$

with $Q = \mathbb{1} - P$, E_0 the ideal doublet energy, and v bounding the frequencies in V . The states removed by the projector still shape the effective Hamiltonian through virtual excursions out of the kept subspace and back. Those processes generate the second-order term in (26.10).

For a physical bond between site i of cluster C and site j of cluster D , equation (26.3) gives the displayed projected form:

$$P \tau_i^\alpha T_{ij}^{\alpha\beta} \tau_j^\beta P = M_\alpha M_\beta \Sigma_C^\alpha T_{ij}^{\alpha\beta} \Sigma_D^\beta, \quad (M_x, M_y, M_z) = \left(\frac{2}{3}, \frac{2}{3}, \frac{1}{3}\right), \quad (26.11)$$

Here repeated indices α, β are summed, T_{ij} is the physical dipolar tensor in hertz, and $\Sigma_C = (X_C, Y_C, Z_C)$. A uniform resonant transverse drive projects as $P_C \Gamma \sum_i \tau_i^x P_C = 2\Gamma X_C$.

A detuning term $\sum_i \delta_i \tau_i^z / 2$ yields $(\sum_i \delta_i) Z_C / 6$ at first order. Unequal detunings additionally mix the doublet with the chiral states, giving corrections of order $\sigma_\delta^2 / (\Delta_C / h)$.

The resulting model is the displayed effective Hamiltonian:

$$\frac{H_{\text{eff}}}{h} = \sum_C (2\Gamma_C X_C + \bar{\delta}_C Z_C) + \sum_{\langle C,D \rangle} \sum_{\alpha,\beta} \tilde{J}_{CD}^{\alpha\beta} \Sigma_C^\alpha \Sigma_D^\beta + \delta H^{(2)}/h, \quad (26.12)$$

This effective Hamiltonian is a driven pseudospin model rather than a string-net Hamiltonian, a constrained many-body model whose edge labels and local recoupling rules can realize topological order. In (26.12), $\tilde{J} = MTM$, and $\delta H^{(2)}/h$ holds shifts and further two-cluster terms. When two bonds share a virtual cluster excitation, it can also hold small multicluster terms with scale at most approximately $u^2/(\Delta_C/h)$.

The geometry and the perturbative energy denominators fix these coefficients. They do not automatically equal the Fibonacci F -symbols, the recoupling coefficients of Fibonacci anyon fusion.

At first order, then, an intercluster bond stays a bilinear bond between cluster Pauli operators, with projection factors $(2/3, 2/3, 1/3)$. The cluster doublet is an encoded pseudospin, an effective two-level degree of freedom built from several microscopic spins.

A periodically driven array of these pseudospins forms an analog emulator. Neither the pseudospin encoding nor the analog-emulator description, alone or combined, demonstrates emergent anyons or equilibrium topological order.

For a concrete parameter estimate, take $B_0 = 20$ mT. With $D = 2.87$ GHz and $\gamma_e = 28$ GHz/T, the transition frequencies become $\omega_- \approx 2.31$ GHz and $\omega_+ \approx 3.43$ GHz.

For an illustrative triangle side $r = 5$ nm, equation (26.6) gives $c \approx 52/5^3$ MHz = 0.416 MHz and therefore $\Delta_C/h \approx 0.624$ MHz. At a representative intercluster separation of 10 nm, the unprojected dipolar scale is $u \lesssim 52$ kHz before angular and projection factors enter.

A weak drive with $\Gamma \sim 20$ kHz would satisfy $u, \Gamma \ll \Delta_C/h \ll \omega_\pm$.

This dimensional hierarchy is plausible as a theoretical estimate. The full operating regime is narrower, as the displayed hierarchy shows:

$$\omega_\pm \gg c \gg u, \Gamma, \sigma_\delta, \Gamma_{\text{decoh}}, \quad k_B T/h \ll \Delta_{\text{desired}}/h, \quad (26.13)$$

The nuclear-spin states must also be fixed and the rotating-frame phases synchronized. Here Γ_{decoh} is the relevant decoherence rate, k_B is Boltzmann's constant, T is temperature, and Δ_{desired} is the desired many-body energy gap. The final inequality points to a many-body gap not yet shown to exist for (26.12).

Nanoscale aperture masks with sub-10 nm features have confined implanted NV clusters, including clusters of up to three centers [Experiment] [R195]. Ion straggle and conversion statistics keep this result from establishing deterministic placement of a specified 5 nm equilateral triangle, and it does not establish fabrication of a honeycomb-edge array.

These numerical values define a falsifiable design point [Proposal]; they do not claim that such a device has been demonstrated.

The leading controlled corrections are the displayed operator-norm bound:

$$\|\delta H^{(2)}/h\| = O\left(\frac{u^2 + \Gamma^2 + \sigma_\delta^2}{\Delta_C/h} + \frac{c^2 + \Omega^2}{\min(\omega_-, \omega_+)} + \frac{c^2 + \Omega^2 + E_\perp^2}{\Delta_{\text{leak}}/h}\right), \quad (26.14)$$

Here Ω is a physical microwave Rabi frequency, $E_\perp^2 = E_x^2 + E_y^2$, and Δ_{leak} is the relevant separation from the $m_s = +1$ state. The first fraction in (26.14) corrects for cluster leakage, the second for the nonsecular or Bloch–Siegert shift, and the third for spin-1 leakage. A Bloch–Siegert correction is the frequency shift from counter-rotating drive terms dropped by the rotating-wave approximation.

Equation (26.14) gives the scaling of these corrections with the energy denominators above. A device-specific calculation must diagonalize the actual interaction tensors fixed by its geometry.

When u is comparable to Δ_C/h , the Schrieffer–Wolff truncation fails: transitions out of the kept doublet stop being perturbative, and the doublet stops defining the slow effective subspace.

Missing operators required for topological order

The proposed sequence runs from NV electron spins through coupled three-NV clusters and cluster pseudospins to a honeycomb-edge model and finally topological order. The last step in this chain has not been established.

A low-energy projector carries over the microscopic interactions already in the Hamiltonian. Choosing an architecture does not by itself add further interactions.

The doubled-Fibonacci Levin–Wen target holds a local branching constraint Q_v at every trivalent vertex and a plaquette operator B_p that inserts a loop and recouples edge labels with the Fibonacci F -symbols [Ro18]. For the honeycomb-edge architecture, equation (26.12) lacks the following:

- a three-edge vertex projector realizing the allowed Fibonacci fusion channels;
- the conditional many-edge off-diagonal plaquette action with the correct relative amplitudes and phases;
- a mechanism cancelling the larger one-body and two-body terms while keeping the intended gap open;
- evidence placing the residual terms of (26.14) inside the stability region of the doubled-Fibonacci phase;
- a passive laboratory-frame ground manifold, since the present doublet is resonantly dressed.

Higher-order perturbative gadgets could be proposed for the first three items. They would need mediator degrees of freedom and would yield amplitudes suppressed by products of u/Δ_C . Those mediators are absent from the selected microscopic Hamiltonian. Hence [Theory] the derived model is an anisotropic, driven pseudospin model.

[Speculation] It might contribute to a separately engineered gadget construction, but it does not presently derive Fibonacci order.

The derivation settles three separate questions. Diamond supplies known interactions; the kept subspace retains calculable interactions; and the surviving interactions lack the structure the target topological model requires.

Experimentally established components and unresolved requirements

[Experiment] Single-NV zero-field, Zeeman, hyperfine, electric/strain, microwave, optical, and phonon-relaxation physics are well characterized [R074]; [R191]; [R192]; [R193]; [R194]. [Experiment] Dipolar coupling has entangled two individually addressed NV electron spins [R080]. These results justify the corresponding terms in (26.4).

The unestablished elements are equally specific. References [R074]; [R191]; [R192]; [R193]; [R080]; [R194]; [R018]; [R195] report no mesoscopic array of identical 5 nm three-NV triangles with fixed orientation, initialized nuclear states, uniform detunings, and individually phased drives.

These references also report no measurement of the cluster gap in (26.2), a many-body topological gap, topological degeneracy, a string operator, or an anyonic excitation in this material architecture. The established microscopic ingredients and the proposed architecture therefore remain separated by unresolved fabrication and control requirements.

A measured pair coupling is a physical bond, and a projected cluster doublet is an encoded residual degree of freedom. Neither acts as a vertex projector, a plaquette recoupling operator, or a topological gap.

Common modeling errors

- Quasienergy degeneracy depends on an applied drive and a clock or phase reference. It gives no thermal ground-state protection. If the drive is turned off, the doublet in (26.2) is lost because the laboratory-frame splitting between $|W_1\rangle$ and $|W_2\rangle$ returns.
- Isotropic antiferromagnetic exchange gives a convenient encoded-spin algebra, but it is not a supported coupling between NV centers 5–10 nm apart in this architecture. The term J_{ij}^{ex} should stay visible in (26.4) and then be set to zero explicitly. Restoring it without justification changes the assumed host Hamiltonian.
- An unresolved hyperfine manifold is large compared with a sub-megahertz cluster gap, so it cannot be treated as a negligible correction.
- The same spin–phonon coupling also opens relaxation and dephasing channels. Any coherent-mode proposal must therefore include loss and thermal occupation.
- First-order projection preserves operator locality. Multibody terms need virtual processes with energy denominators and competing corrections.
- The labels “honeycomb,” “cluster,” and “plaquette” do not establish a physical phase. The deciding evidence is the spectrum, gap, ground-state degeneracy, and quasiparticle properties.
- The values five and ten nanometres are assumed separations rather than demonstrated placement distributions or fabrication yields.

Consistency checks

- **Exchange term.** Exchange, a spin–spin interaction transferring excitations between NV centers, enters the microscopic Hamiltonian and is then set to zero, making the resource audit explicit. Long-range tunable NV–NV exchange is not a supported resource at the chosen spacing. Keeping it implicitly would change the assumed host system.
- **Cluster leakage gap.** The cluster leakage gap Δ_C is the energy separation between the encoded cluster subspace and the states outside it. For the ideal triangle,

$$\Delta_C/h = 3c/2.$$

In the one-excitation basis, the symmetric W -state has energy $-5c/4$, while the chiral states with opposite phase winding around the triangle have energy $+c/4$. Their separation is therefore $3c/2$. The two-excitation partner $|W_2\rangle$ gives the same result.

- **Replacement by isotropic antiferromagnetic exchange.** Isotropic antiferromagnetic exchange is a rotationally invariant spin coupling favoring antiparallel spin alignment. Replacing the dipolar interaction with this coupling changes the model: the secular pair energy, keeping the resonant or energy-conserving interaction terms, loses the XXZ form of (26.1). Here, XXZ means an anisotropic spin interaction with equal transverse couplings and a generally different longitudinal coupling. The abstract three-spin Heisenberg encoding of Chapter 24 therefore does not follow from this NV geometry.
- **First-order form of an intercluster dipolar bond.** A bilinear interaction holds one operator from each of two coupled clusters. Equation (26.3) maps each physical Pauli operator onto a cluster Pauli operator times one of the projection factors $(2/3, 2/3, 1/3)$. Hence (26.11) stays a two-cluster bilinear coupling,

$$\tilde{J} = MTM,$$

where T is the physical dipolar coupling tensor and M holds the projection factors.

- **Absent target interactions.** The construction produces neither the Fibonacci branching projector nor the conditional F -symbol plaquette recoupling operator, and it does not demonstrate a passive topological gap.
- **Removal of the resonant drive.** Quasienergy is the energy-like eigenvalue of a periodically driven system. If the resonant drive is turned off, the quasienergy degeneracy of $|W_1\rangle$ and $|W_2\rangle$ disappears. The laboratory-frame energy splitting returns, and the encoded doublet stops forming a passive ground-space pair.

A same-orientation NV array has all NV symmetry axes aligned and obeys (26.4). Spin-1 projection, the restriction of the physical NV spin-1 Hilbert space to the selected effective subspace, gives (26.9) and a secular dipolar tensor.

An ideal resonantly driven three-NV triangle holds the doublet (26.2) of two encoded states. Local operators project into this doublet according to (26.3).

Intercluster bonds give (26.11), with corrections scaling as (26.14). These results provide no static encoded doublet, no isotropic exchange, no vertex projector, no plaquette recoupling operator, and no topological gap.

Sources

- [R074] M. W. Doherty, N. B. Manson, P. Delaney, et al., “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* 528, 1–45 (2013). DOI: [10.1016/j.physrep.2013.02.001](https://doi.org/10.1016/j.physrep.2013.02.001).
 - [R191] F. Dolde, H. Fedder, M. W. Doherty, et al., “Electric-field sensing using single diamond spins,” *Nature Physics* 7, 459–463 (2011). DOI: [10.1038/nphys1969](https://doi.org/10.1038/nphys1969).
 - [R192] A. Barfuss, J. Teissier, E. Neu, A. Nunnenkamp, and P. Maletinsky, “Strong mechanical driving of a single electron spin,” *Nature Physics* 11, 820–824 (2015). DOI: [10.1038/nphys3411](https://doi.org/10.1038/nphys3411).
 - [R193] E. R. MacQuarrie, T. A. Gosavi, A. M. Moehle, N. R. Jungwirth, S. A. Bhawe, and G. D. Fuchs, “Coherent control of a nitrogen-vacancy center spin ensemble with a diamond mechanical resonator,” *Optica* 2, 233–238 (2015). DOI: [10.1364/OPTICA.2.000233](https://doi.org/10.1364/OPTICA.2.000233).
 - [R080] F. Dolde, I. Jakobi, B. Naydenov, et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* 9, 139–143 (2013). DOI: [10.1038/nphys2545](https://doi.org/10.1038/nphys2545).
 - [R194] A. Jarmola, V. Acosta, K. Jensen, S. Chemerisov, and D. Budker, “Temperature- and magnetic-field-dependent longitudinal spin relaxation in nitrogen-vacancy ensembles in diamond,” *Physical Review Letters* 108, 197601 (2012). DOI: [10.1103/PhysRevLett.108.197601](https://doi.org/10.1103/PhysRevLett.108.197601).
 - [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110).
 - [R195] T.-Y. Hwang, J. Lee, S.-W. Jeon, et al., “Sub-10 nm precision engineering of solid-state defects via nanoscale aperture array mask,” *Nano Letters* 22, 1672–1679 (2022). DOI: [10.1021/acs.nanolett.1c04699](https://doi.org/10.1021/acs.nanolett.1c04699).
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PART X — QUANTITATIVE SCALES, NOISE, AND EXPERIMENTAL ANALYSIS

This part compares the proposed device’s energy scales and distances with measured values, keeping measurement conditions and units explicit. It then uses noise and measurement limits to assess what those scales can support.

Chapter 27 — Unit conversion and interpretation of measured quantities

A coherence time is meaningless as a bare number like 13 ms. The number must stay attached to the conditions of its measurement.

The reported 13 ms coherence time belongs to one negatively charged silicon-vacancy center, SiV^- , specifically its electron spin near 100 mK after a 32-pulse Carr–Purcell–Meiboom–Gill sequence (CPMG-32) in diamond with $10^{-3}\%$ ^{13}C [Ro83]. Changing the temperature, dropping the pulse sequence, or restoring a higher carbon-13 concentration changes the physical system or the measurement protocol, and with it the quantity under study.

In this chapter, a numerical value plus all conditions needed to read it is a measurement-condition record.

A frequency, time, or fidelity without such a record cannot serve as a parameter in a Hamiltonian, the operator fixing the energy and dynamics of a quantum system. The value may stay qualitatively informative, but it cannot support a quantitative engineering comparison across systems or operating conditions.

Three physical quantities expressed in a common unit

The values 2.87 GHz, 4.93 kHz, and 12.7 kHz all read in hertz, yet they name three physically different kinds of quantity.

The first kind is a static level splitting, an energy difference between stationary states. Such a splitting fixes a local basis. One example is the zero-field splitting of the negatively charged nitrogen-vacancy center, NV^- , in its electronic ground state. It separates the $m_s = 0$ spin projection from the $m_s = \pm 1$ projections before any external magnetic field is applied.

The second kind is a coupling, a Hamiltonian interaction that can move quantum information or generate entanglement. The fitted dipolar matrix element for two NV centers 25 nm apart is an example.

The third kind is a decay rate or linewidth, a measure of phase loss or spectral broadening. Ramsey linewidths, spin-echo linewidths, and phonon-broadened optical linewidths are examples. They are not interchangeable because each protocol samples different noise processes.

All three kinds can be quoted in megahertz, yet the shared unit gives them no shared physical meaning.

An on-site splitting sets local level structure, whereas intercluster coupling sets the interaction scale; increasing the former does not strengthen the latter. Spin-echo duration measures refocused phase memory, so it does not predict undriven many-body evolution. Room-temperature operation likewise leaves thermal populations determined by $(k_{\text{B}}T)$ and the level splittings, not by the operating label.

Two comparisons recur through the rest of the book:

useful coupling > uncontrolled broadening and drift,
effective gap > thermal and disorder scales.

The numbers on both sides of these inequalities keep their measurement conditions. Satisfying either inequality does not by itself establish a topological gap, a property of the many-body energy spectrum not calculated on this page.

Each numerical entry should therefore be stored as a record with the following mandatory fields:

quantity | platform and charge state | sample | temperature
| field/strain/drive protocol | measurement definition | source

If any field is missing, the value can be misapplied in an inequality for a different physical regime.

Conversion between frequency and temperature units

Experiments commonly report frequencies, temperatures, or energies. A Hamiltonian carries dimensions of energy, in joules in the International System of Units. These descriptions cover the same energy interval only once the conversion convention is stated.

Let ν be ordinary cyclic frequency in hertz, one cycle per second. Let $\omega = 2\pi\nu$ be angular frequency in radians per second. Let E be energy in joules or electronvolts. They connect through the displayed relations:

$$E = h\nu = \hbar\omega, \quad T_E = \frac{E}{k_B} = \frac{h\nu}{k_B},$$

where h is Planck's constant, $\hbar = h/(2\pi)$ is the reduced Planck constant, and k_B is Boltzmann's constant. The quantity T_E is the equivalent temperature belonging to the energy E .

The equivalent temperature need not equal the cryostat temperature. It is the temperature at which the thermal energy scale $k_B T$ equals E .

The constants $h = 6.62607015 \times 10^{-34}$ J s, $k_B = 1.380649 \times 10^{-23}$ J K⁻¹, and the elementary charge are exact in the SI [R196]. The conversion table therefore reads as displayed:

$$\begin{aligned} 1 \text{ GHz} &\leftrightarrow 4.135667696 \text{ } \mu\text{eV} \leftrightarrow 47.9924307 \text{ mK}, \\ 1 \text{ MHz} &\leftrightarrow 0.004135667696 \text{ } \mu\text{eV} \leftrightarrow 47.9924307 \text{ } \mu\text{K}, \\ 1 \text{ kHz} &\leftrightarrow 4.135667696 \times 10^{-6} \text{ } \mu\text{eV} \leftrightarrow 47.9924307 \text{ nK}, \\ 1 \text{ K} &\leftrightarrow 86.1733326 \text{ } \mu\text{eV} \leftrightarrow 20.8366191 \text{ GHz}. \end{aligned}$$

The arrows convert numerically between representations of one energy interval. They do not make gigahertz a unit of energy.

Gigahertz measures cyclic frequency; microelectronvolts measure energy.

Millikelvin in these conversions means equivalent temperature. A Hamiltonian coefficient should be written explicitly as $E = h\nu$ or $E/h = \nu$.

Checking units gives $(\text{J s})(\text{s}^{-1}) = \text{J}$, and $\text{J}/(\text{J K}^{-1}) = \text{K}$.

In this chapter, Hamiltonian coefficients read as cyclic frequencies. When a Hamiltonian term has energy J , the table entry is J/h in hertz.

If a paper writes $H = \hbar g(\dots)$, then g normally means an angular frequency. The matching cyclic frequency is $g/(2\pi)$. Dropping the factor of 2π shifts every later numerical comparison by about a factor of six.

The equivalent temperature T_E must also be kept apart from the cryostat set point. At a cryostat temperature of 100 mK, the thermal scale $k_B T$ far exceeds the energy of a 5 kHz splitting, whose equivalent temperature is a fraction of a microkelvin.

Conversion of three measured quantities into common units

For the NV zero-field splitting, take the $-$ ground-state spin triplet near room temperature. Its splitting is $D/h \approx 2.87 \text{ GHz}$ [R197]. The matching energy and equivalent temperature are the displayed values:

$$E_D = h(2.87 \text{ GHz}) = 11.87 \text{ } \mu\text{eV}, \quad T_D = 0.1377 \text{ K}.$$

At 300 K, $k_B T/h \approx 6.251 \text{ THz}$, far above D/h . Room-temperature optical polarization therefore runs through a driven optical cycle out of thermal equilibrium; the $m_s = 0$ state is not thermally isolated from the $m_s = \pm 1$ states.

For a measured NV–NV coupling, take an implanted pair $25 \pm 2 \text{ nm}$ apart at room temperature. The fitted secular dipolar coupling, the component of the magnetic dipole interaction kept under the relevant rotating-frame and energy-conservation approximation, was $4.93 \pm 0.05 \text{ kHz}$ [Ro80]. Its energy and equivalent temperature are the displayed values:

$$E_{dd} = 2.039 \times 10^{-5} \text{ } \mu\text{eV}, \quad T_{dd} = 2.366 \times 10^{-7} \text{ K}.$$

This small equivalent temperature still allows a driven gate at 300 K. It means thermal equilibrium cannot preferentially fill states split only by this bare pair coupling.

For comparison, at a cryostat set point of $T = 100 \text{ mK}$, the thermal energy reads as displayed:

$$k_B T = 8.617 \text{ } \mu\text{eV}, \quad \frac{k_B T}{h} = 2.084 \text{ GHz}.$$

An SiV orbital splitting with cyclic frequency 46 GHz has an equivalent temperature of 2.21 K. Its upper orbital branch is therefore strongly suppressed at 100 mK. A 5 kHz dipolar splitting is not thermally resolved under the same conditions. A single cryostat temperature can consequently suppress excitations of one degree of freedom while producing nearly equal thermal populations for another degree of freedom, corresponding to an effectively infinite-temperature regime for the latter.

The refrigerator temperature alone does not say which energy splittings thermal equilibrium resolves.

Distinction between decay times and energy gaps

Let T_2 be a coherence time, the decay time of phase coherence under a specified protocol. Under a purely exponential envelope $\exp(-t/T_2)$, the Lorentzian full width at half maximum is the displayed value:

$$\Delta\nu_{\text{FWHM}} = \frac{1}{\pi T_2}.$$

A Lorentzian is the spectral line shape from exponential temporal decay. This conversion depends on that model and does not define linewidth in general. Gaussian Ramsey decay, spectral diffusion, stretched-exponential decay, and pulse-filtered noise obey different relations. The measurement protocol and fitted envelope therefore belong with the reported value.

A gate duration t_g sets an inverse operation timescale $1/t_g$. This rate is not by itself a Hamiltonian gap.

$t_g = 20$ ns gives $1/t_g = 50$ MHz. The resonant Rabi frequency for a π rotation can differ from this inverse duration by factors set by the pulse convention. Keep the quantity labeled as an inverse duration rather than an energy.

Without the exponential-decay assumption, one reported T_2 can cover physically different noise processes. A quasi-static, calibratable detuning and an irreversible Markovian linewidth can give the same Ramsey decay time, although a later control pulse can reverse the former and not the latter.

Upper-level population

For two nondegenerate levels split by energy Δ , thermal equilibrium gives the displayed Boltzmann factor:

$$\frac{p_1}{p_0} = \exp\left(-\frac{\Delta}{k_{\text{B}}T}\right),$$

Here p_0 and p_1 are the occupation probabilities of the lower and upper levels. The dimensionless thermal-resolution ratio is then the displayed quotient:

$$R_T = \frac{\Delta}{k_{\text{B}}T}.$$

This ratio decides whether thermal equilibrium resolves the splitting.

When $R_T \ll 1$, the two levels are nearly equally populated in equilibrium unless a drive, measurement, or engineered reservoir pushes the system out of equilibrium. When $R_T \gg 1$, thermal excitation to the upper level is exponentially suppressed. When either level is degenerate, state multiplicity adds a degeneracy factor, and the displayed Boltzmann relation needs the corresponding modification.

Architecture analysis also requires the coupling-to-noise ratio and coherence-cycle count:

$$R_C = \frac{|J|/h}{\Delta\nu_{\text{noise}}}, \quad N_{\text{coh}} = \frac{|J|T_{2,\text{relevant}}}{h}.$$

Here J is the coupling energy, $\Delta\nu_{\text{noise}}$ is the relevant noise broadening as a cyclic frequency, and N_{coh} is roughly the number of coupling cycles fitting inside the applicable coherence time. The applicable coherence time must come from free-induction evolution or from a protection protocol compatible with the Hamiltonian of interest, rather than the longest available dynamical-decoupling number.

R_T and R_C test different physical comparisons, and neither is a many-body gap. A system can miss thermal resolution of a splitting yet still run driven gates on it. Conversely, a thermally resolved splitting can still be unusable under strong noise.

Experimentally reported quantities

The following table gives conditional ranges from identifiable experiments or, where marked, transparent conversions built on measurements.

A range can code for tensor components, crystallographically inequivalent defect configurations, or different protocols on one sample. The “conditions” column names the applicable reading. No such range counts as a measured distribution over a whole defect population.

The selected rows cover samples relevant to interacting-defect architectures: bulk and nanostructured hosts, implanted pairs with measured couplings, and protocols with stated temperatures. Strong dynamical-decoupling results remain in the table but are separated from coherence during undriven analog evolution. Where a comparable single-defect metric is missing, the table leaves that gap explicit rather than substituting an ensemble measurement for a different task.

Every frequency-valued Hamiltonian entry reads as E/h . “Equivalent T_E ” means E/k_B , not the operating temperature. Uncertainties and ranges keep the meanings of their original sources.

Quantity	Platform and defensible value	Explicit conversion	Conditions and what the range means	Status/source
Ground-state zero-field splitting	NV [−] , $D/h \approx 2.87$ GHz	11.87 μeV ; $T_E = 0.1377$ K	The value applies to single NV centers in bulk type-IIa diamond during room-temperature optical and microwave work. The final digits vary with	[Experiment] [R197]; [R198]

Quantity	Platform and defensible value	Explicit conversion	Conditions and what the range means	Status/source
			temperature, strain, and sample. Therefore, $D/h = 2.87$ GHz is a representative value rather than a universal exact constant.	
Inequivalent zero-field splittings	Neutral divacancies in 4H-SiC, about 1.30–1.34 GHz	5.38–5.54 μeV ; $T_E = 62.4\text{--}64.3$ mK	These values were measured for single c-axis divacancy configurations in high-purity semi-insulating 4H-SiC at 20 K. The range represents crystallographically inequivalent configurations rather than temperature drift [R096].	[Experiment] [R096]
Ruby zero-field transition	$\text{Cr}^{3+}:\text{Al}_2\text{O}_3$, 11.493 ± 0.004 GHz	47.53 ± 0.02 μeV ; $T_E = 551.6 \pm 0.2$ mK	A dilute Cr^{3+} ensemble in ruby was measured in a dilution refrigerator. The optically inferred local temperature reached a minimum of 143 ± 7 mK, although the cryostat reached 20 mK. This result concerns an ensemble transition rather than a demonstrated array of single-defect qubits [R088].	[Experiment] [R088]
Hyperfine tensor	$\text{NV}^- - ^{14}\text{N}$, $ A /h \approx 2.14\text{--}2.70$ MHz	$0.00885\text{--}0.01117$ μeV ; $T_E = 0.103\text{--}0.130$ mK	The values were obtained by room-temperature electron paramagnetic resonance and electron–nuclear double resonance, EPR/ENDOR, on NV ensembles. The interval gives the magnitudes of axial and transverse components of the hyperfine tensor, which describes the anisotropic interaction between electron and nuclear spins. It does not represent variation among nominally equivalent centers. Nearby ^{13}C sites can have very different hyperfine couplings [R197].	[Experiment] [R197]
Direct electron-spin coupling	NV^- pair, 4.93 ± 0.05 kHz	$(2.039 \pm 0.021) \times 10^{-5}$ μeV ; ; $T_E = 0.2366 \pm 0.0024$ μK	The sample contained two ^{15}N -implanted NV centers separated by 25 ± 2 nm and operated at room temperature. The reported value is the fitted secular dipolar interaction for that	[Experiment] [R080]

Quantity	Platform and defensible value	Explicit conversion	Conditions and what the range means	Status/source
			orientation. The same experiment measured double-quantum phase accumulation four times faster, but this does not imply a fourfold increase in the microscopic dipole coupling [R080].	
Dipolar engineering range	Electron spins at 10–50 nm: roughly 0.4–100 kHz before angular cancellation	1.7×10^{-6} – $4.1 \times 10^{-4} \mu\text{eV}$; $T_E \approx 0.02$ – $4.8 \mu\text{K}$	[Theory] The interval follows from the electron dipolar r^{-3} distance scaling and order-unity angular factors, constrained by the measured 25 nm pair above. It is a geometry-conditioned design range rather than a survey of fabricated pairs. Exchange interactions at atomic separations are excluded.	[Theory] based on [R080]
Static strain response	SiV [−] orbital transitions, order 0.1–1 PHz per unit strain; demonstrated optical tuning 150 GHz	The observed tuning is $620 \mu\text{eV}$, or 7.20 K equivalent	The measurements used individual SiV centers in an actuated diamond cantilever at 4 K. The susceptibility interval distinguishes spin-sensitive channels from orbital or symmetry channels. The 150 GHz value is the observed tuning range of one optical line, not a coherent spin–spin coupling [R200].	[Experiment] [R200]
Phonon-induced broadening	SiV [−] optical D line about 0.10–0.48 GHz over 4–20 K	0.41 – $1.99 \mu\text{eV}$; $T_E = 4.8$ – 23 mK	The values were measured in low-strain high-pressure, high-temperature and chemical-vapor-deposition, HPHT/CVD, bulk samples. Before a crossover, the low-temperature linewidth approximately followed $\Gamma = (-1.05 + 24.26T/\text{K}) \text{ MHz}$, and it saturated near the lifetime limit around 4 K. This quantity is a measured phonon-sensitive linewidth rather than a single-mode coupling constant [R201].	[Experiment] [R201]
Optical linewidth	Same low-strain SiV [−] D transition: near 100 MHz at 4 K, increasing with temperature	$0.414 \mu\text{eV}$; $T_E = 4.80 \text{ mK}$ at 100 MHz	This value is a single-site photoluminescence-excitation linewidth measured in bulk material. Nanostructure strain, spectral diffusion, power	[Experiment] [R201]

Quantity	Platform and defensible value	Explicit conversion	Conditions and what the range means	Status/source
			broadening, and the selected optical line can increase the linewidth. The bulk value therefore cannot be assigned directly to an implanted nanodevice [R201].	
Coherence without/with refocusing	Coupled NV pair: $T_2^* = 22.6\text{--}27.8\ \mu\text{s}$; double-quantum echo $T_2 = 150\text{--}514\ \mu\text{s}$	Exponential-equivalent widths: 11.5–14.1 kHz for T_2^* , and 0.62–2.12 kHz for echo	The two values in each interval correspond to the two implanted centers in the same 25 nm pair at room temperature. Ramsey and echo measurements are distinct protocols and cannot be combined into a single undifferentiated coherence value [R080].	[Experiment] [R080]
Coherence under strong protection	SiV ⁻ : $T_2^* \approx 1.5\text{--}13\ \mu\text{s}$ in $10^{-3}\%$ ^{13}C material; $T_2 = 13 \pm 1.7\ \text{ms}$ with CPMG-32	For 13 ms, exponential-equivalent width $\approx 24\ \text{Hz}$	The measurement used a single SiV center near 100 mK. Natural-abundance diamond in the same study gave $T_2^* \sim 0.3\ \mu\text{s}$. The 13 ms result characterizes pulse-protected memory and does not demonstrate 13 ms of unperturbed analog dynamics [R083].	[Experiment] [R083]
Coherence in another host	4H-SiC divacancy, Hahn-echo $T_2 \approx 1.2\text{--}1.25\ \text{ms}$	Exponential-equivalent width $\approx 255\text{--}265\ \text{Hz}$	The measurements involved isolated neutral divacancies in high-purity semi-insulating 4H-SiC at 20 K. A Hahn echo is a refocusing sequence containing one inversion pulse. The interval covers reported single centers or configurations under Hahn echo rather than shallow implanted arrays [R096].	[Experiment] [R096]
MW gate-duration and optical-initialization examples	NV ⁻ : a 20 ns near-inversion MW pulse in the measured Rabi sequence; a $1.3\ \mu\text{s}$, 532 nm polarization pulse	Inverse durations: 50 MHz and 0.77 MHz; these are rates, not gaps	The experiment used a single NV center in type-IIa HPHT diamond at 300 K. The study did not associate an average gate fidelity with the 20 ns pulse. It is therefore an example of duration rather than a platform-wide gate benchmark [R198].	[Experiment] [R198]

Quantity	Platform and defensible value	Explicit conversion	Conditions and what the range means	Status/source
Readout and initialization fidelity	NV ⁻ : electron initialization $\geq 99.7 \pm 0.1\%$; single-shot electron readout $93.2 \pm 0.5\%$ using a $40 \mu\text{s}$ window	Readout inverse duration 25 kHz, not an energy splitting	The measurements used a single NV center in bulk diamond below 10 K with resonant, spin-selective optical excitation. Room-temperature off-resonant fluorescence normally requires repeated averaging and cannot be assigned this cryogenic fidelity [R199].	[Experiment] [R199]
Slow cryogenic optical operations	SiV ⁻ : optical pumping timescale about 30 ms; single-shot readout about 89% with 20 ms integration	Inverse durations about 33 Hz and 50 Hz	The experiment used a single SiV center near 100 mK with the magnetic field aligned to the defect axis. The long-coherence experiment used still longer preparation and readout pulses. These durations characterize optical cycling and photon collection rather than microwave gate speed [R083].	[Experiment] [R083]

Interpretation requires fixing each quantity’s physical meaning and measurement conditions before comparing rows.

At room temperature, the NV control cycle is fast, while the demonstrated dipolar interaction for a 25 nm pair remains in the kilohertz range. SiV centers have narrow, stable optical transitions, but phonon coupling to their orbital states means that the longest reported spin memory requires millikelvin operation.

SiC shows millisecond Hahn-echo coherence at 20 K for isolated defects in high-quality material; a densely implanted interacting lattice is a different system still to be demonstrated.

Ruby offers a robust ensemble transition at $E/h = 11.493 \text{ GHz}$. Modern evidence for arrays of individually initialized and coupled Cr³⁺ defects with full initialization, control, coupling, and readout is still absent [Ro88].

Three representations of a single pair interaction

The measured room-temperature NV pair keeps coupling, coherence, and temperature data inside one experimental context. Let the coupling be $J/h = 4.93 \text{ kHz}$. For the two centers, take $T_2^* = 22.6\text{--}27.8 \mu\text{s}$ and echo times of 150–514 μs , all from the same sample and protocol family [Ro80]. The free-induction coupling-cycle count is then the displayed number:

$$N_{\text{coh}}^* = \frac{J}{h} T_2^* = 0.11\text{--}0.14,$$

The echo-based counterpart is the displayed number:

$$N_{\text{coh}}^{\text{echo}} = 0.74\text{--}2.53.$$

The free-induction value thus completes less than one coupling cycle. Echo refocusing lengthens the corresponding coherence measure, and the reported entangling sequence combined tailored double-quantum phase accumulation with refocusing [Ro80].

A static many-body Hamiltonian does not inherit the same pulse sequence automatically. Refocusing pulses can average away the very interaction the analog evolution should preserve.

For thermal equilibrium at 300 K, the thermal-resolution ratio is the displayed value:

$$R_T = \frac{h(4.93 \text{ kHz})}{k_B(300 \text{ K})} \approx 7.9 \times 10^{-10}.$$

The experiment thus demonstrates driven entanglement while staying far from passive thermal polarization in the interaction eigenbasis. Driven quantum control and equilibrium many-body protection are distinct physical capabilities, so these observations agree.

Under an assumed exponential $T_2^* = 25 \text{ } \mu\text{s}$, $\Delta\nu = 1/(\pi T_2^*) = 12.7 \text{ kHz}$ follows. Against a 4.93 kHz coupling this gives $R_C \approx 0.39$.

This estimate stays conservative only for the stated line shape and protocol. A quasi-static, calibratable detuning and an irreversible Markovian linewidth can share one Ramsey timescale yet imply substantially different control consequences.

Only a measured noise spectrum separates these processes; extra digits on a single decay time cannot.

Common analytical errors

- Comparing D directly with the noise does not show that the useful coupling is protected. Here, D is the on-site zero-field-splitting parameter. For example, suppose $D/h = 2.87 \text{ GHz}$ and the linewidth, the spectral width of the transition, is a kilohertz.

This comparison shows only that the on-site splitting is well resolved. It says nothing about the interaction meant to generate a topological stabilizer, an operator defining the protected subspace of a topological code. The quantities that matter are the projected intercluster interaction and the projected noise, with projection meaning restriction of the physical Hamiltonian and noise operators to the chosen low-energy or encoded subspace.

- An optical linewidth becomes a spin linewidth only after the corresponding noise operators are identified. For example, suppose the optical D line is 100 MHz wide.

This width touches photon indistinguishability and resonant optical readout. Assigning it directly to the spin mistakes the optical noise spectrum for the spin noise spectrum. Optical and microwave transitions couple to different operators and generally sample different noise processes.

- An inverse time counts as an energy only after a Hamiltonian and frequency convention are fixed. For example, suppose a pulse lasts 20 ns.

The inverse duration,

$$1/(20 \text{ ns}) = 50 \text{ MHz},$$

is a useful characteristic frequency. Standing alone it is not a 50 MHz energy gap, because no Hamiltonian coefficient has been named. The resonant Rabi frequency, the driven oscillation frequency between two quantum states, needed for a π rotation can also differ with the pulse convention.

- The cryostat temperature need not match the local sample temperature. For example, suppose the refrigerator reaches 20 mK.

In the ruby experiment, that refrigerator reading left the optically illuminated spin ensemble warmer than 20 mK. At low optical power, the inferred local temperature was $143 \pm 7 \text{ mK}$ [Ro88]. A complete parameter record must therefore include laser heating, imperfect thermal anchoring, and microwave dissipation.

- A protection protocol can also reshape or suppress the intended Hamiltonian. Dynamical decoupling extends memory coherence by reversing selected couplings.

When the toggling-frame average, the Hamiltonian averaged in the control-defined rotating frame, cancels J , then the interaction J needed by the architecture goes away with the noise. The resulting longer T_2 , the transverse coherence time, is numerically large but irrelevant to that architecture.

High-fidelity resonant NV readout below 10 K does not imply passive low-temperature order [R199]. Conversely, weak thermal polarization still permits active optical initialization.

- A driven entangled pair is a different object than a topological qubit. A pulse sequence entangling two NV centers is an experiment on driven physical qubits.

Such an experiment demonstrates no encoded topological qubit, no emergent anyon, and no topological order. An anyon is a quasiparticle excitation with exchange statistics specific to two-dimensional many-body systems. A future projection onto a cluster subspace would generate new effective couplings and new effective noise operators. Those effective scales must be derived rather than copied from a table of physical-spin parameters.

- Values measured under different experimental conditions cannot be combined into one parameter set. A record coherence time often requires an unusually pure host, a deep defect, low defect density, low temperature, and many refocusing pulses.

A dense implanted array generally degrades several of these conditions. Combining the best coherence from one sample, the best optical linewidth from another, and the closest defect pair from a third produces a parameter set that has not been realized in any single crystal. T_2^* , the inhomogeneous dephasing time; echo T_2 , the coherence time measured using a spin-echo sequence; and dynamically decoupled T_2 must therefore remain in separate columns. The recorded temperature must also be the local temperature under illumination rather than only the refrigerator temperature.

[Proposal] Until a candidate cluster has been fabricated, proposals should use scenario ranges rather than a single “best” input. A pessimistic case should be tied to implanted or shallow-defect data, a base case should be tied to reproducible bulk or device data, and an optimistic case should state every required enabling condition. The best coherence from one sample, the best optical linewidth from another, and the closest pair from a third must not be treated as properties of one crystal. Such a combined sample has not been experimentally realized.

[Experiment] Across demonstrated defect platforms, local cyclic frequencies E/h , defined as energies E expressed in cycles per second by division by Planck’s constant h , span roughly 1–12 GHz in the specific NV, 4H-SiC divacancy, and ruby cases tabulated here. By contrast, directly measured electron dipolar interactions over tens of nanometres can occur at kilohertz scales [R197]; [Ro80]; [Ro96]; [Ro88]. An optical strain susceptibility quantifies the change in an optical transition frequency caused by strain and can be very large, but static optical tuning is not itself a coherent interaction between defects [R200]. For silicon-vacancy (SiV) centers, phonon processes can strongly change linewidths between 4 and 20 K [R201].

These observations establish a hierarchy of energy and frequency scales. Whether an architecture works requires projection into an effective subspace and a calculation of its effective interactions and noise. Perturbative gadgets, which generate desired effective interactions through higher-order virtual processes, usually reduce useful intercluster scales further, while disorder and leakage remain. Leakage is population transfer out of the intended computational or encoded subspace. Any viable proposal must calculate the effective J_{eff} , the effective linewidth, and the many-body gap from the same conditional parameter set.

[Speculation] Strain-enhanced, phonon-mediated, exchange-assisted, or cavity-mediated designs may produce interactions larger than bare dipolar kilohertz scales. This chapter provides no experimental basis for assigning a specific topological gap to such a design. A proposed coupling must remain in a separate row until it has been measured.

Verification calculations and parameter checks

- For

$$H = \hbar g S_x$$

with

$$g = 2\pi \times 5 \text{ MHz},$$

g is an angular frequency and S_x is the spin operator along the x axis. The cyclic frequency used in this parameter record is

$$g/(2\pi) = 5 \text{ MHz}.$$

The corresponding energy is

$$\hbar g = h(5 \text{ MHz}).$$

- A cyclic frequency of 1 MHz corresponds to an energy of

$$0.0041357 \text{ } \mu\text{eV}$$

and an equivalent temperature of

$$47.992 \text{ } \mu\text{K}.$$

The equivalent temperature is defined by $T = E/k_{\text{B}}$, where k_{B} is the Boltzmann constant.

- An NV splitting with

$$D/h = 2.87 \text{ GHz}$$

does not thermally polarize the spin at 300 K. Its equivalent temperature is only

$$0.1377 \text{ K}.$$

The useful spin polarization is instead produced by optical pumping, a nonequilibrium initialization process.

- The 13 ms SiV coherence time cannot be inserted directly into a static analog-simulation parameter set. It was obtained using CPMG-32, a Carr–Purcell–Meiboom–Gill dynamical-decoupling sequence containing 32 refocusing pulses, near 100 mK in isotopically purified diamond. The pulse sequence and isotope conditions must be compatible with the intended Hamiltonian because the protection sequence can average away the interaction that the simulation is intended to retain.
- A reported readout fidelity must be accompanied by the platform, sample, temperature, optical protocol, integration time, state definition, and a statement of whether the measurement is single-shot.
- Comparing the zero-field splitting with the noise does not establish that the useful coupling is protected. D is an on-site splitting. The relevant comparison is between the projected useful coupling and the projected noise, and ultimately between the many-body gap and the projected noise.

A numerical parameter without experimental and theoretical provenance is not sufficient for architectural analysis. Hamiltonian coefficients are reported here as cyclic frequencies E/h .

The equivalent temperature is E/k_{B} ; it is not necessarily equal to the cryostat temperature. The laboratory local cyclic frequencies span roughly 1–12 GHz in the tabulated NV, 4H-SiC divacancy, and ruby cases, while a measured electron dipolar coupling at 25 nm is 4.93 kHz.

Driven entanglement at 300 K and thermal resolution of the same coupling are distinct experimental capabilities. The next chapter examines the actual separations between defects.

Sources

- [R196] Bureau International des Poids et Mesures, The International System of Units (SI Brochure), 9th ed., version 3.01,
24. Stable URL: <https://www.bipm.org/en/publications/si-brochure>.
- [R197] S. Felton et al., “Hyperfine interaction in the ground state of the negatively charged nitrogen vacancy center in diamond,” *Physical Review B* 79, 075203 (2009). DOI: [10.1103/PhysRevB.79.075203](https://doi.org/10.1103/PhysRevB.79.075203).
 - [Ro80] F. Dolde et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* 9, 139–143 (2013). DOI: [10.1038/nphys2545](https://doi.org/10.1038/nphys2545); arXiv: [1212.2804](https://arxiv.org/abs/1212.2804).
 - [R198] L. Robledo et al., “Spin dynamics in the optical cycle of single nitrogen-vacancy centres in diamond,” *New Journal of Physics* 13, 025013 (2011). DOI: [10.1088/1367-2630/13/2/025013](https://doi.org/10.1088/1367-2630/13/2/025013); arXiv: [1010.1192](https://arxiv.org/abs/1010.1192).
 - [R199] L. Robledo et al., “High-fidelity projective read-out of a solid-state spin quantum register,” *Nature* 477, 574–578 (2011). DOI: [10.1038/nature10401](https://doi.org/10.1038/nature10401); arXiv: [1301.0392](https://arxiv.org/abs/1301.0392).
 - [Ro83] D. D. Sukachev et al., “Silicon-vacancy spin qubit in diamond: a quantum memory exceeding 10 ms with single-shot state readout,” *Physical Review Letters* 119, 223602 (2017). DOI: [10.1103/PhysRevLett.119.223602](https://doi.org/10.1103/PhysRevLett.119.223602); arXiv: [1708.08852](https://arxiv.org/abs/1708.08852).
 - [R200] S. Meesala et al., “Strain engineering of the silicon-vacancy center in diamond,” *Physical Review B* 97, 205444 (2018). DOI: [10.1103/PhysRevB.97.205444](https://doi.org/10.1103/PhysRevB.97.205444); arXiv: [1801.09833](https://arxiv.org/abs/1801.09833).
 - [R201] K. D. Jahnke et al., “Electron–phonon processes of the silicon-vacancy centre in diamond,” *New Journal of Physics* 17, 043011 (2015). DOI: [10.1088/1367-2630/17/4/043011](https://doi.org/10.1088/1367-2630/17/4/043011); arXiv: [1411.2871](https://arxiv.org/abs/1411.2871).
 - [Ro96] D. J. Christle et al., “Isolated electron spins in silicon carbide with millisecond coherence times,” *Nature Materials* 14, 160–163 (2015). DOI: [10.1038/nmat4144](https://doi.org/10.1038/nmat4144); arXiv: [1406.7325](https://arxiv.org/abs/1406.7325).
 - [Ro88] V. K. Sewani et al., “Spin thermometry and spin relaxation of optically detected Cr³⁺ ions in Al₂O₃ (ruby),” *Physical Review B* 102, 104114 (2020). DOI: [10.1103/PhysRevB.102.104114](https://doi.org/10.1103/PhysRevB.102.104114); arXiv: [2007.07493](https://arxiv.org/abs/2007.07493).
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Chapter 28 — Separation and successful atom placement

Two nitrogen-vacancy (NV) centers separated by exactly ten nanometres would have a point-dipole coefficient of 52 kHz if both occupied their intended positions. A fabrication procedure can define target coordinates, implant nitrogen through apertures, anneal the crystal, and detect fluorescence. Two target coordinates start the process; a usable pair exists only after implantation, annealing, charge stabilization, and spin/coherence screening each succeed at both sites.

The interaction energy depends on the actual displacement between the centers and on whether the intended charge-stable spin forms at each endpoint. If the mean separation is 10 nm but individual positions vary by several nanometres, the resulting couplings have a broad distribution. If only a small fraction of target sites contain the intended spin, most nominal pairs are incomplete.

The Hamiltonian — the operator describing the system's energy and dynamics — depends on the measured positions of centers that pass optical, spin, charge-state, and coherence screening. Earlier fabrication coordinates (beam aim, stopping estimates) predict that final configuration approximately; the accepted-center positions define it.

Distinguishing beam position from final defect position

An implantation system assigns a coordinate to an incoming ion, but the fabrication process moves through at least five distinct spatial or probabilistic stages. Each stage answers a different question — where the ion entered, where it stopped, where the vacancy traveled, whether the right charge state formed, where the surviving usable centers sit — so a number from one stage cannot stand in for another.

- The first quantity is the position at which the incoming ion crosses the crystal surface. It is determined by the beam spot or by the aperture in a mask.
- The second is the ion's stopping position after transport through the solid.
- The third is the displacement of a mobile vacancy during annealing before it is captured, recombines, forms a cluster, or is lost at a surface.
- The fourth is the probability that the delivered ion or a created vacancy forms the specified structural defect in the specified charge state.
- The fifth is the measured position after retaining only those centers that also satisfy optical, spin, charge-state, and coherence acceptance tests.

The lateral and depth-dependent spread associated with the second quantity is called implantation straggle: the random spread from collisions inside the crystal. Instrument specifications quote the beam that arrives at the surface; straggle adds inside the crystal, after the instrument's control ends.

For example, a beam narrower than 10 nm can produce a stopping distribution with 20 nm straggle. Conversely, even if an implanted nitrogen atom has a narrowly distributed stopping position, it can capture a vacancy that migrated during annealing.

A center can sit at the intended position yet stay optically dark and fail screening. Instrument documentation quotes the surface-entry distribution, the quantity the instrument sets; the coupling Hamiltonian uses the fifth quantity, the positions of centers that survive screening.

The sequence can be represented schematically as follows:

```
commanded site
|
+-- surface entry: beam/aperture distribution
|
+-- stopping point: entry convolved with ion straggle
|
+-- annealed complex: vacancy migration and capture
|
+-- created center: structural/charge conversion succeeds
|
+-- usable center: position measured; spin passes acceptance tests
```

A convolution is the probability distribution obtained when independent random displacements are added as random variables. For independent Gaussian one-dimensional errors with standard deviations $\sigma_1, \sigma_2, \dots$, the total standard deviation is

$$\sigma_{\text{total}} = \sqrt{\sigma_1^2 + \sigma_2^2 + \dots}.$$

Thus, independent random errors combine in quadrature. Systematic registration offsets are persistent biases rather than random scatter. Calibration measures and subtracts each bias; the quadrature sum covers only the random parts.

A beam-distribution report therefore characterizes surface entry only. Stopping positions, vacancy capture, and the screened-center positions each need their own measurement before any coupling claim follows.

Dipolar interaction between two localized spins

Consider two localized electron spins separated by a displacement vector $\mathbf{r}$. Its magnitude is r , and the corresponding unit vector is $\hat{\mathbf{r}} = \mathbf{r}/r$. Let $\mathbf{S}_1$ and $\mathbf{S}_2$ denote dimensionless spin operators, whose eigenvalues are therefore pure numbers. For electron Landé factors g_1 and g_2 , the point-dipole interaction energy is

$$H_{dd} = h\nu_d(r) [\mathbf{S}_1 \cdot \mathbf{S}_2 - 3(\mathbf{S}_1 \cdot \hat{\mathbf{r}})(\mathbf{S}_2 \cdot \hat{\mathbf{r}})] ,$$

where h is Planck's constant and

$$\nu_d(r) = \frac{\mu_0}{4\pi} \frac{g_1 g_2 \mu_B^2}{hr^3}.$$

Here, μ_0 is the vacuum permeability, μ_B is the Bohr magneton, and ν_d is the dipolar coefficient. The equation defines the long-range magnetic interaction in the point-dipole approximation.

The dipolar coefficient sets the distance scale of the interaction. An observed transition splitting or gate rate multiplies that scale by the angular factor, the spin-state matrix elements, and the detuning and pulse-sequence conventions of the measurement.

For two electron-like spins with $g_1 = g_2 = 2.0023$, using rounded 2022 CODATA values [\[R2o8\]](#),

$$\frac{\mu_0}{4\pi} \approx 10^{-7} \text{ N A}^{-2}, \qquad \mu_B = 9.274010 \times 10^{-24} \text{ J T}^{-1},$$

$$h = 6.62607015 \times 10^{-34} \text{ J s},$$

the coefficient is

$$\nu_d(r) \approx \frac{52.04 \text{ MHz nm}^3}{r^3}.$$

A tesla is $\text{T} = \text{N (A m)}^{-1}$. Therefore,

$$(\text{N A}^{-2}) \frac{(\text{J T}^{-1})^2}{(\text{J s}) \text{ m}^3} = (\text{N A}^{-2}) \frac{(\text{A m}^2)^2}{(\text{N m s}) \text{ m}^3} = \text{s}^{-1}.$$

This dimensional analysis confirms that ν_d has units of frequency. Substituting $r = 1 \text{ nm} = 10^{-9} \text{ m}$ gives $5.204 \times 10^7 \text{ s}^{-1} = 52.04 \text{ MHz}$. The other values follow from the r^{-3} scaling.

Separation r	$r^3 \text{ (nm}^3\text{)}$	Dipolar coefficient ν_d	Square-lattice density proxy $1/r^2 \text{ (sites/}\mu\text{m}^2\text{)}$
1 nm	1	52.0 MHz	1,000,000
2 nm	8	6.50 MHz	250,000
5 nm	125	416 kHz	40,000
10 nm	1,000	52.0 kHz	10,000
20 nm	8,000	6.50 kHz	2,500
50 nm	125,000	416 Hz	400

The density column is only the geometric value for one site per square of side r . Honeycomb, kagome, cluster, routing, and optical-access layouts have different areas per site.

The dipolar values are consistent in scale with an [Experiment] demonstrating coherent coupling between two NV electron spins whose inferred separation was 9.8 ± 0.3 nm. After the geometry was included, the observed coupling was in the tens-of-kilohertz range [R181]. That experiment established one coupled pair. A deterministic array at that pitch — 10,000 working sites per square micrometre — remains undemonstrated.

The angular dependence is essential. If both spin moments are quantized along a unit vector $\hat{z}$, the secular approximation keeps the energy-conserving part of the interaction in this chosen quantization basis. Its common coefficient contains

$$A(\theta) = 1 - 3 \cos^2 \theta,$$

where θ is the angle between $\hat{r}$ and $\hat{z}$. The factor ranges from -2 to $+1$ and vanishes at the magic angle

$$\theta = \cos^{-1}(1/\sqrt{3}) \approx 54.7^\circ.$$

A 10 nm separation therefore sets a 52 kHz coefficient scale. The secular coupling multiplies that scale by the angular factor, which vanishes at the magic angle: a silent pair there reflects a canceling geometry, not missing spins. Read with the angle included, 52 kHz is the geometric envelope of the coupling, not a delivered gate rate.

Breakdown of the point-dipole approximation at nanometre separations

At separations of 1–2 nm, representing each defect as a point magnetic dipole is not a complete microscopic model. Electronic wavefunctions can overlap, and the interaction can depend on charge configuration, relative crystallographic orientation, strain, and the exact atomic realization of the pair.

The remaining short-range interaction includes exchange, which arises from wavefunction overlap and the fermionic symmetry of the electrons. Exchange is not generally a small correction to the tabulated dipolar values. Its magnitude often varies approximately exponentially with separation rather than following a universal power law. [Numerics] Atomistic calculations for diamond spin centers found strongly orientation-dependent exchange interactions that can exceed dipolar coupling for some NV-pair geometries below roughly 3 nm [R107].

The computed exchange varies pair by pair — strong for some orientations, weak for others — so the 6.50 MHz dipolar entry stands as the long-range contribution only, with the site-specific short-range part still to be calculated or measured for each pair.

At these separations, the actual pair must be calculated or measured. The analysis must verify that two recognizable defects and their desired charge states remain stable and must include exchange, tunnelling, altered orbital levels, and implantation damage. An assumed exchange entry belongs in the table only with a microscopic calculation or measurement behind it.

Keeping only the r^{-3} term below 2 nm keeps the universal long-range part and drops a nonuniversal short-range part that can dominate it.

Propagation of placement uncertainty into coupling uncertainty

Let the intended separation be r_0 , and let δr be a small error in separation along the bond direction. Because $\nu_d = Cr^{-3}$, where $C = 52.04 \text{ MHz nm}^3$ for the electron-like case,

$$\frac{d\nu_d}{dr} = -3Cr^{-4} = -\frac{3\nu_d}{r}.$$

A first-order expansion about r_0 gives

$$\frac{\delta\nu_d}{\nu_d} \approx -3\frac{\delta r}{r_0}.$$

This equation relates small radial placement errors to fractional variations in the dipolar coefficient.

Suppose that each endpoint has an independent one-axis placement standard deviation σ_{site} along the bond. The standard deviation of the relative displacement is then

$$\sigma_r = \sqrt{2}\sigma_{\text{site}}, \quad \frac{\sigma_\nu}{\nu_d} \approx \frac{3\sqrt{2}\sigma_{\text{site}}}{r_0}.$$

This is an optimistic estimate that includes only radial disorder. It omits disorder in the angular factor, non-Gaussian tails, missing sites, and variation in exchange interactions. Requiring radial dipolar disorder below 10% gives

$$\sigma_{\text{site}} < \frac{0.10}{3\sqrt{2}}r_0 \approx 0.0236r_0.$$

Target spacing	Maximum independent per-site σ_{site} for 10% radial coupling scatter
1 nm	0.024 nm
2 nm	0.047 nm
5 nm	0.118 nm
10 nm	0.236 nm
20 nm	0.471 nm
50 nm	1.18 nm

These values are tolerance targets derived from the 10% uniformity goal; they describe what fabrication must reach, not what any cited process has reached.

For example, a 2 nm per-site standard deviation would produce an estimated fractional radial coupling scatter of

$$3\sqrt{2}(2/10) \approx 0.85,$$

or 85%, for a design with 10 nm spacing. Increasing the spacing to 50 nm relaxes the relative placement tolerance, but the bare dipolar coefficient decreases to 416 Hz. Spatial design therefore involves a trade-off between placement tolerance and interaction strength.

The linear approximation is invalid when σ_r/r_0 is not small. If a scalar Gaussian model is applied without suitable constraints, it can also assign nonzero probability to unphysical negative distances.

A more complete disorder model samples three-dimensional endpoint positions, computes $\mathbf{r}$, evaluates the full dipolar tensor and any short-range terms, and rejects configurations that are atomically impossible or correspond to merged defects. The resulting distribution should be reported using quantiles of J , where J denotes the interaction strength, rather than only J evaluated at the mean distance. Because r^{-3} is a convex function for positive r , rare close pairs produce a long tail of strong couplings.

Coupling reports therefore quote quantiles of the sampled J distribution rather than the value at the mean distance: the close-distance tail lifts the mean above the typical value, and only the distribution shows both.

Conditional probability of obtaining a usable site

A target site is usable only if several successive conditions are satisfied. Define the following events for one target site:

- A : the intended ion or vacancy is delivered;
- C : the desired structural complex forms;
- Q : the complex has the usable charge state;
- O : the center can be initialized and read out;
- S : its spin and coherence satisfy the specification; and
- G : its final geometry lies within tolerance.

The exact usable-site probability is

$$p_{\text{use}} = P(A)P(C | A)P(Q | A, C)P(O | A, C, Q)P(S | A, C, Q, O)P(G | A, C, Q, O, S).$$

This product defines a yield as a chain of conditional probabilities rather than as one undifferentiated success probability.

The factors need not be statistically independent. Detection of a single implanted ion addresses event A , but it does not establish the remaining events. A reported “20% conversion” addresses a version of event C under specified processing conditions; it does not directly give p_{use} . Similarly, “33 nm placement” is meaningful only after specifying whether it refers to lateral position, depth, or one standard deviation,

If N sites were independent and every site were required, the defect-free array yield would be p_{use}^N . At $p_{\text{use}} = 0.96$, an array of 100 sites would have

$$0.96^{100} \approx 1.69\%$$

defect-free yield. At 1,000 sites, the value is about 1.9×10^{-18} . Repair, repeated implantation, movable qubits, redundant layouts, or tolerance of missing sites can change this conclusion. Site yield nevertheless cannot be substituted for array yield without explicitly accounting for array size and architecture.

A related limitation occurs when the number of created centers per target follows a Poisson distribution, a model for independent creation events whose count fluctuates around a mean λ . The probability of creating exactly one center is

$$P(1) = \lambda e^{-\lambda},$$

which is maximized at $\lambda = 1$, where

$$P(1) = e^{-1} \approx 36.8\%.$$

Under a Poisson model, dose tuning alone cannot exceed this ceiling. Feedback or postselection changes the physical or statistical process and therefore changes the applicable model.

Every reported percentage therefore names its denominator, conditioning events, and array size; without those three, it cannot serve as a device yield.

Experimental measures of placement and creation yield

The following rows intentionally retain different definitions. They should not be reduced to a single ranking called “resolution.” Full width at half maximum (FWHM) is the width of a distribution measured between the two points where its value is half its maximum. PMMA is poly(methyl methacrylate), used here as a lithographic mask. SRIM is a numerical ion-transport simulation package used to estimate stopping and straggle distributions.

Host and process	Surface-entry control	Straggle/stopping distribution	Vacancy motion	Reported creation yield	Final measured placement or usability statement
Diamond, 20 keV N component of	80 nm diameter aperture	[Numerics] SRIM stopping depth	800 °C, 2 h anneal; diffusion contribution	[Experiment] mean 3.5 NV centers from	Emission sites followed the mask;

Host and process	Surface-entry control	Straggle/stopping distribution	Vacancy motion	Reported creation yield	Final measured placement or usability statement
implanted CN [−] through a PMMA mask		30 ± 10 nm and about 9 nm straggle	not separately measured	about 50 N atoms per aperture, approximately 7% N-to-NV conversion, over 49 sites [R202]	the paper did not establish a sub-10-nm final usable-position distribution [R202]
Diamond, focused Si implantation for SiV	Typical beam FWHM below 40 nm	[Numerics] 19 nm lateral straggle; combining a 40 nm FWHM beam ($\sigma \approx 17$ nm) and straggle predicted $\sigma \approx 26$ nm	High-temperature anneal; vacancy capture controls conversion	[Experiment] up to about 2.5% at 100 keV; electron irradiation plus re-anneal raised a tested bulk-sample value to about 20% [R190]	[Experiment] created single SiVs had 32 nm one-axis standard deviations; radial offset 40 ± 20 nm. Cavity targeting gave 48 ± 21 nm [R190]
Diamond, femtosecond vacancy writing plus local annealing and optical feedback	Diffraction-scale optical focus, not an ion beam	No implanted-ion straggle	Vacancy creation and local annealing are the mechanism; feedback stops once emission is detected	[Experiment] approximately 96% single-NV creation yield under the reported feedback protocol [R203]	[Experiment] about 33 nm in-plane positioning deviation; this is not equivalent to 33 nm three-dimensional nearest-neighbour control [R203]
Diamond, nitrogen delta-doping during growth plus irradiation/anneal	No deterministic lateral coordinate	[Experiment] a roughly 1–2 nm doped layer; cap thickness selected nominal NV depths from about 5 to 100 nm	Vacancies supplied after growth and annealed into the N layer	Not reported as a deterministic per-site useful yield in this study	[Experiment] tight depth engineering, but no lateral array placement; shallow nominal 5 nm layers retained $T_2 > 100 \mu\text{s}$ under reported conditions [R204]

Diamond, delta-doped N plus localized electron irradiation used a 200 keV electron beam with a 20 nm spot.

Electrons create vacancies along a narrow path; the N layer, rather than ion stopping, sets depth.

850 °C, 11 min anneal; [Numerics] diffusion-capture modeling accompanied the measurement.

[Exper

Host and process	Surface-entry control	Straggle/stopping distribution	Vacancy motion	Reported creation yield	Final measured placement or usability statement
4H-SiC, 30 keV carbon implantation through a lithographic mask	Mask-defined target arrays	[Numerics] implantation selected shallow silicon vacancies at roughly 40 nm depth	Host- and defect-specific activation; diamond diffusion numbers do not transfer	[Experiment] $19 \pm 4\%$ silicon-vacancy generation efficiency and $34 \pm 4\%$ probability of a single emitter at optimized dose [R205]	Predetermined arrays were demonstrated, but those yield figures do not by themselves establish sub-10-nm final pair-spacing distributions [R205]

The SiV row provides an explicit uncertainty calculation. For a beam with FWHM 40 nm, conversion to a Gaussian standard deviation gives

$$40/2.355 \approx 17 \text{ nm}.$$

Combining this standard deviation with 19 nm straggle gives

$$\sqrt{17^2 + 19^2} \approx 25.5 \text{ nm},$$

which is consistent with the paper’s predicted value of 26 nm.

The measured endpoint distribution—the distribution of final center locations—was broader, with a standard deviation of 32 nm per axis [R190]. Thus, a beam reported as narrower than 40 nm did not produce a final location distribution narrower than 40 nm. This difference demonstrates why the endpoint distribution must be measured.

The CN-mask row supports another uncertainty propagation, although it does not provide a direct endpoint measurement. If entry points were uniformly distributed across a circular aperture of radius $a = 40$ nm, one Cartesian coordinate would have

$$\sigma_x = a/2 = 20 \text{ nm}.$$

Combining this idealized aperture distribution with 9 nm straggle gives

$$\sqrt{20^2 + 9^2} \approx 22 \text{ nm}$$

before including alignment error, diffusion, channeling tails (ions guided along crystal axes or planes), or optical-localization uncertainty. This calculation illustrates the use of the reported inputs; it is not a measured final NV standard deviation [R202].

A small pitch between adjacent apertures sets the target spacing, while the stopping, vacancy-capture, and screening distributions within each aperture still determine the spread of center separations.

Vacancy diffusion can either enable center formation or degrade the intended geometry. [Experiment] In nitrogen-rich type-Ib diamond implanted with focused Ar ions, Racke and colleagues measured 0.04–0.79 created NV centers per implanted Ar ion over 12–132 keV [R206].

[Numerics] Their model of vacancy loss to the surface placed an upper bound of about 300 nm on the single-vacancy diffusion length for an 800 °C anneal [R206]. This 300 nm figure bounds how far a vacancy can travel before surface loss; most vacancies travel less, and the bound cannot serve as a Gaussian placement error.

The implanted nitrogen is comparatively immobile. Formation of the complex requires a mobile vacancy to reach a neighboring lattice site. The final position distribution is therefore a reaction-and-capture distribution conditioned on nitrogen density, depth, surfaces, damage, charge state, temperature, and time.

The available fabrication methods control different quantities. Delta doping can confine the depth distribution while leaving lateral positions random. A focused beam can target lateral coordinates while ion scattering broadens both depth and lateral position. Laser feedback can increase center-creation yield without providing atomic-scale three-dimensional registration.

[Experiment] By 2025, delta doping combined with localized electron-created vacancies had demonstrated about 4 nm depth confinement and 46(1) nm lateral confinement in 280 nm diamond pillars, but not the sub-nanometre relative placement required for a uniform 10 nm dipolar graph [R207].

Such hybrid methods graduate from [Proposal] to demonstrated lattice fabrication when the final pair distribution and usable-array yield are measured.

A regularly spaced fluorescence image establishes an array of physical emitters. Coherently coupled qubits need measured couplings at each link; an encoded cluster needs the doublet spectroscopy of Chapter 41; a topological phase needs gap and nonlocal diagnostics. Spatial fabrication supplies the positions entering the microscopic Hamiltonian — the first of those inputs.

Common errors in interpreting placement data

- **Reporting the beam distribution as the final defect distribution.** An ion-optical spot of 10 nm specifies the surface-entry distribution; collisions in the crystal broaden the stopping distribution. In the reported SiV experiment, the nominal beam was smaller than 40 nm, the calculated lateral straggle was 19 nm, and the measured one-axis spread of created centers was 32 nm [R190]. Each of these numbers describes a different physical quantity; omitting any one of them changes the interpretation.
- **Treating a transport simulation as an endpoint measurement.** Transport simulations are valuable [Numerics] because they model ion trajectories and damage under assumptions about an initially ideal target. They supply transport predictions, while charge-state stability, the position of the annealed complex, channeling tails in a particular crystal, and usable-spin yield require separate measurements. Simulated distributions must be validated using microscopy, spectroscopy, or coupling-based localization.
- **Equating conversion yield with deterministic placement.** Detecting a single implanted ion confirms event A in the yield chain; formation of the requested center still depends on the later conversion and screening events. Conversely, a 96% laser-feedback creation result does not imply that 96% of sites lie within a 1 nm tolerance or satisfy a coherence threshold [R203]. A percentage cannot be interpreted without its denominator, conditioning events, and acceptance tests.
- **Ignoring distribution tails and the nearest unintended neighbors.** The mean and standard deviation hide rare close pairs, whose r^{-3} interaction can dominate. Damage-related paramagnetic defects can also lie closer than the intended qubit and couple more strongly. Because the convexity of r^{-3} gives disproportionate weight to the close-distance tail, analyses should report a three-dimensional point-process distribution—a statistical description of the locations of all relevant emitters and defects—tail quantiles, and the identities of dark spins where possible.

- **Transferring diamond statistics to another host material.** A host changes stopping distributions through ion species, energy, crystal orientation, density, channeling, and target composition. Vacancy migration and complex formation depend on the host's defect chemistry. The 300 nm diamond upper bound [R206], the 7% CN-to-NV result [R202], and the SiC yield [R205] are distinct conditioned experimental results rather than interchangeable constants. For sapphire in particular, a proposed array has no defensible spatial-error budget until the selected defect, charge state, implantation species, annealing procedure, and final spin-acceptance measurement are specified.
- **Assuming that a stronger interaction is necessarily preferable.** At 1–2 nm, exchange and tunnelling can strengthen the coupling while altered defect identity and implantation damage invalidate the intended isolated-spin model [R107]. At 50 nm, placement is less demanding, but a bare dipolar coupling of 416 Hz may be smaller than the linewidth or decoherence rate. The viable range is therefore the range in which the measured distribution of useful couplings exceeds the Chapter 27 noise rates while remaining within the domain of validity of the intended Hamiltonian model.

Technical assessment

- A focused beam with a 10 nm spot does not establish a 10 nm final defect-position uncertainty. The beam spot specifies the transverse extent of the incident beam, whereas ion straggle—the stochastic displacement of an implanted ion from its nominal trajectory—remains. Registration bias, defined as a systematic offset between the intended and realized coordinates, also remains. Additional uncertainties arise from annealing and capture physics and from the conditional final-position distribution of successfully created usable centers.
- For two electron spins separated by 20 nm, the electron-spin dipolar coefficient is 6.50 kHz:

$$52.04 \text{ MHz nm}^3 / 20^3 = 6.50 \text{ kHz},$$

before angular and matrix-element factors are included. Angular factors account for the orientation of the displacement vector relative to the spin quantization axes, while matrix-element factors account for the spin states coupled by the interaction.

- A 1 nm table entry is not a complete coupling prediction. At this separation, wavefunction overlap and site-specific exchange, tunnelling, charge, and structural changes can rival or dominate the point-dipole term. Exchange is the spin coupling produced by wavefunction overlap and fermionic antisymmetry, while tunnelling is the quantum-mechanical transfer amplitude between sites.
- Independent 1 nm bond-direction placement errors at a 10 nm pitch produce approximately 42% radial dipolar scatter. Here, pitch is the nominal center-to-center site separation, and the bond direction is the direction of the vector connecting the two sites. In the small-error approximation,

$$3\sqrt{2}(1/10) \approx 0.42.$$

Angular disorder contributes additional variation.

- A 20% conversion yield does not imply a 20% usable-site yield. Conversion yield is the fraction of starting implantation or creation events that produce the target defect. Usable-site yield additionally depends on the probabilities of obtaining the required charge state, readout performance, coherence, and geometric tolerance. These selection conditions may be correlated.
- A fabrication paper for this architecture must report three-dimensional final-position distributions, non-Gaussian tails and biases, the exact denominator used to define creation yield, charge and spin acceptance criteria, missing and extra defects, the pair-coupling distribution, and the array-level yield under the stated processing conditions.

A nominal 10 nm design spacing does not establish a realized 10 nm defect geometry. For electron-like spins, meaning localized spins with electron magnetic moments, the point-dipole coefficient is $52.04 \text{ MHz nm}^3/r^3$, where r is the spin separation. This expression gives 52.0 MHz, 6.50 MHz, 416 kHz, 52.0 kHz, 6.50 kHz, and 416 Hz at 1, 2, 5, 10, 20, and 50 nm.

Angular factors and short-range exchange must be evaluated separately from this radial point-dipole coefficient. For small radial placement errors, the fractional standard deviation of the coupling is

$$\sigma_J/J \approx 3\sqrt{2}\sigma_{\text{site}}/r,$$

where J is the dipolar coupling, σ_J is its standard deviation, and σ_{site} is the per-site bond-direction position uncertainty. Under this small-error approximation, achieving 10% coupling uniformity requires a per-site bond-direction error below about $0.0236r$.

Existing laboratory results provide several relevant capabilities: coupled 10 nm-class pairs, nanometre-scale stopping calculations, measured final placement at the tens-of-nanometres scale, and process-dependent creation yields. A dense, defect-free interaction graph with sub-nanometre placement tolerance still requires a combined demonstration, so a nominal spacing alone cannot establish a functional device.

Sources

- [R181] P. Neumann et al., “Quantum register based on coupled electron spins in a room-temperature solid,” *Nature Physics* 6, 249–253 (2010). DOI: [10.1038/nphys1536](https://doi.org/10.1038/nphys1536); [arXiv:1004.5090](https://arxiv.org/abs/1004.5090).
- [R107] V. R. Kortan, C. Şahin, and M. E. Flatté, “Nanometer-scale exchange interactions between spin centers in diamond,” *Physical Review B* 93, 220402(R) (2016). DOI: [10.1103/PhysRevB.93.220402](https://doi.org/10.1103/PhysRevB.93.220402); [arXiv:1603.03485](https://arxiv.org/abs/1603.03485).
- [R202] P. Spinicelli et al., “Engineered arrays of NV color centers in diamond based on implantation of CN[−] molecules through nanoapertures,” *New Journal of Physics* 13, 025014 (2011). DOI: [10.1088/1367-2630/13/2/025014](https://doi.org/10.1088/1367-2630/13/2/025014); [arXiv:1008.1483](https://arxiv.org/abs/1008.1483).
- [R190] T. Schröder et al., “Scalable focused ion beam creation of nearly lifetime-limited single quantum emitters in diamond nanostructures,” *Nature Communications* 8, 15376 (2017). DOI: [10.1038/ncomms15376](https://doi.org/10.1038/ncomms15376); stable full text: [PMC5458551](https://www.nature.com/articles/ncomms15376).
- [R203] Y.-C. Chen et al., “Laser writing of individual nitrogen-vacancy defects in diamond with near-unity yield,” *Optica* 6, 662–667 (2019). DOI: [10.1364/OPTICA.6.000662](https://doi.org/10.1364/OPTICA.6.000662).

- [R204] K. Ohno et al., “Engineering shallow spins in diamond with nitrogen delta-doping,” Applied Physics Letters 101, 082413 (2012). DOI: [10.1063/1.4748280](https://doi.org/10.1063/1.4748280); [arXiv:1207.2784](https://arxiv.org/abs/1207.2784).
 - [R205] J. Wang et al., “Efficient generation of an array of single silicon-vacancy defects in silicon carbide,” Physical Review Applied 7, 064021 (2017). DOI: [10.1103/PhysRevApplied.7.064021](https://doi.org/10.1103/PhysRevApplied.7.064021).
 - [R206] P. Racke, L. Pietzonka, J. Meijer, D. Spemann, and R. Wunderlich, “Vacancy diffusion and nitrogen-vacancy center formation near the diamond surface,” Applied Physics Letters 118, 204003 (2021). DOI: [10.1063/5.0046031](https://doi.org/10.1063/5.0046031).
 - [R207] S. Kim et al., “Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures,” Nature Communications 16, 9803 (2025). DOI: [10.1038/s41467-025-64758-4](https://doi.org/10.1038/s41467-025-64758-4).
 - [R208] P. J. Mohr et al., “CODATA recommended values of the fundamental physical constants: 2022,” Reviews of Modern Physics 97, 025002 (2025). DOI: [10.1103/RevModPhys.97.025002](https://doi.org/10.1103/RevModPhys.97.025002); stable constants database: [NIST Reference on Constants, Units, and Uncertainty](https://physics.nist.gov/cuu/Reference).
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Chapter 29 — Distinguishing microscopic coupling from topological gap

A coupling J between two defects measures how strongly neighboring degrees of freedom interact. A topological gap Δ_{topo} measures the energy of the lowest excitation above the ground-state family of the full lattice. The symbol Δ_{topo} is justified only after the relevant many-body spectrum has been established.

In a concrete device, that pairwise strength J , expressed as a frequency, might be fifty megahertz. The gap claim, by contrast, concerns the lowest excitation outside the ground-state family of the entire lattice.

Projection into an effective low-energy subspace, a fourth-order perturbative gadget, finite temperature, disorder, and decoherence each narrow the usable margin between these scales. None of them turns J into a topological gap by renaming it.

All energy scales will therefore be expressed in common units. No experimentally measured material value of Δ_{topo} is currently available for insertion at the end of the analysis.

Consistent units for energy-scale comparisons

Several energy scales can compete within a single spectrum. A quantitative scale budget requires all of them to be written in the same units, counting each contribution exactly once.

Energies are not consumed, but comparing scales requires ordering their magnitudes consistently.

An energy E will usually be quoted as the ordinary frequency E/h , where h is Planck's constant. Thus, a table entry of 1 MHz denotes the energy $h \times 10^6 \text{ s}^{-1}$, not an angular frequency. Temperature is converted to frequency units using

$$\frac{k_B T}{h} = 20.8366 \frac{\text{GHz}}{\text{K}} T = 20.8366 \frac{\text{MHz}}{\text{mK}} T.$$

Here k_B is Boltzmann's constant, and T is the thermodynamic temperature in kelvin. The numerical conversion follows from the exact SI values of k_B and h [R196]. Expressing energies as ordinary frequencies avoids unintended factors of 2π .

Consequently, 1 mK corresponds to 20.8366 MHz in these units, while a kilohertz coupling corresponds to 4.8×10^{-5} mK. A calculation becomes inconsistent if it mixes cyclic frequency with angular frequency or identifies the cryostat temperature with the sample temperature without justification.

Local interaction terms and topological phases

Consider an ideal effective Hamiltonian term

$$H_p = -KB_p,$$

where $K > 0$ is an energy and B_p is a dimensionless operator with eigenvalues $+1$ and -1 . The index p may label a plaquette, meaning a local elementary face of a lattice. A single plaquette term does not by itself establish a topological phase.

The two eigenenergies of this term are $-K$ and $+K$. Reversing the eigenvalue of B_p therefore costs

$$\Delta_p = (+K) - (-K) = 2K.$$

This energy difference, rather than the coefficient K alone, is the excitation energy of the isolated term. In a closed lattice, constraints may require excitations to occur in pairs. Under those conditions, the lowest allowed bulk process may cost $4K$.

Boundaries can alter this numerical factor. A valid value of Δ_{topo} must therefore be derived from the spectrum and boundary conditions of the specified model.

In many effective models, K is not a fundamental microscopic parameter. In the strongly anisotropic limit of Kitaev's honeycomb model, fourth-order perturbation theory generates a plaquette coefficient of the form [\[Ro17\]](#)

$$K = \frac{J_x^2 J_y^2}{16\Lambda^3}.$$

Here J_x and J_y are weak bond energies, while Λ denotes the strong bond scale, which is called J_z in the original model. Setting $J_x = J_y = J$ gives

$$K = \frac{J^4}{16\Lambda^3} = \frac{J}{16} \left(\frac{J}{\Lambda} \right)^3.$$

The dimensional relation is consistent because $[J^4/\Lambda^3] = \text{energy}$. If $J/\Lambda = 0.1$, then the coefficient is $J/16,000$, before disorder or other corrections are included.

[Theory] This fourth-order result is established for the specified honeycomb limit [\[Ro17\]](#). It is not a derived coefficient for a diamond, SiC, or sapphire defect array, and the honeycomb phase in this limit is not doubled Fibonacci order.

For the conservative illustrative scale budget below, define

$$\Delta_{\text{topo}} = 2K.$$

[Assumption] The factor of two represents one violated effective term. This choice is deliberately less favorable than a periodic pair threshold of $4K$. It remains an illustrative assumption rather than a claim about a material gap.

Identifying K directly with the gap would confuse the coefficient of a local Hamiltonian term with the first allowed excitation of the full lattice. Pair-creation constraints and boundary conditions can change the numerical factor. In addition, a crystal does not acquire a value of K from an abstract model in which that coefficient was chosen freely.

Spectral definition of the topological gap

Let $E_0 \leq E_1 \leq E_2 \leq \dots$ denote the eigenenergies of a finite effective Hamiltonian. A topologically ordered system may contain several nearly degenerate ground states. Let $\mathcal{G}$ denote this low-energy ground-state family. The topological gap is defined by

$$\Delta_{\text{topo}} = E_{\text{first outside } \mathcal{G}} - E_{\text{top of } \mathcal{G}}.$$

Thus, Δ_{topo} is a many-body spectral quantity: it measures the separation between the highest-energy state in the ground-state family and the lowest-energy state outside that family. It is not an alternative notation for J .

The finite-size width within $\mathcal{G}$ is a separate quantity:

$$\delta_{\text{fs}} = E_{\text{top of } \mathcal{G}} - E_{\text{bottom of } \mathcal{G}}.$$

Both Δ_{topo} and δ_{fs} have units of joules. They will be quoted as Δ_{topo}/h and δ_{fs}/h , respectively, in hertz.

A cluster leakage gap Λ , an optical transition energy, and a bare nearest-neighbor coupling J are not equivalent to Δ_{topo} . Instead, these quantities constrain the derivation of the effective Hamiltonian.

The gap of that effective Hamiltonian can be obtained only through diagonalization, a controlled analytic solution, or defensible many-body numerical calculations. Exactly solvable commuting-projector string-net models demonstrate that a local Hamiltonian can exhibit topological order [R018]. However, the overall energy scale of such an abstract model is chosen freely, so a physical crystal does not inherit that normalization.

A defect spin is a physical degree of freedom, while a cluster doublet — a selected pair of low-energy cluster states — is the encoded local one.

A circuit that prepares a string-net wavefunction constitutes digital emulation unless the undriven hardware Hamiltonian already realizes the corresponding phase. The quantity Δ_{topo} is a property of the many-body Hamiltonian and its spectrum. Realizing a logical qubit additionally requires controlled initialization, operations, and readout within the ground-state family.

A measurement of a two-spin avoided crossing does not, by itself, establish any of those three logical-qubit capabilities.

The distinction between δ_{fs} and Δ_{topo} is essential. If it is omitted, a small splitting within the ground-state family may be misidentified as a small excitation gap. A small δ_{fs} is desirable, whereas a small Δ_{topo} is undesirable.

Perturbative suppression and uncertainty propagation

Write the microscopic Hamiltonian as

$$H = H_0 + V,$$

where H_0 separates a retained cluster subspace from leakage states by an energy Λ , and V contains weaker intercluster couplings with characteristic local strength J . The retained subspace is the set of low-energy states used to define the effective model, while leakage states are states outside that subspace. The dimensionless ratio

$$\epsilon = \frac{J}{\Lambda}$$

is the perturbative expansion parameter. Each additional power of ϵ suppresses a higher-order contribution.

For an interaction that first appears at perturbative order n , dimensional analysis gives

$$K_n = c_n \frac{J^n}{\Lambda^{n-1}} = c_n J \epsilon^{n-1},$$

where c_n is a dimensionless coefficient determined by the actual interaction graph, matrix elements, energy denominators, and interference among virtual paths. A virtual path is a sequence of intermediate transitions through states outside the retained low-energy subspace. Perturbative gadgets can generate higher-body interactions from two-body couplings, but the intended terms and the associated errors must be bounded together [R209]. [Theory] The scaling law is a bookkeeping relation; neither n nor c_n can be selected retrospectively to obtain a desired result.

Suppose that the first omitted contribution is approximately

$$R_{n+1} = c_{n+1} J \epsilon^n.$$

Then

$$\frac{|R_{n+1}|}{|K_n|} \approx \left| \frac{c_{n+1}}{c_n} \right| \epsilon.$$

A small ϵ improves perturbative control but reduces K_n . A large ϵ increases the nominal target term while weakening the validity of the expansion. The choice of ϵ is therefore an optimization problem. A value such as $\epsilon = 0.9$ cannot be assumed to lie in a controlled perturbative regime without an explicit remainder analysis.

Variations in a microscopic bond produce variations in the effective coefficient K . For $K = cJ^n\Lambda^{1-n}$, logarithmic differentiation gives

$$\frac{\delta K}{K} = n\frac{\delta J}{J} - (n-1)\frac{\delta \Lambda}{\Lambda} + \frac{\delta c}{c}.$$

A spatially common drift in J is amplified by the factor n . By contrast, if n independent bond factors each have the same small fractional standard deviation s , the product has fractional standard deviation approximately $\sqrt{n}s$ to leading order.

The resulting distribution of K therefore depends on the correlations among microscopic variations. A single standard deviation for implantation position does not determine that distribution.

Fourth-order generation is consequently not an unqualified improvement. Relative to J , it introduces a suppression of order ϵ^3 , requires adequate perturbative control, and can amplify the effects of bond variation.

Thermal, disorder, decoherence, and finite-size scales

Temperature. Thermal excitation is controlled by the energy scale $k_B T$. In a dilute, noninteracting estimate, the occupation of an excitation with energy Δ_{topo} contains the Boltzmann factor

$$p_1 \sim e^{-\Delta_{\text{topo}}/k_B T}.$$

[Theory] This expression captures thermal activation but does not include diffusion, entropy, boundaries, or interactions among anyons. Anyons are quasiparticle excitations with topological exchange and fusion properties. Reviews of finite-temperature quantum memories emphasize that a nonzero gap alone does not make a two-dimensional memory self-correcting [R169]. If there are N approximately independent locations at which an excitation can occur, the crude expected number of excitations is Np_1 . Requiring this expected count to remain below a target p_* gives

$$\frac{\Delta_{\text{topo}}}{k_B T} \gtrsim \ln\left(\frac{N}{p_*}\right).$$

This system-size-dependent condition is more restrictive than the inequality $\Delta_{\text{topo}} > k_B T$ alone.

Disorder. Define σ_J as the root-mean-square energy variation of the relevant projected local terms after static calibration. Root-mean-square variation characterizes the typical magnitude of fluctuations around a reference value.

This definition is important because raw microscopic coupling variation can partially renormalize K , generate random effective fields at lower perturbative order, or mix the retained and leakage subspaces.

The relevant local comparison is

$$\sigma_J \ll \Delta_{\text{topo}}.$$

A total operator norm that increases with the number of sites is not the appropriate local comparison scale. An RMS value is also insufficient when the underlying distributions have long tails or spatial correlations.

[Theory] Stability theorems establish persistence of spectral bands for certain topologically ordered commuting-projector Hamiltonians under sufficiently weak, bounded, short-range local perturbations [R142]. These theorems do not yield a universal percentage tolerance for a proposed defect device.

Decoherence broadening. Define the effective decoherence rate as

$$\Gamma = \frac{1}{T_{2,\text{eff}}},$$

where $T_{2,\text{eff}}$ is the decay time, in seconds, of the projected degree of freedom under the specified control sequence. The associated energy scale is $\hbar\Gamma$, where $\hbar = h/(2\pi)$. In ordinary frequency units,

$$\frac{\hbar\Gamma}{h} = \frac{1}{2\pi T_{2,\text{eff}}}.$$

This convention is distinct from a spectroscopic full width at half maximum, which may differ by a numerical factor. In addition, a single scalar T_2 cannot characterize leakage, non-Markovian noise, or correlated errors. The condition $\hbar\Gamma \ll \Delta_{\text{topo}}$ is necessary to resolve coherent many-body dynamics, but it is not sufficient for fault tolerance.

Finite size. In a gapped local topological phase, virtual processes that wind around the sample can split the nominal ground-state family. A common asymptotic form is

$$\delta_{\text{fs}} \sim A\Delta_{\text{topo}}e^{-L/\xi},$$

where L is the shortest noncontractible linear size, ξ is a correlation length expressed in the same units, and A is dimensionless. A noncontractible path is one that cannot be continuously shrunk to a point within the sample geometry. [Theory] Stability results support exponentially narrow low-energy bands under appropriate weak local perturbations [R142], but the prefactor and the regime in which the asymptotic expression is useful depend on the model. Exactly solvable commuting-projector points can have zero splitting, whereas generic perturbations restore a nonzero splitting.

The condition $\Delta_{\text{topo}} > k_B T$ alone omits entropy, system size, local disorder, linewidth, and splitting within the ground-state family. Any one of these effects can eliminate the remaining scale margin.

Combined scale-ordering criterion

A compact screening criterion is

$$\Delta_{\text{topo}} \gg k_B T, \quad \sigma_J, \quad \hbar \Gamma, \quad \delta_{\text{fs}}$$

subject to two additional requirements:

- Unwanted projected terms and perturbative remainders must also be locally much smaller than Δ_{topo} .
- Thermal performance must be evaluated using $\Delta_{\text{topo}}/k_B T \gtrsim \ln(N/p_\star)$, rather than only the comparison with one.

For a numerical screening test, define

$$\mathcal{M} = \frac{\Delta_{\text{topo}}}{\max(k_B T, \sigma_J, \hbar \Gamma, \delta_{\text{fs}})}.$$

A value $\mathcal{M} < 1$ fails even the basic scale-ordering test. A value $\mathcal{M} > 1$ passes only that test. The condition represented by $\gg$ requires a quantitative margin determined by the target error probability and system size; it is stronger than a simple greater-than relation.

Accordingly, a ratio of 4.86 does not by itself constitute a sufficient gap margin. It must still be compared with $\ln(N/p_\star)$.

Three illustrative scale budgets

The following calculation evaluates sensitivity to assumed parameters; it is not a prediction. Every uncited number in the input table is explicitly treated as an assumption.

[Proposal; Assumptions] The calculation uses the fourth-order illustrative coefficient $K = J^4/(16\Lambda^3)$, motivated by the specified honeycomb limit [Ro17]. It sets $\Delta_{\text{topo}} = 2K$, adopts the finite-size prefactor $A = 1$, and assumes that the tabulated quantities apply after projection:

Input	Optimistic	Base	Pessimistic
cluster separation Λ/h	1000 MHz	250 MHz	50 MHz
weak coupling J/h	300 MHz	50 MHz	5 MHz
expansion ratio $\epsilon = J/\Lambda$	0.30	0.20	0.10
physical temperature T	0.010 mK	1.0 mK	100 mK

Input	Optimistic	Base	Pessimistic
residual effective disorder σ_J/h	0.100 MHz	0.025 MHz	0.001 MHz
projected $T_{2,\text{eff}}$	1 ms	0.10 ms	0.010 ms
linear size L/ξ	10	5	2

The optimistic temperature is 10 μ K. This value is an aggressive assumed spin temperature, not a cited capability of a dense defect array.

The other temperatures, couplings, disorder values, and projected coherence times are also scenario inputs. Experiments have separately demonstrated coherent coupling between individual diamond defect spins [Ro80] and long electronic-spin coherence under specialized material and decoupling conditions [R211]. Those results do not establish that all parameters in this table can be achieved simultaneously.

Using ordinary frequency units eliminates repeated factors of h :

$$\frac{K}{h} = \frac{1}{16} \frac{(J/h)^4}{(\Lambda/h)^3}, \qquad \frac{\Delta_{\text{topo}}}{h} = 2 \frac{K}{h}.$$

For the base scenario,

$$\frac{K}{h} = \frac{1}{16} \frac{(50 \text{ MHz})^4}{(250 \text{ MHz})^3} = 0.025 \text{ MHz},$$

so $\Delta_{\text{topo}}/h = 0.050 \text{ MHz} = 50 \text{ kHz}$. The units reduce according to $\text{MHz}^4/\text{MHz}^3 = \text{MHz}$.

Applying the same calculation to all three scenarios gives:

Derived quantity	Optimistic	Base	Pessimistic
K/h	0.50625 MHz	0.025 MHz	0.0003125 MHz
Δ_{topo}/h	1.0125 MHz	0.050 MHz	0.000625 MHz
$k_B T/h$	0.20837 MHz	20.8366 MHz	2083.66 MHz
σ_J/h	0.100 MHz	0.025 MHz	0.001 MHz
$\hbar\Gamma/h$	0.000159 MHz	0.001592 MHz	0.015915 MHz
assumed δ_{fs}/h	0.0000460 MHz	0.000337 MHz	0.0000846 MHz
scale margin $\mathcal{M}$	4.86	2.40×10^{-3}	3.00×10^{-7}

Displayed digits are arithmetic outputs, not claims about measurement precision. The temperature conversion uses the SI constants [R196].

The decoherence calculation uses $1/(2\pi T_{2,\text{eff}})$. The finite-size splitting uses the explicit assumption $A = 1$ together with the listed values of L/ξ .

Assessment. [Proposal] The base and pessimistic scenarios fail because $k_B T$ exceeds the illustrative gap. In the pessimistic scenario, decoherence broadening and effective disorder exceed it as well. Temperature sets $\mathcal{M} = 4.86$ in the optimistic scenario, so that case passes only the weakest scale-ordering test. For an illustrative array with $N = 1000$ possible excitation locations and target $p_\star = 0.01$, the dilute estimate requires

$$\frac{\Delta_{\text{topo}}}{k_B T} \gtrsim \ln(1000/0.01) = 11.51,$$

The optimistic ratio is 4.86, below this thermal target, so the optimistic scenario fails it as well.

These conclusions use only the displayed illustrative formula, exact unit conversion, and labeled assumptions. They do not assign any of the calculated gaps to a real defect architecture.

The calculation exposes a design tradeoff. Reducing J/Λ from 0.30 to 0.10 improves fourth-order control but, with the assumed absolute scales, lowers the illustrative gap from megahertz to hundreds of hertz. A feasibility proposal must optimize gap magnitude and perturbative validity together, while including leading unwanted terms.

Current experimental status

[Experiment] Coherent coupling between individual defect spins and long coherence have each been demonstrated in diamond [R080]; [R211]. These are important components of a possible architecture. They do not constitute measurements of a bulk topological gap, a correlation length, or a topological ground-state family.

For the defect-cluster architecture considered here, the gap analysis currently lacks four experimentally connected elements:

- A fabricated lattice with the required interaction graph.
- Spectroscopy that validates the projected many-body Hamiltonian and characterizes its unwanted terms.
- Finite-size scaling that distinguishes ground-family splitting from the excitation gap.
- Evidence that the fitted phase has the claimed topological data, rather than only a similar low-energy spectrum.

Consequently, no supported material value of Δ_{topo} is available for insertion into the analysis. [Proposal] A defensible experimental and computational program would first measure small-cluster spectra and parameter distributions, fit a microscopic Hamiltonian without omitting unfavorable terms, compute the resulting phase diagram and gap with uncertainty propagation, and then compare progressively larger patches. Any reported gap should specify the host, defect species, charge state, geometry, field, strain, temperature, boundary conditions, and model-fitting procedure.

Temperature must also be characterized carefully. The refrigerator temperature, phonon-bath temperature, and effective temperature of driven spins need not be equal.

Similarly, a single-defect echo T_2 measured under a pulse sequence is not necessarily the $T_{2,\text{eff}}$ of a continuously interacting cluster lattice. Combining the best coupling measured in one device, the best coherence measured in another, and the lowest cryostat temperature obtained in a third does not produce a physically consistent scale budget.

Common analytical errors

- **Identifying the largest energy scale as the gap.** Suppose that the zero-field splitting is a gigahertz while the fourth-order topological term is a kilohertz. The largest energy scale may safely separate some microscopic levels, but it does not specify the gap of the phase. The relevant gap is spectral and phase-specific.
- **Using the normalized gap of the target Hamiltonian as a material energy.** Writing a string-net Hamiltonian with unit coefficients establishes a normalization convention. It does not demonstrate that a defect implementation supplies one joule, one kelvin, or one megahertz [R018]. Without an explicit conversion, the calculation incorrectly assigns the abstract model's unit normalization to the crystal.
- **Ignoring lower-order unwanted terms.** A desired fourth-order plaquette term is ineffective if a symmetry-breaking field remains at first or second order and dominates it. Local norms of projected remainder terms must be compared with Δ_{topo} .
- **Treating RMS disorder as a complete disorder model.** Rare defective bonds can nucleate low-energy excitations. Correlated drift and heavy-tailed distributions differ from independent Gaussian variation. An RMS value describes a typical bond but does not characterize the most unfavorable bond present in the lattice.
- **Equating a nonzero gap with a passive memory.** Thermally created anyons can diffuse and implement a logical operator without paying an energy proportional to the distance traveled. A two-dimensional topological phase can remain stable while providing poor self-correcting memory at nonzero temperature [R169].
- **Confusing ground-state splitting with the bulk gap.** A small δ_{fs} is desirable, whereas a small Δ_{topo} is undesirable. A finite patch may exhibit both quantities. Interchanging their labels misidentifies a narrow ground-state family as a small bulk gap, giving a wrong assessment in either direction.
- **Using T_2 as an energy gap.** The quantity T_2 determines a linewidth scale only after a convention and noise model have been specified. By itself, it provides no evidence of topological order.
- **Interpreting digital evidence as an analog gap.** A programmable device can prepare and manipulate states with anyonic fusion rules even when its native Hamiltonian is an ordinary qubit Hamiltonian. An analog-material claim requires the low-energy spectrum and phase of the material itself to realize the topological order without continuous synthesis by a gate sequence.
- **Extrapolating a perturbation series beyond its controlled regime.** Increasing J/Λ raises both the target coefficient and every omitted perturbative order. The remainder must be calculated or bounded.
- **Reporting only one favorable parameter point.** A phase must occupy a finite region of parameter space. Because fabrication produces a distribution of parameters, uncertainty propagation and finite-size scaling are necessary components of the claim.

Verification of definitions and estimates

- **What defines Δ_{topo} ?**

The topological gap Δ_{topo} is the energy difference between the highest-energy state in the finite-size ground-state family and the first state outside that family. The finite-size ground-state family is the set of low-energy states associated with the ground-state sector of a finite system. This definition applies only after specifying the Hamiltonian, which determines the system's energies, and the boundary condition.

- **Show that if $K = J^4/(16\Lambda^3)$ and $J/\Lambda = 0.1$, then $K = J/16,000$.**

Here, J is the bare coupling, Λ is the energy scale appearing in the perturbative denominator, and K is the resulting effective coupling. Direct substitution gives

$$K = \frac{J}{16} \left(\frac{J}{\Lambda} \right)^3 = \frac{J}{16} \times 10^{-3} = \frac{J}{16,000}.$$

Projection, meaning restriction to a selected low-energy subspace, and the perturbative order n can reduce an effective coefficient to

$$c_n J (J/\Lambda)^{n-1},$$

where c_n is the coefficient at order n . The spectral gap, defined as an energy difference between specified spectral sectors, also includes a model-dependent factor. Consequently, the bare coupling J is not itself the gap.

- **Show that 1 mK is 20.8366 MHz in these units.**

Thermal energy is converted to frequency using $k_B T/h$, where k_B is the Boltzmann constant, T is the temperature, and h is the Planck constant. At $T = 1$ mK,

$$k_B T/h = 20.8366 \text{ MHz},$$

using exact SI constants [\[R196\]](#).

- **Show that if $T_{2,\text{eff}} = 100 \mu\text{s}$ and $\Gamma = 1/T_{2,\text{eff}}$, then $\hbar\Gamma/h = 1.59$ kHz.**

The effective coherence time is $T_{2,\text{eff}}$, and Γ is the corresponding decoherence rate under the stated assumption $\Gamma = 1/T_{2,\text{eff}}$. Since $\hbar/h = 1/(2\pi)$,

$$\frac{\hbar\Gamma}{h} = \frac{1}{2\pi T_{2,\text{eff}}} = 1.59 \text{ kHz}.$$

- **What is omitted if the only requirement is $\Delta_{\text{topo}} > k_B T$?**

This condition does not account for entropy or system size. In a dilute-excitation estimate, the required ratio is approximately

$$\Delta_{\text{topo}}/k_B T \gtrsim \ln(N/p_\star),$$

where N is the relevant system-size measure and $p_\star$ is the stated target probability. Disorder, decoherence, finite-size splitting, and residual terms left by projection must also be considered.

- **What evidence is required to convert a budgeted gap into a gap claim?**

The effective Hamiltonian must first be derived or fitted with quantified errors. Controlled analytic or numerical spectroscopy must then determine its spectrum, and finite-size scaling must establish how the result changes with system size. Experimental spectroscopy provides stronger evidence.

A coupling is a coefficient in a Hamiltonian, whereas a gap is an energy difference in the Hamiltonian's spectrum.

The screening condition is

$$\Delta_{\text{topo}} \gg k_B T, \sigma_J, \hbar\Gamma, \delta_{\text{fs}},$$

where $k_B T$ is the thermal energy scale, σ_J is the disorder scale in the coupling, $\hbar\Gamma$ is the decoherence energy scale, and δ_{fs} is the finite-size splitting. The symbol $\gg$ requires Δ_{topo} to be substantially larger than each competing scale. Small projected remainders and the logarithmic thermal penalty must also be included. The three quantitative budgets in this chapter represent scenarios only.

There is no measured material Δ_{topo} to insert into the budget.

Sources

- [R017] Alexei Kitaev, “Anyons in an exactly solved model and beyond,” *Annals of Physics* 321, 2–111 (2006). DOI: [10.1016/j.aop.2005.10.005](https://doi.org/10.1016/j.aop.2005.10.005); arXiv: [cond-mat/0506438](https://arxiv.org/abs/cond-mat/0506438).
- [R018] Michael A. Levin and Xiao-Gang Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110); arXiv: [cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617).
- [R209] Roberto Oliveira and Barbara M. Terhal, “The complexity of quantum spin systems on a two-dimensional square lattice,” *Quantum Information & Computation* 8, 900–924 (2008). arXiv: [quant-ph/0504050](https://arxiv.org/abs/quant-ph/0504050).
- [R142] Sergey Bravyi, Matthew B. Hastings, and Spyridon Michalakis, “Topological quantum order: Stability under local perturbations,” *Journal of Mathematical Physics* 51, 093512 (2010). DOI: [10.1063/1.3490195](https://doi.org/10.1063/1.3490195); arXiv: [1001.0344](https://arxiv.org/abs/1001.0344).
- [R169] Benjamin J. Brown, Daniel Loss, Jiannis K. Pachos, Chris N. Self, and James R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* 88, 045005 (2016). DOI: [10.1103/RevModPhys.88.045005](https://doi.org/10.1103/RevModPhys.88.045005); arXiv: [1411.6643](https://arxiv.org/abs/1411.6643).

- [R196] Bureau International des Poids et Mesures, The International System of Units (SI Brochure), 9th ed., updated 2026. DOI: [10.59161/AUEZ1291](https://doi.org/10.59161/AUEZ1291); [stable publication page](#).
 - [R080] F. Dolde et al., “Room-temperature entanglement between single defect spins in diamond,” Nature Physics 9, 139–143 (2013). DOI: [10.1038/nphys2545](https://doi.org/10.1038/nphys2545).
 - [R211] N. Bar-Gill, L. M. Pham, A. Jarmola, D. Budker, and R. L. Walsworth, “Solid-state electronic spin coherence time approaching one second,” Nature Communications 4, 1743 (2013). DOI: [10.1038/ncomms2771](https://doi.org/10.1038/ncomms2771).
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Chapter 30 — Encoding-dependent response to identical perturbations

A spatially correlated magnetic perturbation can act on three nearby defects at the same time. The three physical spins may encode a two-level logical subspace, while the remaining states stay outside the intended dynamics.

The perturbation acts on the three physical spins, so its effect must first be projected into the logical encoding. Projection produces a 2×2 operator on the logical states. The complementary component couples the logical subspace to excluded states and can cause leakage, meaning population transfer out of that subspace.

A cluster reshapes how microscopic perturbations act on encoded states; it does not remove those perturbations. Adding physical spins assigns different signs and amplitudes to noise sources already present in the crystal.

Perturbation channels in a cluster encoding

Consider a linear combination of three site-dependent signals. Equal coefficients partly suppress independent fluctuations because their cross-correlations vanish and their noise powers add incoherently. A signal identical at all three sites survives this combination. Reversing one coefficient can cancel a uniform signal while leaving a response to spatial gradients.

A cluster encoding implements such a linear combination through the matrix elements of physical operators within the selected logical subspace. The sites are microscopic defect sites, and the coefficients are determined by the encoded states.

After projection, a perturbation can produce three kinds of dynamics.

- It can shift the two retained energy levels by different amounts. The resulting random relative phase causes dephasing, defined as the loss of coherence between the logical basis states. This is the same process that Chapter 4 characterized through decay of an off-diagonal density-matrix element, now applied to the encoded pair.
- It can mix the two retained levels. This transfers population within the encoded subspace and produces relaxation or excitation of the logical qubit.
- It can transfer population out of the retained pair. This process is leakage into the Q subspace introduced in Chapter 11.

Static fabrication errors add a separate issue: nominally equivalent clusters can realize different Hamiltonians, and slow temporal drift makes that disorder time dependent.

An encoding can reject a spatially uniform magnetic field yet remain sensitive to a gradient, a fluctuating exchange bond, or a pulse that addresses several defects at once. When bath fluctuations are correlated rather than independent, a common perturbation can remain fully visible to the encoded states. The matrix elements of the encoding determine the response.

Three spins and a local field

Consider three spin-1/2 defects with strong, approximately isotropic exchange coupling. Isotropic exchange is a spin-spin interaction with equal strength along all Cartesian spin directions. Let $|\uparrow\rangle$ and $|\downarrow\rangle$ denote eigenstates of the single-spin operator S^z , with eigenvalues $+1/2$ and $-1/2$, respectively. The spin operators are dimensionless. An energy-valued local field $\xi_i(t)$, measured in joules (J), couples to the spins through

$$\delta H_z(t) = \sum_{i=1}^3 \xi_i(t) S_i^z.$$

Here $\delta H_z(t)$ is the longitudinal perturbation Hamiltonian, and i labels the physical defect site.

Restrict attention to the total-magnetization sector $S_{\text{tot}}^z = +1/2$. Choose the logical basis states

$$|0_L\rangle = \frac{|\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle}{\sqrt{2}},$$

$$|1_L\rangle = \sqrt{\frac{2}{3}} |\uparrow\uparrow\downarrow\rangle - \frac{|\uparrow\downarrow\uparrow\rangle + |\downarrow\uparrow\uparrow\rangle}{\sqrt{6}}.$$

These states are the same type of three-spin pair retained in Chapter 11. Define the projector onto this logical doublet by

$$P = |0_L\rangle\langle 0_L| + |1_L\rangle\langle 1_L|.$$

Define logical Pauli operators within the doublet as

$$\tau_z = |0_L\rangle\langle 0_L| - |1_L\rangle\langle 1_L|, \quad \tau_x = |0_L\rangle\langle 1_L| + |1_L\rangle\langle 0_L|.$$

Direct evaluation of the matrix elements gives

$$\begin{aligned} PS_1^z P &= \frac{1}{6}I - \frac{1}{6}\tau_z - \frac{\sqrt{3}}{6}\tau_x, \\ PS_2^z P &= \frac{1}{6}I - \frac{1}{6}\tau_z + \frac{\sqrt{3}}{6}\tau_x, \\ PS_3^z P &= \frac{1}{6}I + \frac{1}{3}\tau_z. \end{aligned}$$

These equations express each physical spin operator as an effective operator within the logical subspace. Substitution into the perturbation Hamiltonian gives

$$P\delta H_z P = h_0 I + h_x \tau_x + h_z \tau_z,$$

where

$$h_0 = \frac{\xi_1 + \xi_2 + \xi_3}{6}, \quad h_x = \frac{\sqrt{3}}{6}(-\xi_1 + \xi_2),$$

$$h_z = \frac{-\xi_1 - \xi_2 + 2\xi_3}{6}.$$

The identity contribution $h_0 I$ gives both logical basis states the same phase and therefore does not alter the encoded logical state. Low-frequency fluctuations in h_z produce logical dephasing. Fluctuations in h_x near the logical angular frequency ω_L , measured in radians per second, drive transitions between the logical states.

For a perfectly common longitudinal fluctuation, set $\xi_1 = \xi_2 = \xi_3 = \xi_c$. Both h_x and h_z then vanish because

$$P(S_1^z + S_2^z + S_3^z)P = \frac{1}{2}I.$$

Thus, a perfectly common longitudinal fluctuation is rejected to first order. [Theory] This is decoherence-free behavior with respect to one specified operator; it is not general immunity to magnetic noise. If spatial uniformity is removed, for example by allowing ξ_3 to differ from ξ_1 and ξ_2 , the longitudinal logical field h_z becomes nonzero immediately.

For equal and mutually uncorrelated site noise, define the two-sided energy-noise spectrum by

$$S_\xi(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} \langle \xi_i(t) \xi_i(0) \rangle.$$

A two-sided spectrum includes both positive and negative angular frequencies. Its units are J^2 s, because the correlation function has units of J^2 and dt contributes seconds. Under the assumptions of equal spectra and zero cross-correlations between sites, the projected spectra are

$$S_{h_x}(\omega) = S_{h_z}(\omega) = \frac{1}{6} S_\xi(\omega).$$

The factor of six represents a reduction in these projected noise spectra under the stated assumptions. It does not imply a universal sixfold increase in lifetime. Dephasing depends on the low-frequency filter function, which describes the frequency-dependent response of a control sequence to noise. Relaxation depends on the spectrum near ω_L , whereas leakage depends on higher-frequency spectral components near the cluster excitation gap.

The state

$$|Q\rangle = \frac{|\uparrow\downarrow\uparrow\rangle + |\downarrow\uparrow\uparrow\rangle + |\uparrow\uparrow\downarrow\rangle}{\sqrt{3}}$$

belongs to the total-spin-3/2 manifold rather than the encoded doublet. Local field gradients have nonzero matrix elements $\langle Q|S_i^z|0_L\rangle$ and $\langle Q|S_i^z|1_L\rangle$. Consequently, noise with spectral weight at the cluster excitation energy can cause leakage.

A uniform longitudinal field is proportional to total S^z and cannot drive this transition. The cluster encoding therefore suppresses one spatial-correlation channel while retaining sensitivity to gradient and leakage channels.

Projection of microscopic bath operators

Let the ideal cluster Hamiltonian be H_C . Assume that its logical doublet is separated from every other cluster state by a minimum energy Δ_C , measured in J. Define

$$Q = I - P$$

as the projector onto the unwanted states outside the logical subspace. Write a general microscopic disturbance as

$$\delta H(t) = \sum_a \xi_a(t) O_a.$$

The index a specifies the site, Cartesian component, and physical mechanism. For example, O_a may be a spin component, an orbital or strain operator, a local charge projector, or a bond operator $\mathbf{S}_i \cdot \mathbf{S}_j$. The coefficient $\xi_a(t)$ has the units required for $\xi_a O_a$ to have units of energy.

Any projected operator acting within a two-dimensional logical subspace can be expanded in the identity and logical Pauli operators:

$$PO_aP = c_{a0}I + \frac{1}{2} \sum_{\mu=x,y,z} g_{a\mu} \tau_\mu.$$

The coefficient c_{a0} is a common energy shift and does not affect the logical state. The coefficients $g_{a\mu}$ are the cluster's noise susceptibilities. A noise susceptibility quantifies how strongly the projected physical operator contributes to the logical Pauli component τ_μ .

The resulting effective logical fields are

$$b_\mu(t) = \sum_a g_{a\mu} \xi_a(t), \quad P\delta H P = \text{common shift} + \frac{1}{2} \sum_\mu b_\mu \tau_\mu.$$

If the unperturbed logical splitting is oriented along the logical z axis, slow fluctuations in b_z produce phase fluctuations. Spectral components of b_x and b_y near ω_L drive logical transitions. Therefore, whether a physical disturbance is longitudinal or transverse is determined after projection into the logical basis, not solely by its laboratory description.

The frequency-dependent covariance of two bath variables is represented by the cross-spectrum

$$S_{ab}(\omega) = \int dt e^{i\omega t} \langle \delta \xi_a(t) \delta \xi_b(0) \rangle.$$

Diagonal elements describe the noise power of each bath variable at frequency ω . Off-diagonal elements describe correlations between distinct bath variables. The corresponding logical spectral matrix is

$$S_{\mu\nu}^{(L)}(\omega) = \sum_{a,b} g_{a\mu} g_{b\nu} S_{ab}(\omega).$$

This equation defines the projected noise model. The diagonal entries give the logical noise power along each Pauli axis, while the off-diagonal entries retain correlations between different logical axes. In a Markovian weak-noise calculation, where bath correlations decay sufficiently rapidly to justify memoryless dynamics, the logical relaxation rate is proportional to

$$[S_{xx}^{(L)}(\omega_L) + S_{yy}^{(L)}(\omega_L)]/\hbar^2.$$

The dimensions are consistent:

$$\text{J}^2\text{s}/(\text{J}^2\text{s}^2) = \text{s}^{-1}.$$

Exact numerical prefactors depend on whether one-sided or two-sided spectral conventions are used. A simulation must therefore state its spectral convention explicitly to avoid an unintended factor of two.

A complete characterization requires the full frequency-dependent logical spectral matrix together with the couplings to states in Q , rather than a single coherence time. Neglecting the off-diagonal blocks of S_{ab} can reverse the predicted effect of the encoding.

Effects of spatial noise correlations

Suppose the same type of operator acts at N sites with logical coefficients g_i . If the site noises are independent and have the same spectrum S_0 , the logical spectrum is

$$S_L = S_0 \sum_i g_i^2.$$

If the site noises are perfectly correlated, the logical spectrum is instead

$$S_L = S_0 \left(\sum_i g_i \right)^2.$$

For a delocalized average with $g_i = 1/N$, independent noise decreases as $1/N$, whereas common noise remains unchanged. For a decoherence-free sign pattern satisfying $\sum_i g_i = 0$, common noise cancels. Conversely, if all g_i have the same order-one sign, the common-noise power grows as N^2 .

Noise correlation is therefore not intrinsically beneficial or harmful. Its effect must be calculated by contracting the bath covariance matrix with the projected susceptibility vector. [Theory] A prediction based only on $\sum_i g_i^2$ fails for common-mode noise: an encoding that suppresses independent noise through equal-weight averaging can respond strongly to correlated noise.

Leakage outside the logical subspace

Projection onto P describes only dynamics within the logical subspace. The discarded operator block QO_aP determines transitions from the logical subspace into unwanted states.

For an unwanted eigenstate $|m\rangle$ with energy difference $E_m - E_L = \hbar\omega_{mL}$, the weak-noise transition rate contains

$$\Gamma_{L \rightarrow m} \propto \frac{1}{\hbar^2} \sum_{a,b} \langle L|O_a|m\rangle \langle m|O_b|L\rangle S_{ab}(\omega_{mL}).$$

The matrix elements $\langle m|O_a|L\rangle$ are the components of QO_aP . A large cluster gap Δ_C suppresses leakage only if the environment has little spectral weight at $\Delta_C/\hbar$ and if applied controls are slow or spectrally narrow relative to that frequency.

Strong pulses, resonant phonons, and charge-switching events need not satisfy these spectral conditions. Every microscopic noise operator must therefore be characterized through both PO_aP , which determines its action inside the logical subspace, and QO_aP , which determines leakage. A single quoted T_2 does not retain either complete set of information.

Classification of microscopic noise sources

The microscopic source, coupling mechanism, projected channel, and conditions for suppression or enhancement must be distinguished. The source names below denote physical mechanisms rather than interchangeable descriptions of decoherence.

Microscopic source	Typical microscopic coupling	Main channel after projection	Clustering can suppress when...	Clustering can worsen when...
Host nuclear spins, including ^{13}C in diamond	Hyperfine fields and nuclear flip-flops	Quasistatic dephasing, spectral diffusion; sometimes resonant relaxation	the nuclear field is common across a fixed-magnetization encoding, or isotopic purification removes most bath spins	each constituent samples a different Overhauser field, and the cluster contains more strongly coupled nuclei
Paramagnetic impurities and unintended defects	Magnetic dipolar fields, bath flip-flops, cross-relaxation	Dephasing, relaxation, correlated magnetic bursts	symmetry rejects common field; detuning avoids bath resonances	high defect density supplies both desired coupling and a denser spin bath
Surface spins	Fluctuating magnetic dipoles at interfaces	Dephasing and relaxation, often depth dependent	the cluster is deep and compact compared with the field's correlation length	shallow fabrication exposes every constituent and a nearby fluctuator acts nonuniformly
Charge noise	Electric/Stark shifts; modulation of orbital energies and exchange	Dephasing, bond noise, optical spectral diffusion	a clock point makes the logical splitting first-order insensitive	one trap modulates several bonds or shifts the cluster through an avoided crossing
Strain and electric-field noise	Crystal-field, spin-orbit, Stark, and spin-strain terms	Dephasing, relaxation, coupling disorder	symmetry makes uniform strain an identity operation	long-wavelength strain drives all sites coherently with non-cancelling matrix elements
Phonons	Dynamic strain and orbital/spin-phonon coupling	T_1 , orbital relaxation, leakage, temperature-dependent dephasing	the encoded transition has a vanishing matrix element or no resonant phonon density	collective matrix elements add, or a phonon matches Δ_C
Implantation/processing damage	Vacancies, interstitials, strain fields, traps, paramagnetic complexes	Static disorder plus magnetic, charge, and strain noise	annealing and materials processing remove or passivate damage	adding more implanted constituents multiplies nearby damage and increases the distribution widths of J and Δ_C
Charge-state instability	Ionization/recombination of the active defect	Leakage or erasure; abrupt Hamiltonian change	redundant heralding detects the event	one constituent changing charge destroys the encoded Hamiltonian and perturbs neighbors

Nuclear and paramagnetic baths are environments formed by surrounding nuclear spins or electron spins. In diamond NV systems, coherent evolution coupled to ^{13}C nuclei and nitrogen-related electron-spin baths has been observed directly. Dynamical decoupling, meaning pulse sequences designed to produce a frequency-dependent response to environmental fluctuations, can modify the effective bath spectrum rather than eliminate it [Experiment] [R214]; [R215].

The same distinction applies to a cluster. Isotopic purification reduces the abundance of nuclear spins, whereas pulse filtering suppresses selected frequency components. Pulse filtering is the reduction of sensitivity to specified spectral components through the filter function of a control sequence. Neither procedure guarantees small leakage matrix elements.

A deliberately dense array can also convert spectator defects, residual implantation products, and unintended charge states into a correlated electron-spin bath.

Surface spins are fluctuating magnetic moments located at or near an interface. Measurements using shallow NV centers support a substantial contribution from surface-related magnetic noise [Experiment] [R216].

A compact cluster whose dimensions are much smaller than its distance from a source can experience an approximately common magnetic field and may reject that field. If one defect is substantially closer to the interface, the common-mode approximation fails.

Nanostructuring introduced for optical access can therefore alter both the noise magnitude and its spatial covariance. The T_2 of a single defect does not determine the encoded T_2 .

Charge noise consists of temporal fluctuations in local electric potentials or charge configurations. Optical spectral diffusion is the resulting time-dependent fluctuation of an optical transition frequency. NV experiments have identified regimes dominated by magnetic noise and regimes dominated by electric-field noise [Experiment] [R217], and electric coupling to an individual NV spin has been measured directly [Experiment] [R191].

Charge traps can shift optical transition frequencies and change a defect's charge state. In a cluster, electric noise may shift individual logical levels, modulate the intra-cluster exchange J_{ij} , or modulate inter-cluster coupling.

These effects correspond to three distinct operators and therefore require three sets of coefficients $g_{a\mu}$. Charge conversion is more severe than ordinary phase noise. If a constituent no longer occupies the required charge or spin manifold, the event is leakage or erasure rather than a small Pauli error.

Near-surface NV ensembles exhibit coupled charge dynamics and density-dependent charge-state behavior [Experiment] [R218].

Static strain is a time-independent spatial distortion and therefore contributes disorder. Time-dependent strain constitutes noise.

Spin-strain coupling has been driven and measured in NV–mechanical systems [Experiment] [R219]. Phonons, which are quantized lattice vibrations, can also relax orbital branches and spins.

For inversion-symmetric group-IV vacancy centers, orbital phonon processes are central to models of temperature-dependent coherence [Theory/Experiment] [R201]. A long-wavelength acoustic mode can be strongly correlated across a cluster with nanometre-scale dimensions.

Such correlation is beneficial only if the projected uniform-strain operator is proportional to I . Otherwise, coherent addition of the matrix elements can amplify logical driving or leakage.

Ion implantation enables spatial placement of defects but also produces collision cascades, vacancies, interstitials, local strain, and charge traps. Annealing converts only part of this damaged region into the desired centers [Experiment/Review] [R212].

Implantation damage is therefore not a single decay channel. It generates static disorder in J_{ij} and produces magnetic, electric, and strain fluctuators.

A model of a fabricated array must correlate damage statistics with placement statistics instead of sampling them independently.

Spectral diffusion is the temporal fluctuation of a transition frequency as surrounding spins or charges change state.

Spectral diffusion is not a distinct microscopic mechanism. Slow nuclear flip-flops, electron-spin flips, and telegraph switching of charge traps can all cause spectral diffusion [R213]; [R214]; [R217].

Gaussian white noise, defined by Gaussian statistics and a frequency-independent spectrum, is often an inadequate model. A single nearby trap can produce discrete, non-Gaussian jumps, whereas an ensemble with a broad distribution of switching rates can generate an approximate $1/f$ spectrum. Echo sequences may refocus slow frequency fluctuations during idle evolution, but the same fluctuations can detune optical initialization, readout, or cluster-mediated gates.

Static and dynamic bond disorder

Let the intended microscopic coupling on bond ij be J_{ij} , measured in J. Decompose it as

$$J_{ij}(t) = \bar{J}_{ij} + \delta J_{ij}^{\text{static}} + \delta J_{ij}^{\text{dyn}}(t).$$

Here $\bar{J}_{ij}$ is the intended mean coupling, $\delta J_{ij}^{\text{static}}$ is a time-independent fabrication error, and $\delta J_{ij}^{\text{dyn}}(t)$ is a time-dependent fluctuation. The static term produces distributions of cluster splittings, leakage gaps, and projected inter-cluster couplings. The dynamic term is bond noise.

Dipolar coupling depends on both separation and orientation, so positional disorder changes it multiplicatively. Exchange coupling is generally even more sensitive to atomic-scale geometry.

After projection, a fluctuating bond operator $\delta J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$ can produce a logical τ_z field, a transverse logical term, or leakage. The result depends on the encoding and the relevant matrix elements.

An array-level model should retain at least four correlation classes:

- **Local independent noise:** Separate nuclei or traps near each constituent produce fluctuations that are approximately uncorrelated between sites.
- **Intra-cluster common-mode noise:** Long-wavelength magnetic, electric, or strain fluctuations act coherently across one cluster.
- **Inter-cluster correlations:** A surface region, electrode, laser, or acoustic mode affects multiple encoded qubits.

- **Control correlations:** Pulse-amplitude error, detuning, crosstalk, and global-drive phase noise act coherently because of the control design.

Control correlations can be particularly important in a cluster. A global pulse intended to rotate every constituent may have a transition matrix element that scales as N , producing error power proportional to N^2 .

Fast control can also populate the Q subspace. A symmetry that cancels passive common-mode noise may suppress the desired control interaction at the same order. Implementing control can then require gradient fields or symmetry-breaking pulses, which restore sensitivity to previously suppressed perturbations.

A cluster doublet is an encoded qubit constructed from physical defect spins. Projection of microscopic noise onto logical Pauli operators does not create topological order.

Even if a subsequent many-body Hamiltonian supports emergent anyons, charge-state changes and transitions out of a cluster doublet occur in microscopic degrees of freedom not represented by the anyon description. Such errors cannot be assumed to produce only local anyon pairs that do not cause logical errors.

Similarly, a digital simulation that imposes a selected Pauli channel does not demonstrate that the material realizes the corresponding native noise correlations.

Experimental characterization status

For individual color centers and ensembles, experiments can measure Ramsey decay, echo decay, relaxation, noise spectra under pulse sequences, optical spectral diffusion, charge-state switching, strain response, and, in some cases, spatial variation near surfaces [R213]; [R214]; [R215]; [R216]; [R217]; [R191]; [R218]; [R219]; [R201]. Ramsey measurements probe free phase evolution, echo measurements refocus sufficiently slow fluctuations, and relaxation measurements determine population-decay times. [Experiment] These measurements establish that the listed mechanisms occur physically. They do not yet provide a complete covariance matrix $S_{ab}(\omega)$ for a fabricated, strongly coupled cluster containing 5–20 defects, and still less for a large array of such clusters.

A characterization procedure sufficient to evaluate the encoding would include the following steps:

- Measure each constituent while individual addressability remains available. The measurements should include resonance frequency, charge state, T_1 , Ramsey decay, echo decay, and optical stability.
- Identify the logical doublet and every nearby leakage level spectroscopically, thereby determining ω_L and $\Delta_C/\hbar$.
- Apply calibrated common-mode and gradient magnetic, electric, and strain perturbations to determine the coefficients $g_{a\mu}$.
- Measure simultaneous time traces or cross-spectra for multiple constituents or clusters rather than assuming statistical independence.
- Fit static bond disorder separately from dynamic bond noise.
- Incorporate measured spectra and leakage matrix elements into many-body simulations rather than replacing them with a single average T_2 .

[Proposal] No cited experiment demonstrates that clustering color centers produces a topologically ordered array with a measured passive-protection advantage. The projection framework provides a falsifiable test: measured susceptibilities and noise spectra must predict logical-error and leakage rates below the relevant interaction and topological-gap scales. Until such a comparison is available, noise resilience from cluster encoding remains a design hypothesis.

Common analytical errors

- Using only T_2 , the measured transverse coherence time, does not specify a complete noise model. Its value depends jointly on the applied pulse sequence, the noise spectrum, and the operating point. It does not characterize charge erasure, optical instability, correlated errors, or leakage.
- Treating all fluctuations as statistically independent neglects correlations produced by common electrodes, surfaces, laser fields, strain waves, and fabrication damage. Here S_{ab} denotes the cross-spectral density between noise sources a and b . Setting $S_{ab} = 0$ for $a \neq b$ can reverse the predicted benefit of an encoding.
- Treating every common-mode fluctuation as harmless is also incorrect. Cancellation of a microscopic operator μ requires $\sum_a g_{a\mu} = 0$, where $g_{a\mu}$ is the projected susceptibility of constituent a to that operator. Common exchange fluctuations or transverse fields need not satisfy this condition.
- Projecting into the logical subspace while neglecting Q , the complementary nonlogical subspace, can invalidate the effective model. An effective Pauli model represents dynamics within the logical subspace P using Pauli operators. It is incomplete when noise or control has spectral weight near $\Delta_C/\hbar$, where Δ_C is the cluster excitation gap and $\hbar$ is the reduced Planck constant. Under these conditions, transitions out of P can occur, and leakage can increase with the number of constituent levels.
- Static disorder and decoherence are distinct effects. Static disorder is time-independent variation of device parameters. It does not by itself destroy phase coherence in one isolated device, but it can change Hamiltonian parameters, close local gaps, impede calibration, and cause ensemble dephasing. Drift is dynamic disorder because the relevant parameters vary in time.
- Echo sequences do not resolve all architectural noise mechanisms. An echo is a control sequence designed to average selected noise contributions. It can reject selected low-frequency terms, but it can also average away desired interactions. It does not reverse irreversible T_1 relaxation, charge conversion, or unobserved leakage.
- Optimizing isolated defects does not necessarily optimize a cluster. A site with the best individual coherence may produce an unsuitable interaction graph, whereas the cluster with the strongest couplings may be located in the most damaging implantation-induced bath. Feasibility therefore depends on the joint distribution of coherence, coupling, geometry, and fabrication-induced noise.
- Topological protection is not a universal suppression mechanism. Topological order can suppress particular local processes under particular energy and temperature conditions. This argument can be bypassed by correlated faults that span multiple effective sites, leakage outside the effective model, or time-dependent disorder in the Hamiltonian.

Independent and correlated noise components contract differently with the cluster susceptibilities, which quantify the response of the logical degrees of freedom to microscopic perturbations. Static bond disorder, static onsite disorder, dynamic dephasing, relaxation, charge erasure, and leakage are distinct error channels.

Clustering suppresses a noise source only when the relevant symmetry and spatial or temporal correlation pattern make the projected logical matrix element small. This projected-noise description, rather than a single-center lifetime taken from another setting, is the minimum model required to test whether an apparent many-body gap remains effective in a realistic defect array.

Verification of noise-model derivations

- A microscopic perturbation dephases or relaxes the cluster qubit according to its coefficients after projection into the logical subspace. A logical τ_z component, where τ_z is the longitudinal logical Pauli operator, causes dephasing when its noise is concentrated near zero frequency. Logical τ_x or τ_y components, which are transverse logical Pauli operators, drive transitions when their noise has spectral weight near the logical transition frequency ω_L .
- A perfectly common longitudinal field is invisible within the three-spin doublet because both logical states have total spin projection $S_{\text{tot}}^z = +1/2$. Consequently,

$$P(S_1^z + S_2^z + S_3^z)P = \frac{1}{2}I.$$

Here P projects onto the logical doublet, S_i^z is the longitudinal spin operator for site i , and I is the identity operator within that subspace. The projected field therefore contributes only a common phase and does not distinguish the logical states.

- Perfectly correlated site noise invalidates the expression for independent noise. The independent-source result

$$S_L = S_0 \sum_i g_i^2$$

is replaced by

$$S_L = S_0 \left(\sum_i g_i \right)^2,$$

where S_L is the logical noise spectral density, S_0 is the common site-noise spectral density, and g_i is the logical susceptibility to noise at site i . Statistical averaging then disappears. A decoherence-free sign pattern still produces cancellation, whereas same-sign coefficients produce N^2 amplification for N constituents.

- Leakage is a transition from the logical doublet P into another cluster state in Q . It is driven by microscopic noise or control with spectral weight at the corresponding transition energy. For a microscopic operator O_a , the relevant transition matrix elements are contained in QO_aP .
- Spectral diffusion is not a separate elementary bath. It is the observed wandering of a transition frequency and can result from nuclear, paramagnetic, or charge dynamics.
- Charge-state switching cannot generally be modeled as a small Stark shift. A Stark shift, defined as an energy shift produced by an electric field, may project into the logical subspace as dephasing. A change of charge state can instead remove the spin required from a constituent and modify the entire cluster Hamiltonian. The

resulting event can produce leakage or erasure rather than a Pauli error confined to P .

Sources

- [R212] S. Pezzagna and J. Meijer, “Quantum computer based on color centers in diamond,” *Applied Physics Reviews* 8, 011308 (2021). DOI: [10.1063/5.0007444](https://doi.org/10.1063/5.0007444).
 - [R213] C. L. Degen, F. Reinhard, and P. Cappellaro, “Quantum sensing,” *Reviews of Modern Physics* 89, 035002 (2017). DOI: [10.1103/RevModPhys.89.035002](https://doi.org/10.1103/RevModPhys.89.035002).
 - [R214] L. Childress et al., “Coherent dynamics of coupled electron and nuclear spin qubits in diamond,” *Science* 314, 281–285 (2006). DOI: [10.1126/science.1131871](https://doi.org/10.1126/science.1131871).
 - [R215] G. de Lange, Z. H. Wang, D. Ristè, V. V. Dobrovitski, and R. Hanson, “Universal dynamical decoupling of a single solid-state spin from a spin bath,” *Science* 330, 60–63 (2010). DOI: [10.1126/science.1192739](https://doi.org/10.1126/science.1192739).
 - [R216] T. Rosskopf et al., “Investigation of surface magnetic noise by shallow spins in diamond,” *Physical Review Letters* 112, 147602 (2014). DOI: [10.1103/PhysRevLett.112.147602](https://doi.org/10.1103/PhysRevLett.112.147602).
 - [R217] P. Jamonneau et al., “Competition between electric field and magnetic field noise in the decoherence of a single spin in diamond,” *Physical Review B* 93, 024305 (2016). DOI: [10.1103/PhysRevB.93.024305](https://doi.org/10.1103/PhysRevB.93.024305).
 - [R191] F. Dolde et al., “Electric-field sensing using single diamond spins,” *Nature Physics* 7, 459–463 (2011). DOI: [10.1038/nphys1969](https://doi.org/10.1038/nphys1969).
 - [R218] S. Dhomkar, H. Jayakumar, P. R. Zangara, and C. A. Meriles, “Charge dynamics in near-surface, variable-density ensembles of nitrogen-vacancy centers in diamond,” *Nano Letters* 18, 4046–4052 (2018). DOI: [10.1021/acs.nanolett.8b01739](https://doi.org/10.1021/acs.nanolett.8b01739).
 - [R219] J. Teissier, A. Barfuss, P. Appel, E. Neu, and P. Maletinsky, “Strain coupling of a nitrogen-vacancy center spin to a diamond mechanical oscillator,” *Physical Review Letters* 113, 020503 (2014). DOI: [10.1103/PhysRevLett.113.020503](https://doi.org/10.1103/PhysRevLett.113.020503).
 - [R201] K. D. Jahnke et al., “Electron–phonon processes of the silicon-vacancy centre in diamond,” *New Journal of Physics* 17, 043011 (2015). DOI: [10.1088/1367-2630/17/4/043011](https://doi.org/10.1088/1367-2630/17/4/043011).
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Chapter 31 — Limitations of topological protection

Consider a single Pauli error that flips one edge of a lattice. The two stabilizer checks touching that edge change sign, creating two anyonic excitations separated by one lattice step.

The event is not yet a logical failure. Local operations can move the pair together and annihilate it, restoring every local syndrome check while preserving the logical state.

The pair can instead follow a path that winds around a noncontractible cycle of the torus before the excitations meet and annihilate. The final syndrome is again trivial, but the accumulated operation can act on the encoded state.

A local fault therefore need not become a logical fault. Topological encoding makes a logical error depend on an extended history of local events; it does not forbid that history.

Four concepts require separate checks: the energy gap, the decoder, the error threshold, and exact error-free operation. The discussion follows a local fault from creation to possible logical failure.

Local errors and topologically nontrivial paths

A single local event contains no global winding information. A continuous path around a periodic direction is global: no short segment reveals its winding, while the complete path can change the logical state.

A topological code follows the same pattern. A local operator generally creates nearby excitations, and additional local operators can move them.

When the excitations meet and annihilate, every local syndrome check can return to its initial value. The full trajectory may nevertheless wind around a hole or handle of the encoded surface.

Such winding acts nontrivially on the logical state.

The central limitation is that topology makes logical faults extended in space or time, while leaving such histories possible. Locality supplies code distance, defined as the minimum weight of a physical operator that implements a nontrivial logical operation.

An energy gap provides a Boltzmann suppression of excitation creation at finite temperature. It does not, by itself, provide an energy barrier that grows with system size and inhibits excitation motion.

The corresponding spacetime description is particularly useful. A data error produces a spatial segment in an error history.

Persistence of that error extends the segment through time. An incorrect syndrome measurement produces a time-like segment.

A decoder is a computational procedure that infers which collection of spacetime segments most plausibly produced the observed syndrome endpoints. A logical failure occurs when the inferred error history and the actual error history differ by a topologically nontrivial loop.

A trivial final syndrome does not establish that no error occurred. It is compatible both with a short, contractible path and with a path that winds around the system.

Logical error chain in the toric code

Consider the toric code on a square lattice with periodic boundary conditions in both spatial directions. Place one physical qubit on each edge. Define the stabilizer operators

$$A_v = \prod_{e \ni v} X_e, \qquad B_p = \prod_{e \in \partial p} Z_e,$$

where v labels a vertex, p labels a plaquette, e labels an edge, and X_e and Z_e are Pauli operators acting on the qubit at edge e . The ground space is the simultaneous $+1$ eigenspace of every vertex operator A_v and plaquette operator B_p . A measurement result of -1 is a syndrome defect, interpreted in the ideal model as an anyonic excitation [Ro30].

Take a periodic row containing four horizontal edges, e_1, e_2, e_3, e_4 . Apply consecutive Z errors defined by

$$E_j = Z_{e_1} Z_{e_2} \cdots Z_{e_j}, \qquad j = 1, 2, 3, 4.$$

A single operator Z_e anticommutes with the two vertex checks at the endpoints of e . Therefore, E_1 creates two violated vertex checks.

Multiplication by Z_{e_2} restores the shared vertex check and violates the next vertex check. This operation moves one endpoint of the error chain. The same process occurs when Z_{e_3} is applied.

After application of Z_{e_4} , the moving endpoint returns to the first endpoint, and both syndrome defects disappear.

Step	Physical action	Visible syndrome	Topological result
0	identity	none	identity
1	Z_{e_1}	two endpoints	pair created
2	$Z_{e_1} Z_{e_2}$	two endpoints	one endpoint moved
3	$Z_{e_1} Z_{e_2} Z_{e_3}$	two endpoints	one endpoint moved again
4	$\bar{Z} = \prod_{i=1}^4 Z_{e_i}$	none	noncontractible logical loop

The operator in the last row is not the identity. To establish this, choose a noncontractible logical X loop $\bar{X}$ that crosses the row once. At the single crossing, $XZ = -ZX$, whereas all other factors commute. Consequently,

$$\bar{X}\bar{Z} = -\bar{Z}\bar{X}.$$

An operator that anticommutes with a logical observable changes the encoded state. Although the final syndrome is empty, the logical qubit has undergone the logical operation $\bar{Z}$. This sequence is the basic local-error-chain mechanism [R030]; [R141].

For an $L \times L$ periodic lattice, the shortest noncontractible loop contains L edges. The code distance is therefore $d = L$, and ideal decoding can guarantee correction of at most $\lfloor (d - 1)/2 \rfloor$ adversarial Pauli errors.

This correction bound is a worst-case combinatorial guarantee. A threshold, discussed later in this chapter, is a different statement: it concerns the asymptotic behavior of a probability distribution over faults under a specified decoder.

The edge spins in this construction may represent physical qubits, cluster-encoded pseudospins, or digitally emulated variables. The algebra of the error chain is unchanged among these implementations.

Their physical protection is not equivalent. A simulator can implement the code Hamiltonian while the underlying hardware remains protected only by conventional active error correction.

An analog material provides passive topological protection only if its own low-energy spectrum, excitations, and response to local perturbations realize the required structure.

Energy gaps, excitation creation, and excitation motion

A Hamiltonian can suppress some errors without syndrome measurement and feedback. This mechanism is called passive suppression because it arises from the device spectrum rather than from a decoder or recovery pulse.

Two mechanisms are relevant.

First, a many-body energy gap makes the creation of excitations energetically costly. Let Δ_a denote the energy required to create one well-separated anyon, measured in joules (J), and let T denote absolute temperature, measured in kelvin (K). In the ideal dilute model, creation of an anyon pair costs approximately $2\Delta_a$. Its thermal weight contains the factor

$$\exp\left(-\frac{2\Delta_a}{k_B T}\right),$$

where k_B is Boltzmann's constant in J/K. The exponent is dimensionless because J divided by (J/K)K is one.

Excitation creation is therefore suppressed when $k_B T \ll \Delta_a$ [R169]. This suppression applies to creation, not necessarily to subsequent motion.

Second, nonlocal encoding makes a weak local perturbation ineffective at distinguishing logical sectors. In suitable gapped, topologically ordered Hamiltonians, sufficiently weak local perturbations preserve the phase, and the finite-size splitting between logical states can scale schematically as

$$\delta E(L) \sim E_0 e^{-L/\xi}.$$

Here δE and E_0 are energies, L is a linear system size, and ξ is a correlation length expressed in the same length unit. [Theory] This stability requires locality, a spectral gap, the relevant topological-order assumptions, and perturbations weak enough that the system remains in the same phase [R142]. It is not a general theorem for an arbitrary fabricated, driven, open array.

Passive suppression neither infers an error chain nor applies a recovery operation. Once an anyon exists, moving it may require little or no additional energy.

For example, the toric-code string described above has no more than two endpoints at any stage. Its maximum energy above the ground space therefore remains constant as L increases. Under the assumptions of the Bravyi–Terhal no-go theorem, two-dimensional local stabilizer Hamiltonians consequently lack a macroscopic energy barrier [R168].

Thermal analyses of the ideal two-dimensional Kitaev model similarly show that topological order of the ground state alone does not make the encoded information a self-correcting memory [R143]. [Theory] The theorem applies specifically to two-dimensional stabilizer-code Hamiltonians with local interactions and does not establish the same conclusion for every non-Abelian phase.

The relevant physical criterion is therefore the energy barrier along an error path, which must be calculated rather than inferred from the existence of a gap.

Syndrome extraction and decoding

The four-error loop returns every stabilizer check to its initial value. Detecting and correcting the error before the loop closes requires syndrome measurement, inference of an error history, and application or tracking of a recovery operation.

A decoder is the algorithm or circuit that performs this inference. The hardware does not intrinsically determine which syndrome endpoints originated from the same error process; the decoder must select a candidate history.

Active error correction measures local checks, repeats measurements when those measurements are noisy, decodes the resulting spacetime syndrome, and either applies or classically tracks a recovery. Its operational requirements are:

- extraction of sufficient syndrome information without uncontrolled backaction;
- assignment of time stamps to repeated measurement rounds;
- use of a decoder adequately matched to the actual noise process;
- completion of recovery before accumulated errors exceed the decoder’s capacity.

Active correction can remove thermally created anyon pairs before their worldlines become topologically nontrivial. Comparison of successive measurement rounds can also distinguish a transient detector fault from a persistent data fault.

These operations are not passive and require explicit measurement and computation.

Without a decoder, the four-error loop illustrates the failure mechanism directly: after the loop closes, the syndrome is trivial even though the logical observable has changed sign.

Conditional meaning of a fault-tolerance threshold

A fault-tolerance threshold is a conditional property of a specified model. It marks a transition between a regime in which increasing code size improves logical performance and one in which increasing code size can worsen it.

More precisely, for a family of codes with increasing distance and for a specified fault model, syndrome circuit, decoder, and operation schedule, there may be a critical physical-fault parameter p_{th} such that, when $p < p_{\text{th}}$, increasing the distance reduces the logical error per operation [R141]. This statement does not imply that every reported component “fidelity” can be compared with a universal percentage.

In the canonical independent stochastic model studied by Dennis and collaborators, the toric/surface-code transition is about 10.9% when syndrome measurements are perfect. In a phenomenological model with repeated noisy measurements and equal data and measurement error rates, the corresponding value is about 2.9% [R141]. [Theory/Numerics] These values characterize those specific models; they are not general acceptable-error specifications for a defect device. Leakage, correlations, geometry, boundaries, decoder choice, and circuit design can all change the threshold.

A threshold statement must specify:

- what constitutes a circuit location and what event constitutes a fault;
- whether faults are independent, local stochastic, biased, leaking, or correlated;
- whether state preparation and measurement are noisy;
- whether qubits can disappear or change charge state;
- the decoder and its latency;
- the logical task and error metric.

An energy-scale condition such as $\Delta_{\text{topo}} > k_{\text{B}}T$ is a gap margin, not an error-correction threshold. Similarly, the adiabatic condition for a braid is not a threshold theorem. These terms describe distinct physical and computational requirements.

Exact error-free operation

Error-free operation means that the logical channel is exactly equal to the intended logical channel. It is a mathematical identity rather than a qualitative description of a small error probability.

An ideal commuting-projector model at zero temperature, initialized exactly in its ground space and acted on by exact operators, can have zero error by assumption. A finite experimental run instead has a nonzero logical-error probability, even when that probability is very small.

Operation below threshold means that errors can be suppressed by increasing resources; it does not mean zero error. Passive protection means reduced sensitivity to selected local faults; it also does not mean zero error.

The four concepts can therefore be distinguished as follows:

- An energy gap imposes an energetic cost on excitation creation. It need not impose a cost on excitation motion.
- A decoder infers an error history from syndrome information and attempts to reverse or compensate for it.
- A threshold is a transition defined for a specified noise model, circuit, and decoder.
- Error-free operation means exact realization of the intended logical channel.

Topological protection is therefore not indefinite preservation. The remaining sections describe mechanisms by which a local fault can still produce a logical error.

Thermal creation and subsequent diffusion

Thermal errors involve two distinct stages. Excitation creation incurs an energy cost, whereas subsequent diffusion may not.

Let a be the lattice spacing in metres, and suppose that the dilute equilibrium anyon density per site scales as $n_a \sim e^{-\Delta_a/(k_B T)}$. In two dimensions, the characteristic separation between anyons then scales as

$$\ell_a \sim \frac{a}{\sqrt{n_a}} \sim a e^{\Delta_a/(2k_B T)}.$$

Both ℓ_a and a have units of metres, and the exponential is dimensionless. [Theory] This estimate neglects species degeneracy, interactions, boundaries, nonequilibrium baths, and kinetic bottlenecks. It nevertheless demonstrates that system size and mean thermal separation are distinct length scales [R169].

For a hypothetical anyon energy $\Delta_a/h = 10$ GHz at $T = 0.10$ K, use $k_B T/h \approx 2.08$ GHz. It follows that $\Delta_a/(k_B T) \approx 4.8$. The ideal single-anyon Boltzmann factor is therefore about 8×10^{-3} , whereas the pair factor is about 7×10^{-5} .

This calculation is illustrative rather than a prediction for a defect array. A quantitative forecast also requires the attempt rate, the number of possible creation sites, the bath spectrum, and the actual many-body gap. For example, ten thousand opportunities can make a small per-site thermal weight operationally significant.

After creation, an anyon can undergo a random walk, move under disorder gradients, or be displaced by control crosstalk. A boundary can absorb one member of a pair and leave a topological charge whose preceding history is difficult to reconstruct.

For non-Abelian anyons, diffusion has consequences beyond classical displacement: unrecorded exchanges and fusion events act on the fusion space, which is the Hilbert space associated with the possible collective fusion channels [R015]. Active syndrome extraction and decoding must operate faster than this kinetic process, rather than addressing only the initial Boltzmann factor.

Consequently, the condition that the temperature is below the gap addresses only excitation creation. The Boltzmann factor does not localize excitations after they have formed.

Competing effects of increasing system size

A physical code patch is finite. Its behavior depends on at least three competing lengths: its linear size L , the correlation length ξ , and the characteristic separation or travel length of unwanted quasiparticles. Increasing L/ξ can exponentially suppress virtual tunnelling between boundaries or around a handle.

The corresponding coherent phase rate is $\delta E/\hbar$, because energy divided by $\hbar$ has units s^{-1} . However, increasing area also creates more possible sites for thermal-pair production and introduces more hardware components that can fail. Increasing size therefore improves one protection mechanism while degrading another.

Disorder cannot be characterized by a single scalar parameter. Weak, bounded, static local perturbations may deform a stable topological phase without destroying it [R142].

[Theory] Fabrication disorder can nevertheless produce spatial variations in local gaps, couplings, cluster projectors, and measurement responses. A rare region with a small local gap can become a preferred site for thermal excitation.

A sign error or a missing interaction can alter the effective Hamiltonian rather than merely perturb it. Sufficiently strong disorder can close the mobility gap or spectral gap and drive the system out of the target phase.

Localization of unwanted quasiparticles can reduce diffusion and improve memory performance in some models [R169]. [Theory/Numerics] This benefit is not unconditional.

The same localization can obstruct intended braids, trap poisoning quasiparticles near computational anyons, broaden spectroscopic features, and invalidate a decoder based on homogeneous quasiparticle motion. A protection claim must therefore specify the disorder distribution, including its tails, and model both equilibrium behavior and dynamics.

Quasiparticle poisoning and leakage

Quasiparticle poisoning and leakage can both take the device outside its intended computational description, but they are physically distinct processes.

Quasiparticle poisoning occurs when an uncontrolled topological charge enters, leaves, or changes the computational region. The resulting fault is an additional anyon rather than a small control-pulse error.

Thermal creation is one source of poisoning. Other sources include injection from an edge, a reservoir, a driven hotspot, or a nonequilibrium defect. In a non-Abelian device, a stray anyon can fuse with a computational anyon and change its fusion channel.

Subsequent nominal braids then implement an incorrect unitary [R015]. Local charge measurements may not reveal this error until information has already escaped from the intended computational space.

Leakage occurs when a physical constituent leaves the two-level subspace used by the code. The affected site can no longer be modeled as the qubit assumed by the decoder.

For a defect cluster, let P be the projector onto the intended low-energy doublet, and let $Q = I - P$ project onto all other cluster states. An ordinary encoded error has support within P .

A leakage event transfers population into Q . The projected Pauli model then ceases to describe the site, and coupling to other sites can propagate errors with amplitudes absent from the decoder's noise model.

Topological distance does not suppress an individual leakage event solely because the intended code is topological. Active leakage-reduction procedures instead remove, reset, swap, or explicitly detect leaked population at regular intervals.

Surface-code studies show that untreated leakage can propagate through syndrome-extraction circuits and requires dedicated mitigation [R220]. [Theory/Numerics] This result concerns an active code architecture and does not prove that the same mitigation protocol applies to an interacting defect phase.

In an analog cluster array, resetting one cluster can itself abruptly change local couplings and create anyons.

Charge-state switching can have several simultaneous effects. It can constitute leakage from the cluster doublet, remove or alter a coupling in the many-body Hamiltonian, and cause optical-readout failure. Representing the entire event as one Pauli error can therefore underestimate its effect.

Initialization, control, and readout errors

Topological protection applies after the system has entered the intended phase and logical sector. Cooling below the gap does not guarantee preparation of a known ground state. A finite system can retain trapped quasiparticles, freeze into an incorrect flux sector, or traverse a small gap nonadiabatically. Initialization therefore requires a verified preparation procedure, such as cooling, pumping, syndrome measurement, boundary-charge removal, or a combination of these methods, together with a measurement of total topological charge.

Control operations must satisfy a finite operating window. A pulse with spectral weight near a leakage transition can populate the Q subspace.

A local control field sufficiently strong to close the gap along a path can nucleate excitations. An intended adiabatic braid must proceed slowly relative to the relevant inverse-gap timescales, but a slower operation increases exposure to poisoning, drift, and dephasing.

The approximate timescale $\hbar/\Delta$ has units of seconds because $(Js)/J = s$. It provides only a necessary scale, not a complete adiabatic bound. A complete bound also depends on transition matrix elements and the minimum gap encountered along the path [R015]. Topology can make the ideal braid unitary insensitive to smooth geometric deformations, but arbitrary pulse errors, unintended braids, and dynamical phases remain harmful.

Readout has two distinct levels: local syndrome readout determines where excitations are present.

Logical readout determines a nonlocal loop value, boundary parity, or fusion outcome. A local detector can exhibit assignment error, measurement backaction, crosstalk, and dead time.

Repeating a noisy stabilizer measurement converts decoding into a three-dimensional spacetime inference problem; repetition does not make the measurement exact [R141]. The interpretation of a readout result also requires a confidence model conditioned on leakage and charge state.

Destructive optical readout may be sufficient at the end of a computation while remaining unsuitable for repeated error correction.

A useful experimental report should therefore provide separate confusion matrices for local checks and logical readout, statistics conditioned on leakage, measurements of backaction, the correction-cycle duration, and decoder latency. Reporting only the highest single-defect readout fidelity does not characterize the full many-body protocol.

Spatially and temporally correlated faults

Threshold estimates often rely strongly on locality assumptions. One formal local-stochastic condition requires that the probability that every location in a set S is faulty be bounded by a quantity that decreases exponentially with $|S|$. Correlations may be present, but high-weight fault events must become rapidly less probable.

A global magnetic transient, laser-intensity excursion, strain wave, shared microwave phase slip, or calibration error can affect many clusters simultaneously. If one physical event creates a length- L error string with probability proportional to a single-event rate, rather than to that rate raised to the L th power, increasing code distance may provide little improvement.

Long temporal correlations can likewise cause repeated syndrome rounds to agree because the same persistent fault affects each round. Studies of correlated environments show that fault-tolerance behavior depends on the spatial and temporal decay of correlations, rather than on a single per-qubit error probability [R221].

[Theory] Correlations do not necessarily eliminate fault tolerance: short-range and sufficiently weak correlations can remain consistent with threshold results under appropriate assumptions.

The required analysis is to measure or bound the correlation function, map the correlated noise through the cluster projector, and incorporate the resulting model into decoding and logical-level simulations. Reducing a covariance matrix to one averaged “error rate” removes spatial and temporal information that is essential to evaluating topological protection.

Current experimental capabilities

For a defect-cluster proposal, where a defect cluster is a local collection of engineered defects used to encode a qubit, the protection stack must be demonstrated sequentially. The protection stack is the set of distinct physical and computational mechanisms required to obtain and maintain a reliable logical qubit. Evidence for one layer does not by itself establish the properties associated with later layers.

Layer	Required evidence	What it does not establish
Cluster encoding	An isolated doublet, meaning two spectrally separated states used as the encoded	Topological order

Layer	Required evidence	What it does not establish
	subspace; a measured leakage gap, meaning the energy separation from states outside that subspace; and projected controls that act within the encoded subspace	
Many-body phase	A many-body gap, correlation measurements, and evidence for topological sectors or anyons, where anyons are quasiparticle excitations with topologically nontrivial exchange or fusion properties	Useful logical lifetime
Passive suppression	Scaling of error suppression against local perturbations, temperature, and system size	Active correction or zero error
Syndrome system	Repeated check or fusion measurements, where a syndrome is the set of measurement outcomes used to diagnose errors, with measurement backaction characterized	Operation below threshold
Active logical memory	Logical-error scaling versus code distance and measurement rounds under experimentally measured noise	Universal protected computation
Braiding/control	The correct process on the fusion space, meaning the state space associated with the possible fusion outcomes of anyons, together with bounds on poisoning and leakage	Error-free gates

[Proposal] A fabricated defect array will contain placement disorder, missing or incorrect charge states, cluster leakage, local drives, and imperfect optical or microwave readout. These effects are absent from the ideal Hamiltonian unless they are included explicitly.

Exact diagonalization, which numerically computes the eigenvalues and eigenstates of a finite Hamiltonian, can therefore establish a property of a clean model patch but cannot establish passive protection under laboratory conditions. Similarly, successful decoding of synthetically injected, independent Pauli errors provides a benchmark for the decoder. It does not demonstrate that the physical noise is independent or describable by Pauli operators.

The decisive test is a scaling experiment. Such an experiment varies a physically meaningful code distance or quasiparticle separation while keeping the relevant physical conditions fixed, characterizes the microscopic noise, and measures the resulting change in logical-error rate.

Repeating the measurement as a function of temperature distinguishes thermally activated errors from control-limited error floors. Repeating it as a function of syndrome cadence, the rate at which syndrome measurements are performed, distinguishes passive suppression from the additional benefit of active correction. If increasing the sample size introduces additional poisoning sources, the scaling data will reveal that effect.

A scaling result remains scientifically informative even when the logical-error rate worsens with increasing system size or distance.

Common analytical errors

- Exponentially small ground-state splitting does not imply a self-correcting memory. Small virtual splitting suppresses coherent mixing between topological sectors, not the distinct processes of thermal excitation creation and subsequent diffusion.
- A below-gap temperature does not imply zero anyons. Boltzmann suppression remains finite, the sample contains many possible excitation sites, and nonequilibrium injection can dominate the anyon population.
- The statement that the physical error rate is below the surface-code threshold is incomplete unless the relevant error rate, code, circuit, leakage model, correlations, decoder, and measurement model are specified. A threshold is the error-rate boundary below which logical error can decrease as computational resources increase, subject to a particular set of assumptions.
- The inference that topological braiding eliminates all control errors is incorrect. Smooth deformations of a braiding path can be harmless when the evolution remains within the same gapped sector. Protection does not apply when the process leaves that sector, braids the wrong topological charge, or poisons the fusion space.
- The inference that weak disorder is beneficial merely because it localizes anyons is incomplete. Disorder may improve one kinetic channel while degrading energy gaps, controls, readout, or the intended motion of quasiparticles.
- An empty syndrome does not imply the absence of a logical error. A noncontractible loop can produce an empty syndrome while implementing a logical operation.
- Repeated measurement is not passive protection. It constitutes active error correction even when software records or tracks the recovery operation instead of applying it physically.
- Operation below threshold does not imply error-free operation. It means that logical error can decrease with increasing resources under the stated assumptions. Every finite implementation retains a residual probability of failure.

Verification of topological-protection criteria

- **Four local Z errors on a noncontractible row can leave no syndrome while flipping the logical qubit.**

The endpoints of the error chain annihilate when the chain closes. The resulting closed chain winds nontrivially and anticommutes with a crossing logical X loop. Consequently, $\bar{X}\bar{Z} = -\bar{Z}\bar{X}$.

- **Treating the gap as a barrier to quasiparticle motion fails because the gap controls creation rather than diffusion.**

The gap sets the energy cost of creating excitations. Once two endpoints exist, moving one endpoint around the lattice need not require additional energy. The two-dimensional stabilizer no-go result follows from the absence of this additional energy barrier.

- **Passive protection and active correction are distinct mechanisms.**

Passive protection follows from the Hamiltonian and encoding without syndrome feedback. Active correction measures syndromes, decodes the measurement outcomes, and either applies recovery operations or tracks the inferred errors.

- **A published threshold is not automatically a hardware specification.**

The threshold applies to a particular code, noise model, syndrome circuit, decoder, and definition of an operation. Leakage and correlated noise can alter or invalidate that threshold.

- **Poisoning and leakage describe different departures from the intended computational model.**

Poisoning changes the uncontrolled topological charge in the computational region. Leakage moves a constituent outside its intended local encoded subspace. A single physical event can produce both effects.

- **Calling a finite device error-free merely because it operates below threshold is incorrect.**

Below-threshold operation describes favorable logical-error scaling under stated assumptions. A finite noisy run still has a nonzero logical-error probability. Error-free operation requires the logical channel to be exactly the intended channel.

A logical fault is a topologically nontrivial spacetime history. An energy gap suppresses excitation creation but does not automatically suppress excitation diffusion.

Finite system size produces a tradeoff between exponentially small virtual splitting and an extensive number of possible fault locations. Disorder can preserve the many-body phase while creating local regions with dangerously weak protection.

Poisoning and leakage take the system outside its intended computational description. Initialization, control, and readout therefore require separate error models.

Correlations can prevent logical errors from being suppressed by increasing code distance. Passive suppression, active correction, threshold operation, and error-free operation are four distinct claims.

Protection does not imply an infinite lifetime.

Sources

- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* 303, 2–30 (2003). DOI: [10.1016/S0003-4916\(02\)00018-0](https://doi.org/10.1016/S0003-4916(02)00018-0). [arXiv:quant-ph/9707021](https://arxiv.org/abs/quant-ph/9707021).
- [R141] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, “Topological quantum memory,” *Journal of Mathematical Physics* 43, 4452–4505 (2002). DOI: [10.1063/1.1499754](https://doi.org/10.1063/1.1499754). [arXiv:quant-ph/0110143](https://arxiv.org/abs/quant-ph/0110143).
- [R168] S. Bravyi and B. Terhal, “A no-go theorem for a two-dimensional self-correcting quantum memory based on stabilizer codes,” *New Journal of Physics* 11, 043029 (2009). DOI: [10.1088/1367-2630/11/4/043029](https://doi.org/10.1088/1367-2630/11/4/043029). [arXiv:0810.1983](https://arxiv.org/abs/0810.1983).
- [R143] R. Alicki, M. Fannes, and M. Horodecki, “On thermalization in Kitaev’s 2D model,” *Journal of Physics A: Mathematical and Theoretical* 42, 065303 (2009). DOI: [10.1088/1751-8113/42/6/065303](https://doi.org/10.1088/1751-8113/42/6/065303). [arXiv:0810.4584](https://arxiv.org/abs/0810.4584).

- [R169] B. J. Brown, D. Loss, J. K. Pachos, C. N. Self, and J. R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* 88, 045005 (2016). DOI: [10.1103/RevModPhys.88.045005](https://doi.org/10.1103/RevModPhys.88.045005). [arXiv:1411.6643](https://arxiv.org/abs/1411.6643).
 - [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* 51, 093512 (2010). DOI: [10.1063/1.3490195](https://doi.org/10.1063/1.3490195). [arXiv:1001.0344](https://arxiv.org/abs/1001.0344).
 - [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083). [arXiv:0707.1889](https://arxiv.org/abs/0707.1889).
 - [R220] A. G. Fowler, “Coping with qubit leakage in topological codes,” *Physical Review A* 88, 042308 (2013). DOI: [10.1103/PhysRevA.88.042308](https://doi.org/10.1103/PhysRevA.88.042308). [arXiv:1308.6642](https://arxiv.org/abs/1308.6642).
 - [R221] E. Novais, E. R. Mucciolo, and H. U. Baranger, “Resilient quantum computation in correlated environments: A quantum phase transition perspective,” *Physical Review Letters* 98, 040501 (2007). DOI: [10.1103/PhysRevLett.98.040501](https://doi.org/10.1103/PhysRevLett.98.040501). [arXiv:quant-ph/0607155](https://arxiv.org/abs/quant-ph/0607155).
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Chapter 32 — Distinguishing spectroscopic peaks from braiding operations

Two other research programs pursue the same objective as defect-engineered topological qubits: they seek localized excitations whose exchange changes the shared quantum state according to a reproducible rule. Moving one excitation around another in this way is a braid.

The first program uses a thin semiconductor wire coupled to a conventional superconductor and exposed to a magnetic field. The objective is to produce localized electronic modes at the two ends of the wire that do not behave as ordinary electrons. If these end modes have the properties predicted by the relevant theory, their braiding statistics are those of Ising anyons. An anyon is a quasiparticle or defect in two spatial dimensions whose exchange can produce a phase or, in the non-Abelian case, a noncommuting unitary transformation within a degenerate state space.

The second program uses a two-dimensional electron system at very low temperature and in a magnetic field strong enough to quantize orbital motion into Landau levels. A Landau level is a discrete orbital-energy level formed by charged particles moving in a perpendicular magnetic field. At a particular filling factor, the aim is to realize an electron liquid whose vortex-like excitations braid as Fibonacci anyons.

Neither platform consists of a crystal containing an engineered array of defects, which is the platform considered in this book.

These programs have received more experimental study than defect-based proposals in several areas. Their theories and experiments therefore supply three checks: the predicted braid operations, what measurements actually establish, and separation of those observations from the interpretations built on them.

A zero-bias conductance peak does not demonstrate a braid. A resistance plateau does not establish the existence of a Fibonacci anyon. Experimental observations must be distinguished from the theoretical interpretations they may support.

Distinguishing allowed phases, identified excitations, and controlled operations

Evidence for a topological platform separates into three questions. Has the model entered an allowed phase? Have the relevant excitations been identified? Can the device perform a controlled operation on an encoded state?

- **An allowed phase.** A Hamiltonian, which is the operator governing the system's energy and dynamics, contains a topological region in an idealized phase diagram.
- **An identified excitation.** Several independent measurements exclude ordinary states and determine the excitation's electric charge, fusion rules, and exchange statistics. Fusion rules specify the possible total topological charges obtained when excitations are combined, while exchange statistics specify the quantum transformation produced when they are exchanged.

- **A controlled computational operation.** Initialization, a braid or parity-measurement sequence, and readout implement a reproducible unitary transformation on an encoded state space. Parity here denotes whether the relevant fermion occupation number is even or odd.

This three-level distinction supplies the standard used in the comparisons that follow.

Majorana nanowires have substantial theoretical support at the first level and partial evidence at the second. Fractional quantum Hall fluids have conclusive demonstrations of Abelian fractional charge and statistics at some filling factors. Abelian statistics produce only a scalar phase under exchange, whereas non-Abelian statistics act by matrices on a multidimensional fusion space.

These results do not identify Fibonacci quasiparticles at filling factor $12/5$. Evidence for one excitation type or filling factor cannot be substituted for evidence concerning another.

The relevant excitations also have physically distinct origins.

A mode localized at the end of a one-dimensional superconducting segment is an extrinsic defect mode. An extrinsic defect is introduced by fabrication or by an externally imposed boundary; in this case, the segment ends are deliberately created.

A vortex in an appropriate two-dimensional topological superconductor can behave as an Ising anyon. A Majorana operator implemented by digital control is an emulation rather than an emergent material excitation.

A quasihole in an intrinsic quantum Hall fluid is an emergent excitation of the correlated electron liquid. “Intrinsic” means that the topological order is a property of the many-body phase itself rather than of a fabricated boundary or programmed operation. These different objects can obey the same braid algebra without having the same experimental evidence or the same degree of passive protection.

Computational capability of Ising braiding

The comparison starts with ideal anyon theories. The matrices below describe what the theories permit; they do not assess whether a device has realized those operations.

The Ising theory has three topological charges: 1, denoting the vacuum sector; ψ , denoting a fermion; and σ , denoting the non-Abelian Ising anyon. Two σ anyons obey the fusion rule

$$\sigma \times \sigma = 1 + \psi.$$

The plus sign indicates that the pair can have either total charge 1 or total charge ψ .

One qubit can be encoded in four σ anyons constrained to have total charge 1. Define $|0\rangle$ as the state in which the first pair fuses to 1, and define $|1\rangle$ as the state in which that pair fuses to ψ . Let B_1 represent a counterclockwise exchange of the first two anyons, and let B_2 represent an exchange of the middle two. In a common phase convention, and after removing a common global phase that has no observable effect on the encoded state,

$$B_1 = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad B_2 = \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}.$$

Multiplying these braid matrices gives

$$B_1 B_2 B_1 = \frac{1+i}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = e^{i\pi/4} H,$$

where

$$H = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

is the Hadamard gate. Thus, up to global phases, Ising braids provide the phase gate $S = \text{diag}(1, i)$ and the Hadamard gate H . These operations belong to the Clifford gate set, the set of quantum gates that maps Pauli operators to Pauli operators under conjugation [Ro15].

A required non-Clifford gate can be written as

$$T = \text{diag}(1, e^{i\pi/4}).$$

No braid of Ising anyons can approximate T arbitrarily accurately. Ising braiding generates only a finite projective Clifford image, rather than a dense subset of all one-qubit rotations. A dense gate set is one whose finite sequences can approximate arbitrary target unitaries to any desired accuracy. Universal computation with Ising anyons therefore requires an additional non-topological resource, commonly a specially prepared magic state, a non-topological phase operation, or an appropriate measurement protocol [Ro15]. A magic state is an ancillary quantum state that enables a non-Clifford operation when combined with Clifford gates and measurements.

Non-Abelian statistics therefore do not by themselves imply computational universality. Ising anyons are non-Abelian, but they do not provide the gate set available from ideal Fibonacci braiding.

Computational capability of Fibonacci braiding

The Fibonacci theory has two topological charges, 1 and τ , with fusion rule

$$\tau \times \tau = 1 + \tau.$$

One qubit can be encoded in three τ anyons constrained to have total charge τ . Define $|0\rangle$ and $|1\rangle$ according to whether the first pair fuses to 1 or to τ . Let

$$\varphi = \frac{1 + \sqrt{5}}{2}$$

denote the golden ratio. A standard choice of the basis-change matrix F and exchange matrix R is

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad R = \begin{pmatrix} e^{-4\pi i/5} & 0 \\ 0 & e^{3\pi i/5} \end{pmatrix}.$$

The F matrix changes the fusion basis, whereas R gives the exchange phase when the pair has definite fusion charge. Thus the first exchange is $B_1 = R$. For the second and third anyons, transform to the appropriate fusion basis, perform the exchange, and transform back:

$$B_2 = F^{-1} R F = F R F,$$

because this convention has $F^{-1} = F$. The relative phase between the two eigenvalues of B_1 is

$$\frac{e^{3\pi i/5}}{e^{-4\pi i/5}} = e^{7\pi i/5} = e^{-3\pi i/5},$$

which is not a Clifford Z -rotation. In addition, B_1 and B_2 produce rotations about different axes of the encoded Bloch sphere.

Braid words are ordered products of elementary exchange generators. In the Fibonacci theory they form a dense representation on the computational space, so longer words can approximate any one-qubit unitary; suitable words in larger encodings also produce entangling gates [Ro15]. An entangling gate creates correlations between qubits. Phase conventions can alter signs and common phases without altering this gate-set conclusion.

The comparison is:

Anyon model	Elementary fusion	Braiding supplies	Braiding alone universal?	Missing engineering
Ising/Majorana	$\sigma \times \sigma = 1 + \psi$	Clifford gates such as H and S	No; no dense gate set	non-Clifford resource, parity control, leakage control
Fibonacci	$\tau \times \tau = 1 + \tau$	dense single-qubit rotations and entangling braids	Yes, ideally	compilation, initialization, fusion readout, thermal and leakage control

Universality is a mathematical property of a braid representation. It does not demonstrate that the corresponding anyons exist in a physical sample.

Majorana endpoint modes in a semiconductor wire

Consider a one-dimensional semiconductor with strong spin–orbit coupling, coupled to a conventional s -wave superconductor and placed in a magnetic field. Spin–orbit coupling links a particle’s momentum to its spin. An s -wave superconductor has an isotropic pairing amplitude in momentum space. Cooper pairing, the pairing of electrons into superconducting correlations, enters the semiconductor through the interface.

The wire is described as proximitized because its superconducting pairing is induced by an adjacent superconductor rather than originating within the semiconductor itself.

A minimal Bogoliubov–de Gennes Hamiltonian is

$$H(p) = \left(\frac{p^2}{2m^*} - \mu \right) \tau_z + \alpha p \sigma_y \tau_z + V_Z \sigma_x + \Delta \tau_x.$$

A Bogoliubov–de Gennes Hamiltonian describes superconducting quasiparticles in a particle–hole basis. Here p is momentum in kg m s^{-1} ; m^* is the electron effective mass in kg; μ is chemical potential in joules or electronvolts; α is the spin–orbit coefficient in energy-times-length; V_Z is Zeeman energy; and Δ is the induced superconducting pairing energy. Zeeman energy is the spin-energy splitting produced by the magnetic field. The Pauli matrices σ_a act on spin, while τ_a act on particle–hole space. Every term has units of energy. For example, αp is energy-times-length multiplied by inverse length.

For the ideal, uniform, single-band model, the bulk excitation gap closes and reopens at

$$V_Z^2 = \Delta^2 + \mu^2.$$

A bulk gap is the minimum energy required to create an excitation in the extended system. The region

$$V_Z > \sqrt{\Delta^2 + \mu^2}$$

is topological and supports a localized mode near each end of a sufficiently long segment [Theory] [R222]. In this model, spin–orbit coupling, Zeeman splitting, and conventional pairing together produce an effective spinless p -wave channel. A p -wave pairing channel has a pairing amplitude that changes sign under reversal of relative momentum.

The endpoint mode is represented by a Majorana operator γ_j , defined by

$$\gamma_j^\dagger = \gamma_j, \quad \{\gamma_i, \gamma_j\} = 2\delta_{ij},$$

where $\delta_{ij} = 1$ for $i = j$ and zero otherwise. The first relation states that a Majorana operator is Hermitian, and the second gives its fermionic anticommutation algebra. Two Majorana operators define one ordinary fermion:

$$f = \frac{\gamma_1 + i\gamma_2}{2}, \quad i\gamma_1\gamma_2 = 2f^\dagger f - 1.$$

The occupation number $f^\dagger f$ is encoded nonlocally between the two ends. Four Majoranas with fixed total fermion parity provide a qubit. Subject to the chosen orientation convention, an ideal counterclockwise exchange is represented by

$$U_{ij} = \exp\left(\frac{\pi}{4}\gamma_i\gamma_j\right).$$

These are the Ising operations used in the preceding calculation.

If the end modes decay over a localization length ξ , meaning the characteristic distance over which their wavefunctions decrease, their overlap in a clean wire of length L is approximately proportional to $e^{-L/\xi}$, often multiplied by an oscillatory prefactor. Increasing L/ξ can therefore suppress coherent energy splitting caused by end-mode overlap.

This geometric suppression does not eliminate quasiparticle poisoning, in which an unwanted quasiparticle changes fermion parity. It also does not suppress a stray low-energy state near one end, a poorly transmitting tunnel barrier, or a control pulse that closes the excitation gap. Nonlocal encoding protects against a specified class of local perturbations, not against every experimental error mechanism.

Alternative platforms for localized endpoint modes

Semiconductor nanowires are not the only proposed platform. Two-dimensional semiconductor–superconductor heterostructures can form gate-defined networks. A heterostructure is an interface or layered structure composed of different materials. Magnetic-atom chains on superconductors can produce effective topological bands, and vortices or boundaries in candidate intrinsic topological superconductors can host zero-energy states [Proposal/Theory] [\[R223\]](#).

The distinction between intrinsic and engineered superconducting topology is important. In an intrinsic topological superconductor, the superconducting bulk itself carries the relevant topological invariant, a quantity that remains unchanged under continuous deformations that do not close the bulk gap.

In a hybrid device, individually conventional components collectively realize an effective topological phase under tuned conditions. A sharp zero-energy spectroscopic feature is compatible with a Majorana mode in either setting. However, ordinary Andreev bound states, disorder, and inhomogeneous potentials can reproduce important signatures [\[R223\]](#). An Andreev bound state is a localized subgap state produced by repeated electron–hole conversion at a superconducting interface.

Spectroscopy can identify a candidate state, but it does not by itself establish the state’s fusion rules or exchange statistics.

Fractional quantum Hall systems

A two-dimensional electron gas in a perpendicular magnetic field B forms Landau levels. Define the dimensionless filling factor

$$\nu = \frac{nh}{eB},$$

where n is the areal electron density in m^{-2} , h is Planck's constant in J s, and e is the elementary charge in C. The filling factor gives the number of occupied Landau levels, including fractional occupation when interactions are important. The units cancel because eB/h has units of inverse area.

Electron interactions can stabilize incompressible fractional quantum Hall liquids at rational values of ν . “Incompressible” means that changing the particle density requires a finite excitation energy. The $k = 3$ Read–Rezayi state occurs at partial filling $3/5$ in its simplest spin-polarized form and contains a Fibonacci topological sector [Theory] [R139]. A topological sector is a class of excitations characterized by a particular topological charge and associated fusion and braiding data.

With two lower Landau levels filled, partial filling $3/5$ corresponds to total filling $13/5$. Particle–hole conjugation within the active Landau level maps this state to partial filling $2/5$, and therefore to total filling $12/5$. Particle–hole conjugation interchanges occupied and unoccupied orbitals within the specified Landau level.

The conjugate state reverses the appropriate chiral data but retains Fibonacci fusion content. Chirality describes the propagation direction of edge modes and related handedness-dependent topological data. The phrase “Read–Rezayi at $12/5$ ” is therefore commonly used as shorthand for this candidate family rather than as an experimentally established identification.

The microscopic Coulomb interaction does not require the system to realize this candidate family. Landau-level mixing, finite quantum-well thickness, disorder, spin polarization, and charge-density order can favor competing phases. Landau-level mixing is the interaction-induced admixture of states from different Landau levels, while charge-density order is a spatial modulation of electron density.

Exact-diagonalization and density-matrix-renormalization studies have identified parameter regimes consistent with the Read–Rezayi state or its particle–hole conjugate [Numerics] [R225]; [R227]; [R228]. Exact diagonalization computes eigenstates of a finite many-body Hamiltonian directly. Density-matrix renormalization is a variational numerical method that approximates low-energy many-body states using restricted entanglement. Finite system sizes and model-dependent corrections leave room for competing Abelian or symmetry-broken interpretations.

Agreement with a trial-state overlap or an entanglement spectrum is substantial numerical evidence. A trial-state overlap measures the similarity between a computed state and a proposed wavefunction, while an entanglement spectrum characterizes the eigenvalue structure obtained by partitioning the state. Neither quantity constitutes an experimental quasiparticle braid.

Significance and identification requirements for Fibonacci quasiholes

A quasihole in the Fibonacci sector carries both electromagnetic charge and topological charge. Its electromagnetic charge affects transport and interferometric phase, while its topological charge determines the relevant fusion space. Exchanging quasiholes acts on that fusion space through the F and R matrices given above. A decisive experimental program must therefore connect several observables:

- a robust incompressible plateau and activated longitudinal resistance;
- fractional quasiparticle charge;
- edge or thermal measurements compatible with the candidate topological order;
- fusion-channel-dependent interference or projective fusion outcomes;
- noncommuting, order-dependent braid operations consistent with the predicted matrices.

Activated longitudinal resistance is a temperature-dependent resistance consistent with thermally excited charge carriers across an energy scale. Projective fusion outcomes are measurements that resolve the total topological charge produced when quasiparticles are combined.

No single item replaces the complete set of evidence. A plateau without fusion evidence, or an interferometer signal obtained before the phase is identified, remains evidence at the first level or only part of the second.

Current experimental evidence

The table summarizes peer-reviewed evidence through 2026. Each verdict states the strongest conclusion supported by the listed measurement or calculation.

Claim	Status	Platform	What was measured or established	Verdict
The ideal spin-orbit nanowire has a topological phase and endpoint Majoranas above the phase boundary	[Theory] established	semiconductor–superconductor	BdG invariant and gap closing/reopening in model Hamiltonians [R222]	candidate mechanism, not material identification
Hybrid and candidate intrinsic superconductors show zero-energy and related signatures	[Experiment] partial	nanowires, planar hybrids, chains, vortex systems	tunnelling and spatial spectroscopy; trivial states can imitate key features [R223]	compatible evidence, platform-dependent and not a braid
Single-shot interferometric parity readout is feasible in an InAs–Al device	[Experiment] demonstrated operation	gate-defined InAs–Al hybrid	up to 1 fF capacitance shift, signal-to-noise ratio 1 in 3.6 μ s, state dwell time above 1 ms near 2 T, and 1% assignment error [R224]	a strong readout primitive; the paper analyzes both trivial and topological origins
Majorana fusion rules and non-Abelian exchange have been conclusively	not established	Majorana candidates	no accepted controlled exchange establishing the	remains a required experiment

Claim	Status	Platform	What was measured or established	Verdict
demonstrated in these solid-state devices			Ising braid representation [R223]; [R224]	
An incompressible $12/5$ fractional quantum Hall state exists in high-quality GaAs	[Experiment] demonstrated	GaAs quantum well	activated transport; one 30-nm-well study reported a zero-tilt activation gap near 30 mK [R226]	establishes the plateau, not its topological order
The $12/5$ state is the particle-hole-conjugate $k = 3$ Read-Rezayi phase	[Numerics/Theory] supported candidate	second Landau level	finite-size spectra, entanglement data, and competing-order studies [R225]; [R227]; [R228]	plausible but not uniquely identified experimentally
Fractional statistics can be measured in an electronic quantum Hall interferometer	[Experiment] demonstrated for Abelian anyons	$\nu = 1/3$ Laughlin state	phase slips consistent with quasiparticles encircling localized fractional charge [R131]	validates an important method, but not Fibonacci statistics
Fibonacci fusion and braiding have been observed at $12/5$	not established	fractional quantum Hall	no conclusive quasiparticle-identification-and-braid sequence [R139]; [R225]; [R226]; [R227]; [R228]	theoretical candidate, not demonstrated Fibonacci hardware

The reported transport scale also shows why operating the quantum Hall state is demanding. An activation gap of 30 mK at $12/5$ corresponds to

$$k_B T = (8.617 \times 10^{-5} \text{ eV K}^{-1})(0.030 \text{ K}) \approx 2.6 \text{ }\mu\text{eV},$$

where k_B is Boltzmann’s constant [R226]. The product of energy per kelvin and temperature in kelvin has units of energy.

This value is an activation scale extracted from transport and is not necessarily equal to the intrinsic many-body gap of a disorder-free system. In either interpretation, electrical wiring, filtering, and local electrostatic control must avoid heating a fluid whose relevant energy scale is very small.

Required advances for each platform

Requirements for the wire platform

- Identification before scaling.** Disorder and smooth confinement can produce low-energy Andreev states that resemble endpoint Majoranas. Replicating an ambiguous device element only produces a larger system with the same ambiguity.
- A hard, uniform gap.** A hard gap is a superconducting density of states with strongly suppressed subgap spectral weight. Gate operation, magnetic field, semiconductor carrier density, and superconductivity must coexist across every segment and junction. The ideal inequality $V_Z > \sqrt{\Delta^2 + \mu^2}$ is not by itself an experimental acceptance criterion.

- **Networks and effective exchange.** Endpoints in one dimension cannot physically pass one another. Braiding therefore requires junction networks or measurement-only parity sequences. Both approaches require calibrated tunnel couplings and paths that preserve the excitation gap.
- **Parity lifetime and readout.** Quasiparticle poisoning changes the encoded parity sector. Fast, repeatable, quantum-nondemolition parity measurements are therefore integral to the qubit. A quantum-nondemolition measurement is designed to preserve the measured observable so that repeated measurements return the same value in the absence of physical transitions.
- **Resources beyond non-universal braids.** Even ideal Ising braiding requires a high-quality non-Clifford resource. This resource can dominate the computational overhead and limits claims that all gates are topologically protected.

Requirements for the quantum Hall fluid

- **Phase identification.** A $12/5$ plateau must be distinguished from competing ordered phases before its quasiparticles can be treated as computational resources.
- **A small operating energy scale.** The fragile state requires ultralow electron temperature, extremely clean material, and low-noise electrostatic gates [\[R226\]](#).
- **Controlled quasiparticles.** Gates must create, position, move, and fuse individual quasiholes without closing the gap or unintentionally reconstructing the edge. Edge reconstruction is a change in the spatial structure and mode content of the boundary caused by confinement and interactions.
- **Stable interferometry.** Coulomb charging, changes in device area, localized quasiparticles, and edge equilibration can modify measured phases. A scalable processor would require simultaneous calibration of many such elements.
- **Readout and leakage control.** Fusion charge must be measured without inadvertently creating additional quasiparticles. Leakage is an unwanted transition from the encoded computational subspace into other physical states. Universal braid mathematics does not specify the required control wiring.

Required milestones for a defect array

A defect array could offer site-specific fabrication and conventional spin readout, capabilities that differ from those of a quantum Hall fluid. The central open question is whether its microscopic Hamiltonian produces the required topological phase and excitations. Comparison with the two established programs gives five milestones:

- Derive a many-body topological phase from measured microscopic defect parameters rather than only from a proposed interaction graph.
- Measure the bulk gap and topological diagnostics while excluding a deliberately prepared but unprotected state.
- Create localized excitations and determine their fusion channels.
- Demonstrate order-dependent braid operations or an equivalent measurement-only protocol.

- Show that the logical error rate decreases as separation, code distance, or another theoretically predicted protection parameter increases. Code distance is the minimum number or spatial extent of physical errors required to produce an undetectable logical error.

The relevant benchmark is not merely the observation of a two-level system, which Majorana devices achieved long ago. The necessary standard is whether the entire sequence from microscopic Hamiltonian to protected operation remains valid after plausible alternative explanations are tested.

Common errors in interpreting evidence

- **“A zero-bias peak is a Majorana.”** A zero-bias peak is a spectroscopic observation. Identifying a Majorana requires correlated evidence concerning spatial localization, the excitation gap, parity, fusion, and ultimately exchange statistics. Trivial bound states can produce similar peaks [R223].
- **“Parity readout proves what carries the parity.”** The 2025 InAs–Al experiment demonstrated a high-quality measurement primitive and explicitly considered both topologically trivial and non-trivial interpretations [R224]. Instrument performance does not determine quasiparticle identity.
- **“The 12/5 plateau proves Fibonacci order.”** Quantized Hall response and activated transport establish an incompressible phase. They do not uniquely determine its fusion category, meaning the mathematical structure that specifies topological charges and their fusion rules [R225]; [R226]; [R227]; [R228].
- **“Quantum Hall anyons have been braided, therefore Fibonacci anyons have been braided.”** The direct interferometric result in Ref. [R131] concerns Abelian Laughlin anyons at $\nu = 1/3$, not the non-Abelian candidate at 12/5.
- **“Non-Abelian means universal.”** Ising anyons are non-Abelian, but their braids produce only Clifford operations. Fibonacci anyons are computationally universal by braiding in the ideal theory [R015].
- **“Universal means scalable.”** Universality means that a gate set is dense in the required unitary space. It does not determine the excitation gap, poisoning rate, control-wiring requirements, braid duration, readout fidelity, fabrication yield, or heat load.
- **“A simulated braid settles the material question.”** A processor can implement the same matrices without hosting the corresponding emergent quasiparticles. Such a result validates the control protocol and the mathematical theory, but it does not establish intrinsic topological order.

Four terms describe different evidentiary stages. A **candidate phase** is an ideal or numerically studied Hamiltonian that supports the proposed topological order.

A **compatible signature** is an observation that agrees with a prediction but also permits alternative explanations. **Quasiparticle identification** requires independent observables that establish charge, fusion, and statistics.

A **topological qubit** is an encoded state that can be initialized, operated, and read out with an error advantage attributable to topology. Evidence at one of these levels must not be represented as evidence for a later level.

Technical checks

- **Majorana zero modes and Ising computation.** A Majorana zero mode is a zero-energy quasiparticle excitation represented by a self-adjoint fermionic operator. They are associated with Ising rather than Fibonacci computation because their fusion space and exchange operators realize the Ising braid representation. This representation generates Clifford gates, but it does not generate a dense universal gate set.
- **Hadamard operation from Ising braids.** Let B_1 and B_2 denote the exchange operators for adjacent Ising anyons. Direct multiplication gives $\frac{1+i}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = e^{i\pi/4} H$. Thus, the Ising braid product $B_1 B_2 B_1$ implements the Hadamard gate H up to the global phase $e^{i\pi/4}$. Braiding alone still cannot supply the non-Clifford T gate.
- **Fibonacci interest at filling $12/5$.** At total filling $12/5$, the active, partially occupied Landau level has filling $2/5$, the particle–hole conjugate of the $k = 3$ Read–Rezayi state at $3/5$. The topological order of the $k = 3$ Read–Rezayi state contains a Fibonacci sector, whose quasiparticles obey Fibonacci fusion rules. This relationship motivates interest in Fibonacci physics at $12/5$.
- **Result of the 2025 InAs–Al experiment.** The experiment established fast single-shot interferometric measurement of fermion parity with 1% assignment error under the reported conditions. It did not conclusively establish Majorana identity, fusion, or braiding [R224].
- **Limitations of interpreting an activated $12/5$ plateau as a Fibonacci qubit.** An activated plateau is a quantized Hall plateau accompanied by thermally activated transport behavior. It establishes an incompressible phase, meaning a gapped phase that resists changes in particle density. It does not uniquely establish the phase’s fusion content, controlled quasiparticle creation and manipulation, fusion-channel readout, a non-Abelian braid test, or scaling of protected logical operations. Therefore, the plateau alone does not establish a Fibonacci qubit.
- **Criterion for defect-engineered proposals.** Each link in the proposed chain of inference must be measured: the microscopic Hamiltonian, the resulting phase, its excitations, their exchange statistics, and the corresponding logical operation. Alternative explanations must remain under consideration until an experiment excludes them.

The semiconductor–superconductor wire is described by a simple effective Hamiltonian, yet the identity of its residual experimental feature remains disputed. The quantum Hall system is experimentally established as an incompressible liquid, while its specific topological order remains disputed.

Neither platform has demonstrated a controlled non-Abelian braid in its intended solid-state device. A defect array is not experimentally more advanced than these platforms and has not yet reached the corresponding stage of experimental discrimination.

Sources

- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083). [arXiv:0707.1889](https://arxiv.org/abs/0707.1889).

- [R222] R. M. Lutchyn, J. D. Sau, and S. Das Sarma, “Majorana fermions and a topological phase transition in semiconductor-superconductor heterostructures,” *Physical Review Letters* 105, 077001 (2010). DOI: [10.1103/PhysRevLett.105.077001](https://doi.org/10.1103/PhysRevLett.105.077001). [arXiv:1002.4033](https://arxiv.org/abs/1002.4033).
 - [R223] K. Flensberg, F. von Oppen, and A. Stern, “Engineered platforms for topological superconductivity and Majorana zero modes,” *Nature Reviews Materials* 6, 944–958 (2021). DOI: [10.1038/s41578-021-00336-6](https://doi.org/10.1038/s41578-021-00336-6). [arXiv:2103.05548](https://arxiv.org/abs/2103.05548).
 - [R224] Microsoft Azure Quantum, M. Aghaee, A. Alcaraz Ramirez, Z. Alam, et al., “Interferometric single-shot parity measurement in InAs–Al hybrid devices,” *Nature* 638, 651–655 (2025). DOI: [10.1038/s41586-024-08445-2](https://doi.org/10.1038/s41586-024-08445-2). [arXiv:2401.09549](https://arxiv.org/abs/2401.09549).
 - [R139] N. Read and E. Rezayi, “Beyond paired quantum Hall states: Parafermions and incompressible states in the first excited Landau level,” *Physical Review B* 59, 8084–8092 (1999). DOI: [10.1103/PhysRevB.59.8084](https://doi.org/10.1103/PhysRevB.59.8084). [arXiv:cond-mat/9809384](https://arxiv.org/abs/cond-mat/9809384).
 - [R225] E. H. Rezayi and N. Read, “Non-Abelian quantized Hall states of electrons at filling factors $12/5$ and $13/5$ in the first excited Landau level,” *Physical Review B* 79, 075306 (2009). DOI: [10.1103/PhysRevB.79.075306](https://doi.org/10.1103/PhysRevB.79.075306). [arXiv:cond-mat/0608346](https://arxiv.org/abs/cond-mat/0608346).
 - [R226] C. Zhang, C. Huan, J. S. Xia, N. S. Sullivan, W. Pan, K. W. Baldwin, K. W. West, L. N. Pfeiffer, and D. C. Tsui, “Spin polarization of the $\nu = 12/5$ fractional quantum Hall state,” *Physical Review B* 85, 241302(R) (2012). DOI: [10.1103/PhysRevB.85.241302](https://doi.org/10.1103/PhysRevB.85.241302). [arXiv:1204.0903](https://arxiv.org/abs/1204.0903).
 - [R227] W. Zhu, S. S. Gong, F. D. M. Haldane, and D. N. Sheng, “Fractional quantum Hall states at $\nu = 13/5$ and $12/5$ and their non-Abelian nature,” *Physical Review Letters* 115, 126805 (2015). DOI: [10.1103/PhysRevLett.115.126805](https://doi.org/10.1103/PhysRevLett.115.126805). [arXiv:1505.03050](https://arxiv.org/abs/1505.03050).
 - [R228] R. S. K. Mong, M. P. Zaletel, F. Pollmann, and Z. Papić, “Fibonacci anyons and charge density order in the $12/5$ and $13/5$ quantum Hall plateaus,” *Physical Review B* 95, 115136 (2017). DOI: [10.1103/PhysRevB.95.115136](https://doi.org/10.1103/PhysRevB.95.115136). [arXiv:1505.02843](https://arxiv.org/abs/1505.02843).
 - [R131] J. Nakamura, S. Liang, G. C. Gardner, and M. J. Manfra, “Direct observation of anyonic braiding statistics,” *Nature Physics* 16, 931–936 (2020). DOI: [10.1038/s41567-020-1019-1](https://doi.org/10.1038/s41567-020-1019-1). [arXiv:2006.14115](https://arxiv.org/abs/2006.14115).
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Chapter 33 — Programmable competitor platforms

A superconducting chip responds to a short sequence of microwave pulses and then reports a measurement. A cold-atom apparatus holds atoms in focused laser traps, lifts them into large electronic orbits, and can reassemble the same atoms into a different interaction graph before the next day of measurements. Each platform has prepared states described by models developed in the preceding chapters.

A lattice of implanted spins could in principle host a Fibonacci anyon, a quasiparticle whose pairwise exchanges apply transformations fixed by Fibonacci fusion rules. Reaching that excitation would require growing the right host material, placing defects accurately, and opening a many-body energy gap that separates the relevant low-energy states from higher excitations. Effort directed toward that material goal is effort unavailable for building processors that already run exchange sequences, so the comparison carries an opportunity cost: the forgone value of the route left unfunded.

This chapter declares no overall winner. It places the same four-spin square on three platforms and accounts for the physical and computational resources of each implementation on equal terms.

State simulation and protected hardware

The chapter separates two goals because each calls for different evidence and different resources.

The first objective is model investigation: preparing states, creating excitations, measuring braids or correlations, and observing the dynamics of a small system.

The second goal is protected hardware: arranging the low-energy sector of the physical system so that it carries topological order. Such a system stores global information that weak local disturbances leave nearly unchanged, so a small local shake-up has only a small effect on what is encoded.

A circuit can prepare the exact ground-state wavefunction of a topological model while the chip itself keeps following an unrelated Hamiltonian, the operator that fixes the energies and time evolution of the hardware. At the measurement time the prepared state can show the entanglement structure characteristic of topological order. That pattern in the state still leaves later local errors energetically unpenalized when the governing Hamiltonian assigns them no cost, so preparing the pattern does not by itself establish a protected phase.

Superconducting processors and Rydberg arrays already provide strong evidence for the first objective. [Experiment] Superconducting processors have prepared toric-code states and moved their excitations [R125], and have digitally simulated Fibonacci string-net states, fusion, and braiding [R138].

[Experiment] A Rydberg array has shown signatures of a toric-code-type spin liquid in a programmable analog simulator [R126]. None of these observations, by itself, establishes that a stored logical qubit received passive topological protection, meaning protection supplied continuously by the physical Hamiltonian rather than by active error-correction operations.

A defect design contributes to the second goal only when its nearly static microscopic Hamiltonian produces the target model at a usable interaction strength. Flexible simulation tests dynamics while passive protection stores information against local noise, and a braid matrix produced by gates cannot stand as evidence for an anyon living in the material.

A common four-spin target

Place four two-level systems on the edges of one square and label them 1, 2, 3, 4. Each system uses the computational basis $|0\rangle_i$ and $|1\rangle_i$. For system i , the Pauli operator Z_i returns $+1$ on $|0\rangle_i$ and -1 on $|1\rangle_i$. The plaquette operator $P_\square$ records the combined Z -parity of the four spins:

$$P_\square = Z_1 Z_2 Z_3 Z_4.$$

The target Hamiltonian is

$$H_\square = -K P_\square,$$

with $K > 0$ an energy in joules or electronvolts. The coefficient K fixes the strength of the four-body plaquette interaction.

Since $P_\square^2 = I$, with I the identity, the plaquette operator $P_\square$ takes only the eigenvalues $p = \pm 1$: the $p = +1$ sector has energy $-K$ and the $p = -1$ sector has energy $+K$. The gap between the sectors is $2K$: one square enforces a parity constraint with an energy cost.

One square carries no topological order: it is a single cell with a four-spin Hilbert space, only local paths around one face, and sectors that local measurement distinguishes.

The square still serves as a shared benchmark because the same plaquette term appears as one component of stabilizer Hamiltonians, whose mutually commuting terms select the target subspace through their specified eigenvalues. Each platform must implement this identical small target, account for the same controls and auxiliary systems, and meet the same criterion for protection.

Leaving out those shared requirements makes a six-gate sequence look equivalent to a Hamiltonian that realizes a phase. The gate sequence remains a legitimate implementation, yet it does not establish that the phase exists in the hardware.

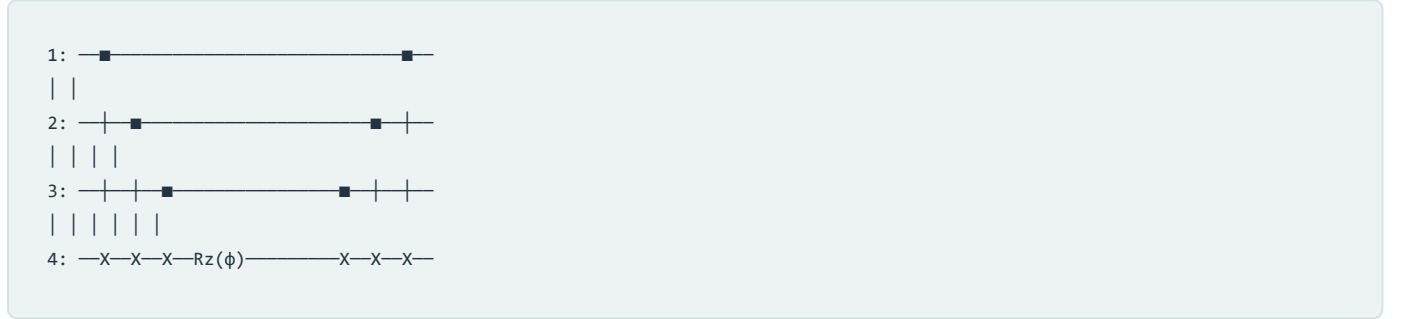
Digital synthesis of plaquette evolution

For an evolution time δt in seconds the target time-evolution operator is

$$U_\square(\delta t) = \exp\left(-\frac{i H_\square \delta t}{\hbar}\right) = \exp\left(+i \frac{K \delta t}{\hbar} P_\square\right),$$

with $\hbar$ Planck's reduced constant in joule-seconds. The combination $K\delta t/\hbar$ is dimensionless, as the argument of an exponential must be.

Write $R_z^{(4)}(\phi) = \exp(-i\phi Z_4/2)$ for a rotation of qubit 4 about its z -axis through angle ϕ in radians. A controlled-NOT (CNOT) flips its target conditionally on its control, and a CNOT with control j and target 4 sends Z_4 to $Z_j Z_4$ under conjugation. The displayed sequence therefore works as follows:



implements $\exp(-i\phi P_\square/2)$. The choice

$$\phi = -\frac{2K\delta t}{\hbar}$$

which gives $U_\square(\delta t)$ exactly in the ideal gate model: six two-qubit gates plus one single-qubit rotation per plaquette-evolution step.

Hardware with sparse connectivity, where direct gates join only selected pairs, may need extra routing gates. A parity-measurement construction can instead recruit an ancilla, an auxiliary system that assists the operation while the target stores the information. Ancilla reset, readout, and feedforward, the use of a measurement result to choose a later operation, then belong in the resource count. Even a short circuit carries nonzero physical cost.

Suppose the target Hamiltonian is $H = A + B$, where A and B contain terms that do not commute. A first-order product formula approximates the full evolution as

$$e^{-iH\delta t/\hbar} \approx e^{-iA\delta t/\hbar} e^{-iB\delta t/\hbar}.$$

The leading operator error per step is proportional to $\delta t^2[A, B]/\hbar^2$, where $[A, B] = AB - BA$ is the commutator.

Shortening each step shrinks this digitization error, the price of replacing continuous evolution by a product of simpler evolutions, at the cost of running more imperfect gates. Commuting stabilizer terms remove the product-formula error while leaving hardware errors in place.

Generation of four-body interactions in defect systems

A defect-spin array ordinarily supplies one-spin terms plus two-spin interactions of dipolar, exchange, superexchange, strain-mediated, or photon-mediated type. It does not ordinarily supply a clean $Z_1 Z_2 Z_3 Z_4$ interaction, so two routes present themselves.

The digital route compiles the CNOT sequence into the entangling gates the defects support. That route needs all four spins initialized, selective control and readout, and a combined error from the six entangling operations that stays inside the error budget.

A fixed sparse graph may require a central electron spin, nuclear ancillas, or swap gates that exchange the quantum states of two sites. [Experiment] Pairwise entanglement between individual diamond defect spins has been demonstrated [Ro80], and a seven-qubit diamond register has implemented fault-tolerant logical operations and flagged stabilizer measurements [R234].

Those demonstrations establish control and measurement building blocks. They do not produce a dense, static four-body topological Hamiltonian.

The analog route adds mediator levels or a small cluster of defects through which the target spins interact. Brief virtual excursions into those mediator states, intermediate transitions permitted in perturbation theory without leaving the mediator populated, can generate an effective four-body interaction in the low-energy description. For a fourth-order gadget with microscopic coupling J , mediator penalty Δ , and dimensionless coefficient c , dimensional analysis permits

$$K_{\text{eff}} = c \frac{J^4}{\Delta^3}.$$

Here K_{eff} is the effective four-body coupling; both sides carry units of energy. Keeping the gadget under perturbative control needs $|J/\Delta| \ll 1$, which forces $K_{\text{eff}} \ll |J|$: a well-controlled gadget yields a gap far below the microscopic coupling scale.

Lower-order shifts and stray couplings then need cancellation or an explicit tolerance budget. [Proposal] Defects compete in passive Hamiltonian engineering only when the resulting K_{eff} clears the decoherence, disorder, temperature, leakage, and finite-size scales set in earlier chapters. No pulse sequence repairs a static gap that starts out too small.

Digital implementation with superconducting circuits

A superconducting version turns the parity circuit into calibrated microwave pulses and native two-qubit gates. Planar chips offer a fixed or tunable local coupling graph, so the all-to-one connectivity of the square may be native, coupler-mediated, or assembled from swap gates. Nanofabrication reproduces a chosen layout reliably, while frequency crowding, crosstalk, wiring, leakage out of the computational doublet, and calibration remain system-level costs [R229]. The doublet names the two energy levels chosen for $|0\rangle$ and $|1\rangle$, and leakage moves population outside that pair.

That control stack gives the most direct digital route to plaquette evolution: software can change K , flip its sign, drop a plaquette, or measure a different string operator without rebuilding the chip.

[Experiment] This control method prepared and probed topologically ordered states on a superconducting processor [R125], and later realized Fibonacci fusion and braid action digitally [R138]. In the Fibonacci demonstration ordinary superconducting qubits encoded the string-net vector space of quantum states and ran gates for the braids.

The chip itself contained no intrinsic material Fibonacci anyons [R138].

A tunable coupler can also produce driven effective interactions, yet a driven superconducting analog is not automatically passive: once the periodic drives, active calibration, or finite-depth preparation circuit switches off, a different native Hamiltonian generally returns. [Experiment] The related below-threshold surface-code scaling on a 105-qubit processor separately demonstrates active quantum error correction, with its repeated measurement and feedback cycles, rather than a self-correcting material [R230].

A superconducting processor that runs the model therefore serves model investigation well. Running the model leaves the protected-hardware question open.

Rydberg blockade and plaquette implementation

A neutral atom sits in an optical tweezer, a tightly focused laser field shaped into a trapping potential. A second laser can lift the outer electron into an orbit of large spatial extent, and an atom with its electron held in such an orbit is a Rydberg atom whose large orbit supports strong interactions across comparatively long distances.

In the standard model the laser couples the computational state $|1\rangle_i$ toward a Rydberg level. Write $n_i = (I - Z_i)/2$ for its occupation projector, Ω_i for the laser Rabi rate in radians per second, δ_i for the laser-transition detuning in radians per second, and V_{ij} for the interaction rate. The Rabi rate fixes the coherent transition speed and the detuning measures how far the laser sits from resonance. A rotating-frame Hamiltonian, a description with the rapid drive phase removed, reads

$$\frac{H_R}{\hbar} = \sum_i \left(\frac{\Omega_i}{2} X_i - \delta_i n_i \right) + \sum_{i < j} V_{ij} n_i n_j, \quad V_{ij} = \frac{C_6}{r_{ij}^6}.$$

Here X_i flips the two states, r_{ij} is the atom separation in metres, and C_6 carries units $\text{rad s}^{-1} \text{m}^6$. The steep r^{-6} interaction causes Rydberg blockade: nearby atoms cannot both be resonantly excited at once. Geometry, detuning, and drive strength thus act as programmable Hamiltonian parameters [R231].

The native interaction is pairwise, $n_i n_j$, rather than the four-body $Z_1 Z_2 Z_3 Z_4$. The plaquette term enters digitally by compiling the same parity circuit into blockade-mediated entangling gates.

A tailored multiqubit blockade pulse or an ancilla constraint can compress the circuit depth, the count of sequential gate layers, though its phase and leakage still need calibration. [Experiment] High-fidelity parallel two-qubit gates on up to 60 atoms, alongside low-error three-qubit gates, show this route running in the laboratory [R232].

Those gate results do not create a native four-body plaquette term.

Rydberg arrays nevertheless offer the most readily reconfigurable analog simulator of the three platforms: the atoms move into new positions while the Hamiltonian above runs continuously instead of being broken into gates.

[Experiment] A constrained 219-atom array in this mode reported signatures of a topological spin liquid [R126]: a finite, driven atomic system with engineered blockade physics.

That evidence goes beyond classical simulation of the same Hamiltonian, yet it does not show the apparatus holding a passive topological qubit once the lasers switch off. Removing the lasers generally removes the engineered Hamiltonian, which limits claims about quantum memory without detracting from the experiment.

Resource-based platform comparison

Physical and programmable connectivity

Connectivity names which pairs of sites interact strongly and selectively enough, including while other operations run in parallel. The count of lines in a hardware diagram does not settle that question by itself.

A long-range tail makes two sites connected in the Hamiltonian, yet the pair counts as independently programmable only when that tail can be switched off or absorbed by compilation. Physical coupling and on-demand control therefore need separate names.

Defect positions freeze at fabrication. Dipolar couplings reach across distance, fall as r^{-3} , and vary with orientation, while exchange couplings stay short-ranged and depend exponentially on wavefunction overlap. An average graph can therefore look suitable while individual fabricated samples fail.

Superconducting chips print reproducible nearest-neighbour or coupler graphs, with bus resonators and tunable couplers adding further links. Distant logical links still cost routing operations, chip area, frequency allocation, or extra electromagnetic modes.

Rydberg tweezers rearrange atoms in two and three dimensions and the blockade radius couples several neighbours at once, but interaction tails and limits on simultaneous addressing persist. The graph is programmable without amounting to an arbitrary software-defined adjacency matrix.

Coherence-normalized operation counts

Coherence time, the interval over which phase or population information survives, does not rank platforms by itself. A slower qubit with proportionally cleaner gates can win, and a long-lived memory matters little when initialization dominates the cycle. The informative figures are dimensionless: operations per coherence time, error per circuit layer, leakage, and the fraction of cycle time spent on useful computation, the duty cycle.

Superconducting gates run fast on electronic timescales and repeat rapidly, while relaxation, dephasing, leakage, and calibration drift share the same chip [R229]. Neutral-atom gates exploit strong Rydberg interactions, while atom loss, Doppler shifts, laser noise, spontaneous emission, rearrangement, and imaging consume cycle time [R231]; [R232]. Defect electron or nuclear spins hold excellent memories, while entangling distant defects, collecting photons, or resolving dense spectra can set the actual pace.

Setting one platform's best memory time against another platform's full algorithm therefore proves nothing. Without duty-cycle accounting the longest-lived spin wins by default, even where memory lifetime never limits the computation.

Initialization, reset, and measurement

Superconducting qubits are electrical circuits read through microwave resonators and amplifiers inside a dilution refrigerator. Fabrication repeats well enough for large processors, while every added control and readout channel raises cryogenic and calibration costs [R229]; [R230].

Rydberg machines assemble identical atoms from a reservoir instead of fabricating each atom, and rearrangement can fill vacancies before a run, though atoms can still be lost mid-run.

State-selective fluorescence reads many atoms in parallel and usually destroys the detected atom, so reloading and sorting belong in the cycle budget. [Experiment] Reconfigurable arrays have run an encoded processor with up to 280 physical qubits and 48 logical qubits in sampling circuits [R233].

Those numbers describe one logical experiment; they do not promise 280 permanently error-free atoms.

Defect hosts pack spins into a compact solid with possible optical readout, but defect identity, charge state, position, orientation, and spectral line each carry their own yield. The seven-spin diamond logical experiment [R234] built a local register around one optically active centre, which does not establish a wafer-scale lattice of equivalent electronic defect clusters.

Distinct definitions of scalability

"Scalable" carries at least four meanings: more physical sites, more simultaneously high-quality sites, deeper executable circuits, and better logical performance after overhead. An atom count demonstrates assembly capability.

A chip count demonstrates integration capability. Neither count demonstrates protected computation.

Superconducting processors need dilution refrigeration and dense classical control. Rydberg processors need ultrahigh vacuum, laser cooling, optical access, stable lasers, and repeated atom handling, but no millikelvin solid-state stage [R231]. Defect behaviour depends on the chosen regime: some spins work at room temperature, while high-quality optical interfaces or particular defect species need cryogenic operation.

A blanket claim that defects need no cryogenics is therefore invalid at platform level; each claim must name the species and interface.

Direct Hamiltonian programmability

Analog programmability asks which coefficients of H change without breaking time evolution into gates.

Rydberg arrays provide tunable geometry, local or global detuning, drive amplitude, and strong finite-range constraints. Their native family of Hamiltonians is extensive but not arbitrary.

Superconducting circuits tune frequencies, drives, and couplers with precise waveform control, usually at the price of heavy calibration and often through digital compilation. Defects allow the least post-fabrication change: local fields and drives adjust behaviour while microscopic exchange paths and defect positions stay put.

Frozen couplings hinder model discovery yet could stabilize a device once the correct static Hamiltonian is in hand. A frozen wrong H gains nothing from that stability.

Current experimental status

Evidence below is assessed through August 2026. The cited experiments are research milestones, not purchasing recommendations, and all three platforms continue to develop.

The compact comparison is:

Criterion	Crystalline defects	Superconducting processor	Rydberg array
Native interactions	Dipolar, exchange, and optical- or phonon-mediated interactions; strongly materials-dependent	Circuit-mediated two-body gates and tunable couplers	Laser drive plus strong distance-dependent blockade interactions
Graph	Fixed and affected by fabrication disorder	Lithographic and usually local; routing or couplers extend it	Reconfigurable geometry with interaction tails
Control style	Local microwave or optical control; static interactions are possible	Highly developed, fast digital pulse control	Strong digital gates and unusually direct analog control
Readout	Species-dependent optical or electrical interface	Multiplexed microwave readout	Parallel fluorescence imaging, with atom-loss and reload costs
Environment	Dependent on the host and transition; room-temperature operation is not universal	Dilution refrigerator and cryogenic wiring	Ultrahigh vacuum, laser cooling, and optical infrastructure
Present topological evidence cited here	Defect entanglement and small logical registers, but not a material topological phase [R080]; [R234]	Digital toric-code and Fibonacci simulations [R125]; [R138]	Analog spin-liquid signatures in a finite driven array [R126]
Plaquette route	Gate compilation or weak gadget-generated K_{eff}	Short digital circuit; driven analog options	Blockade-compiled gate or constrained analog construction
Passive-protection opportunity	A static host Hamiltonian could qualify if all scale tests are passed	Usually active QEC or driven simulation	An emergent analog phase may be possible during the drive; passive memory is not automatic

The table ranks no universal winner; the preferred platform follows from the chosen goal.

[Experiment] For programmable exploration, superconducting processors have executed topological-state circuits and digital non-Abelian braid protocols directly [R125]; [R138]. Rydberg arrays join flexible geometry to an analog Hamiltonian that has shown topological-spin-liquid signatures [R126]. [Experiment] Defect platforms have advanced local control, entanglement, networking, and small registers [R080]; [R234], while the cited evidence includes no defect lattice with an emergent non-Abelian phase.

[Proposal] Defects keep one logically distinct option: a static material Hamiltonian whose interaction network stays active after fabrication without a long gate schedule. Testing that option means measuring K_{eff} , disorder, and thermal stability. The option itself is not evidence that passive protection has been realized.

Common category errors

1. Misidentifying a prepared state as a protected phase

A circuit can prepare the exact ground-state wavefunction of a topological model while the hardware keeps following an unrelated Hamiltonian, the operator fixing the physical energies and time evolution. The prepared state can show the nonlocal entanglement of the target model at measurement time, yet later local errors go energetically unpunished unless the hardware Hamiltonian itself exacts that cost.

2. Conflating active error correction with passive protection

Surface-code cycles can suppress logical errors through repeated syndrome extraction, decoding, and feedback. Syndrome extraction measures error information without directly measuring the encoded logical state; decoding infers the likely errors from those measurements; and feedback applies the corresponding correction or updates the interpretation of later measurements. This procedure provides genuine active protection and is a major engineering achievement [R230].

Passive protection works differently: a static gapped Hamiltonian suppresses local transitions on its own, without repeated syndrome cycles. Active and passive protection therefore name different physics and need different words.

3. Misclassifying a Rydberg analog experiment as purely digital

In a Rydberg analog run the laser-driven many-atom Hamiltonian generates the continuous-time dynamics directly. When that Hamiltonian supports a phase inside its coherent window, the measured correlations and excitations emerge from the analog Hamiltonian itself rather than replaying stored answers.

Memory claims still face the finite lifetime, the reliance on external drives, boundary effects, preparation, and readout.

4. Neglecting the physical cost of connectivity

Connectivity counts physical couplings. Long-range terms add graph edges whether wanted or not, and moving an atom consumes transport time.

A resonator bus adds electromagnetic modes and crowded spectra. A dipolar tail adds disorder and crosstalk, unintended coupling between operations or components meant to stay separate.

Each target graph therefore needs compilation into simultaneous calibrated operations or into a static Hamiltonian whose deviations from the target carry quantitative bounds.

5. Comparing proposed defect arrays directly with demonstrated systems

A projected defect gap belongs under [Proposal] or [Speculation], never in the same table cell as a measured gate or correlation. Programmable hardware that displayed a Fibonacci braid matrix has not thereby shown passive Fibonacci protection.

The superconducting Fibonacci experiment [R138] used physical superconducting qubits to digitally emulate Fibonacci string-net states and braids. The Rydberg spin-liquid experiment [R126] measured correlations that emerged from a driven analog Hamiltonian in a finite atomic array.

A defect spin is one material two-level degree of freedom: a physical qubit. A register code such as [R234] spreads its quantum information across several physical degrees of freedom: an encoded logical qubit. A static defect lattice earns the name topologically ordered only when its many-body Hamiltonian, spectrum, correlations, and excitations meet the criteria for a topological phase. The first three experimental facts listed here do not imply the fourth.

Conceptual checks and derivations

- **Spectral gap implied by $P_{\square}^2 = I$.**

A plaquette is one local face of the interaction lattice and I is the identity. Since the Hermitian plaquette operator obeys $P_{\square}^2 = I$, its eigenvalues are ± 1 . The term

$$H_{\square} = -KP_{\square},$$

with positive energy scale K then gives energies $-K$ and $+K$, separated by the spectral gap $2K$.

- **Absence of topological order in a single plaquette.**

One plaquette lacks an extended many-body phase, nonlocal logical sectors, and the size scaling that damps local perturbations.

- **Difference between a digitally displayed Fibonacci braid and a material anyon.**

A Fibonacci braid run on ordinary qubits needs no Fibonacci anyon, a physical excitation with non-Abelian exchange statistics, inside the hardware Hamiltonian. The qubits span the model vector space and the gates produce the matrix, so the braid matrix leaves the circuit as an output rather than escaping the chip as an excitation.

- **Ideal implementation of $U_{\square}(\delta t)$ using six CNOT gates.**

A CNOT flips its target conditionally on its control. In the ideal gate model the three CNOTs on each side turn the single-qubit rotation $R_z^{(4)}(\phi)$ into

$$\exp(-i\phi P_{\square}/2).$$

Setting

$$\phi = -2K\delta t/\hbar$$

yields

$$\exp(+iK\delta t P_{\square}/\hbar),$$

the plaquette evolution operator $U_{\square}(\delta t)$. The six-CNOT sequence therefore realizes $U_{\square}(\delta t)$ in the ideal gate model.

- **Conditions for a scientifically distinct defect implementation.**

A defect implementation stands apart once it verifies a static low-energy Hamiltonian quantitatively, with a many-body gap and perturbative stability, so the physics arrives without recomputing the dynamics as a circuit. Long-lived isolated spins alone do not meet that bar.

- **Consistent performance comparisons between platforms.**

Comparing one platform's best memory time against another's complete algorithm misleads. The informative figures are operations per coherence time, error per circuit layer, leakage from the computational space, and duty cycle, since a long-lived spin that cannot be entangled or read on schedule gives no algorithmic gain.

Sources

- [R229] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson, and W. D. Oliver, “A quantum engineer’s guide to superconducting qubits,” *Applied Physics Reviews* 6, 021318 (2019). DOI: [10.1063/1.5089550](https://doi.org/10.1063/1.5089550).
 - [R125] K. J. Satzinger et al., “Realizing topologically ordered states on a quantum processor,” *Science* 374, 1237–1241 (2021). DOI: [10.1126/science.abi8378](https://doi.org/10.1126/science.abi8378).
 - [R138] S. Xu et al., “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* 20, 1469–1475 (2024). DOI: [10.1038/s41567-024-02529-6](https://doi.org/10.1038/s41567-024-02529-6).
 - [R230] Google Quantum AI and Collaborators, “Quantum error correction below the surface code threshold,” *Nature* 638, 920–926 (2025; published online 2024). DOI: [10.1038/s41586-024-08449-y](https://doi.org/10.1038/s41586-024-08449-y).
 - [R231] A. Browaeys and T. Lahaye, “Many-body physics with individually controlled Rydberg atoms,” *Nature Physics* 16, 132–142 (2020). DOI: [10.1038/s41567-019-0733-z](https://doi.org/10.1038/s41567-019-0733-z).
 - [R126] G. Semeghini et al., “Probing topological spin liquids on a programmable quantum simulator,” *Science* 374, 1242–1247 (2021). DOI: [10.1126/science.abi8794](https://doi.org/10.1126/science.abi8794).
 - [R232] S. J. Evered et al., “High-fidelity parallel entangling gates on a neutral-atom quantum computer,” *Nature* 622, 268–272 (2023). DOI: [10.1038/s41586-023-06481-y](https://doi.org/10.1038/s41586-023-06481-y).
 - [R233] D. Bluvstein et al., “Logical quantum processor based on reconfigurable atom arrays,” *Nature* 626, 58–65 (2024). DOI: [10.1038/s41586-023-06927-3](https://doi.org/10.1038/s41586-023-06927-3).
 - [Ro80] F. Dolde et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* 9, 139–143 (2013). DOI: [10.1038/nphys2545](https://doi.org/10.1038/nphys2545).
 - [R234] M. H. Abobeih et al., “Fault-tolerant operation of a logical qubit in a diamond quantum processor,” *Nature* 606, 884–889 (2022). DOI: [10.1038/s41586-022-04819-6](https://doi.org/10.1038/s41586-022-04819-6).
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Chapter 34 — Commanded coordinates and realized defect positions

A specified implantation coordinate steers the beam; the implanted nitrogen ion comes to rest somewhere nearby, after colliding with the lattice, shedding energy to electrons and nuclei, and stopping short of or wide of the aim point. Later heating, called annealing, lets vacancies — unoccupied lattice sites — migrate. A mobile vacancy may then bind to the implanted nitrogen, escape at the surface, or react elsewhere.

When an optically active center forms, its position reflects the whole chain — implantation, diffusion, and chemical conversion — on top of the coordinate fed to the machine. A list of commanded positions therefore cannot serve as a Hamiltonian, the operator that specifies the energies and interactions of the physical system actually built.

The built Hamiltonian depends instead on missing and extra centers, unintended charge states, optically silent emitters, damaged spins, and defects sitting several nanometres from their targets. Chapter 28 showed why such offsets matter: the dipolar coupling J between electron spins falls with separation r as $J \propto r^{-3}$, so to first order a small fractional error in bond length triples into the coupling. A 20 nm graph that allows 20% coupling spread therefore works with very little positional slack.

Patterning emitters, growing thin impurity layers, and counting incident ions each exist as working processes. None of the cited work has combined them into a dense, uniformly coupled, defect-complete cluster lattice at the sub-nanometre relative tolerance this graph demands.

Distinct Fabrication and Acceptance Metrics

Fabrication runs through seven successive stages, and each stage poses its own question. The stages and their metrics are:

Beam accuracy specifies the position at which an ion crosses the crystal surface. Straggle is the statistical spread in the ion's final stopping position after it undergoes collisions within the crystal.

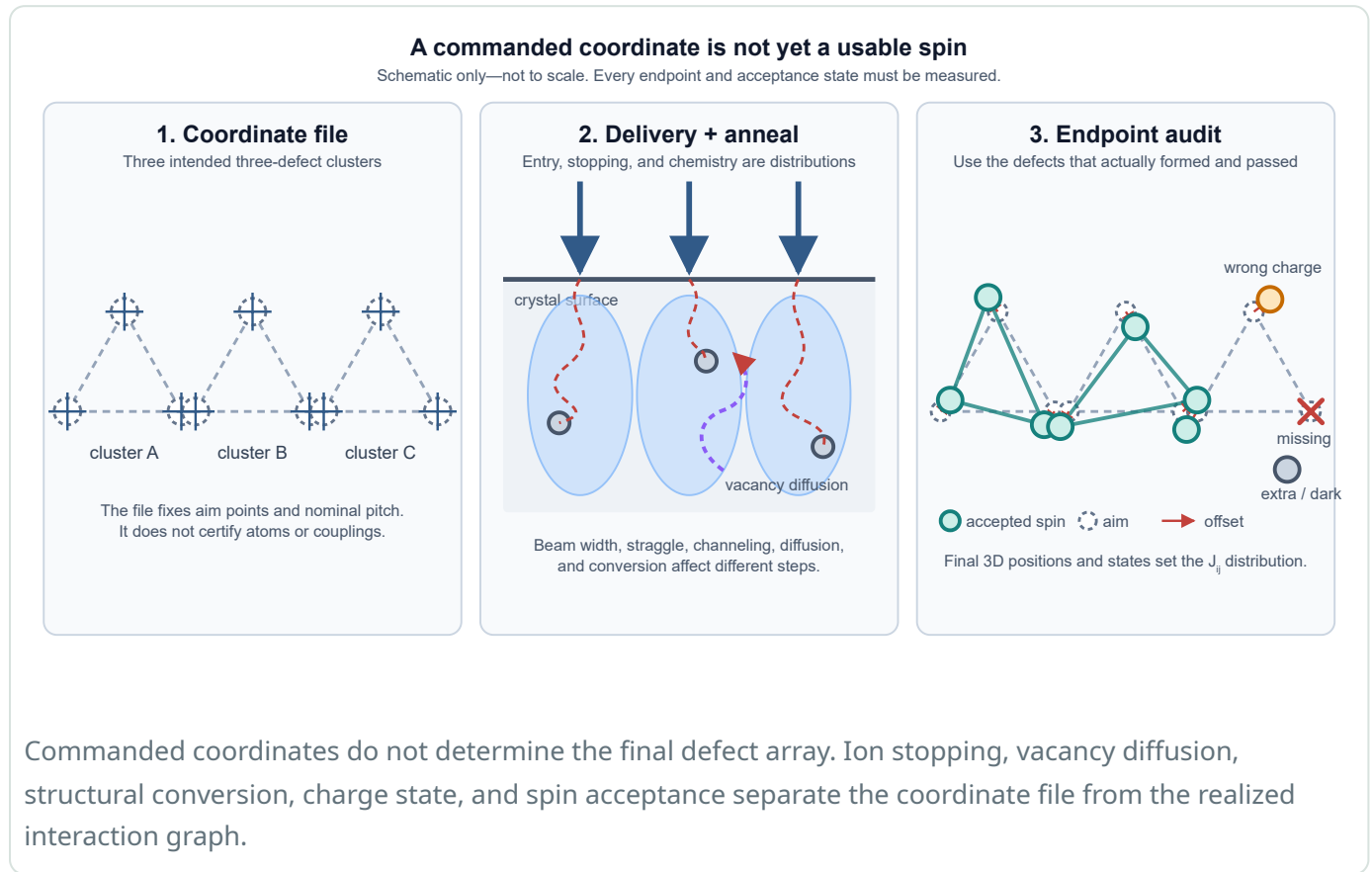
Diffusion specifies the displacement of a mobile vacancy during annealing. Creation yield is the probability that the intended structural defect complex forms.

Charge-state yield is the probability that the complex occupies the required electronic charge state. Optical usability is the probability that optical initialization and readout satisfy the specified criteria.

Spin usability is the probability that the spin can be initialized, remains coherent for the required interval, and responds to control operations within specification.

These metrics do not substitute for one another. A bright optical feature at the intended coordinate can fail the spin-usability test, while a defect with excellent spin properties can sit outside the allowed positional tolerance. Without explicitly built-in spare sites and repair procedures, a complete array needs every indispensable site to pass every one of these tests.

The following image tracks a nine-site coordinate file through ion delivery, annealing, and endpoint acceptance. Dashed rings indicate commanded positions, green circles indicate accepted spin-active centers, and positional offsets or failed sites remain part of the realized device specification.



Commanded coordinates do not determine the final defect array. Ion stopping, vacancy diffusion, structural conversion, charge state, and spin acceptance separate the coordinate file from the realized interaction graph.

Each fabrication and acceptance stage carries its own conditional probability. A beam-position figure, measured in units of length, cannot simply be multiplied by a dimensionless conversion yield and quoted as a complete performance measure. A working device must meet the spatial and the probabilistic requirements together.

Positional Tolerance for a Nine-Spin Test Structure

Consider [Proposal] three triangular encoded clusters of three electron-spin defects each. This nine-site patch is a fabrication coupon: a test structure that asks whether the same intra-cluster interactions reproduce across multiple clusters. It is not itself a topological phase.

Take the intended nearest-neighbour separation within each cluster as $r = 20 \text{ nm}$, and require the root-mean-square disorder in the dipolar coupling — the standard deviation of the coupling fluctuations relative to the intended value — to stay below 20%.

Let σ_{site} denote the independent one-dimensional standard deviation of each bond endpoint along the bond direction. For the dipolar-coupling relation $J = C/r^3$, where C contains the distance-independent coupling factors, first-order uncertainty propagation gives

$$\frac{\sigma_J}{J} \approx \frac{3\sqrt{2}\sigma_{\text{site}}}{r}.$$

This equation turns independent endpoint-position uncertainty into relative coupling uncertainty. Both σ_{site} and r carry dimensions of length, so the ratio is dimensionless, and the placement allocation follows as

$$\sigma_{\text{site}} \leq \frac{0.20(20 \text{ nm})}{3\sqrt{2}} = 0.94 \text{ nm}.$$

The value 0.94 nm is a derived requirement, not a reported fabrication capability. The derivation leaves out angular-factor disorder in the dipolar coupling, the discrete crystallographic sites, exchange variations, and correlated registration errors; restoring those contributions tightens the allowed placement uncertainty further.

This requirement can be compared with a measured endpoint distribution. Focused Si implantation followed by annealing produced individual SiV centers, or silicon-vacancy centers, with a one-axis standard deviation of approximately 32 nm about the target coordinates [R190]. [Numerics] For an order-of-magnitude comparison only, assume that this endpoint distribution is an unbiased Gaussian distribution and that the acceptance interval is $|x| < 0.94 \text{ nm}$. The resulting one-dimensional acceptance probability is

$$P(|x| < 0.94 \text{ nm}) = \text{erf}\left(\frac{0.94}{\sqrt{2}(32)}\right) \approx 0.0235.$$

Here, erf is the error function used to integrate a centered Gaussian probability distribution. If the same acceptance interval is imposed independently in two lateral coordinates, the per-site probability is approximately $(0.0235)^2 = 5.5 \times 10^{-4}$ before the depth coordinate is tested. Under an additional assumption of independent site outcomes, raising this illustrative value to the ninth power gives approximately 4.8×10^{-30} .

This calculation forecasts nothing. The measured endpoint distribution may depart from Gaussian inside its central nanometre, and the cited SiV process was never tuned for this coupon. The calculation measures the gap: endpoint spreads of tens of nanometres against a sub-nanometre allocation derived from the coupling-disorder requirement.

Conditional Yield Accounting

Placement is only one part of the fabrication budget. Define the events

- A : the requested ion or vacancy is delivered;
- C : the desired structural complex forms;
- Q : the desired charge state is occupied;
- O : initialization and optical readout pass;

- S : spin coherence and control pass;
- G : the final three-dimensional position passes.

The probability that a site is usable is the conditional product

$$p_{\text{use}} = P(A)P(C|A)P(Q|A, C)P(O|A, C, Q)P(S|A, C, Q, O)P(G|A, C, Q, O, S).$$

This identity applies the probability chain rule with no independence assumption: each factor is conditioned on acceptance at the preceding stages.

A single published yield therefore seldom equals p_{use} . It typically covers only an initial stretch of this product, scored against the acceptance criteria of that particular experiment.

In a deterministic ^{15}N implantation experiment, counted molecular ions were focused with 121(35) nm lateral resolution and produced about 0.6% optically identified $^{15}\text{NV}^-$ centers after annealing [R258]. [Experiment] The reported 0.6% is already a compound result that includes delivery, structural conversion, negative-charge occupation, and optical detection. It does not provide separate measurements of each factor. Even if 0.006 were treated as an optimistic upper bound on p_{use} , the nine-site yield would satisfy

$$Y_9 \leq (0.006)^9 \approx 1.0 \times 10^{-20},$$

before the 0.94 nm geometric requirement was imposed.

For planning purposes, consider a [Proposal] future fabrication line using the same process but achieving the following explicitly hypothetical conditional values: 0.995 for delivery, 0.50 for structural conversion, 0.90 for the desired charge state, 0.85 for optical acceptance, 0.90 for spin acceptance, and 0.80 for geometric acceptance. Then

$$p_{\text{use}} = 0.995(0.50)(0.90)(0.85)(0.90)(0.80) = 0.274,$$

and, if site outcomes are independent, $Y_9 = p_{\text{use}}^9 = 8.7 \times 10^{-6}$. Correlated damage across a wafer or systematic annealing drift could reduce the array yield further. Conversely, experimentally measured correlations, spare sites, repeated implantation, or post-fabrication selection would require a different yield model. In the absence of repair, a 50% yield for the complete nine-site array requires

$$p_{\text{use}} \geq 0.5^{1/9} = 0.926.$$

The per-site fabrication target is therefore 92.6%. Deterministic delivery of one ion settles only the first factor of the conditional product; dropping the later factors equates a counted incident ion with a completed, usable spin, which the chain rule forbids.

Conventional Ion Implantation and Annealing

Conventional implantation accelerates ions of a selected species into a crystal, often through a lithographically defined aperture. The distribution of surface-entry positions is determined by the aperture geometry, alignment, beam divergence, and mechanical vibration.

Within the crystal, the ion loses energy through electronic collisions, which transfer energy primarily to electrons, and nuclear collisions, which displace lattice nuclei. The resulting distributions of stopping depth and lateral displacement are called longitudinal straggle and lateral straggle, respectively.

Straggle names the residual statistical spread left after beam alignment, physically distinct from targeting error. Crystal channeling — ions running preferentially along open crystallographic directions — can add non-Gaussian tails to the stopping distribution.

An implanted atom does not automatically form the intended defect center. Implanted nitrogen must acquire a neighbouring vacancy to form a nitrogen-vacancy, or NV, center.

An implanted silicon atom must attain the appropriate split-vacancy configuration to form a SiV center. Annealing increases defect mobility and repairs some collision damage, but it can also cause vacancies to migrate to surfaces, form aggregates, or react with unintended impurities.

In one masked CN^- implantation experiment, roughly 50 nitrogen atoms per 80 nm aperture produced a mean 3.5 NV centers — about 7% N-to-NV conversion under an 800 °C, two-hour anneal [R202]. [Experiment] This experiment demonstrated patterned ensembles rather than exactly one usable NV center at each commanded atomic coordinate.

Implantation energy introduces a recurring trade-off. Lower implantation energy produces shallower stopping and usually reduces straggle, but near-surface charge noise, vacancy loss, and implantation damage can reduce structural conversion and spin coherence.

Higher implantation energy can raise vacancy production and conversion while broadening the depth and lateral distributions [R260]. [Experiment] Implantation precision is therefore not a host-independent quantity. Every reported value must be associated with the ion mass, implantation energy, crystal orientation, target composition, and annealing procedure.

Focused-Ion-Beam Implantation

A focused ion beam writes coordinates by rastering across the sample, which registers each spot against fabricated pillars, waveguides, and optical cavities. Registration improves; ion-stopping straggle remains.

For focused silicon implantation into diamond, the beam full width at half maximum, or FWHM, was typically below 40 nm, the calculated lateral straggle was 19 nm, and the measured created-center distribution had a one-axis standard deviation of 32 nm [R190]. [Experiment] The measured Si-to-SiV conversion reached approximately 2.5% at 100 keV. Electron irradiation followed by a second anneal increased the measured conversion in a tested bulk sample to approximately 20% [R190]. These values apply to that specific SiV process and cannot be transferred directly to NV centers, SiC, or sapphire.

Focused writing is useful for sparse photonic devices because each coordinate can be registered independently. In dense interacting lattices, however, beam tails, dose calibration, redeposition, accumulated damage, stage drift, and writing time become array-level variables. A nominal beam spot specifies the intended creation position; it does not specify the covariance matrix of the final center positions.

Even an idealized beam of zero entry width leaves positional uncertainty in place: the ion still stops along a collision-induced distribution and the vacancy still diffuses during annealing. A vanishing surface-entry width therefore still leaves a nonzero error in the realized Hamiltonian.

Delta-Doped Depth Confinement

Delta doping is the introduction of impurities during a short interval of crystal growth, producing a thin impurity-containing layer. It provides depth confinement but does not assign individual lateral lattice coordinates.

In diamond, nitrogen layers approximately 1–2 nm thick were grown, after which vacancies were introduced by irradiation and annealing. The cap-layer thickness selected nominal NV depths from approximately 5 to 100 nm [R204]. [Experiment] This method provides strong depth control and avoids the stopping distribution associated with implanted nitrogen. It does not select a lateral lattice site because dopants remain randomly distributed within the plane.

Delta doping therefore complements a spatially localized vacancy source: it pins down one coordinate while a second process must select x and y .

A 2025 experiment combined a nitrogen delta layer with a localized 200 keV electron beam and reported approximately 4 nm depth confinement and a lateral standard deviation of 46(1) nm in 280 nm diamond pillars [R207]. [Experiment] The characterized NV centers had a mean Hahn-echo $T_2 = 98(37) \mu\text{s}$ under those conditions [R207]. A Hahn echo is a spin-refocusing sequence, and T_2 is the associated coherence time.

This experiment provides both endpoint-position data and a spin-coherence metric. It does not, however, demonstrate a coupled lattice with 20 nm pitch.

Without lateral localization, delta doping yields a depth-confined layer of randomly positioned sites — a useful sample that realizes no prescribed interaction graph.

Electron-Induced Vacancy Generation

Energetic electrons can transfer sufficient momentum to lattice atoms to create vacancies while introducing substantially less foreign chemical material than a heavy-ion beam. Uniform electron irradiation can increase the probability that pre-existing or implanted dopants encounter vacancies.

A focused electron beam can localize vacancy generation laterally, while a delta-doped layer provides depth selection [R207]. The vacancy-generation region nevertheless remains three-dimensional, and the generated vacancies can migrate and react during annealing.

Electron irradiation therefore adjusts $P(C|A)$, the conditional probability of structural-complex formation, and reshapes the damage environment. On its own it guarantees neither A for a selected impurity, Q for the final charge state, nor S for the spin properties.

In the SiV study, irradiation followed by re-annealing improved conversion in a separately tested sample [R190]. It would be invalid to combine the improved yield from that sample with the best placement result from another sample as if both results had been obtained simultaneously.

Femtosecond-Laser Vacancy Writing with Feedback

A tightly focused femtosecond laser pulse can generate vacancies within diamond without implanting a foreign atom. Subsequent local heating or furnace annealing allows a vacancy to bind to native nitrogen. Optical feedback can terminate the repeated writing and annealing sequence after a single emitter is detected, thereby avoiding the Poisson-limited probability of producing one center at a site that receives a fixed dose without feedback.

Chen and colleagues reported approximately 96% single-NV creation yield using feedback, with an in-plane positioning deviation of approximately 33 nm [R203]. [Experiment] This result demonstrates a high probability of producing a single optically detected NV center.

The 96% value does not represent sub-nanometre three-dimensional placement, 96% NV[−] charge-state occupation under all optical conditions, or 96% spin acceptance. The process also requires suitable pre-existing nitrogen near the laser-generated vacancy.

Watching emitter formation and terminating the run raises $P(C|A)$ for a bright center, yet leaves the covariance matrix of the final spin position undetermined. Creation yield and positional covariance stay as separate entries in the fabrication ledger.

Scanning-Probe Aperture Alignment

A scanning probe can incorporate a nanometre-scale aperture into its cantilever. The probe images surface markers, aligns the aperture locally, and transmits ions through that aperture.

Persaud and colleagues integrated a piezoresistive scanning probe with an ion beam and demonstrated single highly charged-ion impacts in resist. The terminal aperture limited the spot size [R259]. [Experiment] This method addresses local registration and permits a stencil to be positioned close to the surface.

Aperture-profile convolution, ion straggle, vacancy diffusion, and stochastic structural conversion all survive this method, and the cited instrument study demonstrated no complete lattice of charge-stable, coherent color centers. A scanning probe therefore specifies alignment and collimation without establishing the usable-site probability.

Vacancy Diffusion During Annealing

Vacancy engineering controls the number and initial spatial distribution of vacancies, together with the competing sinks that can capture them. Heavy ions, electrons, neutrons, and lasers produce different damage profiles. Annealing temperature and duration affect vacancy mobility, recombination, aggregation, loss at surfaces, and capture by dopants.

In nitrogen-rich diamond implanted with focused Ar ions, Racke and colleagues measured 0.04–0.79 created NV centers per implanted Ar ion over 12–132 keV [R206]. [Experiment] Their model of diffusion and surface loss placed an upper bound of approximately 300 nm on the single-vacancy diffusion length during an 800 °C anneal [R206].

[Numerics] This upper bound does not displace every final NV center by 300 nm: nitrogen already filled the crystal, and each final position results from reaction and capture. The bound still rules out modeling annealing as damage removal with all relevant positions held fixed.

Annealing also reshapes charge compensation and optical linewidth by changing the residual-defect population. The planning target is therefore the full joint distribution — final position, desired charge state, optical linewidth and stability, spin coherence, and nearby optically dark paramagnetic defects — since maximum luminescence alone scores only one entry of that distribution.

Treating annealing as a complete positional reset classifies a bright center as a well-positioned spin while ignoring the vacancy migration that produced it. The 300 nm diffusion-length bound falsifies that classification.

Deterministic Ion Delivery and Endpoint Acceptance

An implantation apparatus can detect or prepare a known number of incident ions and stop once the requested count arrives. This is deterministic delivery — control over the incident-ion number, not over their subsequent physical fate.

Deterministic delivery removes count uncertainty from the incident dose. Collision trajectories stay stochastic, and no implanted atom is forced into a particular structural complex or charge state.

Groot-Berning and colleagues extracted individual laser-cooled $^{15}\text{N}_2^+$ ions at 5.9 keV, implanted them without a mask, and verified $^{15}\text{NV}^-$ centers after annealing using optically detected magnetic resonance, or ODMR [R258]. [Experiment] ODMR detects spin resonances through changes in optical emission. The lateral resolution was 121(35) nm, and the conversion was approximately 0.6% [R258].

Delivery of the ion number was deterministic here; creation of usable centers stayed stochastic. Fabrication analysis must keep those two outcomes distinct.

Experimentally Demonstrated Capabilities

The table reports physically distinct metrics in separate columns. A dash indicates that the cited experiment did not establish the corresponding quantity as an array-level probability.

Process and conditions | Beam or creation accuracy | Straggle or diffusion | Structural/charge result | Optical/spin usability | Compound-array verdict | |

CN[−] implanted through 80 nm apertures; diamond; 800 °C, 2 h [R202] | Surface entry defined by an 80 nm mask aperture | Approximately 9 nm simulated straggle for the 20 keV N component [R202] | Approximately 7% N-to-NV conversion | Fluorescent NV arrays; no distribution of complete spin-acceptance outcomes | Patterned sites rather than deterministic individual centers | |

Focused Si implantation in diamond, up to 100 keV [R190] | Beam FWHM below 40 nm | 19 nm calculated lateral straggle; 32 nm measured one-axis endpoint spread | Up to approximately 2.5% SiV conversion; approximately 20% in a separate irradiation and re-annealing test | Nearly lifetime-limited emitters occurred in nanostructures; not every site qualified | Useful targeting for photonic structures; insufficient evidence for dense, complete clusters | |

Nitrogen delta layer combined with focused electrons in diamond [R207] | 20 nm electron-beam spot | Approximately 4 nm depth confinement; 46(1) nm lateral standard deviation | Dose-dependent NV count rather than absolute deterministic conversion | Mean Hahn-echo $T_2 = 98(37) \mu\text{s}$ for characterized NVs | Depth-confinement and spin evidence were obtained together; the lateral graph remained broad | |

Femtosecond vacancy writing with feedback in diamond [R203] | Diffraction-limited writing focus | Vacancy generation and local annealing determined the endpoint | Approximately 96% single-NV creation under feedback | Optical detection provided the feedback signal; the complete charge-state and spin-acceptance rate remained separate | High creation yield with approximately 33 nm in-plane deviation | |

Counted $^{15}\text{N}_2^+$ implanted at 5.9 keV in diamond [R258] | 121(35) nm lateral resolution | Included in the measured resolution rather than eliminated | Approximately 0.6% $^{15}\text{NV}^-$ conversion | ODMR and coherence were characterized for the centers that formed | Deterministic incident-ion counts with a very low compound yield for nine sites | |

30 keV C implantation through a mask in 4H-SiC [R205] | Predetermined array coordinates | Simulated shallow V_{Si} depth of approximately 40 nm [R205] | $19 \pm 4\%$ generation efficiency; $34 \pm 4\%$ probability of a single emitter at the optimized dose | Individual emitters identified optically | Host-specific patterned array; no demonstrated sub-10-nm pair-position distribution | |

The demonstrated capabilities complement rather than accumulate. Delta doping confines depth narrowly; focused beams and scanning apertures register laterally; electron beams and lasers localize vacancy generation; feedback raises the odds of exactly one bright center; counted-ion implantation fixes the incident-ion number; annealing activates defect complexes. No cited process stack delivers all of these at once — sub-nanometre relative placement, near-unity formation of the desired center, near-unity occupation of the desired charge state, array-wide optical and spin qualification, and a complete interaction graph.

A counted ion records a fabrication event; a fluorescent color center is a physical optical emitter; an ODMR-active center is a candidate physical spin qubit. Several coupled spins count as an encoded qubit only once the relevant low-energy subspace has been demonstrated.

A regular image of bright points does not establish an interaction graph, and an interaction graph does not establish an emergent topological phase or a logical topological qubit. Fabrication produces microscopic degrees of freedom with associated disorder distributions; terminology alone adds no topological order.

Common analytical errors

- Do not identify beam width with placement accuracy. The entry width of the ion beam, the simulated stopping straggle of implanted ions, the spatial spread of defect endpoints after annealing, and the registration error between defects and device structures quantify different contributions to positional uncertainty. Each quantity must be reported with its spatial axis, statistical measure, and experimental or simulation conditions.
- Do not identify deterministic implantation with deterministic defect creation. Deterministic implantation, meaning delivery in which individual ion arrivals are prepared or counted, controls A , the number of delivered ions. The 0.6% $^{15}\text{NV}^-$ result [R258], where $^{15}\text{NV}^-$ denotes a negatively charged nitrogen-vacancy center containing the nitrogen-15 isotope, directly demonstrates that C , complex formation, Q , realization of the required charge state, and defect detection remain stochastic.

- Do not infer charge-state or spin yield from optical brightness alone. A fluorescence threshold can exclude optically dark sites, but this measurement does not by itself establish charge stability during control, the spin-coherence time T_2 , gate fidelity, or the absence of nearby optically dark spins.
- Do not treat annealing as restoring an undamaged system without changing defect positions. Annealing, meaning thermal processing that activates defect diffusion and reactions, mobilizes the vacancies required for defect creation and modifies implantation damage. It can also broaden the distribution of capture positions, cause vacancies to be lost at a surface, or produce unwanted defect aggregates [R206].
- Do not combine record values obtained from incompatible samples as though they characterized a single process. The 96% laser-feedback yield [R203], the 1–2 nm delta layer [R204], the 32 nm SiV endpoint spread [R190], where SiV denotes a silicon-vacancy center, and a separate long-coherence sample do not describe one fabrication line.
- Do not substitute single-site yield for full-array yield. If all N sites are indispensable and statistically independent, the array yield is

$$Y_N = p_{\text{use}}^N.$$

Here, p_{use} is the probability that an individual site is usable, and Y_N is the probability that all N sites are usable. If site outcomes are correlated, the array yield requires a measured joint probability model; the exponent cannot simply be omitted.

- Do not represent all spatial disorder by a Gaussian standard deviation. A Gaussian standard deviation describes the width of a normal distribution, but ion channeling and defect diffusion can produce non-Gaussian tails. Because dipolar coupling scales as r^{-3} , where r is the separation between two centers, rare pairs with small separations can dominate the coupling distribution. A complete disorder specification should include three-dimensional endpoint coordinates, systematic biases, covariance, tail quantiles, missing and additional centers, and statistics for optically dark defects.
- Do not transfer diamond-processing results directly to another host material. The result for 4H-SiC, the 4H polytype of silicon carbide, in [R205] depends on its specific implanted ion, implantation energy, vacancy dynamics, charge-state physics, and annealing behavior. Sapphire or hBN, hexagonal boron nitride, requires a new conditioned accounting of these quantities rather than a relabeling of diamond data.

A lithography file records commanded coordinates. The Hamiltonian model — the mathematical representation of the system's energies and interactions — must instead be built from the joint distribution of the centers actually created, brought into the required charge state, seen in fluorescence, qualified for spin properties, and positionally stable. Commanded coordinates and realized physical system are distinct data sets.

Conceptual and quantitative checks

- **What quantity is determined by deterministic implantation?**

Deterministic implantation fixes the number of delivered ions when individual arrivals are prepared or detected. Stopping point, defect-complex formation, charge state, optical usability, and spin quality stay undetermined by that count.

- **Derive the site-position requirement for a 20 nm dipolar graph with 20% radial coupling scatter.**

A dipolar graph carries dipolar couplings on its edges. First-order uncertainty propagation for $J \propto r^{-3}$ gives

$$\sigma_J/J \approx 3\sqrt{2} \sigma_{\text{site}}/r.$$

Here, J is the coupling strength, σ_J is its standard deviation, σ_{site} is the standard deviation of each site position, and r is the nominal separation. Set the fractional coupling scatter on the left side to 0.20 and use $r = 20$ nm. Then

$$\sigma_{\text{site}} \leq 0.20 \times 20/(3\sqrt{2}) = 0.94 \text{ nm}.$$

Thus, under the assumptions of independent site-position errors and first-order propagation, the site-position standard deviation must satisfy $\sigma_{\text{site}} \lesssim 0.94$ nm.

- **What is the principal spatial advantage of delta doping?**

Delta doping incorporates dopants within a very thin layer during growth, which confines them strongly in depth. By itself it assigns no lateral coordinates.

- **What physical effects are omitted when annealing is treated as restoring the initial positions?**

Vacancies turn mobile during annealing, so capture positions broaden, surfaces drain vacancies from the active region, and unwanted aggregates form. The 300 nm single-vacancy diffusion bound for an 800 °C anneal proves by existence that “damage erased” and “position unchanged” cannot both hold.

- **Calculate the nine-site array yield for independent $p_{\text{use}} = 0.90$, and determine the single-site yield required for a 50% array target.**

Under the assumption of statistically independent sites,

$$Y_9 = 0.90^9 \approx 0.387.$$

The probability that all nine indispensable sites are usable is therefore about 38.7%. Inverting the same relation for a target array yield of 0.5 gives

$$p_{\text{use}} \geq 0.5^{1/9} = 0.926.$$

Thus, the required single-site usability is at least 92.6%.

- **What data are required as inputs to a Hamiltonian simulation?**

A Hamiltonian simulation should start from sampled three-dimensional center coordinates and species, flags for missing and extra sites, charge-state and spin-acceptance variables, covariance and distribution tails, and the resulting spreads of intended and unintended couplings. A nominal lattice pitch alone leaves out the disorder the simulation must resolve.

Sources

- [R202] P. Spinicelli et al., “Engineered arrays of NV color centers in diamond based on implantation of CN[−] molecules through nanoapertures,” *New Journal of Physics* 13, 025014 (2011). DOI: [10.1088/1367-2630/13/2/025014](https://doi.org/10.1088/1367-2630/13/2/025014); [arXiv:1008.1483](https://arxiv.org/abs/1008.1483).
 - [R190] T. Schröder et al., “Scalable focused ion beam creation of nearly lifetime-limited single quantum emitters in diamond nanostructures,” *Nature Communications* 8, 15376 (2017). DOI: [10.1038/ncomms15376](https://doi.org/10.1038/ncomms15376); [stable full text](#).
 - [R204] K. Ohno et al., “Engineering shallow spins in diamond with nitrogen delta-doping,” *Applied Physics Letters* 101, 082413 (2012). DOI: [10.1063/1.4748280](https://doi.org/10.1063/1.4748280); [arXiv:1207.2784](https://arxiv.org/abs/1207.2784).
 - [R207] S. Kim et al., “Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures,” *Nature Communications* 16, 9803 (2025). DOI: [10.1038/s41467-025-64758-4](https://doi.org/10.1038/s41467-025-64758-4).
 - [R203] Y.-C. Chen et al., “Laser writing of individual nitrogen-vacancy defects in diamond with near-unity yield,” *Optica* 6, 662–667 (2019). DOI: [10.1364/OPTICA.6.000662](https://doi.org/10.1364/OPTICA.6.000662).
 - [R206] P. Räcke, L. Pietzonka, J. Meijer, D. Spemann, and R. Wunderlich, “Vacancy diffusion and nitrogen-vacancy center formation near the diamond surface,” *Applied Physics Letters* 118, 204003 (2021). DOI: [10.1063/5.0046031](https://doi.org/10.1063/5.0046031).
 - [R205] J. Wang et al., “Efficient generation of an array of single silicon-vacancy defects in silicon carbide,” *Physical Review Applied* 7, 064021 (2017). DOI: [10.1103/PhysRevApplied.7.064021](https://doi.org/10.1103/PhysRevApplied.7.064021).
 - [R258] K. Groot-Berning, G. Jacob, C. Osterkamp, F. Jelezko, and F. Schmidt-Kaler, “Fabrication of ¹⁵NV[−] centers in diamond using a deterministic single ion implanter,” *New Journal of Physics* 23, 063067 (2021). DOI: [10.1088/1367-2630/ac0753](https://doi.org/10.1088/1367-2630/ac0753); [arXiv:2101.01979](https://arxiv.org/abs/2101.01979).
 - [R259] A. Persaud, J. A. Liddle, T. Schenkel, J. Bokor, T. Ivanov, and I. W. Rangelow, “Ion implantation with scanning probe alignment,” *Journal of Vacuum Science & Technology B* 23, 2798–2800 (2005). DOI: [10.1116/1.2062628](https://doi.org/10.1116/1.2062628).
 - [R260] S. Pezzagna, B. Naydenov, F. Jelezko, J. Wrachtrup, and J. Meijer, “Creation efficiency of nitrogen-vacancy centres in diamond,” *New Journal of Physics* 12, 065017 (2010). DOI: [10.1088/1367-2630/12/6/065017](https://doi.org/10.1088/1367-2630/12/6/065017).
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Chapter 35 — Addressing and readout in dense defect arrays

A focused green laser excites a diamond defect, and a detector records the red fluorescence that follows. For one isolated defect, a specified preparation-and-readout protocol converts the photon-count statistics into a spin-state estimate.

Once defects sit close enough to interact, one optical spot gathers light from several neighbors at once, and a microwave field aimed at one target also reaches into neighboring defects and drives their transitions.

Shifting one site's transition frequency apart from the rest gives spectral selectivity: the drive couples resonantly at that site and remains detuned elsewhere. Any shift left over after the control remains as static disorder, a time-independent site-to-site variation of the Hamiltonian. The architecture therefore needs a quantitative way to pick out one cluster while keeping the dense coupling that justifies building the array.

Distinct requirements for initialization, selection, control, measurement, and calibration

Control of one isolated defect under a microscope does not establish control of an array. In a dense region each laser spot covers several neighboring defects, each microwave line spills its field past the intended site, and each imposed frequency shift rewrites the Hamiltonian whose uniformity the proposal relies on.

Five functions must be distinguished.

- Initialize the charge state, physical spin state, and encoded-cluster state.
- Select a site or cluster by position or frequency.
- Drive a specified physical or encoded transition.
- Measure the selected degree of freedom with a stated degree of destructiveness.
- Calibrate the controls without replacing the many-body model under investigation with a different effective model.

Whatever frequency differences distinguish the sites turn into on-site disorder once the selecting control switches off. The architecture must carry that tradeoff explicitly.

Previous chapters assembled localized defect states, coherent coupling, a low-energy cluster doublet, a suitable interaction graph, effective multi-spin terms, and a many-body gap. That whole inventory still leaves open which cluster a given control pulse actually acts on.

In an analog realization of a topological Hamiltonian the control knobs reach directly into the physical system. A magnetic-field gradient shifts the Zeeman energies, the spin-dependent part of each spin's energy in the field.

Strain and electric fields move optical and spin transition frequencies and can reshape encoded wavefunctions. A focused optical reset aimed at one site can ionize its neighbors, warm the structure, or project their quantum states.

Local control thus disturbs the very symmetry and spatial uniformity meant to support the phase.

[Experiment] Optical initialization, coherent microwave control, and optical readout are mature techniques for isolated diamond nitrogen-vacancy (NV) centers, and multi-spin registers associated with individual centers have been controlled [R074]; [R119]. [Proposal] Combining those capabilities with dense defect clusters and an intrinsic topological Hamiltonian remains an architectural proposal rather than a demonstrated machine.

Redrawing a successful single-center register as a repeated schematic does not make it a lattice. It stays a register until lattice-scale interactions and controls are demonstrated.

Spatial, spectral, internal-state, and temporal selection

Picking one subsystem out of a globally driven array needs a label that sets it apart: position, transition frequency, timing, polarization, or a dedicated control line.

In a quantum array that label usually enters the Hamiltonian as an energy term. Giving cluster i its own frequency ω_i makes its energy site-dependent, which breaks the site equivalence the model assumes.

Four common selection mechanisms are used.

- Spatial selection focuses light or a near field on one position.
- Spectral selection drives only the transition resonant with a chosen frequency.
- Internal-state selection uses polarization or selection rules to address a particular orientation or transition.
- Temporal selection uses pulse sequences to refocus unwanted evolution.

Practical devices stack these mechanisms: a moderate gradient spreads the transition frequencies, a shaped microwave pulse narrows the driven bandwidth, and an optical channel handles readout.

Each selection method needs a quantitative error budget. Calling a device “individually addressable” specifies nothing technical until the associated errors are reported.

When two sites fall inside the same driven bandwidth, the spectator excites even under a field concentrated near the target. A selector therefore must be quoted as a quantitative spatial or spectral window, since a beam profile alone leaves the driven bandwidth unspecified.

Off-resonant excitation of a neighboring spectator

Give target cluster C an encoded two-level transition at angular frequency ω_C in radians per second: the two states chosen to stand for the cluster's effective low-energy degree of freedom. Give the undriven neighboring spectator S the transition frequency ω_S . Their detuning, the spectator-minus-target frequency difference, is

$$\Delta_S = \omega_S - \omega_C.$$

Drive at resonance with C , keeping only the drive component that stays phase-matched to the spin transition. Call the target Rabi rate Ω , again in radians per second: the speed at which a resonant coherent drive rotates the target two-level state. A square pulse lasting

$$t_\pi = \frac{\pi}{\Omega}$$

rotates the target by π . The dimensions are consistent because radians are dimensionless, so $(\text{s}^{-1})^{-1} = \text{s}$.

For an ideal two-level spectator initially in its lower state, the maximum off-resonant excitation probability under a constant drive is bounded by

$$P_{S,\max} = \frac{\Omega^2}{\Omega^2 + \Delta_S^2} \approx \left(\frac{\Omega}{\Delta_S} \right)^2 \quad \text{when } |\Delta_S| \gg \Omega.$$

This unintended excitation of the neighbor is crosstalk — an operation error suffered by a spectator, quantified by the driven transition probability rather than by the beam width alone.

Suppose the design permits at most 10^{-3} spectator excitation from this mechanism. The ideal bound then requires

$$\frac{|\Delta_S|}{\Omega} \gtrsim \sqrt{10^3} \approx 31.6.$$

This dimensionless ratio sets an ideal requirement, not a guaranteed device performance level.

Further errors come from pulse edges, extra energy levels, inhomogeneous broadening — the spread of transition frequencies across nominally equivalent systems — calibration drift, and interactions. Smooth pulses trim the spectral sidelobes, the unwanted frequency content outside the main pulse bandwidth, at the price of longer pulse durations.

The cluster also hosts unwanted excited states. Write Δ_{leak} for the energy gap in joules from the encoded doublet to the nearest leakage state, a state outside the intended encoded two-level subspace. Its angular frequency is $\omega_{\text{leak}} = \Delta_{\text{leak}}/\hbar$ in radians per second. Selective control needs a parameter window of the form

$$\Gamma_C \ll \Omega \ll \min(|\Delta_S|, \omega_{\text{leak}}),$$

where Γ_C is the target linewidth in radians per second, the spectral width of that transition. The left inequality forces the pulse to outrun and out-resolve the linewidth; the right inequalities hold down spectator excitation and leakage from the encoded subspace.

This interval is the available selectivity bandwidth. Where no Ω satisfies the inequalities, no pulse optimization opens a valid control window.

Faster gates require larger Ω ; spectral selectivity and leakage suppression require smaller Ω . Longer control times accumulate more decoherence, the loss of quantum phase coherence into uncontrolled degrees of freedom. A viable gate needs all three inequalities at once.

Charge-state, physical-spin, and encoded-state initialization

A defect cluster can demand three distinct initialization layers.

Charge state and structural configuration. The intended optical and spin levels exist only when each defect occupies the required charge state and structural configuration.

Optical illumination prepares the charge state and disturbs it. A bright optical spot therefore does not establish that every cluster member sits in the correct charge state.

Physical-spin state. Spin-dependent intersystem crossing — a nonradiative jump between electronic manifolds of different spin character — allows nonresonant optical pumping of an NV electronic spin.

Resonant cryogenic protocols can provide more selective preparation [R074]. These initialization mechanisms depend on the physical platform.

A group-IV vacancy or a silicon carbide (SiC) defect answers to a different level structure with different temperature requirements.

Encoded cluster state. Polarizing every constituent spin need not prepare any chosen eigenstate of the interacting cluster.

Preparation can call for physical-spin rotations, dissipative pumping — controlled coupling to an environment that favors the desired state — measurement and feedback, or an adiabatic ramp from an easier Hamiltonian, changed slowly enough for the system to track an instantaneous eigenstate. Success then means overlap with the desired encoded state, not fluorescence brightness of one constituent.

Initialization must also fit inside an allotted slot of the experimental sequence, because resetting one constituent of a prepared correlated many-body phase is a local dissipative act. That act can create excitations and strip entanglement, so a global reset before preparation intrudes far less than an arbitrary local reset mid-run.

Polarizing every physical spin and stopping there leaves a product state, which can overlap the encoded doublet barely at all. Optical brightness does not measure that overlap.

Limits of optical spatial and spectral selection

Optical control selects by position, transition frequency, polarization, or coupling to a photonic mode. Ordinary far-field focusing is diffraction-limited — diffraction fixes a minimum spot size — so several defects spaced by nanometres share one optical spot. Super-resolution methods exploit a nonlinear optical response to shrink the effective point-spread function, the spatial response of the imaging system to a point emitter.

[Experiment] Pezzagna and colleagues combined implantation with stimulated-emission-depletion microscopy to optically distinguish closely spaced NV centers [R235]. This result provides evidence for nanoscale optical selection. It does not establish simultaneous low-crosstalk control of a many-body cluster lattice.

Spectral optical addressing tells defects apart by their zero-phonon-line frequencies — optical transitions that neither create nor annihilate lattice vibrations. At cryogenic temperature, where the resonant lines narrow, that separation supports spin-selective excitation and single-shot protocols.

Spectral addressing also answers to spectral diffusion — the time-dependent wander of a transition frequency — plus strain variation, electric-field noise, and charge rearrangement. Nanophotonic fabrication that improves photon collection can simultaneously shift and broaden the emitter lines.

[Experiment] Integrated diamond nanophotonics has coupled multiple silicon-vacancy centers to optical modes, and two emitters in a cavity have shown photon-mediated interactions after spectral tuning [Ro86]; [R238]. Separately, a 2025 platform optically resolved and manipulated more than 100 NV centers in parallel while using shared coherent control [R239].

These results advance routing and multiplexing. A uniform, strongly interacting cluster lattice stays undemonstrated.

Optical frequency differences that label emitters for photon routing turn into undesirable disorder wherever the corresponding orbital states mediate spin interactions.

Tuning emitters into mutual resonance serves protocols needing indistinguishable photons; tuning them apart serves frequency-selective local addressing. The operating schedule must name the required configuration at each stage.

Resolving an optical line need not resolve a cluster: one cluster can carry several transitions and several defects can share one line. Assigning a line to a cluster therefore takes controlled perturbations that test each candidate's response.

Global microwave fields and local frequency selection

Microwave wavelengths dwarf typical defect separations, so a conventional antenna bathes the array in a nearly global magnetic field. Individual selection must then come from frequency differences, local near-field conductors, magnetic-field gradients, or pulse refocusing.

[Experiment] A microcoil-generated gradient of about 0.1 G nm^{-1} frequency-encoded four NV sites separated by about 100 nm , enabling site-selective Rabi control; each site contained multiple NVs at smaller separations [R236]. Here G denotes gauss, with $1 \text{ G} = 10^{-4} \text{ T}$, and nm denotes nanometre. These values and this geometry were obtained in a sensing-oriented device rather than a demonstrated strongly coupled topological patch.

A local static gradient shifts Zeeman splittings. A local wire delivers a stronger near field at the price of wiring density, dissipation, fabrication variation, and possible magnetic noise. Frequency multiplexing — controlling multiple sites through distinct frequency channels — trims the wire count, provided each channel spacing clears the linewidths, drive bandwidth, frequency drift, and interaction-induced shifts.

For an encoded cluster the microwave field acts through a physical operator projected into the encoded subspace. With P_C the projector onto the encoded doublet and S_j^x the transverse spin operator of constituent j , the encoded drive follows

$$P_C \left(\sum_j g_j S_j^x \right) P_C,$$

where g_j is the local field coupling and has units of energy if the drive Hamiltonian is written directly in joules. For a highly symmetric cluster under a spatially uniform field, the matrix element of this operator between the logical states can vanish.

A vanishing matrix element shields the encoded states from uniform magnetic noise and simultaneously bars a uniform microwave field from driving the desired gate. Local gradients or deliberately asymmetric couplings reopen a nonzero control matrix element, at the risk of fresh noise and leakage channels. The same vanishing element suppresses the noise coupling and the control coupling together, so a protocol must compare both rates quantitatively.

Measurement back-action and readout volume

Room-temperature NV readout generally converts spin-dependent optical dynamics into different fluorescence statistics and estimates the spin state by averaging over many repetitions [Ro74]. Cryogenic resonant excitation can provide single-shot, projective electron-spin readout, as demonstrated for an NV center [R199]. A projective readout maps the system onto an eigenstate associated with the measured outcome. Spin-to-charge conversion maps spin information onto a longer-lived charge-state distinction and can improve the photon budget under suitable conditions [R237].

Each method carries a measurement volume — the region the detector learns from — and a back-action volume — the region the measurement can disturb. Excitation light can pump neighboring defects, flip their charge states, and heat a nanostructure.

Photons from several centers can land on the same detector. Spectral filters, confocal or super-resolution collection, optical cavities, and separate waveguides shrink that overlap without removing measurement back-action.

Three distinct measurement claims must be separated.

- **Destructive local readout:** The target state can be lost, while effects on neighboring systems are quantitatively bounded.
- **Repeatable local readout:** The target can be measured repeatedly with a stated quantum-nondemolition fidelity. A quantum-nondemolition measurement is designed to preserve the measured observable so that it can be measured again.
- **Phase-preserving many-body readout:** The measurement extracts the intended observable without uncontrolled projection or excitation of the surrounding phase.

Strong single-defect precedents support the first claim. The third claim stays an architectural requirement, [Proposal] for the defect-cluster topological system here: ordinary fluorescence readout of one spin is not automatically a stabilizer measurement — an operator whose eigenvalue diagnoses an encoded state — and it does not directly report a logical topological charge.

A bright photon stream alone does not establish a nondemolition loop measurement. The measured operator and the repeatability need specifying independently of the detector signal.

Calibration quantities and scaling

A scalable calibration record holds far more than one resonance frequency per defect and encoded cluster:

- charge-state preparation and survival probabilities;
- spin initialization and readout confusion matrices;
- microwave and optical transition frequencies and linewidths;
- Rabi-rate response versus control amplitude;
- pulse phase and timing offsets;
- leakage spectra and encoded-state matrix elements;
- pairwise interaction shifts;
- optical and microwave crosstalk matrices; and
- drift versus time, temperature, and preceding illumination.

A readout confusion matrix tabulates the conditional probability of each reported outcome for each prepared state. An unconstrained N -cluster crosstalk matrix holds $N(N - 1)$ off-diagonal entries, so exhaustive measurement grows quadratically with cluster count. Verified locality and repeated device geometry can lighten that calibration load.

Calibration still consumes experimental repetitions: practical runs spend sacrificial and interleaved reference shots tracking common drift alongside local deviations.

A crosstalk value is no beam diameter. It must name the affected subsystem, the unwanted observable or operation, the control sequence, the neighboring state, and the measurement conditions; without all five, the reported value defies independent audit.

Global and local control architectures

Global pulses need fewer control channels and can respect lattice symmetries, suiting them to polarization, echo protocols, and repeated bulk sequences. They cannot pick out individual defects or correct local errors unless the Hamiltonian maps those errors onto collective signals.

Local controls add routing and calibration flexibility together with disorder, wiring, heat, and crosstalk. A hybrid architecture of global bulk operations plus sparse local controls is [Proposal]: specifying it does not establish that sparse controls suffice.

Applying the same programmed pulse to every cluster can digitally or Floquet-engineer an effective evolution — periodic driving shaped into an effective time-averaged Hamiltonian. That driven evolution does not demonstrate an intrinsic topological phase in the undriven material.

Local optical readout of physical defect spins likewise does not report a logical topological charge by itself. A complete control specification names the physical operator, the encoded operator after projection, and whether the phase survives only under active control.

Competing requirements for uniformity and addressability

Let σ_ω denote the root-mean-square spread of cluster transition angular frequencies when all addressing controls are off. Let J_{eff} be a characteristic intended intercluster coupling energy and Δ_{mb} the many-body energy gap, both measured in joules. A rough uniformity requirement is

$$\hbar\sigma_\omega \ll \min(|J_{\text{eff}}|, \Delta_{\text{mb}}).$$

The left-hand side has units $(\text{J s})(\text{s}^{-1}) = \text{J}$, so the comparison is dimensionally valid. Spectral addressability instead requires neighboring transition-frequency separations to exceed the driven bandwidth and the relevant linewidths:

$$|\omega_i - \omega_j| \gg \max(\Omega, \Gamma_i, \Gamma_j).$$

Permanent random detunings generally fail both inequalities together wherever the transition energy enters the target Hamiltonian. Candidate resolutions exist, each conditional:

- apply a switchable local shift only during control and return the system to a uniform idle point;
- encode the logical transition so that its addressing frequency changes while the relevant coefficient of the static Hamiltonian does not;
- compensate known offsets in a rotating frame or with echo pulses, thereby accepting active rather than passive operation; or
- use spatial selection without introducing large spectral disorder.

Switching a control field is itself nonideal: turning a local shift on and off can excite leakage states, pile up unknown phases, or close the local many-body gap. A symmetry-protected encoded state can further go dark to the control operator — its control matrix element vanishes.

The test that matters is a verified round trip: from a uniform idle Hamiltonian out to a selective operation and back to the same idle Hamiltonian, with errors below the phase's tolerance. Skipping the return leg strands the system under a different Hamiltonian than the one under investigation.

Nine-stage prepare–probe–read schedule for one cluster

Consider a target cluster C within a dense region of spectator clusters. The following schedule describes a prepare–probe–read experiment. It does not assume that fluorescence can reset or measure C nondestructively during an established topological state.

Stage	Control action	Required observation or bound	Phase status
0. Sacrificial map	In separate calibration shots, sweep weak microwave and optical probes and fit ω_C , the linewidth, leakage transitions, neighboring detunings, and the readout confusion matrix.	Obtain stable confidence intervals over the planned run time. Measure the crosstalk matrix rather than inferring it from the beam size.	The target phase is not claimed.
1. Global reset	Prepare the charge states and optically polarize all clusters with the platform-specific sequence. Where possible, herald or reject runs containing incorrect charge states.	Report the per-cluster and whole-patch initialization yields separately.	No many-body phase has yet been prepared.
2. Encoded preparation	Apply calibrated global and local pulses, or an adiabatic cluster ramp, to place each cluster in its low-energy encoded state.	Bound leakage from C and representative spectators by spectroscopy.	The cluster encoding is prepared, but topology has not yet been established.
3. Hamiltonian ramp	Turn on the intended intercluster Hamiltonian with a globally specified ramp.	Justify the ramp time using the measured finite-patch spectrum and coherence window.	A candidate many-body state is prepared.
4. Select C	Apply a reversible local Stark, strain, or Zeeman shift $\delta_{\text{sel}}(t)$, or activate a calibrated gradient. Use a smooth temporal envelope.	During the pulse, require $ \delta_{\text{sel}} \gg \Omega, \Gamma$ for spectators. After the pulse, require the residual shift to satisfy the uniformity budget.	The local Hamiltonian is perturbed, and induced excitations must be counted.
5. Drive C	Apply a shaped encoded microwave pulse with area $\int \Omega(t)dt = \pi$ for a π rotation, or use the smaller angle required by the probe.	Require $\Omega \ll \omega_{\text{leak}}$. Verify that the measured spectator error and target leakage satisfy the experiment's error budget.	The operation produces an intentional local excitation or rotation but is not automatically a logical topological gate.
6. Restore and refocus	Reverse the selection envelope and use a calibrated echo only if it preserves the intended interaction terms. Track the dynamical phase $\phi_C = \int \delta_{\text{sel}}(t)dt$.	Verify that residual frequency and interaction changes return within idle tolerances. Compensate ϕ_C or include it in the model.	The candidate bulk Hamiltonian is restored.
7. Evolve	Allow the patch to evolve for the protocol's dwell time. Use global refocusing only when its effective Hamiltonian has been derived.	Keep the total schedule shorter than the relevant encoded coherence time, and bound control-induced heating and drift.	The target dynamics are under test.
8. Read at endpoint	Stop or reverse the phase-preparation ramp if required. Then map the encoded observable onto a readable physical spin and perform optical or spin-to-charge readout.	Report the target confusion matrix and changes in neighboring states. Repeat the procedure over many shots.	Readout can be destructive, and no survival of the phase is claimed afterward.

Each invasive operation sits either before phase preparation or on the record as an explicit perturbation. Claiming nondestructive control inside the phase would take evidence that stages 4–6 neither close the local gap nor breed uncontrolled quasiparticles — emergent excitations of the interacting many-body system — and that stage 8 reads the intended encoded observable through an ancilla, an auxiliary quantum subsystem mediating the measurement, or an equivalent channel. No such evidence exists for this architecture yet.

Experimentally demonstrated capabilities and remaining integration requirements

The strongest available evidence is modular: individual capabilities demonstrated separately, not yet integrated into one device. A nitrogen-vacancy (NV) center is a diamond point defect — a substitutional nitrogen atom beside a vacant lattice site — whose spin state optical preparation initializes and fluorescence readout infers from emitted light.

Capability	Status through August 2026	What it establishes	What it does not establish
Optical spin preparation and fluorescence readout of single NV centers [R074]	[Experiment]	A single defect can be initialized and observed.	Dense-cluster, phase-preserving reset/readout, in which initialization or measurement retains the relevant quantum phase relations.
Nanoscale optical distinction of implanted NVs [R235]	[Experiment]	Far-field super-resolution, meaning optical resolution beyond the ordinary diffraction limit, can distinguish selected centers.	Parallel control with low measurement or control back-action at the spacing required for a strongly interacting lattice.
Four-site gradient/frequency encoding [R236]	[Experiment]	A field gradient can assign site-dependent resonance frequencies, enabling microwave site selection and coherent control in a small NV array.	Uniform interacting clusters or topological dynamics, meaning dynamics governed by the system’s intended topological many-body structure.
Entanglement of two separated NV electronic spins [R080]	[Experiment]	Selective control can coexist with a measured two-spin interaction in a small device. Entanglement is a nonseparable quantum correlation between the two spins.	Scaling to a dense, calibrated many-body patch.
Cryogenic single-shot NV readout and spin-to-charge conversion [R199] ; [R237]	[Experiment]	Measurement primitives stronger than averaged fluorescence exist. Single-shot readout determines a state from one experimental realization, while spin-to-charge conversion maps spin information onto a charge state before detection.	Neighbor-safe logical topological measurement.
Ten-qubit register around one NV [R119]	[Experiment]	Sophisticated calibration and control have been demonstrated for a local electron–nuclear register, which combines an electronic spin with nearby nuclear spins.	A lattice of equivalent defect clusters with intercluster topology.
Multi-emitter nanophotonics [R086] ; [R238]	[Experiment]	Optical routing, spectral tuning, and integration of multiple	Intrinsic topological order in a defect array.

Capability	Status through August 2026	What it establishes	What it does not establish
		emitters into nanoscale photonic structures are advancing.	
Parallel control and readout of more than 100 resolved NVs [R239]	[Experiment]	Spatially selective optical manipulation and shared coherent control scale beyond a few sites.	Independent local microwave control or a strongly interacting uniform lattice.

The missing integrated demonstration matters. Methods tuned for isolated emitters tend to space defects too widely for strong direct coupling, while dense defect creation broadens spectral lines until assigning observed transitions to specific defects or clusters turns ambiguous.

Frequency gradients sharpen site selection while splitting sites meant to be identical. Fabrication tolerances, interaction strengths, optical performance, microwave-control constraints, thermal requirements, and calibration resources therefore need joint modeling rather than independent optimization: the target Hamiltonian — the operator specifying the system's energies and interactions — must stay compatible with all of them at once.

Common interpretive and design errors

- **A spectrally resolved line does not imply a resolved cluster.** Several transitions from one cluster may be visible, and several defects may contribute to the same spectral line. Reliable assignment therefore requires controlled perturbations that test how each candidate transition responds.
- **Disorder is not a reliable addressing mechanism.** Accidental site-dependent frequency shifts can label individual sites, but those labels drift and modify the target Hamiltonian.
- **Optical spot size does not quantify crosstalk.** Crosstalk is the unintended effect of a control operation on a neighboring subsystem. It must be measured through unwanted rotations, phase shifts, charge-state changes, or Hamiltonian shifts on neighboring sites.
- **Optical readout is not necessarily nondestructive.** A readout protocol must specify both its repeatability and the observable being measured. Nondestructive measurement requires that repeated measurements preserve the relevant state or observable to the stated accuracy.
- **Control of physical spins does not imply control of encoded states.** The applied drive must be projected into the logical doublet, meaning the two-dimensional encoded subspace used as a logical qubit, and leakage out of that subspace must be bounded.
- **Compensation constitutes active operation rather than passive protection.** Continuous echo sequences or site-specific corrections may synthesize useful dynamics, but their success does not demonstrate that the device is intrinsically protected without active intervention.
- **Symmetry breaking can remove protection.** A gradient that enables addressing may also lift a required degeneracy or eliminate cancellation of uniform noise. A complete gate protocol must therefore include restoration of the symmetric idle point after the addressed operation.

Analytical consistency checks

- **Permanent site-dependent frequencies generally cannot provide both selective addressing and a uniform Hamiltonian.**

Selective addressing requires

$$|\omega_i - \omega_j| \gg \max(\Omega, \Gamma_i, \Gamma_j).$$

Here, ω_i and ω_j are the transition angular frequencies of sites i and j , Ω is the drive's Rabi frequency, and Γ_i and Γ_j are the corresponding linewidths. This inequality states that the frequency separation must greatly exceed both the drive scale and the spectral widths.

Uniform many-body physics may instead require

$$\hbar\sigma_\omega \ll \min(|J_{\text{eff}}|, \Delta_{\text{mb}}).$$

Here, $\hbar$ is the reduced Planck constant, σ_ω characterizes the spread of site frequencies, J_{eff} is the effective interaction energy, and Δ_{mb} is the many-body energy gap. This condition requires disorder energy to remain much smaller than the relevant interaction and gap scales. Permanent random detunings generally cannot satisfy both requirements when the transition energy appears in the target Hamiltonian.

- **The ideal two-level leakage bound $P_{S,\text{max}} \leq 10^{-3}$ requires $|\Delta_S|/\Omega \gtrsim 31.6$.**

Let $P_{S,\text{max}}$ denote the maximum population transferred to a spectator transition, and let Δ_S be that transition's detuning from the applied drive. Under the far-detuned assumption $|\Delta_S| \gg \Omega$,

$$P_{S,\text{max}} \approx (\Omega/\Delta_S)^2.$$

Setting the right-hand side to 10^{-3} and inverting gives

$$|\Delta_S|/\Omega \gtrsim \sqrt{10^3} \approx 31.6.$$

This result applies to the stated ideal two-level bound.

- **Initialization of every physical spin does not necessarily initialize the encoded state.**

The encoded state of an interacting cluster is a particular collective eigenstate or subspace of the cluster Hamiltonian. A product state in which the constituent spins are individually polarized may have poor overlap with that collective encoded state.

- **A crosstalk specification requires more than beam geometry.**

A quantitative crosstalk value must identify the victim subsystem, the unwanted observable or operation, the applied control sequence, the state of the neighboring subsystem, and the measurement conditions. Beam geometry alone does not determine the resulting operation error.

- **A uniform field may fail to drive the logical transition of a highly symmetric encoded cluster.**

Let P_C be the projector onto the encoded cluster subspace, g_j the coupling of physical spin j to the drive, and S_j^x the x -component spin operator for that site. The drive restricted to the encoded subspace is

$$P_C(\sum_j g_j S_j^x)P_C.$$

If all g_j are equal and the logical states transform oppositely under a symmetry that changes the sign of the summed operator, the logical transition matrix element vanishes. The same symmetry that protects the encoded states from uniform noise can therefore make the logical transition inaccessible to a uniform microwave field.

- **Driving one encoded cluster does not generally implement a topological logical gate.**

A local operator may create quasiparticles, which are collective excitations of the many-body system, or may probe a local degree of freedom. A logical topological operation instead requires the appropriate nonlocal or braided process, together with evidence that the topological phase survives the operation.

Control operations charge physical costs and constraints. Initialization must separately establish charge state, physical-spin state, and encoded-cluster state. A selective drive needs a nonempty bandwidth interval between the linewidth scale and the nearest spectator-transition or leakage scale. Crosstalk means error in the implemented operation, not a geometric property read off a beam image. Optical, microwave, spectral, and local selectors all perturb the device: global protocols preserve symmetry and save wiring without providing arbitrary routing, while local protocols gain routing flexibility at the cost of added disorder and calibration overhead. Reading a local physical spin still does not constitute logical topological readout.

Sources

- [R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* 528, 1–45 (2013). DOI: [10.1016/j.physrep.2013.02.001](https://doi.org/10.1016/j.physrep.2013.02.001); arXiv: [1302.3288](https://arxiv.org/abs/1302.3288).
- [R235] S. Pezzagna, D. Wildanger, P. Mazarov et al., “Nanoscale Engineering and Optical Addressing of Single Spins in Diamond,” *Small* 6, 2117–2121 (2010). DOI: [10.1002/sml.201000902](https://doi.org/10.1002/sml.201000902).
- [R236] H. Zhang, K. Arai, C. Belthangady, J.-C. Jaskula, and R. L. Walsworth, “Selective addressing of solid-state spins at the nanoscale via magnetic resonance frequency encoding,” *npj Quantum Information* 3, 31 (2017). DOI: [10.1038/s41534-017-0033-3](https://doi.org/10.1038/s41534-017-0033-3); arXiv: [1701.01154](https://arxiv.org/abs/1701.01154).
- [R080] F. Dolde, I. Jakobi, B. Naydenov et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* 9, 139–143 (2013). DOI: [10.1038/nphys2545](https://doi.org/10.1038/nphys2545); arXiv: [1212.2804](https://arxiv.org/abs/1212.2804).
- [R199] L. Robledo, L. Childress, H. Bernien et al., “High-fidelity projective read-out of a solid-state spin quantum register,” *Nature* 477, 574–578 (2011). DOI: [10.1038/nature10401](https://doi.org/10.1038/nature10401); arXiv: [1108.1196](https://arxiv.org/abs/1108.1196).

- [R237] B. J. Shields, Q. P. Unterreithmeier, N. P. de Leon, H. Park, and M. D. Lukin, “Efficient Readout of a Single Spin State in Diamond via Spin-to-Charge Conversion,” *Physical Review Letters* 114, 136402 (2015). DOI: [10.1103/PhysRevLett.114.136402](https://doi.org/10.1103/PhysRevLett.114.136402); arXiv: [1410.0370](https://arxiv.org/abs/1410.0370).
 - [R119] C. E. Bradley, J. Randall, M. H. Abobeih et al., “A Ten-Qubit Solid-State Spin Register with Quantum Memory up to One Minute,” *Physical Review X* 9, 031045 (2019). DOI: [10.1103/PhysRevX.9.031045](https://doi.org/10.1103/PhysRevX.9.031045); arXiv: [1905.02094](https://arxiv.org/abs/1905.02094).
 - [Ro86] R. E. Evans, A. Sipahigil, D. D. Sukachev et al., “Photon-mediated interactions between quantum emitters in a diamond nanocavity,” *Science* 362, 662–665 (2018). DOI: [10.1126/science.aau4691](https://doi.org/10.1126/science.aau4691); arXiv: [1807.04265](https://arxiv.org/abs/1807.04265).
 - [R238] A. Sipahigil, R. E. Evans, D. D. Sukachev et al., “An integrated diamond nanophotonics platform for quantum-optical networks,” *Science* 354, 847–850 (2016). DOI: [10.1126/science.aah6875](https://doi.org/10.1126/science.aah6875); arXiv: [1608.05147](https://arxiv.org/abs/1608.05147).
 - [R239] M. Cambria, S. Chand, C. M. Reiter, and S. Kolkowitz, “Scalable Parallel Measurement of Individual Nitrogen-Vacancy Centers,” *Physical Review X* 15, 031015 (2025). DOI: [10.1103/jdzq-jbfz](https://doi.org/10.1103/jdzq-jbfz); arXiv: [2408.11715](https://arxiv.org/abs/2408.11715).
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Chapter 36 — Measurement of topological information

One spin is measured by flashing an optical pulse and collecting the spin-dependent fluorescence. The photon record then classifies the spin under its conventional “up” or “down” label. That procedure reads a specific microscopic degree of freedom with a physical detector.

A proposed topological bit encodes differently. No individual spin holds the whole message; the message lives in a nonlocal sector, one of two globally distinct many-body configurations that agree on every local measurement in a specified set.

Reading such information takes a ladder of operators and protocols. Single-spin readout comes first, then tests of four-neighbor local constraints. Spectroscopy maps the energies a drive can reach. Open string operators ask whether excitations sit at their endpoints. Noncontractible closed loops separate the global sectors, and fusion experiments report the total charge when two excitations combine.

Each rung of that ladder is a different operator. An experimental claim stays incomplete until it names the operator measured.

Measurement hierarchy

A single-site measurement names a local physical state; a four-site parity measurement reports whether a local constraint holds. Neither reveals whether the configuration winds nontrivially around the whole system. That winding takes an extended path measurement, or a comparison of globally distinct configurations.

The hierarchy can be represented as follows:

```
one-site readout -> physical spin or edge label
local parity -> stabilizer / nearby excitation
spectral response -> energy and transition matrix element
open string endpoints -> created or moved charges
closed loop -> enclosed charge or logical sector
fusion experiment -> possible total-charge channels
many-basis data -> tomography or selected correlators
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A quantum measurement pairs an operator with the probability distribution over its outcomes. Put the data system in state ρ , where ρ is the density operator — a positive, unit-trace operator covering pure states and statistical mixtures. For an observable M with eigenvalues m , the ideal projective outcome probabilities are

$$P(m) = \text{Tr}(\rho \Pi_m),$$

where Π_m projects onto the eigenspace of M with eigenvalue m . That is the Born-rule probability of outcome m . A readout protocol carries physical meaning only once a calibrated error model maps the detector record onto these ideal outcomes.

One distinction recurs: local stabilizers versus nonlocal logical operators. A stabilizer is an operator whose specified eigenvalue fixes a local constraint of a quantum code, so a local stabilizer tests whether a small neighborhood obeys one term of the code Hamiltonian.

A nonlocal logical loop separates globally different code states that all satisfy every local stabilizer constraint. Its circuit can repeat the same parity-gate pattern in hardware, but one operator is a local term of the code Hamiltonian and the other is a logical observable.

Every claimed signal needs its measured operator, preparation assumptions, measurement circuit, calibration procedure, error model, and plausible competing explanations on the record. Without them a data set does not identify the underlying physical phenomenon.

Four-edge stabilizer measurement

Follow the toric-code convention: one physical two-level system on every edge of a square lattice with periodic boundary conditions — a two-dimensional Hilbert space per edge, representable as a spin or qubit, with opposite boundaries identified so the lattice forms a torus. For each edge e , write X_e and Z_e for the Pauli operators and define the plaquette operator

$$B_p = \prod_{e \in \partial p} Z_e,$$

where ∂p collects the four edges bounding plaquette p . Its eigenvalues are $b_p = +1$ and -1 . At every vertex s , the star operator is

$$A_s = \prod_{e \ni s} X_e,$$

where $e \ni s$ collects the four edges meeting at s . The ideal toric-code ground space is the joint $+1$ eigenspace of all star and plaquette operators: $A_s = +1$ and $B_p = +1$ for every s and p [Ro30]. [Theory]

The operator B_p is measured with an extra two-level system a , an ancilla initialized in $|0\rangle_a$. It stands outside the encoded data system; its job is recording the value of one parity operator.

For each of the four data edges in ∂p , apply a controlled-NOT gate with the data edge as the control and the ancilla as the target. Then measure the ancilla operator Z_a . In the computational basis, the ancilla records the parity of the four data bits:

$$m_a = z_1 z_2 z_3 z_4 = b_p,$$

where each $z_j = \pm 1$ is the corresponding Z -eigenvalue and m_a is the ancilla outcome. In the ideal circuit, $m_a = -1$ indicates a violated plaquette constraint.

For a coherent superposition the circuit measures the collective operator B_p : the data land in one B_p eigenspace while the four individual Z_e values stay unrevealed, rather than four classical bits read out from pre-existing definite values.

Repeated ancilla-mediated stabilizer extraction is established practice in superconducting surface-code experiments [R230]. [Experiment] This result demonstrates a measurement primitive: repeated measurement of stabilizer operators. It does not establish passive topological order in the processor.

Omitting the ancilla and measuring the four edges individually still yields the plaquette parity on multiplying the outcomes. The individual readouts expose the factors of that product, though, and generally erase coherence in bases incompatible with Z . The circuit thus fixes both the information gained and the back-action inflicted.

Noncontractible logical-loop measurement

Take a closed path C_x winding once around the torus in the x direction. Its logical Z operator is

$$\overline{Z}_x = W_Z(C_x) = \prod_{e \in C_x} Z_e.$$

Here $W_Z(C_x)$ is the Z -type loop supported on the edges of C_x . An ancilla measures it with the same parity-circuit form as for B_p , coupled to every edge along C_x before readout. The outcome $\overline{z}_x = \pm 1$ distinguishes two logical sectors. Since the loop commutes with every stabilizer, states of different $\overline{z}_x$ can agree on all local stabilizer outcomes.

The path topology decides what the loop measures. A contractible closed path C — one that deforms continuously to a point — carries a Z -type loop equal to the product of its enclosed plaquette stabilizers in the ideal code, so it adds no independent logical measurement. Only a noncontractible loop, which no continuous deformation shrinks to a point, defines a logical operator on the torus; in a planar surface code the corresponding logical path must join the right boundaries or encircle a puncture. Length is irrelevant here: topology decides whether the operator stands independent of the local stabilizers.

Measurement	Support	Ideal question	A -1 result means
B_p	four nearby edges	Is this local constraint satisfied?	a local syndrome/excitation is present
$\overline{Z}_x$	a system-spanning noncontractible path	Which logical sector is occupied?	logical eigenvalue -1 , not necessarily a local defect

The two operators take similar hardware, yet B_p is a local term of the code Hamiltonian while $\overline{Z}_x$ is a logical observable. Finding $B_p = +1$ on every plaquette leaves $\overline{Z}_x$ undetermined; measuring $\overline{Z}_x$ once leaves the local constraints unchecked.

Measuring every data edge along the path in the Z basis and multiplying the outcomes estimates the same parity destructively in that shot, at the price of exposing each individual value and generally erasing coherence in incompatible bases. That simpler procedure serves as an effective final readout, but not where the state must stay coherent for a later braid. Length alone never promotes a correlator to a logical loop: a contractible product of plaquette operators stays a stabilizer product, not an independent sector label.

Physical detector records and Pauli observables

A local readout maps a microscopic state onto a classical detector record such as photon number, electric charge, current, resonator phase, or another measured signal. In diamond nitrogen-vacancy centers, confocal fluorescence combined with magnetic resonance enabled detection of individual defect centers [R240], and spin-selective optical protocols later achieved projective readout of a solid-state spin register [R199]. [Experiment] Related platforms use spin-dependent tunnelling, spin-to-charge conversion, cavity transmission, or dispersive frequency shifts.

What the detector reports depends on the platform: a fluorescence count is not itself the Pauli observable Z .

Calibration determines the conditional distributions $P(r|z = +1)$ and $P(r|z = -1)$, where r is the detector record and $z = \pm 1$ denotes the inferred Z -eigenvalue. The inference must account for preparation errors, state changes during readout, background counts, crosstalk, and the chosen decision threshold.

Quoting only the thresholded binary result discards the information needed to evaluate and reproduce that inference.

Local spin readout reports local quantities — occupation, polarization, a component after a basis rotation, or correlations built from repeated shots. Multiplying local outcomes reconstructs an extended operator, without making that extended measurement robust against the mounting count of local readout errors.

A calibrated transducer is therefore necessary without being sufficient: the measured operator still needs naming, and the total measurement error generally grows with the operator support.

Ancilla measurement of Pauli products

Consider a Pauli product

$$S = P_1 P_2 \cdots P_w,$$

where each P_j is X , Y , or Z acting on one data system, and w is the operator weight, defined as the number of data systems on which the product acts nontrivially. An ancilla can be used to acquire the eigenvalue of S . Basis rotations transform measurements of X - or Y -parity into a Z -parity circuit. Measuring the ancilla then reveals $s = \pm 1$ while ideally preserving coherent superpositions within the same eigenspace of S .

The gate schedule governs fault propagation: one ancilla fault can pass through several two-body gates and seed a correlated error across multiple data systems.

Fault-tolerant layouts therefore constrain gate ordering, add flag ancillas that detect dangerous fault propagation, repeat measurement rounds, and decode the complete measurement history rather than accepting one bit as conclusive. Modern surface-code experiments demonstrate repeated stabilizer extraction and below-threshold logical-error scaling in engineered superconducting circuits [R230].

[Experiment] These results demonstrate active error correction, not a naturally occurring topological phase.

Stabilizer expectation values can also estimate the energy of an ideal commuting Hamiltonian such as

$$H = -J \sum_s A_s - J \sum_p B_p,$$

where J is an energy, A_s is a star operator, and B_p is a plaquette operator. Because the Hamiltonian is a sum of these observables, measurements of $\langle A_s \rangle$ and $\langle B_p \rangle$ estimate $\langle H \rangle$. This energy estimate does not, by itself, determine the spectral gap, long-range entanglement, or robustness against perturbations that were not measured.

An energy estimate therefore does not identify a phase. A stabilizer value near its ideal target certifies nothing topological without the remaining preparation, measurement, calibration, error, and alternative-model evidence.

Spectroscopic measurement of energy gaps

Spectroscopy drives at angular frequency ω and records absorption, fluorescence, resonator response, or state transfer. A resonance obeying

$$\hbar\omega = E_n - E_0$$

connects an initial state of energy E_0 to an excited state of energy E_n , provided that the drive operator has a nonzero transition matrix element between those states. Both sides of the equation have units of joules. By varying momentum, position, polarization, and the drive operator, an experiment can probe energy gaps, dispersion relations, selection rules, bound states, and continuum thresholds.

Many protection arguments require a spectral gap, yet a gap does not imply a topological phase: topologically trivial magnets and molecules also show nonzero excitation gaps. A spectrum grows persuasive as the measured states add the predicted charge, fusion behavior, nonlocal response, and perturbation dependence.

A gap measurement fixes an energy difference under the tested conditions and drive operators. Nonlocal sectors, anyonic statistics, and logical-sector robustness stay outside what it establishes.

Open-string and closed-loop measurements

An open string operator multiplies local operators along a path with distinct endpoints. In the toric code, applying an open string creates, moves, or annihilates endpoint excitations while commuting with stabilizers elsewhere along the path [R030]. Comparing endpoint syndromes before and after applying the string tests this predicted behavior. A

syndrome is the set of measured stabilizer eigenvalues used to locate constraint violations.

A closed Wilson loop $W_a(C)$ carries a charge type a around a closed contour C . Contour and charge label a jointly define the observable.

Depending on the model, a Wilson-loop expectation value diagnoses confinement, names enclosed topological charge, or reads a logical sector. Contractible and noncontractible logical loops need separate identification.

There are three practical measurement methods:

- **Ancilla accumulation:** One ancilla is coherently coupled to every term in the operator product, after which its accumulated phase or parity is measured.
- **Destructive multiplication:** Every site is measured in an appropriate local basis, and the outcomes are multiplied separately for each experimental shot.
- **Interferometry:** A probe is coherently placed in a superposition of two propagation paths, one path is made to encircle a region, and the path amplitudes are recombined. The output probabilities are used to infer an acquired Abelian phase or non-Abelian action.

Interferometry probes the enclosed charge directly. In an Abelian theory one charge carried around another contributes a scalar phase. In a non-Abelian theory braiding applies a matrix on the fusion space, the Hilbert space of possible collective fusion channels, so one output intensity is generally insufficient: the experiment also needs controlled state preparation plus a final fusion-channel measurement [Ro15]. [Theory] Reduced interference visibility may result from dephasing, path distinguishability, leakage, thermal quasiparticles, or a statistical mixture of charge types.

Visibility loss is therefore not automatically evidence of non-Abelian behavior. Reduced fringe contrast does not fix a braid matrix: competing contrast-loss mechanisms need testing before the signal is attributed to particle statistics.

Fusion-channel measurement

A fusion-rule measurement prepares identified quasiparticles, moves them toward one another or deforms the code equivalently, and measures the total topological charge in the enclosing region. For Fibonacci charge τ ,

$$\tau \times \tau = 1 + \tau$$

means two τ charges can fuse to vacuum charge 1 or to charge τ . A useful experiment goes beyond two histogram peaks: it maps those peaks onto charge projectors with calibration, shows repeatability or specifies controlled destructiveness, and tracks dependence on preparation and braid history.

The total charge of a region can be measured using a Wilson loop around its boundary, by interferometry, or by applying a known fusion circuit that maps the charge onto local degrees of freedom and then measuring those degrees of freedom. The final procedure is often called destructive fusion readout: the anyons are combined, the resulting local syndrome is measured, and the original encoded state is not preserved.

Digital superconducting-processor experiments have run code deformations and non-Abelian graph-vertex braiding [R132], and a 27-transmon experiment prepared a Fibonacci string-net state and measured creation, fusion, and braiding signatures [R138]. [Experiment: digital simulation] These experiments demonstrate protocols in controlled Hilbert spaces. They did not observe intrinsic Fibonacci quasiparticles emerging from an unprogrammed defect material; instead, the operations were compiled into gates acting on ordinary processor qubits [R132]; [R138].

Logical-charge measurement applies the same principle at the encoding scale. If logical information is stored in the total charge of a set of anyons or holes, the measurement must use a loop enclosing exactly that set or must fuse the objects according to the encoding tree and measure the final channel.

Loop geometry, orientation, charge label, and fusion basis jointly define the observable: naming “the anyons” without them names no measurement circuit. Two spectral peaks alone likewise establish no fusion rule — without calibrated charge projectors and controlled history-dependent tests the data fit two ordinary resonances.

State tomography and selected observables

Full state tomography measures enough observables to reconstruct ρ . Tomography determines a quantum state without by itself identifying a phase of matter.

For N two-level systems ρ is a $2^N \times 2^N$ matrix, so unrestricted reconstruction costs resources exponential in N . Full tomography stays practical only for small patches.

Larger arrays require more restricted methods, including selected correlators, stabilizer sampling, reduced density matrices, randomized measurements, or classical-shadow protocols [R241]. [Theory/Protocol] A reduced density matrix describes a subsystem after the remaining degrees of freedom have been traced out. Classical-shadow protocols use randomized measurements and classical post-processing to estimate selected properties without reconstructing the complete density matrix.

Reconstructing a small state near an ideal code state verifies preparation on that finite patch, without independently establishing thermodynamic ground-state degeneracy or stability under growing system size. Observables chosen to probe nonlocal structure or entanglement, by contrast, can test a specific hypothesis more cheaply than full reconstruction.

Complete tomography of four spins certifies those four spins. A many-body phase in the thermodynamic regime takes scaling tests the density matrix alone cannot supply.

Existing experimental capabilities

The required measurement techniques exist separately, though not all integrated into one platform. Individual defects can be optically located and their spins read out [R240]; [R199].

Superconducting processors repeatedly measure high-weight parity checks using ancillas [R230]. A superconducting experiment prepared a toric-code state and probed its excitations and topological observables [R125].

[Experiment: digital state preparation] A programmable Rydberg array measured signatures consistent with a toric-code-type spin liquid [R126]. [Experiment: analog quantum simulation] Non-Abelian and Fibonacci protocols have been implemented digitally on superconducting processors [R132]; [R138].

These experiments pose different physical questions. In a gate-based processor the measured operator is usually known from its explicit compilation into a gate sequence, so fidelity, scaling, and survival of the predicted structure in the prepared state are the live issues. In an analog simulator no gate program specifies the Hamiltonian as directly, so phase identification leans harder on a combined set of observables.

A defect lattice needs further interfaces on top: local optical or microwave addressability, ancillas or mediator modes for parity readout, measurement cycles gentle enough to repeat, and a way to lay out extended loops without drowning in accumulated errors.

A credible defect experiment reports the raw and corrected outcomes, detector calibration, leakage and loss, control sequence, operator support, error bars, blind or held-out model comparisons where practical, and results versus system size, temperature, preparation time, and controlled perturbations — with and without readout correction. Otherwise a large corrected Wilson-loop value may mostly encode the calibration model's assumptions.

Shared measurement vocabulary covers physically distinct situations: a stabilizer circuit on programmed qubits performs active syndrome extraction. A prepared code state counts as digital emulation unless it forms an equilibrium or dynamically stable phase of the physical Hamiltonian under study. An analog simulator may realize an engineered many-body Hamiltonian, yet signatures from one finite patch do not automatically demonstrate thermodynamic topological order.

An emergent anyon is an excitation of such a many-body phase; a logical qubit is information held in a nonlocal sector of that phase or code. The physical setting needs stating explicitly, not inferring from shared vocabulary.

Common interpretation errors

- **Identifying one stabilizer with topology.** A local constraint can be satisfied by a trivial product state or by a state prepared specifically to satisfy that check. The complete constraint pattern and independent nonlocal observables must also be measured.
- **Identifying a long correlator with a logical loop.** Length is insufficient. The path must have the appropriate topology and operator content. A contractible loop may be only a product of stabilizers.
- **Using postselection to produce idealized results.** Discarding shots with leakage or unwanted syndromes changes the sampled ensemble. Acceptance rates and conclusions obtained without postselection must be reported.
- **Applying readout mitigation that assumes an incorrect noise model.** Independent single-site confusion matrices can fail in the presence of correlated crosstalk. The correction should be validated on states with known multi-site parity.
- **Identifying an energy gap with a phase.** Spectroscopy can demonstrate an energy gap. Topological order additionally requires nonlocal structure and robustness.
- **Identifying fusion-like populations with fusion rules.** Evidence for fusion requires controlled charges, a defined fusion basis, calibrated charge projectors, and history-dependent tests. Two peaks alone establish only spectroscopic structure.

- **Interpreting finite prepared-state signatures as passive protection.** The toric-code and Fibonacci processor demonstrations [R125]; [R132]; [R138] measured deliberately prepared or digitally evolved states. [Experiment] They do not show that the same hardware passively relaxes into that phase or passively protects it.
- **Treating one Wilson-loop value as universal evidence.** Loop behavior depends on the contour, state, temperature, boundaries, matter content, and noise. It must be combined with local syndromes, gap data, entanglement diagnostics, sector structure, and perturbation tests.
- **Calling destructive parity measurement nondemolition.** Multiplying individually measured spin outcomes can estimate a loop value but destroys coherence in incompatible bases. A quantum-nondemolition claim requires repeated agreement beyond what can be explained by state re-preparation or detector memory.

A cumulative evidence hierarchy is:

Evidence	What it supports	What remains open
Local spin contrast	controllable/readable constituents	interactions and collective phase
Stabilizers near target values	local code constraints	nonlocal sector and long-range entanglement
Gap spectroscopy	energetic isolation	topology of the gapped state
Open-string endpoint rule	candidate quasiparticle motion	statistics and deconfinement
Noncontractible loops	global sector information	robustness and thermodynamic scaling
Fusion and braid matrices	anyonic process within calibrated space	intrinsic emergence versus compiled simulation
Size/perturbation/temperature trends plus entanglement data	coherent phase-identification dossier	ultimate scalability and protection

No rung of this hierarchy excuses the rungs below it. Strong phase identification stacks local constraint measurements, independent nonlocal observables, spectral and entanglement information, controlled excitation creation and fusion, scaling tests, and exclusion of plausible topologically trivial models.

Satzinger et al. and Semeghini et al. provide complementary examples of digital and analog evidence sets rather than a single decisive measurement [R125]; [R126]. [Experiment]

Conceptual assessment

- **Ancilla-based stabilizer measurement.** An ancilla is an auxiliary quantum system used to extract information from the data system. A stabilizer is an operator whose eigenvalue specifies a code-space constraint. Measuring a stabilizer with an ancilla reveals the eigenvalue of a specified local product of operators while, ideally, revealing no individual data values within the corresponding eigenspace, which is the subspace associated with that eigenvalue.
- **Dependence of a contractible Z loop on the torus.** A contractible loop is a closed path that can be continuously reduced to a point on the surface. In the ideal code, a contractible Z loop is the product of the enclosed plaquette operators B_p , where each B_p is a stabilizer associated with a plaquette. Therefore, any two

ground states satisfying every $B_p = +1$ have the same eigenvalue for that loop, so the loop is not an independent logical observable. By contrast, a noncontractible loop, which cannot be continuously reduced to a point on the torus, cannot be expressed as that product and can distinguish the ground states.

- **Destructive and nondestructive loop readout.** Destructive readout measures the individual sites that constitute a loop and multiplies their measurement outcomes to obtain the loop eigenvalue. Nondestructive ancilla readout instead extracts only the loop eigenvalue and ideally preserves coherence between states within the corresponding eigenspace.
- **Limitations of identifying a spectral gap with topological order.** A spectral gap is an energy difference between the ground-state sector and an excited-state sector. Its measurement establishes only an energy difference under the conditions being probed. Because ordinary magnets can also be gapped, a gap alone does not demonstrate topological order. In particular, it does not establish nonlocal entanglement, anyonic statistics, or robustness of logical sectors.
- **Limitations of interpreting two histogram peaks as a fusion rule.** A fusion rule specifies the possible total charges obtained by combining identified input charges. Evidence for such a rule requires identified input charges, a defined fusion basis, calibrated projectors onto charge sectors, and tests that depend on the preceding operation history. Without these elements, two histogram peaks constitute spectroscopic data rather than evidence of a fusion rule.
- **Appropriate description of a gate-compiled Fibonacci experiment.** A gate-compiled experiment implements a target model through a sequence of programmed quantum gates. The minimum justified description is a digital simulation or emulation of Fibonacci states and operations unless independent evidence establishes intrinsic emergent Fibonacci order in the hardware Hamiltonian.

Local spins report through calibrated transducers that convert the spin-dependent response into a detectable signal. Pauli products yield either to ancilla readout or to destructive snapshots of the constituent sites. Spectroscopy fixes energy differences. Open strings, laid along paths with distinct endpoints, test the charges at those endpoints. Closed loops and interferometers probe enclosed or logical charge. Fusion readout projects onto a defined total-charge channel. State tomography reconstructs selected state information, with full tomography costing exponentially in system size. No single method here measures topology directly. Every physical claim therefore attaches to a specific observable, documented with its definition, calibration, assumptions, and limitations.

Sources

- [R240] A. Gruber, A. Dräbenstedt, C. Tietz, L. Fleury, J. Wrachtrup, and C. von Borczyskowski, “Scanning Confocal Optical Microscopy and Magnetic Resonance on Single Defect Centers,” *Science* 276, 2012–2014 (1997). DOI: [10.1126/science.276.5321.2012](https://doi.org/10.1126/science.276.5321.2012).
- [R199] L. Robledo, L. Childress, H. Bernien, B. Hensen, P. F. A. Alkemade, and R. Hanson, “High-fidelity projective read-out of a solid-state spin quantum register,” *Nature* 477, 574–578 (2011). DOI: [10.1038/nature10401](https://doi.org/10.1038/nature10401).
- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* 303, 2–30 (2003). DOI: [10.1016/S0003-4916\(02\)00018-0](https://doi.org/10.1016/S0003-4916(02)00018-0). arXiv: [quant-ph/9707021](https://arxiv.org/abs/quant-ph/9707021).
- [R230] Google Quantum AI and Collaborators, “Quantum error correction below the surface code threshold,” *Nature* 638, 920–926 (2025). DOI: [10.1038/s41586-024-08449-y](https://doi.org/10.1038/s41586-024-08449-y). arXiv: [2408.13687](https://arxiv.org/abs/2408.13687).

- [R125] K. J. Satzinger et al., “Realizing topologically ordered states on a quantum processor,” *Science* 374, 1237–1241 (2021). DOI: [10.1126/science.abi8378](https://doi.org/10.1126/science.abi8378). arXiv: [2104.01180](https://arxiv.org/abs/2104.01180).
 - [R126] G. Semeghini et al., “Probing topological spin liquids on a programmable quantum simulator,” *Science* 374, 1242–1247 (2021). DOI: [10.1126/science.abi8794](https://doi.org/10.1126/science.abi8794). arXiv: [2104.04119](https://arxiv.org/abs/2104.04119).
 - [R132] Google Quantum AI and Collaborators, “Non-Abelian braiding of graph vertices in a superconducting processor,” *Nature* 618, 264–269 (2023). DOI: [10.1038/s41586-023-05954-4](https://doi.org/10.1038/s41586-023-05954-4). arXiv: [2211.09802](https://arxiv.org/abs/2211.09802).
 - [R138] S. Xu et al., “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* 20, 1469–1475 (2024). DOI: [10.1038/s41567-024-02529-6](https://doi.org/10.1038/s41567-024-02529-6). arXiv: [2305.14028](https://arxiv.org/abs/2305.14028).
 - [R241] H.-Y. Huang, R. Kueng, and J. Preskill, “Predicting many properties of a quantum system from very few measurements,” *Nature Physics* 16, 1050–1057 (2020). DOI: [10.1038/s41567-020-0932-7](https://doi.org/10.1038/s41567-020-0932-7). arXiv: [2002.08953](https://arxiv.org/abs/2002.08953).
 - [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083). arXiv: [0707.1889](https://arxiv.org/abs/0707.1889).
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PART XI — ASSESSMENT

The final assessments collect the evidence needed to judge the proposal. They identify the claims that experiments support and the questions that still require a decisive test.

Assessment I — The proposal and its limits

A cluster that behaves as one controllable two-state object is still far from a memory whose energy gap suppresses its own errors. This unit first constructs the most defensible version of the defect-cluster proposal, then checks whether the required energy scales, stability conditions, and fabrication tolerances can hold simultaneously. The four steps follow that dependency: the mechanism, the numbers it produces, and the disorder and scaling tests that decide whether those numbers survive in hardware.

Original chapter framing and supporting arguments

The favorable case starts from one physical step: several defects grouped into a cluster can behave, at low energy, as a single controllable two-state object. The chapter derives that reduction first, then estimates the collective energy scale the resulting inter-cluster interactions could generate. The first calculation shows what the mechanism allows in principle; the estimates afterward show what a fabricated device would still have to demonstrate.

Background, scope, and supporting argument

Several relevant capabilities have been measured. Two nearby defect spins have been entangled. A chosen crystal plane has imposed a common sign on the dipolar couplings within a dense layer. One molecular ion has delivered three nitrogen atoms in a single implantation event. A neighboring nuclear spin has held a state for more than a minute. Each of these is a property of a few defects or of an averaged ensemble. A topological phase is a different kind of claim: a property of the whole many-body system, such as excitations that can only be created and moved nonlocally, that no single defect possesses. None of the cited measurements demonstrates such a phase.

The assessment therefore examines the strongest version of the proposal that stays consistent with the evidence: the version a careful advocate would defend. The word *steelman* marks that choice. The point is to test the proposal at its best, so that any failure reflects the physics rather than a weak presentation of it.

The architectural feature under test is the grouping of defects into clusters. Several interacting spins occupy many joint states; a well-designed cluster splits that spectrum so that two states sit well below the rest. At operating energies the cluster then moves only within that pair, and the pair functions as one effective two-state object. An isolated defect offers a fixed spin and

fixed couplings, while the cluster's spectrum, wavefunctions, and symmetry become adjustable parameters that shape the effective object and its interactions.

Neighboring clusters interact through their physical spins, and those interactions project onto couplings between the effective two-state objects. Whether the array reaches a collective phase then depends on two concrete conditions: the projected couplings must have the operator structure the target model requires, and the cluster spectra and couplings must repeat closely enough from cell to cell that one Hamiltonian describes the whole array.

The argument keeps the evidential status of each statement explicit with the labels [Experiment], [Theory], [Proposal], and [Speculation] introduced below.

[Experiment] Several required components have been demonstrated separately: coherent coupling of nearby defect spins, small multi-defect motifs, dense two-dimensional dipolar ensembles, local electron–nuclear registers, controlled optical interfaces, and improving three-dimensional placement. [Theory] Local two-body Hamiltonians, which contain interactions between pairs of degrees of freedom, can support emergent topological phases, and projection onto a cluster's low-energy subspace can generate interactions that are not evident in the microscopic model. [Speculation] No defect-cluster array has been shown to realize a topologically ordered phase, doubled-Fibonacci order, or intrinsic Fibonacci quasiparticles as of August 2026.

Chapter 38 — The strongest case against

The preceding derivation showed that a cluster can supply an adjustable effective two-state object. The objection developed here is that every later requirement — the size of the useful couplings, the time available for preparation, and the placement precision — draws on the same limited budgets of interaction strength, coherence time, and fabrication accuracy. The chapter traces those shared dependencies and checks whether they can be satisfied together in one sample.

Background, scope, and supporting argument

A color center is a point defect in a crystal whose internal states can be addressed optically or electronically. Such a defect can sense fields, link optical photons to spins, or serve as a physical qubit, and still be unsuitable

for the proposal considered here. That proposal asks for a topologically ordered phase generated directly by a many-body Hamiltonian that acts continuously, rather than by programmed gate sequences. It requires the following sequence of physical and engineering steps:

defects $\rightarrow$ clusters $\rightarrow$ pseudospins $\rightarrow$ target graph
 $\rightarrow$ many-body terms $\rightarrow$ gapped phase $\rightarrow$ logical protection.

Here a pseudospin is the effective two-level object encoded in one cluster, and the target graph lists which pseudospins interact. Each arrow must hold simultaneously in the same sample and parameter regime: the clusters must form, encode clean doublets, couple in the right pattern, generate the intended many-body terms, open a gap, and protect the logical states, all at once.

Demonstrating each arrow in a different sample leaves the sequence unproven, because the operating conditions that favor one step can spoil another. The central difficulty is that the steps share one energy budget: the procedures that convert physical two-body couplings into the desired many-body terms divide the useful energy scale by powers of a small ratio, while every later requirement must still be paid from that reduced scale.

Four requirements constrain everything downstream: the strength of the physical interactions, the isolation of the cluster doublet, the precision of defect placement, and the fidelity of the interaction graph to the target model.

The effective plaquette term, the local many-body interaction around one face of the target graph, is computed from those four quantities, and the topological gap — the energy cost of the cheapest excitation above the ground-state manifold — is a fraction of that term.

State preparation must finish while coherence lasts, measured against that same gap; thermal occupation is set by the ratio of temperature to that gap; and scaling asks for the whole structure to be reproduced hundreds or thousands of times with the gap intact in every cell.

In this unit:

1. [What must the proposal establish?](#)

2. [How could clusters produce useful effective physics?](#)
3. [Do the energy scales leave a usable operating window?](#)
4. [What survives disorder, scaling, and the evidence?](#)

What must the proposal establish?

This step states the claim precisely and lays out the chain of requirements, so that each later section can be read as testing one link.

Introduction: The Steelman Argument

A **steelman** is the version of the proposal a careful advocate would defend: the strongest statement that stays consistent with the measurements. The assessment tests that version, so that any failure belongs to the physics rather than to a weak presentation.

Four measured capabilities support it:

- Two nearby defect spins can become entangled.
- A suitable crystal plane can impose a common sign on dipolar interactions.
- A molecular ion can implant three nitrogen atoms in a correlated event.
- A neighboring nuclear spin can retain a state for more than a minute.

None of these four is a topological phase. Each concerns a few spins or an averaged ensemble; none shows collective nonlocal excitations of a many-body system.

Introduction: The Conjunctive Chain

The proposal requires every arrow in the following sequence to hold in the same sample at the same time:

defects → clusters → pseudospins → target graph
→ many-body terms → gapped phase → logical protection

Each arrow depends on shared parameters — the same couplings, coherence times, and placement precision — so demonstrating each arrow in a different sample leaves the joint claim untested. The rest of the unit checks the arrows together rather than one at a time.

Two Criteria for Ranking Obstacles

Obstacles differ along two independent axes, and the ranking below keeps them separate. The first axis is evidential: a claim may have been measured in a relevant defect array, seen only in an isolated component, derived for an idealized model, or proposed without direct demonstration. A theorem about an ideal Hamiltonian is then not a measurement of a fabricated patch. The second axis is structural: some failures cost a workaround while others remove a requirement on which every later step depends.

Evidence proximity records how directly the available evidence applies to the intended hardware.

Reach leverage, scored from 1 for a localized cost to 5 for a failure that blocks the central claim, records how many later requirements fail if the obstacle stands.

Rank	Obstacle	Evidence proximity	Leverage	Basis for ranking
1	Generate the required high-order terms	strong theory; architecture-specific implementation is a proposal	5	Without the stabilizer/plaquette algebra, there is no target phase
2	Retain a usable topological gap	rigorous stability theory starts from an already gapped target; defect value unmeasured	5	Every noise, preparation, and thermal inequality is paid from this one scale
3	Scale yield and calibration	direct component evidence; no phase-scale array	5	Per-site and per-link imperfections compound with system size
4	Place the right defects at the right coordinates	direct fabrication evidence	5	Position controls coupling, graph, cluster spectrum, and disorder
5	Make interactions coherent and strong enough	direct pair-level evidence	5	$J \lesssim \hbar\Gamma$ kills coherent Hamiltonian engineering
6	Control disorder	direct evidence for inhomogeneity sources	4	Disorder perturbs denominators and effective terms
7	Realize the right interaction graph	microscopic interactions known; graph conversion is a proposal	4	Extra long-range and angular couplings are Hamiltonian terms
8	Survive finite temperature	strong theory; thermalization rates platform-dependent	4–5	Existential for passive memory, but can be deferred for a proof-of-principle
9	Prepare the phase before it decoheres	strong adiabatic theory; no integrated demonstration	4	A small minimum gap can make a formally valid Hamiltonian unreachable
10	Initialize, address, drive, and read the array	direct small-register evidence; array extrapolation uncertain	3–4	Severe, but global protocols and improved interfaces offer more workaround space

Detailed treatment: two criteria for ranking obstacles

Two criteria for ranking obstacles

The two axes from the main text work as follows. Evidence proximity distinguishes a measurement in a relevant array from a component demonstration, an idealized-model derivation, or a proposal without direct demonstration. Reach leverage distinguishes a costly workaround from the failure of an inequality on which the rest of the proposal depends.

A low rank marks an obstacle whose solution would matter only after a higher-ranked obstacle is cleared. Control and readout sit last for that reason: they admit partial workarounds, while a missing plaquette algebra leaves no phase to control. The ordering is a decision order for experiments, not a measure of difficulty.

Ranking of ten obstacles

Ranking of ten obstacles

The ranking below assesses only the defect-cluster route to an analog, emergent, gapped topological Hamiltonian: a phase produced by a continuously acting many-body Hamiltonian and diagnosed by its excitations, not defect qubits in general. “Direct” marks evidence for the relevant ingredient in isolation; no cited experiment integrates all ingredients in one array.

Rank	Obstacle	Evidence proximity	Leverage	Basis for ranking
1	Generate the required high-order terms	strong theory; architecture-specific implementation is a proposal	5	Without the stabilizer/plaquette algebra, there is no target phase to protect.
2	Retain a usable topological gap	rigorous stability theory starts from an already gapped target; defect value unmeasured	5	Every noise, preparation, and thermal inequality is paid from this one scale.
3	Scale yield and calibration	direct component evidence; no phase-scale array	5	Per-site and per-link imperfections

Rank	Obstacle	Evidence proximity	Leverage	Basis for ranking
				compound with system size.
4	Place the right defects at the right coordinates	direct fabrication evidence	5	Position controls coupling, graph, cluster spectrum, and disorder at once.
5	Make interactions coherent and strong enough	direct pair-level evidence	5	$J \lesssim \hbar\Gamma$ kills coherent Hamiltonian engineering before topology enters.
6	Control disorder	direct evidence for inhomogeneity sources; phase-level tolerance model-dependent	4	Disorder perturbs denominators and effective terms and can close or fill the gap.
7	Realize the right interaction graph	microscopic interactions known; graph conversion is a proposal	4	Extra long-range and angular couplings are Hamiltonian terms, not merely wiring inconvenience.
8	Survive finite temperature	strong theory; thermalization rates platform-dependent	4–5	It is existential for passive memory, but can be deferred for a short proof-of-principle state.
9	Prepare the phase before it decoheres	strong adiabatic theory; no integrated demonstration	4	A small minimum gap can make a formally valid Hamiltonian experimentally unreachable.
10	Initialize, address, drive, and read the array	direct small-register evidence; array extrapolation uncertain	3–4	Severe, but global protocols and improved interfaces offer more workaround space than missing terms do.

Ranks 3–7 are close. A host material with mediator-enhanced coupling could lower the ranking of interaction strength while introducing additional control or loss problems.

If the objective is an equilibrium passive memory rather than a finite-time demonstration of a phase, thermal stability rises to rank 3, because the ranking orders which experiment to attempt first for the stated objective.

Required many-body interaction algebra

String-net Hamiltonians require local constraints together with plaquette recoupling operations that obey a precise algebra [Ro18]. The available defect interactions are predominantly two-body interactions.

Perturbative gadgets can generate many-body terms [Theory]: auxiliary couplings combined with a large energy penalty leave a weaker effective many-body coupling in the low-energy subspace. Higher order then exacts two costs: the desired coefficient shrinks by another power of the small expansion ratio, and the same processes generate lower-order energy shifts and unwanted operators [R174]. Symmetry can cancel some of those byproducts. That cancellation holds only where the symmetry holds, so strain, misorientation, and coupling imbalance bring the cancelled terms back.

A nonzero four-body coefficient is therefore not the condition that matters. The useful term must dominate everything that competes with it throughout a manufacturable patch:

$$\|H_{\text{unwanted}}\|, \sigma_K, \hbar\Gamma \ll K$$

throughout a manufacturable patch. Here, H_{unwanted} denotes unwanted Hamiltonian terms, σ_K denotes disorder in the effective plaquette coupling, and Γ is the decoherence rate. No defect-cluster experiment cited here has measured this hierarchy. [Proposal] Until a patch-scale measurement shows that hierarchy, generating the required terms in a real array remains an engineering hypothesis rather than a demonstrated step.

A term can therefore appear in the perturbative expansion and still lose to a larger parasitic term in the physical Hamiltonian. Dominance, not occurrence, is what the experiment must show.

Limits of topological-gap stability theorems

Topological-order stability theorems start from an ideal Hamiltonian that is already local and gapped, and show that its separated spectral bands survive sufficiently weak local perturbations [R142]. [Theory] This result constrains only a device that has already reached the target phase with perturbations below the threshold. A weak approximate plaquette term accompanied by larger parasitic terms falls outside those hypotheses.

Define Δ_{topo} as the many-body energy gap above the relevant ground-state manifold. The required hierarchy is

$$\Delta_{\text{topo}} \gg \max(k_B T, \sigma_K, \|H_{\text{unwanted}}\|, \hbar\Gamma, \delta_{\text{fs}}),$$

where σ_K is effective-coupling disorder, T is temperature in kelvin, k_B is Boltzmann's constant, Γ is a decoherence rate in inverse seconds, and δ_{fs} is a finite-size splitting in joules. An experimentally observed spectral gap is not by itself evidence of topological order. The excitations, nonlocal operators, and robustness must also agree with those of the target phase.

Invoking the theorem before dominance is shown therefore assumes the conclusion the experiment is supposed to establish.

Yield reduction under scaling

Consider the strictest yield model: every required site works independently with probability p , and the patch counts as good only when every site works. The yield of an N -site patch is then

$$Y_N = p^N.$$

The numbers show why per-site probabilities near unity are still demanding at scale: for $N = 1000$, $p = 0.99$ gives $Y_N \approx 4.3 \times 10^{-5}$, whereas $p = 0.999$ gives $Y_N \approx 0.368$. This model treats sites as independent and allows no repair; correlated errors change the distribution, and links, orientations, charge states, optical usability, and cluster spectra each add conditions of their own unless the architecture tolerates vacancies and reroutes around them.

Calibration compounds in the same way. Each fabricated cell has its own frequencies and couplings, so the calibration data grows with system size. A design that cancels each parasitic bond with its own analog tone may use a scalable number of components and still require an impractical volume of calibration. Scaling therefore outranks several limitations that look more severe in a single device.

Correlated fabrication or repair would need its own yield model, with explicit success criteria for the repaired patch. Within the independence model, $p = 0.99$ leaves essentially no all-good thousand-site patches.

Required defect-placement precision

[Experiment] A 2025 diamond experiment combined nitrogen delta doping with localized electron irradiation in prefabricated nanopillars. It reported approximately 4 nm depth precision and 46(1) nm lateral precision in 280 nm-diameter pillars, with mean Hahn-echo $T_2 = 98 \mu s$ for the created single NV centers [R207]. These results are

important fabrication advances under the conditions of that experiment. They do not demonstrate arbitrary three-dimensional, few-nanometre, orientation-selected interacting lattices.

The relevant comparison is between demonstrated precision and the tolerance the architecture requires. For a dipolar design with separation r that permits a relative coupling spread q , radial placement alone requires approximately

$$\sigma_r/r \lesssim q/3.$$

Exchange-mediated couplings tighten this requirement further, because exchange strength typically falls exponentially with wave-function overlap. Mapping the fabricated structure afterward identifies what was built, but the mapped graph must still lie in the desired phase or admit repair.

The reported 46 nm lateral precision therefore sets a measured starting point: that 2025 process had not reached a few-nanometre interacting lattice, and any proposal that needs one must specify the fabrication advance that closes the gap.

Pair interactions compared with lattice-scale requirements

[Experiment] Two engineered NV electron spins separated by approximately 25 nm were entangled at room temperature using their roughly 5 kHz magnetic dipolar coupling [Ro80]. This experiment establishes coherent defect–defect interaction for a selected pair driven by control sequences. Extending that pair coupling to a clean high-order gap across a large lattice is a separate requirement the experiment leaves untested.

Dipolar coupling ties strength, angle, and range together in one formula:

$$J_{ij} \propto \frac{1 - 3 \cos^2 \theta_{ij}}{r_{ij}^3},$$

where θ_{ij} is the angle between the separation vector and the quantization axis. Bringing defects closer raises J through the denominator, and at the same time raises fabrication damage, spectral crowding, unwanted exchange, and the difficulty of addressing one defect without disturbing its neighbors.

Optical, phononic, or cavity mediators extend the interaction range through an intermediate mode, which brings its own loss, fabrication, and mode-crowding constraints. The limiting requirement moves from direct-coupling strength to mediator performance.

Satisfying $J/\hbar = 10\Gamma$ for one pair therefore compares the bare coupling J against decoherence for that pair. The protection requirement compares the much smaller topological gap against the same decoherence, together with disorder and temperature.

Nonlinear amplification of disorder

Microscopic disorder enters twice. It shifts bare frequencies, orientations, and pair couplings J_{ij} , and it shifts the energy denominators of the perturbative expansion, so the effective coefficients vary nonlinearly with the underlying disorder. A cluster can lose its isolated doublet at this stage, before the array-level phase is ever evaluated.

Local disorder below the stability threshold of an already established phase is tolerable [R142]. The demanding regime for this proposal is disorder acting on a gap that projection has already shrunk: the comparison scale is the reduced effective gap, not the bare physical couplings.

Echo sequences refocus selected single-spin frequency offsets by reversing their phase accumulation. A static error in the engineered couplings accumulates through the same Hamiltonian whose dynamics the proposal needs, so refocusing it would cancel the target evolution as well.

Reading disorder as a set of local fields misses this effect, because local fields alone leave the doublet intact. Once the denominator shifts are included, the doublet itself can dissolve before anyonic excitations become the relevant question.

Unintended graph edges as Hamiltonian terms

A string-net or bond-directional model specifies two things at once: which degrees of freedom interact, and which operator acts on each edge or plaquette [Ro18]. A three-dimensional crystal supplies fixed crystallographic orientations, long-range dipolar tails, and surfaces instead. Shaping those physical couplings into the specified pattern is itself part of the engineering task.

Clusters, frequency selection, pulse sequences, or mediators can reshape the physical couplings toward the target graph. Each method exacts a price: weaker effective energy scales, time-dependent control, or additional hardware. [Proposal] A digitally toggled

average Hamiltonian is one such method; the resulting protection then rests on continued control pulses, which places it outside the autonomous-material claim assessed here.

Every extra dipolar tail that survives the reshaping enters H as an operator with its own dynamics, on the same footing as the intended couplings.

Finite-temperature limitations

At nonzero temperature the bath creates excitations at rates set jointly by the energy gap and the bath's spectral density. Those excitations can move through the array, and a path that winds around the system applies a logical operator: the stored information changes without any local signal marking the event.

The finite-temperature quantum-memory literature therefore treats the zero-temperature gap as one input among several: dimensionality, energy barriers, excitation kinetics, and decoding all enter the lifetime [R169]. [Theory] Under the assumptions of the Bravyi–Terhal no-go theorem, two-dimensional local stabilizer Hamiltonians cannot provide a self-correcting quantum memory with a macroscopic energy barrier [R168].

That theorem assumes a two-dimensional local stabilizer Hamiltonian, so architectures that are non-Abelian, driven, long-range, or actively corrected fall outside its hypotheses. Within its scope it rules out indefinite passive storage: no two-dimensional local stabilizer Hamiltonian of that class acquires a macroscopic energy barrier from its gap alone.

A short experiment can still probe ground-state properties when $k_B T > \Delta_{\text{topo}}$ by cooling actively and measuring before excitations accumulate. That outcome demonstrates properties of the model; passive protection — survival without intervention — requires a separate measurement at equilibrium.

The quantity that decides thermal occupation is the ratio $k_B T / \Delta_{\text{topo}}$. A millikelvin refrigerator temperature alongside a hertz-scale emergent gap gives a ratio of order a million, which means the excited states are heavily populated at equilibrium.

Preparation within the coherence window

Reaching a Hamiltonian in the desired phase still requires traversing a path to it, and the gap along that path can dip below the final gap. Call the smallest gap encountered along the path Δ_{min} . Adiabatic bounds depend on the full schedule — the rate of change of the Hamiltonian and the matrix elements connecting ground and excited states — with that gap setting the dominant timescale [R245]. [Theory] As system size

grows, $\Delta_{\min}$ typically shrinks near the phase transition the path must cross, while decoherence and calibration drift accumulate over the lengthened preparation. Both trends squeeze the same operating window from opposite sides.

Measurement-assisted or dissipative preparation can shorten the preparation by replacing slow adiabatic following with ancilla measurements, engineered dissipation, reset, and verification. Each replacement brings its own hardware requirements. A prepared wavefunction then still needs its equilibrium character established: the subsequent continuously acting Hamiltonian and its excitation spectrum supply that evidence.

A Hamiltonian that is valid on paper but unreachable within coherence time and calibration drift is not an operational device proposal.

Control and readout as comparatively tractable constraints

[Experiment] A seven-spin diamond processor demonstrated fault-tolerant protocol primitives for a five-qubit-code logical qubit, including flagged stabilizer measurements and real-time processing [R234]. The authors reported that fidelity and qubit number still needed improvement before logical errors fell below physical errors. Their result shows sophisticated control in one selected register; extending that control to simultaneous analog operation across an extensive defect lattice remains untested.

Reviews of color-center networks list the same simultaneity at the control layer: spin coherence, optical interfaces, spectral stability, and integrated fabrication must all perform at once [R246]. Local tuning that compensates disorder also shifts the cluster spectrum and can break the symmetry the projection relies on. Resolving optical lines for addressability introduces detunings between cells. Packing defects densely strengthens couplings and simultaneously strengthens crosstalk. Each control knob thus moves several requirements at once.

Control ranks last because workarounds exist — global pulses, improved interfaces, additional calibration — that soften a control shortfall while leaving the target phase intact. No analogous workaround exists for missing plaquette algebra: without those terms there is no target phase to control.

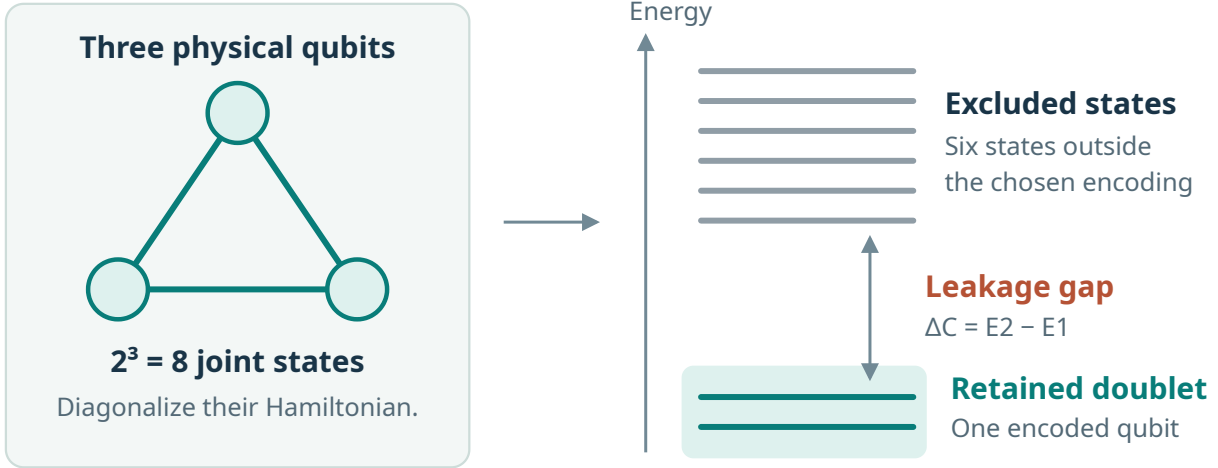
How could clusters produce useful effective physics?

This step constructs the mechanism: selecting the low-energy doublet, projecting the physical interactions into it, and identifying which fabrication choices set the outcome.

Low-Energy Cluster Degree of Freedom

“Low-energy” describes the outcome of diagonalizing the cluster Hamiltonian H_C : its eigenstates are the stationary states, the splitting within the chosen pair sets the encoded local field, and the distance to the nearest excluded state sets the leakage gap. The encoding is that pair of eigenstates, not two convenient basis states picked in advance.

One cluster, two different gaps



The splitting inside the doublet, $\delta C = E_1 - E_0$, is a separate quantity.

The encoding follows from the computed spectrum: the splitting within the retained pair sets an encoded local field, and the gap above the pair suppresses leakage into the six excluded states.

When the operating temperature, drives, and noise carry energies well below the cluster gap, the higher cluster states stay unoccupied and the cluster moves within its lowest pair of states.

For cluster C :

$$H = \sum_C H_C + \sum_{\langle C,D \rangle} V_{CD}$$

The internal energy H_C defines a two-state manifold with projector P_C . The energy difference between that manifold and the nearest excluded cluster state is the cluster gap Δ_C .

Write v for a representative intercluster matrix element, an energy. That ratio compares the coupling that could drive the cluster out of its doublet against the energy cost of leaving it. Where $v/\Delta_C \ll 1$, those transitions are perturbatively suppressed and each cluster contributes one pseudospin to the array description.

Two independent quantities come out of the spectrum. The splitting within the doublet acts as an encoded local field on the pseudospin. The gap above the doublet sets the energy cost of leakage. A large leakage gap is compatible with any internal splitting, large or small, because they are different energy differences.

For eigenvalues ordered as $E_0 \leq E_1 \leq E_2 \leq \dots$, these quantities are

$$\delta_C = E_1 - E_0, \quad \Delta_C = E_2 - E_1.$$

A small static ratio v/Δ_C justifies the perturbative form of the couplings, and initialization must separately place the system in the retained doublet. Temperature, resonant drives, and noise near a leakage transition can populate the excluded states even where the static ratio is small.

Detailed treatment: low-energy cluster degree of freedom

Low-energy cluster degree of freedom

Long coherence times and bright fluorescence make defects good sensors and qubits. Neither property shapes the cluster spectrum, so neither explains how grouping defects could produce collective topological order. That explanation starts from the spectrum and the projected operators.

Assume that the high-energy excitations of a cluster are energetically inaccessible under the relevant operating conditions. The remaining low-energy subspace can then be a two-state degree of freedom separated from other states by an energy gap.

The microscopic description uses physical spins and orbitals. The reduced description keeps one pair of cluster eigenstates — the doublet — and discards the rest.

The cluster contributes three ingredients to that reduction: the choice of retained pair, the energy cost that keeps the system inside it, and the projected form the physical interactions take within it.

For cluster C , write

$$H = \sum_C H_C + \sum_{\langle C,D \rangle} V_{CD}.$$

Here H is the full Hamiltonian, which represents the system's energy and generates its time evolution. The term H_C is the internal Hamiltonian of cluster C , while V_{CD} is the interaction between neighboring clusters C and D , with $\langle C, D \rangle$ denoting the relevant interacting pairs.

The internal energy H_C defines a two-state manifold with projector P_C , where a projector is an operator that restricts states and observables to a specified subspace. The energy difference between that manifold and the nearest excluded cluster state is the cluster gap Δ_C , measured in joules or electronvolts.

The intercluster interaction V_{CD} has a characteristic matrix element v , also measured as an energy. If $v/\Delta_C \ll 1$, transitions out of the two-state manifold are perturbatively suppressed. The array may then be described by effective two-state objects, called pseudospins, within the total projected subspace $P = \prod_C P_C$.

The resulting pseudospin can have a different magnetic moment, anisotropy, and coupling pattern from any physical spin in the cluster, because it inherits its properties from the cluster wavefunctions rather than from a single defect.

Projection changes the operator structure rather than strengthening the couplings. The cluster spectrum, wavefunctions, symmetry, and placement become adjustable parameters, and a microscopic term with an unsuitable physical-spin form can project onto a useful effective operator:

$$PV_{CD}P = J_x \tau_C^x \tau_D^x + J_y \tau_C^y \tau_D^y + J_z \tau_C^z \tau_D^z + \text{local terms},$$

where $\tau_C^{x,y,z}$ are Pauli operators acting on cluster C 's doublet, and J_x , J_y , and J_z are effective coupling energies. Pauli operators are the standard matrices representing the three independent nontrivial observables of a two-state system. Virtual leakage, meaning intermediate transitions through excluded cluster states without permanent occupation of those states, can add terms proportional to v^2/Δ_C or higher powers. The projected Hamiltonian defines the available effective interactions.

This reduction describes one cluster; an array needs it to repeat. If spectra and projected operators vary randomly from cell to cell, the result is a disordered collection of few-body systems with no shared Hamiltonian. The proposal therefore requires cluster-to-cluster reproducibility tight enough that one calibration strategy covers the array.

Effective Coupling Between Two Clusters

Given two clusters that stay in their doublets, projection computes what a physical bond between them does to the two encoded qubits. With $P = P_A \otimes P_B$, each operator restricted to a retained doublet becomes a 2×2 encoded matrix, and the projected bond is read directly from those matrices.

Consider two identical three-spin clusters, each with a low-energy doublet separated by Δ_C from the other cluster states.

Suppose a physical dipolar or exchange interaction couples one spin in A to one spin in B :

$$V_{AB} = v S_{A,3}^z S_{B,1}^z$$

Within the encoded doublets:

$$P_A S_{A,3}^z P_A = a_0 I_A + a_z \tau_A^z, \quad P_B S_{B,1}^z P_B = b_0 I_B + b_z \tau_B^z$$

The projected coupling is:

$$P V_{AB} P = v(a_0 I_A + a_z \tau_A^z)(b_0 I_B + b_z \tau_B^z)$$

The resulting two-cluster coupling has coefficient:

$$J_{zz}^{(1)} = v a_z b_z$$

Expanding the product shows every term the bond generates, including the single-cluster fields that accompany the two-qubit coupling:

$$P V_{AB} P = v a_0 b_0 I_A I_B + v a_z b_0 \tau_A^z I_B + v a_0 b_z I_A \tau_B^z + v a_z b_z \tau_A^z \tau_B^z.$$

The first term shifts all encoded energies equally, the next two act as local fields on each cluster, and only the final term couples the encoded qubits. A physical two-spin bond thus always produces a package of effective terms, and using the two-qubit part requires handling the accompanying fields.

The $I + \tau^z$ form above is a simplification for this example. A projected Hermitian spin operator generally carries I , X , Y , and Z components, and while one such operator can be diagonalized by choosing the doublet basis, the remaining projected operators then keep transverse components in that basis.

At second order, virtual cluster excitations add terms with characteristic scale:

$$J^{(2)} \sim \alpha \frac{v^2}{\Delta_C}$$

A second-order process applies the physical interaction twice with one intermediate excitation energy in the denominator: the system mixes briefly with an excluded cluster state and returns to the encoded sector. That virtual excursion shifts the effective dynamics even where lasting leakage stays small. The coefficient α sums the matrix elements, signs, and available paths, and interfering paths can shrink it or cancel it. Appendix F develops this mechanism.

Isolation and virtual-process strength share the denominator Δ_C : raising it at fixed v suppresses leakage and simultaneously shrinks the generated interactions.

Detailed treatment: effective coupling between two clusters

Effective coupling between two clusters

Consider two identical three-spin clusters, A and B . Assume that each has a low-energy doublet $\{|\tilde{0}\rangle, |\tilde{1}\rangle\}$ separated by Δ_C from the other cluster states.

That doublet is the encoded effective degree of freedom: one building block for the array. Topological protection would additionally require the whole array to enter the target phase, which the doublet alone leaves untested.

Suppose that a physical dipolar or exchange interaction couples one spin in A to one spin in B :

$$V_{AB} = v S_{A,3}^z S_{B,1}^z.$$

A dipolar interaction couples spin magnetic moments through space; an exchange interaction couples spins through the symmetry of the shared electronic wavefunction. Here S^z is the dimensionless spin operator and v carries the energy scale. Within the encoded doublets the cluster wavefunctions set the matrix elements, taken here as

$$P_A S_{A,3}^z P_A = a_0 I_A + a_z \tau_A^z, \quad P_B S_{B,1}^z P_B = b_0 I_B + b_z \tau_B^z.$$

Here I_A and I_B are identity operators in the respective doublets. The dimensionless coefficients a_0, a_z, b_0, b_z are determined by the cluster eigenstates. Projection to first order in v gives

$$PV_{AB}P = v(a_0 I_A + a_z \tau_A^z)(b_0 I_B + b_z \tau_B^z).$$

The resulting two-cluster coupling has coefficient

$$J_{zz}^{(1)} = v a_z b_z.$$

The same expansion also produces the local fields $v a_0 b_z \tau_B^z$ and $v a_z b_0 \tau_A^z$ plus a constant offset. The microscopic interaction stays fixed while the projected coefficients follow the cluster wavefunctions, so redesigning the internal couplings or symmetry reshapes the effective operator without requiring a different fundamental interaction.

Where a symmetry forces $a_0 = b_0 = 0$, the corresponding local fields vanish. Without such a symmetry they remain in the Hamiltonian with the magnitudes the wavefunctions dictate. A useful encoding therefore needs each undesired operator either forbidden by symmetry or pushed to higher perturbative order.

At second order, virtual cluster excitations add terms with characteristic scale

$$J^{(2)} \sim \alpha \frac{v^2}{\Delta_C},$$

where the dimensionless coefficient α is a signed sum over matrix elements and energy denominators. [Theory] Schrieffer–Wolff perturbation theory, a systematic method for eliminating high-energy states from a Hamiltonian, makes this approximation controlled when $v/\Delta_C \ll 1$ [Ro25].

[Proposal] Repeating such clusters across a lattice could synthesize an interaction graph that bare defects cannot supply. That graph is a prerequisite for topological order in the array, and the two-cluster calculation leaves the array-level question open.

Five Structural Design Variables

Each design variable below controls a different part of the reduction just derived. Cluster structure sets the retained spectrum and the projected operators. Geometry sets the physical matrix elements between clusters. Auxiliary spins and optical channels decide which states can be prepared and read. The favorable case needs all of these working at one shared operating point.

The five variables are:

1. **Cluster isolation:** A cluster can reduce a complicated microscopic state space to a designed low-energy degree of freedom.
2. **Crystal orientation:** Crystal orientation and patterned density can control the sign and effective dimensionality of dipolar interactions.
3. **Interaction hierarchy:** Short-range and mediated couplings can occupy different energy or length-scale levels.
4. **Auxiliary degrees:** Nearby nuclear spins and optical transitions can provide local control and measurement.
5. **Analog Hamiltonian:** A static solid-state array could implement an always-on analog Hamiltonian.

Detailed treatment: five structural design variables enabled by clustering

Five structural design variables enabled by clustering

Each item below is a design variable the proposal can adjust, not a general property of diamond.

- A cluster can reduce a complicated microscopic state space to a designed low-energy degree of freedom.
- Crystal orientation and patterned density can control the sign and effective dimensionality of dipolar interactions.
- Short-range and mediated couplings can, in principle, occupy different energy or length-scale levels within the interaction hierarchy.
- Nearby nuclear spins and optical transitions can provide local control and measurement without serving as the encoded topological degree of freedom.

- A static solid-state array could implement an always-on analog Hamiltonian, rather than reproducing its time evolution only through a sequence of digital gates.

The five sections that follow examine one variable each. Because the variables must perform together, a demonstration of one of them in a separate sample leaves the integrated requirement untested.

Cluster isolation, energy gap, and functional separation

A bare spin offers its fixed states and couplings. A cluster adds adjustable structure: an isolated doublet to encode in, an energy gap that exacts a cost for leaving it, symmetry-imposed selection rules that permit or forbid particular transitions, and projected operators shaped by the cluster wavefunctions. The reduction of the microscopic Hamiltonian is controlled only where all four are characterized.

That structure pays off in two places.

First, strong interactions within a cell and weaker interactions between cells take on separate jobs: the intracell couplings create the effective degree of freedom and open its gap, while the intercell couplings generate the many-body model. Geometric separation supplies a natural hierarchy for dipolar coupling, which scales as r^{-3} , where r is the separation. Exchange interactions provide an even sharper short-range dependence.

Second, a symmetry can forbid a selected local field or matrix element outright, so the unwanted operator first appears at higher perturbative order while the desired operator survives at lower order.

[Theory] Encoding into a subspace to reshape operators is standard practice, not a defect-specific discovery.

[Speculation] No measured color-center cluster — a point defect with characteristic optical transitions — has been shown to project realistic couplings onto the complete operator set of a doubled-Fibonacci string-net Hamiltonian, the local many-body model whose collective states would carry that topological order.

The proposal therefore starts with spectroscopy and exact diagonalization of one repeatable cluster: measuring its energy levels and transitions, and computing its eigenvalues and eigenstates directly from a finite-dimensional Hamiltonian. Where the doublet is not isolated, the projection has no controlled starting point.

Crystal orientation as a control of dipolar sign

Dipolar coupling depends on the separation between spins and on the angle between their displacement vector and the quantization axes. That angular dependence spreads couplings across random orientations, and a crystal cut that aligns the axes turns the same dependence into a design variable.

[Experiment] Hughes and collaborators grew dense, preferentially aligned, two-dimensional NV ensembles in (111)-oriented diamond and showed that this geometry gives the in-plane dipolar interactions a common positive sign in their convention [R242]. An NV center is a nitrogen-vacancy defect in diamond consisting of a substitutional nitrogen atom adjacent to a vacant lattice site. The observed interaction-dependent line-shape asymmetry changed with spin polarization, supporting the dipolar interpretation. A 2026 erratum corrected density and sensitivity calculations, so those corrected metrology numbers should not be transferred without accounting for the revised analysis. The relevant structural result is the orientation-dependent interaction sign [R242].

The structural result is that the host crystal plane, defect axis, and layer dimensionality jointly set the interaction-sign pattern. The experiment showed this in an ensemble rather than an individually patterned cluster graph, and a common dipolar sign is still one ingredient of a string-net Hamiltonian, not the Hamiltonian itself.

A common orientation is what promotes the sign pattern from accident to design variable. The ensemble average leaves open whether individually patterned clusters would each encode a clean doublet with that sign pattern intact.

Correlated fabrication of a multi-defect cluster

A cluster that needs three defects within a few nanometres is expensive to build by placing each defect independently: every placement adds its own yield loss and registration error. Correlated implantation delivers several dopant atoms in one molecular ion, so the three atoms share one implantation event and their relative positions spread only by implantation straggle — the statistical scatter produced as the atoms slow in the host material. The motif becomes one fabrication unit instead of three.

[Experiment] Haruyama and collaborators used a nitrogen-bearing molecular ion to create triple NV centers. Their calculated implanted-nitrogen separation was 9 ± 4 nm; among 7,116 implantation events they identified nine optical triples, one strongly coupled triple and one weakly coupled triple. The total NV creation yield was 7.4%, and the longest reported Hahn-echo T_2 in the measured population was $428 \mu\text{s}$ [R243]. A Hahn echo is the pulse sequence that refocuses static and slowly varying dephasing to extract that T_2 .

One strongly coupled triple in 7,116 events is a proof that the motif can form, not a yield that populates a lattice.

The process therefore optimizes as one unit: the yield of the motif and the energy spectrum the motif produces are tuned together.

That hypothesis is testable on isolated motifs before any lattice is attempted: fabricate the motif repeatedly and check whether its spectrum repeats. Treating each defect as a separate placement instead multiplies the placement errors and leaves the spectrum at the mercy of three uncorrelated outcomes.

Auxiliary degrees of freedom distinct from the encoded state

A dense analog array cannot be initialized, controlled, and read through its interacting electron spins alone: touching one spin disturbs its neighbors. Defect platforms divide the labor across physically adjacent systems: electron spins supply the faster interactions, nearby nuclear spins hold states and assist control, optical transitions initialize and read the electron spin, and photonic structures route the optical signals.

[Experiment] A diamond NV register containing one electron spin, one nitrogen nuclear spin, and eight ^{13}C nuclear spins demonstrated pairwise control across ten qubits and protected an arbitrary single-qubit state for more than 75 s at about 3.7 K under dynamical-decoupling control [R119]. Dynamical decoupling — repeated pulses that average away environmental noise — sustained that memory. The experiment used one defect neighborhood rather than a cluster of interacting electron defects, and the protection came from active pulses rather than from a static spectrum. Its message for this proposal is that a single neighborhood already contains the auxiliary and memory resources a repeated cell would draw on.

[Experiment] Optical photon-mediated interactions have also been induced between two silicon-vacancy centers in a diamond nanocavity [Ro86].

A silicon-vacancy center — a silicon impurity with associated lattice vacancies — coupled to a wavelength-scale optical resonator, a nanocavity that concentrates light–matter interaction, lets photons carry coupling between emitters. That mediation can supply a missing graph edge or separate the interaction channel from the readout channel. It brings requirements of its own: the emitters must match spectrally, the cavity must be fabricated around them, external driving must be supplied, and photon loss must stay low. Meeting those requirements is separate work from realizing an equilibrium short-range Hamiltonian.

The repeated cell is therefore functionally heterogeneous: the encoded degree of freedom, a leakage monitor, a readout helper, and a photonic port sit side by side and perform distinct tasks. A nuclear memory driven by active pulses belongs on the control side of that ledger; passive protection must come from the static many-body spectrum.

Lower-complexity topological targets before Fibonacci order

The research program earns each step by passing through simpler topological targets first.

[Theory] Kitaev’s honeycomb model proves that local two-body, bond-dependent spin interactions can yield emergent Majorana fermions and a non-Abelian Ising phase [Ro17]. Majorana fermions are the self-adjoint fermionic excitations of that solution, and its non-Abelian phase exchanges them by noncommuting operations on a degenerate state space. Levin–Wen string-net Hamiltonians provide commuting-projector targets — sums of mutually compatible local projectors — for doubled topological orders, including doubled non-Abelian ones [Ro18]. A commuting-projector Hamiltonian is a sum of mutually commuting local projection operators. These results are existence proofs for Hamiltonians, not demonstrations in materials.

A defect-cluster platform can climb that ladder: first a projected bond-dependent model, then a generated stabilizer — an operator whose fixed eigenvalue constrains the many-body state — then a small topological patch. Passing an Abelian or Ising-type stage validates the Hamiltonian engineering while leaving Fibonacci order untested, because Abelian exchange operations commute and Fibonacci exchange operations do not. Doubled-Fibonacci order is therefore its own further target, not a corollary of the simpler ones.

Attempting Fibonacci order first reverses the diagnostic order: a failure could come from the cluster projection, the interaction graph, or the recoupling coefficients, with no measurement distinguishing them.

Do the energy scales leave a usable operating window?

This step puts numbers to the mechanism: an optimistic parameter set, the perturbative suppression that shrinks its useful scale, and the stability-theorem hypotheses that decide whether the remainder counts as protection.

Quantitative Assumptions

The table below is an optimistic calculation assembled from separate experiments and design targets, not a measured device. Its rows read top to bottom, each derived quantity building on the assumptions above it. The perturbative order q counts how many times the physical interaction enters the process that generates the useful term, giving the characteristic scaling

$$J_{\text{eff}} \sim \alpha v \left(\frac{v}{\Delta_C} \right)^{q-1}.$$

The scaling alone predicts no gap: the actual operator and its coefficient must still be computed for the target lattice. The table carries that last step as c , the assumed fraction of the effective interaction energy that becomes a topological gap.

Layer	Best-case input	Status	Depends on
Cluster leakage gap	$\nu_\Delta = 1 \text{ GHz}$	[Proposal] design target	reproducible cluster spectrum
Intercluster matrix element	$\nu_v = 100 \text{ MHz}$	[Speculation] as an array-wide value	distance, orientation, mediator, disorder
Expansion ratio	$\epsilon = \nu_v / \nu_\Delta = 0.10$	derived	two rows above
Desired perturbative order	$q = 2$	[Proposal]	selection rules and projected matrix elements
Path coefficient	$\alpha = 1$	optimistic assumption	interference among virtual paths
Effective interaction	$\nu_{\text{eff}} = \alpha \nu_v \epsilon^{q-1} = 10 \text{ MHz}$	derived	all rows above
Generic next-order scale	$\nu_{\text{corr}} \sim \nu_{\text{eff}} \epsilon = 1 \text{ MHz}$	scaling estimate	no enhanced coefficients
Gap fraction	$c = \Delta_{\text{topo}} / J_{\text{eff}} = 0.2$	[Speculation] model assumption	verified many-body phase
Topological gap	$\nu_{\text{topo}} = c \nu_{\text{eff}} = 2 \text{ MHz}$	derived, not measured	correct Hamiltonian and thermodynamic extrapolation

$$\nu_{\text{eff}} = \frac{\nu_v^2}{\nu_\Delta} = \frac{(100 \text{ MHz})^2}{1000 \text{ MHz}} = 10 \text{ MHz}$$

$$\nu_{\text{topo}} T_2 = (2 \times 10^6 \text{ s}^{-1})(10^{-4} \text{ s}) = 200$$

$$\frac{k_B T}{h} \approx 208 \text{ MHz} \quad \text{at } T = 10 \text{ mK}$$

The thermal frequency is therefore about a hundred times the assumed 2 MHz gap.

The calculation thus splits. Against decoherence the optimistic point looks comfortable: with $T_2 = 100 \mu\text{s}$ the gap-frequency product is 200, an energy ratio of about $2\pi \times 200 = 1257$ against $\hbar/T_2$. Against temperature it fails by about a factor of 104. Fast coherent dynamics and low thermal occupation are separate comparisons, and passing one leaves the other open.

The table carries a second warning: the generic next-order correction is 1 MHz against the assumed 2 MHz gap. An expansion parameter of 0.1 guarantees the series converges; it does not guarantee the next term is negligible beside the collective scale. Whether that 1 MHz correction competes with the gap depends on its operator form, which must be computed.

Detailed treatment: quantitative assumptions not yet demonstrated together

Quantitative assumptions not yet demonstrated together

Frequency units keep the numerical comparison readable: for any energy X , $\nu_X = X/h$ with Planck's constant h in joule-seconds, so ν_X is the frequency corresponding to the energy X . The design point below is deliberately optimistic — every entry chosen to favor the proposal:

Layer | Best-case input | Status | Depends on | |

Cluster leakage gap | $\nu_\Delta = 1$ GHz | [Proposal] design target | reproducible cluster spectrum | |

Intercluster matrix element | $\nu_v = 100$ MHz | [Speculation] as an array-wide value | distance, orientation, mediator, disorder | |

Expansion ratio | $\epsilon = \nu_v/\nu_\Delta = 0.10$ | derived | two rows above | |

Desired perturbative order | $q = 2$ | [Proposal] | selection rules and projected matrix elements | |

Path coefficient | $\alpha = 1$ | optimistic assumption | interference among virtual paths | |

Effective interaction | $\nu_{\text{eff}} = \alpha\nu_v\epsilon^{q-1} = 10$ MHz | derived | all rows above | |

Generic next-order scale | $\nu_{\text{corr}} \sim \nu_{\text{eff}}\epsilon = 1$ MHz | scaling estimate | no enhanced coefficients or small denominators | |

Gap fraction | $c = \Delta_{\text{topo}}/J_{\text{eff}} = 0.2$ | [Speculation] model assumption | verified many-body phase | |

Topological gap | $\nu_{\text{topo}} = c\nu_{\text{eff}} = 2$ MHz | derived, not measured | correct Hamiltonian and thermodynamic extrapolation | |

Comparator coherence | $T_2 = 100 \mu\text{s}$, so $1/T_2 = 10 \text{ kHz}$ | [Experiment, borrowed]
 approximately the mean Hahn-echo scale of positioned NVs in one 2025 process [R207]
 | incompatible sample and control conditions may apply | |

Disorder target | $\sigma_J/h \lesssim 0.2 \text{ MHz}$ | [Proposal] ten-percent-of-gap rule | placement,
 strain, charge, calibration | |

The dimensional check is

$$\nu_{\text{eff}} = \frac{\nu_v^2}{\nu_\Delta} = \frac{(100 \text{ MHz})^2}{1000 \text{ MHz}} = 10 \text{ MHz}.$$

Under these assumptions the proposed topological gap runs 200 times faster than the borrowed decoherence rate:

$$\nu_{\text{topo}} T_2 = (2 \times 10^6 \text{ s}^{-1})(10^{-4} \text{ s}) = 200.$$

This is the central quantitative point in favor: a low-order generated interaction can in principle outrun single-defect decoherence. It is not a device prediction, because its inputs — the 1 GHz cluster gap, the 100 MHz array-wide coupling, $\alpha = 1$, $c = 0.2$, the disorder target, and the borrowed T_2 — have never been realized together in one array.

Perturbative order decides whether that margin survives. At the same $\epsilon = 0.1$, a target that first appears at fourth order gives

$$\nu_{\text{eff}}^{(4)} \sim \nu_v \epsilon^3 = 0.1 \text{ MHz}, \quad \nu_{\text{topo}}^{(4)} \sim 20 \text{ kHz},$$

That 20 kHz gap is only twice the borrowed $1/T_2$, before disorder or control errors. The favorable case therefore needs a parent model whose useful terms appear at first or second order, or a symmetry mechanism that produces them exactly, rather than a high-order perturbative route to a Fibonacci plaquette.

Temperature restricts the proposal more tightly than decoherence. Since $k_B/h \approx 20.84 \text{ GHz/K}$, where k_B is Boltzmann's constant,

$$\frac{k_B T}{h} \approx 208 \text{ MHz} \quad \text{at } T = 10 \text{ mK}.$$

That thermal frequency is roughly a hundred times the assumed 2 MHz gap. Passive suppression at equilibrium would need a substantially larger gap or an effective temperature below

$$T_{\text{gap}} = \frac{h\nu_{\text{topo}}}{k_B} \approx 96 \mu\text{K}.$$

No cited defect-cluster experiment reaches this combined thermal and interaction regime. The near-term objective supported by the numbers is preparation and spectroscopy of an analog phase within coherence time. A passively self-correcting memory is a distinct objective with its own thermal inequality, and results against the first do not transfer to the second.

The architecture has the following dependency chain:

```
repeatable motif and placement
|
v
cluster spectrum: Delta_C, matrix elements
|
v
uniform intercluster coupling v and correct graph
|
v
low perturbative order q + bounded corrections
|
v
many-body gap Delta_topo and disorder tolerance
|
+-----> preparation faster than decoherence
|
+-----> k_B T below the relevant gap for passive protection
|
v
local and nonlocal observables identify the phase
```

Each stage of that chain is independently testable by experiment or numerics, so the architecture can fail early: a stage that fails removes the conclusions drawn from every stage below it, without waiting for a full array.

Current experimental and theoretical evidence

The available evidence covers many relevant components in separate systems. The integration step — all of them working in one array — is untested:

Structurally relevant claim | Status | What was actually shown | What remains missing | |

Nearby defect electron spins can interact coherently | [Experiment] | Room-temperature entanglement between two diamond defect spins [Ro80] | repeatable large graph and uniform couplings | |

A multi-defect motif can be fabricated | [Experiment] | one strongly coupled triple NV motif found in a 7,116-event molecular-implantation data set [R243] | deterministic yield, target spectrum, replication | |

Crystal orientation can shape dipolar sign | [Experiment] | aligned 2D (111) NV ensembles with same-sign in-plane dipolar geometry [R242] | individually defined cluster lattice and topological Hamiltonian | |

Defects can be registered to nanostructures | [Experiment] | 2025 δ -doping plus electron irradiation gave about 4 nm depth precision, 46(1) nm lateral precision in 280 nm pillars, and mean Hahn-echo $T_2 = 98(37) \mu\text{s}$ [R207] | nanometre-scale relative placement required for a chosen direct-coupling graph | |

One defect neighborhood can host control helpers | [Experiment] | ten-qubit electron–nuclear register and long protected nuclear memory [R119] | parallel operation inside an interacting analog array | |

A cavity can mediate emitter interactions | [Experiment] | two SiV emitters interacted through one nanocavity [Ro86] | low-loss, uniform, many-cell Hamiltonian regime | |

Cluster projection can synthesize effective operators | [Theory] | controlled effective-Hamiltonian formalism [R025] | measured defect-cluster coefficients | |

Local spin models can host non-Abelian order | [Theory] | solvable honeycomb and string-net models [R017]; [R018] | mapping from available defect terms with a usable gap | |

Defect clusters realize doubled-Fibonacci order | [Speculation] | no experimental or numerical defect-specific demonstration cited here | essentially the entire integrated chain | |

A separate proposal showed how NV registers and optical links could be organized into a scalable solid-state processor [R244]. That work supports modular engineering concepts; whether its material Hamiltonian is topologically ordered is a separate question it leaves open.

A coupled defect triple is a physical spin cluster, and an isolated low-energy doublet inside it can serve as the encoded effective degree of freedom.

A lattice of those encoded objects can emulate a model through gate sequences or realize it directly as a continuously acting analog Hamiltonian. Emergent anyons appear only where the realized Hamiltonian lies in the appropriate topological phase, so the choice between emulation and realization decides what the experiment can claim.

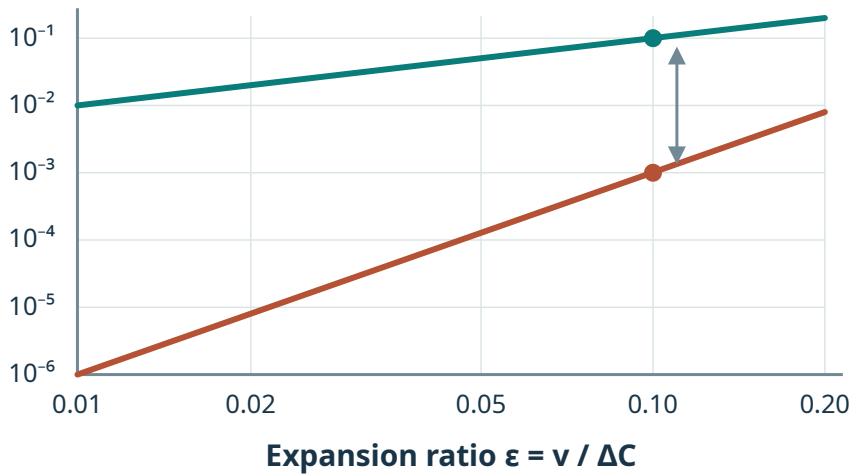
A topological logical qubit additionally requires a demonstrated nonlocal encoding with measured error suppression. The physical cluster, the encoded doublet, the simulated model, the emergent anyons, and the logical qubit form a ladder of successive tests: passing one qualifies the experiment for the next, and the names must not be used interchangeably.

Perturbative Suppression of the Useful Energy Scale

The second-order example above was chosen to favor the proposal. A connected four-body process typically first appears at fourth order: three applications of the interaction plus the initial coupling, each contributing one power of the small expansion ratio. The name “four-body” alone does not fix that order; the allowed virtual processes do, and the worked example below takes the fourth-order case explicitly.

Isolation has a cost in interaction strength

Relative effective scale, $J_{\text{eff}} / v \cdot \text{unit path coefficient} \cdot \text{logarithmic axes}$



Second order

$J_{\text{eff}} / v \approx \epsilon$

Fourth order

$J_{\text{eff}} / v \approx \epsilon^3$

At $\epsilon = 0.1$

100× between the
two effective scales

Stronger isolation $\leftarrow$ smaller ϵ

larger $\epsilon \rightarrow$ stronger virtual mixing

Both curves hold the microscopic interaction v fixed with path coefficient one. The slope — one power of the expansion ratio per additional perturbative order — sets the scale; the name of the target operator does not.

Note the reused symbol: here c is the virtual-path coefficient of this example, while Section 37.5 used c for the gap fraction. The two play different roles and take different values.

Each cluster has a low-energy residual doublet separated from unwanted cluster states by an energy Δ_c . Neighboring defects interact through a two-body energy J . The dimensionless perturbative parameter is:

$$\epsilon = \frac{J}{\Delta_c}$$

For a four-body plaquette coefficient:

$$K \sim cJ\epsilon^3 = c \frac{J^4}{\Delta_c^3}$$

$$\frac{\Delta_c}{h} = 5 \text{ GHz}, \quad \frac{J}{h} = 100 \text{ MHz}, \quad c = 1$$

$$\epsilon = 0.02, \quad \frac{K}{h} = 100 \text{ MHz} \times (0.02)^3 = 800 \text{ Hz}$$

For $T_2 = 1 \text{ ms}$:

$$\frac{\hbar\Gamma}{h} = \frac{\Gamma}{2\pi} = 159 \text{ Hz}$$

$$\frac{K}{\hbar\Gamma} \approx 5$$

A factor of 5 against decoherence leaves no room for the disorder, calibration, and coefficient uncertainties still to come, well short of the robust $\gg 1$ hierarchy the proposal needs.

The 800 Hz figure is the estimated coefficient K/h ; using it as the gap frequency below is an added toy-model simplification, since a real many-body spectrum yields its own gap. The intended ratio $K/(\hbar\Gamma)$ with $\Gamma = 1/T_2 = 1000 \text{ s}^{-1}$ is $800/(1000/2\pi) \approx 5.0$, not $800/1000$ when the intended ratio uses matched frequency units.

Preparation adds a separate timescale. Parametrize a fixed interpolation from an easy initial Hamiltonian to the desired one by $s = t/t_f$. A common adiabatic estimate requires the total time to exceed the rate of Hamiltonian change divided by the square of the smallest gap along the whole path:

$$t_f \gg \frac{\hbar \|dH/ds\|}{\Delta_{\min}^2} = \frac{1}{2\pi} \frac{10^8}{800^2} \text{ s} \approx 25 \text{ s}$$

The assumed path takes $\|dH/ds\|/h = 10^8 \text{ Hz}$ with $\Delta_{\min}/h = 800 \text{ Hz}$. The resulting 25 s is a sufficient-scale estimate for that path — matrix elements, path shape, smoothness, and error target all shift it — and it exceeds the assumed millisecond coherence time by four orders of magnitude.

At the same $T = 10 \text{ mK}$ used above:

$$\frac{k_B T}{h} = (20.84 \text{ GHz/K})(0.010 \text{ K}) \approx 208 \text{ MHz}$$

That thermal frequency exceeds an 800 Hz equilibrium gap by a factor of about 260,000.

Detailed treatment: perturbative suppression of the useful energy scale

Perturbative suppression of the useful energy scale

The worked example uses a toy model chosen to favor the proposal: each cluster contributes its low-energy doublet as the encoded degree of freedom, separated from the unwanted cluster states by Δ_c , while neighboring defects or clusters couple through a two-body energy J . Whether the unwanted states can be eliminated perturbatively is decided by the dimensionless ratio

$$\epsilon = \frac{J}{\Delta_c}.$$

Controlled projection needs $\epsilon \ll 1$. The example takes the desired four-body plaquette coefficient K to first appear at fourth order, following the general perturbative-gadget pattern in which a k -body term built from two-body couplings first arises at order k [R174]. If the dimensionless geometry coefficient c is retained but its detailed value is not specified, then

$$K \sim c J \epsilon^3 = c \frac{J^4}{\Delta_c^3}.$$

The same small ratio therefore does double duty: it protects the encoded doublet and simultaneously shrinks the interaction meant to open the topological gap.

The dependencies among the relevant quantities are as follows.

```
fabrication and charge-state yield
|
+-- positions and orientations -----> pair couplings J_ij and graph
| |
| +--> coupling disorder sigma_J
|
+-- cluster spectrum -----> isolated doublet, gap Delta_c
|
+-- require epsilon = J/Delta_c K ~ J epsilon^3
|
+-----+-----+
| | |
unwanted terms topological gap preparation gap
```

```

delta H Delta_topo Delta_min
| | |
+----- require -----+-----+-----+
|
Delta_topo >> sigma_K, ||delta H||, hbar Gamma, k_B T
|
finite patch and logical sector
|
repeat with adequate yield and control

```

Assign hypothetical values, chosen optimistically rather than measured in any array:

$$\frac{\Delta_c}{h} = 5 \text{ GHz}, \quad \frac{J}{h} = 100 \text{ MHz}, \quad c = 1,$$

where h is Planck's constant. These values give $\epsilon = 0.02$, and therefore

$$\frac{K}{h} = 100 \text{ MHz} \times (0.02)^3 = 800 \text{ Hz}.$$

The projection condition is comfortably satisfied at those values, and the resulting many-body scale is 800 Hz. For a hypothetical coherence time $T_2 = 1 \text{ ms}$, define the decoherence rate $\Gamma = 1/T_2 = 10^3 \text{ s}^{-1}$. The corresponding energy linewidth, expressed in frequency units, is

$$\frac{\hbar\Gamma}{h} = \frac{\Gamma}{2\pi} = 159 \text{ Hz}.$$

The ratio is $K/(\hbar\Gamma) \approx 5$, short of a robust $\gg 1$ hierarchy, and the thin margin is exposed on three sides: a realistic path coefficient below the optimistic $c = 1$, one additional perturbative order, or correlated noise each suffice to erase it.

Placement errors grow through the same fourth power. For a magnetic dipolar coupling,

$$J \propto r^{-3},$$

where r is the separation in metres. A small radial error δr produces

$$\frac{\delta J}{J} \approx -3 \frac{\delta r}{r}.$$

Because $K \propto J^4$ along this toy path,

$$\frac{\delta K}{K} \approx 4 \frac{\delta J}{J} \approx -12 \frac{\delta r}{r}.$$

In this simplified estimate with one shared coupling, a 5% spread in separation becomes a 60% spread in K . That arithmetic measures sensitivity along this path, not a universal disorder law; independent bonds and angular errors need the actual Hamiltonian.

State preparation pays the same small gap. Take the minimum many-body gap along an interpolation optimistically as $\Delta_{\min} = \hbar \times 800$ Hz, and let the norm of the Hamiltonian derivative with respect to the dimensionless interpolation coordinate s be $\|dH/ds\| = \hbar \times 100$ MHz. A common sufficient adiabatic timescale is [\[R245\]](#)

$$t_f \gg \frac{\hbar \|dH/ds\|}{\Delta_{\min}^2} = \frac{1}{2\pi} \frac{10^8}{800^2} \text{ s} \approx 25 \text{ s}.$$

That duration is about 2.5×10^4 hypothetical coherence times. Schedule and theorem details shift the prefactor; the inverse-square gap dependence is what makes the mismatch structural rather than numerical.

At $T = 10$ mK,

$$\frac{k_B T}{\hbar} = (20.84 \text{ GHz/K})(0.010 \text{ K}) \approx 208 \text{ MHz},$$

which exceeds an 800 Hz equilibrium gap by five orders of magnitude. Driven preparation can outrun thermalization temporarily, and that speed changes what is being demonstrated: a controlled nonequilibrium preparation rather than a passive thermal phase.

The 800 Hz scale fails as a shared budget: it must simultaneously clear disorder, decoherence, and temperature while also allowing preparation within its own inverse-gap timescale. An 800 Hz coupling is usable in isolation; it is the simultaneity that

defeats it here.

Omitting any one inequality makes the toy model look promising; the assessment stands or falls on imposing them together.

Limits of Topological-Gap Stability Theorems

A stability result protects a phase already established under its hypotheses. Deriving the target phase from a fabricated interaction graph is prior work the theorem assumes complete.

Topological-order stability theorems start from ideal Hamiltonians that are already local and gapped, and show their separated spectral bands surviving sufficiently weak local perturbations.

The result therefore applies to an engineered Hamiltonian already inside the target phase with perturbations below the threshold. A weak approximate plaquette term beside larger parasitic terms falls outside those hypotheses.

$$\Delta_{\text{topo}} \gg \max(k_B T, \sigma_K, \|H_{\text{unwanted}}\|, \hbar\Gamma, \delta_{\text{fs}})$$

Here σ_K is the disorder in the useful couplings and δ_{fs} the splitting of candidate ground sectors at finite size. The displayed hierarchy is a conservative engineering checklist assembled from the sections above, distinct from the statement of any one stability theorem.

In particular, the global norm $\|H_{\text{unwanted}}\|$ accumulates with the number of sites even where every local perturbation is weak. A size-independent stability claim therefore uses the theorem's local interaction norm or local strength bound. A total operator norm suits a small-cluster screening calculation; carrying it to a large array without adjustment overstates the constraint.

What survives disorder, scaling, and the evidence?

The final step moves beyond the ideal cluster: disorder, scaling, and the assembled evidence each test whether the mechanism survives contact with fabrication.

Failure Modes of the Evidence-Based Case

The favorable argument assumes compatible parameters across all steps and a complete effective Hamiltonian. The five points below name where that assumption commonly breaks: each states the error and the physical reason it matters.

Point 1: Optimal parameters from incompatible samples cannot be combined into one device claim.

- **Explanation:** A device claim must demonstrate its coherence time, interaction strength, placement precision, and optical linewidth under one shared set of conditions. Assembling the best value of each from different samples — a dilute bulk spin, a damaged close pair, a nanostructure process, a cryogenic center — describes four devices, not one.

Point 2: Disorder tolerance still requires the right interaction graph.

- **Explanation:** A topological phase tolerates bounded local perturbations around the target Hamiltonian. An interaction graph that misses the target Hamiltonian is a different defect, outside that tolerance.

Point 3: Auxiliary systems supply active control, not passive protection.

- **Explanation:** Nuclear-spin memories and repeated control operations improve preparation and readout as active resources: they cost pulses and calibration. Passive protection means the static many-body spectrum suppresses errors without that intervention.

Point 4: A generated term counts only beside all competing terms.

- **Explanation:** The desired v^q/Δ^{q-1} term competes with lower-order local fields, direction-dependent anisotropies, leakage corrections, disorder, and decoherence, and must exceed their sum to set the physics.

Point 5: A pair demonstration fixes one separation, not the array scaling.

- **Explanation:** Dipolar coupling falls steeply with distance, so a 10% coupling tolerance needs roughly 3% relative distance control, before angular errors tighten it further.

Detailed treatment: failure modes of the evidence-based case

Failure modes of the evidence-based case

- Optimal parameters from incompatible samples cannot be combined into one device claim. The parameter stack above performs that combination once, explicitly labeled, to show the gap between its borrowed inputs. A device claim needs coherence — the retention of quantum phase information — interaction strength, placement precision, and optical linewidth — the spectral width of the optical transition — demonstrated under one shared set of conditions.

- Disorder tolerance assumes the right graph. A topological phase tolerates bounded local perturbations — spatially local changes of limited magnitude — around its target Hamiltonian, the operator defining the intended dynamics and spectrum. An interaction graph that misses that target is a different Hamiltonian, outside the tolerance the theorems grant.
- Auxiliary systems supply active control rather than passive protection. Nuclear-spin memories and repeated control operations improve preparation and readout at the cost of pulses and calibration. Passive protection means the time-independent energy spectrum suppresses the relevant errors without repeated intervention.
- A generated interaction term counts only beside its competitors. The desired v^q/Δ^{q-1} term, where v is the perturbative coupling scale, Δ is the relevant excitation-energy denominator, and q is the perturbative order, must exceed lower-order local fields, direction-dependent anisotropies, leakage corrections arising from transitions outside the intended low-energy subspace, disorder, and decoherence. The existence of a nonzero coefficient is therefore insufficient to establish useful operation.
- A pair demonstration fixes one separation; array scaling needs the coupling law. For $J \propto r^{-3}$ with strength J and separation r , a small radial error gives $\delta J/J \approx -3\delta r/r$. A 10% coupling tolerance therefore needs roughly 3% relative distance control, before angular errors tighten it further.
- “Fibonacci-like” names no phase by itself. Ising non-Abelian order exchanges particles by noncommuting operations; doubled-Fibonacci string-net order is the nonchiral topological order described by a string-net model; digital braid emulation reproduces braid operations with programmed gate sequences; genuine chiral Fibonacci order supports Fibonacci anyons with a preferred propagation chirality. The four have different excitations and different computational power, so evidence for one leaves the others open.
- Long coherence and low thermal occupation are separate requirements. T_2 , the transverse coherence time under specified control, measures phase memory; thermal excitation follows $k_B T/\Delta_{\text{topo}}$ with Boltzmann constant k_B , temperature T , and topological gap Δ_{topo} . Passing the coherence comparison leaves the thermal comparison still to be passed.
- The structural advantage claimed is the coordinated design of cluster motif, energy spectrum, interaction, auxiliary systems, and measurement protocol together. The case rests on that coordination, not on any host material being universally superior.

Five testable claims

The strongest version of the proposal is a staged program of five independently testable hypotheses, each checkable before any large array is fabricated:

- Correlated fabrication can produce a reproducible cluster motif, meaning a repeated spatial arrangement of defects.
- At least one motif has an isolated doublet, meaning a pair of low-energy states separated from other states, with useful operators after projection into that doublet.
- Crystal orientation, together with either local or mediated coupling, can realize the required low-order interaction graph.
- Under one shared set of conditions, the resulting many-body gap — the energy separation between the target low-energy sector and its excitations — exceeds the disorder and decoherence scales.
- Available ancillas — auxiliary quantum degrees of freedom for control and measurement — can read the local and nonlocal observables that identify the phase, while the phase itself arises from the continuously acting Hamiltonian rather than from programmed gate sequences.

The proposal stands as a staged, falsifiable research program: each hypothesis above can fail on its own terms. The next chapter presses on the dependencies where failure costs the most — coupling uniformity, perturbative order, thermal scale, and fabrication yield.

Self-assessment

Self-assessment

- **Strongest cluster-specific advantage.**

A cluster's spectrum and wavefunctions can convert available microscopic couplings into designed residual operators within the low-energy subspace while placing leakage states behind a measurable energy gap.

- **First-order projection of $V_{AB} = vS_{A,3}^z S_{B,1}^z$.**

Let P_A and P_B project onto the doublets of clusters A and B , respectively. Let I_A and I_B denote identity operators in those doublets, and let τ_A^z and τ_B^z denote effective Pauli- z operators. Inside the doublets,

$$P_A S_{A,3}^z P_A = a_0 I_A + a_z \tau_A^z$$

and

$$P_B S_{B,1}^z P_B = b_0 I_B + b_z \tau_B^z.$$

Expanding the projected product shows that the $\tau^z \tau^z$ coefficient is $va_z b_z$. Therefore, the first-order residual coupling is $J_{zz}^{(1)} = va_z b_z$. Local-field terms remain unless symmetry forces $a_0 = b_0 = 0$.

- **Consequence of using the same $\epsilon = 0.1$ at fourth rather than second order.**

Here, ϵ is the dimensionless perturbative ratio. At fourth order,

$$\nu_{\text{eff}}^{(4)} \sim \nu_v \epsilon^3 = 0.1 \text{ MHz}$$

and

$$\nu_{\text{topo}}^{(4)} \sim 20 \text{ kHz},$$

which is only twice the adopted $1/T_2$ estimate before disorder is included. Thus, the optimistic parameter stack exceeds the coherence rate only at low perturbative order.

- **Parameter with the greatest leverage in the worked stack.**

The perturbative order q has the greatest effect. At $v/\Delta_C = 0.1$, where Δ_C is the cluster excitation gap, increasing the process from second to fourth order suppresses the effective term by 10^{-2} .

- **Evaluation and limitation of $\nu_{\text{topo}} T_2 = 200$ at the optimistic point.**

$$(2 \times 10^6 \text{ s}^{-1})(10^{-4} \text{ s}) = 200.$$

At most, this result indicates that coherent dynamics might resolve a gap under matched conditions. It does not establish thermal protection, realization of the correct phase, or scalable readout.

- **Remaining speculative claim.**

The unverified claim is that a fabricated defect-cluster array can realize a gapped doubled-Fibonacci Hamiltonian and host its emergent quasiparticles, which are collective excitations of the many-body system.

Grouping defects is a specific design strategy whose five elements — the residual two-state object, the crystal plane, the single-fabrication motif, the nearby auxiliaries, and the continuously acting analog Hamiltonian — have each been investigated separately.

Those five elements have never operated together in one array. The optimistic parameter stack outruns the borrowed coherence rate only at low perturbative order, and even there its topological scale sits about two orders of magnitude below the thermal scale at 10 mK.

The defensible case is therefore a staged research program that must still earn each step; a passively self-correcting Fibonacci crystal is the endpoint such a program would have to reach, not its starting point.

Sources

Sources

- [R242] L. B. Hughes, S. A. Meynell, W. Wu, S. Parthasarathy, L. Chen, Z. Zhang, Z. Wang, E. J. Davis, K. Mukherjee, N. Y. Yao, and A. C. Bleszynski Jayich, “Strongly Interacting, Two-Dimensional, Dipolar Spin Ensembles in (111)-Oriented Diamond,” *Physical Review X* 15, 021035 (2025). DOI: [10.1103/PhysRevX.15.021035](https://doi.org/10.1103/PhysRevX.15.021035); arXiv: [2404.10075](https://arxiv.org/abs/2404.10075). See also the 2026 [publisher erratum](#).
- [R243] M. Haruyama et al., “Triple nitrogen-vacancy centre fabrication by C₅N₄H_n ion implantation,” *Nature Communications* 10, 2664 (2019). DOI: [10.1038/s41467-019-10529-x](https://doi.org/10.1038/s41467-019-10529-x).
- [Ro80] F. Dolde et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* 9, 139–143 (2013). DOI: [10.1038/nphys2545](https://doi.org/10.1038/nphys2545).
- [R207] S. Kim, P. London, D. Yang, L. B. Hughes, J. Ahlers, S. Meynell, W. J. Mitchell, K. Mukherjee, and A. C. Bleszynski Jayich, “Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures,” *Nature*

Communications 16, 9803 (2025). DOI: [10.1038/s41467-025-64758-4](https://doi.org/10.1038/s41467-025-64758-4); arXiv: [2502.01198](https://arxiv.org/abs/2502.01198).

- [R119] C. E. Bradley et al., “A Ten-Qubit Solid-State Spin Register with Quantum Memory up to One Minute,” *Physical Review X* 9, 031045 (2019). DOI: [10.1103/PhysRevX.9.031045](https://doi.org/10.1103/PhysRevX.9.031045); arXiv: [1905.02094](https://arxiv.org/abs/1905.02094).
- [Ro86] R. E. Evans et al., “Photon-mediated interactions between quantum emitters in a diamond nanocavity,” *Science* 362, 662–665 (2018). DOI: [10.1126/science.aau4691](https://doi.org/10.1126/science.aau4691).
- [Ro25] S. Bravyi, D. P. DiVincenzo, and D. Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” *Annals of Physics* 326, 2793–2826 (2011). DOI: [10.1016/j.aop.2011.06.004](https://doi.org/10.1016/j.aop.2011.06.004); arXiv: [1105.0675](https://arxiv.org/abs/1105.0675).
- [Ro17] A. Kitaev, “Anyons in an exactly solved model and beyond,” *Annals of Physics* 321, 2–111 (2006). DOI: [10.1016/j.aop.2005.10.005](https://doi.org/10.1016/j.aop.2005.10.005); arXiv: [cond-mat/0506438](https://arxiv.org/abs/cond-mat/0506438).
- [Ro18] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110); arXiv: [cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617).
- [R244] N. Y. Yao, L. Jiang, A. V. Gorshkov, P. C. Maurer, G. Giedke, J. I. Cirac, and M. D. Lukin, “Scalable architecture for a room temperature solid-state quantum information processor,” *Nature Communications* 3, 800 (2012). DOI: [10.1038/ncomms1788](https://doi.org/10.1038/ncomms1788); arXiv: [1012.2864](https://arxiv.org/abs/1012.2864).

Yield Reduction Under Scaling

Fabrication compounds like calibration: the yield model depends on what counts as a working device. An array that needs every site differs from one that tolerates vacancies or reroutes around them. The calculation starts from the strict all-sites-required case; any repair mechanism then needs its own stated model.

Take every required site to work independently with probability p :

$$Y_N = p^N$$

For $N = 1000$, $p = 0.99$ gives $Y_N \approx 4.3 \times 10^{-5}$. $p = 0.999$ gives $Y_N \approx 0.368$.

Taking logarithms shows the scaling: $\ln Y_N = N \ln p \approx -N(1 - p)$ when p is close to one, so a small per-site failure probability multiplies into a large patch-level one. Where independent bonds must also work with probability p_b , the model extends to $Y = p_s^{N_s} p_b^{N_b}$.

Those formulas assume independence with no repair or defect tolerance. Correlated errors, postselection, modular assembly, or vacancy tolerance each need their own yield model, with their own definition of a successful device.

Calibration grows the same way: every cell brings its own frequencies and couplings. A design needing one analog cancellation tone per parasitic bond can field a scalable component count alongside an impractical calibration volume.

Current Experimental and Theoretical Evidence

The evidence base is asymmetric — strong on components and theory, thin on integration:

- **[Experiment]** Individual defects, small registers, coherent pair coupling, logical-protocol primitives, and improving registered fabrication have been demonstrated.
- **[Theory]** Perturbative generation, stability of ideal topological phases, adiabatic conditions, and finite-temperature limitations are well developed.
- **[Proposal]** Mapping a particular manufacturable defect-cluster Hamiltonian to a clean non-Abelian string-net model remains a proposal.
- **[Speculation]** Extrapolating present defect components to a scalable, passively protected Fibonacci-like material is scientifically coherent but remains unsupported as an integrated hardware claim.

Detailed treatment: current experimental and theoretical evidence

Current experimental and theoretical evidence

The evidence available through 2026 is asymmetric.

- [Experiment] Individual defects, small registers, coherent pair coupling, logical-protocol primitives, and improving registered fabrication have been demonstrated [\[R207\]](#); [\[R080\]](#); [\[R234\]](#).
- [Theory] Perturbative generation, stability of ideal topological phases, adiabatic conditions, and finite-temperature limitations are well developed [\[R174\]](#); [\[R245\]](#); [\[R018\]](#); [\[R142\]](#); [\[R169\]](#); [\[R168\]](#).

- [Proposal] Mapping a particular manufacturable defect-cluster Hamiltonian to a clean non-Abelian string-net model remains a proposal.
- [Numerics] Small-model numerical calculations can validate a specified effective Hamiltonian, but they cannot determine fabrication distributions or decoherence parameters that have not been measured.
- [Speculation] Extrapolating present defect components to a scalable, passively protected Fibonacci-like material is scientifically coherent but remains unsupported as an integrated hardware claim.

The resulting negative assessment is not that the proposal is impossible. It is that the central conjunction of requirements has no demonstrated margin. The strongest experimental results establish lower-level components of the dependency structure, whereas the proposed topologically protected phase depends on the complete structure.

Common Analytical Errors

Point 1: Components must be tested against all inequalities together.

- **Explanation:** A long coherence time, a strong pairwise coupling, accurate implantation, and realization of a topological model in four different samples do not constitute a single functional device.

Point 2: Only a vetted operator counts as a plaquette term.

- **Explanation:** A four-spin spectral shift is one entry on the checklist. The operator structure, coefficient sign, competing Hamiltonian terms, spatial pattern, and validity of the perturbative approximation must each match the target Hamiltonian before the observed shift counts as its plaquette term.

Point 3: A digital implementation tests control, not passive emergence.

- **Explanation:** Pulse synthesis cancels unwanted interaction edges and generates multi-body average Hamiltonians through continued driving. Where continuous calibration and periodic driving are essential, the demonstrated protection comes from that control, and an autonomous equilibrium Hamiltonian remains untested.

Point 4: Stability results apply after dominance is shown.

- **Explanation:** The relevant theorem covers a topological phase already established, with perturbations below its threshold. Invoking it before the target term is shown to dominate its competitors assumes the premise the experiment must establish.
-

Detailed treatment: common analytical errors

Common analytical errors

- Testing components one at a time leaves the joint claim open. A long coherence time T_2 , where T_2 is the transverse dephasing time; a strong pairwise coupling J ; accurate implantation, meaning controlled placement of defects; and realization of a topological model in four different samples do not constitute a single functional device. The relevant quantity is the joint distribution—the correlated statistical variation—of J , the cluster-isolation energy Δ_c , disorder, coherence, fabrication yield, and control performance within the same fabricated patch.
- An arbitrary effective interaction needs vetting before it counts as a plaquette term — a Hamiltonian operator supported on the degrees of freedom around an elementary face of a lattice. Observation of a four-spin spectral shift alone is insufficient. The operator structure, coefficient sign, competing Hamiltonian terms, spatial pattern, and validity of the perturbative approximation must all agree with the target Hamiltonian, which is the model intended for physical realization. Otherwise, the experiment demonstrates a higher-order interaction but not the required operator algebra.
- A digital implementation tests control rather than passive emergence. Pulse synthesis cancels unwanted interaction edges and generates multi-body average Hamiltonians through continued driving. Such an implementation may provide an effective digital or Floquet simulation, where Floquet simulation uses periodic driving to produce a time-averaged Hamiltonian. However, if continuous calibration and periodic driving are essential, the protection mechanism cannot be attributed solely to an autonomous equilibrium Hamiltonian.
- Topological-stability results apply once dominance is shown. The relevant theorem covers a topological phase already established, with perturbations below its threshold. It cannot be invoked until the target term has been shown to dominate the competing terms and the unperturbed system has been established to lie within that phase.
- Postselection — retaining only realizations that satisfy a chosen success criterion — must be reported alongside the yield it conceals. Selecting one successful pair from a large implanted field demonstrates pair-level physics. A scalable architecture must also account for failed sites, unusable charge states, missing interaction links, calibration time, and the effects of graph repair on the interaction network.

- Refrigerator temperature and effective-Hamiltonian temperature play different roles. The dimensionless comparison that decides thermal occupation is $k_B T / \Delta_{\text{topo}}$, where k_B is Boltzmann's constant, T is temperature, and Δ_{topo} is the topological excitation gap. Refrigerator temperature alone is not the relevant criterion. A millikelvin temperature does not provide strong thermal protection if the emergent gap is on the hertz scale.
- Several physically distinct claims form a ladder, not an equivalence. A physical defect qubit can exhibit coherence. A cluster can encode a residual degree of freedom. A controlled array can digitally emulate a string-net Hamiltonian, where a string-net Hamiltonian is a many-body model that can support emergent topological order. An analog material can possess emergent topological order. An encoded state can also be protected through active error correction. Failure on the analog defect-cluster rung leaves the other four claims standing, each testable on its own terms.

Consistency checks

Consistency checks

- **Why high-order-term generation outranks control complexity.** Control shortfalls admit architectural workarounds — different pulse sequences, added calibration, improved interfaces — that leave the target phase intact. Missing plaquette or stabilizer algebra — the mutually compatible constraint operators defining the encoded subspace — leaves no target phase for control to address. High-order synthesis additionally sets the small energy scale that every later performance test inherits.
- **Check of $K/h = 800$ Hz and $K/(\hbar\Gamma) \approx 5$.** Here, K is the effective high-order coupling energy, h is Planck's constant, $\hbar = h/2\pi$ is the reduced Planck constant, Γ is the decoherence rate, and ϵ is the perturbative expansion parameter. With $J/h = 100$ MHz and $\epsilon = 0.02$,

$$K/h = 100 \text{ MHz} \times (0.02)^3 = 800 \text{ Hz}.$$

For $T_2 = 1$ ms, $\Gamma = 10^3$ s⁻¹ and $\hbar\Gamma/\hbar = \Gamma/2\pi = 159$ Hz. The ratio is therefore about 5 rather than $\gg 1$: the effective coupling exceeds the decoherence scale by a small factor that later uncertainties can erase.

- **Consequence of increasing cluster isolation Δ_c/J to protect the doublet.**

The doublet is the intended two-dimensional low-energy subspace of a cluster. A fourth-order coefficient scales as

$$J(J/\Delta_c)^3.$$

Raising Δ_c/J tightens the projection into the doublet through the same denominator that sets the effective interaction, so better isolation directly shrinks the topological gap.

- **Effect of a 5% radial spread on K for a fourth-order dipolar path.** For a dipolar interaction, the fractional coupling variation satisfies

$$\delta J/J \approx -3\delta r/r,$$

where r is the separation and δr is its variation. Because

$$K \propto J^4,$$

the corresponding variation is

$$\delta K/K \approx -12\delta r/r.$$

Thus,

$$12 \times 0.05 = 0.60.$$

A 5% radial spread therefore becomes a 60% spread in K along this fourth-order dipolar path. Independent bonds and angular variations need the actual Hamiltonian, where each bond carries its own error.

- **What 46 nm lateral precision means for a few-nanometre architecture.**

One advanced high-yield 2025 process reached 46 nm lateral precision against a few-nanometre requirement. Later processes may improve on that figure; any

architectural assessment must name its required tolerance and compare it against a directly comparable process.

- **What $k_B T > \Delta_{\text{topo}}$ leaves open.** Fast, driven, cooled, measured, or postselected experiments can still reveal the model's dynamics under that inequality. A passive equilibrium memory at that temperature is the outcome they leave untested.

Sources

Sources

- [R174] S. P. Jordan and E. Farhi, “Perturbative Gadgets at Arbitrary Orders,” *Physical Review A* 77, 062329 (2008). DOI: [10.1103/PhysRevA.77.062329](https://doi.org/10.1103/PhysRevA.77.062329); [arXiv:0802.1874](https://arxiv.org/abs/0802.1874).
- [R245] T. Albash and D. A. Lidar, “Adiabatic quantum computation,” *Reviews of Modern Physics* 90, 015002 (2018). DOI: [10.1103/RevModPhys.90.015002](https://doi.org/10.1103/RevModPhys.90.015002); [arXiv:1611.04471](https://arxiv.org/abs/1611.04471).
- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110); [arXiv:cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617).
- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: Stability under local perturbations,” *Journal of Mathematical Physics* 51, 093512 (2010). DOI: [10.1063/1.3490195](https://doi.org/10.1063/1.3490195); [arXiv:1001.0344](https://arxiv.org/abs/1001.0344).
- [R207] S. Kim et al., “Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures,” *Nature Communications* 16, 9803 (2025). DOI: [10.1038/s41467-025-64758-4](https://doi.org/10.1038/s41467-025-64758-4); [arXiv:2502.01198](https://arxiv.org/abs/2502.01198).
- [R080] F. Dolde et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* 9, 139–143 (2013). DOI: [10.1038/nphys2545](https://doi.org/10.1038/nphys2545); [arXiv:1212.2804](https://arxiv.org/abs/1212.2804).
- [R169] B. J. Brown, D. Loss, J. K. Pachos, C. N. Self, and J. R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* 88, 045005 (2016). DOI: [10.1103/RevModPhys.88.045005](https://doi.org/10.1103/RevModPhys.88.045005); [arXiv:1411.6643](https://arxiv.org/abs/1411.6643).

- [R168] S. Bravyi and B. M. Terhal, “A no-go theorem for a two-dimensional self-correcting quantum memory based on stabilizer codes,” New Journal of Physics 11, 043029 (2009). DOI: [10.1088/1367-2630/11/4/043029](https://doi.org/10.1088/1367-2630/11/4/043029); [arXiv:0810.1983](https://arxiv.org/abs/0810.1983).
 - [R234] M. H. Abobeih et al., “Fault-tolerant operation of a logical qubit in a diamond quantum processor,” Nature 606, 884–889 (2022). DOI: [10.1038/s41586-022-04819-6](https://doi.org/10.1038/s41586-022-04819-6); [arXiv:2108.01646](https://arxiv.org/abs/2108.01646).
 - [R246] M. Ruf, N. H. Wan, H. Choi, D. Englund, and R. Hanson, “Quantum networks based on color centers in diamond,” Journal of Applied Physics 130, 070901 (2021). DOI: [10.1063/5.0056534](https://doi.org/10.1063/5.0056534); [arXiv:2105.04341](https://arxiv.org/abs/2105.04341).
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Assessment II — Turn requirements into decisive tests

The preceding assessment located the points where the proposal could fail. This chapter converts each of those points into a numerical rejection test and lays out the staged calculation that supplies the numbers. The decision rules come first, so a favorable-looking result is judged against thresholds fixed in advance.

Original chapter framing and supporting arguments

The previous chapter listed the ways the architecture could fail. A workable research plan gives each one a numerical consequence and a pass-fail value. Each value follows from the job the device must do: how long it must store information, how much failure is tolerable, how large the array must be, and what control resources are available.

Background, scope, and supporting argument

A stable bit in a cooled crystal can absorb years of experimental work. Before committing cryogenic infrastructure and fabrication effort, the project writes down the numerical result that would invalidate its central claim.

That number describes the unfavorable outcome precisely enough to end the specific claim. It is recorded early, while the interpretation of later data is still open, and it stays fixed once measurement begins.

A predeclared threshold of this kind is called a kill criterion. It pairs a measurable condition with a concrete response: change direction or terminate the claim.

A design target says that a larger energy gap is desirable. A kill criterion says that, if the experimentally supported upper bound on the activation gap stays below a stated value after measurement X , work stops on a memory meant to hold information at temperature T .

Every constraint applies at once.

Low cluster yield makes a complete patch harder to fabricate. Weak coupling shrinks the energy gap.

A smaller gap stretches the preparation time. A longer preparation exposes the system to decoherence for longer.

Viability therefore depends on all factors acting together. Strength in one factor leaves the others in force.

Writing the rule down in advance commits the project to a quantitative decision procedure before any result is known.

Chapter 40 — Calculations with rejection criteria

The criteria are in place; the calculations that supply their inputs come next. Each stage hands a defined object to the following one: a validated defect model, a cluster encoding, an effective interaction, and finally a many-body phase assessment. An early failure is informative because it points to the assumption that needs revision.

Background, scope, and supporting argument

The calculation runs through tests of increasing cost. It starts with two defects, each a localized departure from the host structure that supplies microscopic degrees of freedom. Diagonalizing the corresponding four-by-four Hamiltonian gives its eigenvalues and eigenstates. If the two lowest-energy states fail to form a usable encoded pair, the architecture is rejected at this stage.

The intended sequence is

two defects → one cluster leftover → three-to-six-cluster plaquette → small patch.

Here, a cluster is a finite group of coupled defects; a leftover is the retained low-energy degree of freedom used as an encoded two-state system; a plaquette is a local arrangement of clusters associated with a candidate many-body interaction; and a patch is a finite portion of the proposed lattice model. The computational cost increases along this sequence.

At each stage the inputs are specified, the relevant observables are computed, and convergence is tested by tightening approximations or solver tolerances. The result is compared with acceptance criteria fixed before the calculation. An architecture that misses the criteria stops there, without consuming the more expensive stage. This ordering removes infeasible architectures before large-scale computing time is spent.

A plausible defect pair still leaves open whether an isolated cluster doublet exists: a pair of low-energy states separated from higher states. A plausible doublet still leaves open whether the intended plaquette operator appears.

A recognizable plaquette term still leaves open whether the system sits in a topological phase. Likewise, two closely spaced eigenvalues on a small finite patch do not by themselves establish topological degeneracy.

Each later calculation builds on quantitative outputs of the preceding stage: calibrated matrices, projectors, uncertainty ranges, effective couplings, and estimated residual errors. A qualitative sense of confidence has no place among these outputs.

In this unit:

1. [Define the decision before running the model](#)
2. [Translate physical limits into quantitative tolerances](#)
3. [Calculate from the smallest system to the patch](#)
4. [Check that the test really addresses the claim](#)

Define the decision before running the model

Start by naming what counts as failure, which observable reveals it, and how the decision redirects the next step.

Introduction: The Kill Criterion

A **kill criterion** is a measurable condition that commits a project to change direction or terminate a claim once the condition is met. A general list of design objectives states aims; a kill criterion states a predeclared threshold together with the action it triggers for the claim under test. That action may end the project, narrow its objective, or require a redesign.

A stable bit in a cooled crystal can take years of experimental work. Before that commitment, the project writes down the numerical result that would invalidate its central claim.

Components of a Useful Rejection Criterion

1. A specific claim that stays identifiable after the measurement.
2. A metric with stated units and a stated measurement or calculation protocol.
3. A threshold derived from that claim.
4. An uncertainty rule fixed before the result is observed.
5. A predetermined action: continue, redesign, pivot, or stop.

The threshold stays fixed once the result is known. A failed threshold with no predeclared action attached dissolves into further discussion rather than producing a decision.

Detailed treatment: components of a useful rejection criterion

Components of a useful rejection criterion

A workable kill criterion contains five parts:

- a specific claim that remains identifiable after the measurement;
- a metric with defined units and a specified measurement or calculation protocol;
- a threshold derived from that claim;
- an uncertainty rule established before the result is observed;
- a predetermined action: continue, redesign, pivot, or stop.

The fourth part is preregistration: the decision rule is fixed before the data are examined, and the threshold stays in place once the result is known.

A ratio such as $J/\Gamma > 10$, with J an interaction energy and Γ a decay rate on an angular-frequency scale, cannot serve every protocol. The required ratio depends on how many interaction times the protocol consumes and what fraction of the total error budget goes to decoherence.

The condition $\Delta > k_B T$, with Δ an energy gap, k_B Boltzmann’s constant, and T temperature, sets too weak a bar for a thousand-site sample when the allowed probability of even one thermally generated harmful excitation is very small. A usable criterion combines the physical model with the operational task.

A failed threshold attached to a predeclared action produces a decision directly.

Example Decision Table

Read the table as an illustration of a decision process. The bracketed ranges are hypothetical uncertainty intervals; the source leaves their confidence level and statistical construction unspecified. They are not measured confidence intervals.

“Resolve” means the uncertainty straddles the decision boundary, so a targeted measurement or calculation must settle the row. “Redesign” means the proposed implementation misses a requirement that a changed design might still meet. “Stop/pivot” drops the stated objective at the assumed operating point. A pass in one row leaves a failure in an essential row in force.

The proposal assumes that one thousand residual cluster degrees of freedom form a two-dimensional analog topological memory with $N_s = 1000$ required sites and $N_b = 1500$ required bonds.

Gate	Hypothetical estimate	Preregistered threshold	Decision	Required action
Coherent interaction $R_J = J_{\text{eff}}/(\hbar\Gamma)$	1.26×10^4 [$3.1 \times 10^3, 5.0 \times 10^4$]	$R_J \geq 2.0 \times 10^4$	Resolve	Measure T_2 under the complete pulse schedule
Thermal activation $x_T = E_{\text{act}}/(k_B T)$	0.0048 [0.0010, 0.015] at 10 mK	$x_T \geq \ln(N_s/10^{-3})$ $= 13.8$	Stop/pivot	Abandon passive equilibrium memory for this scale
Coupling-disorder quantile q_J/J	0.08 [0.04, 0.15]	model phase boundary $\eta_c = 0.05$ [0.03, 0.08]	Resolve	Simulate measured disorder tails and missing bonds
Placement spread for direct dipolar bonds	$\sigma_r = 1.0$ nm [0.5, 2.0] at $r = 10$ nm	$\sigma_r \leq 0.17$ nm for 5% coupling spread	Redesign	Add tunability or change coupling mechanism
Unhandled leakage per control location p_u	2×10^{-4} [$5 \times 10^{-5}, 8 \times 10^{-4}$]	$p_u \leq 10^{-6}$ for 1000 locations	Redesign	Demonstrate detection/reset before accepting more locations

Detailed treatment: example decision table

Example decision table

Consider a deliberately hypothetical target, not a claim about an existing material. [Proposal] The proposal takes one thousand residual cluster degrees of freedom to form a two-dimensional analog topological memory with $N_s = 1000$ required sites and $N_b = 1500$ required bonds.

The effective interaction frequency is estimated as $\nu_{\text{eff}} = J_{\text{eff}}/h = 20$ MHz, with a credible interval of 10–40 MHz. A credible interval is an interval assigned a specified posterior probability under the adopted statistical model. The measured dephasing time under the intended controls is $T_2 = 100 \mu\text{s}$, with an interval of 50–200 μs .

Here J_{eff} is an energy in joules (J), h is Planck's constant in J·s, and T_2 is in seconds.

Before any measurement, the project defines four possible decisions:

- Continue: the conservative credible bound satisfies the threshold.
- Resolve: the credible intervals straddle the threshold, so the project must obtain the measurement or larger simulation that controls the decision.
- Redesign: the current implementation fails, but a stated modification would change the controlling equation.
- Stop/pivot: even the optimistic credible bound fails. The stated target is dropped; a different target, such as active digital simulation, starts a new project.

The numbers below are invented to demonstrate the calculations. A real project replaces them with its parameter ledger and model calculations. The table illustrates a decision procedure; it evaluates no real crystal.

Gate | Hypothetical estimate, including uncertainty | Preregistered threshold | Decision
| Required action | |

Coherent interaction $R_J = J_{\text{eff}}/(\hbar\Gamma)$ | 1.26×10^4 [3.1×10^3 , 5.0×10^4] |
 $R_J \geq 2.0 \times 10^4$ | Resolve | Measure T_2 under the complete pulse schedule; do not
scale the array yet | |

Thermal activation $x_T = E_{\text{act}}/(k_B T)$ | 0.0048 [0.0010, 0.015] at 10 mK |
 $x_T \geq \ln(N_s/10^{-3}) = 13.8$ | Stop/pivot | Abandon passive equilibrium memory for this
scale unless the gap or temperature changes by orders of magnitude | |

Coupling-disorder quantile q_J/J | 0.08 [0.04, 0.15] | model phase boundary $\eta_c = 0.05$ [0.03, 0.08] | Resolve | Simulate measured disorder tails and missing bonds; RMS alone is insufficient | |

Placement spread for direct dipolar bonds | $\sigma_r = 1.0$ nm [0.5, 2.0] at $r = 10$ nm | $\sigma_r \leq 0.17$ nm for 5% coupling spread | Redesign | Add tunability or change coupling mechanism; direct fixed dipolar layout fails this tolerance | |

Unhandled leakage per control location p_u | 2×10^{-4} [5×10^{-5} , 8×10^{-4}] | $p_u \leq 10^{-6}$ for 1000 locations and 10^{-3} budget | Redesign | Demonstrate detection/reset before accepting more locations | |

Preparation window | $t_{\text{ad}} = 100 \mu\text{s}$ [30, 500] | $t_{\text{ad}} \leq 0.01T_2 = 0.5\text{--}2 \mu\text{s}$ | Stop/pivot | Change preparation path or use dissipative/active preparation; present ramp has no window | |

Site-only patch yield $Y_s = y_s^{N_s}$ | 0.0067 [4.3×10^{-5} , 0.368] for $y_s = 0.995$ [0.990, 0.999] | $Y_s \geq 0.10$, before bond losses | Resolve | Measure correlated yield and test repairable layouts; monolithic scale-up is not justified | |

Logical scaling, $p_L(d+2)/p_L(d)$ | 0.80 [0.60, 1.10] | upper credible bound < 1 at fixed physical conditions | Resolve | Add distances and blind decoder analysis; do not extrapolate an improving lifetime yet | |

Two rows illustrate how these decisions are obtained. Define $\Gamma = 1/T_2$, in s^{-1} , as the dephasing rate, and define the reduced Planck constant as $\hbar = h/(2\pi)$. Then

$$R_J = \frac{J_{\text{eff}}}{\hbar\Gamma} = 2\pi\nu_{\text{eff}}T_2 = 2\pi(20 \times 10^6 \text{ s}^{-1})(100 \times 10^{-6} \text{ s}) \approx 1.26 \times 10^4.$$

The units cancel, so R_J is dimensionless. A protocol that consumes $\alpha = 20$ inverse-interaction times against a decoherence budget $\epsilon_J = 10^{-3}$ has exposure $\Gamma t \approx \alpha/R_J$, which demands $R_J \geq \alpha/\epsilon_J = 2.0 \times 10^4$.

The central estimate misses this threshold while the credible interval extends past it. The predeclared action is then a sharper measurement, rather than a continue or stop verdict from the central value alone.

For the thermal row, suppose a small-patch calculation gives activation energy $E_{\text{act}}/h = c_{\text{act}}\nu_{\text{eff}}$ with $c_{\text{act}} = 0.05$ [0.02, 0.08]. The central activation frequency is 1 MHz.

At $T = 10$ mK, $k_B T/h \approx 208$ MHz, so $x_T \approx 1/208 = 0.0048$. Even pairing the optimistic endpoints gives only $3.2/208 \approx 0.015$.

The optimistic estimate sits nearly three orders of magnitude below 13.8. Sharper parameter fitting cannot rescue the stated passive-memory target.

This table tests an analog memory meant to hold information with no active intervention. A digital simulator can operate with $E_{\text{act}} < k_B T$ because calibrated gates and active error correction, not thermal equilibrium, sustain its effective Hamiltonian.

A digital-circuit benchmark therefore leaves the analog thermal criterion untested. Failure as a passive memory still leaves open other uses of the same hardware for physical qubits, sensors, or emulation.

One hardware platform can thus carry two distinct claims, each with its own rejection criterion.

Translate physical limits into quantitative tolerances

Work out the numerical limits for coherent operation, thermal errors, coupling disorder, and placement uncertainty, keeping the model assumptions that each number rests on.

Coherent Interaction and Decoherence Criterion

A threshold starts from the time the operation requires. A large interaction helps exactly insofar as it shortens that time relative to decay. The dimensionless protocol cost α records how much evolution the operation needs; its value need not be one.

Take J_{eff} as the coefficient of the effective Hamiltonian term that generates the required low-energy dynamics, and Γ as the measured decay rate at the operating point.

$$R_J = \frac{J_{\text{eff}}}{\hbar \Gamma}$$

A protocol requiring time $t_{\text{req}} = \alpha \hbar / J_{\text{eff}}$ accumulates first-order Markovian exposure $\epsilon_{\text{dec}} \approx \Gamma t_{\text{req}} = \alpha / R_J$.

Reserving at most ϵ_J for this decoherence channel gives:

$$R_J \geq \frac{\alpha}{\epsilon_J}$$

With $\alpha = 20$ and an allocated error $\epsilon_J = 10^{-3}$, the threshold used in the table is $R_J \geq 2 \times 10^4$. A central estimate of 1.26×10^4 misses it, and an uncertainty interval reaching above it calls for “Resolve” rather than a confident pass.

This approximation assumes small exposure and Markovian exponential decay. In that model the exact decay probability is $1 - e^{-\Gamma t_{\text{req}}}$, and the exact bound reads $R_J \geq \alpha/[-\ln(1 - \epsilon_J)]$. Correlated noise, nonexponential decay, and leakage each need their own protocol-specific calculation, and coherence must be measured under the actual control schedule.

Detailed treatment: coherent interaction and decoherence criterion

Coherent interaction and decoherence criterion

J_{eff} , in J, is the coefficient of the effective Hamiltonian term that generates the required low-energy dynamics — distinct from the bare intra-cluster splitting and from the largest microscopic coupling. Γ , in s^{-1} , is the measured decay rate at the operating point under the control schedule for that term. Their dimensionless ratio is

$$R_J = \frac{J_{\text{eff}}}{\hbar\Gamma}.$$

If a protocol requires a time

$$t_{\text{req}} = \alpha \frac{\hbar}{J_{\text{eff}}},$$

where α is the dimensionless path or circuit cost from simulation. The first-order Markovian exposure is then $\epsilon_{\text{dec}} \approx \Gamma t_{\text{req}} = \alpha/R_J$; a Markovian model treats the relevant noise as having negligible memory on the dynamical timescale. Reserving at most ϵ_J for this decoherence channel gives the falsifiable criterion

$$R_J \geq \frac{\alpha}{\epsilon_J}.$$

The bare inequality $J_{\text{eff}} > \hbar\Gamma$ permits only order-one dynamics before decay, so it tests little. An isolated-defect T_2 value needs remeasurement in the device: clustering, continuous drives, optical cycling, and correlated noise can all shift Γ .

The kill rule stops the specified protocol when the credible upper bound on R_J stays below the credible lower bound on α/ϵ_J after the effective coupling and the driven coherence have both been measured in the same device. A different protocol with smaller α counts as a redesign and needs a fresh criterion.

An isolated-spin echo value pairs J_{eff} with a decay rate from different physical conditions, so the resulting inequality does not test the proposed device.

Thermal-Seed Probability Criterion

A thermal excitation can seed a logical error. The activation energy E_{act} is the energy cost of the excitation process that matters; it may differ from the cluster gap or any single Hamiltonian coefficient. The factor g counts how many excitation varieties the simplified model includes.

With N_c approximately independent opportunities to create an excitation:

$$P_{\text{seed}} \lesssim N_c g \exp\left(-\frac{E_{\text{act}}}{k_B T}\right)$$

Requiring $P_{\text{seed}} \leq \epsilon_T$ yields:

$$\boxed{\frac{E_{\text{act}}}{k_B T} \geq \ln\left(\frac{N_c g}{\epsilon_T}\right)}$$

With $N_c = 1000$, $g = 1$, and $\epsilon_T = 10^{-3}$, the requirement is $\ln(10^6) \approx 13.82$. A ratio barely above one leaves far too many excitation opportunities across the full device.

This estimate holds in the dilute-activation regime, equivalently as a union bound once the per-opportunity probability is justified. If its right-hand probability estimate exceeds one, the bound carries no information; probabilities never exceed one. A storage-time claim needs more than a spatial site count. A simple rate model instead gives an expected count $\Lambda = N_s g \gamma_0 t_{\text{store}} e^{-E_{\text{act}}/(k_B T)}$, with attempt rate γ_0 , and Poisson probability $1 - e^{-\Lambda}$. Subsequent diffusion, annihilation, and correction determine whether a seed becomes a logical failure.

Detailed treatment: thermal-seed probability criterion

Thermal-seed probability criterion

E_{act} , in J, is the lowest energy barrier among locally accessible processes that create a harmful excitation. It may differ from the clean spectral gap Δ_{topo} , which is the energy difference between the relevant low-energy sector and the next excited state in an ideal disorder-free model. Boundaries, weak bonds, disorder, and multistep processes can reduce E_{act} . For N_c approximately independent opportunities to create an excitation, a dilute-equilibrium estimate gives

$$P_{\text{seed}} \lesssim N_c g \exp\left(-\frac{E_{\text{act}}}{k_B T}\right),$$

where g is a dimensionless factor that counts relevant species or channels, and P_{seed} is the probability of at least one harmful seed. Requiring $P_{\text{seed}} \leq \epsilon_T$ yields

$$\boxed{\frac{E_{\text{act}}}{k_B T} \geq \ln\left(\frac{N_c g}{\epsilon_T}\right)}.$$

The exponent is dimensionless: joules divided by (J/K)K. Finite-temperature topological memories need a kinetic analysis alongside this equilibrium-occupancy estimate. Under the no-go theorem's assumptions, a constant energy barrier in a two-dimensional local stabilizer Hamiltonian rules out passive self-correction [R168]; [R169]. [Theory]

For a passive equilibrium memory, the kill rule stops the project when the optimistic disorder-renormalized E_{act} , the lowest credible temperature, and the smallest useful N_c jointly fail the logarithmic bound.

Proposed active excitation removal replaces this criterion with a measured creation rate, a diffusion model, a syndrome-measurement cadence, and a logical-error target. The resulting system is then described as active, not passive.

The condition $\Delta > k_B T$ tests something different and weaker. The required ratio grows as $\ln(N_c g / \epsilon_T)$, and excitation diffusion can dominate even when the equilibrium occupancy is small.

Coupling-Disorder Threshold

A large array is sensitive to rare bad bonds as well as to the spread of typical bonds. Let O_b be the specified dimensionless bond operator with fixed normalization, so δJ_b carries energy units. Renormalizing the operator changes what a numerical coefficient threshold means.

Write the realized Hamiltonian as:

$$H = H_{\text{target}} + \delta H, \quad \delta H = \sum_b \delta J_b O_b$$

The fabrication metric should be a high quantile:

$$q_J = Q_{1-\alpha/N_b}(|\delta J_b|)$$

Here Q_p is a p -quantile and α is the allowed probability that at least one required bond exceeds the stated limit; this symbol is distinct from the protocol cost in Section 39.4. If every bond has tail probability at most α/N_b , the union bound caps the probability that any of the N_b bonds exceeds the limit at α . The bound needs no independence assumption, but the assumed tail bounds must hold for every bond.

With $\alpha = 0.01$ and $N_b = 1500$, the relevant quantile is approximately 0.9999933. Ordinary sample averages say little about so distant a tail; estimating it takes adequate data or a defensible distribution model that includes spatial correlations and missing bonds.

Monolithic scale-up stops when even the credible lower tail of q_J/J_{eff} exceeds the credible upper model tolerance η_c .

Detailed treatment: coupling-disorder threshold

Coupling-disorder threshold

Write the realized Hamiltonian as

$$H = H_{\text{target}} + \delta H, \quad \delta H = \sum_b \delta J_b O_b,$$

Here H_{target} is the intended Hamiltonian, δH the perturbation, b the bond or local-term label, δJ_b a coupling error in J , and O_b a dimensionless operator. Topological phases can survive sufficiently weak local perturbations, but the stability theorem assumes a gapped target Hamiltonian and bounded local perturbations. It supplies no conversion from a root-mean-square (RMS) fabrication error — the square root of the mean squared error — into a universal tolerance [R142]. [Theory]

The threshold η_c must come from the disordered phase diagram of the candidate model or from a conservative bound on the local gap. At minimum, a simulation sweeps the measured disorder distribution, including missing terms and spatial correlations, and locates where the gap, a topological diagnostic, or logical scaling fails. The fabrication metric for that comparison is the high quantile,

$$q_J = Q_{1-\alpha/N_b}(|\delta J_b|),$$

rather than only the standard deviation. Here Q_p is the p -quantile, meaning the value below which a fraction p of the distribution lies; N_b is the number of required bonds; and α is the permitted probability that any bond exceeds the quoted value. The factor $1/N_b$ causes the required control of the distribution tail to become stricter as the system size increases.

The kill rule stops monolithic scale-up when even the credible lower tail of q_J/J_{eff} exceeds the credible upper model tolerance η_c , or when missing or sign-reversed terms move the realized Hamiltonian into a different phase. A static offset with an available calibration still needs proof: the calibration must preserve the many-body gap and add no fresh control noise.

An average coupling error cannot reveal a single critically weak bond in a large patch. That question belongs to the high quantile; the RMS describes a different property of the distribution.

Position-to-Coupling Error Propagation

A placement tolerance gains physical meaning once translated into a coupling tolerance. The calculation below holds the angular part of the dipolar interaction fixed and treats radial errors as small against the separation.

For a dipolar interaction $J(r) = Cr^{-3}$:

$$\left| \frac{\sigma_J}{J} \right| \approx 3 \frac{\sigma_r}{r}$$

Differentiating gives the factor of three: $dJ/dr = -3Cr^{-4} = -3J/r$, so $\delta J/J \approx -3\delta r/r$. The minus sign records that larger separation weakens the coupling; the spread uses the magnitude.

A full dipolar bond also depends on angle. For a secular factor $f(\theta) = 1 - 3\cos^2\theta$, a small-error expansion gives $\delta J/J \approx -3\delta r/r + [f'(\theta)/f(\theta)]\delta\theta$ away from zeros of f . Near an angular zero the relative error can grow very large, and the linear estimate needs reexamination.

An allowed fractional coupling spread η_J then requires:

$$\sigma_r \leq \frac{\eta_J r}{3}$$

At $r = 10$ nm and $\eta_J = 0.05$:

$$\sigma_r \leq \frac{0.05 \times 10}{3} = 0.17 \text{ nm}$$

Detailed treatment: position-to-coupling error propagation

Position-to-coupling error propagation

Positioning errors matter through a model-dependent transfer function from spatial uncertainty to coupling uncertainty. Let r be the nominal separation and σ_r a small radial uncertainty, both in metres. For a dipolar interaction $J(r) = Cr^{-3}$, differentiation gives

$$\left| \frac{\sigma_J}{J} \right| \approx \left| \frac{d \ln J}{dr} \right| \sigma_r = 3 \frac{\sigma_r}{r}.$$

Thus, an allowed fractional coupling spread η_J requires

$$\sigma_r \leq \frac{\eta_J r}{3}.$$

At $r = 10$ nm and $\eta_J = 0.05$, the required position uncertainty is 0.17 nm. The fractional error is dimensionless as a ratio of lengths. Angular uncertainty enters as well, because the dipolar coupling carries an orientation-dependent factor.

For an exchange-like coupling $J(r) = J_0 e^{-r/\lambda}$, where λ is a decay length in metres, the corresponding result is $\sigma_J/J \approx \sigma_r/\lambda$. The required position tolerance can then be much smaller than a nanometre. Reviews of defect platforms show that defect creation, charge-state control, coherence, and placement have separate yields. Consequently, a nominal beam diameter does not specify the complete position distribution [Ro74].

[Experiment]

The kill rule stops the fixed-coupling layout when the best demonstrated final-position distribution, propagated through the full angular and radial coupling law, exceeds η_c . Tunable couplers, spectroscopy-based calibration, or a less position-sensitive geometry count as redesigns, and their added noise and wiring costs enter the accounting.

A beam diameter differs from σ_r . Straggle, diffusion, conversion, charge selection, and registration all shape the realized coupling distribution.

Unhandled leakage criterion

Let p_ℓ be the leakage probability per relevant control or syndrome location. Leakage means population outside the local Hilbert-space subspace used by the computational or effective model. Let $p_u \leq p_\ell$ be the probability that leakage stays undetected and unremoved long enough to propagate. For N_{loc} locations in a task, the dilute approximation gives

$$P_{\text{leak}} \approx N_{\text{loc}} p_u.$$

Allocating ϵ_ℓ to unhandled leakage requires

$$\boxed{p_u \leq \frac{\epsilon_\ell}{N_{\text{loc}}}}.$$

A reset every m cycles changes p_u while possibly interrupting the analog Hamiltonian. Surface-code studies show that long-lived leakage produces time-correlated, propagating faults, and that dedicated leakage-reduction circuitry can restore threshold behavior in particular active circuits [R220]. [Theory] [Numerics] This result identifies a method to test in those circuits; the same threshold cannot be assumed for clusters.

The kill rule stops scaling when the optimistic estimate of the unhandled leakage probability exceeds the task allocation and no compatible detection or reset operation has been demonstrated. Charge-state switching, departure from the cluster doublet, and loss of a constituent enter as separate contributions before they are combined.

Treating leakage as a Pauli error misses the physical point: the system has left the modeled local Hilbert space.

Compatibility of preparation and decoherence times

Let the preparation path be $H(s)$ with dimensionless schedule coordinate $s \in [0, 1]$. Define $A = \max_s \|\partial_s H\|$, in J, as the largest norm of the Hamiltonian derivative along the path, and $\Delta_{\min}$, in J, as the smallest many-body gap met on the actual path. A schematic leading adiabatic estimate is

$$\epsilon_{\text{ad}} \sim \left(\frac{\hbar A}{t_p \Delta_{\min}^2} \right)^2,$$

where t_p is the preparation time in seconds and ϵ_{ad} is the diabatic error. The combination $\hbar A / \Delta_{\min}^2$ has units $(\text{J s})\text{J}/\text{J}^2 = \text{s}$. Rigorous adiabatic bounds also depend on endpoint smoothness, higher derivatives, matrix elements, and system-size scaling [R247]. [Theory] Solving for a specified diabatic-error allocation ϵ_{ad}^* gives the lower bound

$$t_p \geq t_{\min} = \frac{\hbar A}{\Delta_{\min}^2 \sqrt{\epsilon_{\text{ad}}^*}}.$$

In the small-error Markovian approximation, decoherence sets an upper bound $t_p \leq t_{\max} \approx \epsilon_{\text{dec}}^* / \Gamma$, where ϵ_{dec}^* is the allocated decoherence error. A viable ramp therefore requires

$$\boxed{t_{\min} < t_{\max}}.$$

The kill rule stops the adiabatic preparation route when the credible bounds leave no overlap, after including the system-size dependence of $\Delta_{\min}$. A dissipative or measurement-assisted route counts as a new protocol, with its own mixing or convergence time, residual excitation density, and separate decoherence and readout budgets.

Slowing the ramp without limit fails as a remedy: smaller diabatic error comes with larger open-system exposure. When $t_{\min} > t_{\max}$, no preparation time satisfies both constraints.

Full-patch fabrication yield

Let y_s be the probability that a required site has the correct position, species, charge state, local spectrum, and addressability, and y_b the probability that a required bond lies within tolerance. If these events are independent, the yield of an unrepairable patch is

$$Y_{\text{patch}} = y_s^{N_s} y_b^{N_b}.$$

This quantity is dimensionless. At $y_s = 0.995$ and $N_s = 1000$, the site-only yield is $0.995^{1000} \approx 0.0067$.

Bond failures can only lower this number. Independence is not guaranteed: implantation damage, annealing, strain, and optical collection can correlate failures.

Known-loss thresholds for topological codes can be high in specially designed active codes. The idealized surface-code analysis by Stace and collaborators, for instance, relates tolerable heralded loss — loss whose location is known — to lattice percolation [R248]. [Theory] [Numerics] That threshold does not transfer directly to a string-net Hamiltonian whose interaction graph and Hamiltonian terms vanish with a lost site. A reroutable design passes only when the defective graph is shown to stay in the target phase and the readout reliably identifies losses.

The kill rule starts from an economic or experimental minimum yield $Y_{\min} = 1/N_{\text{attempt,max}}$.

Monolithic fabrication stops when the credible upper bound on Y_{patch} falls below $Y_{\min}$. A repairable architecture continues only when measured defect maps, an explicit repair algorithm, and phase simulations all pass their criteria.

Advertised yields cannot be multiplied without denominators and conditioning information; bare multiplication treats distinct conditional events as identical and independent.

Finite-size and logical-scaling criteria

Small patches can mimic the signatures of a gap or topological degeneracy without establishing their persistence in larger systems. The extrapolation model must therefore be stated. For linear size L , one gapped finite-size ansatz is

$$\Delta(L) = \Delta_{\infty} + ae^{-L/\xi},$$

where $\Delta(L)$, Δ_{∞} , and a are in J, while L and the correlation length ξ have the same length units. A phase claim requires a credibly positive $\Delta_{\infty} > 0$ and comparison with competing finite-size fits. A logical-memory claim under fixed physical noise may instead use

$$p_L(d) = A_0 e^{-\alpha d},$$

where $p_L(d)$ is the logical-error probability and the code distance d , prefactor A_0 , and exponent α are dimensionless. Improvement with distance requires $\alpha > 0$, equivalently $p_L(d + \delta d)/p_L(d) < 1$.

The kill rule drops the scalable-protection claim when the credible upper bound on the improvement ratio reaches at least one across preregistered larger sizes under the same noise distribution, or when the extrapolation excludes a nonzero thermodynamic gap.

Postselection, better control on larger samples, a different decoder, or a lower temperature each define a different scaling curve; none of them counts as distance scaling. Such changes may still support useful experiments.

Failure here invalidates the stated scaling claim under the stated noise model, while leaving other uses of the hardware open.

Required experimental and computational records

A kill criterion is reliable only with an explicit uncertainty model. Each input should carry both a probability distribution and the conditions under which it was obtained:

- J_{eff} : inferred from the low-energy spectrum, including corrections and fit covariance;
- Γ : measured under the complete operating schedule rather than copied from an isolated-spin echo measurement;
- E_{act} and Δ_{min} : minimized over boundaries, disorder samples, leakage sectors, and the preparation path;
- placement and coupling disorder: represented by joint spatial distributions and confidence in their tails, rather than only RMS values;
- leakage: characterized by the state-resolved rate, dwell-time distribution, propagation, and reset efficacy;
- yield: reported with denominators for creation, correct species, charge state, coherence, addressability, and bonds;
- scaling: reported through raw logical-event counts, a decoder fixed before unblinding, and uncertainty in both fit parameters and model choice.

[Numerics] Exact diagonalization establishes these quantities only for the Hamiltonian, system sizes, boundary conditions, and disorder ensemble actually simulated. [Experiment] Spectroscopy establishes a gap in a finite device; thermodynamic topological order needs more than spectra. [Proposal] Combining numerical and experimental evidence requires posterior predictive checks: measured parameter distributions enter the model, and its predictions are tested against held-out spectra, dynamics, and logical observables.

Decisions use one-sided intervals. A quantity that must be large to pass is compared through its lower bound.

A quantity that must be small to pass is compared through its upper bound. A hard stop is justified when even the optimistic bound fails after a measurement with enough statistical power to resolve the decision.

An imprecise experiment generally yields “resolve” rather than “continue” or “stop.”

Calculate from the smallest system to the patch

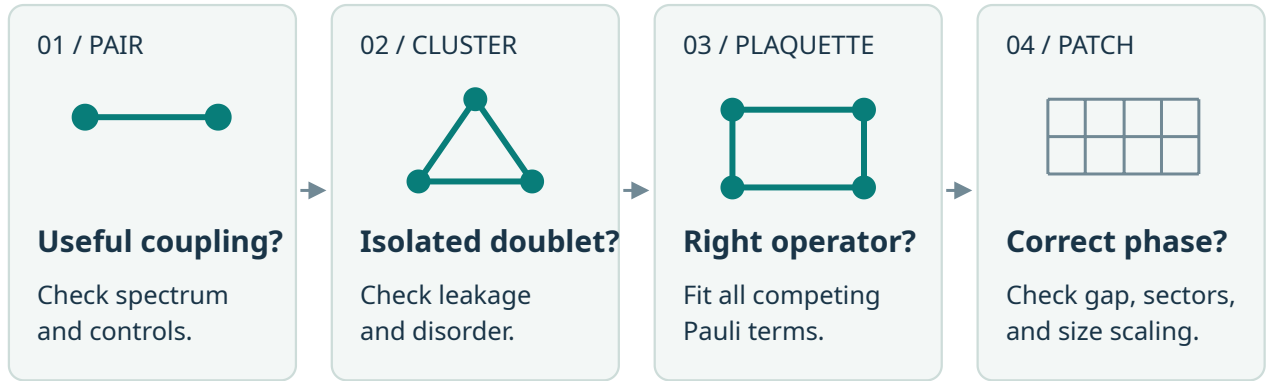
Answer each open question with the cheapest model that can settle it. The diagram below gives the sequence; the calculations that follow explain what each stage must establish.

Introduction: The Staged Calculation

The calculation runs through tests of increasing cost:

Each calculation earns the next step

Increase system size only after the earlier model passes its own checks.



A positive result validates one link in the argument. It does not skip the remaining links.

The simulation sequence isolates the source of a failure. A cluster or plaquette result is a prerequisite for the next stage, not a many-body phase demonstration.

two defects → one encoded cluster → three-to-six-cluster plaquette → small patch

This ordering removes infeasible architectures before large-scale computing time is spent. A plausible defect pair still leaves the isolated cluster doublet unresolved; a plausible doublet still leaves the intended plaquette operator unresolved; and a recognizable plaquette term still leaves the topological phase unresolved.

Two-Defect Screening Calculation

This four-dimensional model is a first screening tool. It shows how local splittings and anisotropic couplings shape a pair spectrum. It omits electronic levels that a full NV-center model keeps; Chapter 41 returns to the physical spin-1 description.

The coefficients below multiply Pauli matrices directly. If a physical interaction is written as $J_{\text{phys}} S_1^x S_2^x$ with dimensionless $S^x = X/2$, its coefficient of $X \otimes X$ is $J_{\text{phys}}/4$. Keep this normalization when comparing fitted parameters.

Each defect is represented by a spin-1/2. In units with $\hbar = 1$:

$$H_2 = \frac{\omega_1}{2} Z \otimes I + \frac{\omega_2}{2} I \otimes Z + J_x X \otimes X + J_y Y \otimes Y + J_z Z \otimes Z$$

```

using LinearAlgebra

I2 = Matrix{ComplexF64}(I, 2, 2)
X = ComplexF64[0 1; 1 0]
Y = ComplexF64[0 -im; im 0]
Z = ComplexF64[1 0; 0 -1]

function pair_spectrum( $\omega_1$ ,  $\omega_2$ , Jx, Jy, Jz)
    H = 0.5 *  $\omega_1$  * kron(Z, I2) + 0.5 *  $\omega_2$  * kron(I2, Z)
    H += Jx * kron(X, X) + Jy * kron(Y, Y) + Jz * kron(Z, Z)

    decomposition = eigen(Hermitian(H))
    E = decomposition.values
    U = decomposition.vectors
    P = U[:, 1:2]
     $\Delta_{\text{leak}}$  = E[3] - E[2]

    physical_ops = Dict{
        "Z1" => kron(Z, I2),
        "Z2" => kron(I2, Z),
        "X1" => kron(X, I2),
    }
    projected_ops = Dict{
        name => P' * op * P for (name, op) in physical_ops
    }

    scale = max(norm(H), 1.0)
    residual = norm(H * U - U * Diagonal(E)) / scale
    return E,  $\Delta_{\text{leak}}$ , projected_ops, residual
end

```

The outputs feed three immediate tests:

1. The two lowest states must form a useful doublet.
2. With zero-based eigenvalue labels, the leakage gap $\Delta_{\text{leak}} = E_2 - E_1$ must be large relative to intercluster coupling, drive bandwidth, disorder, and decoherence.
3. The projected physical operators must generate useful controls within the encoded pair.

The code uses Julia's one-based indexing, so $E[3] - E[2]$ is the leakage gap written $E_2 - E_1$ in zero-based notation. The internal doublet splitting is $E[2] - E[1]$; the two gaps answer different questions.

The variable P in this short program holds two eigenvector columns. Mathematically it is the isometry W from the notation key, not the square projector $WW^\dagger$. Hence $P' * \text{op} * P$ is the correctly sized 2×2 encoded operator.

The returned residual checks the numerical eigenvalue equation. A tiny residual confirms accurate diagonalization of the supplied matrix. It leaves three further questions untouched: whether the microscopic model is valid, whether the doublet is robust, and whether the controls are useful. If all available projected controls commute, they cannot generate arbitrary single-qubit rotations even though their matrices are well defined.

With $\hbar = 1$, time evolution reads e^{-iHt} and parameters are angular-frequency coefficients when time is in seconds. If the input matrix is instead H/h in hertz, use $e^{-i2\pi(H/h)t}$. An overall unit conversion leaves eigenvectors unchanged and rescales predicted times.

Detailed treatment: two-defect screening calculation

Two-defect screening calculation

The first calculation screens candidate models; it makes no claim about a specific material.

Represent each defect by a spin-1/2, meaning a two-level quantum degree of freedom. Let X , Y , and Z denote the dimensionless Pauli matrices, let I denote the two-dimensional identity matrix, and let $\otimes$ denote the tensor product between the two defect Hilbert spaces. In units where $\hbar = 1$, use the Hamiltonian

$$H_2 = \frac{\omega_1}{2} Z \otimes I + \frac{\omega_2}{2} I \otimes Z + J_x X \otimes X + J_y Y \otimes Y + J_z Z \otimes Z.$$

The coefficients $\omega_1, \omega_2, J_x, J_y, J_z$ must all be expressed in the same angular-frequency unit, such as radians per second. Every eigenvalue of H_2 then has the same unit. Multiplication by $\hbar$ converts an angular frequency to an energy.

The Julia program below constructs the four-by-four Hamiltonian and computes all its eigenvalues and eigenvectors. It keeps the two lowest eigenvectors as the columns of a matrix P , which maps the candidate two-state encoded basis into the full two-defect Hilbert space. It also computes the leakage gap — here the energy separation between the retained pair and the next state — projects three physical control operators into the retained subspace, and evaluates the residual of the eigendecomposition. The program carries no benchmark parameters of its own; its inputs must come from cited sources under stated physical conditions.

```
using LinearAlgebra

I2 = Matrix{ComplexF64}(I, 2, 2)
X = ComplexF64[0 1; 1 0]
Y = ComplexF64[0 -im; im 0]
Z = ComplexF64[1 0; 0 -1]

function pair_spectrum(ω1, ω2, Jx, Jy, Jz)
```

```

H = 0.5 * ω1 * kron(Z, I2) + 0.5 * ω2 * kron(I2, Z)
H += Jx * kron(X, X) + Jy * kron(Y, Y) + Jz * kron(Z, Z)

decomposition = eigen(Hermitian(H))
E = decomposition.values # ascending
U = decomposition.vectors
P = U[:, 1:2] # candidate leftover basis
Δleak = E[3] - E[2]

physical_ops = Dict(
  "Z1" => kron(Z, I2),
  "Z2" => kron(I2, Z),
  "X1" => kron(X, I2),
)
projected_ops = Dict(
  name => P' * op * P for (name, op) in physical_ops
)

scale = max(norm(H), 1.0)
residual = norm(H * U - U * Diagonal(E)) / scale
return E, Δleak, projected_ops, residual
end

```

Constructing the complete matrix and computing every eigenpair is called exact diagonalization (ED). For the stated finite matrix, diagonalization introduces no model-space truncation; numerical error comes from finite-precision arithmetic and the eigensolver implementation.

This program tests the computational pipeline. It does not model a specific color center — an optically active localized defect in a solid. Real defects may need a larger basis with higher spin states, orbital degrees of freedom, hyperfine coupling to nuclear spins, strain-dependent states, or distinct charge configurations. A material-specific calculation replaces the illustrative Hamiltonian with the cited microscopic Hamiltonian developed in Chapters 7–10 and 26.

The outputs support three immediate tests. First, the two lowest states must form a useful doublet. Second, the leakage gap

$$\Delta_{\text{leak}} = E_2 - E_1$$

must be large relative to the intercluster coupling, drive bandwidth, disorder scale, and decoherence rate, with all quantities expressed in the same units. Here, drive bandwidth is the frequency range occupied by the applied control, disorder denotes static sample-to-sample parameter variation, and decoherence is the loss of quantum

coherence through coupling to uncontrolled degrees of freedom. Third, the projected physical operators must generate useful controls within the encoded pair rather than reducing to multiples of the identity.

A low-energy pair counts as an encoded degree of freedom only with a physically addressable transition.

Parameter specification before diagonalization

Before diagonalization, a single machine-readable record stores all parameters:

- defect species, charge state, spin and retained orbitals;
- magnetic-field vector, strain tensor, electric field, and temperature;
- pair positions and crystal orientations;
- every Hamiltonian coefficient, unit, sign convention, source, sample conditions, and uncertainty interval;
- which interactions are measured, calculated, fitted, or assumed;
- the architecture's predeclared kill criteria from Chapter 39.

One internal unit system governs the whole record. If the Hamiltonian coefficients are angular frequencies, every linewidth and inverse coherence time is converted to angular frequency before comparison.

If the coefficients are energies, temperature is converted to an energy scale through $k_B T$, with k_B the Boltzmann constant. Megahertz and microelectronvolts need explicit unit conversion before any quantitative comparison.

Stop rule 0 terminates the material-specific calculation when an interaction term essential to the target Hamiltonian has neither a cited physical mechanism nor a bounded parameter range. An unconstrained fitting parameter can reproduce many small spectra while predicting nothing.

The parameter record defines the Hamiltonian actually studied. Without a coefficient's provenance, a later claim of agreement with experiment names no specified quantity for comparison.

Two-defect uncertainty and convergence tests

Dense ED, which stores and diagonalizes the full Hamiltonian, serves the smallest basis. When the same small model later needs time-dependent pulses, collapse operators for open-system processes, or correlation functions, QuTiP supplies the corresponding numerical tools [R249]. For a closed system described by a Hermitian matrix, standard linear algebra in NumPy, SciPy, or Julia is sufficient.

The calculation samples the full fabrication uncertainty, not only a best-fit parameter point. For every sampled position, orientation, field, and coupling, it stores:

- the low-energy spectrum E_n ;

- level splittings and avoided crossings;
- eigenstate composition in the microscopic basis;
- derivatives $\partial E_n / \partial p$ with respect to uncertain parameter p ;
- matrix elements of physically available drives and readout operators;
- leakage under a representative pulse or weak interpair coupling;
- solver residuals and symmetry quantum numbers.

An avoided crossing is a near approach of energy levels that stay apart because they are coupled. A symmetry quantum number labels an eigenspace of an operator that commutes with the Hamiltonian.

The convergence test enlarges the local basis — adding the nearest omitted orbital or hyperfine state, for example — and compares both the retained eigenvalues and the projected operators.

Energy shifts alone cannot certify convergence: a spectrum that barely moves while its control matrix elements stay unstable is still unconverged.

Stop rule 1 rejects the pair design when no connected region of the cited uncertainty range preserves the required state ordering, or when the required coupling appears only at an isolated fine-tuned point narrower than fabrication and control tolerances. The calculation also stops when the leakage gap fails the model-specific inequality established in Chapter 39.

A best-fit-point calculation establishes only that one Hamiltonian with the desired behavior exists, possibly in a set of measure zero. The physical architecture is characterized by a parameter distribution.

Complete-Cluster Calculation

A cluster holds m defects, each with retained local Hilbert-space dimension d . The untruncated product Hilbert space has dimension d^m .

Let the two lowest cluster eigenstates be $|\tilde{0}\rangle$ and $|\tilde{1}\rangle$. The rectangular encoding isometry is:

$$W = |\tilde{0}\rangle\langle 0| + |\tilde{1}\rangle\langle 1|$$

It maps a two-component encoded state into the d^m -dimensional physical cluster space. Its columns are orthonormal, so $W^\dagger W = I_2$ and $P = WW^\dagger$ is the physical-space projector. Keeping the isometry and the projector distinct prevents dimension errors.

Every microscopic operator O has an encoded-subspace representation:

$$O_{\text{eff}} = W^\dagger O W$$

The resulting 2x2 matrix can be expanded:

$$O_{\text{eff}} = c_0 I + c_x X + c_y Y + c_z Z$$

with coefficients:

$$c_\mu = \frac{1}{2} \text{Tr}(\sigma_\mu O_{\text{eff}})$$

Here $\sigma_0 = I$ and $\sigma_{x,y,z} = X, Y, Z$. Multiplying the expansion by σ_μ and tracing, with $\text{Tr}(\sigma_\mu \sigma_\nu) = 2\delta_{\mu\nu}$, removes every other component. For a Hermitian operator these coefficients are real, up to numerical roundoff.

The calculation is repeated across the intended fabrication and operating distributions, tracking the gap, internal splitting, and projected operators together. Eigenvectors carry arbitrary phases and can rotate within a degenerate subspace, so comparing raw matrix entries between samples needs a consistent encoded-basis convention.

Detailed treatment: complete-cluster calculation

Complete-cluster calculation

Consider a cluster containing m defects, each represented by a retained local Hilbert-space dimension d . The dimension of the untruncated product Hilbert space is

$$d^m.$$

Every exact symmetry block-diagonalizes the Hamiltonian: the matrix separates into independent sectors. Later perturbations still need checking, since they may break the symmetries used in the reduction.

Dense ED computes every eigenpair and suits the regime where the largest symmetry block fits comfortably in memory. When the full matrix grows too large, sparse Lanczos or Davidson eigensolvers compute a selected low-energy spectral window without the complete spectrum.

A custom sparse implementation suits Hamiltonians with repeated local terms and exact bit-level structure. In Julia, sparse matrices combined with KrylovKit give a direct implementation. ITensors.jl helps when the state and operators admit a natural matrix-product representation [R250].

Let the two lowest cluster eigenstates be $|\tilde{0}\rangle$ and $|\tilde{1}\rangle$. Construct the encoding isometry (the rectangular matrix of retained eigenvectors)

$$P = |\tilde{0}\rangle\langle 0| + |\tilde{1}\rangle\langle 1|,$$

where P is understood as a map from the two-state leftover basis $\{|0\rangle, |1\rangle\}$ into the microscopic cluster Hilbert space. For every microscopic operator O , compute its representation in the encoded subspace:

$$O_{\text{eff}} = P^\dagger O P.$$

The resulting two-by-two matrix can be expanded in the Pauli basis as

$$O_{\text{eff}} = c_0 I + c_x X + c_y Y + c_z Z,$$

with coefficients

$$c_\mu = \frac{1}{2} \text{Tr}(\sigma_\mu O_{\text{eff}}),$$

where $\sigma_\mu \in \{I, X, Y, Z\}$. This decomposition determines how microscopic local fields, noise operators, readout operators, and intercluster couplings act within the encoded subspace.

The cluster splitting, leakage gap, projected control strengths, and projected noise operators are evaluated across the uncertainty ensemble. Two complete clusters are then coupled through the microscopic interaction, and their exact low-energy spectrum is compared with the spectrum predicted by the encoded interaction.

[Theory] A Schrieffer–Wolff expansion, which perturbatively eliminates high-energy states to obtain a low-energy effective Hamiltonian, is controlled only when the coupling between retained and discarded sectors is small relative to the relevant energy

denominators. Direct low-energy diagonalization supplies the required finite-system benchmark [Ro25].

Convergence tests increase the number of retained cluster states, tighten sparse-eigensolver residuals, compare calculations with and without each nominally small microscopic term, and verify that arbitrary phase choices within the doublet leave basis-invariant conclusions unchanged.

Stop rule 2 rejects the cluster architecture when the doublet disappears under realistic disorder, when required controls project nearly to the identity while noise projects strongly onto Pauli operators, or when the exact two-cluster spectra cannot be reproduced within the predeclared effective-model error budget.

A leftover counts as characterized when it is spectrally isolated, carries usable projected controls, shows tolerable projected noise, and reproduces exact coupled-cluster spectra through the effective model. An isolated-cell doublet alone meets none of these tests beyond the first.

Three-to-Six-Cluster Plaquette Calculation

This stage asks whether the microscopic couplings generate the required encoded operator. A Pauli string has *weight* equal to its number of nonidentity factors; $Z \otimes I \otimes X \otimes Z$ has weight three. A desired weight-four term must be separated from local fields and lower-weight couplings.

If each cluster is truncated to one two-state degree of freedom, six clusters span only 2^6 states.

The calculation retains additional cluster states and the microscopic intercluster couplings. A bare projection PHP keeps only the direct retained-space action and generally omits interactions from virtual excursions into $Q = I - P$.

Those effects enter through a controlled effective Hamiltonian — a Schrieffer–Wolff transformation, for example — or through a consistent map of a selected exact low-energy band into the encoded basis. The resulting H_{fit} must reproduce low-energy dynamics or spectral data to a stated accuracy. Eigenvalue matching alone leaves its operator content undetermined without a basis map.

Once virtual-excitation effects are included, the effective Hamiltonian in the encoded Hilbert space expands as:

$$H_{\text{fit}} = \sum_{\alpha} c_{\alpha} \sigma_{\alpha_1} \otimes \cdots \otimes \sigma_{\alpha_n}$$

Because Pauli strings are orthogonal:

$$c_{\alpha} = 2^{-n} \text{Tr}(\sigma_{\alpha}^{\dagger} H_{\text{fit}})$$

There are 4^n Pauli strings in the complete operator basis. For four encoded qubits, an explicit fit exposes every competing coefficient. Reconstructing H_{fit} from the coefficients and checking the residual tests the fit; retaining additional leakage states then tests whether the desired coefficient survives.

A four-spin spectral shift alone does not identify a connected four-body interaction, since lower-body terms also shift four-spin levels. The operator decomposition and its coupling-strength dependence distinguish the mechanisms.

Detailed treatment: three-to-six-cluster plaquette calculation

Three-to-six-cluster plaquette calculation

The plaquette calculation decisively tests the proposed interaction gadget and stays within reach of ED. If each cluster is truncated to one two-state leftover, six clusters span only 2^6 states. Two calculations run side by side:

- the effective calculation, using one leftover per cluster;
- the leakage-aware calculation, retaining additional cluster states and the microscopic intercluster couplings.

Let n denote the number of clusters around the candidate plaquette. Project the leakage-aware Hamiltonian into the encoded Hilbert space and expand it in the complete basis of Pauli strings:

$$H_{\text{fit}} = \sum_{\alpha} c_{\alpha} \sigma_{\alpha_1} \otimes \cdots \otimes \sigma_{\alpha_n},$$

where each α_j is 0, x , y , or z , and $\sigma_0 = I$. A Pauli string is a tensor product containing one Pauli operator or identity on each encoded cluster. Because Pauli strings are orthogonal under the trace inner product, their coefficients are

$$c_{\alpha} = 2^{-n} \text{Tr}(\sigma_{\alpha}^{\dagger} H_{\text{fit}}).$$

Beyond asking whether the target plaquette coefficient K differs from zero, the calculation identifies every generated one-body, two-body, and higher-body coefficient. The full exact low-energy spectrum and its eigenvectors are compared with those of the fitted operator; comparing K alone is insufficient.

The intercluster scale λ is swept over a stated range. A useful interval meets two conditions together: the target term clears the relevant noise and temperature thresholds, and the effective description stays accurate. If perturbation theory predicts the plaquette interaction first at a specified order in λ , the scaling fit is restricted to the region where exact and effective spectra agree. The fitting interval is fixed before the fit, never chosen retrospectively to produce the desired exponent [R025].

Reported quantities include the spectrum, plaquette gap, target coefficient, norm of unwanted terms, leakage weight, and response to every physically allowed local perturbation. QuTiP suits small driven or open-system plaquettes [R249].

For closed-system spectra, sparse ED is generally simpler. At this system size, tensor-network methods add approximation without usefully extending the accessible size.

Stop rule 3 rejects the gadget when no parameter interval satisfies the hierarchy, leakage, disorder, and decoherence criteria together; when cancelling unwanted terms demands unattainable precision; or when the target operator appears only after projection removes states that mix strongly in the exact calculation.

A nonzero K alone does not pass. Acceptance requires K , all competing terms, and quantitative agreement with leakage-aware ED over the same parameter interval.

Finite-Patch Calculation

The first finite-patch calculation uses the least ambitious topological target that can still test the interaction architecture — typically an Abelian stabilizer model or a gauge-theory proxy, before any doubled-Fibonacci model.

ED serves while the symmetry-reduced Hilbert space and the required Krylov vectors fit in memory. For N encoded spins, one complex state vector holds 2^N entries. If each entry is stored as a 16-byte complex number:

$$\text{storage} = 16 \times 2^N \text{ bytes}$$

For long strips and cylinders, the state is represented as a chain of tensors. A controlled DMRG calculation increases the bond dimension systematically until the discarded weight and observables converge. DMRG is a variational optimization method commonly formulated with matrix product states; the bond-dimension sweep is one convergence check within it.

Genuinely two-dimensional patches beyond ED call for finite or infinite PEPS where the target state admits a tractable bond dimension.

Detailed treatment: finite-patch calculation

Finite-patch calculation

The first finite-patch calculation uses the least ambitious topological target that can still test the interaction architecture — typically an Abelian stabilizer model or a gauge-theory proxy, before any doubled-Fibonacci model. An Abelian stabilizer model is defined by mutually commuting constraint operators with Abelian excitation statistics; a gauge-theory proxy is a simpler model that tests the intended local constraints and interactions. The aim is to validate the emergence of the target behavior from the proposed couplings before taking on the most complex candidate phase.

ED serves while the symmetry-reduced Hilbert space and the required Krylov vectors fit in memory. Krylov vectors span the iterative subspace used by sparse eigensolvers. For N encoded spins, one complex state vector holds 2^N entries.

If each entry is stored as a 16-byte complex number, the storage required for one state is

$$16 \times 2^N$$

bytes. An iterative solver requires several such vectors in addition to operator workspace. This memory estimate must be performed before selecting ED.

For long strips and cylinders, the state becomes a chain of tensors joined by bonds that encode correlations across each cut. Increasing the bond dimension systematically until the discarded weight and observables converge defines the density-matrix renormalization group (DMRG) in its matrix-product-state form. The bond dimension is the size of the auxiliary index joining neighboring tensors, and the discarded weight quantifies the truncated part of the Schmidt decomposition.

ITensor and TeNPy implement this method [R035]; [R250]; [R251]. DMRG is most effective when the entanglement across the chosen one-dimensional ordering remains manageable.

Mapping a two-dimensional lattice onto a cylinder grows harder as the circumference increases. Results from one narrow cylinder do not reach the two-dimensional thermodynamic limit [R035]; [R252].

For genuinely two-dimensional patches beyond ED, finite or infinite projected entangled-pair states—PEPS—can be used when the target state admits a tractable bond dimension and the tensor contractions can be converged [R252]; [R253]. PEPS are tensor-network states designed for higher-dimensional lattices. In Julia, TensorKit.jl with MPSKit.jl or PEPSKit.jl provides symmetry-aware tensor infrastructure; ITensors.jl is a mature option for matrix-product calculations [R250].

Package versions and contraction algorithms are part of the report. Naming tensor networks alone does not define a reproducible method.

Quantum Monte Carlo suits cases where the statistical weights are nonnegative or where a demonstrated treatment of the sign problem applies. The sign problem is the loss of a nonnegative sampling distribution through negative or complex weights. Frustrated, complex, or non-Abelian effective interactions commonly remove the usual Monte Carlo advantage. A large lattice-site count alone is no reason to recommend Monte Carlo.

The name of a numerical method alone guarantees nothing about entanglement representation. A wide, highly entangled two-dimensional cluster may challenge DMRG more than a smaller ED calculation.

Ordered Observables and Diagnostics

Diagnostics are used in combination because each one admits competing explanations. A small gap can come from a phase transition, a boundary mode, or a poor numerical approximation. Near-degeneracy can come from symmetry breaking. Entanglement appears in ordinary correlated states too. The aim is a mutually consistent pattern with controlled finite-size behavior.

Several of the lowest-energy states are computed. Define Δ as the separation between the candidate ground-state manifold and the first excited state, and δ as the splitting within that manifold.

Δ and δ are tracked as the patch grows and as boundary conditions change. A candidate topological ground sector separates its internal splitting from the excitation gap at the relevant sizes. Boundary geometry decides which sectors to expect at all: torus degeneracy cannot be demanded of an arbitrary open patch.

Local expectation values and reduced density matrices are compared across the candidate ground states. A useful mathematical target is $P_{\text{gs}} O P_{\text{gs}} \approx c_O P_{\text{gs}}$ for operators O supported on small regions. This tests both diagonal distinguishability and off-diagonal mixing. Finite patches may show small corrections; the question is whether they shrink with size as expected or persist as an order-one local distinction.

For a region A :

$$S(A) = -\text{Tr}[\rho_A \log \rho_A]$$

For suitable large adjacent regions, a Kitaev–Preskill combination cancels boundary contributions:

$$S_{\text{topo}} = S(A) + S(B) + S(C) - S(AB) - S(AC) - S(BC) + S(ABC).$$

In the appropriate gapped two-dimensional setting it approaches $-\gamma$, with $\gamma = \ln \mathcal{D}$ and $\mathcal{D}$ the total quantum dimension. The calculation uses the prescribed geometry with correlation and region-size scales well separated; an arbitrary subtraction on tiny regions does not reach this limit. The original derivation explains the cancellation and its assumptions: [Kitaev and Preskill, *Topological entanglement entropy*](#). Levin–Wen subtraction offers another geometry.

Contractible loops of several shapes are measured, plus noncontractible loops where boundary conditions allow. The loop operator for the target model is defined first. Contractible loops probe local excitation or gauge structure, while noncontractible loops can act within global ground sectors. A single loop expectation value does not identify an anyon theory.

Candidate excitations are created at spatial separation, combined along alternative paths, and projected onto charge sectors. The allowed outcomes and their probabilities are compared with the proposed fusion rules at controlled separation with unwanted excitations suppressed.

Modular data are attempted only with a resolved quasi-degenerate ground-state manifold on a torus in hand. Extraction also requires a controlled ground-state basis and phase conventions. Appendix D gives an Abelian example, and Appendix E explains why recoupling and exchange data must agree with each other.

Detailed treatment: ordered observables and diagnostics

Ordered observables and diagnostics

Observables are calculated in an order that keeps advanced diagnostics from hiding a failure of the basic spectrum.

Spectrum, gap, and degeneracy. For every system size and boundary condition, compute several of the lowest-energy states.

Δ is the separation between the candidate ground-state manifold and the first excited state; δ is the splitting within that manifold. A ground-state manifold is the set of lowest-energy states expected to become degenerate in the relevant large-system limit. Both Δ and δ are tracked as functions of linear system size.

A small δ at one size may be accidental.

Local indistinguishability and perturbation response. Local indistinguishability means that candidate ground states cannot be distinguished by observables supported on sufficiently small regions. Compare local expectation values and reduced density matrices across the candidate ground states.

Each realistic local perturbation is added separately and then in sampled combinations, tracking Δ , δ , mixing between candidate sectors, leakage, and changes in observables.

The phase claim fails when a generic weak local perturbation distinguishes or mixes the proposed sectors at a scale incompatible with the architecture's kill criteria.

Entanglement entropy. For a region A , compute

$$S(A) = -\text{Tr}[\rho_A \log \rho_A],$$

where ρ_A is the reduced density matrix from tracing out the complement of A . Convergence covers the Schmidt spectrum — the singular values, equivalently the reduced-density-matrix eigenvalues, for the bipartition — and goes beyond the scalar entropy.

Topological entanglement entropy. Use the Kitaev–Preskill or Levin–Wen subtraction geometries to cancel boundary contributions [\[R254\]](#); [\[R255\]](#).

Selected regions are large against the correlation length — the characteristic decay length of connected correlations — and small against the complete system. On a small patch the subtraction is a finite-size diagnostic, not a measurement of a universal constant.

Wilson loops. A Wilson loop is a closed string operator of the model's gauge or string algebra. Its explicit form is derived from that algebra, never assumed.

Contractible loops of several shapes are measured, plus noncontractible loops where boundary conditions allow. Their expectation values are tested for perimeter versus area scaling and compared between sectors.

One loop expectation value cannot carry this test alone.

Fusion. Fusion specifies which charge sectors can result when excitations combine. Candidate excitations are created at separation with open-string operators, combined along alternative paths, and projected onto charge sectors.

Allowed channels, state counts, and path dependence are recorded. On a small patch this tests the intended operator algebra; asymptotic quasiparticles need larger systems.

Modular S and T . These matrices encode transformations from exchanging noncontractible cycles and from topological twists. They are attempted only with a resolved quasi-degenerate ground-state manifold on a torus or an equivalent geometry.

Construct minimally entangled states, implement or infer the relevant modular transformations, fix the arbitrary phases of the ground states, and estimate uncertainty by bootstrapping over numerical truncation errors. Ground-state entanglement can in principle reveal quasiparticle statistics [R256], but an inadequately resolved manifold makes the extracted S and T matrices dependent on arbitrary gauge and basis choices.

Modular data from an accidental degeneracy carry no valid interpretation. Any nearly degenerate finite matrix permits arbitrary basis rotations within the corresponding subspace, so the resulting matrices hinge on an unjustified phase and basis convention.

Numerical controls and convergence requirements

Every reported result is tested against more than one numerical control parameter.

Method	Must vary	Must report	Invalid use
Dense/sparse ED	basis truncation, eigensolver tolerance, symmetry sectors	residuals, number of eigenpairs, omitted levels	claiming a phase from one size
MPS-DMRG	bond dimension, sweeps, initial states, cylinder length/circumference	energy variance, discarded weight, sector, observable drift	treating an unconverged wide cylinder as exact
PEPS/iPEPS	bond dimension, environment dimension, optimizer starts, unit cell	energy/observable convergence and contraction error	reporting one bond dimension
Open-system QuTIP	local basis, time step/solver tolerance, collapse operators	trace preservation, positivity checks, parameter provenance	scaling to a many-body patch by dense matrices

Method	Must vary	Must report	Invalid use
Monte Carlo	size, temperature, sampling length, autocorrelation	sign, effective sample size, finite-size scaling	ignoring a sign problem

DMRG uses multiple initial states and explicit sector constraints, since a variational optimizer may return repeatedly to the easiest sector instead of resolving the complete ground-state manifold. PEPS varies both the state bond dimension and the contraction environment dimension.

Every approximate method is compared with ED at one overlapping system size at least. Agreement in the overlap validates the transfer from ED to the larger-scale method.

A single PEPS calculation does not demonstrate convergence. The bond dimension, environment dimension, unit cell, and initial state are all varied [R252]; [R253]. A low variational energy alone does not establish that the calculation has found the correct sector.

Computational workflow

The workflow below runs at the standard expected of a research repository. It specifies the required calculations and data products without asserting any numerical outcome.

Run A — two-defect audit. Load the versioned parameter record and construct the complete two-defect Hamiltonian matrix. Dense diagonalization computes the full eigensystem directly. Export the sorted eigenvalues, eigenvectors, operator matrix elements, eigenpair residuals, and derivatives with respect to the model parameters.

The calculation is repeated over the fabrication-uncertainty ensemble — the sampled parameter distribution consistent with fabrication variability. The passage from the simplified model to the full model is tested by progressively including omitted local energy levels.

Decision A. The model passes only with a connected, robust parameter region surviving Stop rule 1. After this decision the retained microscopic basis is fixed; later retuning to improve the plaquette calculation is excluded.

Run B — cluster projection. Construct the complete intra-cluster Hamiltonian.

Enough low-energy states are computed to include both the candidate doublet — the proposed two-dimensional encoded subspace — and the relevant leakage multiplets, the groups of states outside that subspace. Export the projector P onto the candidate doublet and the projected operators $P^\dagger O_i P$ for every drive, noise, readout, and coupling operator O_i . Also export the corresponding results for all uncertainty samples.

Two clusters are coupled and the spectral discrepancy between the exact model and its projected effective model is exported.

Decision B. The model passes only by avoiding Stop rule 2 and by keeping the exact-versus-effective error within the projection-error budget declared before the calculation.

Run C — plaquette decomposition. Candidate systems of three, four, five, and six clusters are constructed wherever the corresponding geometry can be physically implemented.

Both the encoded and the leakage-aware Hamiltonians are diagonalized. Every coefficient in the Pauli-string decomposition, the low-energy spectra, the leakage weights, and the perturbation-sweep results are exported.

Designs are ranked by a vector with components for target-interaction strength, unwanted-term norm, robustness, and fabrication cost. These tradeoffs stay separate rather than merging into a single hand-tuned score.

Decision C. At most one primary plaquette and one fallback plaquette are selected. If no candidate passes Stop rule 3, the topological-patch branch of the study terminates and the strongest result established at a lower level is reported.

Run D — exact-diagonalization patch. Exact diagonalization (ED) computes eigenvalues and eigenvectors directly in the finite-system Hilbert space. The selected plaquette is tiled into the smallest open patch, cylinder, and periodic cluster for which each geometry addresses a distinct physical question.

Export the low-energy manifold, the gap Δ , the sector splitting δ , local reduced states, Wilson loops, excitation and fusion tests, and responses to perturbations. A Wilson loop is a nonlocal loop operator used here as a diagnostic of topological sectors. Repeat these calculations for every system size and boundary condition accessible to ED.

Decision D. The study continues only when size and boundary trends show a stable gap, decreasing sector splitting, and local indistinguishability together. A single favorable system size is classified as “unresolved,” not as a pass.

Run E — size extension. For cylindrical geometries, the largest ED result is first reproduced with density-matrix renormalization group (DMRG) calculations in ITensor or TeNPy. DMRG is a matrix-product-state variational method. Once the ED result is reproduced, cylinder length, circumference, and bond dimension increase [R035]; [R250]; [R251].

Where genuinely two-dimensional scaling is essential, an ED result is first reproduced with finite projected entangled-pair states (PEPS), a two-dimensional tensor-network ansatz, before the patch grows [R252]; [R253]. Compute entanglement diagnostics and Wilson-loop diagnostics before attempting more advanced quantities.

Modular matrices are attempted only after meeting the prerequisites for identifying and controlling the ground-state manifold [R256].

Decision E. The result is classified as positive, negative, or numerically unresolved. Failure to converge argues neither for nor against the phase; it establishes only that the calculation did not determine the answer.

All inputs, package versions, random seeds, symmetry sectors, solver logs, and immutable outputs are preserved. Results from direct microscopic ED, projected ED, and tensor-network approximations are labeled separately.

Changing a stop rule after examining the results is goalpost relocation, not calibration. Thresholds may be revised when new experimental evidence changes the input contract, never solely because the preferred model failed them.

Experimentally constrained inputs

The simulation starts from parameters conditioned on laboratory measurements, not symbols chosen for computational convenience. Defect positions are distributed variables, orientations may take discrete values, charge states can change, couplings depend on sample conditions, and coherence data depend on temperature and pulse sequence.

[Experiment] supplies parameter distributions and bounds, including spectra, linewidths, placement statistics, coupling measurements, control matrix elements, and noise correlations.

[Theory] supplies the microscopic Hamiltonian, the projection procedure, the perturbative expansion, and diagnostics of the ideal phase.

[Numerics] supplies finite-system spectra, fitted effective operators, wavefunctions, response curves, and controlled extrapolations.

A numerical phase from a Hamiltonian outside the experimentally allowed uncertainty range describes that Hamiltonian; it is no evidence that the defect platform realizes the phase.

Conversely, failure of a simplified Hamiltonian excludes only the declared model within the declared parameter domain, not every possible defect architecture. The scope of the exclusion is part of the result.

Any open-system simulation waits until the closed-system Hamiltonian is shown to have the required low-energy structure. Lindblad evolution — Markovian open-system dynamics through a master equation — can accurately compute decay in a model with no topological sector, but it cannot generate topology absent from the Hamiltonian's relevant state structure.

Check that the test really addresses the claim

Before a go/no-go decision, inspect the threshold and the calculation for shortcuts that could flatter an unfavorable architecture.

Common Analytical Errors

Point 1: Applying a model-specific threshold universally.

- **Explanation:** The roughly 11% perfect-syndrome toric-code threshold assumes perfect syndrome measurements, a stochastic Pauli error model, and specific decoder assumptions.

Point 2: Comparing quantities in consistent units.

- **Explanation:** The effective interaction frequency J_{eff}/h is measured in hertz, whereas the decoherence rate $\Gamma = 1/T_2$ is measured in s^{-1} . The corresponding dimensionless energy ratio is $2\pi(J_{\text{eff}}/h)T_2$.

Point 3: Treating $\Delta/k_B T > 1$ as a sufficient binary test.

- **Explanation:** The required ratio grows as $\ln(N_c g/\epsilon_T)$, and diffusion can remain the dominant failure process.

Point 4: Characterizing a large patch by average disorder.

- **Explanation:** Failure is governed by rare weak bonds, missing sites, and spatially correlated regions.

Point 5: Modeling leakage as an ordinary Pauli error.

- **Explanation:** A leakage error transfers a state outside the modeled local Hilbert space, and the leaked state can persist or spread.

Detailed treatment: common analytical errors

Common analytical errors

- A model-specific threshold does not transfer automatically. The roughly 11% perfect-syndrome toric-code threshold assumes perfect syndrome measurements, a stochastic Pauli error model — errors as randomly sampled Pauli operators — and specific decoder assumptions, with the decoder the algorithm that infers and corrects errors from syndrome data [R141]. This value is not a threshold for disorder, leakage, or fabrication defects.
- Comparisons must use consistent units. The effective interaction frequency J_{eff}/h is measured in hertz, whereas the decoherence rate $\Gamma = 1/T_2$ is measured in s^{-1} , with T_2 denoting the coherence time. The corresponding dimensionless energy ratio is

$$\frac{J_{\text{eff}}}{\hbar\Gamma} = 2\pi \left(\frac{J_{\text{eff}}}{h} \right) T_2.$$

Omitting the factor 2π can change the outcome of a criterion that lies close to its acceptance threshold.

- The condition $\Delta/k_{\text{B}}T > 1$ is not a sufficient binary test. Here, Δ is the relevant energy gap, k_{B} is the Boltzmann constant, and T is temperature. The required ratio grows as $\ln(N_{\text{c}}g/\epsilon_T)$, and diffusion can remain the dominant failure process.
- Average disorder is not sufficient to characterize a large patch. Disorder denotes spatial variation in physical parameters. Failure is governed by rare weak bonds, missing sites, and spatially correlated regions rather than only by the mean disorder.
- Nominal implantation precision differs from the final placement error. The realized coupling distribution also reflects implantation straggle, diffusion, conversion, charge-state selection, and registration.
- Leakage needs its own model, separate from ordinary Pauli errors. A leakage error moves a state outside the modeled local Hilbert space — the state space assigned to the local qubit — where the leaked population can persist or spread.

- Preparation needs both time bounds together. Adiabaticity — the state following the intended instantaneous eigenstate during a sufficiently slow change — sets a minimum preparation time. Decoherence sets a maximum preparation time.
- Advertised yields multiply only with stated denominators and conditioning events. Creation yield, correct-charge yield, optical usability, cluster completion, and bond tolerance are distinct conditional events.
- A scaling claim holds comparison conditions fixed. Logical improvement with code distance is demonstrated at fixed physical noise and fixed analysis rules.
- Failure of one criterion closes only the claim it tests. Failure of a passive-memory criterion can motivate an active simulator, sensor, network node, or smaller proof of principle — each a new claim with new acceptance criteria.

Quantitative consistency checks

Quantitative consistency checks

- For the hypothetical parameter values, define the dimensionless interaction-to-decoherence ratio as

$$R_J = 2\pi\nu_{\text{eff}}T_2,$$

with $\nu_{\text{eff}} = J_{\text{eff}}/h$ the effective interaction frequency. Its value is

$$2\pi(20 \times 10^6 \text{ s}^{-1})(100 \times 10^{-6} \text{ s}) \approx 1.26 \times 10^4.$$

This value permits many interaction cycles, yet the condition $R_J > 1$ is usually too weak. A protocol requiring α interaction times against an interaction-error budget ϵ_J must satisfy

$$R_J \geq \alpha/\epsilon_J.$$

For these parameters, $\alpha/\epsilon_J = 2.0 \times 10^4$, which exceeds R_J .

- Treating $\Delta > k_B T$ as a binary acceptance test omits system size and the allowed thermal-error budget. The required ratio grows as

$$\ln(N_c g / \epsilon_T),$$

where ϵ_T is the thermal-error budget. For $N_s = 1000$ and $\epsilon_T = 10^{-3}$, this logarithm is 13.8. Diffusion can still dominate even after the thermally excited occupancy appears small.

- A 5% dipolar-coupling tolerance at $r = 10$ nm requires $\sigma_r \leq 0.17$ nm. For dipolar coupling,

$$J \propto r^{-3},$$

where J is the coupling strength and r is the separation. For small placement fluctuations, the relative coupling variation is

$$\frac{\sigma_J}{J} \approx \frac{3\sigma_r}{r},$$

where σ_J and σ_r are the standard deviations of coupling and separation, respectively. Therefore, for a fractional coupling tolerance η_J ,

$$\sigma_r \leq \eta_J r / 3 = 0.05 \times 10 / 3 \approx 0.17 \text{ nm}.$$

- If $t_{\min} > t_{\max}$, no valid preparation-time interval exists, even with arbitrarily slow ramps technically available. The adiabatic lower bound $t_{\min}$ then exceeds the decoherence upper bound $t_{\max}$. A longer ramp reduces diabatic error — failure to remain in the intended instantaneous eigenstate — while increasing open-system error from coupling to the environment.
- A 99.5% site yield can still fail a large, unrepairable patch. Combined over a thousand-site patch, the site-only yield is

$$0.995^{1000} \approx 0.0067,$$

before bond failures are included.

- A result definitively rejects the stated route when a preregistered threshold is missed even by the optimistic bound that remains credible after a measurement or calculation capable of resolving the acceptance criterion. Any other miss calls for more data.

The eight inequalities are go/no-go decision criteria, not target values. A stage advances only when the conservative bounds satisfy the corresponding criteria.

When an uncertainty interval straddles a decision threshold, the next step is the cheapest calculation capable of resolving the criterion. When even the optimistic bound fails, the stated route terminates or requires an explicit redesign.

Sources

Sources

- [R168] S. Bravyi and B. Terhal, “A no-go theorem for a two-dimensional self-correcting quantum memory based on stabilizer codes,” *New Journal of Physics* 11, 043029 (2009). DOI: [10.1088/1367-2630/11/4/043029](https://doi.org/10.1088/1367-2630/11/4/043029). [arXiv:0810.1983](https://arxiv.org/abs/0810.1983).
- [R169] B. J. Brown, D. Loss, J. K. Pachos, C. N. Self, and J. R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* 88, 045005 (2016). DOI: [10.1103/RevModPhys.88.045005](https://doi.org/10.1103/RevModPhys.88.045005). [arXiv:1411.6643](https://arxiv.org/abs/1411.6643).
- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* 51, 093512 (2010). DOI: [10.1063/1.3490195](https://doi.org/10.1063/1.3490195). [arXiv:1001.0344](https://arxiv.org/abs/1001.0344).
- [R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* 528, 1–45 (2013). DOI: [10.1016/j.physrep.2013.02.001](https://doi.org/10.1016/j.physrep.2013.02.001). [arXiv:1302.3288](https://arxiv.org/abs/1302.3288).
- [R220] A. G. Fowler, “Coping with qubit leakage in topological codes,” *Physical Review A* 88, 042308 (2013). DOI: [10.1103/PhysRevA.88.042308](https://doi.org/10.1103/PhysRevA.88.042308). [arXiv:1308.6642](https://arxiv.org/abs/1308.6642).
- [R247] S. Jansen, M.-B. Ruskai, and R. Seiler, “Bounds for the adiabatic approximation with applications to quantum computation,” *Journal of Mathematical Physics* 48, 102111 (2007). DOI: [10.1063/1.2798382](https://doi.org/10.1063/1.2798382). [arXiv:quant-ph/0605080](https://arxiv.org/abs/quant-ph/0605080).

[ph/0603175](#).

- [R248] T. M. Stace, S. D. Barrett, and A. C. Doherty, “Thresholds for topological codes in the presence of loss,” *Physical Review Letters* 102, 200501 (2009). DOI: [10.1103/PhysRevLett.102.200501](#). [arXiv:0904.3556](#).
 - [R141] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, “Topological quantum memory,” *Journal of Mathematical Physics* 43, 4452–4505 (2002). DOI: [10.1063/1.1499754](#). [arXiv:quant-ph/0110143](#).
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Common Methodological Errors

Point 1: Beginning with the largest patch.

- **Explanation:** This procedure prevents identification of whether a failure originates in the defect model, cluster truncation, perturbative gadget, boundary condition, or numerical solver.

Point 2: Selecting DMRG solely because the system is large.

- **Explanation:** DMRG is most effective for one-dimensional and quasi-one-dimensional geometries. A wide two-dimensional cluster with high entanglement may be less suitable.

Point 3: Treating one PEPS calculation as evidence of convergence.

- **Explanation:** PEPS optimization and contraction are approximate procedures. Convergence tests must vary the bond dimension, environment dimension, unit cell, and initial state.

Point 4: Measuring topological entropy before reaching the required length scales.

- **Explanation:** Topological-entanglement-entropy subtraction formulas assume scale separation among the lattice spacing, correlation length, subsystem size, and total system size.

Point 5: Changing stop rules after examining the results.

- **Explanation:** Thresholds may be revised when new experimental evidence changes the input contract. They may not be revised solely because the preferred model failed to meet them.
-

Common methodological errors

- **Beginning with the largest patch.** A largest-first calculation tests the defect model, cluster truncation, perturbative gadget, boundary condition, and numerical solver all at once, so the source of a failure stays hidden. The staged overlap calculations localize the failed modeling step.
- **Selecting DMRG solely because the system is large.** DMRG is a matrix-product variational method, most effective for one-dimensional and quasi-one-dimensional geometries [R035]. A wide two-dimensional cluster with high entanglement may be less suitable for DMRG than a smaller ED calculation. The selected method must provide sufficient entanglement capacity for the geometry and state being studied.
- **Treating one PEPS calculation as evidence of convergence.** PEPS optimization and contraction are approximate procedures. Convergence tests vary the bond dimension, environment dimension, unit cell, and initial state [R252]; [R253]. Obtaining a low energy alone does not certify that the calculation has reached the correct sector.
- **Measuring topological entropy before reaching the required length scales.** Topological-entanglement-entropy subtraction formulas assume scale separation among the lattice spacing, correlation length, subsystem size, and total system size [R254]; [R255]. A value obtained on a very small patch may be suggestive, but it is not asymptotic evidence.
- **Extracting modular data from accidental degeneracy.** Any nearly degenerate finite-dimensional matrix permits arbitrary basis rotations within its nearly degenerate subspace. Modular matrices are physically meaningful only after the candidate ground-state manifold, system topology, symmetry action, and phase conventions have been controlled [R256].
- **Conflating fitted interactions with emergent interactions.** A manually inserted target plaquette term turns the simulation into a study of the consequences of that imposed term.

A target term derived from two-body defect couplings and validated against leakage-aware ED turns the simulation into a test of whether that interaction emerges from the microscopic model. The two cases are plotted and reported as distinct models.

- **Discarding unfavorable uncertainty samples.** A design that works only at nominal coupling values and fails across measured placement or strain distributions misses its engineering objective. Sampling over uncertainty belongs to the Hamiltonian analysis; it is not an optional source of cosmetic error bars.
- **Changing stop rules after examining the results.** Thresholds may be revised when new experimental evidence changes the input contract.

Failure of the preferred model alone never justifies a revision. Setting thresholds before a run is calibration; moving them afterward is goalpost relocation.

Technical checks

Technical checks

- **Rationale for beginning with two defects rather than a topological patch.**

The two-defect calculation tests the microscopic interaction, retained basis, and uncertainty range at low cost. Every later projection depends on all three.

- **Derivation of the Pauli-string coefficients on n retained degrees of freedom.**

For a fitted Hamiltonian H_{fit} , the coefficient of the Pauli string σ_α is

$$c_\alpha = 2^{-n} \text{Tr}(\sigma_\alpha^\dagger H_{\text{fit}}).$$

A Pauli string is a tensor product of single-spin Pauli operators and identities. Pauli strings are orthogonal under the trace inner product, and $\text{Tr}(I) = 2^n$. The factor 2^{-n} is therefore fixed by inner-product normalization rather than chosen during fitting. The purpose of the expansion is to expose unwanted terms, not merely to report the target coefficient K .

- **Consequence of beginning with the largest patch.**

A largest-first calculation tests the defect model, cluster truncation, perturbative gadget, boundary condition, and solver simultaneously, so the origin of a failure cannot be identified. The staged overlap calculations identify the specific failed step.

- **Memory requirement for one complex state vector on N encoded spins.**

The Hilbert space holds 2^N amplitudes. At 16 bytes per complex amplitude, one state vector occupies

$$16 \times 2^N$$

bytes before accounting for additional vectors required by the eigensolver. This estimate must be performed before selecting ED.

- **Conditions under which DMRG is appropriate.**

DMRG suits one-dimensional or quasi-one-dimensional geometries with efficient matrix-product-state representations. Convergence is tested against bond dimension, number of sweeps, symmetry sector, and cylinder dimensions.

- **Failure of modular S and T extraction under accidental degeneracy.**

Modular S and T matrices from an accidental degeneracy hinge on arbitrary gauge and basis choices and carry no controlled physical interpretation. The calculation first needs a resolved quasi-degenerate ground-state manifold on a suitable geometry, controlled truncation, minimally entangled states, and fixed basis phases.

At the end of the workflow, the established data products are a versioned parameter record, a two-defect spectrum, a validated projector, and an accepted or rejected plaquette candidate. Each decision stage returns positive, negative, or numerically unresolved. The result follows from the final decision stage that passed its declared criteria, not from the most ambitious calculation attempted.

Sources

Sources

- [R249] J. R. Johansson, P. D. Nation, and F. Nori, “QuTiP: An open-source Python framework for the dynamics of open quantum systems,” *Computer Physics Communications* 183, 1760–1772 (2012). DOI: [10.1016/j.cpc.2012.02.021](https://doi.org/10.1016/j.cpc.2012.02.021).

- [R035] U. Schollwöck, “The density-matrix renormalization group in the age of matrix product states,” *Annals of Physics* 326, 96–192 (2011). DOI: [10.1016/j.aop.2010.09.012](https://doi.org/10.1016/j.aop.2010.09.012); arXiv: [1008.3477](https://arxiv.org/abs/1008.3477).
 - [R250] M. Fishman, S. R. White, and E. M. Stoudenmire, “The ITensor Software Library for Tensor Network Calculations,” *SciPost Physics Codebases* 4 (2022). DOI: [10.21468/SciPostPhysCodeb.4](https://doi.org/10.21468/SciPostPhysCodeb.4); arXiv: [2007.14822](https://arxiv.org/abs/2007.14822).
 - [R251] J. Hauschild and F. Pollmann, “Efficient numerical simulations with Tensor Networks: Tensor Network Python (TeNPy),” *SciPost Physics Lecture Notes* 5 (2018). DOI: [10.21468/SciPostPhysLectNotes.5](https://doi.org/10.21468/SciPostPhysLectNotes.5); arXiv: [1805.00055](https://arxiv.org/abs/1805.00055).
 - [R252] J. I. Cirac, D. Pérez-García, N. Schuch, and F. Verstraete, “Matrix product states and projected entangled pair states: Concepts, symmetries, theorems,” *Reviews of Modern Physics* 93, 045003 (2021). DOI: [10.1103/RevModPhys.93.045003](https://doi.org/10.1103/RevModPhys.93.045003); arXiv: [2011.12127](https://arxiv.org/abs/2011.12127).
 - [R253] J. Jordan, R. Orús, G. Vidal, F. Verstraete, and J. I. Cirac, “Classical simulation of infinite-size quantum lattice systems in two spatial dimensions,” *Physical Review Letters* 101, 250602 (2008). DOI: [10.1103/PhysRevLett.101.250602](https://doi.org/10.1103/PhysRevLett.101.250602); arXiv: [cond-mat/0703788](https://arxiv.org/abs/cond-mat/0703788).
 - [R025] S. Bravyi, D. P. DiVincenzo, and D. Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” *Annals of Physics* 326, 2793–2826 (2011). DOI: [10.1016/j.aop.2011.06.004](https://doi.org/10.1016/j.aop.2011.06.004); arXiv: [1105.0675](https://arxiv.org/abs/1105.0675).
 - [R254] A. Kitaev and J. Preskill, “Topological entanglement entropy,” *Physical Review Letters* 96, 110404 (2006). DOI: [10.1103/PhysRevLett.96.110404](https://doi.org/10.1103/PhysRevLett.96.110404); arXiv: [hep-th/0510092](https://arxiv.org/abs/hep-th/0510092).
 - [R255] M. Levin and X.-G. Wen, “Detecting topological order in a ground state wave function,” *Physical Review Letters* 96, 110405 (2006). DOI: [10.1103/PhysRevLett.96.110405](https://doi.org/10.1103/PhysRevLett.96.110405); arXiv: [cond-mat/0510613](https://arxiv.org/abs/cond-mat/0510613).
 - [R256] Y. Zhang, T. Grover, A. Turner, M. Oshikawa, and A. Vishwanath, “Quasiparticle statistics and braiding from ground-state entanglement,” *Physical Review B* 85, 235151 (2012). DOI: [10.1103/PhysRevB.85.235151](https://doi.org/10.1103/PhysRevB.85.235151); arXiv: [1111.2342](https://arxiv.org/abs/1111.2342).
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Assessment III — A worked four-cluster assessment

Earlier chapters set out how to judge a proposed topological memory: separate each physical step, name the evidence each step needs, and design a measurement that can reject the mechanism early. This chapter runs that procedure on the defect-cluster proposal developed in this book. The full proposal is a crystal of defect clusters whose collective interactions would enter a doubled-Fibonacci phase and protect a logical qubit. The worked calculation here covers only the first physical step toward that goal.

Background, scope, and supporting argument

The full proposal calls for a crystal of defect clusters whose collective interactions enter a doubled-Fibonacci phase: a non-Abelian topological phase described by Fibonacci fusion data together with its time-reversed counterpart. The protected logical qubit would live in that collective phase. This construction agrees with known physical law, so the assessment turns on evidence rather than on possibility.

That evidence is thin. The required interaction and the required cluster geometry have never appeared together in one measured system, and no calculation has derived them together from an established microscopic model. Nor does the proposal extend an existing device platform: several components would each need independent demonstration. Showing that no law forbids the device leaves the feasibility question open, because feasibility needs positive evidence at each step.

The calculation that can supply the first piece of evidence uses four triangular defect clusters, twelve spins in total. It asks whether known pairwise couplings between those spins can combine into a usable interaction shared by all four clusters, and how large that four-body term is beside the ordinary pairwise terms that accompany it.

Introduction: The Overall Assessment

The assessment of the full proposal is **highly speculative but physically coherent**. The proposal asks a crystal of defect clusters to enter a doubled-Fibonacci phase and protect a logical qubit there. Nothing in that construction contradicts known physics, which is why the verdict concerns evidence rather than possibility.

Measured and calculated support for the full device is missing: the interaction, the cluster geometry, and the surrounding control have never been shown working together, and the proposal reaches beyond any single existing device. The unresolved parts are now named explicitly in this chapter, and the four-cluster calculation below gives

the first measurement-like test capable of rejecting the mechanism.

Detailed treatment: initial operation requiring active control in the architecture

Initial operation requiring active control in the architecture

The architecture consists of the following sequence of physical and computational reductions:

```
defects -> cluster doublet -> interacting encoded spins  
-> many-body projector -> topological phase -> useful logical qubit
```

A cluster doublet is a pair of low-energy cluster states used as an effective two-level system. An encoded spin is the effective spin-1/2 degree of freedom represented by that doublet. A many-body projector is an operator that selects a specified subspace through a simultaneous interaction among several encoded spins. Testing the final link first would waste effort and obscure the cause of failure: a large device can fail through fabrication, initialization, control, readout, or the missing phase itself, and one failed run cannot distinguish those causes. The informative experiment isolates the first step that pairwise physics does not supply on its own.

A three-defect cluster in a controlled model can hold an isolated pair of low-energy states, separated in energy from the remaining cluster states. That isolated pair matters because it supplies the two-level system on which the rest of the proposal builds. It records nothing about topological order: projecting a physical pairwise coupling onto the isolated pairs of two clusters ordinarily yields another pairwise coupling between the two effective spins.

A toric-code stabilizer, which is a commuting operator defining the local constraints of the toric-code model, acts as a product on four encoded spins. A doubled-Fibonacci plaquette operator demands more: its amplitudes must implement specified recoupling data, which determine how different fusion orderings are related [R175]; [R018]; [R030].

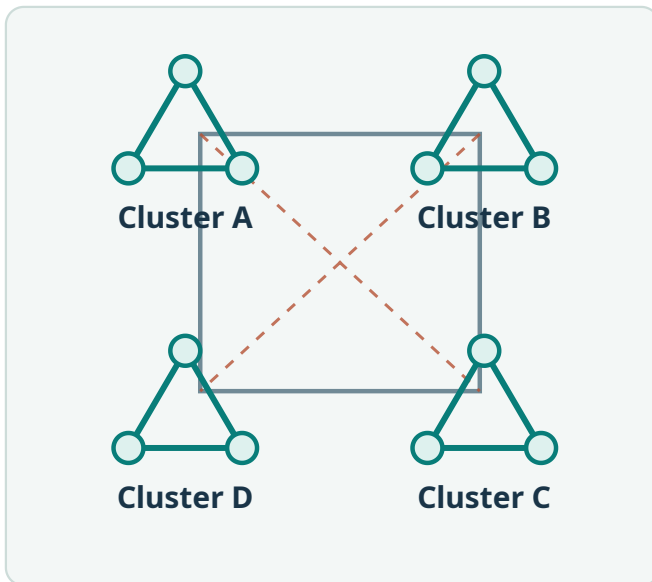
The first study therefore measures one thing: whether the four-cluster loop generates a plaquette interaction large enough to use, while every accompanying unwanted operator stays small enough to tolerate. This study tests the interaction mechanism. If the required coupling stays out of reach, the Hamiltonian behind the proposal is ruled

out and larger patches would test nothing new. If the coupling appears at a usable scale, a larger finite patch becomes worth analyzing. Building the large patch first would leave any failure stranded among the many steps listed above.

The Initial Operation: Four Triangular Clusters

The triangle enters for a practical reason: with all three sides equal, the cluster spectrum can be found in closed form. That closed-form spectrum then serves as the clean reference against which disorder and omitted levels are measured. An equation can be symmetric while the fabricated triangles vary, so the symmetric result calibrates the calculation without predicting device uniformity.

Four triangles test the interaction mechanism



FULL TWO-LEVEL MODEL

12 physical qubits

$2^{12} = 4,096$ basis states



ONE DOUBLET PER TRIANGLE

4 encoded qubits

$2^4 = 16$ encoded states

Candidate: K4 ZA ZB ZC ZD

Compare with all other Pauli terms.

Illustrative connections only: the microscopic calculation must include all physical dipolar bonds.

Each triangle is one local encoded degree of freedom. The four-cluster calculation can identify a weight-four interaction and its competitors; it is too small to establish a thermodynamic phase.

The immediate device is four triangular defect clusters holding twelve spins. The question it answers is whether known pairwise couplings can combine into a usable four-body interaction, or whether the ordinary pairwise terms stay dominant. A resonant microwave drive dresses the symmetric triangle so that its effective cluster Hamiltonian takes the form:

$$\frac{H_C}{h} = \frac{c}{4} \sum_{i < j \in C} (\tau_i^z \tau_j^z - \tau_i^x \tau_j^x - \tau_i^y \tau_j^y)$$

Its two lowest states are the symmetric one- and two-excitation superpositions:

$$|W_1\rangle = \frac{|100\rangle + |010\rangle + |001\rangle}{\sqrt{3}}, \quad |W_2\rangle = \frac{|011\rangle + |101\rangle + |110\rangle}{\sqrt{3}}$$

The gap from this pair to the nearest other cluster states is:

$$\Delta_C/h = \frac{3c}{2}$$

The doublet sits lowest when $c > 0$, with c measured in hertz in H_C/h . In the one-excitation sector, each basis state has diagonal entry $-c/4$. The $XX + YY$ terms exchange the excitation between each pair of sites with off-diagonal entry $-c/2$. The symmetric vector $|W_1\rangle$ therefore has eigenvalue $-c/4 - 2(c/2) = -5c/4$.

Two orthogonal vectors in that sector have eigenvalue $c/4$. The two-excitation sector has the same structure and contains $|W_2\rangle$ at $-5c/4$. The fully polarized states have energy $3c/4$. Thus the ground doublet lies at $-5c/4$, the next four states at $c/4$, and the last two at $3c/4$, giving the stated gap.

For the encoded basis $|0_L\rangle = |W_1\rangle$, $|1_L\rangle = |W_2\rangle$, the projection of any one physical Pauli Z_i is $Z_L/3$: the average Z_i is $+1/3$ in W_1 and $-1/3$ in W_2 . Similarly the encoded matrices of X_i and Y_i are $2X_L/3$ and $2Y_L/3$. Ordinary physical bonds therefore generate encoded pair interactions already at first order, with appreciable coefficients ($1/3$ and $2/3$). A usable four-body term must outcompete these pairwise terms, which enter at lower order in the same expansion.

Four encoded spins form the smallest loop that supports a weight-four coefficient together with every lower-weight competitor, which is why the study starts there.

Detailed treatment: four triangular clusters containing twelve spins

Four triangular clusters containing twelve spins

The calculation uses negatively charged nitrogen-vacancy centers, NV^- , in diamond, the defect whose spin Hamiltonian, optical initialization and readout, and microwave control are characterized in the most detail [\[Ro74\]](#). Four nominal clusters are introduced and labeled $C = 1, 2, 3, 4$.

Each cluster contains three NV electron spins with the same crystallographic orientation, arranged as a triangle. The retained physical levels are $|0\rangle = |m_s = 0\rangle$ and $|1\rangle = |m_s = -1\rangle$, where m_s is the spin projection quantum number along the NV quantization axis.

The omitted $m_s = +1$ level is a leakage state, meaning a physical state outside the intended two-level computational subspace.

For a symmetric triangle under resonant microwave dressing, define physical-defect Pauli operators τ_i^α , where i labels a defect and $\alpha \in \{x, y, z\}$. Resonant dressing means that a microwave field is applied near a relevant transition frequency so that the rotating-frame dynamics implement an effective interaction. One useful ideal cluster Hamiltonian is

$$\frac{H_C}{h} = \frac{c}{4} \sum_{i < j \in C} \left(\tau_i^z \tau_j^z - \tau_i^x \tau_j^x - \tau_i^y \tau_j^y \right), \quad (41.1)$$

where h is Planck's constant and c is the intracluster dipolar interaction frequency. Division by h expresses the Hamiltonian in frequency units. The low-energy states of this model are

$$|W_1\rangle = \frac{|100\rangle + |010\rangle + |001\rangle}{\sqrt{3}}, \quad |W_2\rangle = \frac{|011\rangle + |101\rangle + |110\rangle}{\sqrt{3}},$$

and the separation between this pair and the nearest ideal cluster states is $\Delta_C/h = 3c/2$. Denote this pair as an encoded residual degree of freedom and define its encoded Pauli operators as X_C, Y_C, Z_C . These two states form a qubit encoded across three defects: a local two-level degree of freedom, while an anyon would be a delocalized excitation of a many-body phase with fusion and braiding rules.

The four triangular clusters are placed on a loop. The model keeps every physical dipolar bond, including the long diagonal bonds across the loop, because dropping inconvenient couplings would undercount the competing pairwise terms the plaquette interaction must beat. Twelve two-level defects span a Hilbert space of dimension $2^{12} = 4096$, which is small enough for exact diagonalization, meaning direct numerical computation of the Hamiltonian eigenvalues and eigenvectors.

A full spin-1 calculation has dimension $3^{12} = 531,441$. Sparse-matrix methods can still access a few low-energy states at this dimension. These values are exact Hilbert-space dimensions, not predictions of computational performance.

The target observable is an encoded operator coefficient of weight four, where operator weight is the number of encoded clusters on which the operator acts nontrivially. One example is

$$K_4 Z_1 Z_2 Z_3 Z_4,$$

where each Z_C acts on one cluster doublet and K_4 has units of frequency. Observing one weight-four term establishes that the microscopic mechanism can build a many-body interaction out of pairwise physical couplings. Establishing topological order takes more: dominance over competitors, a stable gap, and nonlocal diagnostics on larger patches.

Three clusters support at most a weight-three term, so four encoded spins are the smallest loop that can host the weight-four candidate together with its lower-weight rivals.

Assumptions permitted in the initial study

A defensible title is:

Can dipolar NV triangles generate a useful encoded plaquette interaction? A four-cluster microscopic benchmark.

The study should report positive and negative outcomes using the same standards. Its title should avoid “Fibonacci phase”: the study derives neither the branching constraint that restricts locally allowed fusion labels nor a Fibonacci plaquette operator, so the title must describe the plaquette benchmark actually performed.

All assumptions should be declared before fitting numerical parameters.

- **Host and charge state.** Use same-axis NV^- electron spins in isotopically enriched diamond, meaning diamond in which the abundance of spin-carrying isotopes has been reduced. Include the measured zero-field, Zeeman, strain/electric, hyperfine, drive, and magnetic dipole terms summarized by the established NV Hamiltonian [Ro74]. The Zeeman term describes coupling to a magnetic field, and the hyperfine term describes coupling between electron and nuclear spins.
- **Geometry.** Use nominally equilateral three-NV clusters with intracluster side length r_C and arrange the four clusters in a loop with separation R . Both quantities must remain variable, and position errors must be sampled. The value $r_C = 5$ nm is a screening parameter for the calculation; no cited experiment has fabricated repeated triangles at that side length.
- **No assumed exchange interaction.** Set long-range, lithographically tunable NV–NV exchange to zero unless an independent microscopic calculation or measurement establishes it. Exchange is a spin interaction arising from quantum-mechanical wavefunction overlap or an explicitly identified mediator. The supported resources are dipolar coupling and microwave control. Entanglement through direct dipolar coupling between two NV electron spins has been demonstrated [Experiment] [Ro80].
- **Driven character of the model.** Equation (41.1) is a rotating-frame model sustained by a microwave frequency reference. A rotating frame is a representation that follows the phase of the applied drive. The resulting system is a driven analog emulator rather than a passive equilibrium ground-state Hamiltonian.
- **Nuclear spins and disorder.** Either include the nitrogen and selected ^{13}C nuclear spins explicitly or fix their states and state the corresponding preparation assumption. Detuning, strain, angular, amplitude, and position disorder must be sampled separately so that their effects can be distinguished.

- **No result-dependent basis selection.** Fix the cluster basis using isolated-cluster spectroscopy. If local basis rotations are optimized, restrict them to calibrated single-cluster controls and also report the coefficients in the unrotated basis.
- **Separate treatment of two protocols.** Analyze both (a) continuously applied couplings that are static in the rotating frame and (b) a periodic refocusing sequence specified in advance. Refocusing uses controlled pulses to average or cancel selected interactions. The second protocol is Floquet engineering, meaning the design of an effective Hamiltonian through periodic driving; it runs on continuous pulses and an external clock, unlike a passive ground-state gap.

No cited fabrication result combines all these assumptions in one device. Nanoaperture implantation localized NV clusters on approximately the 10-nm scale and reported separations as small as about 40 nm [Experiment] [R257].

A later deterministic single-ion source produced a 5×5 pattern at $2 \mu\text{m}$ pitch, with 121(35) nm lateral resolution and about 0.6% NV conversion in that shallow-implantation experiment [Experiment] [R258].

Each cited experiment demonstrates some of these capabilities. Repeated triangles with 5-nm side lengths combining localization, exact ion count, conversion efficiency, charge stability, crystallographic orientation, and coherent-device yield remain undemonstrated.

Fitting a fresh encoded basis to every disorder realization inflates the reported target coefficient: each sample gets its own tailor-made frame, which no single fixed control protocol can implement. A realizable device is tested by fixing the basis from isolated-cluster spectroscopy and grading every sample in that shared frame.

The Microscopic Hamiltonian

The ideal triangle keeps two levels per defect for tractability. A physical NV electron spin carries three levels, so the calculation must verify that the omitted level stays unoccupied and uncoupled at the operating parameters. In the following model $\mathbf{S}_i$ denotes dimensionless spin-1 matrices; the explicit constants supply energy units. The labels describe zero-field splitting, Zeeman coupling, hyperfine interaction, strain, drive, and magnetic dipolar interaction, respectively.

The calculation should start from the physical spin-1 operators $\mathbf{S}_i$:

$$H(t) = \sum_i [D_i(S_i^z)^2 + g_e \mu_B \mathbf{B}_i \cdot \mathbf{S}_i + H_{\text{hf},i} + H_{\text{strain},i} + H_{\text{drive},i}(t)] \\ + \sum_{i < j} \frac{\mu_0 (g_e \mu_B)^2}{4\pi r_{ij}^3} [\mathbf{S}_i \cdot \mathbf{S}_j - 3(\mathbf{S}_i \cdot \hat{\mathbf{r}}_{ij})(\mathbf{S}_j \cdot \hat{\mathbf{r}}_{ij})]$$

Project onto the four cluster doublets and expand in the complete encoded Pauli basis:

$$\frac{H_{\text{eff}}}{h} = \sum_{P_a \in \{I, X, Y, Z\}^{\otimes 4}} k_a P_a, \quad k_a = \frac{1}{16} \text{Tr} \left(P_a \frac{H_{\text{eff}}}{h} \right)$$

For the proposed values $r_C = 5 \text{ nm}$ and $R = 10 \text{ nm}$:

$$c \approx 0.416 \text{ MHz}, \quad \Delta_C/h \approx 0.624 \text{ MHz}, \quad u \lesssim 52 \text{ kHz}$$

Assuming the first nonzero connected process occurs at fourth order:

$$K_4 \sim \frac{u^4}{(\Delta_C/h)^3} \sim 30 \text{ Hz}$$

$$hK_4/k_B \approx 1.4 \text{ nK}$$

At millikelvin operating temperatures, thermal energy exceeds this estimated interaction by many orders of magnitude, so the estimate argues against a passive equilibrium memory at these parameters. Driven or actively corrected uses face a different comparison — coupling strength against coherence and control precision — and need their own calculation.

Here both u and K_4 are ordinary frequencies, so the expansion parameter is $u/(\Delta_C/h)$. With the stated values it is at most $52/624 \approx 0.0833$, and $52,000(0.0833)^3 \approx 30 \text{ Hz}$. The estimate assumes a fourth-order connected path with a coefficient of order one. The full calculation must still derive that path and bound the first-, second-, and third-order competitors, which can each exceed it.

Detailed treatment: microscopic Hamiltonian and effective-model extraction

Microscopic Hamiltonian and effective-model extraction

The calculation starts from the physical spin-1 operators $\mathbf{S}_i$, so that every energy scale enters from measured defect physics; the stabilizer form, if present, comes out of the projection. The microscopic model is

$$H(t) = \sum_i \left[D_i (S_i^z)^2 + g_e \mu_B \mathbf{B}_i \cdot \mathbf{S}_i + H_{\text{hf},i} + H_{\text{strain},i} + H_{\text{drive},i}(t) \right] + \sum_{i < j} \frac{\mu_0 (g_e \mu_B)^2}{4\pi r_{ij}^3} [\mathbf{S}_i \cdot \mathbf{S}_j - 3(\mathbf{S}_i \cdot \hat{\mathbf{r}}_{ij})(\mathbf{S}_j \cdot \hat{\mathbf{r}}_{ij})]. \quad (41.2)$$

Here D_i is the zero-field splitting energy, g_e is the electron g factor, μ_B is the Bohr magneton, and μ_0 is the vacuum permeability. The displacement vector $\mathbf{r}_{ij} = r_{ij} \hat{\mathbf{r}}_{ij}$ connects defects i and j , with magnitude r_{ij} and direction $\hat{\mathbf{r}}_{ij}$. The terms $H_{\text{hf},i}$, $H_{\text{strain},i}$, and $H_{\text{drive},i}(t)$ describe hyperfine, strain/electric, and microwave interactions,

respectively. Every term in (41.2) has units of joules. The experimentally established NV^- ground-state value is $D/h \approx 2.87$ GHz, and the corresponding spin Hamiltonian is well characterized [Ro74].

The next step is to transform into the declared rotating frame and verify the rotating-wave approximation against the full model. The rotating-wave approximation neglects rapidly oscillating terms whose net influence is expected to average to a small value. The calculation then projects onto the four cluster doublets. Let P denote the 2^4 -dimensional projector onto this encoded subspace and let $Q = I - P$ project onto its complement.

A Schrieffer–Wolff transformation, which perturbatively removes coupling between separated energy subspaces, yields an effective Hamiltonian. Exact low-energy eigenvectors provide a nonperturbative comparison. The effective Hamiltonian is expanded in the complete encoded Pauli basis:

$$\frac{H_{\text{eff}}}{h} = \sum_{P_a \in \{I, X, Y, Z\}^{\otimes 4}} k_a P_a, \quad k_a = \frac{1}{16} \text{Tr} \left(P_a \frac{H_{\text{eff}}}{h} \right). \quad (41.3)$$

This expansion contains all $4^4 = 256$ Pauli strings on four encoded spins. The coefficient k_{ZZZZ} is K_4 . Every other coefficient k_a represents a potentially competing interaction; each stays in the published table with its fitted uncertainty unless a symmetry argument removes it. Perturbative gadgets can generate many-body interactions from two-body resources, but the desired terms contain suppressing energy denominators and are accompanied by unwanted corrections [Theory] [R175].

An illustrative scale estimate shows why the complete coefficient table is necessary. The characteristic interaction frequency of electron dipoles is approximately $52 \text{ MHz} (1 \text{ nm}/r)^3$.

For the proposed but undemonstrated values $r_C = 5 \text{ nm}$ and $R = 10 \text{ nm}$, this relation gives $c \approx 0.416 \text{ MHz}$, $\Delta_C/h \approx 0.624 \text{ MHz}$, and an unprojected intercluster interaction scale $u \lesssim 52 \text{ kHz}$ before geometric factors are included.

The bare coefficients are therefore approximately 416 kHz within each triangle and 52 kHz between triangles. These quantities are microscopic inputs, not the substantially smaller effective plaquette scale generated perturbatively.

Assuming that the first nonzero connected process around the four-cluster loop occurs at fourth order, dimensional analysis gives

$$K_4 \sim \frac{u^4}{(\Delta_C/h)^3} \sim 30 \text{ Hz.} \quad (41.4)$$

The dimensional consistency is $\text{Hz}^4/\text{Hz}^3 = \text{Hz}$. Both the numerical coefficient and the leading perturbative order depend on symmetry and pulse design. Consequently, 30 Hz is a [Proposal] scale estimate rather than a prediction.

The equivalent temperature is $\hbar K_4/k_B \approx 1.4 \text{ nK}$, where k_B is Boltzmann's constant. A passive phase governed by an energy scale this small would be implausible.

A driven spectroscopic signal could nevertheless be measurable if coherence time, calibration accuracy, and statistical averaging are sufficient. The calculation must determine whether constructive processes occur at lower perturbative order or whether larger pairwise interactions dominate.

No demonstrated fabrication process repeatedly produces charge-stable, coherent NV triangles with 5-nm side lengths and acceptable compound yield. Linewidth, disorder, temperature, and perturbative suppression must be evaluated together.

The geometry and probability distributions specified above remain explicit study parameters within the screening ranges used in this book; they are not demonstrated fabrication capabilities.

Thus, observing $K_4 \neq 0$ would verify only the proposed interaction-generation mechanism. It would not establish a phase.

Required Calculations and Decision Criteria

1. **Calibrate one cluster:** Diagonalize the full 3^3 -dimensional spin-1 Hamiltonian.
2. **Analyze two clusters:** Diagonalize the six-spin system.
3. **Fit four clusters:** Perform exact diagonalization of the 4096-state physical-qubit model.
4. **Evaluate the driven protocol:** Calculate the one-period propagator and compare its stroboscopic dynamics with the proposed effective model.
5. **Sample fabrication distributions:** Vary every defect position, orientation, detuning, strain, and drive amplitude.
6. **Analyze a patch only after the interaction criterion is met.**

The Hilbert-space counts refer to different truncations. Three physical spin-1 defects span $3^3 = 27$ electronic states; six span $3^6 = 729$. Twelve full spin-1 defects span $3^{12} = 531441$ states, whereas twelve retained physical qubits span $2^{12} = 4096$. Projecting those twelve qubits into four cluster doublets leaves $2^4 = 16$ encoded states. Nuclear-spin levels enlarge the full space further. State which space is used whenever quoting computational cost or claiming that leakage has been checked.

For a drive period T_d , the propagator is $U(T_d) = \mathcal{T} \exp[-(i/\hbar) \int_0^{T_d} H(t) dt]$. An effective stroboscopic Hamiltonian satisfies $U(T_d) = e^{-iH_F T_d/\hbar}$, but quasienergies are defined modulo $2\pi\hbar/T_d$. A consistent branch choice and within-period leakage analysis are needed, because a matrix logarithm alone records stroboscopic phases while passive protection requires continuous confinement to the encoded subspace.

Write the encoded Hamiltonian in frequency units, $H_{\text{eff}}/\hbar = \sum_a k_a \sigma_a$. Let K_4 be the desired weight-four coefficient and $\mathcal{U}$ a predeclared set of unwanted nonidentity terms. The identity shift is omitted because it has no effect on relative dynamics. Define:

For the scoped four-cluster study, the unwanted set contains symmetry-allowed, noncommuting residual terms after declared, physically implementable single-cluster calibration. A sum over all unwanted terms is a more conservative additional diagnostic; keep the chosen denominator fixed when comparing results with the thresholds.

$$R_4 = \frac{|K_4|}{\sum_{a \in \mathcal{U}} |k_a|}$$

The minimum theoretical success criteria are:

$$R_4 > 1, \quad \epsilon_{\text{spec}} < 0.1, \quad p_{\text{leak}} < 0.01$$

To make these proposed thresholds operational, define ϵ_{spec} before fitting. One possible convention is the maximum discrepancy between matched retained-band energies after removing a common offset, divided by a preselected nonzero useful energy scale. If that scale is $\hbar|K_4|$, a vanishing K_4 itself fails the usefulness test. Eigenvalue matching must respect the chosen low-energy band.

For the static spectral screen, define $p_{\text{leak}}^{\text{spec}}$ as the largest excluded-space probability among the sixteen retained-band eigenstates. A driven protocol also needs $p_{\text{leak}}^{\text{dyn}}$: the maximum of $1 - \langle \psi(t) | P | \psi(t) \rangle$ over specified initial states and the protocol duration. Report these separately. The detailed static criterion below uses the spectral quantity; small eigenstate admixture alone does not establish small leakage throughout a driven evolution.

The values 0.1 and 0.01 are proposed screening gates, not universal phase boundaries. Likewise, the coefficient sum in R_4 is a convenient small-system bound because each Pauli string has norm one. Whether particular residual terms destabilize a phase still depends on their structure and spatial arrangement.

A promising result additionally requires $R_4 > 10$ and $|K_4|$ greater than independently measured encoded-decoherence and quasistatic-disorder scales in consistent frequency units. For an exponential decay rate Γ , compare with $\Gamma/(2\pi)$ when K_4 is in hertz. At the four-cluster stage this tests encoded dynamics on one plaquette; a measured topological logical lifetime needs a larger patch with its own gap and error tracking.

The project should stop or be redesigned if:

- no isolated cluster doublet survives realistic hyperfine and geometric disorder;
- suppressing leakage requires $u/(\Delta_C/h)$ to be so small that K_4 falls below the relevant disorder or decoherence scale.
- Lower-weight noncommuting terms remain larger than the desired four-body coefficient after physically available calibration.
- A refocusing sequence requires pulse errors below demonstrated control capabilities.
- The fitted patch lacks a stable gap or the expected topological-sector structure.
- Success requires exchange interactions, mediators, placement precision, or cooling resources outside the declared platform.

Detailed treatment: required calculations and decision criteria

Required calculations and decision criteria

The initial paper is a numerical and spectroscopic study: it needs no twelve-defect device, only cluster spectroscopy data and the microscopic parameters cited above.

- **Calibrate one cluster.** Diagonalize the full 3^3 -dimensional spin-1 Hamiltonian over magnetic field, drive parameters, strain, hyperfine state, and triangular distortion. Record the doublet splitting, the gap Δ_C , the leakage matrix elements, and the projected single-spin operators.
- **Analyze two clusters.** Diagonalize the six-spin system and compare its exact low-energy spectrum with first- and second-order Schrieffer–Wolff results. This comparison identifies the parameter regime in which the encoded Hamiltonian is controlled and determines the dominant one-cluster and two-cluster Pauli terms.
- **Fit four clusters.** Perform exact diagonalization of the 4096-state physical-qubit model, supplemented by selected full-spin-1 calculations. Construct H_{eff} using a specified quasi-degenerate projection, meaning a projection method appropriate to a group of nearby low-energy levels. Expand all 256 encoded Pauli strings using (41.3), and publish the complete coefficient table.
- **Evaluate the driven protocol.** For a periodic sequence of duration T_d , calculate the one-period propagator $U(T_d)$ and its quasienergies. Quasienergies are the phase-derived spectral values associated with one period of a driven system. Track micromotion, leakage, sensitivity to pulse errors, and the logarithm branch used in $H_F = (i\hbar/T_d) \log U(T_d)$, where H_F is the Floquet Hamiltonian. A quasienergy gap

records phase windings per drive period, while a thermal gap records Boltzmann-suppressed excitation energy; the driven gap supports coherent control claims, and the thermal gap supports passive-storage claims.

- **Sample fabrication distributions.** Vary every defect position, orientation, detuning, strain value, and drive amplitude. Report medians, distribution tails, failure fractions, and correlations across the sampled ensemble, since one ideal geometry cannot represent fabrication spread.
- **Analyze a patch only after the interaction criterion is met.** If a dominant and robust K_4 is found, place the fitted operators on the smallest periodic stabilizer patch and compute its spectrum, low-energy degeneracy, Wilson loops, and response to local perturbations. A Wilson loop is a nonlocal closed-string operator used to diagnose topological sectors. If the isolated plaquette fails the interaction criterion, the study stops there: topological diagnostics on a larger patch presuppose the microscopic interaction this step was meant to supply.

Define

$$R_4 = \frac{|K_4|}{\sum_{a \in \mathcal{U}} |k_a|},$$

where $\mathcal{U}$ is the predeclared set of symmetry-allowed, noncommuting unwanted terms after calibratable one-cluster fields have been removed. Also define ϵ_{spec} as the maximum error among the effective model's sixteen low-energy levels, divided by $|hK_4|$, and define p_{leak} as the largest Q -space probability weight among those states.

The minimum theoretical success criteria are $R_4 > 1$, $\epsilon_{\text{spec}} < 0.1$, and $p_{\text{leak}} < 0.01$, with all three inequalities remaining valid across a declared fabrication distribution. A promising result additionally requires $R_4 > 10$ and $|K_4|$ greater than independently measured logical-decoherence and quasistatic-disorder rates. The numerical values 1, 0.1, 0.01, and 10 are project decision thresholds rather than universal constants.

The project should stop or be redesigned if any of the following conditions occurs:

- no isolated cluster doublet survives realistic hyperfine and geometric disorder;
- suppressing leakage requires $u/(\Delta_C/h)$ to be so small that K_4 falls below the relevant linewidths;

- lower-weight noncommuting terms remain larger than K_4 after all physically allowed calibration;
- a refocusing sequence succeeds only when pulse errors are smaller than demonstrated control capabilities;
- the fitted patch lacks either a stable gap or the expected topological-sector structure;
- success requires exchange interactions, mediators, placement accuracy, or cooling resources not included in the declared platform.

The numerical study is difficult but computationally tractable because the relevant Hilbert spaces are moderate in size.

The principal theoretical challenges are a gauge-consistent low-energy projection, statistically adequate disorder sampling, and avoidance of post hoc parameter cancellations. Gauge consistency here requires that phase and basis conventions be maintained across the projected low-energy states. A two-person theory effort could reproduce the study using public Hamiltonian parameters without requiring a new fabrication facility.

A physical demonstration using twelve defects would be substantially more difficult. It would require nanometre-scale geometry, sufficient same-axis and charge-state yield, nuclear-state preparation, individual or patterned microwave phases, spectral discrimination, and sufficiently long coherence.

Existing experiments establish important components but not their simultaneous realization [R074]; [R080]; [R257]; [R258]. Realizing a doubled-Fibonacci material would additionally require the correct vertex constraints, conditional recoupling amplitudes, a many-body gap, and phase diagnostics [R018].

The next experimental target should therefore be the four-cluster plaquette benchmark, since a doubled-Fibonacci material cannot yet be specified from a validated microscopic interaction.

If the four-cluster coefficient analysis fails, further work should determine whether a modified geometry, an explicit mediator, or a different host material changes the relevant scaling before increasing the system size. If the analysis succeeds, the physical experimental sequence should be spectroscopy of one triangle, followed by two coupled triangles, and only then the twelve-defect plaquette.

A larger topological patch is justified only after the desired interaction dominates its competitors. A Fibonacci claim requires demonstrated branching constraints, recoupling amplitudes, a stable gap, and non-Abelian diagnostics.

A Floquet quasienergy gap depends on an external clock, drive stability, and continuous power. It does not thermalize in the same way as a static ground-state gap. Describing such a gap as passive protection would therefore constitute a different physical claim.

Current Experimental and Theoretical Status

Claim	Status	Platform	Evidence	Reference
The single-NV spin Hamiltonian, initialization, readout, and microwave control are understood.	experimentally demonstrated	Diamond NV ⁻	Extensive spectroscopy and control literature	[R074]
Two separated NV electron spins can be entangled through their interaction.	experimentally demonstrated	Diamond NV ⁻	Room-temperature two-spin entanglement experiment	[R080]
Nanometre-localized implanted NV clusters exist.	experimentally partial	Diamond NV ⁻	Nanoaperture localization; repeated deterministic 5-nm triangles were not demonstrated	[R257]
Joint exact ion delivery, NV conversion, and nanoscale coherent-array yield have been achieved.	experimentally partial	Diamond NV ⁻	Individual fabrication ingredients exist, but the deterministic-delivery experiment retained coarse focus and low conversion	[R257]; [R258]
Two-body interactions can perturbatively produce encoded many-body terms.	theoretically established	Abstract spin gadgets	Controlled expansions with denominators and error bounds	[R175]
The required Fibonacci string-net projectors define doubled-Fibonacci order.	theoretically established	Ideal lattice model	Exact commuting-projector construction	[R018]
Digital processors can prepare and braid Fibonacci-model excitations.	experimentally demonstrated	Superconducting processors	State preparation, fusion, and digital braiding simulate the target model	[R138]; [R165]
A defect lattice could host genuine Fibonacci quasiparticles.	speculative	Diamond, sapphire, SiC, others	The cited literature establishes defect qubits, ideal models, and digital simulations, but no defect-material realization	[R074]; [R018]; [R138]; [R165]

Detailed treatment: current experimental and theoretical status

Current experimental and theoretical status

Claim	Status	Platform	Evidence	Reference
The single-NV spin Hamiltonian, initialization, readout, and microwave control are understood.	experimentally demonstrated	Diamond NV [−]	Extensive spectroscopy and control literature	[R074]
Two separated NV electron spins can be entangled through their interaction.	experimentally demonstrated	Diamond NV [−]	Room-temperature two-spin entanglement experiment	[R080]
Nanometre-localized implanted NV clusters exist.	experimentally partial	Diamond NV [−]	Nanoaperture localization; repeated deterministic 5-nm triangles were not demonstrated	[R257]
Joint exact ion delivery, NV conversion, and nanoscale coherent-array yield have been achieved.	experimentally partial	Diamond NV [−]	Individual fabrication ingredients exist, but the deterministic-delivery experiment retained coarse focus and low conversion under its stated conditions	[R257] ; [R258]
Two-body interactions can perturbatively produce encoded many-body terms.	theoretically established	Abstract spin gadgets	Controlled expansions with denominators and error bounds	[R175]
The required Fibonacci string-net projectors define doubled-Fibonacci order.	theoretically established	Ideal lattice model	Exact commuting-projector construction	[R018]
Digital processors can prepare and braid Fibonacci-model excitations.	experimentally demonstrated	Superconducting processors	State preparation, fusion, and digital braiding simulate the target model; they do not establish an autonomous material phase	[R138] ; [R165]
A defect lattice could host genuine Fibonacci quasiparticles.	speculative	Diamond, sapphire, SiC, others	The cited literature establishes defect qubits, ideal models, and digital simulations, but no defect-material realization	[R074] ; [R018] ; [R138] ; [R165]

Claim	Status	Platform	Evidence	Reference
Ruby supplies some control ingredients relevant to defect clusters.	experimentally partial	$\text{Cr}^{3+}:\text{Al}_2\text{O}_3$	Ensemble spin-cavity coupling has been measured; deterministic single-ion arrays have not	[R089]

Sources of invalid inference

Sources of invalid inference

- **Result-dependent basis optimization.** Selecting a different encoded basis for every disorder realization can produce a large apparent target coefficient that cannot be implemented by any fixed control protocol.
- **Interpreting $K_4 \neq 0$ as a phase.** One local interaction is evidence only for a microscopic mechanism. Topological order requires either a thermodynamic phase or convincing finite-size scaling with the appropriate degeneracy and nonlocal diagnostics.
- **Confusing Floquet quasienergy with passive protection.** A driven frame runs on a clock, stable drive, and continuous power, and its quasienergy spectrum governs stroboscopic phases. Passive protection additionally needs a thermal gap with Boltzmann-suppressed excitations.
- **Mixing placement results from incompatible fabrication processes.** Beam focus, implantation straggle, conversion yield, charge yield, and coherent yield measure different steps; each needs its own reported number before any combined yield claim.
- **Treating digital Fibonacci demonstrations as evidence for a material phase.** Experiments have digitally prepared and braided Fibonacci-model anyons [Experiment] [\[R138\]](#); [\[R165\]](#). These results establish control protocols and simulated model behavior, not native Fibonacci quasiparticles in the processor substrate.
- **Omitting perturbative suppression.** A high-order perturbative gadget carries energy denominators that shrink the desired coefficient; the calculation must compare that shrunken scale directly against the measured decoherence and disorder scales.

Fifteen Assessment Questions

The questions below revisit the full argument in four groups: what is encoded, which material hosts it, what limits the interactions and fabrication, and what an experiment must demonstrate. Read them after the detailed calculation: each answer tests one level of the chain from defects to logical qubit.

Encoding, simulation, and emergence: questions 1–6

1. Encoding one effective qubit in multiple defects.

Several defects can share one effective qubit when two of their joint states sit apart in energy from all the rest. The ideal driven three-NV triangle produces such a doublet. Whether the isolation survives fabrication spread and disorder is the open experimental point.

2. Emulation of anyons with effective defect qubits.

Effective defect qubits can emulate anyons: programmed gates or driven couplings reproduce string operators, fusion measurements, and braiding statistics. The emulator inherits the error rates of its defect controls, since the statistics come from the program rather than from a protected phase.

3. Digital simulation of Fibonacci anyons.

A defect processor with a universal gate set, initialization, and readout can run the same digital Fibonacci-braiding circuits already demonstrated on superconducting processors, so the open question is fabrication scale and gate fidelity rather than algorithm availability.

4. Native Fibonacci topological order in a defect Hamiltonian.

Known physics permits a defect Hamiltonian with Fibonacci topological order, yet the cited defect couplings have not been shown to produce the required projectors and gap. The native-order claim therefore stays at the speculation stage.

5. Distinction between simulated and emergent anyons.

Simulation programs physical qubits to reproduce chosen anyonic observables, while emergence means the many-body Hamiltonian itself hosts anyons as its low-energy excitations with phase-wide stability. Matching circuit output therefore demonstrates control; only gap and stability measurements demonstrate a phase.

6. Protection obtained from defect clustering.

Clustering helps against specific noise: a symmetric encoding averages out a uniform field shared by its members. That averaging trades against steeper sensitivity to field gradients across the cluster, extra leakage levels, and more ways for fabrication to fail.

Choice of host material: questions 7–8

7. Assessment of diamond as a substrate.

Diamond leads on one combined metric: optically addressable individual color-center spins with mature control. Dense arrays of identical, interacting centers are still hard to place and activate, so the best host depends on which bottleneck dominates.

8. Assessment of corundum and sapphire.

Corundum may be competitive as a low-loss host with mature ensemble spin spectroscopy. The evidence is weaker for dense, individually controlled defect clusters.

Interaction and fabrication bottlenecks: questions 9–10

9. Primary interaction bottleneck.

The principal interaction bottleneck is the realization of a strong, controllable, correctly signed many-body encoded interaction that exceeds ordinary one- and two-body terms.

10. Primary fabrication bottleneck.

The principal fabrication bottleneck is repeated three-dimensional placement and activation of the correct defect species, orientation, charge state, and local environment.

Operating conditions and decisive tests: questions 11–15

11. Required hierarchy of energy and frequency scales.

$$\frac{\Delta_C}{h} \gg u, \nu_\Omega, \sigma, \frac{\Gamma_{\text{decoh}}}{2\pi}, \quad \frac{\Delta_{\text{topo}}}{h} \gg \frac{k_B T}{h}, \frac{\Gamma_{\text{decoh}}}{2\pi}, \sigma_{\text{eff}}.$$

Here ν_Ω is the relevant drive coupling or bandwidth in ordinary-frequency units, and $\sigma, \sigma_{\text{eff}}$ are frequency spreads. For a resonant drive built into the dressed cluster Hamiltonian, leakage is counted among the dressed eigenstates; the bare carrier frequency says nothing about that dressed leakage.

12. Operating-temperature constraint.

A passive realization requires $k_B T$ to remain well below Δ_{topo} . A tenfold Boltzmann margin gives $T < \Delta_{\text{topo}}/(10k_B)$.

13. Number of defects per effective degree of freedom.

The scoped triangle model uses three physical defects for each effective degree of freedom. No current evidence identifies an optimum number.

14. Smallest falsification experiment.

Spectroscopy of one fabricated three-NV triangle can falsify the assumed cluster doublet. Falsifying the proposed many-body mechanism requires four such clusters (twelve defects).

15. Evidence required for a promising result.

A reproducible cluster doublet must be followed by a measured or microscopically validated many-body coefficient that dominates noncommuting residual terms, exceeds disorder and decoherence, remains stable under realistic position distributions, and produces the expected gap and nonlocal diagnostics.

Detailed treatment: fifteen assessment questions

Fifteen assessment questions

- 1. Encoding one effective qubit in multiple defects.** Multiple defects can, in principle, encode one effective qubit. This encoding requires a spectrally isolated two-dimensional manifold: a pair of quantum states separated in energy from all other accessible states. Control and readout operations must act predictably within this encoded subspace. For the specific three-NV triangle, where NV denotes a nitrogen-vacancy center in diamond, the ideal driven model produces a doublet, or pair of relevant states. Robustness to fabrication variation and disorder has not been established. The concept is therefore plausible for this specific platform but has not been demonstrated as a property of an array.
- 2. Emulation of anyons with effective defect qubits.** Multiple effective defect qubits can emulate anyons, which are quasiparticle excitations characterized by nontrivial exchange and fusion rules. Gates or driven interactions can reproduce string operators, fusion measurements, and braiding statistics. A string operator is an extended operator associated with creating, moving, or detecting quasiparticle excitations. A fusion measurement determines the combined topological charge of anyons. Braiding statistics describe the transformation produced by exchanging anyons. Such a system is an emulator, and its errors remain those of the underlying defect controls.
- 3. Digital simulation of Fibonacci anyons.** Defect systems can, in principle, digitally simulate Fibonacci anyons if they support a sufficiently universal gate set together with initialization and readout. Fibonacci anyons are non-Abelian anyons whose fusion rules support universal topological quantum computation. Superconducting processors have already demonstrated digital Fibonacci-model braiding [R138]; [R165], so no known principle excludes implementation on a defect processor. The unresolved issue is whether defect arrays can achieve competitive scale and fidelity, rather than whether the algorithm exists.
- 4. Native Fibonacci topological order in a defect Hamiltonian.** A static or driven defect Hamiltonian could, within known physics, possess Fibonacci topological order, but the cited defect interactions do not derive such an order. A Hamiltonian is the operator that specifies a system's energy and dynamics. A static realization would require the appropriate low-energy projectors, which energetically

select the target local subspaces, together with a stable excitation gap. A driven realization would additionally require Floquet stability, meaning stability under periodic driving, and controlled energy absorption or heating. This possibility remains speculative.

5. **Distinction between simulated and emergent anyons.** Anyon simulation programs physical qubits to reproduce selected anyonic observables. Emergent anyons are instead the system's intrinsic many-body low-energy excitations, with fusion and braiding structure that remains robust throughout a phase of matter. Exact circuit output can reproduce selected anyonic behavior without providing autonomous protection from the hardware Hamiltonian.
6. **Protection obtained from defect clustering.** Clustering defects can provide additional protection in some cases. A symmetric encoding can reject a common-mode field, which acts similarly on all constituents, or reduce a selected transition matrix element. The same encoding can increase sensitivity to spatial gradients, introduce leakage from the encoded subspace, and create additional fabrication failure channels. Protection must therefore be evaluated by projecting the relevant noise operators into the encoded subspace rather than inferred from the number of defects.
7. **Assessment of diamond as a substrate.** Diamond is not categorically the best substrate. It currently provides the strongest combined evidence for optically addressable individual color-center spins, where a color center is a point defect with addressable electronic or spin states. However, dense arrays of identical, interacting centers remain difficult to place and activate. Platform ranking depends on the limiting metric, such as coherence, deterministic fabrication, coupling strength, optical networking, or passive interaction strength.
8. **Assessment of corundum and sapphire.** Corundum, including sapphire and chromium-doped ruby, may be competitive as a low-loss host with mature ensemble spin spectroscopy. An ensemble measurement probes many nominally similar spins collectively. The evidence is weaker for the specific platform considered here: dense, individually controlled defect clusters. The 2025 ruby result demonstrates Cr^{3+} ensemble–cavity physics, not deterministic single-defect plaquettes [Ro89]. Sapphire should therefore remain a comparison platform rather than being treated as equivalent by material analogy.
9. **Primary interaction bottleneck.** The principal interaction bottleneck is the realization of a strong, controllable, correctly signed many-body encoded interaction that exceeds ordinary one- and two-body terms. A many-body interaction acts jointly on more than two effective degrees of freedom. Dipolar coupling is the

magnetic interaction between spin dipoles and is present in the proposed system. The unresolved task is to transform that coupling into the required projector without reducing the target interaction below the noise scale.

10. Primary fabrication bottleneck. The principal fabrication bottleneck is repeated three-dimensional placement and activation of the correct defect species, orientation, charge state, and local environment at the relevant interaction length scale, with high coherent yield. Coherent yield is the fraction of fabricated defects that retain the required quantum coherence and control properties. Deterministic ion arrival alone does not satisfy these requirements [R257]; [R258].

11. Required hierarchy of energy and frequency scales. The design requires the hierarchy

$$\frac{\Delta_C}{h} \gg u, \Omega, \sigma, \Gamma_{\text{decoh}}, \quad \frac{\Delta_{\text{topo}}}{h} \gg \frac{k_B T}{h}, \Gamma_{\text{decoh}}, \sigma_{\text{eff}},$$

where all compared quantities have units of frequency. Here h is Planck's constant, u is the intercluster coupling, Ω is a drive scale, σ is disorder, and Γ_{decoh} is a decoherence rate. The quantity Δ_C/h is the cluster-gap frequency, where Δ_C is the energy separating the encoded cluster manifold from other cluster states. The quantity Δ_{topo}/h is the many-body-gap frequency, where Δ_{topo} is the energy gap protecting the intended many-body phase. In the second inequality, k_B is Boltzmann's constant, T is temperature, and σ_{eff} is the effective disorder within the low-energy description.

The first inequality states that the cluster gap must exceed couplings, driving, disorder, and decoherence sufficiently to suppress leakage from the encoded manifold. The second states that the many-body gap must exceed thermal, decoherence, and effective-disorder scales. The illustrative NV geometry produces sub-megahertz cluster scales and may generate only hertz-scale high-order terms. These values indicate the need for quantitative calculation; they are not device requirements.

12. Operating-temperature constraint. No reliable operating temperature can be specified before the relevant gap is measured. A passive realization requires $k_B T$ to remain well below the energy Δ_{topo} . A tenfold Boltzmann margin gives

$$T < \Delta_{\text{topo}}/(10k_B).$$

Thus, $\Delta_{\text{topo}}/h = 1 \text{ MHz}$ implies $T < 4.8 \mu\text{K}$, while $\Delta_{\text{topo}}/h = 1 \text{ GHz}$ implies $T < 4.8 \text{ mK}$. These are design inequalities rather than forecasts. A driven quasienergy gap, which is a gap in the effective spectrum of a periodically driven system, does not automatically satisfy these equilibrium thermal conditions.

13. **Number of defects per effective degree of freedom.** The scoped triangle model uses three physical defects for each effective degree of freedom. Additional defects may be required to obtain more elaborate symmetry, suppress leakage, or introduce gadget mediators. A gadget mediator is an auxiliary degree of freedom used to generate an effective interaction perturbatively. No current evidence identifies an optimum number, so a universal estimate of “5–20” would not be supported by calculation.
14. **Smallest falsification experiment.** Spectroscopy of one fabricated three-NV triangle can falsify the assumed cluster doublet by testing its predicted energy-level structure. Falsifying the proposed many-body mechanism requires four such clusters, corresponding to twelve defects, because this is the smallest device from which a weight-four coefficient and its competing terms can be extracted. A weight-four term acts jointly on four encoded degrees of freedom. The four-cluster numerical experiment should precede fabrication and can already falsify the proposed Hamiltonian mechanism.
15. **Evidence required for a promising result.** The research direction would become genuinely promising if a reproducible cluster doublet were followed by a measured or microscopically validated many-body coefficient that dominates noncommuting residual terms, exceeds disorder and decoherence, remains stable under realistic position distributions, and produces the expected gap and nonlocal diagnostics on a larger array. Noncommuting residual terms are unwanted Hamiltonian terms that cannot be simultaneously diagonalized with the target interaction and can therefore disrupt its phase. A geometrically plausible lattice or a digitally programmed braid does not meet these criteria.

Confidence Assessment

Proposition	Confidence	Explanation
Defect clusters can form encoded two-dimensional subspaces	Medium	Few-spin encoding is established theory, and the ideal NV triangle is solvable. However, the specified symmetric cluster array has not been demonstrated.

Proposition	Confidence	Explanation
Dense interacting defect arrays are manufacturable	Low	Placement, conversion, charge, orientation, and coherent yield have not been combined at the required repeated nanometre geometry.
Effective many-body interactions can be engineered	Low	Gadget theory permits such interactions, but in this platform the desired term is suppressed by perturbative energy denominators.
Toric-code-like phases are plausible	Medium	These phases require simpler Abelian stabilizer data than Fibonacci order. A passive defect realization has not been demonstrated.
Non-Abelian phases are plausible	Low	No derived defect Hamiltonian, gap, or phase diagram supports such a phase here.
Doubled-Fibonacci phase is physically realistic	Low	The ideal target is established, but no supported microscopic defect derivation produces its branching and recoupling projectors.
Passive topological protection improves coherence	Low	The conditional theory is valid, but this architecture has neither a demonstrated passive topological gap nor evidence that thermal and leakage errors remain below that gap.
Diamond is superior to competing platforms	Low	Diamond leads in several single-defect capabilities but not in every system-level metric.

Detailed treatment: confidence assessment

Confidence assessment

Proposition	Confidence	Explanation
Defect clusters can form encoded two-dimensional subspaces	Medium	Few-spin encoding is established theory, and the ideal NV triangle is solvable. However, the specified symmetric cluster array has not been demonstrated.
Dense interacting defect arrays are manufacturable	Low	Placement, conversion, charge, orientation, and coherent yield have not been combined at the required repeated nanometre geometry [R257]; [R258].
Effective many-body interactions can be engineered	Low	Gadget theory permits such interactions [R175], but in this platform the desired term is suppressed by perturbative energy denominators and competes with larger native terms.

Proposition	Confidence	Explanation
Toric-code-like phases are plausible	Medium	These phases require simpler Abelian stabilizer data than Fibonacci order and can be emulated digitally or through Floquet driving. A passive defect realization has not been demonstrated.
Non-Abelian phases are plausible	Low	No derived defect Hamiltonian, gap, or phase diagram supports such a phase here. Digital non-Abelian behavior does not increase the assessment of the underlying material system.
Doubled-Fibonacci phase is physically realistic	Low	The ideal target is established [R018], but no supported microscopic defect derivation produces its branching and recoupling projectors.
Passive topological protection improves coherence	Low	The conditional theory is valid, but this architecture has neither a demonstrated passive topological gap nor evidence that thermal and leakage errors remain below that gap.
Diamond is superior to competing platforms	Low	Diamond leads in several single-defect capabilities but not in every system-level metric. Fabrication and coupling requirements may favor another host or a hybrid platform.

The following five classifications specify the evidential status of each claim:

- **established** — measured or derived in the relevant system.
- **extension** — a direct subsequent device implementation using existing technology.
- **proposal** — a credible but difficult research program with an initial measurement capable of falsifying the proposed mechanism.
- **speculation** — physically coherent, but dependent on indispensable components that have not been demonstrated.
- **incompatible** — in conflict with known physics.

For the complete proposal—dense defect clusters producing a genuinely topologically ordered, potentially doubled-Fibonacci material and a passively improved logical qubit—the only justified classification is

highly speculative but physically coherent.

Each ingredient is individually allowed by known physics: local Hamiltonians with doubled-Fibonacci order exist, perturbative gadgets generate effective interactions, and defect spins are controllable. The full architecture still falls short of a credible-but-difficult program, because the central interaction and a manufacturable geometry have yet to appear together in measurement or in a validated microscopic calculation. The four-cluster study earns the narrower proposal label: even a negative result teaches whether the mechanism works.

The Final Verdict

For the complete proposal—dense defect clusters producing a genuinely topologically ordered, potentially doubled-Fibonacci material and a passively improved logical qubit—the only justified classification is:

highly speculative but physically coherent

Each ingredient is individually allowed by known physics: local Hamiltonians with doubled-Fibonacci order exist, perturbative gadgets generate effective interactions, and defect spins are controllable. The full architecture still falls short of a credible-but-difficult program, because the central interaction and a manufacturable geometry have yet to appear together in measurement or in a validated microscopic calculation.

The four-cluster study earns the narrower proposal label: even a negative result teaches whether the mechanism works.

Closest alternative approach

The nearest technically justified alternative is a digitally controlled defect-spin network with active error correction. Such an architecture would use individual defects or small clusters only when they provide a measured improvement in control or noise. It would prepare stabilizer states or small string-net states by applying gates, measure error syndromes, and implement anyon operations as quantum simulations. A stabilizer state is defined as the simultaneous eigenstate of a commuting set of operators. A syndrome is a set of measurements used to identify errors without directly measuring the encoded logical information. A string-net state is a many-body state described by configurations of labeled extended degrees of freedom that satisfy specified local branching and recoupling rules. Optical links or a cavity could connect spatially separated modules and thereby reduce the requirement for a dense, nearly perfect crystal lattice.

This alternative gives up autonomous Fibonacci order and passive self-protection. It keeps what defects already deliver: long-lived local memories, distributed entanglement, active logical encoding, and a working platform for anyon protocols.

If dense NV fabrication stalls, the same measured criteria — coherence, placement yield, coupling strength, and control fidelity — decide among SiC defects, donors, rare-earth ions, trapped ions, neutral atoms, and superconducting circuits.

Existing digital Fibonacci experiments demonstrate that scientifically significant anyon studies do not require a native Fibonacci material [R138]; [R165].

Digital simulation, driven emulation, and equilibrium emergence must be assessed as distinct categories. Combining evidence from these categories would produce an unjustifiably favorable evaluation.

Review questions and answers

Review questions and answers

- **Basis for selecting the four-cluster calculation as the initial study.**

Four encoded spins constitute the smallest system that can reveal a weight-four plaquette coefficient together with every lower-weight competitor. A plaquette coefficient is the strength of an interaction associated with a local face of the effective lattice.

- **Derivation of the ideal-triangle gap and estimate of the fourth-order interaction.**

For the ideal triangle in (41.1), the isolated-cluster splitting is $3c/2$, and therefore

$$\Delta_C/h = 3c/2.$$

With $c \approx 0.416$ MHz, one obtains $\Delta_C/h \approx 0.624$ MHz. At the proposed 5 nm / 10 nm geometry, $u \lesssim 52$ kHz. The fourth-order coefficient is consequently estimated as

$$K_4 \sim u^4/(\Delta_C/h)^3 \sim 30 \text{ Hz},$$

or $hK_4/k_B \approx 1.4$ nK. This is a calculation indicating the expected scale, not a measured coupling.

- **Limitation of treating $K_4 \neq 0$ as evidence for a phase.**

A single nonzero local term tests only the proposed interaction mechanism. Establishing a phase additionally requires dominance over competing terms, robustness, a finite gap, and topological diagnostics. A nonzero coefficient alone

does not establish topological order.

- **Reason for not beginning with doubled Fibonacci.**

The vertex and recoupling projectors of the doubled-Fibonacci model require more microscopic structure than has been established for the proposed Hamiltonian. Testing the simpler stabilizer interaction provides an earlier opportunity to falsify the proposed mechanism.

- **Failure caused by choosing a different encoded basis for every disorder sample.**

Sample-dependent basis selection can produce a large apparent target coefficient that cannot be realized by any fixed control protocol. The cluster basis must therefore be fixed using isolated-cluster spectroscopy or restricted to calibrated single-cluster controls. Results in the unrotated basis must also be reported.

- **Technically justified alternative.**

Defects can be used as controllable physical qubits or cluster-encoded qubits in an actively corrected digital architecture. Anyon experiments in such a system should be classified as simulations unless the hardware Hamiltonian itself produces the phase.

Sources

Sources

- [Ro74] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* 528, 1–45 (2013). DOI: [10.1016/j.physrep.2013.02.001](https://doi.org/10.1016/j.physrep.2013.02.001).
- [Ro80] F. Dolde, I. Jakobi, B. Naydenov, et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* 9, 139–143 (2013). DOI: [10.1038/nphys2545](https://doi.org/10.1038/nphys2545).
- [R257] D. Scarabelli, M. Trusheim, O. Gaathon, D. Englund, and S. J. Wind, “Nanoscale engineering of closely-spaced electronic spins in diamond,” *Nano Letters* 16, 4982–4990 (2016). DOI: [10.1021/acs.nanolett.6b01692](https://doi.org/10.1021/acs.nanolett.6b01692).

- [R258] K. Groot-Berning, T. Kornher, G. Jacob, et al., “Fabrication of $^{15}\text{NV}^-$ centers in diamond using a deterministic single ion implanter,” *New Journal of Physics* 23, 063067 (2021). DOI: [10.1088/1367-2630/ac0753](https://doi.org/10.1088/1367-2630/ac0753).
 - [R175] S. Bravyi, D. P. DiVincenzo, D. Loss, and B. M. Terhal, “Quantum simulation of many-body Hamiltonians using perturbation theory with bounded-strength interactions,” *Physical Review Letters* 101, 070503 (2008). DOI: [10.1103/PhysRevLett.101.070503](https://doi.org/10.1103/PhysRevLett.101.070503). arXiv: [0803.2686](https://arxiv.org/abs/0803.2686).
 - [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110). arXiv: [cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617).
 - [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* 303, 2–30 (2003). DOI: [10.1016/S0003-4916\(02\)00018-0](https://doi.org/10.1016/S0003-4916(02)00018-0). arXiv: [quant-ph/9707021](https://arxiv.org/abs/quant-ph/9707021).
 - [R138] S. Xu et al., “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* 20, 1469–1475 (2024). DOI: [10.1038/s41567-024-02529-6](https://doi.org/10.1038/s41567-024-02529-6).
 - [R165] Z. K. Mineev et al., “Realizing string-net condensation: Fibonacci anyon braiding for universal gates and sampling chromatic polynomials,” *Nature Communications* 16, 6225 (2025). DOI: [10.1038/s41467-025-61493-8](https://doi.org/10.1038/s41467-025-61493-8).
 - [R089] Z. Velluire-Pellat, E. Maréchal, C. Feuillet-Palma, et al., “Spin-photon interaction between a ruby crystal and a high-critical-temperature superconducting microwave cavity,” *Communications Physics* 8, 236 (2025). DOI: [10.1038/s42005-025-02159-1](https://doi.org/10.1038/s42005-025-02159-1).
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Mathematical appendices

The main chapters cite a compact mathematical reference: complex vector spaces and linear operators (Appendix A) and tensor products for composite systems (Appendix B). Read these appendices alongside the chapters that invoke them; each construction is motivated by the cluster and lattice calculations that need it.

Appendix A — Complex vector spaces and linear operators

This appendix supplies the language needed to diagonalize a cluster and describe its retained states. The progression is from vectors, to inner products, to operators, to spectra. Each construction answers a physical modeling question rather than introducing notation for its own sake.

Background, scope, and supporting argument

Consider two arrows drawn on paper. Their sum is defined geometrically by placing the tail of the second arrow at the tip of the first and drawing the arrow corresponding to the resulting path. Scalar multiplication changes an arrow by stretching it by 2, shrinking it by $1/2$, or reversing its direction by multiplying it by -1 .

These operations are already part of elementary geometry. They can be extended by allowing the scalar multiplier to be a complex number. After identifying the plane with the complex numbers, multiplication by i rotates an arrow through one quarter turn. Multiplication by $3 + 4i$ stretches it by 5, the modulus of $3 + 4i$, and also rotates it. Vector addition remains the same tip-to-tail operation.

This construction carries no physical assumption yet: it enlarges the allowed scalars from real to complex, and the physical meaning of the new phases arrives with the Born rule and interference in later chapters.

A.1 Introduction: From Arrows to State Vectors

Consider two arrows drawn on paper. Their sum is defined geometrically by placing the tail of the second arrow at the tip of the first and drawing the arrow corresponding to the resulting path. Scalar multiplication changes an arrow by stretching it by 2, shrinking it by $1/2$, or reversing its direction by multiplying it by -1 .

These operations can be extended by allowing the scalar multiplier to be a complex number. After identifying the plane with the complex numbers, multiplication by i rotates an arrow through one quarter turn. Multiplication by $3 + 4i$ stretches it by 5 (the modulus of $3 + 4i$) and also rotates it.

The plane shows what complex scaling means geometrically: a stretch combined with a rotation. A general complex vector space keeps the same two operations — addition and scalar multiplication — while dropping the restriction to arrows in a plane. Quantum mechanics then assigns physical meaning to the components: each component is a probability amplitude, and phase differences between components determine interference.

A.2 Vector Addition and Complex Scalar Multiplication

The same two operations govern objects far beyond arrows in physical space: columns of amplitudes, functions, and abstract states all qualify as vectors when their addition and scalar multiplication obey the vector-space laws.

A vector space is a set of objects equipped with vector addition and multiplication by complex scalars. These operations include a zero vector, an additive opposite for every vector, and the identity rule that multiplication by 1 leaves every vector unchanged. They also satisfy the required associative, commutative, and distributive laws.

Let $\mathbb{C}$ denote the complex numbers, and let $V = \mathbb{C}^2$ denote the vector space of columns of height two. Addition and scalar multiplication act entrywise.

For vectors $u, v, w \in V$ and scalars $a, b \in \mathbb{C}$, vector addition is associative and commutative, and scalar multiplication and vector addition distribute over one another.

For example, with $u = (1, i)^T$ and $v = (2, -1)^T$,

$$u + v = (3, i - 1)^T, \quad iu = (i, -1)^T.$$

The second calculation uses $i^2 = -1$. These are vector operations; they do not automatically preserve the normalization required of a physical pure-state representative.

Detailed treatment: vector addition and complex scalar multiplication

Vector addition and complex scalar multiplication

A vector space is a set of objects equipped with vector addition and multiplication by complex scalars. These operations include a zero vector, an additive opposite for every vector, and the identity rule that multiplication by 1 leaves every vector unchanged. They also satisfy the required associative, commutative, and distributive laws. This text takes these properties as the definition of a complex vector space.

The vectors may be geometric arrows, columns containing two complex numbers, or the internal states of a silver atom. The vector-space structure consists entirely of the two operations and their algebraic properties.

Let $\mathbb{C}$ denote the complex numbers, and let $V = \mathbb{C}^2$ denote the vector space of columns of height two. Addition and scalar multiplication act entrywise.

For vectors $u, v, w \in V$ and scalars $a, b \in \mathbb{C}$, vector addition is associative and commutative, and scalar multiplication and vector addition distribute over one another. At this stage, the vector-space axioms do not define lengths or angles.

A.3 Abstract Vectors and Coordinate Representations

A state and its coordinate column play different roles: the column lists components against a chosen basis, while the abstract vector is the object those components describe. Cluster calculations compare encoded operators from different triangles, so keeping the object separate from its per-basis description prevents mixing a physical change with a change of reference frame.

The ket $|\psi\rangle$ denotes an abstract vector. A ket is Dirac notation for a vector in a complex vector space. Its coordinate column depends on the selected basis.

The standard basis fixes the reference columns:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

A normalized state — one whose probabilities sum to one — then reads:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

The diagonal basis, used throughout the cluster chapters for states with definite relative phase, is:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Substituting the diagonal-basis definitions and collecting terms gives the same state in new coordinates:

$$|\psi\rangle = \frac{\alpha + \beta}{\sqrt{2}}|+\rangle + \frac{\alpha - \beta}{\sqrt{2}}|-\rangle$$

If U is the matrix whose columns contain the new orthonormal basis vectors expressed in the old orthonormal basis:

$$[\psi]_{\text{new}} = U^\dagger [\psi]_{\text{old}}$$

Why the adjoint? The j th new coordinate is the overlap $\langle e_j^{\text{new}} | \psi \rangle$. The rows of $U^\dagger$ are precisely those bras. Operators transform consistently as $A_{\text{new}} = U^\dagger A_{\text{old}} U$.

For $\psi = |0\rangle$, the new coordinates in the $|+\rangle, |-\rangle$ basis are $(1/\sqrt{2}, 1/\sqrt{2})^T$. The physical state is unchanged; only its coordinate column is new.

A common phase $e^{i\phi}$ multiplying every component leaves all measurement probabilities unchanged, so it labels the same isolated pure state. A relative phase between components changes interference outcomes: $|+\rangle$ and $|-\rangle$ give identical probabilities in the computational basis yet are orthogonal states, distinguished by a diagonal-basis measurement.

Detailed treatment: abstract vectors and coordinate representations

Abstract vectors and coordinate representations

The ket $|\psi\rangle$ denotes an abstract vector. A ket is Dirac notation for a vector in a complex vector space. Its coordinate column depends on the selected basis, which is a reusable set of reference directions.

In the standard basis

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

a normalized state has the form

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Here $\alpha, \beta \in \mathbb{C}$, and $|\alpha|$ denotes the complex modulus of α . Normalization means that the state has unit length with respect to the inner product introduced below. Define a second basis by

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Direct substitution, without any additional physical assumption, gives

$$|\psi\rangle = \frac{\alpha + \beta}{\sqrt{2}}|+\rangle + \frac{\alpha - \beta}{\sqrt{2}}|-\rangle.$$

The abstract state is unchanged; only its coordinates differ. If U is the matrix whose columns contain the new orthonormal basis vectors expressed in the old orthonormal basis, then the coordinate columns satisfy

$$[\psi]_{\text{new}} = U^\dagger [\psi]_{\text{old}}.$$

The symbol $\dagger$ denotes the conjugate transpose in orthonormal coordinates, as defined below. For this basis transformation, U preserves lengths, so normalization is unchanged.

A finite list $v_1, \dots, v_k$ is linearly independent if the equation

$$a_1 v_1 + \dots + a_k v_k = 0$$

implies $a_1 = \dots = a_k = 0$. The span of the list is the set of all weighted sums formed from its vectors.

A basis is a linearly independent list whose span equals all of V . Every basis of a finite-dimensional vector space contains the same number of vectors. This number is the dimension, denoted $\dim V$.

A subspace $W \subseteq V$ is a subset that contains 0 and is closed under vector addition and complex scalar multiplication. The logical doublet of a defect cluster is a two-dimensional subspace of a much larger space. The designation as a subspace does not imply that the energy operator or noise processes preserve that subspace.

The direct-sum relation $V = W \oplus W'$ means that every $v \in V$ has exactly one decomposition $v = w + w'$, where $w \in W$ and $w' \in W'$. Projection methods use such a decomposition to separate a desired low-energy subspace from the remaining degrees of freedom.

Linear independence is a property of an entire list of vectors, not of an individual vector. If a spanning list is not linearly independent, the same vector can have multiple coordinate columns relative to that list.

A.4 Inner Products, Norms, and Orthogonality

The inner product turns a vector space into a space with lengths and overlaps. In an orthonormal coordinate basis, $\langle u|v\rangle = \sum_j u_j^* v_j$. The complex conjugation makes $\langle v|v\rangle$ real and nonnegative.

An inner product is a map that assigns a complex number $\langle u|v\rangle$ to each ordered pair of vectors. This text uses the physics convention: the inner product is conjugate-linear in its first argument and linear in its second argument.

$$\langle au + bw|v\rangle = a^* \langle u|v\rangle + b^* \langle w|v\rangle$$

$$\langle u|av + bw\rangle = a \langle u|v\rangle + b \langle u|w\rangle$$

The norm of a vector is:

$$\|v\| = \sqrt{\langle v|v\rangle}$$

Two vectors are orthogonal if their inner product vanishes. A list $\{|e_j\rangle\}$ is orthonormal if:

$$\langle e_j|e_k\rangle = \delta_{jk}$$

For any orthonormal basis $\{|e_j\rangle\}_{j=1}^d$:

$$I = \sum_{j=1}^d |e_j\rangle\langle e_j|, \quad |v\rangle = \sum_{j=1}^d |e_j\rangle\langle e_j|v\rangle$$

Each term $|e_j\rangle\langle e_j|v\rangle$ extracts one component of v and restores it along its basis direction. Summing over a complete basis reconstructs the vector. Summing only over selected orthonormal states gives a projector onto their span, which is exactly the retained-cluster construction used in Chapter 37.

Detailed treatment: inner products, norms, and orthogonality

Inner products, norms, and orthogonality

Addition and scalar multiplication fix the algebra but assign no lengths or angles. An inner product is a map that assigns a complex number $\langle u|v\rangle$ to each ordered pair of vectors. This text uses the physics convention: the inner product is conjugate-linear in its first argument and linear in its second argument. Thus,

$$\langle au + bw|v\rangle = a^*\langle u|v\rangle + b^*\langle w|v\rangle,$$

and

$$\langle u|av + bw\rangle = a\langle u|v\rangle + b\langle u|w\rangle.$$

The inner product also satisfies conjugate symmetry, $\langle u|v\rangle = \langle v|u\rangle^*$, and positive definiteness: $\langle v|v\rangle \geq 0$, with equality only for $v = 0$. The star denotes complex conjugation.

From the inner product come length, angle, orthogonality, and — once states are normalized — the probabilities of the Born rule.

The length or norm of a vector is

$$\|v\| = \sqrt{\langle v|v\rangle}.$$

Two vectors are orthogonal if their inner product vanishes. A list $\{|e_j\rangle\}$ is orthonormal if

$$\langle e_j|e_k\rangle = \delta_{jk},$$

where the Kronecker delta δ_{jk} equals one for $j = k$ and zero otherwise.

A ket $|v\rangle$ is a vector. Its associated bra $\langle v|$ is the linear functional that maps $|w\rangle$ to $\langle v|w\rangle$. A linear functional is a linear map from the vector space to its scalar field. In an orthonormal coordinate basis, the bra corresponding to a ket column is obtained by transposing the column and complex-conjugating each entry.

For any orthonormal basis $\{|e_j\rangle\}_{j=1}^d$,

$$I = \sum_{j=1}^d |e_j\rangle\langle e_j|, \quad |v\rangle = \sum_{j=1}^d |e_j\rangle\langle e_j|v\rangle.$$

The first equation is the resolution of the identity: the sum of the one-dimensional basis projectors equals the identity operator I . The second equation shows that the coordinates of a vector in an orthonormal basis are the inner products $\langle e_j|v\rangle$.

Consequently, after an orthonormal basis has been selected, the abstract vector uniquely determines its coordinate column, and the coordinate column uniquely determines the abstract vector.

The Cauchy–Schwarz inequality is

$$|\langle u|v\rangle| \leq \|u\| \|v\|.$$

It implies that the overlap of two normalized states has modulus at most one. One proof evaluates the nonnegative squared length of $|u\rangle - z|v\rangle$ and selects the complex number z that minimizes it. This inequality provides the algebraic reason that Born probabilities lie between zero and one.

A complex vector space with a positive-definite inner product is called a Hilbert space when it is complete in the induced norm — when every Cauchy sequence converges to a vector in the space. Finite-dimensional spaces are automatically complete, so matrix calculations use the inner product directly with no extra completeness check.

If positive definiteness is omitted, the pairing may still be a Hermitian form. It then does not define a length that vanishes only at the zero vector, and the final step in the Cauchy–Schwarz argument fails.

A.5 Linear Maps and Matrix Representations

Recording where each basis vector lands fixes a linear operator everywhere: linearity extends the basis images to every superposition. That is why each matrix column reads as the image of one basis vector.

A map $A : V \rightarrow W$ is linear if:

$$A(au + bv) = aA(u) + bA(v)$$

After bases have been chosen for V and W , the linear map A is represented by a matrix. In orthonormal bases, its entry in row j and column k is:

$$A_{jk} = \langle e_j | A | e_k \rangle$$

For a linear map with finite-dimensional domain V :

$$\dim V = \dim(\ker A) + \dim(\operatorname{im} A)$$

The kernel contains inputs sent to zero, while the image contains reachable outputs. A rank-two projector on an eight-dimensional cluster space preserves a two-dimensional image and annihilates a six-dimensional kernel. It is not invertible on the full space, even though it acts as the identity on the retained subspace.

Detailed treatment: linear maps and matrix representations

Linear maps and matrix representations

A map $A : V \rightarrow W$ is linear if

$$A(au + bv) = aA(u) + bA(v).$$

After bases have been chosen for V and W , the linear map A is represented by a matrix. In orthonormal bases, its entry in row j and column k is

$$A_{jk} = \langle e_j | A | e_k \rangle.$$

Matrix multiplication represents composition of linear maps:

$$(AB)|v\rangle = A(B|v\rangle),$$

so the rightmost operator acts first.

The kernel $\ker A$ is the set of vectors mapped to zero. The image $\text{im } A$ is the set of outputs $A|v\rangle$. For a linear map with finite-dimensional domain V , the rank–nullity theorem states

$$\dim V = \dim(\ker A) + \dim(\text{im } A).$$

Thus, the input directions mapped to zero contribute to the kernel, while the remaining independent directions span the image.

A.6 Adjoints and Operator Classes

In orthonormal coordinates the adjoint is the conjugate transpose. Observables, time evolution, and projection each take a one-line form through it: Hermiticity for the first, unitarity for the second, and idempotence for the third.

For a linear map A , its adjoint $A^\dagger$ is the unique linear map defined by:

$$\langle u|Av\rangle = \langle A^\dagger u|v\rangle$$

- A is Hermitian if $A = A^\dagger$.
- U is unitary if $U^\dagger U = UU^\dagger = I$.
- P is an orthogonal projector if $P = P^\dagger = P^2$.
- A is positive semidefinite if $\langle v|A|v\rangle \geq 0$ for every $|v\rangle$.

$$\text{Tr } A = \sum_j \langle e_j|A|e_j\rangle$$

For example, the outer product $|u\rangle\langle v|$ is an operator, while $\langle v|u\rangle$ is a scalar. In a finite-dimensional space, $\text{Tr}(|u\rangle\langle v|) = \langle v|u\rangle$. A normalized pure-state projector therefore has trace one.

For finite-dimensional products:

$$\text{Tr}(AB) = \text{Tr}(BA)$$

Cyclicity moves a factor from one end of a product to the other — and that is the full extent of the freedom. Factors keep their cyclic order, so $AB \neq BA$ in general, while the trace values agree.

Detailed treatment: adjoints and operator classes

Adjoint and operator classes

For a linear map A , its adjoint $A^\dagger$ is the unique linear map defined by

$$\langle u | Av \rangle = \langle A^\dagger u | v \rangle$$

for all vectors u, v for which the expressions are defined. In orthonormal bases, the matrix representing $A^\dagger$ is the conjugate transpose of the matrix representing A . Comparisons between A and $A^\dagger$ determine important properties related to lengths, measurement averages, and time evolution.

Several operator classes occur repeatedly.

- A is Hermitian if $A = A^\dagger$. Observables and energy operators are represented by Hermitian operators.
- U is unitary if $U^\dagger U = UU^\dagger = I$. Unitary operators preserve inner products.
- P is an orthogonal projector if $P = P^\dagger = P^2$. Its image is the retained subspace.
- A is positive semidefinite, written $A \geq 0$, if $\langle v | A | v \rangle \geq 0$ for every $|v\rangle$.
- A is normal if $AA^\dagger = A^\dagger A$. Hermitian and unitary operators are normal.

The trace of an operator is the basis-independent number

$$\text{Tr } A = \sum_j \langle e_j | A | e_j \rangle.$$

For finite-dimensional products that are defined, the trace satisfies cyclicity:

$$\text{Tr}(AB) = \text{Tr}(BA).$$

Cyclicity permits cyclic permutations but not arbitrary reorderings. In general,

$$\text{Tr}(ABC) = \text{Tr}(BCA),$$

but not

$$\text{Tr}(ACB).$$

The adjoint is defined through the inner product. A bare matrix transpose remains available, but it tracks the chosen basis rather than a geometric operation.

A.7 Eigenvectors and Spectral Decompositions

Eigenvectors identify directions on which an operator acts by a scalar. For a Hamiltonian they are stationary energy states, so its spectral decomposition is the natural starting point for selecting a low-energy band.

A nonzero vector $|v\rangle$ is an eigenvector of A with eigenvalue $\lambda \in \mathbb{C}$ if:

$$A|v\rangle = \lambda|v\rangle$$

If $A = A^\dagger$, $A|v\rangle = \lambda|v\rangle$, and $\|v\| = 1$:

$$\lambda = \langle v|A|v\rangle = \langle Av|v\rangle = \lambda^*$$

A Hermitian operator has the spectral decomposition:

$$A = \sum_r \lambda_r P_r$$

where λ_r are its distinct real eigenvalues and P_r are mutually orthogonal projectors.

A function of A is defined spectrally:

$$f(A) = \sum_r f(\lambda_r) P_r$$

For a Hamiltonian H , unitary evolution is:

$$U(t) = e^{-iHt/\hbar} = \sum_r e^{-i\lambda_r t/\hbar} P_r$$

Each energy component thus accumulates its own phase, and for time-independent Hermitian H every phase factor has unit magnitude, giving $U^\dagger U = I$. Leakage and adiabatic estimates combine two ingredients: the gap, which compares eigenvalues, and the perturbing observable's matrix elements, which set whether the perturbation connects the corresponding states.

Detailed treatment: eigenvectors and spectral decompositions

Eigenvectors and spectral decompositions

A nonzero vector $|v\rangle$ is an eigenvector of A with eigenvalue $\lambda \in \mathbb{C}$ if

$$A|v\rangle = \lambda|v\rangle.$$

The action of A on an eigenvector is therefore multiplication by a scalar, with no component generated in another direction. An eigenvalue is degenerate if its eigenspace, the set of eigenvectors with that eigenvalue together with the zero vector, has dimension greater than one.

Suppose $A = A^\dagger$, $A|v\rangle = \lambda|v\rangle$, and $\|v\| = 1$. Then

$$\lambda = \langle v|A|v\rangle = \langle Av|v\rangle = \lambda^*,$$

so λ is real. Therefore, every eigenvalue of a Hermitian operator is real.

The finite-dimensional spectral theorem states that a normal operator has an orthonormal eigenbasis [Roo1]; [Roo2]. In particular, a Hermitian operator has the spectral decomposition

$$A = \sum_r \lambda_r P_r,$$

where λ_r are its distinct real eigenvalues and P_r are mutually orthogonal projectors satisfying $P_r P_s = \delta_{rs} P_r$ and $\sum_r P_r = I$. If an eigenvalue is degenerate, its projector P_r projects onto the corresponding multidimensional eigenspace.

A function of A is defined spectrally by

$$f(A) = \sum_r f(\lambda_r) P_r.$$

For a Hamiltonian H , which is an energy operator measured in joules (J), unitary evolution through a time t , measured in seconds (s), is

$$U(t) = e^{-iHt/\hbar} = \sum_r e^{-i\lambda_r t/\hbar} P_r.$$

The exponent is dimensionless because (J)(s)/(J s) = 1. Here $\hbar$ is the reduced Planck constant.

The operator $U(t)$ is unitary. In the spectral basis, each real eigenvalue of H is mapped to a complex phase of modulus one, and therefore $U(t)^\dagger U(t) = I$.

Over $\mathbb{C}$, the characteristic polynomial of a finite matrix has a root, so a normal operator has at least one eigenvector. This eigenvector can be normalized.

Normality ensures that the orthogonal complement of the eigenvector is invariant under both A and $A^\dagger$. The operator A can then be restricted to this lower-dimensional subspace, and the argument can be repeated.

Induction yields an orthonormal eigenbasis. This outline uses no additional physical assumptions, although a complete proof must establish the invariant-complement step carefully [Roo1]; [Roo2].

If normality is omitted, a finite matrix need not have an orthonormal eigenbasis. It may also be defective, meaning that it has too few linearly independent eigenvectors to span the space. In that case, the stated spectral formula for $f(A)$ is unavailable.

General two-level Hamiltonian

Define the Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Every 2×2 Hermitian Hamiltonian can be written as

$$H = cI + h_x\sigma_x + h_y\sigma_y + h_z\sigma_z = cI + \mathbf{h} \cdot \boldsymbol{\sigma},$$

where $c, h_x, h_y, h_z \in \mathbb{R}$ have units of J, $\mathbf{h} = (h_x, h_y, h_z)$, and $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. Define the magnitude

$$h = \sqrt{h_x^2 + h_y^2 + h_z^2}.$$

For $h \neq 0$, define the unit vector $\mathbf{n} = \mathbf{h}/h$.

The Pauli matrices satisfy the multiplication rule

$$\sigma_j\sigma_k = \delta_{jk}I + i \sum_{\ell} \varepsilon_{jkl}\sigma_{\ell},$$

where ε_{jkl} is the antisymmetric Levi-Civita symbol. Because the product $n_j n_k$ is symmetric under interchange of j and k , its contraction with the antisymmetric term vanishes. Therefore,

$$(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I.$$

An operator whose square equals I has eigenvalues ± 1 . The corresponding projectors are

$$P_{\pm} = \frac{1}{2}(I \pm \mathbf{n} \cdot \boldsymbol{\sigma}),$$

because $P_{\pm}^2 = P_{\pm}$, $P_+P_- = 0$, and $P_+ + P_- = I$. The spectral decomposition of H is consequently

$$H = (c + h)P_+ + (c - h)P_-.$$

The two energies are $E_{\pm} = c \pm h$, and their separation is $2h$. Applying the exponential function spectrally gives

$$\begin{aligned} e^{-iHt/\hbar} &= e^{-i(c+h)t/\hbar}P_+ + e^{-i(c-h)t/\hbar}P_- \\ &= e^{-ict/\hbar} \left[\cos\left(\frac{ht}{\hbar}\right) I - i \sin\left(\frac{ht}{\hbar}\right) \mathbf{n} \cdot \boldsymbol{\sigma} \right]. \end{aligned}$$

This formula covers Zeeman precession, a driven qubit in its rotating frame, and an effective cluster pseudospin: in each case the state rotates about the axis $\mathbf{n}$ at the rate set by the magnitude defined above. At $h = 0$ there is no distinguished axis; $H = cI$, and the evolution contributes only a common phase.

A two-dimensional subspace has the mathematics of a qubit. Whether it functions as a controllable physical qubit, an error-corrected logical qubit, or a topological ground space depends on further dynamical and operational conditions — control access, error rates, gap stability — that linear algebra alone cannot settle.

Distinct transformations and conventions

Some authors represent vectors as columns without ket notation, adopt the mathematics convention in which the inner product is linear in its first argument, or set $\hbar = 1$. The convention must be identified before moving a complex scalar through an inner product. The following operations must also be distinguished:

- A basis change alters the coordinates used to represent a fixed abstract vector.
- A unitary physical evolution changes the state relative to fixed measurement operators.
- A similarity transformation $A' = U^\dagger A U$ represents the same abstract operator in a new orthonormal basis.
- A projected operator PAP generally discards information and is not a basis change.

For infinite-dimensional Hilbert spaces, the spectral theorem is formulated in terms of projection-valued measures, which assign projectors to measurable subsets of the spectrum. Unbounded operators also require explicit domains because the expression AB may not be defined on every vector. The finite-dimensional matrix formulation handles the defect clusters and finite lattice models of this book; continuum treatments with unbounded operators call on the more general methods of functional analysis [R003]; [R005].

Common errors

Common errors

- Omitting complex conjugation is incorrect. Bras are conjugate-transposed kets; otherwise, norms need not be real or positive.
- Treating every matrix as though it were Hermitian is incorrect. A non-normal matrix may lack an orthonormal eigenbasis and may even be defective.
- Degeneracy does not determine a unique basis. The eigenspace is fixed, but any orthonormal basis within it is allowed.
- Replacing A by PAP omits coupling through the complementary subspace. Perturbation theory restores some of those virtual effects.
- Trace cyclicity is not commutativity. Operators remain order-sensitive outside the cyclic trace identity.
- Units must be retained in exponentials. The expression e^{-iHt} implicitly assumes $\hbar = 1$; the dimensionless combination $tH/\hbar$ must be restored before numerical work.

Exercises and answers

Exercises and answers

- Identify which object is basis independent. The ket is abstract and basis independent; its coordinate column depends on the basis.
- Demonstrate that every eigenvalue of a Hermitian operator is real. If $A = A^\dagger$ and $A|v\rangle = \lambda|v\rangle$ with $\|v\| = 1$, then $\lambda = \langle v|A|v\rangle = \langle Av|v\rangle = \lambda^*$.
- Identify the additional structure that turns a complex vector space into a Hilbert space in finite dimensions. The required structure is a positive-definite inner product. Finite-dimensional spaces are automatically complete in the induced norm.
- Demonstrate that $e^{-iHt/\hbar}$ is unitary when $H = H^\dagger$. In the spectral basis, every eigenvalue of H is real, so every eigenvalue of the exponential is a phase of modulus one.

- State the consequence of omitting normality. A finite matrix need not have an orthonormal eigenbasis, and the spectral formula for $f(A)$ need not apply.
- Identify the information lost when A is replaced by PAP . The omitted information consists of matrix elements connecting the retained subspace to its complement and all action wholly within that complement.

Later chapters invoke six results without rederivation, each tied to its use: orthonormal bases resolve the identity (to expand cluster states); Hermitian operators possess real spectral decompositions (to diagonalize clusters); projectors isolate possibly degenerate subspaces (to retain doublets); operator functions act eigenvalue by eigenvalue (to exponentiate Hamiltonians); unitary changes of orthonormal basis preserve inner products (to compare encodings); and a projected two-dimensional operator is a linear combination of $I, \sigma_x, \sigma_y, \sigma_z$ (to parametrize encoded pseudospins). Appendix B builds on this foundation with composite systems, incomplete information, and irreversible channels.

Sources

Sources

- [R001] S. Axler, Linear Algebra Done Right, 3rd ed., Springer, 2015. DOI: [10.1007/978-3-319-11080-6](https://doi.org/10.1007/978-3-319-11080-6).
 - [R002] R. A. Horn and C. R. Johnson, Matrix Analysis, 2nd ed., Cambridge University Press, 2013. DOI: [10.1017/CBO9781139020411](https://doi.org/10.1017/CBO9781139020411).
 - [R003] B. C. Hall, Quantum Theory for Mathematicians, Springer, 2013. DOI: [10.1007/978-1-4614-7116-5](https://doi.org/10.1007/978-1-4614-7116-5).
 - [R004] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, 10th anniversary ed., Cambridge University Press, 2010. DOI: [10.1017/CBO9780511976667](https://doi.org/10.1017/CBO9780511976667).
 - [R005] A. Peres, Quantum Theory: Concepts and Methods, Kluwer Academic, 1995. DOI: [10.1007/0-306-47120-5](https://doi.org/10.1007/0-306-47120-5).
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Appendix B — Tensor products and composite quantum systems

A cluster is a composite system, so its state space must describe joint amplitudes. The tensor product supplies that space. Density matrices and the partial trace then describe what can be learned when only part of the cluster is observed.

Background, scope, and supporting argument

Alice records two numbers, one for each possible outcome of her coin:

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Bob independently records two numbers:

$$\begin{pmatrix} \gamma \\ \delta \end{pmatrix}.$$

A description of all pairs of outcomes requires four components rather than two separate pairs of components:

$$\begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix}.$$

When each subsystem is represented by a definite column vector, this four-component column is the combined representation. Quantum mechanics retains these four components and permits linear combinations of such combined columns. Some resulting joint vectors cannot be factored into one vector for Alice and one for Bob.

This construction assumes no crystal and no environmental bath. Its two ingredients are the combination rule that builds the joint space and the partial-trace rule that describes one half when the other is discarded.

B.1 Introduction: Combining State Spaces

Alice has a two-outcome system and Bob has another. Joint outcomes carry two labels: 00, 01, 10, and 11. A general joint distribution requires information about all four possibilities; separate descriptions of Alice and Bob do not determine their correlations. Quantum states similarly require joint amplitudes, with the additional possibility of relative phases and entanglement.

Quantum mechanics keeps all four joint components and superposes such combined columns. For two pure product states,

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle.$$

A general joint pure state carries four independent amplitudes, while a product state is fixed by two pairs of subsystem amplitudes. The extra freedom is where entanglement lives: joint amplitudes with correlations no product assignment reproduces.

B.2 Tensor Products of Two Vector Spaces

Fix the subsystem order before writing matrices. Throughout this example the first label belongs to A and the second to B. The tensor product is linear separately in each input, which explains the product-state expansion above.

For two two-level systems:

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle$$

$$(av + bv') \otimes w = a(v \otimes w) + b(v' \otimes w)$$

$$v \otimes (aw + bw') = a(v \otimes w) + b(v \otimes w')$$

$$\dim(V \otimes W) = (\dim V)(\dim W)$$

For two qubits: $2 \times 2 = 4$.

$$\langle u \otimes x | v \otimes y \rangle = \langle u | v \rangle \langle x | y \rangle$$

$$(A \otimes B)(v \otimes w) = Av \otimes Bw$$

For example, $X \otimes I$ flips the first qubit, so $(X \otimes I)|01\rangle = |11\rangle$. The operator $I \otimes X$ instead produces $|00\rangle$. In Julia, `kron(A, B)` constructs the matrix in this ordered product basis. Exchanging subsystem order requires a consistent permutation of both states and operators.

Detailed treatment: tensor products of two vector spaces

Tensor products of two vector spaces

Let Alice's two possible outcomes be 0 and 1. Represent them by column vectors, written in ket notation as

$$|0\rangle_A = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle_A = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A ket $|\psi\rangle$ denotes a vector in a complex vector space. Give Bob the same two basis states, with the subsystem label B . A joint outcome is an ordered pair. The four joint basis states are

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle.$$

The symbol $\otimes$ denotes the tensor product. For two column vectors, it is computed by multiplying the entire second column by each entry of the first column and stacking the resulting columns. Two vectors with two components therefore produce a vector with four components rather than a vector with $2 + 2$ components.

The tensor-product space is the vector space containing all linear combinations, with complex coefficients, of such product vectors. If Alice's vector space is V and Bob's is W , the tensor product is linear in each argument separately:

$$(av + bv') \otimes w = a(v \otimes w) + b(v' \otimes w),$$

$$v \otimes (aw + bw') = a(v \otimes w) + b(v \otimes w').$$

Here $v, v' \in V$, $w, w' \in W$, and a, b are complex scalars. This separate linearity is a defining property of the tensor product.

Choose a basis $\{|e_j\rangle\}$ for V and a basis $\{|f_k\rangle\}$ for W . A basis is a linearly independent set that spans the vector space. The product vectors $\{|e_j\rangle \otimes |f_k\rangle\}$ then form a basis for $V \otimes W$. Consequently, the dimensions multiply:

$$\dim(V \otimes W) = (\dim V)(\dim W).$$

For N two-level systems, the dimension is 2^N . This exponential dependence counts basis states. A computational speedup needs more: an algorithm whose gate count and error correction scale better than the best classical method on the same task.

An inner product is a scalar-valued operation that determines overlaps, norms, and orthogonality. On product vectors, inner products multiply and then extend to general vectors by linearity:

$$\langle u \otimes x | v \otimes y \rangle = \langle u | v \rangle \langle x | y \rangle.$$

When subsystem labels are unambiguous, the abbreviated notation $|ab\rangle$ denotes $|a\rangle_A \otimes |b\rangle_B$.

If a matrix A acts on V and a matrix B acts on W , their tensor-product operator is defined by its action on product vectors:

$$(A \otimes B)(v \otimes w) = Av \otimes Bw.$$

A local operator is an operator that acts on only one subsystem. An operator acting only on Alice is $A \otimes I_W$, where I_W is the identity operator on W . After the subsystem has been specified, this operator is often abbreviated as A . Operators acting on distinct

tensor factors commute:

$$(A \otimes I_W)(I_V \otimes B) = A \otimes B = (I_V \otimes B)(A \otimes I_W).$$

A vector of the form $|a\rangle \otimes |b\rangle$ is called a product state. In such a state, Alice and Bob each have an individual state vector, and the joint state is their tensor product.

Remark. If linearity in either argument is removed, the resulting operation is not a tensor product. Ordinary multiplication of two numbers of the same length is a different operation and does not produce a four-component vector from two two-component vectors.

B.3 Entangled Bipartite Pure States

A pure state is entangled when it cannot be written as one tensor product across the chosen split. The Bell state gives a small example where this failure of factorization can be proved directly.

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Suppose $|\Phi^+\rangle = (a|0\rangle + b|1\rangle)_A \otimes (c|0\rangle + d|1\rangle)_B$. Expanding gives ac, ad, bc, bd . Agreement requires $ac = bd = 1/\sqrt{2}$ and $ad = bc = 0$, which is impossible.

$$|\psi\rangle_{AB} = \sum_r s_r |u_r\rangle_A |v_r\rangle_B, \quad \sum_r s_r^2 = 1$$

The Schmidt vectors $\{|u_r\rangle\}$ and $\{|v_r\rangle\}$ are orthonormal sets, and the coefficients s_r can be chosen nonnegative. A pure state is a product state exactly when only one Schmidt coefficient is nonzero.

The decomposition comes from a singular-value decomposition of the coefficient matrix ψ_{ab} . Its nonzero singular values are the s_r . For the Bell state they are $1/\sqrt{2}, 1/\sqrt{2}$, so two correlated terms are necessary. This connection to matrix factorization will return in Appendix H.

Detailed treatment: entangled bipartite pure states

Entangled bipartite pure states

Not every four-component vector is the tensor product of two two-component vectors. Consider the Bell state

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

A Bell state is a maximally entangled two-qubit state. Suppose that this state could be factored as $(a|0\rangle + b|1\rangle)_A \otimes (c|0\rangle + d|1\rangle)_B$. Expanding the tensor product would produce the coefficients ac, ad, bc, bd .

Agreement with $|\Phi^+\rangle$ would require $ac = bd = 1/\sqrt{2}$ and $ad = bc = 0$. The nonzero conditions $ac \neq 0$ and $bd \neq 0$ require all four scalars to be nonzero, which contradicts the conditions $ad = bc = 0$.

Therefore, $|\Phi^+\rangle$ has no product decomposition. Such a pure joint state is called entangled: its joint amplitudes resist factorization into subsystem vectors. Two qubits can interact and still sit in a product state, so the factorization test — not the interaction history — decides entanglement.

Every bipartite pure state admits orthonormal sets $\{|u_r\rangle_A\}$ and $\{|v_r\rangle_B\}$, together with nonnegative numbers s_r , such that

$$|\psi\rangle_{AB} = \sum_r s_r |u_r\rangle_A |v_r\rangle_B, \quad \sum_r s_r^2 = 1.$$

This expansion is the Schmidt decomposition, and the numbers s_r are the Schmidt coefficients. A state is a product state exactly when only one Schmidt coefficient is nonzero. The eigenvalues of either reduced density operator defined below are s_r^2 , including zero eigenvalues.

B.4 Density Operators for Composite Systems

A density operator describes both a known pure state and a statistical mixture. It is especially useful when a subsystem is entangled with degrees of freedom that are not observed. The same matrix determines every expectation value through $\langle O \rangle = \text{Tr}(\rho O)$.

For a pure state:

$$\rho = |\psi\rangle\langle\psi|$$

For a mixed state:

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|$$

Here $p_k \geq 0$, $\sum_k p_k = 1$, and each $|\psi_k\rangle$ is normalized. The ensemble description need not be unique. For instance, equal mixtures of $|0\rangle, |1\rangle$ and of $|+\rangle, |-\rangle$ both give $I/2$.

$$\rho \geq 0, \quad \text{Tr } \rho = 1$$

For a qubit:

$$\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma}), \quad \|\mathbf{r}\| \leq 1$$

The vector $\mathbf{r}$ contains the three Pauli expectation values. Its length is one for a pure qubit state and zero for the maximally mixed state. More generally, $\text{Tr}(\rho^2) = 1$ characterizes a pure density matrix; a mixed state has smaller purity.

Detailed treatment: density operators for composite systems

Density operators for composite systems

A single ket completely specifies a known pure state. More general preparation and subsystem-discarding procedures require a broader state representation.

For example, a classical coin may determine which of two kets is prepared. Alternatively, Bob's subsystem may be discarded while only Alice's subsystem is retained. Both cases require an operator that predicts all local measurement probabilities.

For the Bell pair, form an operator by multiplying the ket by its bra:

$$\rho_{AB} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|).$$

The bra $\langle\psi|$ is the conjugate transpose of the ket $|\psi\rangle$. The resulting matrix ρ_{AB} is a density operator. A density operator contains the information required to calculate every measurement probability for a system, whether the state is represented by a single ket or by a statistical mixture of kets.

A density operator ρ on a Hilbert space $\mathcal{H}$, where a Hilbert space is an inner-product vector space, satisfies

$$\rho \geq 0, \quad \text{Tr } \rho = 1.$$

The condition $\rho \geq 0$ means that ρ is positive: $\langle\psi|\rho|\psi\rangle \geq 0$ for every $|\psi\rangle$. In finite dimensions, positivity also implies Hermiticity. The trace $\text{Tr } \rho$, defined as the sum of the diagonal matrix elements in any orthonormal basis, provides normalization.

If a preparation selects normalized pure states $|\psi_r\rangle$ with classical probabilities $p_r \geq 0$, where $\sum_r p_r = 1$, then

$$\rho = \sum_r p_r |\psi_r\rangle\langle\psi_r|.$$

Different ensembles $\{p_r, |\psi_r\rangle\}$ can produce the same ρ . Because measurement probabilities depend only on ρ , these ensembles cannot be distinguished unless an additional record of the preparation is retained.

A state is pure exactly when $\rho^2 = \rho$, equivalently when $\text{Tr}(\rho^2) = 1$. It is mixed when $\text{Tr}(\rho^2) < 1$. The scalar $\text{Tr}(\rho^2)$ is called the purity. By the spectral theorem,

$$\rho = \sum_j q_j |q_j\rangle\langle q_j|,$$

where $q_j \geq 0$ and $\sum_j q_j = 1$. Thus, a density operator is defined as a positive matrix with unit trace.

For a qubit, every density operator has a Bloch representation

$$\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma}),$$

where $\mathbf{r} = (r_x, r_y, r_z) \in \mathbb{R}^3$, $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$, and positivity requires $\|\mathbf{r}\| \leq 1$. Here $\mathbf{r}$ is the Bloch vector and $\sigma_x, \sigma_y, \sigma_z$ are the Pauli matrices. Pure states correspond to points on the unit sphere, whereas mixed states correspond to points inside it.

An observable is a Hermitian operator $M = M^\dagger$ representing a measurable quantity. Its expectation value is

$$\langle M \rangle = \text{Tr}(\rho M).$$

More generally, a positive-operator-valued measure, or POVM, is a set of positive operators $\{E_m\}$ satisfying $\sum_m E_m = I$. The probability of outcome m is $p(m) = \text{Tr}(\rho E_m)$.

A POVM specifies outcome probabilities but does not specify the state remaining after an outcome. A quantum instrument provides the corresponding state-update maps.

Remark. Without positivity, $\text{Tr}(\rho M)$ need not be a probability even when $0 \leq M \leq I$. The nonuniqueness of ensemble decompositions does not imply nonuniqueness of ρ : many ensembles can represent the same density operator.

B.5 Partial Trace and Reduced States

The partial trace produces the unique reduced density matrix that reproduces every local expectation value: $\text{Tr}(\rho_A O_A) = \text{Tr}[\rho_{AB}(O_A \otimes I_B)]$. It summarizes Alice's statistics with B unobserved; the laboratory still holds the full joint state.

$$\rho_A = \text{Tr}_B \rho_{AB} = \sum_b \langle b|_B \rho_{AB} |b\rangle_B$$

For $|\Phi^+\rangle$:

$$\rho_A = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I_A}{2}$$

The joint Bell state is pure, but either subsystem alone is maximally mixed.

To see why, expand its density matrix. The terms $|00\rangle\langle 11|$ and $|11\rangle\langle 00|$ vanish under the partial trace over B because $\langle 1|0\rangle = 0$. The two diagonal terms survive with weight one half.

The mixed state $\frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|$ has the same one-qubit reduced states but lacks the Bell state's joint coherence. It is a separable, classically correlated mixture. A mixed Alice state therefore signals either entanglement with an unobserved partner or ordinary classical uncertainty, and local data alone cannot separate the two.

Detailed treatment: partial trace and reduced states

Partial trace and reduced states

Suppose Bob's half of the Bell pair is inaccessible. Alice's measurement statistics must then be predicted from a local density operator. This reduced density operator is obtained by summing over Bob's degrees of freedom in ρ_{AB} . The corresponding operation is the partial trace:

$$\rho_A = \text{Tr}_B \rho_{AB}.$$

The partial trace acts as a trace on one tensor factor while leaving the other factor unchanged.

For a rank-one product operator, its action is

$$\text{Tr}_B (|i\rangle_A \langle j|_B \langle k|_A \langle \ell|_B) = \langle \ell|j\rangle_B |i\rangle_A \langle k|_A.$$

For the Bell state, the off-diagonal terms vanish because $\langle 1|0\rangle = 0$, while the diagonal terms remain. Therefore,

$$\rho_A = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} I_A.$$

The joint Bell state is pure, but either subsystem alone is maximally mixed, with the same outcome probabilities as a fair classical coin.

Consequently, an experiment performed only on Alice cannot distinguish a classical random preparation from the loss of an entangled partner. The density operator does not identify which preparation procedure produced it.

The partial trace is also the unique linear map satisfying

$$\text{Tr}_B(A \otimes B) = A \text{Tr}(B).$$

For every observable M_A acting only on Alice, this definition guarantees

$$\text{Tr}_{AB}[\rho_{AB}(M_A \otimes I_B)] = \text{Tr}_A(\rho_A M_A),$$

where $\rho_A = \text{Tr}_B \rho_{AB}$. This identity establishes that ρ_A is the correct local state because it reproduces every local expectation value.

A joint density operator is separable if it can be expressed as a classical mixture of product density operators:

$$\rho_{AB} = \sum_r p_r \rho_A^{(r)} \otimes \rho_B^{(r)}.$$

A state that is not separable is entangled. Separable states can nevertheless exhibit strong classical correlations. Therefore, a nonfactorizing probability table or a nonzero covariance does not by itself establish entanglement.

The von Neumann entropy of a density operator is

$$S(\rho) = -\text{Tr}(\rho \log \rho) = -\sum_j q_j \log q_j,$$

with the convention $0 \log 0 = 0$. If the logarithm has base two, entropy is measured in bits.

For a bipartite pure state, $S(\rho_A) = S(\rho_B)$ quantifies the entanglement across that bipartition. For mixed states, the same entropy also includes ordinary statistical mixture.

Local entropy alone is therefore not an entanglement measure.

Remark. If the joint state is not pure, $S(\rho_A)$ does not by itself measure entanglement. It continues to quantify the mixedness of the reduced density operator.

Quantum channels from discarded environments

Alice's atom may interact with an unobserved system and subsequently exhibit apparently irreversible dynamics. The combined evolution can remain unitary even though the reduced evolution of Alice's subsystem is not unitary.

A quantum channel $\mathcal{E}$ is a linear map that sends density operators to density operators, including when the input system is one part of a larger composite system. Trace preservation means

$$\text{Tr } \mathcal{E}(X) = \text{Tr } X.$$

Ordinary positivity keeps positive inputs positive. A laboratory input can be half of an entangled pair, so the physical channel must stay positive on the joint system too.

Complete positivity is the stronger requirement that, for the identity map $\mathcal{I}_R$ on an arbitrary reference system R , the extended map $\mathcal{I}_R \otimes \mathcal{E}$ also preserve positivity. This requirement is necessary because the input may be entangled with an unaffected reference system [Roo7]; [Roo8]. A map that is positive but not completely positive can produce an invalid matrix when applied to one half of a Bell pair.

Every finite-dimensional quantum channel has a Kraus representation

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I.$$

The operators K_k are called Kraus operators. The second equation is the condition for trace preservation. A Kraus representation is not unique: a unitary transformation among the members of one Kraus list produces another list representing the same channel.

A channel can always be realized by introducing an environment E in a fixed state, applying a joint unitary operator U , and tracing out E :

$$\mathcal{E}(\rho) = \text{Tr}_E [U(\rho \otimes |0\rangle_E \langle 0|)U^\dagger].$$

This construction is a finite-dimensional Stinespring dilation [Roo7]. The dilation is an existence claim for analysis: some environment, joint unitary, and trace reproduce the channel. It identifies no particular physical bath as small or memoryless.

The Choi operator provides a diagnostic for complete positivity. If the input dimension is d , define the unnormalized maximally entangled vector

$$|\Omega\rangle = \sum_{j=1}^d |j\rangle_R |j\rangle_A.$$

Then

$$J(\mathcal{E}) = (\mathcal{I}_R \otimes \mathcal{E})(|\Omega\rangle\langle\Omega|).$$

Choi's theorem states that $\mathcal{E}$ is completely positive exactly when $J(\mathcal{E}) \geq 0$ [Roo8]. For this unnormalized convention, trace preservation is equivalent to

$$\text{Tr}_{\text{out}} J(\mathcal{E}) = I_R.$$

Remark. A map that is positive but not completely positive may appear valid on product states. When applied to one half of $|\Phi^+\rangle$, however, its output need not be a density operator.

Amplitude damping and unitary dilation

Amplitude damping is a minimal model of relaxation in which the excited state $|1\rangle$ decays to the ground state $|0\rangle$, while an environment records whether the decay occurred. Let the system be S , the environment be E , and let $p \in [0, 1]$ be the decay probability. Define the action of a unitary operator on the relevant basis states by

$$U|0\rangle_S|0\rangle_E = |0\rangle_S|0\rangle_E,$$

$$U|1\rangle_S|0\rangle_E = \sqrt{1-p}|1\rangle_S|0\rangle_E + \sqrt{p}|0\rangle_S|1\rangle_E.$$

The two output vectors are normalized and orthogonal. This isometry can therefore be extended to a unitary operator on the full four-dimensional space. Prepare E in $|0\rangle_E$. Taking the environment matrix element $K_e = {}_E\langle e|U|0\rangle_E$ for $e = 0, 1$ gives

$$K_0 = |0\rangle\langle 0| + \sqrt{1-p}|1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix},$$

$$K_1 = \sqrt{p}|0\rangle\langle 1| = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}.$$

These Kraus operators satisfy

$$K_0^\dagger K_0 + K_1^\dagger K_1 = I,$$

so the channel is trace preserving. Its Kraus form also makes complete positivity explicit. For an arbitrary input density operator

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix},$$

direct matrix multiplication yields

$$\mathcal{E}_p(\rho) = \begin{pmatrix} \rho_{00} + p\rho_{11} & \sqrt{1-p}\rho_{01} \\ \sqrt{1-p}\rho_{10} & (1-p)\rho_{11} \end{pmatrix}.$$

The population in $|1\rangle$ is transferred to $|0\rangle$, and the coherences are reduced by the factor $\sqrt{1-p}$. For a Markovian relaxation model with time $t \geq 0$ and relaxation time $T_1 > 0$, the decay probability is written as

$$p(t) = 1 - e^{-t/T_1}.$$

Both t and T_1 have units of seconds, so the exponent is dimensionless. The resulting channels form a semigroup: evolution for t_1 followed by evolution for t_2 is equivalent to evolution for $t_1 + t_2$.

The combined $S + E$ evolution is unitary. The reduced map on Alice becomes irreversible after E is discarded.

If the environment retains memory and later interacts with the system again, a one-parameter Markov channel may not be valid. Relaxation of the observed subsystem is consistent with unitary evolution of system plus environment: evolve jointly, then discard the environment record. The reduced map looks irreversible because the record is gone.

Under time-homogeneous Markov assumptions, the density operator can obey the Gorini–Kossakowski–Sudarshan–Lindblad equation [R009]; [R010],

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_j \gamma_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\} \right).$$

Here H is Hermitian, L_j are jump operators, $\gamma_j \geq 0$ have units s^{-1} , $[A, B] = AB - BA$, and $\{A, B\} = AB + BA$. This generator preserves trace and complete positivity. It is a model based on stated assumptions rather than a definition of all open-system dynamics [R011].

Encoded subspaces, leakage, and correlated noise

For a cluster of N physical defects, the microscopic state belongs to the tensor-product Hilbert space

$$\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_N.$$

An encoded doublet, meaning a selected two-dimensional subspace used to represent a qubit, is specified by a projector P .

The encoded state is not obtained merely by tracing out the remaining energy levels. Its definition requires the preparation procedure within P , whether transitions into the complementary subspace $Q = I - P$ are possible, and whether recovery or heralding maps leakage back into the encoded subspace.

Correlated noise acting on two clusters is represented by a joint quantum channel. Such a channel need not factor as $\mathcal{E}_A \otimes \mathcal{E}_B$.

Replacing correlated noise with independent single-cluster channels erases the spatial covariance between clusters. Keeping the joint channel, or at least a many-body reduced density operator spanning both, preserves those correlations for the later gap analysis.

A mixed reduced state fits both a topological ground space and an ordinary noisy state, so the diagnosis waits on the nonlocal loop operators and finite-size scaling of later chapters.

The projection

$$\rho \mapsto P\rho P$$

is trace decreasing when leakage has occurred. The missing trace,

$$1 - \text{Tr}(P\rho P),$$

is the leakage probability. Renormalizing $P\rho P$ describes the state conditioned on detecting no leakage. Reporting this conditional state without its associated success probability conceals device failures.

A tensor product describes a composite system in terms of specified subsystems. Entanglement is a property of a state relative to that subsystem decomposition.

An encoded qubit is a chosen subspace. A noisy channel is a map acting on states.

Each structure supplies one ingredient — joint spaces, chosen subspaces, noise maps — while an emergent quasiparticle, a topological phase, or a fault-tolerant logical qubit needs the full many-body energetics, gap, and diagnostics built on top of them.

B.6 Common Errors

Point 1: “Treating $\otimes$ as ordinary multiplication.”

- **Explanation:** Track dimensions and subsystem order: $A \otimes B$ acts on the product space, so its matrix size and basis ordering differ from either factor.

Point 2: “Not every correlated state is entangled.”

- **Explanation:** A separable mixture of product states produces correlated outcomes with no entanglement; test factorization of the state, not of the statistics.

Point 3: “An ensemble decomposition is not unique.”

- **Explanation:** Measurement statistics fix the density operator, while many preparation lists $\{p_r, |\psi_r\rangle\}$ compile to that same operator; keep the preparation record when the distinction matters.

Point 4: “Positivity cannot replace complete positivity.”

- **Explanation:** Test the map on half of an entangled pair: a merely positive map can return a non-positive joint operator there, which no physical channel permits.

Detailed treatment: common errors

Common errors

- Treating $\otimes$ as ordinary multiplication is incorrect. Track dimensions and subsystem order: $A \otimes B$ acts on the product space, so its matrix size and basis ordering differ from either factor, and exchanging subsystem order requires a consistent permutation of both states and operators.
- Not every correlated state is entangled. A separable mixture of product states produces correlated outcomes with no entanglement; test factorization of the state, not of the statistics.
- An ensemble decomposition is not unique. Measurement statistics fix the density operator, while many preparation lists $\{p_r, |\psi_r\rangle\}$ compile to that same operator; keep the preparation record when the distinction matters.

- Positivity cannot replace complete positivity. Test the map on half of an entangled pair: a merely positive map can return a non-positive joint operator there, which no physical channel permits.
- Trace loss must be included in the analysis. Postselection and leakage projections are trace-nonincreasing, and renormalization changes the claims that can be made about the process.
- Kraus operators should not automatically be interpreted as physical outcomes. They can label outcomes in a chosen dilation or quantum instrument, but different Kraus lists may describe the same unobserved channel.
- A Lindblad fit should not be identified with a unique microscopic model. Non-Markovian baths, slow spectral diffusion, and correlated fluctuators may violate the semigroup assumptions.
- Tracing out a subsystem is not equivalent to asserting that it was measured. Ignoring a subsystem and measuring it without reading the outcome can produce the same reduced state in some experimental arrangements, but their environmental records and the effects of later interventions can still differ.

Exercises and checks

Exercises and checks

- Show that $\dim(V \otimes W) = (\dim V)(\dim W)$. If $\{|e_j\rangle\}$ is a basis of V and $\{|f_k\rangle\}$ is a basis of W , then the product vectors $\{|e_j\rangle \otimes |f_k\rangle\}$ form a basis of $V \otimes W$. The dimensions therefore multiply. For two qubits, $2 \times 2 = 4$.
- Show that $|\Phi^+\rangle$ is not a product state. A product state would have coefficients ac, ad, bc, bd . The required conditions $ac = bd = 1/\sqrt{2}$ and $ad = bc = 0$ cannot hold simultaneously.
- State the two defining conditions for a density operator. They are positivity and unit trace.
- Show that the reduced state of $|\Phi^+\rangle$ is $\frac{1}{2}I$. Apply the partial trace to the four terms of $|\Phi^+\rangle\langle\Phi^+|$. The cross terms vanish by orthogonality, and the diagonal terms contribute $\frac{1}{2}|0\rangle\langle 0|$ and $\frac{1}{2}|1\rangle\langle 1|$.

- Explain the meaning of complete positivity. A completely positive map remains positive when tensored with the identity map on an arbitrary unaffected reference system.
- Explain what is omitted when only $P\rho P / \text{Tr}(P\rho P)$ is reported. This normalized operator is conditioned on the state remaining in P , so reporting it alone conceals the leakage probability $1 - \text{Tr}(P\rho P)$.

The constructions developed here are the tensor product for combining subsystem state spaces, the density operator for states that cannot be represented adequately by a single ket, and the reduced dynamical map obtained after discarding a subsystem.

Later chapters invoke eight results without rederivation, each at its point of use: composite dimensions multiply (to size cluster spaces); pure entangled states can have mixed reduced states (to read single-cluster data); the partial trace is fixed by reproducing local expectation values (to define reduced states); density operators are positive with unit trace (to check candidate states); physical deterministic processes are completely positive and trace preserving (to vet noise models); every finite-dimensional channel has Kraus and unitary-dilation representations (to construct and dilate channels); leakage projections can decrease trace (to account for lost population); and independent-channel models fail when a shared environment correlates two clusters (to keep covariance in the model).

Sources

Sources

- [Roo4] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information: 10th Anniversary Edition, Cambridge University Press (2010). DOI: [10.1017/CBO9780511976667](https://doi.org/10.1017/CBO9780511976667).
- [Roo6] J. Watrous, The Theory of Quantum Information, Cambridge University Press (2018). DOI: [10.1017/9781316848142](https://doi.org/10.1017/9781316848142).
- [Roo7] W. F. Stinespring, “Positive functions on C-algebras,” Proceedings of the American Mathematical Society* 6, 211–216 (1955). DOI: [10.1090/S0002-9939-1955-0069403-4](https://doi.org/10.1090/S0002-9939-1955-0069403-4).

- [Roo8] M.-D. Choi, “Completely positive linear maps on complex matrices,” *Linear Algebra and its Applications* 10, 285–290 (1975). DOI: [10.1016/0024-3795\(75\)90075-0](https://doi.org/10.1016/0024-3795(75)90075-0).
 - [Roo9] G. Lindblad, “On the generators of quantum dynamical semigroups,” *Communications in Mathematical Physics* 48, 119–130 (1976). DOI: [10.1007/BF01608499](https://doi.org/10.1007/BF01608499).
 - [Ro10] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, “Completely positive dynamical semigroups of N-level systems,” *Journal of Mathematical Physics* 17, 821–825 (1976). DOI: [10.1063/1.522979](https://doi.org/10.1063/1.522979).
 - [Ro11] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press (2002). DOI: [10.1093/acprof:oso/9780199213900.001.0001](https://doi.org/10.1093/acprof:oso/9780199213900.001.0001).
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Appendix C — Groups, spin rotations, and braid representations

The main text uses two kinds of operation: rotations of a spin qubit, fixed by choosing an angle, and exchanges of anyons confined to a plane, recorded by crossing data. A group records how operations in each family compose in sequence, and a representation turns that composition into unitary operators on a state space. The resulting matrices can look alike, so this appendix keeps the two underlying structures separate: continuous spin rotations on one side, discrete braids on the other.

Background, scope, and supporting argument

A continuously parameterized transformation is specified by one or more real parameters varying over an interval. For example, a planar rotation may have any angle in an interval, including a half turn or a 37-degree rotation, and the transformation varies continuously with the angle.

A braid crossing records discrete data: at each projected crossing, one trajectory passes over the other, and the opposite choice of overpass records the inverse crossing.

Deforming the trajectories without letting them collide changes their geometric shape and parametrization while leaving the crossing record unchanged, so the appendix omits geometric detail that the crossing record already absorbs. What remains is the ordered list of signed crossings, taken up to the defining relations of the braid group.

This appendix compares the two families side by side. Spin rotations carry a continuous parameter; planar exchanges carry discrete crossing data. Both families act through unitary matrices, and those matrices can look similar, while the groups behind them differ.

C.1 Introduction: Continuous vs. Discrete Transformations

A continuous transformation is fixed by choosing one or more real parameters: a planar rotation, for example, is fixed by choosing its angle, and nearby angles give nearby rotations.

A braid crossing instead records which of two neighboring worldlines passes over the other. Deforming the picture without letting the lines touch changes the geometry while leaving that record unchanged.

C.2 Groups and Invertible Operations

A group is a set equipped with an associative composition law, an identity element, and an inverse for every element.

$$(gh)k = g(hk), \quad gg^{-1} = g^{-1}g = e$$

Detailed treatment: groups and invertible operations

Groups and invertible operations

Take a set of composable transformations, with a distinguished identity transformation that leaves the system unchanged. The product of two transformations means performing them in sequence, and each transformation has an inverse that undoes it. When the composition of three transformations is independent of where the parentheses fall, these data satisfy the group axioms. The set of transformations is denoted by G , and the identity transformation is denoted by e .

In symbols, associativity is $(gh)k = g(hk)$. Every $g \in G$ has an inverse g^{-1} with $gg^{-1} = g^{-1}g = e$.

A group therefore records reversible composition on its own, with no geometric or analytic structure assumed.

Remark. Dropping inverses leaves a monoid: elements still compose, while some operations have no reverse. Dropping associativity leaves the product of three transformations ambiguous until parentheses fix the order.

C.3 Continuous One-Parameter Transformations

A Lie group is a group whose elements also form a smooth manifold, with multiplication and inversion smooth in those coordinates. The rotations of the plane form one example: the circle group $SO(2)$, where each angle names a group element.

A spin-1/2 rotation through angle θ about the z-axis is:

$$U_z(\theta) = \exp\left(-\frac{i\theta Z}{2}\right) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$$

At $\theta = 2\pi$, $U_z = -I$. The spinor returns to its original vector only after a 4π rotation. The minus sign from a 2π rotation is a global phase on an isolated state: it leaves every measurement probability unchanged, and it shows up only when a coherent experiment compares a rotated branch against an unrotated branch, where it enters as a relative phase.

Detailed treatment: continuous one-parameter transformations

Continuous one-parameter transformations

Give the transformations smooth coordinates. Near the identity, each direction in which the coordinates can change defines an infinitesimal transformation, represented by a tangent vector at the identity.

Exponentiating those infinitesimal generators produces finite transformations. The rotations of a plane illustrate the pattern: the group is a circle, and one continuously variable angle selects each element.

A group with this additional smooth structure is a Lie group. $SO(2)$ is the circle group of ordinary plane rotations, and every angle specifies an allowed group element.

More precisely, a Lie group carries a smooth-manifold structure for which multiplication and inversion are smooth maps. That smooth dependence on parameters is what separates a Lie group from a discrete group, where no such continuous variation exists.

Detailed treatment: spinor behavior under (2π) and (4π) rotations

Spinor behavior under 2π and 4π rotations

Let I be the 2×2 identity matrix and let

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is the Pauli Z matrix from Chapter 1. A spin-1/2 rotation through angle θ about the z axis is

$$U_z(\theta) = \exp\left(-\frac{i\theta Z}{2}\right),$$

where $i^2 = -1$ and the matrix exponential is $e^A = \sum_{k=0}^{\infty} A^k/k!$. In this expression, $Z/2$ is the Hermitian generator, and $-iZ/2$ is the corresponding anti-Hermitian Lie-algebra generator, exponentiated as $-i\theta Z/2$ for each angle.

Since $Z^2 = I$, the even and odd powers in the exponential series can be separated:

$$U_z(\theta) = I \cos \frac{\theta}{2} - iZ \sin \frac{\theta}{2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}.$$

Its determinant is one and $U_z^\dagger U_z = I$, so it belongs to $SU(2)$, the Lie group of 2×2 unitary matrices with determinant one. A unitary matrix is a matrix whose conjugate transpose is its inverse. Acting on $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ gives

$$U_z(\theta)|\psi\rangle = e^{-i\theta/2}\alpha|0\rangle + e^{i\theta/2}\beta|1\rangle.$$

The two amplitudes acquire a relative phase $e^{i\theta}$. At $\theta = 2\pi$, $U_z = -I$ rather than I .

A spinor is a vector in the spin-1/2 representation space, and it returns to its original vector only after a 4π rotation. For an isolated state the global phase -1 leaves all probabilities unchanged. This behavior expresses the double-cover relation $SU(2) \rightarrow SO(3)$, under which two elements of $SU(2)$ correspond to each spatial rotation in $SO(3)$ [Ro14].

Remark. If the factor of $1/2$ were removed from the exponent, a 2π rotation would already equal the identity.

The factor of $1/2$ in the exponent is what places the rotation in the double cover, so that a spin-1/2 object needs two full turns to come back to itself.

C.4 Topological Invariance of Braids

Draw particle positions along the horizontal axis and time along the vertical axis, so each particle traces a worldline. Exchanging two neighbors makes their worldlines cross. A generator σ_i denotes one chosen orientation of exchange between neighbors i and $i + 1$; σ_i^{-1} denotes the reverse exchange. In a plane, exchanging the same pair twice can leave the worldlines wound around each other, so doing it twice differs topologically from doing nothing, and the braid group does not impose $\sigma_i^2 = e$.

For three strands:

$$B_3 = \langle \sigma_1, \sigma_2 \mid \sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2 \rangle$$

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, \quad \sigma_i \sigma_j = \sigma_j \sigma_i \quad (|i - j| \geq 2)$$

Either side of the first relation moves three neighboring strands past one another through a different intermediate picture; the relation records that the two pictures deform into each other. The second says that exchanges of disjoint neighboring pairs proceed independently and hence commute. A braid representation assigns operators B_i satisfying these same relations. Matrices B_i that satisfy these relations while failing to commute are the algebraic signature of non-Abelian braiding when they act on a fusion space. Building the same matrices with control pulses on ordinary qubits reproduces the algebra; calling that result topological requires, in addition, the physical excitations and their protection.

Detailed treatment: topological invariance of braids

Topological invariance of braids

Fix n marked points in a plane and move them continuously, never letting two points collide, until the set of occupied positions is the same as at the start.

Two such motions count as the same when a continuous deformation through collision-free motions with the same endpoints connects them. A deformation of this kind is called a homotopy. Multiplying two motions means performing one and then the next.

Identifying motions that differ only by such a deformation leaves an equivalence class, and that class is a braid [Ro12]; [Ro13].

Three strands, time downward

identity σ_1 σ_2
1 2 3 1 2 3 1 2 3

```

| | | \ / | | \ /
| | | x | | x
| | | / \ | | / \

```

In this diagram, each strand represents the trajectory of one marked point as a function of time. Here σ_i denotes a chosen overcrossing that exchanges positions i and $i + 1$ counterclockwise, while σ_i^{-1} denotes the opposite crossing. Orientation conventions differ, so each source must specify which crossing is represented by its generator.

Braids multiply by concatenation: perform one braid, then the next. Generators name the individual crossings, and the defining relations decide when two crossing diagrams name the same braid. The algebraic description admits only integer crossing words.

Noncommutative braid operations on three strands

The braid classes with this multiplication form the braid group. The marked points move continuously, yet the group elements are the discrete classes, not the individual trajectories. For three strands, the group has the presentation

$$B_3 = \langle \sigma_1, \sigma_2 \mid \sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2 \rangle.$$

A presentation lists generators together with the relations they satisfy, with the understanding that anything following from those relations also holds. The displayed equality is the braid relation, also called the Yang–Baxter relation: the two three-crossing diagrams deform into each other through collision-free configurations.

The braid relation still leaves the two orderings distinct: it does not imply $\sigma_1 \sigma_2 = \sigma_2 \sigma_1$, since those two words run the exchanges in different orders.

For a concrete unitary representation, take three identical particles of the Ising σ type, whose combined topological charge is another σ . Equivalently, the three particles fuse jointly to the σ charge sector. The remaining quantum state space is two-dimensional. On that space, define

$$F = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = e^{-i\pi/8} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}.$$

The map $\rho : B_3 \rightarrow U(2)$, where $U(2)$ is the group of 2×2 unitary matrices, is defined by

$$\rho(\sigma_1) = D, \quad \rho(\sigma_2) = FDF^{-1}.$$

Because $F^{-1} = F$, direct multiplication gives, after omitting the common phase $e^{-3i\pi/8}$,

$$D_0(FD_0F)D_0 = (FD_0F)D_0(FD_0F) = \frac{1+i}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

where $D_0 = \text{diag}(1, i)$. Thus $\rho(\sigma_1\sigma_2\sigma_1) = \rho(\sigma_2\sigma_1\sigma_2)$ exactly. Nevertheless, the two generator matrices do not commute.

The assigned matrices live inside a Lie group of unitary matrices. That placement describes the matrices, while the braid group supplying the words stays discrete.

The representation maps braid words to quantum gates while keeping the braid group and the unitary group distinct. In particular, Ising braids alone generate a nonuniversal gate set [\[Ro15\]](#).

Lie algebra generators at the identity

A matrix Lie group is a set of invertible matrices that forms both a group and a smooth manifold, with smooth multiplication and inversion. The unitary, special unitary, and rotation groups are the examples used here:

$$U(n) = \{U \in \mathbb{C}^{n \times n} : U^\dagger U = I\},$$

$$SU(n) = \{U \in U(n) : \det U = 1\},$$

and

$$SO(n) = \{R \in \mathbb{R}^{n \times n} : R^T R = I, \det R = 1\}.$$

Here $\mathbb{C}^{n \times n}$ denotes complex $n \times n$ matrices, $\mathbb{R}$ denotes real numbers, $\dagger$ denotes conjugate transpose, and T denotes transpose.

The Lie algebra is the tangent space at the identity element: $\mathfrak{g}$ of G . At any point of a smooth manifold, the tangent space is the vector space of infinitesimal directions through that point. At the identity of a Lie group, this vector space carries in addition the Lie bracket

$$[X, Y] = XY - YX.$$

For $U(n)$, the tangent generators X are anti-Hermitian, meaning that $X^\dagger = -X$. In physics notation one often writes $X = -iH$, where H is a Hermitian generator satisfying $H = H^\dagger$. Exponentiation then gives a one-parameter subgroup, a curve through the identity compatible with group multiplication:

$$U(t) = e^{tX} = e^{-itH}.$$

If t represents time, H has units of angular frequency when $\hbar = 1$, or units of energy when the exponent is $-itH/\hbar$. In either convention, the combination in the exponent must be dimensionless.

The commutator records the leading failure of two small transformations to commute. For small real parameters s, t ,

$$e^{sX}e^{tY}e^{-sX}e^{-tY} = e^{st[X,Y]+O(s^2t, st^2)}.$$

The symbol $O(s^2t, st^2)$ collects terms of the indicated higher orders in the small parameters. The available Hamiltonian generators together with their nested commutators therefore decide which connected unitary transformations the controls can reach. That reachability question belongs to the Lie algebra, a separate question from which discrete braids a device supports [\[Ro14\]](#).

Remark. Keeping only the vector-space structure of the infinitesimal transformations discards the bracket, and with it the information about which nested commutators the available generators can synthesize.

Commutation and braid relations for braid-group generators

For $n \geq 2$, the Artin braid group B_n has generators

$$\sigma_1, \sigma_2, \dots, \sigma_{n-1}$$

and relations

$$\sigma_i\sigma_j = \sigma_j\sigma_i \quad \text{when } |i - j| \geq 2,$$

$$\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}.$$

Exchanges of separated pairs commute; that is the first relation. The second relation records the local deformation involving three neighboring strands. Each generator has an inverse, and inverse crossings cancel:

$$\sigma_i\sigma_i^{-1} = e.$$

With two strands the group B_2 has one generator and no further relation, so $B_2 \cong \mathbb{Z}$: the corresponding integer counts signed crossings. For $n \geq 3$, B_n is non-Abelian, meaning that some pairs of elements fail to commute. Sending each σ_i to the adjacent transposition $(i \ i + 1)$ defines a surjective homomorphism

$$p : B_n \longrightarrow S_n,$$

where S_n is the permutation group: a homomorphism preserves composition, and surjectivity means every permutation arises from some braid. The braids mapping to the identity permutation form the pure braid group P_n , consisting of braids whose strands return to their individually labeled starting positions. Forgetting the winding is exactly what p does: σ_i^2 is a nontrivial pure braid even though $p(\sigma_i^2)$ is the identity permutation [Ro12]; [Ro13].

A braid can therefore return every labeled strand to its starting position while remaining topologically nontrivial.

Reading a braid word requires a stated time-ordering convention. This book composes operators from right to left: in $\rho(\sigma_1\sigma_2)$, $\rho(\sigma_2)$ acts first on a state vector.

Many diagrammatic sources stack the left factor first instead. Each convention is consistent on its own; mixing them without conversion gives wrong matrices.

Matrix representations of braid-group elements

A unitary braid representation assigns each braid a unitary operator through a homomorphism

$$\rho : B_n \rightarrow U(V),$$

where V is the finite-dimensional fusion space left over after fixing the total charge, and $U(V)$ is its unitary group. The representation sends braid elements to unitary operators while keeping the two structures distinct.

Because $\rho(b_1b_2) = \rho(b_1)\rho(b_2)$, equivalent braid diagrams act identically. In an anyon theory, an F -move changes to a fusion basis in which the pair about to be exchanged has a definite intermediate charge, and an R -move then performs the exchange. The consistency conditions among these moves yield the braid relations [Ro15]; [Ro16].

[Theory] Braiding counts as computationally universal on an encoded subspace when its matrices approximate every desired unitary on that subspace to arbitrary accuracy, where unitaries differing only by an overall phase count as the same gate. Precisely, universality holds when the closure of $\rho(B_n)$ is dense in the relevant projective unitary group.

The closure forms a Lie group although B_n itself is discrete. Fibonacci braids can fill the relevant gate set densely on suitable encodings, while Ising braids cannot supply universality on their own [Ro15].

Configuration space gives a second view of the same group: let $M = \mathbb{R}^2$ be the plane and delete every configuration in which two of n points coincide; for indistinguishable particles, further identify configurations differing only by a permutation. Each surviving point describes one allowed particle arrangement, and the space of all such points is $C_n(M)$. The braid group is its fundamental group:

$$B_n \cong \pi_1(C_n(\mathbb{R}^2)).$$

Here π_1 denotes homotopy classes of loops based at a chosen configuration: each loop records a collision-free exchange history, and homotopic loops record histories deformable into each other. In three spatial dimensions, nearby trajectories generally deform past one another without collision, so only the final permutation survives. In two dimensions the winding survives [Ro13]; [Ro15].

Remark. Allowing collisions fills in the deleted configurations. The resulting space has trivial fundamental group, so it records no braid information.

Control pulses, parameter-space paths, and error mechanisms

Suppose a laboratory can switch between two Hamiltonians, H_x and H_z , to an encoded qubit. Each Hamiltonian generates time evolution during the interval it is switched on. The pulse durations are continuous real parameters, so an ideal control sequence has the form

$$U = e^{-it_k H_{j_k}/\hbar} \dots e^{-it_2 H_{j_2}/\hbar} e^{-it_1 H_{j_1}/\hbar}.$$

Here each t_r is a time, each H_{j_r} carries units of energy, and $\hbar$ carries units of energy times time, so every exponent is dimensionless.

Whether these pulses can generate all of $SU(2)$ is decided by the Lie algebra generated by $-iH_x/\hbar$, $-iH_z/\hbar$, and their commutators. A small calibration error shifts the real pulse parameters and shifts the implemented unitary along with them.

A braid protocol instead specifies an integer word

$$b = \sigma_{i_k}^{s_k} \dots \sigma_{i_2}^{s_2} \sigma_{i_1}^{s_1}, \quad s_r \in \{+1, -1\}.$$

Deforming a strand trajectory without creating a collision leaves b unchanged. That stability under deformation is the topological insensitivity of the ideal anyon model.

A physical protocol still fails through discrete errors: an unintended quasiparticle can enter the region, two worldlines can be exchanged in the wrong order, or leakage can move the state outside the intended fusion space. Topology absorbs small geometric variations of the paths, while a missing or extra braid generator names a different word [Ro15].

A hybrid scheme combines the two structures: braids supply a discrete protected gate set, while calibrated continuous operations, measurements, or ancillary states supply gates outside the braid image.

For Ising anyons the braid image is not dense, so those non-braid resources carry the missing gates and the computation depends on them logically. When the image is dense, each target gate still needs a finite braid approximating it, with approximation error measured by the operator norm or worst-case projective distance.

C.5 Common Errors

Point 1: “Reversing clockwise and counterclockwise conventions without stating the change.”

- **Explanation:** This sends σ_i to σ_i^{-1} .

Point 2: “Implementing the unitary matrices of a braid representation on ordinary qubits constitutes a braid simulation.”

- **Explanation:** It provides evidence of correct control, but it does not by itself provide evidence that the device contains emergent anyons.
-

Detailed treatment: common errors

Common errors

- Reversing clockwise and counterclockwise conventions without stating the change sends σ_i to σ_i^{-1} .
- Relabeling a fusion-tree basis conjugates every representation matrix by the same change of basis. Eigenvalues of closed operations and measurable probabilities stay the same.
- A global phase leaves every computational outcome unchanged. Keeping track of it places gates in $U(d)$ rather than the projective group $PU(d)$, while quotienting by it places them in the projective group; the distinction matters for bookkeeping, and an isolated outcome is unchanged by it.
- Building the unitary matrices of a braid representation on ordinary qubits simulates the braid algebra. Such an experiment demonstrates control over those unitaries; establishing emergent anyons needs evidence about the excitations themselves.
- Continuously tunable gates carry no topological protection by themselves. Protection comes from the physical realization: its energy gap, locality, and error processes.
- A braid group with dense unitary image stays discrete: the domain and the Lie group containing its image remain different structures.

Exercises and checks

Exercises and checks

- **Additional structure required for a Lie group.** A group becomes a Lie group when a compatible smooth-manifold structure makes its multiplication and inversion smooth.
- **Evaluation of $U_z(2\pi)$.** Since $U_z(\theta) = \text{diag}(e^{-i\theta/2}, e^{i\theta/2})$, setting $\theta = 2\pi$ yields $\text{diag}(-1, -1) = -I$.
- **Noncommutativity of B_3 .** The defining relation identifies two three-letter words; it does not identify $\sigma_1\sigma_2$ with $\sigma_2\sigma_1$: the Ising matrices D and FDF^{-1} satisfy the Yang–Baxter relation and do not commute.
- **Difference between a braid and a permutation.** A permutation records only the final rearrangement. A pure braid returns every label to its initial position and can still wind nontrivially in between.
- **Failure of identifying B_n with its unitary image.** The unitary image can be dense in a Lie group although B_n is discrete. A representation sends one structure into the other while keeping them distinct.
- **Relation between a discrete braid group and continuous universal gates.** The unitary image of the braid group can fill a continuous projective unitary group densely.

Sources

Sources

- [R012] E. Artin, “Theory of Braids,” *Annals of Mathematics* 48, 101–126 (1947). DOI: [10.2307/1969218](https://doi.org/10.2307/1969218).
- [R013] C. Kassel and V. Turaev, *Braid Groups*, Graduate Texts in Mathematics 247, Springer (2008). DOI: [10.1007/978-0-387-68548-9](https://doi.org/10.1007/978-0-387-68548-9).

- [R014] B. C. Hall, Lie Groups, Lie Algebras, and Representations: An Elementary Introduction, 2nd ed., Springer (2015). DOI: [10.1007/978-3-319-13467-3](https://doi.org/10.1007/978-3-319-13467-3).
 - [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian Anyons and Topological Quantum Computation,” Reviews of Modern Physics 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083); [arXiv:0707.1889](https://arxiv.org/abs/0707.1889).
 - [R016] E. C. Rowell and Z. Wang, “Mathematics of Topological Quantum Computing,” Bulletin of the American Mathematical Society 55, 183–238 (2018). DOI: [10.1090/bull/1605](https://doi.org/10.1090/bull/1605); [arXiv:1705.06206](https://arxiv.org/abs/1705.06206).
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Appendix D — Four charges and a table of crossings

The toric code has four topological charges whose fusion and braiding fit on one page. Working through its fusion table and its two modular matrices gives concrete content to the S and T matrices of Chapter 40, on an example where every fusion outcome is unique. The later non-Abelian theories keep the same ingredients — charges, fusion spaces, exchange phases — with multi-dimensional fusion spaces added.

Background, scope, and supporting argument

Read the following chain as a memory aid for the order in which structures accumulate, not as a definition. A category consists of objects, morphisms, associative composition, and identity morphisms. A monoidal category adds a product and a unit object. A fusion category is a finite, semisimple, rigid, $\mathbb{C}$ -linear monoidal category with further conditions specified below. Braided and ribbon structures add consistent exchange and twist operations. A modular tensor category is a nondegenerate braided ribbon fusion category in the setting considered here.

$$\begin{aligned} \text{category} &\subset \text{monoidal category} \subset \text{fusion category} \\ &\subset \text{braided/ribbon structures} \subset \text{modular tensor category}. \end{aligned}$$

Each $\subset$ marks added structure — new graphical elements with new axioms — rather than a literal set inclusion. The drawing starts with two elements: a labeled wire and an arrow on a wire.

D.1 Objects Represented by Labeled Wires

A labeled wire represents an object. In an anyon theory, an object is a topological charge type or a direct sum of charge types.

a
|
|
|

Detailed treatment: objects represented by labeled wires

Objects represented by labeled wires

Chapter 14 drew fusion trees whose edges carry charge labels. Each labeled edge stands for an object: one topological charge, or a direct sum of charges. A single isolated edge looks like this:



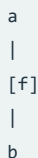
a
|
|
|

The wire labeled a is that object, drawn. Category theory says *object* rather than particle or state, because the wire may stand for a charge sector of a many-body system rather than a localized particle. In an anyon theory the possible labels are topological charge types and their direct sums.

Other categories use different objects: finite-dimensional complex Hilbert spaces with linear maps, or a single unlabeled point whose arrows are the elements of a group.

D.2 Morphisms Represented by Arrows

In a category, a box on a wire stands for a morphism between the two labels — an allowed arrow from the incoming object to the outgoing one. In an anyon theory the allowed arrows are the physical fusion and splitting processes, so the labels must satisfy the fusion rules.



a
|
[f]
|
b

The notation $f : a \rightarrow b$ names this arrow. When the two endpoint labels are distinct simple charges, a superselection rule sets that morphism space to zero: no local operation turns one isolated simple charge into another. Arrows implementing fusion or splitting carry several wires in or out, and the labels on those wires must satisfy the fusion rules.

Detailed treatment: morphisms represented by arrows

Morphisms represented by arrows

A box on a wire stands for a process that can change the wire label from the incoming value to the outgoing value.



a
|
[f]
|
b

With the top-to-bottom convention used in Chapter 14, the box f represents a process that begins at a and ends at b .

The notation $f : a \rightarrow b$ names that process. A *morphism* is any arrow the category allows between its two endpoint objects.

Everything that follows in this appendix is built by composing such arrows: stacking boxes, placing wires side by side, and exchanging them.

For anyons, a morphism is one allowed fusion or splitting channel: one concrete way to pass from the incoming charge, or direct sum of charges, to the outgoing one. The vector space of all morphisms from a to b is denoted by $\text{Hom}(a, b)$.

A *simple charge* is a charge object with no nontrivial decomposition as a direct sum. If a and b are distinct simple charges, they admit no morphisms between them — the superselection rule:

$$\text{Hom}(a, b) = 0 \quad (a \neq b), \quad \text{Hom}(a, a) \cong \mathbb{C}.$$

The second relation says that an endomorphism of a simple charge — a morphism from the object to itself — acts as multiplication by one complex scalar. That is Schur's lemma in categorical form.

Composition and identity morphisms

Take $f : a \rightarrow b$ and $g : b \rightarrow c$. The output label of f matches the input label of g , so the two boxes stack end to end.

```
a a
| |
[f] |
| = [g ∘ f]
b |
| |
[g] c
|
c
```

Stacking defines the composite morphism $g \circ f : a \rightarrow c$: apply the first box first, then the second. Parentheses never matter when three boxes stack, which is associativity. Each object also has an identity morphism, the arrow that leaves the object unchanged:

$$h \circ (g \circ f) = (h \circ g) \circ f, \quad f \circ \text{id}_a = f = \text{id}_b \circ f.$$

The arrow that does nothing to a is written $\text{id}_a : a \rightarrow a$.

A category is exactly these four ingredients: objects, morphisms, associative composition, and identity morphisms. Wires and boxes draw the data; the category is the set of rules saying which drawings are legal and when two drawings name the same process.

The category $\mathbf{Hilb}_{\text{fd}}$ has finite-dimensional complex Hilbert spaces as objects and linear maps as morphisms. A group is a category with one object: the arrows from that object to itself are the group elements, composed by the group law. The category keeps the composition rule and drops the rest.

Without identities, an unadorned wire stops denoting a legal process. Without associativity, three stacked boxes name two composites — one per parenthesization — with nothing saying they agree.

D.3 The Four Simple Charges of the Toric Code

$$\{1, e, m, \varepsilon\}, \quad \varepsilon = e \otimes m$$

Here 1 is the vacuum — no excitation. The charges e and m are the two elementary excitations, and ε is what they form together. Fusing two excitations means asking for their total charge. In this example every pair of simple charges fuses to exactly one outcome, so all fusion spaces are one-dimensional; that uniqueness is what Abelian means here.

Fusion rules. Fusing with the vacuum changes nothing, $1 \otimes a = a$:

$$e \otimes e = m \otimes m = \varepsilon \otimes \varepsilon = 1$$

$$e \otimes m = \varepsilon, \quad e \otimes \varepsilon = m, \quad m \otimes \varepsilon = e$$

Braiding data. With rows and columns ordered as $(1, e, m, \varepsilon)$:

$$S = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

$$T = \text{diag}(1, 1, 1, -1)$$

The off-diagonal signs in S record what happens when one excitation winds around another: carrying e once around m contributes a minus sign, while winding a charge around itself contributes plus one. The diagonal of T records the self-twist phase of each charge in the convention used here: e and m twist trivially, like bosons, while ε picks up a minus sign, like a fermion. The normalization $1/2$ in S sets the rows to unit length; it is the reciprocal of the total quantum dimension $\mathcal{D} = 2$ computed below.

Knowing the degeneracy alone would not fix the exchange phases; the fusion table plus the braiding data together specify how the excitations combine and wind. For the path from these data to topological computation, see [Preskill's lecture notes on topological quantum computation](#).

Detailed treatment: the four simple charges of the toric code

The four simple charges of the toric code

The toric code has four simple charges — four labels with no finer decomposition:

$$\{1, e, m, \varepsilon\},$$

where 1 is the vacuum and $\varepsilon = e \otimes m$ is the composite of the two elementary charges. The symbol $\otimes$ combines charges; Chapter 16 often writes $\times$ for the same operation on charge labels. The fusion rules are

$$e \otimes e = m \otimes m = \varepsilon \otimes \varepsilon = 1,$$

$$e \otimes m = \varepsilon, \quad e \otimes \varepsilon = m, \quad m \otimes \varepsilon = e.$$

Every pair of simple charges fuses to exactly one outcome, so there is never a multi-dimensional fusion space to choose a basis in. Hence every simple charge has quantum dimension one. The *quantum dimension* d_a measures how fast the fusion-state spaces grow when many copies of the charge a fuse together. The *total quantum dimension* $\mathcal{D}$ combines the individual dimensions by summing their squares and taking the square root:

$$\mathcal{D} = \sqrt{d_1^2 + d_e^2 + d_m^2 + d_\varepsilon^2} = 2.$$

The fusion table says what charges combine into; it says nothing about what happens when they move around each other. The braiding data supply that motion: transporting e completely around m contributes a phase of -1 , while transporting e around e or m around m contributes $+1$. In the ordered basis $(1, e, m, \varepsilon)$, the normalized modular matrix is

$$S = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Here each entry S_{ab} is $1/\mathcal{D}$ times the quantum trace of winding a fully around b and back — a closed spacetime loop whose amplitude the ribbon structure evaluates. A *quantum trace* is that closed-loop evaluation. Since every charge here has quantum dimension one, the first row reads $d_a/\mathcal{D} = 1/2$ throughout.

Multiplying the matrix by its Hermitian adjoint gives the identity: with $S^\dagger S = I$, where $S^\dagger$ denotes the conjugate transpose and I the identity matrix, S is unitary and hence invertible. In physical terms, every non-vacuum charge is detected by some other charge: e is detected by m , m by e , and ε by both.

The twist phase of each charge — its topological spin, the phase from a 2π twist of its ribbon — reads

$$\theta_1 = \theta_e = \theta_m = 1, \quad \theta_\varepsilon = -1.$$

So e and m exchange like bosons, while their composite ε exchanges like a fermion. Some authors define the modular T matrix as $T = \text{diag}(\theta_a)$; others include an extra overall phase that depends on the framing and the chiral central charge. The convention here leaves that overall phase out:

$$T = \text{diag}(1, 1, 1, -1).$$

All four charges have quantum dimension one and every fusion outcome is unique, so every anyon in this example is Abelian — yet the example still qualifies as a modular tensor category, because modularity asks whether braiding detects every charge type, as it does here [Ro17]; [Ro18].

Complex-linear and semisimple categories

A category is $\mathbb{C}$ -linear when arrows can be added and scaled by complex numbers: every $\text{Hom}(a, b)$ is a complex vector space and composition respects that linear structure in each argument. It is *semisimple* when every object splits as a finite direct sum of simple objects. Given objects a and b , their direct sum $a \oplus b$ keeps both alternatives present at once, with a and b as its summands.

Isomorphic objects — objects connected by an invertible arrow — carry the same charge even when they are not literally the same object, so the theory keeps one representative label per isomorphism class, collected as $\{a\}$. In an anyon theory those representative labels are the topological charges.

An isomorphism $f : a \rightarrow b$ is a morphism with an inverse $f^{-1} : b \rightarrow a$ going the other way.

Monoidal products and associators

Fusing two charges needs an operation that combines two objects. Graphically, combining is placing two wires side by side.



a b
| |
| |

A *monoidal category* equips a category with a combination operation $\otimes$ — a bifunctor, meaning it combines both objects and morphisms — plus a unit object 1 for the empty combination, plus the regrouping isomorphisms

$$\alpha_{a,b,c} : (a \otimes b) \otimes c \longrightarrow a \otimes (b \otimes c),$$

$$\lambda_a : 1 \otimes a \rightarrow a, \quad \rho_a : a \otimes 1 \rightarrow a.$$

The map α is the *associator*: it identifies the two parenthesizations of a triple product as two bases of the same space. After bases are chosen in the relevant fusion spaces, its matrix entries are the F -symbols. The regrouping is *natural*, meaning it commutes with every compatible morphism — it depends only on the charges, never on how an object was presented.

The associator and unit maps obey the pentagon and triangle coherence equations. Those equations guarantee that every way of removing parentheses, step by step, composes to the same map.

Fusion rules usually suppress parentheses. The associator still matters whenever fusion spaces have dimension above one and bases must be changed there. Appendix E writes the coherence equations out explicitly.

Without an associator, the two parenthesizations of three wires stay two unrelated expressions; nothing yet says they are two bases of one space.

Fusion categories, duals, and dimensions

In this book, a *fusion category* is a $\mathbb{C}$ -linear, finite, semisimple, rigid monoidal category with a simple unit object and finite-dimensional morphism spaces [Ro19]; [Ro20]. The defining conditions have the following meanings:

- finite: there are finitely many simple-object classes;
- semisimple: objects decompose into finite direct sums of simple objects;
- rigid: every object has a dual;
- simple unit: $\text{End}(1) \cong \mathbb{C}$.

Here, $\text{End}(1) = \text{Hom}(1, 1)$ is the endomorphism space of the unit object.

Rigidity assigns a dual object a^* to every object a , together with evaluation and coevaluation morphisms

$$\text{ev}_a : a^* \otimes a \rightarrow 1, \quad \text{coev}_a : 1 \rightarrow a \otimes a^*.$$

These morphisms satisfy the *snake identities*. Algebraically, the identities state that creating a dual pair and then immediately annihilating the appropriate pair leaves the remaining line equal to id_a .

```

a a
| |
| a* |
\ / |
\ / = |
/\ |
/ \ |
| a |
| |
a a

```

In charge language, the dual a^* is written $\bar{a}$.

The number of independent fusion channels is counted by the *fusion coefficients*:

$$N_{ab}^c = \dim \text{Hom}(c, a \otimes b),$$

Here N_{ab}^c counts how many independent channels carry the pair a and b to the outcome c . Equivalently, the fusion decomposes as

$$a \otimes b \cong \bigoplus_c N_{ab}^c c.$$

Some texts write the same count as $\text{Hom}(a \otimes b, c)$: fusing first, then mapping to the outcome, rather than mapping the outcome into the pair. Duality converts between the two, but the index positions move with the convention and must be converted along with it.

Fix the incoming charge a and tabulate all outcomes in the *fusion matrix* $(N_a)^c_b = N_{ab}^c$: rows indexed by the outcome, columns by the partner charge. Its largest nonnegative eigenvalue — the Perron–Frobenius eigenvalue — is the Frobenius–Perron dimension $\text{FPdim}(a)$. In a unitary fusion category this eigenvalue equals the positive categorical dimension, the quantum dimension, d_a used in Chapter 14:

$$d_a d_b = \sum_c N_{ab}^c d_c, \quad d_1 = 1.$$

Quantum dimensions thus multiply when charges fuse and add across the outcomes of a direct sum.

Outside the unitary setting, the categorical dimension of a charge can depend on the chosen pivotal structure — the extra data identifying each object with its double dual — and can differ from the Frobenius–Perron eigenvalue. Every physical anyon model in this book is unitary unless the text says otherwise.

Without rigidity no charge has a canonical antiparticle; without finiteness the sum defining the total quantum dimension can diverge.

Braiding, twists, and ribbon structure

A fusion table lists what charges combine into; it says nothing about exchanging them. A *braiding* supplies that exchange as a natural isomorphism

$$c_{a,b} : a \otimes b \longrightarrow b \otimes a$$

compatible with the associator through two hexagon equations. In chosen fusion bases, $c_{a,b}$ is recorded by R -symbols. Braided here need not mean symmetric: in general, swapping twice need not return the starting state, so

$$c_{b,a} \circ c_{a,b} \neq \text{id}_{a \otimes b}.$$

The composite $c_{b,a} \circ c_{a,b}$ — the double exchange, or monodromy — records that mutual statistic as an operator.

```

a b b a
 \ / \ /
  --> \
 / \ / \
b a a b
```

A *twist* $\theta_a : a \rightarrow a$ turns that ribbon through 2π ; an automorphism is an invertible arrow from an object to itself. On a simple object that arrow is multiplication by a phase, the topological spin of the charge.

A *ribbon category* carries duality, braiding, and twist structures satisfying mutual compatibility. Drawing charges as ribbons rather than lines keeps the framing — the twist count — so self-twists compose consistently, which bare lines would lose [Ro17]; [Ro19].

A *unitary braided fusion category* gives every morphism space an inner product with adjoints, so that all structural maps — associators, braidings, twists, dualities — can be chosen unitary. In that setting an F -move is a change of basis between fusion trees and an R -move is a physical exchange of two charges.

A bare list of R -matrices earns the name braiding only with the hexagon equations: without them the matrices can violate the braid relations. Braiding asks for a swap operation; it asks nothing about squaring to the identity, so anyonic exchange is a swap that need not undo itself.

Nondegenerate braiding and modularity

A charge x is *transparent* when winding any other charge fully around it leaves every state unchanged:

$$c_{a,x} \circ c_{x,a} = \text{id}_{x \otimes a} \quad \text{for all } a.$$

The transparent objects collectively form the *Müger center*. A braided fusion category is *nondegenerate* when that center holds nothing but the vacuum.

With compatible ribbon and unitary structures, a nondegenerate braided fusion category is exactly the modular tensor category used for $(2 + 1)$ -dimensional anyon models. Equivalently, in this setting, the S matrix is invertible [Ro19]; [Ro16].

Here *tensor* abbreviates the monoidal (fusion) product, not a numerical array; *modular* refers to the mapping-class-group action on the torus ground states, which exists exactly when the braiding is nondegenerate — not to modular hardware design.

The matrices S and T act on the torus ground-state space as the two standard large diffeomorphisms — smooth deformations of the torus not deformable to the identity — up to convention-dependent phases. The matrix S packages all mutual-braiding phases, while T packages the topological spins. Together, S and T obey the modular-group relations up to phases.

As identifying data the pair S and T reaches far but not all the way: inequivalent modular categories can share identical modular data. The full F - and R -symbols retain the finer information [Ro19]; [Ro16].

In the toric-code example, S is invertible and the vacuum is the only transparent charge, with $T = \text{diag}(1, 1, 1, -1)$. Those three facts are exactly the modularity check, so the example sits at the top of the opening hierarchy despite having only Abelian charges.

Functors and equivalence of categories

A *functor* $F : \mathcal{C} \rightarrow \mathcal{D}$ maps objects to objects and morphisms to morphisms while preserving identity morphisms and composition:

$$F(\text{id}_a) = \text{id}_{F(a)}, \quad F(g \circ f) = F(g) \circ F(f).$$

A *monoidal functor* comes with coherent isomorphisms identifying $F(a) \otimes F(b)$ with $F(a \otimes b)$ and the image of the unit with the target unit. A *braided monoidal functor* also carries the braiding across. Without those extra maps a functor could match fusion tables while scrambling exchange phases.

Two categories are equivalent when functors back and forth preserve the relevant structure up to natural isomorphism — a relabeling of objects together with compatible maps on arrows. Object names can differ across the two sides. For anyon theories, matching fusion tables alone is too weak; the comparison should be unitary braided or ribbon equivalence.

Swapping the labels e and m in the toric code extends to a full equivalence, because the fusion rules, twists, and mutual braiding map across consistently. A theory and its mirror show the limit of table-matching: they share the same fusion rules with braiding phases complex-conjugated.

As oriented braided theories the two mirrors are generally inequivalent: no structure-preserving relabeling carries one braiding to the other.

The distinction matters when comparing numerics against a reference: a simulation can label its charge sectors differently from the reference convention.

A serious comparison then exhibits a structure-preserving relabeling, the compatible basis changes in each fusion space, and matching amplitudes for closed, gauge-invariant processes. Matching names, or one fusion table, is weaker evidence.

Physical scope and limitations of the categorical description

[Theory] A modular tensor category records the universal long-distance fusion and braiding data of an ideal gapped topological phase: the quantities that survive at distances far beyond the correlation length. The microscopic Hamiltonian, energy gap, quasiparticle mass, coherence time, temperature tolerance, and fabrication route all lie outside it; each needs its own physical analysis [R017]; [R018].

A physical qubit is a controlled two-level subsystem. Labeling it with an anyon name does not place it inside an emergent anyon category. An encoded qubit is a chosen subspace of the many-body Hilbert space.

That subspace earns the name topological only when the system supplies nonlocal storage and protection against local errors. A digital emulator can reproduce the morphisms and modular matrices faithfully while the device sits in no such phase.

Extrinsic symmetry defects — fluxes tied to a global symmetry — can obey generalized fusion rules, often of G -crossed type, where that letter denotes the symmetry group. Such a defect is tied to its symmetry background and is not yet an intrinsic excitation of the host phase. Gauging the symmetry can promote it to one [R021].

A doubled phase from a Drinfeld center contains every sector alongside its opposite-chirality partner. “Doubled Fibonacci” therefore names that doubled theory, not the chiral Fibonacci order.

Review questions and answers

Review questions and answers

- A category rests on four ingredients: objects, morphisms, associative composition, and identity morphisms.

- The toric-code S matrix is invertible by direct multiplication, which gives $S^\dagger S = I$. The same fact reads physically: every non-vacuum charge is detected by braiding with something — e detects m , m detects e , and ε detects both.
- The categorical map recorded by an F -matrix is the associator $\alpha_{a,b,c}$, after bases are chosen in the fusion spaces.
- A fusion category can come without braiding. Fusion and associativity supply $\otimes$ and α ; they provide no exchange map $c_{a,b}$.
- Braiding is modular exactly when it is nondegenerate — when the vacuum is the only transparent charge. In the unitary ribbon setting that condition matches invertibility of the S matrix.
- Modular data and energy gaps live on different levels: topological phases versus microscopic Hamiltonians. The matrices S and T describe topological data but do not specify microscopic energy scales.

Later chapters translate between the two languages: simple objects with charges, $\text{Hom}(c, a \otimes b)$ with fusion spaces, the associator with F , the braiding with R , dual objects with antiparticles, and nondegenerate ribbon braiding with modular structure. Appendix E then fixes bases, tracks how the symbols change under basis changes, and writes the pentagon and hexagon equations that keep the graphical calculus consistent.

Sources

Sources

- [Ro17] A. Kitaev, “Anyons in an Exactly Solved Model and Beyond,” *Annals of Physics* 321, 2–111 (2006). DOI: [10.1016/j.aop.2005.10.005](https://doi.org/10.1016/j.aop.2005.10.005); [arXiv:cond-mat/0506438](https://arxiv.org/abs/cond-mat/0506438).
- [Ro18] M. A. Levin and X.-G. Wen, “String-Net Condensation: A Physical Mechanism for Topological Phases,” *Physical Review B* 71, 045110 (2005). DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110); [arXiv:cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617).
- [Ro19] B. Bakalov and A. Kirillov Jr., *Lectures on Tensor Categories and Modular Functors*, University Lecture Series 21, American Mathematical Society (2001). DOI: [10.1090/ulect/021](https://doi.org/10.1090/ulect/021).

- [R020] P. Etingof, S. Gelaki, D. Nikshych, and V. Ostrik, Tensor Categories, Mathematical Surveys and Monographs 205, American Mathematical Society (2015). DOI: [10.1090/surv/205](https://doi.org/10.1090/surv/205).
 - [R016] E. C. Rowell and Z. Wang, “Mathematics of Topological Quantum Computing,” Bulletin of the American Mathematical Society 55, 183–238 (2018). DOI: [10.1090/bull/1605](https://doi.org/10.1090/bull/1605); [arXiv:1705.06206](https://arxiv.org/abs/1705.06206).
 - [R021] M. Barkeshli, P. Bonderson, M. Cheng, and Z. Wang, “Symmetry Fractionalization, Defects, and Gauging of Topological Phases,” Physical Review B 100, 115147 (2019). DOI: [10.1103/PhysRevB.100.115147](https://doi.org/10.1103/PhysRevB.100.115147); [arXiv:1410.4540](https://arxiv.org/abs/1410.4540).
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Appendix E — Consistency of alternative fusion trees

A fusion tree for several anyons records the order in which pairs combine and the intermediate charge at each step. Two trees with the same incoming charges and the same total charge can span the same state space in different bases. Consistency means the bookkeeping never leaks into the physics: two sequences of basis changes that start and end at the same trees compose to the same linear map.

Background, scope, and supporting argument

Take three particles in a line carrying topological charges a , b , and c : each charge names the superselection sector of its excitation, the sector no local operation can change. Fix their combined charge at d .

Two parenthesizations are possible. Either a and b combine first and the result combines with c , or b and c combine first and a joins the result. Nothing moves; only the grouping on paper changes.

Each parenthesization, together with its allowed intermediate charges, defines one basis of the fixed-total-charge fusion space. Braid calculations may run in either basis, and the translation between them must leave every physical amplitude unchanged.

E.1 Left-Associated and Right-Associated Fusion Bases

The two groupings, drawn with time running downward:

$((a\ b)_e\ c)_d$	$(a\ (b\ c)_f)_d$
$a\ b\ c$	$a\ b\ c$
$\backslash\ /\ $	$\ \ \backslash /\$
$e\ $	$\ \ f$
$\ \backslash /\$	$\ \backslash /\$
d	d

Detailed treatment: left-associated and right-associated fusion bases

Left-associated and right-associated fusion bases

Both diagrams read top to bottom: incoming charges at the top, total charge at the bottom.

Read top to bottom

$((a\ b)_e\ c)_d\ (a\ (b\ c)_f)_d$

```

a b c a b c
 \ / | | \ /
 e | | f
 \ / \ /
 d d

```

The internal labels e and f are the intermediate charges allowed by the fusion rules. When the same three labels can combine in several independent ways, each vertex carries an extra basis label distinguishing the channels. Those vertex bases are orthonormal.

A fusion tree is therefore a parenthesization plus a choice of intermediate charges for fixed incoming and total labels. It names one basis vector, not a collision history: nothing in the tree says one pair of particles physically met before another.

E.2 The F-Move

Above, a, b, c enter and d exits. The left tree groups a with b through e ; the right tree groups b with c through f . The F -move is the change of basis between them:

$$|((ab)_e c)_d; \mu, \nu\rangle = \sum_{f, \alpha, \beta} [F_d^{abc}]_{(e, \mu, \nu)(f, \alpha, \beta)} |(a(bc)_f)_d; \alpha, \beta\rangle$$

The extra indices μ, ν and α, β select among parallel channels when one triple of labels fuses in several independent ways. When every triple fuses in at most one way — a multiplicity-free theory — those indices drop out. The sum visits only channels the fusion rules allow.

An F -move re-labels the same state; the particles stay put. Appendix A changed coordinates on a vector; here the coordinates are the intermediate charge labels.

Detailed treatment: associativity as a change of basis

Associativity as a change of basis

Re-pairing the labels moves no particle. It re-expresses the same fusion state in the other basis:

```
a b c a b c
 \ / | | \ /
  e | --> | f
 \ / \ /
   d d
```

Let μ label the vertex $a, b \rightarrow e$, and let ν label the vertex $e, c \rightarrow d$ in the left-associated tree. Let α label the vertex $b, c \rightarrow f$, and let β label the vertex $a, f \rightarrow d$ in the right-associated tree. The transformation between these bases is

$$|((ab)_e c)_d; \mu, \nu\rangle = \sum_{f, \alpha, \beta} [F_d^{abc}]_{(e, \mu, \nu)(f, \alpha, \beta)} |(a(bc)_f)_d; \alpha, \beta\rangle.$$

Each vertex label enumerates the independent channels of its triple: $\mu = 1, \dots, N_{ab}^e$, $\nu = 1, \dots, N_{ec}^d$, $\alpha = 1, \dots, N_{bc}^f$, and $\beta = 1, \dots, N_{af}^d$. The integer N_{ab}^e counts how many independent ways charges a and b reach total charge e .

Those matrix elements are the F -symbols: the entries of the basis-change matrix for this three-body space, recording no additional physical event.

In a unitary anyon theory — one whose fusion spaces carry inner products preserved by all allowed maps — F_d^{abc} is a unitary matrix on the fixed- d fusion space. Without unitarity the two trees still span the same abstract space, but changing basis can distort norms, so probabilities computed with a naive conjugate transpose come out wrong.

E.3 The R-Move

Once two anyons sit in a definite total channel c , exchanging their positions maps the fusion space for one ordering to the space for the other:

$$R_c^{ab} : V_{ab}^c \rightarrow V_{ba}^c$$

Here V_{ab}^c collects the distinct ways a, b fuse to c . In a larger tree the pair to be exchanged may sit apart; the standard maneuver re-pairs the tree until the pair is adjacent, applies R , and returns with the inverse F . Braiding matrices therefore mix recoupling entries with exchange phases.

Detailed treatment: oriented exchange of two charges

Oriented exchange of two charges

Physically swapping two charges moves them. Fixing the total charge c and choosing the counterclockwise sense of the swap gives the linear map

$$R_c^{ab} : V_{ab}^c \longrightarrow V_{ba}^c.$$

Here V_{ab}^c holds the distinct channels through which a and b reach total charge c .

The entries of that map are the R -symbols. An F -move re-pairs labels on paper; an R -move records a physical swap with a chosen orientation.

In a multiplicity-free theory — every triple fuses in at most one way — each allowed channel c gives a single phase R_c^{ab} . With multiplicities present, the same symbol denotes a unitary matrix mixing the parallel vertex channels.

The conventions fix domains with orientation: the clockwise swap of the same pair is $(R_c^{ba})^{-1}$, with the labels transposed and the map inverted. Complex-conjugating every R -symbol reproduces that inverse only after the charge labels and orientation conventions are matched; applied blindly it gives the wrong operator.

Dropping the orientation labels collapses clockwise and counterclockwise swaps onto the same notation while the operators differ. The hexagon equations below compare specific oriented swaps, so they need those labels kept.

Three Ising σ anyons

The Ising charges 1 , ψ , and σ satisfy the fusion rules

$$\sigma \times \sigma = 1 + \psi, \quad \psi \times \sigma = \sigma, \quad \psi \times \psi = 1.$$

Take three σ anyons with total charge σ . In the left-associated tree the first pair fuses through $e \in \{1, \psi\}$. In the standard gauge — the standard choice of vertex basis phases —

$$F \equiv F_{\sigma}^{\sigma\sigma\sigma} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

$$R_1^{\sigma\sigma} = e^{-i\pi/8}, \quad R_{\psi}^{\sigma\sigma} = e^{3i\pi/8}.$$

Rows and columns of F run in the order $(1, \psi)$. Multiplying the matrix by itself returns the identity, which together with unitarity $F^{\dagger}F = I$ gives $F^{-1} = F$. A state carrying left intermediate channel $e = 1$ therefore reads in the right-associated basis as

$$|1\rangle_L = \frac{|1\rangle_R + |\psi\rangle_R}{\sqrt{2}}.$$

Measuring the right intermediate channel then returns either outcome with probability $1/2$. The fusion rule lists the two possible outcomes; the F -matrix fixes the amplitude of each.

Exchanging the first pair in the left-associated basis gives

$$B_1 = \begin{pmatrix} R_1^{\sigma\sigma} & 0 \\ 0 & R_{\psi}^{\sigma\sigma} \end{pmatrix} = e^{-i\pi/8} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

; the second pair is not adjacent there, so an F -move first re-pairs the tree to expose its channel:

$$B_2 = FB_1F^{-1} = \frac{e^{-i\pi/8}}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}.$$

Writing $D_0 = \text{diag}(1, i)$ for the phase pattern and setting the common phase aside for a line, direct multiplication gives

$$D_0(FD_0F)D_0 = (FD_0F)D_0(FD_0F) = \frac{1+i}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Multiplying the common phase $e^{-3i\pi/8}$ back onto both sides gives

$$B_1 B_2 B_1 = B_2 B_1 B_2.$$

The local F - and R -symbols therefore assemble into matrices satisfying the three-strand braid relation, while $B_1 B_2 \neq B_2 B_1$ keeps the representation non-Abelian. Satisfying the braid relation leaves commutativity open [Ro15]; [Ro17].

The basis-independent associator

Before choosing bases at the fusion vertices, reassociation is the abstract map

$$\alpha_{a,b,c} : (a \otimes b) \otimes c \rightarrow a \otimes (b \otimes c).$$

At the categorical level the tensor product $\otimes$ combines charges, and α is the associator regrouping a triple. Choosing bases in the relevant fusion spaces turns α into a matrix whose entries are the F -symbols.

For fixed a, b, c, d , orthonormal vertex bases turn unitarity of the map into the matrix identity

$$\sum_{f,\alpha,\beta} [F_d^{abc}]_{(e,\mu,\nu)(f,\alpha,\beta)} [F_d^{abc}]_{(e',\mu',\nu')(f,\alpha,\beta)}^* = \delta_{ee'} \delta_{\mu\mu'} \delta_{\nu\nu'},$$

where $*$ is complex conjugation and δ is one on equal indices and zero otherwise. The identity needs orthonormal vertex bases: with non-orthonormal bases the abstract associator α stays an isomorphism, but its matrix need not satisfy this conjugate-transpose relation.

E.4 The Pentagons and Hexagons

Four objects admit five parenthesizations, arranged as the five corners the pentagon visits. The pentagon relation demands that the two routes between the extreme corners compose to the same map. Writing the associator as $\alpha_{a,b,c} : (a \otimes b) \otimes c \rightarrow a \otimes (b \otimes c)$; each composite applies its rightmost map first.

The pentagon equation:

$$\alpha_{a,b,c \otimes d} \circ \alpha_{a \otimes b,c,d} = (\text{id}_a \otimes \alpha_{b,c,d}) \circ \alpha_{a,b \otimes c,d} \circ (\alpha_{a,b,c} \otimes \text{id}_d)$$

Reassociation and braiding meet in the hexagon equations. Writing $\beta_{a,b} : a \otimes b \rightarrow b \otimes a$ for the braiding map, the first hexagon moves one object past a fused pair in one step and past its two constituents in two steps, with associators bridging the two parenthesizations:

$$\alpha_{b,c,a} \circ \beta_{a,b \otimes c} \circ \alpha_{a,b,c} = (\text{id}_b \otimes \beta_{a,c}) \circ \alpha_{b,a,c} \circ (\beta_{a,b} \otimes \text{id}_c)$$

$$\alpha_{c,a,b}^{-1} \circ \beta_{a \otimes b, c} \circ \alpha_{a,b,c}^{-1} = (\beta_{a,c} \otimes \text{id}_b) \circ \alpha_{a,c,b}^{-1} \circ (\text{id}_a \otimes \beta_{b,c})$$

Detailed treatment: four-charge reassociation and the pentagon equation

Four-charge reassociation and the pentagon equation

Reassociating four objects from $((a \otimes b) \otimes c) \otimes d$ to $a \otimes (b \otimes (c \otimes d))$ offers two routes through the intermediate parenthesizations:

```
((a b) (c d))
/ \
(((a b) c) d) (a (b (c d)))
\ /
((a (b c)) d) -> (a ((b c) d))
```

The upper route composes two associators; the lower route composes three. The pentagon equation equates the two composites:

$$\alpha_{a,b,c \otimes d} \circ \alpha_{a \otimes b, c, d} = (\text{id}_a \otimes \alpha_{b,c,d}) \circ \alpha_{a,b \otimes c, d} \circ (\alpha_{a,b,c} \otimes \text{id}_d).$$

Both composites run from the same source tree to the same target tree, each read rightmost-first. Expanding the identity in fusion-tree bases turns it into polynomial equations for the entries of the F -matrices, summing over every allowed intermediate charge and multiplicity label [Ro22]; [Ro19].

The equation is visible even in the simplest case: in a pointed model with a gauge where every allowed fusion space is one-dimensional and every associator is $F = 1$, the pentagon reads $1 \cdot 1 = 1 \cdot 1 \cdot 1$.

Beyond that trivial case, pointed categories can carry phase-valued associators classified by a group 3-cocycle, and the pentagon equation becomes the cocycle condition. The Ising and Fibonacci theories put nontrivial matrix-valued F -symbols through the same test.

The pentagon is a coherence condition in the strict sense: dropping it lets a four-anyon amplitude depend on which sequence of intermediate basis changes produced it, so no single linear map answers all routes.

Braiding an object past a composite

The categorical braiding is the map

$$\beta_{a,b} : a \otimes b \rightarrow b \otimes a$$

Its matrix entries in chosen bases are the R -symbols. Moving a past the composite $b \otimes c$ in one exchange must match moving a past b and then past c in two. Each move starts and ends in a different parenthesization, so associators bridge the exchanges on each side.

Reading composites rightmost-first, the positive-crossing identity reads

$$\alpha_{b,c,a} \circ \beta_{a,b \otimes c} \circ \alpha_{a,b,c} = (\text{id}_b \otimes \beta_{a,c}) \circ \alpha_{b,a,c} \circ (\beta_{a,b} \otimes \text{id}_c).$$

Both composites run from $(a \otimes b) \otimes c$ to $b \otimes (c \otimes a)$. The companion identity moves a composite past c instead:

$$\alpha_{c,a,b}^{-1} \circ \beta_{a \otimes b,c} \circ \alpha_{a,b,c}^{-1} = (\beta_{a,c} \otimes \text{id}_b) \circ \alpha_{a,c,b}^{-1} \circ (\text{id}_a \otimes \beta_{b,c}).$$

Both composites run from $a \otimes (b \otimes c)$ to $(c \otimes a) \otimes b$. Each identity strings three exchanges and three regroupings into a six-sided loop; the hexagon equations demand that loop compose to one unambiguous map.

With the pentagon equation, the hexagon equations guarantee that any two labeled ribbon diagrams differing by allowed moves evaluate to the same operator — F -moves and crossings assign each equivalence class one value [Ro22]; [Ro23]. Each hexagon governs one side of the tensor product: omitting the first leaves exchange past a right-hand composite unconstrained, and omitting the second leaves exchange past a left-hand composite unconstrained.

Written out in components, the first hexagon in a multiplicity-free theory contributes, among others, the relation

$$R_e^{ab}[F_d^{bac}]_{eg}R_g^{ac} = \sum_f [F_d^{abc}]_{ef}R_d^{af}[F_d^{bca}]_{fg}.$$

The relation follows the downward-tree, counterclockwise-exchange conventions of this appendix, with each basis change taken in the fixed direction, and the sum visits only allowed intermediate charges f . Sources using upward-oriented trees, clockwise R -symbols, or the inverse F convention print equations that look different and match after the conventions are translated.

Derivation of the braid relations

For three anyons in the left-associated basis, B_1 applies R to the first pair directly, while B_2 reaches the second pair by the re-pair, exchange, return maneuver: an F -move, then an R -move on the exposed pair, then the inverse F -move. Because the hexagon equations reconcile exchange with fusion and the pentagon equation reconciles the reassociation routes with each other, the two matrices satisfy

$$B_i B_{i+1} B_i = B_{i+1} B_i B_{i+1}, \quad B_i B_j = B_j B_i \quad (|i - j| \geq 2).$$

The Ising calculation above checks the three-strand identity on two-by-two matrices. Those relations qualify matrices as a braid representation; density of the image and computational universality are further properties the relations alone leave undecided.

Double braiding and monodromy

One exchange sends the ordered pair a, b to the opposite order. Winding one charge fully around the other and back — the double braid, or monodromy — restores the order. In a multiplicity-free fusion channel c that closed winding acts as

$$M_c^{ab} = R_c^{ab} R_c^{ba}.$$

The single exchange moves between the two orderings; the double braid starts and ends in the same ordering, which is why interferometry measures the double braid.

In a ribbon theory — braiding and twisting made mutually compatible — the double braid reduces to twist phases through the balancing relation:

$$M_c^{ab} = \frac{\theta_c}{\theta_a \theta_b},$$

where θ_x is the phase of a 2π twist of the ribbon carrying simple charge x . Vertex rephasings of the kind described below cancel out of this ratio, so the monodromy — unlike either individual R phase on its own — matches what an interference experiment extracts [Ro15]; [Ro23].

In the standard gauge the Ising twists read $\theta_1 = 1$, $\theta_\psi = -1$, and $\theta_\sigma = e^{i\pi/8}$ for $a = b = \sigma$. The vacuum channel then gives

$$M_1^{\sigma\sigma} = (e^{-i\pi/8})^2 = e^{-i\pi/4} = \frac{\theta_1}{\theta_\sigma^2},$$

and the ψ channel gives

$$M_\psi^{\sigma\sigma} = (e^{3i\pi/8})^2 = e^{3i\pi/4} = \frac{\theta_\psi}{\theta_\sigma^2}.$$

The two channels differ by that relative sign. Twist phases are quoted up to an overall framing choice, so any table reporting them must name the framing convention it uses.

Gauge transformations of fusion vertices

Each fusion vertex admits its own choice of orthonormal basis. In a multiplicity-free theory that freedom is one phase per vertex:

$$|a, b; c\rangle' = u_c^{ab} |a, b; c\rangle, \quad |u_c^{ab}| = 1.$$

Rephasing the vertices sends the symbols to

$$[F_d^{abc}]'_{ef} = \frac{u_e^{ab} u_d^{ec}}{u_f^{bc} u_d^{af}} [F_d^{abc}]_{ef},$$

$$[R_c^{ab}]' = \frac{u_c^{ab}}{u_c^{ba}} R_c^{ab}.$$

With fusion multiplicities each u widens to a unitary matrix contracting the parallel channels. Either way the transformation is a gauge change: the same physics written in a different vertex basis.

The pentagon and hexagon equations hold in every gauge. Changing the external computational basis conjugates the complete braid operators accordingly.

A single entry such as $[F]_{ef}$ shifts under rephasing and carries no invariant meaning on its own. What survives rephasing is the gauge-invariant content: fusion probabilities stated with their vertex conventions, spectra and traces of closed braids, topological spins, and modular data. Matching those quantities across sources still needs the framing and overall-phase conventions spelled out [\[R015\]](#); [\[R019\]](#).

Mirroring reverses every crossing. In a unitary convention the mirrored braiding data read as inverses or complex conjugates, but the bookkeeping does not end there: orientation labels and framing phases still need tracking. Chiral Fibonacci theory, its mirror, and the doubled theory holding both therefore stay three distinct theories; dropping a minus sign does not convert one into another.

Limits of consistency tables

[Theory] A table of N , F , and R satisfying unitarity and the pentagon and hexagon equations is internally consistent as topological data. Internal consistency leaves two further questions open: whether a local Hamiltonian realizes those data, and whether the corresponding many-body system has a gap [R017]; [R019].

[Numerics] Approximate matrices extracted from a finite system come with four mandatory appendices: the basis conventions, the finite-size scaling, the measured leakage outside the proposed fusion space, and the residual path dependence. Checking one braid word exercises one path; the coherence structure comprises all of them.

[Experiment] A processor programmed with the Ising matrices above demonstrates control over that representation to its measured accuracy — evidence about the gates, not about the medium. Intrinsic anyons need evidence of a different kind: localized excitations, addressable fusion sectors, and braid responses emerging from the device's many-body physics. Writing anyonic labels on the basis vectors of an encoded subspace supplies none of that evidence on its own.

Common sources of error

Common sources of error

- Multiplying braid words using one temporal ordering convention while drawing them using the opposite convention.
- Applying F where F^{-1} is required to return to the original fusion tree.
- Summing over forbidden intermediate labels instead of enforcing the fusion coefficients.
- Dropping the vertex labels that distinguish parallel channels when $N_{ab}^c > 1$.
- Treating gauge-dependent symbol entries from different vertex-phase choices as physical disagreements.
- Calling residual pentagon and hexagon violations harmless numerical noise without checking them against the claimed precision.

- Concluding computational universality from noncommuting braid matrices alone — density of the image is the further property universality needs.

Verification exercises

Verification exercises

Check that $F_{\sigma}^{\sigma\sigma\sigma}$ squares to the identity.

Multiplying the displayed matrix by itself returns I , and with unitarity $F^{\dagger}F = I$ that product is exactly $F^{-1} = F$.

What does one R -symbol record?

An R -symbol records one oriented exchange of two charges fusing to the specified total channel.

Do both sides of the pentagon equation start and end at the same trees?

Each composite starts at $((a \otimes b) \otimes c) \otimes d$ and finishes at $a \otimes (b \otimes (c \otimes d))$: two factorizations of one map.

Why are there two hexagon equations rather than one?

Braiding must clear composites on either side. One equation moves a single object past a composite; the other moves a composite past a single object; together the two equations cover both sides.

Verify that $M_1^{\sigma\sigma} = \theta_1/\theta_{\sigma}^2$ using the listed phases.

$R_1^{\sigma\sigma} = e^{-i\pi/8}$ squares to $e^{-i\pi/4}$, while $\theta_1/\theta_{\sigma}^2 = 1/e^{i\pi/4} = e^{-i\pi/4}$.

State the consequence of omitting the pentagon equation.

A four-anyon amplitude can then depend on the selected sequence of intermediate basis changes. The symbols no longer define a single linear map.

Sources

Sources

- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian Anyons and Topological Quantum Computation,” *Reviews of Modern Physics* 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083); [arXiv:0707.1889](https://arxiv.org/abs/0707.1889).
 - [R017] A. Kitaev, “Anyons in an Exactly Solved Model and Beyond,” *Annals of Physics* 321, 2–111 (2006). DOI: [10.1016/j.aop.2005.10.005](https://doi.org/10.1016/j.aop.2005.10.005); [arXiv:cond-mat/0506438](https://arxiv.org/abs/cond-mat/0506438).
 - [R022] G. Moore and N. Seiberg, “Classical and Quantum Conformal Field Theory,” *Communications in Mathematical Physics* 123, 177–254 (1989). DOI: [10.1007/BF01238857](https://doi.org/10.1007/BF01238857).
 - [R019] B. Bakalov and A. Kirillov Jr., *Lectures on Tensor Categories and Modular Functors*, University Lecture Series 21, American Mathematical Society (2001). DOI: [10.1090/ulect/021](https://doi.org/10.1090/ulect/021).
 - [R023] P. Bonderson, K. Shtengel, and J. K. Slingerland, “Interferometry of Non-Abelian Anyons,” *Annals of Physics* 323, 2709–2755 (2008). DOI: [10.1016/j.aop.2008.01.012](https://doi.org/10.1016/j.aop.2008.01.012); [arXiv:0707.4206](https://arxiv.org/abs/0707.4206).
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Appendix F — Unitary elimination of high-energy subspaces

Chapter 37 explained effective couplings as excursions into high-energy states and back. This appendix turns that picture into an operator calculation: a Schrieffer–Wolff rotation re-bases the full Hilbert space so the low-energy block stands alone through a chosen perturbative order, keeping exactly the return-trip corrections that bare projection drops. The reference for conditions and error bounds is [Bravyi, DiVincenzo, and Loss](#).

Background, scope, and supporting argument

An earlier calculation compressed two low-energy states plus one high-energy state into an effective 2×2 matrix, whose induced coupling came out as $-g^2/\Delta$: coupling squared over gap, with a minus sign from the energy denominator.

No high-energy eigenstate survives in the reduced description. The excitation survives only as an energy denominator suppressing the induced coupling.

That three-state example generalizes to a whole low-energy family — including a degenerate family sharing one unperturbed energy. This appendix derives the rotation in general and then evaluates it in closed form for the two-site Hubbard model.

Deleting the high-energy basis vectors gives a different, cruder answer: truncation keeps only processes staying inside the low sector and drops every excursion that leaves it and returns.

The remedy is a unitary rotation — a norm-preserving change of basis — applied to the full space before truncating the Q block. The rotation decouples the P low block from the Q high block order by order; truncating after the rotation keeps the excursion effects inside the retained block.

before rotation after rotation

P states Q states P block | 0

-----+-----

0 | Q block

When several retained states share one unperturbed energy, single-state perturbation theory encounters zero energy differences. Degenerate perturbation theory therefore rotates the whole P family as one block while the Q family remains eliminated; the

correction acts on an unperturbed eigenspace of dimension greater than one rather than on an individual state.

F.1 Decomposition into Retained and Eliminated Subspaces

Write $H = H_0 + \lambda V$, where H_0 has a known isolated low-energy subspace and λ is a bookkeeping parameter. Let P project onto the low subspace and Q onto everything orthogonal to it:

$$P^2 = P, \quad Q^2 = Q, \quad PQ = 0, \quad P + Q = I$$

Detailed treatment: decomposition into retained and eliminated subspaces

Decomposition into retained and eliminated subspaces

Cut the Hilbert space with two complementary operators P and Q : orthogonal projectors selecting mutually orthogonal subspaces, satisfying

$$P^2 = P, \quad Q^2 = Q, \quad PQ = 0, \quad P + Q = I.$$

The identity I is their sum. The range of P is the retained low sector; the range of Q is the eliminated high sector. The Hamiltonian plus its perturbation reads

$$H = H_0 + \lambda V,$$

where H_0 diagonalizes explicitly, V is Hermitian and carries units of energy, and λ is the dimensionless power-counter for the expansion. Once the series is truncated and the physical small ratio identified, λ returns to one.

Start with the degenerate case: every retained state shares one unperturbed energy E_0 :

$$PH_0P = E_0P,$$

with no eigenvalue of QH_0Q equal to E_0 : the retained level stands clear of the eliminated spectrum. Their separation is the gap

$$\Delta_0 = \min_{\mu \in Q} |E_\mu - E_0| > 0,$$

with E_μ running over the H_0 eigenvalues inside Q . A nonzero gap in H_0 makes the resolvent $(E_0 - QH_0Q)^{-1}$ a well-defined operator on Q . A closing gap destroys that inverse, which signals the cut between P and Q was drawn in the wrong place.

Every operator splits into a block-diagonal piece staying inside P or inside Q , plus an off-diagonal piece crossing between them. For the perturbation that split reads

$$V_d = PVP + QVQ, \quad V_{od} = PVQ + QVP.$$

The off-diagonal piece V_{od} is the bridge between the sectors. Bare projection PHP drops that bridge, keeping only stay-at-home processes. The rotation below folds the leave-and-return excursions back into the retained block.

F.2 The Schrieffer-Wolff Transformation

Pick an anti-Hermitian generator $S - S^\dagger = -S$, so its exponential is unitary — to rotate $e^S H e^{-S}$ toward block-diagonal form, order by order. The retained block then carries the excursion corrections inside ordinary low-energy matrix elements.

The displayed formula covers the degenerate case $PH_0P = E_0P$, with retained energy E_0 . Its resolvent acts only inside Q , where the gap keeps it finite. A retained band of finite width requires the quasi-degenerate extension developed in the detailed treatment below.

Through second order the retained block reads:

$$H_{\text{eff}} = E_0P + \lambda PVP + \lambda^2 PVQ \frac{1}{E_0 - QH_0Q} QVP + O(\lambda^3)$$

Read the second-order term right to left as a three-step excursion: QVP hops from the retained sector into the excluded sector; the resolvent weights the visit by its energy denominator; PVQ hops back. In an excluded eigenbasis that excursion is

$$\langle a | H_{\text{eff}}^{(2)} | b \rangle = \lambda^2 \sum_{m \in Q} \frac{\langle a | V | m \rangle \langle m | V | b \rangle}{E_0 - E_m}.$$

With all excluded levels above E_0 every denominator is negative — but the sign of a given effective coupling still depends on the numerator matrix elements as well. Collapsing all denominators to one gap and all numerators to one scale yields an order-of-magnitude estimate, not the coupling.

Detailed treatment: perturbative block diagonalization

Perturbative block diagonalization

Choose an anti-Hermitian generator S with $S^\dagger = -S$, connecting only the two subspaces — no block-diagonal part:

$$PSP = QSQ = 0.$$

Exponentiation gives the unitary $U = e^S$. The rotated Hamiltonian is

$$\widetilde{H} = e^S H e^{-S}.$$

Conjugation expands in nested commutators — the Baker–Campbell–Hausdorff series:

$$\widetilde{H} = H + [S, H] + \frac{1}{2!} [S, [S, H]] + \dots,$$

with $[A, B] = AB - BA$. Expand the generator in powers of the bookkeeping parameter:

$$S = \lambda S_1 + \lambda^2 S_2 + \dots.$$

Collecting the terms linear in λ , the surviving off-diagonal piece is

$$\lambda(V_{\text{od}} + [S_1, H_0]).$$

It vanishes by the choice

$$[H_0, S_1] = V_{\text{od}}.$$

For $|a\rangle$ in P and $|\mu\rangle$ in Q , the condition reads in components

$$\langle a|S_1|\mu\rangle = \frac{\langle a|V|\mu\rangle}{E_0 - E_\mu}.$$

Anti-Hermiticity fixes the reverse element S_1 . Projecting the rotated Hamiltonian back into P gives the retained block through second order:

$$H_{\text{eff}} = E_0 P + \lambda P V P + \lambda^2 P V Q \frac{1}{E_0 - Q H_0 Q} Q V P + O(\lambda^3).$$

Here $O(\lambda^3)$ collects all terms of third and higher order in λ . The effective Hamiltonian is thus no new physical system: it is one block of the rotated full Hamiltonian, viewed in the rotated basis.

The boxed block is Hermitian, and its second-order term narrates the excursion: out of P via V , dwell in Q weighted by the resolvent, back via V . The units check out as

$$[V][E^{-1}][V] = E E^{-1} E = E.$$

With $E_\mu > E_0$ each denominator $E_0 - E_\mu < 0$ is negative, so a coupled low-energy state typically shifts downward at second order. The word “virtual” alone does not fix that sign; the denominator does. Here “virtual” means one thing: the intermediate visit occupies a state outside the retained subspace. The many-body version with locality and error bounds is developed by Bravyi, DiVincenzo, and Loss [Ro25].

Decoupling P from Q order by order, without touching the exact spectrum, is the Schrieffer–Wolff transformation. The name comes from the original Anderson-to-Kondo application; current usage covers any low-energy block coupled to high-energy

states [Ro24]; [Ro25].

When the retained band has finite width, write $|a\rangle$ and $|b\rangle$ in P with their own unperturbed energies E_a and E_b . The Hermitian second-order form — the average of the two one-sided denominators — is

$$(H_{\text{eff}}^{(2)})_{ab} = \frac{1}{2} \sum_{\mu \in Q} V_{a\mu} V_{\mu b} \left(\frac{1}{E_a - E_\mu} + \frac{1}{E_b - E_\mu} \right).$$

Here $V_{a\mu} = \langle a|V|\mu\rangle$. Setting $E_a = E_b = E_0$ collapses both denominators to the boxed form. Other conventions for the effective Hamiltonian rotate the result by a unitary acting only inside P ; spectra and consistently rotated observables then agree through the retained order [Ro25].

F.3 The Two-Site Hubbard Model

Two sites share two electrons with repulsive on-site cost $U > 0$ per double occupancy and hopping amplitude t between sites. At half filling with $|t| \ll U$ the low sector holds one electron per site. One hop creates a doubly occupied site at cost U ; the second hop returns to the singly occupied sector.

Only the singlet can make that round trip, as the Pauli analysis below shows; the triplets are blocked. With dimensionless spin-1/2 operators the second-order retained block is:

$$H_{\text{eff}}^{(2)} = J \left(\mathbf{S}_1 \cdot \mathbf{S}_2 - \frac{1}{4} n_1 n_2 \right), \quad J = \frac{4t^2}{U} > 0$$

Inside the singly occupied sector $n_1 n_2 = 1$, so the operator reduces to numbers on the two spin sectors: the singlet ($\mathbf{S}_1 \cdot \mathbf{S}_2 = -3/4$) shifts by $-J$; each triplet ($+1/4$) shifts by zero. The positive exchange coefficient thus lowers the singlet below the triplets. That spin splitting from round-trip charge motion is superexchange.

Defect-cluster gadgets borrow this round-trip logic, but each gadget needs its own matrix elements and a full accounting of competing terms: the two-site value fixes no four-body coefficient.

Detailed treatment: two-site Hubbard superexchange

Two-site Hubbard superexchange

Consider two sites, labeled 1 and 2, with one spin-1/2 fermion per site. A fermion is a particle whose creation and annihilation operators obey anticommutation relations. Let $c_{i\sigma}^\dagger$ create a fermion of spin $\sigma \in \{\uparrow, \downarrow\}$ at site i , and define the corresponding occupation-number operator by $n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$. The two-site Hubbard Hamiltonian is divided into

$$H_0 = U \sum_{i=1}^2 n_{i\uparrow} n_{i\downarrow},$$

$$V = -t \sum_{\sigma} \left(c_{1\sigma}^\dagger c_{2\sigma} + c_{2\sigma}^\dagger c_{1\sigma} \right).$$

The on-site repulsion $U > 0$ is the energy cost of placing two fermions on the same site, and the hopping amplitude t controls motion between the sites. Both U and t have units of energy. The retained subspace P contains states with exactly one fermion on each site, so $E_0 = 0$.

The eliminated subspace Q contains states with one doubly occupied site and one empty site. These states have energy U . The perturbative assumption is

$$|t|/U \ll 1.$$

The first-order contribution vanishes because one hop always takes a state out of the one-particle-per-site sector. Thus $PVP = 0$. At second order,

$$H_{\text{eff}}^{(2)} = -\frac{1}{U} PVQVP.$$

Introduce the spin operator on site i ,

$$\mathbf{S}_i = \frac{1}{2} \sum_{\alpha, \beta} c_{i\alpha}^\dagger \boldsymbol{\sigma}_{\alpha\beta} c_{i\beta},$$

where $\sigma = (X, Y, Z)$ collects the three Pauli matrices. Summing the two-hop paths over intermediate sites and spins gives [Ro26]

$$H_{\text{eff}}^{(2)} = \frac{4t^2}{U} \left(\mathbf{S}_1 \cdot \mathbf{S}_2 - \frac{1}{4} n_1 n_2 \right),$$

where $n_i = n_{i\uparrow} + n_{i\downarrow}$. Within P , each site is singly occupied, so $n_1 n_2 = 1$.

Define the exchange coefficient

$$J_{\text{ex}} = 4t^2/U.$$

For the spin singlet, $\mathbf{S}_1 \cdot \mathbf{S}_2 = -3/4$, and the second-order energy shift is $-J_{\text{ex}} = -4t^2/U$.

For any triplet, $\mathbf{S}_1 \cdot \mathbf{S}_2 = +1/4$, and the corresponding shift is zero. The singlet–triplet separation is therefore

$$\Delta_{ST} = \frac{4t^2}{U}.$$

The result is antiferromagnetic superexchange: virtual hops through doubly occupied states leave behind a spin coupling favoring the singlet. The coupling is no new microscopic hopping: it is two hops times one denominator U . Written as $4t(t/U)$ it is visibly smaller than bare hopping by the small ratio $|t|/U \ll 1$.

Pauli exclusion separates the two cases. A triplet carries symmetric spin, so its spatial part is antisymmetric — and that antisymmetry blocks the on-site double occupancy the round trip needs.

The singlet carries the complementary spatial symmetry, so its round trip proceeds and its energy drops. The symmetry argument and the explicit two-hop algebra agree.

Procedure for validating an effective Hamiltonian

On a finite-dimensional problem, build and check the effective Hamiltonian through the following steps. Each step has a failure mode worth naming.

- Choose the projector P by energy scales: include every state whose splittings are comparable to the perturbation. Leaving a near-resonant state outside shrinks a denominator instead of simplifying the theory.
- Diagonalize PVP first: within an exactly degenerate manifold its eigenvectors are the correct zeroth-order combinations.
- Construct the resolvent $(E_0 - QH_0Q)^{-1}$ — the inverse operator packaging all intermediate-state energy denominators — acting only on Q . Each eliminated-state contribution enters divided by its excitation-energy difference.
- Assemble H_{eff} order by order, keeping every operator the order produces — including unwelcome fields and bare constants.
- Rotate observables with the same unitary. For a microscopic observable O the consistent effective operator is

$$O_{\text{eff}} = Pe^S O e^{-S} P.$$

while bare truncation POP drops the virtual-transition corrections.

- Benchmark the effective model against the full Hamiltonian on the smallest tractable cluster: low-energy eigenvalues, spectral subspaces (not just individual eigenvectors), and rotated observables, tracked as $|\lambda|$ varies.

The expansion is controlled when the coupling matrix elements between P and Q are small against the relevant denominators. On a finite cluster that condition is summarized by the dimensionless ratio $\|V_{\text{od}}\|/\Delta_0$, with $\|\cdot\|$ the operator norm.

On an extended lattice the full perturbation norm grows with volume, so the global ratio overestimates the danger. Rigorous bounds use locality and linked-cluster structure instead of the extensive norm alone [Ro25]. A large global norm still allows a controlled local expansion, while one small bond does not certify control.

Existence and accuracy are separate questions. The existence question asks whether some exact unitary maps the perturbed low-energy spectral subspace back onto the reference P .

For a finite-dimensional problem whose isolated band stays isolated under the perturbation, that unitary exists, which is the geometric fact behind exact Schrieffer–Wolff theory [Ro25]; [Ro27]. The accuracy question is whether a short Taylor series captures that unitary well enough at the retained order.

Accuracy needs its own evidence: a small parameter plus a direct truncation check. An exact rotation can exist while the second-order formula still fails.

An effective Hamiltonian carries two redundancies: an overall additive constant and a unitary basis choice inside P . Constants shift all levels together, leaving dynamics and gaps untouched.

Basis rotations inside P shuffle individual matrix entries, so two derivations can print different-looking Hamiltonians for the same physics. Compare in the invariant-first order: spectra first, then observables rotated under the same convention, and only then individual terms.

A many-body effective Hamiltonian can print a term shaped like a toric-code check operator. That resemblance is one term among many: topological order is a property of the full low-energy spectrum — degeneracy, gaps, and excitations — and needs that whole-spectrum analysis.

Gadget constructions that cancel the first several orders deliberately live on a knife-edge: record the surviving coefficient together with the leading omitted term and every symmetry-breaking term the cancellation leaves behind. Before trusting the gadget at scale, check its spectrum against the full Hamiltonian on the smallest exactly diagonalizable cluster.

A controlled order- m approximation improves by one power of the small parameter per added order, over an interval where the parameter is genuinely small. Two data points cannot fit that exponent reliably.

Compare retained subspaces with projectors, not single vectors: individual eigenvectors inside a degenerate manifold rotate freely while the subspace stands still.

The original Schrieffer–Wolff calculation related the Anderson and Kondo descriptions [R024]. Modern rigorous treatments use the same warning in the other direction: the method never licenses keeping only terms judged desirable [R025].

Common errors in Schrieffer–Wolff reductions

Common errors in Schrieffer–Wolff reductions

- Replacing the full Hamiltonian by bare truncation PHP keeps only stay-at-home processes: it drops every excursion through Q and with it the energy shifts and induced couplings those excursions generate.
- Drawing the cut so a near-resonant state falls outside P breaks the assumed scale separation: under the perturbation that state can join the low-energy band, and its denominator collapses toward zero.
- A small typical bond establishes only local information, while an extensive global norm can coexist with a controlled local expansion; lattice-wide control requires local bounds.
- An exact block-diagonalizing unitary guarantees existence, not accuracy: the second-order formula still needs its own truncation check.
- Two effective Hamiltonians compared term by term under different bases inside P can disagree on paper while describing the same physics; align the conventions first.
- Pairing the rotated Hamiltonian with unrotated microscopic observables drops the virtual corrections the same rotation would have added to them.

- A coveted fourth-order term can drown in an unnoticed first- or second-order field. Track every term at the order that generates it.
- Calling the rotated model an encoded code, a topological phase, or an anyon theory claims many-body properties the rotation alone cannot supply. Those properties belong to the resulting Hamiltonian and its states. The rotation builds candidate local degrees of freedom and interactions; the many-body analysis comes after.

Watch for drift on both sides of the fraction: a Q state can approach the retained energies from below, and fabrication shifts can resize the numerator matrix elements and the denominator gaps together.

Verification statements

Verification statements

- Keep the Q states through the rotation: excursions through Q shift energies and generate interactions inside P .
- With $E_\mu > E_0$ each resolvent entry $1/(E_0 - E_\mu) < 0$ is negative, so the sandwich $PVQ(\dots)QVP$ corrects P downward as a negative-semidefinite operator.
- Setting $E_a = E_b = E_0$ makes the two denominators equal, so their average is $1/(E_0 - E_\mu)$ and the sum over μ reassembles the matrix element of $PVQ(E_0 - QH_0Q)^{-1}QVP$.
- The two-site scale $4t^2/U$ assembles from four factors: two hops return the system to the retained sector, amplitudes multiply to t^2 , the virtual double occupancy contributes the denominator U , and spin-plus-path counting supplies the factor of four.
- The same unitary must be applied to observables and states alongside the Hamiltonian.
- A near-resonant state left outside P brings a collapsing denominator: under the perturbation it can join the low-energy band.

Sources

Sources

- [R024] J. R. Schrieffer and P. A. Wolff, “Relation between the Anderson and Kondo Hamiltonians,” *Physical Review* 149, 491–492 (1966). DOI: [10.1103/PhysRev.149.491](https://doi.org/10.1103/PhysRev.149.491).
 - [R025] Sergey Bravyi, David P. DiVincenzo, and Daniel Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” *Annals of Physics* 326, 2793–2826 (2011). DOI: [10.1016/j.aop.2011.06.004](https://doi.org/10.1016/j.aop.2011.06.004); arXiv: [1105.0675](https://arxiv.org/abs/1105.0675).
 - [R026] A. H. MacDonald, S. M. Girvin, and D. Yoshioka, “ t/U expansion for the Hubbard model,” *Physical Review B* 37, 9753–9756 (1988). DOI: [10.1103/PhysRevB.37.9753](https://doi.org/10.1103/PhysRevB.37.9753).
 - [R027] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Springer (1976; corrected printing 1995). DOI: [10.1007/978-3-642-66282-9](https://doi.org/10.1007/978-3-642-66282-9).
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Appendix G — Stabilizer checks without logical-state measurement

A stabilizer encoding stores information in the joint eigenspace of a set of commuting checks. Measuring those checks reports which constraints an error has violated, while leaving the encoded amplitudes unresolved. Syndrome measurement therefore probes error relations; a logical-state measurement probes an observable acting within the encoded subspace.

Background, scope, and supporting argument

Compare coin 1 with coin 2 and coin 2 with coin 3, recording whether each pair agrees. If both comparisons report agreement, all three faces match, but neither result tells whether the common face is heads or tails. The undetermined common face is the information that the checks preserve.

A general pure state of n qubits needs 2^n complex amplitudes. Some important quantum states admit a shorter description as a list of constraints. The condition “all three bits agree,” for example, selects the allowed computational-basis strings directly instead of listing all eight strings and assigning zero amplitude to six of them.

G.1 Single-Qubit Pauli Matrices

These four matrices are the one-qubit factors used in tensor-product checks; their explicit forms are:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Stabilizers and their invariant states

Suppose an allowed state $|\psi\rangle$ is unchanged by an operator S :

$$S|\psi\rangle = |\psi\rangle.$$

The operator S leaves $|\psi\rangle$ unchanged, so we call it a stabilizer; equivalently, $|\psi\rangle$ is a $+1$ eigenstate of S . The eigenvalue $+1$ records that the constraint represented by S is satisfied.

A stabilizer code is the common $+1$ eigenspace of a commuting set of Pauli operators. Its check list specifies the allowed subspace used for error correction, toric-code constructions, and efficient classical simulation of Clifford circuits [R028]; [R031].

Clifford operations preserve a compact stabilizer description, whereas non-stabilizer resources such as magic generally require a larger classical description.

Detailed treatment: single-qubit Pauli matrices

Single-qubit Pauli matrices

The four Pauli matrices on one qubit are:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They satisfy $X^2 = Y^2 = Z^2 = I$. Some pairs anticommute, for example:

$$XZ = -ZX.$$

The n -qubit Pauli group $\mathcal{P}_n$ collects tensor products of these matrices multiplied by phases in $\{+1, -1, +i, -i\}$. Tensor-product notation abbreviates $Z_1 Z_2$ for $Z \otimes Z \otimes I \otimes \dots$.

Any two Pauli strings either commute or anticommute. Count the qubit positions where both strings carry nonidentity factors that anticommute: an even count means the strings commute, an odd count means they anticommute. Pauli commutation is thus a binary parity.

G.2 Conditions on a Stabilizer Group

Choose commuting Hermitian Pauli products with specified $+1$ eigenvalues; closure under multiplication produces the stabilizer group. Commutation permits simultaneous eigenspaces. Excluding $-I$ keeps the defining equation $-|\psi\rangle = |\psi\rangle$ satisfiable by a nonzero code state — the equation as written admits only the zero vector.

A stabilizer group $\mathcal{S} \subset \mathcal{P}_n$ is Abelian and excludes $-I$.

Detailed treatment: conditions on a stabilizer group

Conditions on a stabilizer group

A stabilizer group $\mathcal{S} \subset \mathcal{P}_n$ is Abelian, meaning all its elements commute. If $-I$ is included in $\mathcal{S}$, the constraints require $\mathcal{S}$ and $|\psi\rangle = |\psi\rangle$ simultaneously; only the zero vector can satisfy them.

Suppose $|\psi\rangle = |\psi\rangle$ has $\mathcal{S}$ independent generators r . Independence means no nonempty product of generators equals $S_1, \dots, S_r$. The projector onto the code space $\mathcal{C}$ is then

$\mathcal{C}$

Each factor

$$\Pi_{\mathcal{C}} = \prod_{j=1}^r \frac{I + S_j}{2}.$$

projects onto the $(I + S_j)/2$ eigenspace of $+1$. Because each independent check halves the currently allowed subspace,

S_j

A Hilbert space of dimension

$$\dim \mathcal{C} = 2^{n-r}.$$

represents 2^k logical qubits, so the code encodes

k

logical qubits, denoted

$$k = n - r$$

once the distance $[[n, k, d]]$ is known.

The same dimension follows from the projector trace, because a projector's rank equals its trace. Expanding d gives $\Pi_{\mathcal{C}}$ times a sum of stabilizer elements. Every nonidentity Pauli has zero trace, and independence makes the empty generator product the only term equal to 2^{-r} . Thus,

$$\text{Tr } I = 2^n$$

When the listed generators are not independent, counting all of them overstates the constraints; a redundant check removes no further logical qubit.

G.3 The Dimension Formula

When I has

$$\text{Tr } \Pi_{\mathcal{C}} = 2^{-r} 2^n = 2^{n-r}.$$

independent generators:

$$\mathcal{S}$$

Each independent check imposes one binary eigenvalue condition and halves the allowed state-space dimension. Starting from r dimensions,

$$\dim \mathcal{C} = 2^{n-r}$$

independent checks leave 2^n dimensions, enough for r logical qubits. A redundant check imposes no new condition.

For independent generators 2^{n-r} , the corresponding code projector is $k = n - r$.

G.4 The Three-Qubit Repetition Code

$$S_j$$

The code space is $P_{\mathcal{C}} = \prod_{j=1}^r (I + S_j)/2$, with logical operators:

$$S_1 = Z_1 Z_2, \quad S_2 = Z_2 Z_3$$

An encoded state is $\text{span}\{|000\rangle, |111\rangle\}$. Each component returns the same check outcomes,

$$\overline{X} = X_1 X_2 X_3, \quad \overline{Z} = Z_1$$

, so measuring them for any superposition cannot distinguish $\alpha|000\rangle + \beta|111\rangle$ from $(+1, +1)$. A single α error changes the outcomes as follows:

Error	β	X
None	$Z_1 Z_2$	$Z_2 Z_3$
+1	+1	X_1
-1	+1	X_2
-1	-1	X_3

Under this noise model, the syndrome identifies a single bit flip. A +1 error commutes with both checks and acts as logical -1 , so the repetition code does not protect against phase errors. Protection claims therefore depend on the specified error model.

The logical operators preserve the code space and act nontrivially within it. Measuring Z_1 reveals logical information; measuring a stabilizer does not. A stabilizer Hamiltonian such as $\overline{Z}$ energetically favors the code space for $\overline{Z}$. Repeated syndrome measurement and recovery are active procedures, independent of this passive energy penalty.

Detailed treatment: three-qubit repetition code

Three-qubit repetition code

Consider three physical qubits with two stabilizer generators:

$$H = -\sum_j J_j S_j$$

These generators commute. The simultaneous $J_j > 0$ eigenvalue condition requires qubit 1 to agree with qubit 2 and qubit 2 to agree with qubit 3. The resulting code space is:

$$S_1 = Z_1 Z_2, \quad S_2 = Z_2 Z_3.$$

Here +1 with

$$\mathcal{C} = \text{span}\{|000\rangle, |111\rangle\}.$$

independent generators, so the code encodes $n = 3$ logical qubit with logical computational basis

$$r = 2$$

A physical bit-flip error $k = 1$ anticommutes with

$$|0_L\rangle = |000\rangle, \quad |1_L\rangle = |111\rangle.$$

and commutes with X_1 . Measuring the two stabilizers returns their eigenvalues, and the collection of eigenvalues is called the syndrome: it records which constraints changed sign while leaving the encoded logical value undetermined. Representing S_1 by bit 0 and S_2 by bit 1 gives:

error	+1	−1	syndrome
S_1	S_2	I	00
+1	+1	X_1	10
−1	+1	X_2	11
−1	−1	X_3	01

The syndrome identifies any single +1 error in this model. It does not determine whether the encoded state was −1 or X ; stabilizer measurement extracts error relations, not encoded amplitudes.

A logical Pauli operator preserves the code space while acting nontrivially inside it. One possible choice is:

$$|0_L\rangle$$

The overbar denotes a logical operator. Both operators commute with $|1_L\rangle$ and

$$\overline{X} = X_1 X_2 X_3, \quad \overline{Z} = Z_1.$$

and satisfy the logical Pauli anticommutation relation

S_1

Multiplying a logical operator by a stabilizer changes its physical representative while keeping its action on code states. For example, S_2 ,

$$\overline{XZ} = -\overline{ZX}.$$

, and Z_1 equivalently represent logical Z_2 .

This code corrects one bit flip under noise restricted to Z_3 errors. Against arbitrary single-qubit Pauli errors it has distance one, because $\overline{Z}$ is already a weight-one undetectable logical operation.

Its full label is X [Ro28]. The label “single-error-correcting quantum code” is valid only for a specified restricted noise model; arbitrary single-qubit Pauli noise includes a weight-one undetectable logical operation.

Normalizer and undetectable logical operators

The normalizer of Z_1 within the Pauli group is

$$[[3, 1, 1]]$$

Thus $\mathcal{S}$ is the set of Pauli strings commuting with every stabilizer check. In group language, such a string normalizes

$$N(\mathcal{S}) = \{P \in \mathcal{P}_n : PS = SP \text{ for every } S \in \mathcal{S}\}.$$

because conjugating a stabilizer by it leaves that stabilizer unchanged, apart from a physically irrelevant global phase. Applying a normalizer element therefore leaves every check eigenvalue, and hence the syndrome, unchanged.

A Pauli error $N(\mathcal{S})$ falls into one of three classes:

- If E anticommutes with at least one generator, that check's eigenvalue changes sign and E is detectable.
- If $E \in \mathcal{S}$, it acts trivially on every code state.
- If $E \in N(\mathcal{S}) \setminus \mathcal{S}$, it commutes with every check but acts nontrivially on the encoded state, so it is an undetectable logical Pauli operator.

The weight $\text{wt}(E)$ of a Pauli operator counts the qubits where E acts by an operator other than I . The code distance is then

$$d = \min_{E \in N(\mathcal{S}) \setminus \mathcal{S}} \text{wt}(E).$$

A distance- d code detects all Pauli errors of weight below d and corrects arbitrary errors on at most $\lfloor (d-1)/2 \rfloor$ qubits, assuming ideal syndrome extraction and a suitable decoder [Ro28]; [Ro29]. A code is degenerate when distinct physical errors have the same action on the code because they differ by a stabilizer.

The distance minimizes over $N(\mathcal{S}) \setminus \mathcal{S}$, excluding stabilizers. Stabilizers lie in the normalizer but act as the identity on the code space; without the set difference, they would count as logical operators despite having no logical action.

Syndrome measurement and energetic enforcement

For a generator S_j , the projectors onto its $s_j = \pm 1$ eigenspaces are:

$$\Pi_{s_j} = \frac{I + s_j S_j}{2}.$$

To extract a check in a quantum circuit, an ancilla couples to the parity associated with S_j and is measured. In a static system, the same operator can appear in the stabilizer Hamiltonian

$$H_{\text{stab}} = - \sum_j K_j S_j,$$

Here $K_j > 0$ has units of energy. For stabilizer S_j , a violation changes its Hamiltonian contribution from $-K_j$ to $+K_j$, costing $2K_j$. Passive energetic enforcement and active repeated syndrome extraction use the same operators but different physical mechanisms.

The algebraic stabilizer specifies an allowed subspace. Active correction additionally measures its checks and applies a recovery operation.

A static stabilizer Hamiltonian energetically favors a code space. Topological order is a phase-level claim: checks must be local, logical operators must require support that grows with system size (for example, loops that cannot be shrunk to a point on a torus), and the many-body system must realize the appropriate phase.

Keeping constraint, active code, energetics, and topological order distinct prevents an algebraic description from being mistaken for an error-correction protocol, an energy penalty, or a topological phase.

Binary symplectic representation

Global phases do not affect Pauli commutation, so a Pauli string is encoded by two binary vectors $\mathbf{x}, \mathbf{z} \in \mathbb{F}_2^n$, where $\mathbb{F}_2 = \{0, 1\}$ is the finite field with arithmetic modulo two:

$$P \leftrightarrow (\mathbf{x} \mid \mathbf{z}).$$

At qubit j the correspondence is:

$$(0,0) \leftrightarrow I, \quad (1,0) \leftrightarrow X, \quad (0,1) \leftrightarrow Z, \quad (1,1) \leftrightarrow Y$$

For two Pauli operators $u = (\mathbf{x} \mid \mathbf{z})$ and $v = (\mathbf{x}' \mid \mathbf{z}')$, the symplectic pairing records their commutation parity. They commute exactly when this pairing vanishes:

$$[u, v]_{\text{sp}} = \mathbf{x} \cdot \mathbf{z}' + \mathbf{z} \cdot \mathbf{x}' = 0 \pmod{2}.$$

The pairing is a binary parity: 0 means the Pauli strings commute and 1 means they anticommute. Software evaluates it instead of multiplying the full matrices.

The r generator vectors form the rows of a binary check matrix,

$$H = (H_X \mid H_Z).$$

Pairwise commutation of all generators is equivalent to the matrix condition

$$H_X H_Z^T + H_Z H_X^T = 0 \pmod{2}.$$

For an error vector $e = (\mathbf{x}_e \mid \mathbf{z}_e)$, each syndrome bit is the symplectic product of e with the corresponding generator row. Row reduction over $\mathbb{F}_2$ supplies the rank r and exposes redundant checks. Solving the commutation equations yields logical representatives and flags inconsistent constraints. This representation lets software test the code without constructing full operator matrices.

Toric-code star and plaquette stabilizers

Place one qubit on each edge of a square lattice with periodic boundary conditions. For each vertex v the star operator is

$$A_v = \prod_{e \ni v} X_e,$$

where the product runs over edges incident on v . For each plaquette p the plaquette operator is

$$B_p = \prod_{e \in \partial p} Z_e,$$

where ∂p is the boundary of p .

A star and a plaquette share either zero edges or two edges, and each shared edge contributes one anticommutation $XZ = -ZX$. With two shared edges the two minus signs cancel, so every A_v commutes with every B_p [Ro30].

On a periodic connected square lattice, the product of all star operators is the identity, as is the product of all plaquette operators. Each relation makes one listed generator dependent, so the rank is lower by two.

For an $L \times L$ torus there are $n = 2L^2$ edge qubits and $r = 2L^2 - 2$ independent stabilizers, so $k = 2$. [Theory] The two logical qubits correspond to noncontractible loop operators [Ro30].

A shortest logical loop has weight L , so the ideal periodic square-lattice code has distance $d = L$. Changing the boundaries or lattice defects changes this minimum, so the geometry must be specified.

An open Z string anticommutes with the star checks at its two endpoints, so only those endpoints report violations; these are electric syndromes. Extending the string moves an endpoint while the interior creates no additional violated checks.

A closed string that can shrink to a point is a product of plaquettes and therefore a stabilizer. A closed string that winds around the torus cannot be shrunk in this way and acts as a logical operator. This distinction underlies Chapter 16's error-chain description.

Limitations of stabilizer checks

Limitations of stabilizer checks

Stabilizer algebra determines exact commutation relations, ground-space dimension, syndromes, and logical-loop structure for the ideal toric code. A microscopic defect Hamiltonian must be checked separately to determine whether its low-energy sector realizes those relations.

A perturbative gadget may be designed to generate a term $-K_j S_j$ in a low-energy subspace. The Schrieffer–Wolff calculation eliminates high-energy states perturbatively and produces an effective Hamiltonian; it must identify unwanted terms as well as the intended one because they can alter the low-energy spectrum or logical dynamics.

A perturbation that anticommutes with a stabilizer flips its eigenvalue and creates an excitation. A perturbation in the normalizer but outside the stabilizer group leaves the syndrome unchanged and can split or rotate the encoded subspace. The commutation

relation determines which of these effects occurs; the coefficient then sets its energy or rotation scale.

For a proposed small code patch, software first constructs the binary check matrix and verifies pairwise commutation, rank, encoded-qubit count, logical representatives, and distance where tractable. Sparse exact diagonalization then tests the predicted ground-space dimension and excitation energies. If perturbations are not Pauli operators, exact stabilizer labels need not apply; use expectation values $\langle S_j \rangle$, Wilson loops (loop observables that test string operators), and overlap with the ideal code subspace.

The Gottesman–Knill theorem permits efficient classical simulation of stabilizer states evolving under Clifford gates, Pauli measurements, and classical feed-forward [Ro28]; [Ro31]. [Theory] The theorem applies only to that restricted circuit family; a material with arbitrary interactions requires a different simulation argument.

Common errors in stabilizer analysis

Common errors in stabilizer analysis

- Including $-I$ forces the code space to be empty.
- Counting listed generators rather than independent generators overstates the number of constraints and the inferred rank.
- Binary vectors record commutation data only; the generator signs choose the $+1$ eigenspace. Omitting those signs can select the wrong code space.
- Stabilizers are in the normalizer but act trivially, so treating every normalizer element as a logical operator includes operators with no logical action.
- Distance depends on boundary geometry: a planar patch, torus, puncture, rough boundary, or smooth boundary permits different string endpoints and therefore different logical operators.
- Distance is a support count; an energy gap has energy units, so they are different quantities.

- A syndrome can correspond to several physical errors; a decoder chooses among their equivalence classes using the noise model.

Exercises and solutions

Exercises and solutions

- Show that if $-I \in \mathcal{S}$, then the code space is $\{0\}$.

A stabilized state would have to satisfy both $|\psi\rangle = |\psi\rangle$ and $-|\psi\rangle = |\psi\rangle$, which together imply $|\psi\rangle = 0$.

- Show that r independent stabilizers on n qubits encode $k = n - r$ logical qubits.

The projector $\Pi_{\mathcal{C}}$ has trace 2^{n-r} , equal to $\dim \mathcal{C}$, hence $k = n - r$.

- What condition makes a Pauli error detectable?

A Pauli error is detectable when it anticommutes with at least one stabilizer generator, flipping the corresponding syndrome bit.

- Show that Z_1 , Z_2 , and Z_3 are equivalent logical $\bar{Z}$ representatives for the three-qubit code.

They differ by products of $S_1 = Z_1 Z_2$ and $S_2 = Z_2 Z_3$, so they act identically on every code state.

- Explain why the repetition code is only $[[3, 1, 1]]$ as a full quantum code.

A weight-one Z operator is already an undetectable logical operation.

- Show that toric-code stars and plaquettes commute.

Their supports overlap on an even number of edges, so the minus signs from $XZ = -ZX$ cancel in pairs.

Sources

Sources

- [R028] Daniel Gottesman, Stabilizer Codes and Quantum Error Correction, PhD thesis, California Institute of Technology (1997). DOI: [10.7907/rzr7-dt72](https://doi.org/10.7907/rzr7-dt72); arXiv: [quant-ph/9705052](https://arxiv.org/abs/quant-ph/9705052).
 - [R029] A. R. Calderbank, E. M. Rains, P. W. Shor, and N. J. A. Sloane, “Quantum error correction via codes over $GF(4)$,” IEEE Transactions on Information Theory 44, 1369–1387 (1998). DOI: [10.1109/18.681315](https://doi.org/10.1109/18.681315); arXiv: [quant-ph/9608006](https://arxiv.org/abs/quant-ph/9608006).
 - [R030] Alexei Kitaev, “Fault-tolerant quantum computation by anyons,” Annals of Physics 303, 2–30 (2003). DOI: [10.1016/S0003-4916\(02\)00018-0](https://doi.org/10.1016/S0003-4916(02)00018-0); arXiv: [quant-ph/9707021](https://arxiv.org/abs/quant-ph/9707021).
 - [R031] Scott Aaronson and Daniel Gottesman, “Improved simulation of stabilizer circuits,” Physical Review A 70, 052328 (2004). DOI: [10.1103/PhysRevA.70.052328](https://doi.org/10.1103/PhysRevA.70.052328); arXiv: [quant-ph/0406196](https://arxiv.org/abs/quant-ph/0406196).
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Appendix H — Tensor networks

A general state of many sites needs far more amplitudes than any machine can store, so direct simulation stops early. A tensor network stores a narrower class of states: those whose correlations across each cut pass through a slender internal channel. Its cost follows the widths of those channels. The full Hilbert space keeps its dimension; the network stores cheaply only the states whose widths stay small.

Background, scope, and supporting argument

Picture three boxes in a row. Each box sends one wire upward, and each neighboring pair shares one horizontal wire.

```
s1 s2 s3 physical indices
| | |
[A]---a1---[A]---a2---[A]
| |
boundary boundary

joined line a1: sum over a1
open line s1: coefficient still depends on s1
```

This diagram defines the tensor-network notation used here. Each box holds a table of numbers, called a tensor. A wire leaving a box open marks an index whose value stays visible in the final amplitude. A wire joining two boxes marks an index shared by exactly those two tables, summed over and thus removed from the result. The wiring states the sums to perform and belongs to the definition of the expression, alongside the tables.

H.1 Full Coefficient Arrays and Tensor Decomposition

For local dimension d , the coefficient array holds d^N complex amplitudes. Split the array across one cut and apply the Schmidt decomposition of Appendix B, which separates the two sides into paired components with matching weights. Repeating the split along the chain leaves a string of tensors, with each internal index labeling the paired components across one cut.

A wavefunction for N local degrees of freedom is represented by:

$$\psi_{s_1 s_2 \dots s_N}$$

A tensor-network representation stores smaller multidimensional arrays in place of that single large array, and recovers each amplitude by summing over the indices the small arrays share.

Detailed treatment: full coefficient arrays and tensor decomposition

Full coefficient arrays and tensor decomposition

A wavefunction for N local degrees of freedom is represented by the coefficient array

$$\psi_{s_1 s_2 \dots s_N},$$

where each physical index s_i names a local basis state at site i . When every index takes d values, the array contains d^N complex entries. Direct storage therefore becomes impractical already for chains of modest length.

Each amplitude is recovered by summing over the shared indices of those small arrays. These small arrays are tensors. In this terminology, a scalar has no indices, a vector v_i has one index, a matrix M_{ij} has two indices, and a rank- r tensor $T_{i_1 \dots i_r}$ has r indices.

Here tensor rank counts indices. Matrix rank counts independent rows or columns. The two uses of rank differ, so each passage states which one it means.

Factorization gives a working picture: a large object is held through smaller factors. The tensor factors carry several index positions each, and finding good factors can remain hard. Storage shrinks exactly when the correlations of the state pass through internal indices of moderate width. A tensor network serves two roles at once: a notation for writing such factored amplitudes, and an ansatz family, a restricted class of candidate states used for representation or optimization. Membership in that class guarantees neither a cheap computation nor an accurate approximation.

Contraction of shared indices

When two tensors A_{ij} and B_{jk} share an index j , joining their j wires directs the sum

$$C_{ik} = \sum_j A_{ij} B_{jk}.$$

That summation is called a contraction. The contracted index j is absent from the result.

The remaining indices i and k stay free: the resulting tensor still depends on them. Every term of a tensor equation carries the same free indices.

Within one product, an index normally appears at most twice: once for a free index, twice for a summed index. Where that reading could confuse, the sum is written out explicitly.

With exact arithmetic, the contraction order changes the computational cost and leaves the final scalar or tensor fixed. With approximate contraction, the order and the truncation together shape the numerical error. The diagram then still fixes the network structure, while the number it evaluates to depends on the stated approximation procedure as well.

Physical and virtual indices

For a quantum state, the open indices that name basis states are physical indices. They label the local degrees of freedom the state describes.

Factoring the coefficient array introduces further internal indices, called virtual or auxiliary indices. No local measurement reads them; they record correlation between blocks rather than observable variables.

Bond dimension

When a virtual index takes χ possible values, χ is its bond dimension. That number sets how much virtual information can pass between the two tensors the index joins. It parametrizes the representation; by itself it estimates no numerical error.

H.2 Matrix Product States

For an open chain, $A_i^{s_i}$ is a $\chi_{i-1} \times \chi_i$ matrix with boundary dimensions $\chi_0 = \chi_N = 1$. The product below is a scalar coefficient for each string of physical labels:

$$|\psi\rangle = \sum_{s_1, \dots, s_N} (A_1^{s_1} A_2^{s_2} \dots A_N^{s_N}) |s_1 s_2 \dots s_N\rangle$$

The internal, or bond, dimensions χ_i limit the Schmidt rank across each cut. Keeping every Schmidt value represents any finite state exactly, at the possible price of exponentially large bonds. Dropping small Schmidt values lowers the cost and introduces an approximation; the discarded weight at the truncated bond is recorded, and each claimed observable is converged separately against the numerical controls below, since the energy can settle before a small gap, correlation length, or entanglement diagnostic does.

For roughly uniform bond dimension χ , the stored tensor entries scale as $O(Nd\chi^2)$, much smaller than d^N where the required χ stays moderate. The saving comes from that moderate width, and the Hilbert space itself is unchanged.

Detailed treatment: matrix product states

Matrix product states

Take an open chain of N sites. At each site i and for each physical value s_i , write a matrix $A_i^{s_i}$ of size $\chi_{i-1} \times \chi_i$. Multiplying the matrices chosen by one fixed label sequence yields the amplitude of that sequence. $s_1 \cdots s_N$

That construction is a matrix product state, or MPS:

$$|\psi\rangle = \sum_{s_1, \dots, s_N} (A_1^{s_1} A_2^{s_2} \cdots A_N^{s_N}) |s_1 s_2 \cdots s_N\rangle.$$

With open boundary conditions, $\chi_0 = \chi_N = 1$, so the matrix product collapses to a scalar. In fully indexed notation,

$$\psi_{s_1 \cdots s_N} = \sum_{a_1, \dots, a_{N-1}} (A_1)_{1a_1}^{s_1} (A_2)_{a_1 a_2}^{s_2} \cdots (A_N)_{a_{N-1} 1}^{s_N}.$$

Each physical index s_i has dimension d_i , while the virtual indices beside site i have dimensions χ_{i-1} and χ_i . Site-dependent widths are allowed and often computationally useful.

Gauge freedom and canonical forms

On any MPS bond, insert an invertible matrix G and its inverse as follows:

$$A_i^{s_i} A_{i+1}^{s_{i+1}} = (A_i^{s_i} G)(G^{-1} A_{i+1}^{s_{i+1}}).$$

Since GG^{-1} is the identity, every wavefunction amplitude stays fixed, so the physical state is unchanged. Individual tensor entries therefore carry no direct observable meaning.

That nonuniqueness is called gauge freedom. A canonical form is a gauge choice fixed by stated orthogonality properties, reached by passing an invertible matrix across a virtual bond and absorbing its inverse into the neighboring tensor.

A non-invertible insertion may have no reversal, and the represented state may change with it.

Schmidt rank, entropy, and bond dimension

Cut the chain at one bond. The state then admits a Schmidt decomposition

$$|\psi\rangle = \sum_{\alpha=1}^r \lambda_{\alpha} |\alpha_L\rangle |\alpha_R\rangle,$$

with Schmidt coefficients $\lambda_{\alpha} \geq 0$ obeying $\sum_{\alpha} \lambda_{\alpha}^2 = 1$. The left and right Schmidt states are orthonormal. The Schmidt rank r , the count of nonzero Schmidt coefficients, cannot exceed the MPS bond dimension χ at that cut. The bond sets a ceiling on how many correlated components the state can carry there.

For a pure state, the bipartite von Neumann entropy measures the entanglement between the two sides:

$$S = - \sum_{\alpha} \lambda_{\alpha}^2 \ln \lambda_{\alpha}^2 \leq \ln \chi.$$

The bond dimension thus caps the entanglement the MPS can carry across a cut. That cap certifies no accuracy by itself. A target state may demand a large χ at one cut, while a poorly optimized MPS may leave its available bond dimension partly unused.

H.3 The GHZ State

The GHZ state admits an exact MPS representation with bond dimension $\chi = 2$:

$$A^0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A^1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

The bulk matrices alone leave the boundaries open. A complete specification adds boundary vectors $l^T = (1, 1)/\sqrt{2}$ and $r = (1, 1)^T$, with amplitude $l^T A^{s_1} \dots A^{s_N} r$. Any mixed bit string contains a product $A^0 A^1$ or $A^1 A^0$, which vanishes. The all-zero and all-one strings each carry amplitude $1/\sqrt{2}$, giving the normalized state

$$|\text{GHZ}_N\rangle = \frac{|0 \dots 0\rangle + |1 \dots 1\rangle}{\sqrt{2}}.$$

Under the open-chain convention above, the vector ℓ^T is absorbed into the first tensor and r into the last. The two bulk matrices by themselves specify neither the boundary contraction nor the normalization.

Detailed treatment: exact MPS representation of the GHZ state

Exact MPS representation of the GHZ state

For N qubits, the Greenberger–Horne–Zeilinger state is

$$|\text{GHZ}_N\rangle = \frac{|00 \dots 0\rangle + |11 \dots 1\rangle}{\sqrt{2}}.$$

Its exact MPS representation uses bond dimension $\chi = 2$ and repeats the same two matrices at every site:

$$A^0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A^1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Take the left boundary row and the right boundary column as

$$\ell = \frac{1}{\sqrt{2}}(1, 1), \quad r = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The amplitude of a bit string $s_1 \dots s_N$ is then

$$\psi_{s_1 \dots s_N} = \ell A^{s_1} A^{s_2} \dots A^{s_N} r.$$

When every $s_i = 0$, the matrix product collapses to A^0 , giving amplitude $1/\sqrt{2}$. When every $s_i = 1$, the product collapses to A^1 , again giving amplitude $1/\sqrt{2}$. When the string mixes 0 and 1, the product contains at least one factor

$$A^0 A^1 = A^1 A^0 = 0,$$

so that amplitude vanishes. The MPS thus matches the GHZ state on every bit string.

Its norm follows from the two nonzero amplitudes, with no need to expand all 2^N basis amplitudes:

$$\langle \text{GHZ}_N | \text{GHZ}_N \rangle = \frac{1}{2} + \frac{1}{2} = 1.$$

For two distinct sites i and j ,

$$\langle Z_i \rangle = 0, \quad \langle Z_i Z_j \rangle = 1.$$

Subtracting the product of one-site expectation values leaves a connected correlation of one. Across any cut, the two Schmidt coefficients are $1/\sqrt{2}$, so $S = \ln 2$, consistent with $\chi = 2$.

Long-range correlation coexists here with a small bond dimension. Short-ranged interactions can also sit beside entanglement that demands large bonds. Correlation length and required bond dimension measure different properties.

Matrix product operators and transfer matrices

A local operator carries one input and one output physical index at each site. Collecting those index pairs into a further tensor layer gives

$$O = \sum_{\mathbf{s}, \mathbf{s}'} W_1^{s_1 s'_1} W_2^{s_2 s'_2} \dots W_N^{s_N s'_N} |s_1 \dots s_N\rangle \langle s'_1 \dots s'_N|,$$

where each $W_i^{s_i s'_i}$ is a matrix on virtual indices. The bold symbol $\mathbf{s}$ abbreviates the complete sequence $(s_1, \dots, s_N)$.

That representation is a matrix product operator, or MPO. It follows the MPS construction with two physical indices per site, one input and one output.

Expectation values such as $\langle \psi | O | \psi \rangle$ are evaluated by placing the MPO between the bra and ket MPS and contracting all physical and virtual indices. A one-dimensional slice through the resulting double-layer network carries the virtual data at one cut to the virtual data at the next.

That map is the transfer matrix. Under suitable assumptions, its leading eigenvalues fix normalization properties and correlation lengths.

MPOs store sums of local Hamiltonian terms compactly. That storage saving leaves the later optimization approximate and potentially costly.

H.4 Projected Entangled Pair States

On a square lattice, one PEPS tensor reads:

$$A_{lurd}^s$$

where s is the physical index and l, u, r, d are the left, up, right, and down virtual indices. Each neighboring pair of virtual indices is contracted; the uncontracted physical indices label the state amplitudes.

PEPS matches the network geometry to a two-dimensional lattice. A square-lattice tensor with virtual dimension D holds roughly dD^4 entries. Storing the tensors leaves two further tasks: contracting the network and optimizing the state, both generally approximate. Chapter 40 therefore checks convergence both in the state bond dimension and in the contraction environment.

Storage and search play distinct roles here. MPS and PEPS specify how a state is stored; DMRG and other optimization schemes specify how a state is found. A compact representation can still hold an inaccurate variational state when the optimization and the observable calculations are left uncontrolled.

Detailed treatment: projected entangled pair states on general graphs

Projected entangled pair states on general graphs

An MPS follows a one-dimensional chain. Running the same tensor-network construction on a higher-dimensional graph produces a projected entangled pair state, or PEPS [Ro32]. On a square lattice, a local PEPS tensor reads

$$A_{lurd}^s,$$

where s is the physical index and l, u, r, d are the left, up, right, and down virtual indices. Each neighboring pair of virtual indices is contracted. On a general defect graph, each tensor instead carries one virtual index per incident edge, with no square-lattice direction labels needed.

One construction starts from entangled pairs placed on the graph edges, then applies at each vertex a local map from the virtual spaces meeting there to the physical space. The name “projected entangled pair state” records that two-step build. Boundary tensors either meet fewer neighbors or carry trivial one-dimensional virtual legs.

PEPS respects two-dimensional locality by construction. Still, exact contraction of a generic two-dimensional PEPS is computationally hard, so contraction is normally approximate [Ro33]. [Theory] Chapter 40 therefore reports both the PEPS bond dimension D and contraction controls such as the environment dimension χ_{env} , the size of the approximate surrounding network kept during contraction. Raising only one of the two parameters can hide error attached to the other.

A PEPS represents a quantum state. That state may be an exact fixed-point topological state, a variational approximation to a microscopic ground state, or a fully nontopological state. Running a PEPS algorithm on conventional hardware performs a classical numerical calculation; no anyons are physically created.

Virtual symmetries and topological order

In a topologically ordered PEPS, symmetries can act on the virtual indices. A virtual symmetry string passes through local tensors leaving the physical state fixed, while noncontractible virtual strings can label distinct ground states. For the principal PEPS classes, that structure explains ground-state degeneracy, local indistinguishability, topological entropy, and anyonic sectors [Ro34].

Closed loops drawn in a tensor-network diagram do not establish topological order on their own. The next paragraph lists what does: the virtual symmetry or algebra, the parent Hamiltonian or an independent phase determination, and robustness tests.

A valid identification determines the relevant virtual symmetry or algebra, constructs the parent Hamiltonian or establishes the phase by another method, and tests robustness. A parent Hamiltonian is a Hamiltonian with the tensor-network state as a ground state. Injective PEPS, whose virtual-to-physical map is one-to-one after any required blocking, typically give a unique ground state of a suitable parent Hamiltonian. Topological PEPS need generalized structures such as group symmetries or matrix-product-operator symmetries [Ro34].

Canonicalization, truncation, and variational optimization

For an MPS, sweeps built from QR or singular-value decompositions move the orthogonality center and produce left- or right-canonical tensors. A QR decomposition factors a matrix into an orthogonal or unitary factor and an upper-triangular factor. A singular-value decomposition separates a matrix into left and right orthonormal factors with a diagonal of nonnegative singular values. The orthogonality center names the site or bond relative to which the tensors on either side meet the corresponding canonical conditions.

At one bond, truncating the Schmidt decomposition to the largest χ coefficients discards the weight

$$w_{\text{disc}} = \sum_{\alpha > \chi} \lambda_{\alpha}^2.$$

That discarded weight describes the single truncation. A small value is useful local evidence of accuracy. It does not by itself bound the error of every observable; each claimed observable is converged separately in the validation procedure below.

The density-matrix renormalization group (DMRG) optimizes an MPS variationally, usually one or two sites at a time [Ro35]. It works best for one-dimensional and quasi-one-dimensional systems whose entanglement stays manageable across the chosen site ordering. Wrapping a wide two-dimensional lattice into a one-dimensional snake-like ordering can drive the required bond dimension up rapidly with the lattice width.

PEPS algorithms optimize local tensors while approximating each tensor's environment, the remainder of the network surrounding the tensor or region under update. Infinite PEPS works directly with a repeating unit cell; finite PEPS keeps explicit boundaries.

The numerical method follows the physical problem. Implantation disorder means the randomly varying defect positions, depths, and local couplings left by the implantation process. A translation-invariant infinite ansatz repeats one unit cell everywhere, so it cannot represent that site-to-site variation; capturing it needs a larger unit cell, statistical sampling over realizations, or another suitable representation.

Validation procedure for defect-cluster calculations

A defect-cluster calculation proceeds through the following stages:

- Name every physical index: state whether s_i labels a microscopic spin, a projected cluster pseudospin, or an edge label, and state its dimension d_i . A projected cluster pseudospin is an effective reduced degree of freedom assigned to a chosen low-energy cluster subspace.
- Fix the graph. Every virtual bond stands for an intended tensor contraction. A virtual bond may join tensors whose sites do not interact physically; the bond records shared correlation capacity, not a coupling in the Hamiltonian.
- Choose the ansatz for the geometry and system size: exact diagonalization for the smallest complete clusters, MPS/DMRG for chains and cylinders, and PEPS where two-dimensional connectivity and accessible contraction controls support that choice.
- State all numerical controls: the MPS or PEPS bond dimension, environment dimension, unit-cell size, truncation rule, optimizer tolerance, boundary condition, and initialization strategy.
- Test convergence separately for each observable. The energy can settle before a small gap, correlation length, Wilson loop, or entanglement diagnostic does. A Wilson loop is a nonlocal operator tied to a closed contour. Plot each claimed observable against the numerical controls that govern it.
- Start from competing initial states. Near first-order transitions or barriers between topological sectors, one initialization can stay metastable, trapped in a locally stable state instead of reaching the best available variational state. Compare energies and diagnostics across multiple sectors, since one run can otherwise report a metastable plateau as convergence.
- Validate the effective-model handoff: compare a small microscopic calculation with the projected Hamiltonian before applying tensor-network methods to the effective model alone.

For the observables considered in Chapter 40:

- obtain the gap from excited-state methods or sector-resolved energies, not from convergence of the ground-state energy alone;
- calculate entanglement entropy from Schmidt values for an MPS, stating the geometry and logarithm base;
- estimate topological entanglement entropy with a subtraction or scaling protocol whose finite-size errors have been tested;
- represent Wilson loops as MPOs or tensor insertions, and analyze perimeter or area behavior only over numerically resolved system sizes;
- extract modular data only after constructing and controlling a basis of ground-state sectors. Modular data describe transformations among those sectors.

No single diagnostic establishes topological order in a small noisy region. A stronger case needs the spectrum, degeneracy pattern, loop observables, entanglement structure, and response to perturbations to agree.

Common interpretation and implementation errors

- Unlabeled indices hide which legs are physical and which are virtual, and hide input legs from output legs.
- Tensor rank and matrix rank name different properties. State which definition each passage uses.
- Gauge-dependent tensor entries carry no observable meaning. Insertions of GG^{-1} change the tensors while the represented state stays fixed.
- A bare bond dimension reports no accuracy. Report it with convergence tests: $D = 8$ is an input parameter, not an error bar.
- Contraction error needs explicit control. A PEPS calculation with large D and a small environment can be less reliable than a smaller, well-converged calculation.
- DMRG follows the geometry. A two-dimensional cylinder tractable at one width can mislead at the next.
- A low variational energy does not establish topology. Competing phases can share similar energies while differing in nonlocal structure, so the energy comparison is supplemented by the degeneracy, loop, and entanglement diagnostics above.
- Projection error belongs in the assessment. An accurate solution of an inaccurate effective Hamiltonian stays inaccurate.
- A loop-like tensor-network diagram is not evidence of topological order. Such an order is established by the diagnostics listed above: the virtual symmetry or algebra, the parent Hamiltonian or an independent phase determination, robustness tests, and agreement among spectrum, degeneracy, loop observables, and entanglement structure.

The diagrammatic notation, algebraic identities, and validation criteria above carry the information needed to translate tensor-network diagrams into explicit index contractions.

Verification checks

Verification checks

- A joined tensor-network line directs summation over the shared index.
- Inserting G and G^{-1} on an MPS bond leaves the state unchanged because

$$A_i^{s_i} A_{i+1}^{s_{i+1}} = (A_i^{s_i} G)(G^{-1} A_{i+1}^{s_{i+1}})$$

for any invertible G . Every amplitude therefore stays fixed.

- For an MPS cut with bond dimension χ , at most χ Schmidt coefficients can be nonzero. The entropy $-\sum p_\alpha \ln p_\alpha$ of a distribution supported on at most χ outcomes is at most $\ln \chi$. Hence $S \leq \ln \chi$.
- The GHZ MPS survives only on all-zero and all-one strings. Every mixed string contains one of the zero products $A^0 A^1$ or $A^1 A^0$.
- A PEPS diagram does not prove topological order. The proof uses the virtual symmetries together with physical diagnostics: the degeneracy pattern, loop observables, and entanglement structure.
- Reporting D without an environment dimension leaves the contraction error uncontrolled. A large PEPS bond dimension with a small environment can hide contraction error. D is an input, not an error bar.

Sources

Sources

- [R032] Frank Verstraete and J. Ignacio Cirac, “Renormalization algorithms for quantum-many body systems in two and higher dimensions,” arXiv: [cond-mat/0407066](https://arxiv.org/abs/cond-mat/0407066) (2004).

- [R033] Norbert Schuch, Michael M. Wolf, Frank Verstraete, and J. I. Cirac, “Computational complexity of projected entangled pair states,” *Physical Review Letters* 98, 140506 (2007). DOI: [10.1103/PhysRevLett.98.140506](https://doi.org/10.1103/PhysRevLett.98.140506); arXiv: [quant-ph/0611050](https://arxiv.org/abs/quant-ph/0611050).
 - [R034] Norbert Schuch, J. I. Cirac, and David Pérez-García, “PEPS as ground states: Degeneracy and topology,” *Annals of Physics* 325, 2153–2192 (2010). DOI: [10.1016/j.aop.2010.05.008](https://doi.org/10.1016/j.aop.2010.05.008); arXiv: [1001.3807](https://arxiv.org/abs/1001.3807).
 - [R035] Ulrich Schollwöck, “The density-matrix renormalization group in the age of matrix product states,” *Annals of Physics* 326, 96–192 (2011). DOI: [10.1016/j.aop.2010.09.012](https://doi.org/10.1016/j.aop.2010.09.012); arXiv: [1008.3477](https://arxiv.org/abs/1008.3477).
 - [R036] Román Orús, “A practical introduction to tensor networks: Matrix product states and projected entangled pair states,” *Annals of Physics* 349, 117–158 (2014). DOI: [10.1016/j.aop.2014.06.013](https://doi.org/10.1016/j.aop.2014.06.013); arXiv: [1306.2164](https://arxiv.org/abs/1306.2164).
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Global evidence table

This table inventories the evidence claim by claim. “Experimentally partial” records that some ingredients are supported within the stated platform and scope; it leaves the full wording of an ambitious claim unvalidated. “Theoretically established” also covers a limitation result that rejects the claim, as in GAP-04. The evidence column belongs to every status; the two are read together.

Reference labels point to the annotated bibliography. Each experimental result belongs with its platform, operating conditions, and stated scope.

Each row below judges one claim by the type and scope of its supporting evidence.

Claim	Status	Platform	Evidence	Reference
HOST-01 — Coherent single-NV control	experimentally demonstrated	Diamond NV ⁻	Single-center spectroscopy, optical initialization and readout, and coherent microwave-driven rotations	[R074]
HOST-02 — Diamond is an automatic overall winner	experimentally partial	Diamond color centers	Strongest integrated evidence for single-node optical and spin control; uniform interacting arrays absent	[R074]; [R207]
HOST-03 — SiC supplies controllable defect-spin nodes	experimentally demonstrated	4H-SiC divacancies and silicon vacancies	Individual optical isolation and control, millisecond-scale coherence, patterned 8x8 defect pattern	[R096]; [R099]
HOST-04 — SiC supplies the required electron-spin interaction graph	experimentally partial	4H-SiC	Patterned implantation and integrated spin registers; no nanometre-registered electron-electron couplings	[R099]; [R205]
HOST-05 — Sapphire matches diamond as an individual-defect platform	experimentally partial	Cr ³⁺ :sapphire and sapphire color centers	Ensemble ODMR, long low-temperature T ₁ , collective coupling to a cavity; no single-ion control or T ₂	[R088]; [R089]
COUP-01 — Nearby defects couple coherently	experimentally demonstrated	Diamond NV ⁻ pairs	5 kHz magnetic dipolar interaction and room-temperature entanglement under refocused optimal control	[R106]; [R080]
COUP-02 — Cavity photons mediate defect-defect interaction	experimentally demonstrated	Diamond SiV ⁻ emitters in one nanocavity	Photon-mediated interaction through a shared cavity mode	[R086]
COUP-03 — Nanometre-separated defect pairs can have strong exchange	numerically demonstrated	Closely spaced diamond spin centers	Atomistic calculations predict direct exchange can dominate at very short separations	[R107]
ENC-01 — Encoded defect-associated qubits exist	experimentally demonstrated	NV electron- ¹³ C nuclear-spin registers	Decoherence-protected nuclear-spin subspaces	[R116]; [R117]

Claim	Status	Platform	Evidence	Reference
			and repeated error correction	
TOPO-01 — Local two-body spin models can host a non-Abelian phase	theoretically established	Kitaev honeycomb model	Exactly solved model supports emergent Ising-type non-Abelian phase	[R017]
TOPO-02 — Engineered analog spin models show topological-phase signatures	experimentally partial	Driven Rydberg-atom arrays	Signatures of a topological spin liquid under a driven blockade Hamiltonian	[R126]
TOPO-03 — The Levin-Wen target supports doubled-Fibonacci order	theoretically established	Ideal Fibonacci string-net Hamiltonian	Exact commuting-projector construction	[R018]
SIM-01 — Topologically ordered model states can be digitally prepared	experimentally demonstrated	Superconducting processors	Finite-depth gate circuits prepared toric-code-type states	[R125]
SIM-02 — Digital Fibonacci simulation has been demonstrated	experimentally demonstrated	Superconducting transmon processors	Prepared finite doubled-Fibonacci string-net states and performed fusion and braid operations	[R138]; [R165]
EMERG-01 — Genuine Fibonacci quasiparticles at $12/5$	experimentally partial	Fractional quantum Hall, superconducting, and defect candidates	Incompressible state established; controlled Fibonacci fusion and noncommuting braiding not established	[R226]; [R228]; [R138]
GAD-01 — Two-body interactions can generate many-body effective terms	theoretically established	Abstract perturbative gadgets	Controlled constructions can generate n-body effective terms at scale g^n / Δ^{n-1}	[R175]
GAP-01 — A weak perturbation preserves an already established gapped topological phase	theoretically established	Local topologically ordered Hamiltonians	Stability theorems preserve spectral structure against sufficiently weak local perturbations	[R142]
GAP-04 — A 2D local stabilizer Hamiltonian is automatically a self-correcting memory	theoretically established	Two-dimensional local stabilizer codes	Not automatically self-correcting; energy barrier remains bounded	[R168]; [R169]
FAB-01 — Exact single-ion delivery is possible	experimentally demonstrated	Deterministic nitrogen implantation in diamond	Detected individual delivered ions and produced patterned ^{15}NV centers	[R258]
FAB-06 — The required dense usable defect lattice is manufacturable	experimentally partial	Diamond and SiC implantation platforms	Capabilities demonstrated separately; not jointly in required repeated geometry	[R190]; [R203]; [R207]; [R205]

Complete evidence inventory and scope of every claim

Each entry below judges one claim by the type and scope of its supporting evidence.

An experiment on a pair of defects establishes the behavior of that pair; the behavior of an extended lattice needs separate evidence. A circuit that simulates anyonic operations demonstrates the compiled transformation; a crystalline material whose native excitations are anyons needs separate evidence.

The status categories are used exactly as written: experimentally demonstrated, experimentally partial, numerically demonstrated, theoretically established, theoretically proposed, and speculative.

Claim | Status | Platform | Evidence | Reference | |

HOST-01 — Coherent single-NV control | experimentally demonstrated | Diamond NV⁻ | The diamond NV⁻ center is a negatively charged nitrogen-vacancy point defect. Experiments have established single-center spectroscopy, optical initialization and readout, and coherent microwave-driven rotations. These results cover the capabilities required of a physical qubit. They supply no evidence about an array or a many-body phase. | [\[R074\]](#) | |

HOST-02 — Diamond is an automatic overall winner | experimentally partial | Diamond color centers | A color center is a localized lattice defect with optically accessible electronic states. Among the platforms considered in this survey, diamond color centers hold the strongest integrated evidence for single-node optical and spin control. Uniform interacting arrays with demonstrated nanometre-scale placement are absent, so comparative superiority depends on the chosen system-level performance metric. | [\[R074\]](#); [\[R207\]](#) | |

HOST-03 — SiC supplies controllable defect-spin nodes | experimentally demonstrated | 4H-SiC divacancies and silicon vacancies | In 4H-SiC, a divacancy is an adjacent pair of missing lattice atoms, while a silicon vacancy is a missing silicon atom. Experiments demonstrated individual optical isolation and control of these defects, together with millisecond-scale coherence. Later work combined a masked 8×8 defect pattern with a waveguide-integrated electron–nuclear spin register. | [\[R096\]](#); [\[R099\]](#) | |

HOST-04 — SiC supplies the required electron-spin interaction graph | experimentally partial | 4H-SiC | An interaction graph specifies which electron spins are coupled and the strengths of those couplings. Patterned implantation and integrated spin registers establish several required fabrication and control capabilities. Nanometre-registered electron–electron couplings throughout the patterned array remain unestablished. | [\[R099\]](#); [\[R205\]](#) | |

HOST-05 — Sapphire matches diamond as an individual-defect platform | experimentally partial | Cr^{3+} :sapphire and sapphire color centers | Experiments on ruby, which is Cr^{3+} -doped sapphire, demonstrate ensemble optically detected magnetic resonance (ODMR), long low-temperature longitudinal relaxation times T_1 , and collective coupling to a cavity. ODMR measures spin resonances optically, and T_1 characterizes relaxation of the spin population toward equilibrium. Single-ion control, transverse coherence time T_2 , readout, and two-defect gate capabilities remain undemonstrated, and they are required here. T_2 characterizes loss of phase coherence between superposition components, as distinct from the population relaxation measured by T_1 . | [\[Ro88\]](#); [\[Ro89\]](#) | |

COUP-01 — Nearby defects couple coherently | experimentally demonstrated | Diamond NV^- pairs | A selected pair separated by about 25 nm exhibited an approximately 5 kHz magnetic dipolar interaction and room-temperature entanglement under refocused optimal control. A dipolar interaction is the orientation- and distance-dependent magnetic coupling between two spins. This result establishes a two-defect interaction primitive, and nothing beyond it. | [\[R106\]](#); [\[Ro80\]](#) | |

COUP-02 — Cavity photons mediate defect–defect interaction | experimentally demonstrated | Diamond SiV^- emitters in one nanocavity | The SiV^- center is a negatively charged silicon-vacancy defect in diamond. Two spectrally tuned SiV^- emitters exhibited a photon-mediated interaction through a shared cavity mode, where a cavity mode is a confined electromagnetic field pattern. Uniform, low-loss mediation across a lattice remains undemonstrated. | [\[Ro86\]](#) | |

COUP-03 — Nanometre-separated defect pairs can have strong, geometry-sensitive direct exchange | numerically demonstrated | Closely spaced diamond spin centers | Atomistic calculations predict that direct exchange, the coupling produced by overlap and quantum statistics of electronic wavefunctions, can dominate at very short separations. The predicted exchange varies strongly with lattice site and orientation. An extensive, calibrated exchange-interaction graph lies beyond what this pair-level calculation establishes. | [\[R107\]](#) | |

COUP-04 — Ordinary coupled defects imply topological order | theoretically established | Any pair or finite defect register | Ordinary coupled defects imply no topological order. A measured two-body coupling supplies one term in the Hamiltonian, the operator that determines the system's energy and dynamics. Topological order further requires an extended gapped phase together with nonlocal sectors or observables. | [\[Ro80\]](#); [\[Ro15\]](#) | |

ENC-01 — Encoded defect-associated qubits exist | experimentally demonstrated | NV electron– ^{13}C nuclear-spin registers | Experiments have demonstrated decoherence-protected nuclear-spin subspaces and repeated error correction of a three-spin encoded qubit using NV electron spins as ancillary systems. A decoherence-free subspace (DFS) is an encoded subspace that is insensitive to specified environmental couplings. The observed protection comes from DFS or active encoding; emergent topological order plays no role in it. | [\[R116\]](#); [\[R117\]](#) | |

ENC-02 — A fabricated electronic-defect cluster is an isolated pseudospin | theoretically proposed | Three-color-center electronic-spin cluster | A pseudospin is an effective two-level degree of freedom used to represent a selected low-energy subspace. Theoretical projection onto an isolated doublet, a pair of low-energy states separated from other levels, is controlled. One triple-NV motif has also been observed. No cited experiment demonstrates a reproducible static three-center doublet together with measurements of the operators projected into that doublet. | [\[R025\]](#); [\[R243\]](#) | |

ENC-03 — Encoding a cluster establishes topological protection | theoretically established | Finite spin clusters | Encoding a finite cluster establishes no topological protection. A locally distinguishable cluster code can suppress specified noise or leakage from its computational subspace while lacking a thermodynamic phase, deconfined excitations that separate without growing energy cost, and nonlocal logical operators. | [\[R116\]](#); [\[R015\]](#) | |

TOPO-01 — Local two-body spin models can host a non-Abelian phase | theoretically established | Kitaev honeycomb model | The exactly solved Kitaev honeycomb model uses bond-dependent two-body spin interactions and supports an emergent Ising-type non-Abelian phase. Non-Abelian anyons are quasiparticles whose braiding acts through noncommuting transformations on a degenerate state space. The anyons in this model belong to the Ising type, distinct from Fibonacci anyons. | [\[R017\]](#) | |

TOPO-02 — Engineered analog spin models show topological-phase signatures | experimentally partial | Driven Rydberg-atom arrays | A programmable 219-atom array measured signatures of a topological spin liquid under a driven blockade Hamiltonian. A Rydberg blockade suppresses simultaneous excitation of nearby atoms and thereby generates constrained many-body dynamics. Finite system size, dependence on external driving, and the state-preparation procedure do not establish an autonomous memory. | [\[R126\]](#) | |

TOPO-03 — The Levin–Wen target supports doubled-Fibonacci order | theoretically established | Ideal Fibonacci string-net Hamiltonian | The ideal Fibonacci string-net Hamiltonian is a commuting-projector model whose local terms have compatible eigenspaces. Its construction specifies branching rules for allowed string configurations and plaquette recoupling data for transforming them. The resulting ideal many-body

model has doubled, nonchiral Fibonacci topological order, containing both chiral sectors with no net chirality. This theorem concerns the target model alone; it demonstrates nothing in a defect system. | [\[R018\]](#) | |

SIM-01 — Topologically ordered model states can be digitally prepared | experimentally demonstrated | Superconducting processors | Finite-depth gate circuits on superconducting processors have prepared toric-code-type states and measured stabilizers, entanglement, and string observables. Stabilizers are mutually compatible operators whose common eigenspace defines the code; string observables are extended operators supported along paths. These experiments left the processor's native Hamiltonian unconverted into a passive topological material. | [\[R125\]](#) | |

SIM-02 — Digital Fibonacci simulation has been demonstrated | experimentally demonstrated | Superconducting transmon processors | Experiments using superconducting transmons, which are weakly anharmonic circuit qubits, prepared finite doubled-Fibonacci string-net states and performed fusion and braid operations. The implementations used ordinary transmons, echo sequences, calibration, and/or error mitigation. A native Fibonacci energy gap appeared in neither experiment. | [\[R138\]](#); [\[R165\]](#) | |

SIM-03 — A defect processor could digitally simulate Fibonacci anyons :| theoretically proposed | Gate-controlled defect-spin processor | The simulation algorithm imposes no anyon requirement on the physical qubits. A defect array would still require adequate gate operations, connectivity, initialization, readout, and fidelity. A successful implementation would remain a digital simulation, with no native Fibonacci phase. | [\[R138\]](#); [\[R074\]](#) | |

EMERG-01 — A correct braid circuit establishes an emergent anyon | theoretically established | Any digital simulator | A correct braid circuit establishes no emergent anyon. Emergence requires the hardware many-body Hamiltonian to generate and stabilize localized excitations with the target fusion and braiding data throughout a phase. A compiled unitary operation establishes the simulated transformation alone. | [\[R018\]](#); [\[R138\]](#) | |

EMERG-02 — Genuine Fibonacci quasiparticles at $12/5$ | experimentally partial | Fractional quantum Hall, superconducting, and defect candidates | At fractional quantum Hall filling $12/5$, transport measurements establish an incompressible state, meaning a state with an excitation gap against density changes, and numerical calculations support candidates in the Read–Rezayi family. Controlled Fibonacci fusion and noncommuting braiding remain unestablished. Digital transmon results emulate native quasiparticles without observing them. | [\[R226\]](#); [\[R228\]](#); [\[R138\]](#) | |

EMERG-03 — A defect-based Fibonacci phase exists | speculative | Diamond, SiC, sapphire, and other defect lattices | No cited experiment or defect-specific numerical phase calculation demonstrates, within a single defect Hamiltonian, branching constraints, recoupling terms, a stable many-body gap, topological sectors, and Fibonacci diagnostics together. | [\[R018\]](#); [\[R175\]](#); [\[R207\]](#); [\[R080\]](#) | |

GAD-01 — Two-body interactions can generate many-body effective terms | theoretically established | Abstract perturbative gadgets | A perturbative gadget is a construction in which auxiliary high-energy degrees of freedom generate desired low-energy interactions. Controlled constructions can generate n -body effective terms at a scale roughly g^n/Δ^{n-1} , where g is the perturbative coupling and Δ is the auxiliary-state energy penalty. The errors remain bounded when the assumed perturbative hierarchy holds. | [\[R175\]](#) | |

GAD-02 — Perturbatively generated plaquette interactions exist | theoretically established | Kitaev honeycomb and code-gadget models | A plaquette interaction is a many-body operator associated with the sites or bonds around an elementary lattice cell. Fourth-order virtual processes generate plaquette operators in the anisotropic limit of the Kitaev honeycomb model. Two-body code gadgets likewise generate toric-code or quantum-double terms. These results are model constructions, with no defect-device demonstration. | [\[R017\]](#); [\[R177\]](#) | |

GAD-03 — A defect gadget realizes a Fibonacci plaquette | theoretically proposed | Defect-cluster string-net architecture | No cited mapping derives the complete Fibonacci plaquette recoupling operator from available defect interaction tensors while also bounding lower-order effective fields, leakage, and interaction paths between different gadgets. | [\[R175\]](#); [\[R018\]](#) | |

GAD-04 — The gadget penalty gap is the topological gap | theoretically established | Perturbative Hamiltonian engineering | The gadget penalty gap and the topological gap are distinct scales. The penalty Δ suppresses leakage into mediator states. The desired effective coefficient K and the gap of the extended model are parametrically smaller, and these smaller scales must independently exceed the relevant noise, disorder, and temperature scales. | [\[R175\]](#); [\[R177\]](#) | |

GAP-01 — A weak perturbation preserves an already established gapped topological phase | theoretically established | Local topologically ordered Hamiltonians | Under their stated assumptions, stability theorems preserve the spectral structure of a gapped topological phase against sufficiently weak, bounded, local perturbations. A system that only approximately implements the target Hamiltonian gains no gap from these theorems. | [\[R142\]](#) | |

GAP-02 — A defect-cluster topological gap has been established | speculative | Proposed defect-cluster arrays | No connected experiment or validated defect-specific many-body calculation provides the topological gap Δ_{topo} , defined as the relevant excitation-energy separation protecting the topological phase. Microscopic level splittings, pairwise coupling strengths, and perturbative-gadget penalties cannot substitute for this quantity. | [\[R142\]](#); [\[Ro18\]](#); [\[Ro80\]](#) | |

GAP-03 — Passive feasibility requires a gap budget | theoretically established | Static analog topological memory | For a static analog topological memory, the phase gap must exceed thermal energy, projected disorder, decoherence broadening, finite-size splitting, and unwanted local residual terms by a margin set by the intended task. | [\[R169\]](#); [\[R142\]](#) | |

GAP-04 — A 2D local stabilizer Hamiltonian is automatically a self-correcting memory | theoretically established | Two-dimensional local stabilizer codes | A two-dimensional local stabilizer Hamiltonian is not automatically a self-correcting memory. Under the assumptions of the no-go theorem, the energy barrier against logical errors remains bounded, so a zero-temperature phase gap alone produces no storage lifetime that improves indefinitely with system size at finite temperature. | [\[R168\]](#); [\[R169\]](#) | |

FAB-01 — Exact single-ion delivery is possible | experimentally demonstrated | Deterministic nitrogen implantation in diamond | A deterministic implanter detected individual delivered ions and produced patterned ^{15}NV centers. Exact ion delivery left NV conversion, charge state, coherence, and final relative defect geometry nondeterministic. | [\[R258\]](#) | |

FAB-02 — Focused implantation gives deterministic final defect coordinates | experimentally partial | Focused Si implantation for diamond SiV | For focused silicon implantation used to produce diamond SiV centers, the sub-40-nm beam width, calculated implantation straggle, and measured approximately 32-nm one-axis spread of the resulting centers were three distinct quantities. Straggle is the statistical displacement produced as implanted ions scatter and stop in the material. Defect conversion stayed process dependent. | [\[R190\]](#) | |

FAB-03 — Feedback laser writing gives near-unity usable-site yield | experimentally partial | Laser-written diamond NV^- | Under the reported conditions, feedback-controlled laser writing achieved approximately 96% single-NV creation with an approximately 33 nm in-plane positional deviation. Three-dimensional bond-position tolerance and the yield of a complete coherent array remain unestablished. | [\[R203\]](#) | |

FAB-04 — Delta doping solves three-dimensional placement | experimentally partial | Delta-doped diamond NV^- | Delta doping confined nitrogen during growth to a depth layer approximately 1–2 nm thick, while the lateral positions remained undetermined.

Later localized vacancy creation achieved approximately 4 nm depth and 46(1) nm lateral spread. | [\[R204\]](#); [\[R207\]](#) | |

FAB-05 — A multi-defect motif can be fabricated reproducibly | experimentally partial | Molecularly implanted diamond NV triples | One strongly coupled triple was identified among 7,116 implantation events. The observation shows the motif can occur. It establishes no deterministic repetition, target spectrum, or high fabrication yield. | [\[R243\]](#) | |

FAB-06 — The required dense usable defect lattice is manufacturable | experimentally partial | Diamond and SiC implantation platforms | Experiments have separately demonstrated subsets of the required capabilities, including beam registration, ion stopping and straggle control, structural conversion, charge-state preparation, optical and spin usability, coherence, bond tolerance, and compound array yield. Their joint demonstration in the required repeated geometry is still missing. | [\[R190\]](#); [\[R203\]](#); [\[R207\]](#); [\[R205\]](#) | |

CTRL-01 — Dense-array addressability is integrated with strong uniform coupling | experimentally partial | Diamond NV arrays | Experiments have separately demonstrated nanoscale optical selection, four-site frequency encoding, and parallel measurement of more than 100 resolved NV centers. Low-crosstalk encoded control within a uniformly interacting topological patch remains undemonstrated. | [\[R235\]](#); [\[R236\]](#); [\[R239\]](#) | |

KILL-01 — Cluster-spectrum gate | theoretically proposed | One candidate defect cluster | This decision criterion stops or redesigns the proposed implementation when no two-dimensional low-energy manifold stays isolated from leakage across the measured distributions of hyperfine coupling, strain, and geometry. Hyperfine coupling is the interaction between electronic and nuclear spins. The criterion is platform specific, not a universal threshold. | [\[R025\]](#); [\[R243\]](#) | |

KILL-02 — Desired-operator dominance gate | theoretically proposed | Three-to-six encoded clusters | Development advances under this criterion only when the target plaquette coefficient exceeds the predeclared sum or norm of noncommuting residual terms and the effective-model spectrum reproduces the exact low-energy spectrum with bounded leakage. | [\[R175\]](#); [\[R177\]](#) | |

KILL-03 — Many-body phase gate | theoretically proposed | Small patches followed by scaling | A nonzero plaquette coefficient alone establishes no many-body phase. The required evidence includes a stable excitation gap, the correct low-energy sectors, local indistinguishability of topological states, Wilson-loop or fusion data, robustness under perturbations, and trends with increasing system size. | [\[R018\]](#); [\[R142\]](#) | |

KILL-04 — Thermal-activation gate | theoretically proposed | Passive analog memory | The activation requirement scales with system size and the allowed error budget. For dilute error seeds, the approximate condition is $E_{\text{act}}/k_B T \gtrsim \ln(N_c g/\epsilon)$, where E_{act} is the activation energy, k_B is Boltzmann's constant, T is temperature, N_c is the number of relevant components or locations, g is a multiplicity factor, and ϵ is the error budget. Diffusion kinetics must then be tested separately. The condition $\Delta > k_B T$ alone is insufficient. | [\[R169\]](#); [\[R168\]](#) | |

KILL-05 — Placement/disorder gate | theoretically proposed | Direct :dipolar defect bonds | For direct dipolar defect bonds, $J \propto r^{-3}$, where J is the coupling strength and r is the defect separation. Before angular errors enter, a small radial spread therefore produces $\sigma_J/J \approx 3\sigma_r/r$, where σ_J and σ_r are the spreads in coupling and separation. Measured distribution tails and missing bonds belong in the phase model; nominal beam-spot dimensions cannot replace them. | [\[R106\]](#); [\[R207\]](#) | |

KILL-06 — Preparation-window gate | theoretically proposed | Adiabatic analog preparation | For a specified adiabatic preparation schedule, the proposed ramp is rejected when its lower time bound for adiabaticity exceeds the upper time bound imposed by decoherence. Slowing the ramp then satisfies neither constraint. | [\[R247\]](#) | |

KILL-07 — Compound-yield gate | theoretically proposed | Large fixed defect arrays | The yields for site occupancy, defect species, charge state, coherence, addressability, and bond formation are evaluated conditionally and propagated to the yield of a complete patch. A high yield for one fabrication step justifies no monolithic scale-up without repair mechanisms. | [\[R258\]](#); [\[R190\]](#); [\[R207\]](#) | |

KILL-08 — Logical-scaling gate | theoretically proposed | Encoded topological memory | A protection claim requires the logical error rate to decrease or the memory lifetime to increase with system size while physical conditions and analysis or decoder rules remain fixed. A finite-system prepared-state signature supplies no evidence of logical scaling. | [\[R141\]](#); [\[R125\]](#) | |

KILL-09 — Smallest decisive mechanism test | theoretically proposed | Four encoded clusters; candidate twelve-defect plaquette | After validating one-cluster and two-cluster behavior, a four-cluster system is the smallest proposed device or calculation that can reveal a weight-four coefficient together with every lower-weight competing term. Failure at this scale invalidates the stated mechanism before any attempt at Fibonacci scaling. | [\[R175\]](#); [\[R243\]](#) | |

Glossary

Use these short definitions while reading. The contextual glossary below gives longer explanations with the chapter where each term first appears.

Abelian anyon. An anyon for which exchange multiplies the state by a phase, so successive exchanges within the relevant sector commute.

Active quantum error correction. Protection by repeatedly extracting information about errors and applying a recovery operation.

Adjoint ($A^\dagger$). The conjugate transpose of a matrix or linear operator.

Amplitude. A complex number assigned to a possible quantum-mechanical outcome or to a component of a quantum state.

Analog Hamiltonian engineering. Design of physical couplings so a device evolves under a target Hamiltonian, either statically or within a controlled driven regime.

Anyon. A quasiparticle type possible in two spatial dimensions whose exchanges obey braid statistics beyond the bosonic and fermionic alternatives.

Band. A range of allowed electron energies formed when discrete atomic levels broaden in a periodic crystal.

Band gap. An energy interval with no extended bulk electronic states in an idealized band description.

Bloch sphere. A geometric representation mapping the pure states of a single two-level system to points on a sphere.

Bloch state. A spatially extended single-particle state consistent with lattice periodicity. It shares only a name with the qubit Bloch sphere.

Born rule. The rule converting a quantum amplitude into measurement-outcome probabilities.

Braid. The exchange history of particle positions in two dimensions with worldlines kept from crossing.

Braid group. The mathematical group generated by exchanges σ_i of neighboring particles, subject to the braid relations.

Charge state. The net electron-occupation state of a defect measured against a selected neutral reference.

Chirality. Handedness of propagation or topological response.

Coherence. Preserved phase relationships that permit quantum interference.

Color center. A localized electronic defect complex in a wide-bandgap crystal with characteristic optical absorption or emission.

Commuting-projector Hamiltonian. A Hamiltonian written as a sum of mutually commuting local projector-like terms.

Correlation length (ξ). The characteristic length over which local connected correlations decay appreciably in a gapped phase.

Crystal field. The electrostatic and covalent environment of neighboring atoms.

Decoherence. Loss of observable phase coherence as a system correlates with uncontrolled degrees of freedom.

Decoherence-free subspace (DFS). A subspace where a specified dominant noise interaction acts identically on all states.

Density operator (density matrix, ρ). A positive unit-trace operator representing a pure state or a statistical or marginal mixed state.

Defect. A deviation from a crystal's periodic structure or composition.

Defect cluster. A deliberately chosen set of nearby interacting defects handled as one subsystem.

Digital quantum simulation. Representation of a target evolution with gates on programmable qubits.

Dipolar interaction. The coupling between magnetic dipole moments. Its strength scales as $1/r^3$ and depends on orientation.

Doubled Fibonacci order. A nonchiral topological order combining Fibonacci data with its time-reversed counterpart.

Effective Hamiltonian. A Hamiltonian for selected low-energy degrees of freedom after higher-energy states are projected out or treated perturbatively.

Eigenstate and eigenvalue. An eigenstate $|a\rangle$ of an operator A and its corresponding eigenvalue a satisfy $A|a\rangle = a|a\rangle$.

Emergence. Collective low-energy degrees of freedom or effective laws not identifiable with any single microscopic constituent.

Encoded qubit. A two-dimensional information-bearing subspace inside a larger Hilbert space.

Entanglement. For a pure composite state, the impossibility of writing it as a product across a specified split; for a mixed state, the impossibility of writing it as a mixture of such products.

Exchange interaction. A short-range spin coupling from quantum indistinguishability and overlap of electronic wavefunctions.

F-move (recoupling move). A unitary basis change between different fusion orders of the same anyons.

Fidelity. A quantitative overlap between an actual state, gate, or readout operation and its target.

Fibonacci anyon. The nontrivial topological charge τ in Fibonacci theory, with fusion rule $\tau \times \tau = 1 + \tau$.

Fusion category. The data of particle types, allowed fusion channels, associativity transformations, and their consistency relations.

Fusion channel. One possible total topological charge from combining specified anyons.

Fusion rule. An expression $a \times b = \sum_c N_{ab}^c c$ that lists the total charges c obtainable by combining charges a and b .

Fusion space. The vector space of consistent fusion histories for anyons with fixed total charge.

Gap. An energy separation between specified sectors. The sectors (band, defect-level, cluster-leakage, many-body) differ; the entry below names them.

Ground-state degeneracy. More than one state at the lowest energy.

Hamiltonian (H). The operator giving the energies of a closed system and generating its unitary time evolution.

Hermitian operator. An operator equal to its adjoint, with real eigenvalues.

Hilbert space ($\mathcal{H}$). A complex inner-product vector space whose vectors represent quantum states.

Homotopy. Classification by continuous deformation without cutting, crossing a forbidden region, or violating stated boundary conditions.

Hyperfine interaction. Coupling between electronic and nuclear magnetic moments.

Initialization. Preparation of a qubit or many-body system in a known state or sector.

Interaction graph. A graph whose vertices are degrees of freedom and whose edges are available couplings.

Ket ($|\psi\rangle$). The notation for a vector in Hilbert space.

Leakage. Evolution out of the subspace chosen to represent a qubit or another encoded degree of freedom.

Locality. Hamiltonian terms or operations acting only on nearby degrees of freedom.

Localized state. A state with spatial weight concentrated near a defect or finite region.

Logical qubit. A two-dimensional information-bearing degree of freedom encoded within a larger physical system.

Low-energy doublet. Two cluster eigenstates selected to define an effective qubit, separated from other states by a leakage gap.

Majorana mode. An emergent degree of freedom whose operator equals its own adjoint.

Many-body gap. The energy separation between a many-body ground-state sector and the relevant excitations, for a stated size and limiting procedure.

Measurement (readout). An operation producing a classical outcome with probabilities set by the quantum state.

Mixed state. A quantum state whose density operator has more than one nonzero eigenvalue.

Non-Abelian anyon. An anyon whose exchanges act as generally noncommuting matrices on a multidimensional fusion space. Non-Abelian covers Ising and other theories besides Fibonacci.

Noise. Uncontrolled fluctuations or couplings modifying states, gates, measurements, or Hamiltonian parameters.

NV center. A nitrogen-vacancy complex in diamond. Spin, optical, and coherence properties depend on charge state and environment; see the contextual entry.

Partial trace. The operation on a composite density operator that eliminates an unobserved subsystem, leaving the reduced state of the rest.

Passive protection. Error suppression from the energy spectrum, locality, or Hamiltonian alone, without repeated syndrome measurements and recovery.

Pauli operators. The matrices X , Y , and Z , which describe observables and generate rotations for a two-level quantum system.

Perturbative gadget. A construction where auxiliary states and weak couplings let virtual processes produce a desired effective interaction.

Phonon. A quantized collective vibration of a crystal lattice.

Physical qubit. A directly controlled two-level subsystem of a hardware platform.

Projector (P). An operator satisfying $P^2 = P$ that selects a subspace of the full state space.

Pseudospin. An effective two-level degree of freedom using spin-1/2 mathematics.

Quantum dimension (d_a). The asymptotic growth rate of the fusion space as many anyons of type a are added.

Quasiparticle. A collective excitation treated as a particle in an effective many-body description.

Qubit. A controllable quantum degree of freedom with a selected two-dimensional state space. Usable operation needs initialization, coherent control, and readout, not just two levels.

R-move. The unitary for exchanging two anyons in a specified fusion channel.

Relaxation time (T_1). The characteristic timescale for a state population to return toward equilibrium. A long relaxation time leaves phase coherence a separate question.

Schrieffer-Wolff transformation. A perturbative unitary that block-diagonalizes a Hamiltonian to produce a low-energy Hamiltonian.

Simulated or emulated anyon. A state, defect, code excitation, or gate action deliberately mapped to an anyon model on hardware that may host no intrinsic anyonic phase.

Spin. Intrinsic quantum angular momentum. In a solid the label can absorb orbital and crystal-field character; see the contextual entry.

Spin-orbit coupling. An interaction between spin and orbital motion shaped by crystal symmetry.

Stabilizer. An operator whose specified eigenvalue defines part of a code or model subspace.

Stacking fault. An extended planar defect interrupting the normal stacking order of lattice layers.

Strain. A spatial deformation of a crystal. It shifts or mixes defect levels and can serve as noise, inhomogeneity, tuning control, or coupling channel.

String operator. An operator built as a product of local operators along a path.

String-net. A fluctuating network of labelled strings under local branching and recoupling rules.

Substitutional defect. A lattice site occupied by an atomic species different from the ideal host atom.

Superexchange. An effective localized-spin interaction mediated by virtual processes through intermediate orbitals or sites.

Superposition. A linear combination of quantum states. Its amplitudes interfere, which separates it from classical uncertainty.

Symmetry-protected subspace. A subspace where a symmetry constrains matrix elements so selected transitions are forbidden or suppressed.

T_2 and T_2^* . T_2 is the characteristic timescale for homogeneous phase coherence under a specified refocusing convention. T_2^* characterizes free-induction decay.

Tensor product ($\otimes$). The operation combining quantum state spaces.

Topological charge. A label for an anyon sector giving its fusion and braiding behavior.

Topological error protection. Operational suppression of specified logical errors by nonlocal encoding, an energy gap, braid structure, active correction, or a combination. Protection is never total; each claim states its conditions.

Topological invariant. A quantity unchanged under a specified class of continuous deformations. The winding number is the classical example.

Topological order. Intrinsic many-body order with nonlocal structure: long-range entanglement, topology-dependent ground-state sectors, and anyonic excitations in the standard 2D gapped examples.

Topological phase. A phase of matter with topologically organized low-energy states and excitations. Unqualified use here means intrinsic topological order unless stated otherwise.

Topological quantum computation. Storage and processing of quantum information in nonlocal topological degrees of freedom, often by creating, fusing, measuring, and braiding anyons.

Topology. The study of properties invariant under continuous deformations under stated rules. The mathematics alone establishes no error protection or quantum phase.

Toric code. An exactly solvable spin model with mutually commuting star and plaquette terms. Its Abelian anyons and benchmark role are detailed in the contextual entry.

Unitary evolution. Reversible quantum time evolution of the form $|\psi(t)\rangle = U(t)|\psi(0)\rangle$.

Vacancy. A missing atom at a lattice site. Structure alone does not fix qubit suitability; see the contextual entry.

Virtual excitation. An intermediate high-energy state in perturbation theory, never populated as a long-lived real excitation.

Wilson loop. A closed nonlocal operator diagnosing or manipulating gauge and topological sectors. It differs from a locally measured stabilizer.

Winding number. An integer counting how many times a directed loop encircles a puncture under the relevant conditions.

Zero-field splitting. An energy splitting between spin sublevels with no externally applied magnetic field. It is a spectral parameter, not a coherence time.

Definitions in context and first-appearance references

This glossary serves as a reference aid, not prerequisite material. Entries for terms not yet introduced may be deferred. In each citation, “First” names the chapter where this book first explains the term, not the term's historical introduction date.

Four concepts recur together and need separation:

Ordinary defect qubit. A single crystal defect providing two controllable and measurable levels drawn from its localized spin or charge states. (Chapter 3, then 7.)

Encoded defect cluster. Several defects whose joint low-energy state functions as one encoded bit with reduced sensitivity to specified noise. The constituents stay local. (Chapter 11.)

Digital anyon simulation. A classical computer or gate-based quantum processor reproducing the mathematical behavior of an anyon model. The reproduction leaves open whether the hardware realizes an anyonic material phase. (Chapter 20.)

Emergent topological phase. A many-body phase whose intrinsic energy landscape supports emergent quasiparticles with topological properties. Establishing the phase needs its own diagnostics — gap, ground-state sectors, braiding data — not only a prepared state that resembles one of its states. (Chapters 16 and 21.)

A

Abelian anyon. An anyon for which exchange multiplies the state by a phase.

Successive exchanges within the relevant sector therefore commute. The e and m excitations of the toric code are standard examples.

(First: Chapter 13, “Anyons and braids.”)

Active quantum error correction. A protection method based on repeatedly extracting information about errors and applying a recovery operation. Passive energetic suppression from a Hamiltonian works differently, though one device may employ both. (First: Chapter 16, “Toric code.”)

Adjoint ($A^\dagger$). The conjugate transpose of a matrix or linear operator. An operator for which $A = A^\dagger$ is called Hermitian. (First: Chapter 1, “Quantum mechanics foundations.”)

Amplitude. A complex number assigned to a possible quantum-mechanical outcome or to a component of a quantum state. Outcome probabilities are determined by squared magnitudes, and interference between amplitudes can affect those probabilities. (First: Chapter 1, “Quantum mechanics foundations.”)

Analog Hamiltonian engineering. The design of physical couplings such that a device evolves under a target Hamiltonian, either statically or within a controlled driven regime.

Digital simulation represents the target interaction through a compiled gate sequence; analog engineering implements it directly in the device. Direct implementation still leaves approximations, heating, and unwanted Hamiltonian terms to be bounded.

(First: Chapter 21, “Analog Hamiltonian engineering.”)

Anyon. A type of quasiparticle possible in two spatial dimensions whose exchange operations obey braid statistics more general than the bosonic and fermionic alternatives. The term covers exchange and fusion structure only: not every unusual particle, and not every processor-prepared state, qualifies. (First: Chapter 13, “Anyons and braids.”)

B

Band. A range of allowed electron energies that results when discrete atomic levels broaden in a periodic crystal. The resulting band structure depends on the crystal lattice, chemical composition, and electron interactions. (First: Chapter 5, “Crystals, bands, and localized states.”)

Band gap. An energy interval containing no extended bulk electronic states within an idealized band description. A wide band gap assists in isolating defect levels; long coherence and optical addressability need separate evidence. (First: Chapter 5, “Crystals, bands, and localized states.”)

Bloch sphere. A geometric representation in which the pure states of a single two-level system correspond to points on a sphere and mixed states correspond to points inside the sphere. This representation describes qubit states, a different object from a crystal Bloch state. (First: Chapter 3, “Qubits as controllable systems.”)

Bloch state. A spatially extended single-particle state consistent with lattice periodicity, expressed as the product of a plane-wave factor and a periodic function. The shared name notwithstanding, the crystal state and the qubit sphere describe different objects. (First: Chapter 5, “Crystals, bands, and localized states.”)

Born rule. The rule that converts a quantum amplitude into the probability of a measurement outcome. For a normalized state $|\psi\rangle$ and a projector P , the probability of the corresponding outcome is $\langle\psi|P|\psi\rangle$. (First: Chapter 1, “Quantum mechanics foundations.”)

Braid. The history produced by exchanging particle positions in two dimensions while preventing their worldlines from crossing. Braids that resist continuous deformation into one another may transform an anyonic state space differently. (First: Chapter 13, “Anyons and braids.”)

Braid group. The mathematical group generated by exchanges σ_i of neighboring particles, subject to the braid relations. A non-Abelian anyon model assigns matrices to these generators, defining a representation of the braid group. (First: Chapter 13, “Anyons and braids.”)

C

Charge state. The net electron-occupation state of a defect relative to a selected neutral reference. Examples include the labels NV^- and NV^0 . Different charge states can carry entirely different spin and optical properties. (First: Chapter 7, “Diamond color centers.”)

Chirality. A handedness associated with propagation or topological response. A chiral topological phase differs from the nonchiral doubled theory combining that phase with its time-reversed counterpart. (First: Chapter 19, “Doubled versus chiral Fibonacci.”)

Coherence. The preservation of phase relationships that permit quantum interference. Separate experiment-dependent times characterize it; population lifetime, fidelity, and topological protection measure different quantities. (First: Chapter 2, “Composite quantum systems”; measured in Chapter 4.)

Color center. A localized electronic defect complex in a wide-bandgap crystal that produces characteristic optical absorption or emission. Some color centers also support useful spin degrees of freedom. Optical activity alone leaves open whether a given center functions as a qubit or joins a scalable array. (First: Chapter 7, “Diamond color centers.”)

Commuting-projector Hamiltonian. A Hamiltonian expressed as a sum of mutually commuting local terms that have the form or function of projectors. Such models can be exactly solvable and make topological structure explicit. Their microscopic implementation may require interactions unavailable in a given physical platform. (First: Chapter 16, “Toric code.”)

Correlation length (ξ). The characteristic length scale over which local connected correlations decay appreciably in a gapped phase. Asymptotic topological arguments apply reliably when finite devices are large relative to the relevant correlation lengths. (First: Chapter 21, “Analog Hamiltonian engineering.”)

Crystal field. The electrostatic and covalent environment created by neighboring atoms. This environment splits otherwise degenerate electronic orbitals by local symmetry. (First: Chapter 6, “The defect zoo and its interactions.”)

Crystal lattice. The periodic arrangement used to represent an ideal crystal. It defines sites, directions, and symmetries; a real sample adds boundaries, vibrations, disorder, and defects. (First: Chapter 5, “Crystals, bands, and localized states.”)

Crosstalk. An unintended response of non-target qubits or couplings during control or readout. In a dense array, crosstalk can correlate errors even when each control operation tests accurately in isolation. (First: Chapter 35, “Addressing dense arrays.”)

D

Decoherence. The loss of observable phase coherence when a system becomes correlated with uncontrolled degrees of freedom or undergoes random evolution. Decoherence can turn a pure state into a mixed reduced state with no direct energy relaxation. (First: Chapter 2, “Composite quantum systems.”)

Decoherence-free subspace (DFS). A subspace on which a specified dominant noise interaction acts identically on all states, leaving relative quantum information unaffected by that interaction. The protection assumes its noise model; it is not topological order. (First: Chapter 11, “Defect clusters as encoded qubits.”)

Density operator (density matrix, ρ). A positive operator with unit trace that represents either a pure state or a statistical or marginal mixed state. It suits settings with classical uncertainty or entanglement with an unobserved environment. (First: Chapter 2, “Composite quantum systems.”)

Defect. A deviation from the periodic structure or composition of a crystal. Optical activity, spin, or qubit function needs its own evidence.

(First: Chapter 5, “Crystals, bands, and localized states”; taxonomy in Chapter 6.)

Defect cluster. A deliberately chosen set of nearby interacting defects treated as a single subsystem. The term “cluster” specifies a grouping of microscopic constituents; replacing it with an encoded pseudospin is justified only after an isolated low-energy subspace is demonstrated. (First: Chapter 11, “Defect clusters as encoded qubits.”)

Digital quantum simulation. The representation of a target evolution using gates acting on programmable qubits, generally after discretization and compilation. Agreement between simulated and target observables validates the simulation; it leaves the hardware outside the simulated material phase. (First: Chapter 20, “Digital simulation.”)

Dipolar interaction. The coupling between magnetic dipole moments. Its strength scales as $1/r^3$ and depends on the orientation of the dipoles relative to their displacement vector. Its long range couples distant defects, while its anisotropy and unintended couplings constrain lattice design. (First: Chapter 10, “Defect–defect coupling.”)

Dislocation. An extended line defect defined by a mismatch in lattice registry. A dislocation generates strain and electronic states over distances substantially beyond a point defect's reach. (First: Chapter 6, “The defect zoo and its interactions.”)

Disorder. Spatial variation in on-site energies, couplings, fields, positions, or other parameters relative to an intended model. Disorder can close a gap, localize excitations, broaden transitions, or in some cases stabilize a regime; its consequences require calculation, not qualitative assumption. (First: Chapter 21, “Analog Hamiltonian engineering”; budgeted in Chapter 29.)

Doubled Fibonacci order. A nonchiral topological order that combines Fibonacci topological data with their time-reversed counterpart, as occurs naturally in the corresponding Levin–Wen string-net construction. Doubled Fibonacci order is the nonchiral counterpart of the chiral Fibonacci phase. (First: Chapter 18, “Levin–Wen string nets”; distinction developed in Chapter 19.)

E

Effective Hamiltonian. A Hamiltonian describing selected low-energy degrees of freedom after higher-energy states have been projected out or treated perturbatively. Its validity rests on separation between energy scales and on the size of omitted corrections. (First: Chapter 11, “Defect clusters as encoded qubits”; derived systematically in Chapter 22.)

Eigenstate and eigenvalue. An eigenstate $|a\rangle$ of an operator A and its corresponding eigenvalue a satisfy $A|a\rangle = a|a\rangle$. If A is an observable, a is a possible measurement outcome; if A is a Hamiltonian, a is an energy. (First: Chapter 1, “Quantum mechanics foundations.”)

Emergence. The occurrence of collective low-energy degrees of freedom or effective laws that cannot be identified with any single microscopic constituent. In this book, an emergent anyon belongs to an excitation sector of a many-body phase; a relabeled hardware qubit does not qualify. (First: Chapter 13, “Anyons and braids”; Hamiltonian example in Chapter 17.)

Encoded qubit. A two-dimensional information-bearing subspace embedded in a larger Hilbert space, often distributed across several physical constituents. Encoding suppresses only the noise or leakage process it targets; error correction and topological protection are separate constructions. (First: Chapter 3, “Qubits as controllable systems”; cluster construction in Chapter 11.)

Entanglement. A property of a composite quantum state that prevents it from being represented as a product state or, when mixed-state distinctions are relevant, as an appropriate classical mixture. Observed correlation alone establishes no entanglement. (First: Chapter 2, “Composite quantum systems.”)

Exchange interaction. A short-range spin coupling arising from quantum indistinguishability and overlap between electronic wavefunctions. Its magnitude often varies exponentially with atomic arrangement, which makes it sensitive to placement and chemistry. (First: Chapter 10, “Defect–defect coupling.”)

F

F-move (recoupling move). A unitary basis transformation between different orders of fusing the same anyons, including $(a \times b) \times c$ and $a \times (b \times c)$. An *F*-move changes the fusion basis with no physical exchange of the anyons. (First: Chapter 14, “Fusion categories without the fog.”)

Fidelity. A quantitative measure of overlap or success that compares an actual state, gate, or readout operation with a target. State, process, average-gate, and readout fidelities are distinct quantities; each needs its definition and operating conditions. (First: Chapter 4, “Stability, coherence, and fidelity.”)

Fibonacci anyon. The nontrivial topological charge τ in Fibonacci theory, with fusion rule $\tau \times \tau = 1 + \tau$.

Its fusion spaces grow in dimension by Fibonacci counting. Under standard encodings and assumptions, braiding supports a computationally universal gate set. A circuit signature resembling Fibonacci behavior still needs intrinsic fusion and braiding data to qualify as a genuine Fibonacci quasiparticle.

(First: Chapter 15, “Fibonacci theory.”)

Finite-size splitting. A small energy difference between states that become exactly degenerate only in an infinite or ideal topological system. It is often produced by virtual quasiparticle tunnelling across a finite sample. A small splitting protects in a scale-dependent way; exact degeneracy in a finite device is a separate claim. (First: Chapter 21, “Analog Hamiltonian engineering.”)

Fusion category. Mathematical data specifying particle types, allowed fusion channels, transformations of associativity, and their consistency relations. The structures used in topological quantum computation additionally require braiding and nondegeneracy data. (First: Chapter 14, “Fusion categories without the fog.”)

Fusion channel. A possible total topological charge obtained when specified anyons combine. Multiple fusion channels produce a fusion space for encoding quantum information. (First: Chapter 14, “Fusion categories without the fog.”)

Fusion rule. An expression $a \times b = \sum_c N_{ab}^c c$ that lists the total charges c obtainable by combining charges a and b , with multiplicities N_{ab}^c . A fusion rule lists allowed outcomes; the complete braid theory needs more. (First: Chapter 14, “Fusion categories without the fog.”)

Fusion space. The vector space consisting of consistent fusion histories for a collection of anyons with fixed total charge.

For non-Abelian anyons, braids act as matrices on this space. A digitally encoded fusion space remains a simulator's state space unless intrinsic quasiparticles generate it.

(First: Chapter 13, “Anyons and braids”; formalized in Chapter 14.)

G–H

Gap. An energy separation between specified sectors. This book separates four energy scales with different physical meanings: crystal band gap, defect-level splitting, cluster leakage gap, and many-body topological gap. None substitutes for another. (First: Chapter 5, “Crystals, bands, and localized states”; cluster use in Chapter 11 and topological use in Chapter 16.)

Genuine Fibonacci anyon. An intrinsic emergent quasiparticle whose fusion and braiding data realize the specified Fibonacci theory, rather than a hardware qubit programmed to reproduce those data. Any such claim also specifies whether the host phase is chiral Fibonacci, doubled Fibonacci, or another explicitly defined theory; “Fibonacci-like” does not suffice. (First: Chapter 15, “Fibonacci theory”; phase distinction in Chapter 19.)

Ground-state degeneracy. The presence of more than one state at the lowest energy. In a topologically ordered system, this degeneracy and its dependence on spatial topology are nonlocal properties; an accidental local doublet differs from topological ground-state degeneracy. (First: Chapter 16, “Toric code.”)

Hamiltonian (H). The operator that specifies the energies of a closed system and generates its unitary time evolution. A proposed hardware Hamiltonian includes both the intended terms and the platform's actual corrections. (First: Chapter 1, “Quantum mechanics foundations”; physical defect form in Chapter 26.)

Hermitian operator. An operator equal to its adjoint. Its real eigenvalues let Hermitian operators represent observables such as energy. (First: Chapter 1, “Quantum mechanics foundations.”)

Hilbert space ($\mathcal{H}$). A complex inner-product vector space whose vectors represent quantum states. Its dimension counts independent state amplitudes, not necessarily particles or sites. (First: Chapter 1, “Quantum mechanics foundations.”)

Homotopy. A classification based on continuous deformation without cutting, crossing a forbidden region, or violating specified boundary conditions. Homotopy gives the mathematical language for winding and braiding; quantum topological order needs more. (First: Chapter 12, “Topology for non-mathematicians.”)

Hyperfine interaction. The coupling between electronic and nuclear magnetic moments. Hyperfine interactions can supply useful nuclear memories or spectrally resolved control, and they can also cause dephasing and spectral complexity. (First: Chapter 6, “The defect zoo and its interactions.”)

I–L

Initialization. The preparation of a qubit or many-body system in a known state or sector. High-fidelity initialization of one qubit leaves preparation of a topologically ordered ground state a separate task. (First: Chapter 3, “Qubits as controllable systems.”)

Interaction graph. A graph in which vertices represent degrees of freedom and edges represent available couplings, often annotated by coupling type, strength, and direction. Spatial proximity alone leaves the match between the available interaction graph and the target Hamiltonian's graph unchecked. (First: Chapter 10, “Defect–defect coupling.”)

Interstitial. An atom located between the regular lattice sites. An interstitial is a point defect that may move or combine with other defects into a complex. (First: Chapter 6, “The defect zoo and its interactions.”)

Ket ($|\psi\rangle$). The notation for a vector in Hilbert space. The corresponding bra $\langle\psi|$ is its adjoint, and $\langle\phi|\psi\rangle$ denotes an inner product. (First: Chapter 1, “Quantum mechanics foundations.”)

Leakage. Evolution out of the subspace selected to represent a qubit or another encoded degree of freedom. An energy separation between a cluster's computational states and unwanted states suppresses leakage without eliminating it under strong, noisy, or resonant control. (First: Chapter 4, “Stability, coherence, and fidelity”; cluster leakage in Chapter 11.)

Locality. The condition that Hamiltonian terms or operations act only on nearby degrees of freedom or on a small number of degrees of freedom. Topological stability statements concern sufficiently weak local perturbations, not arbitrary global errors. (First: Chapter 12, “Topology for non-mathematicians”; many-body use in Chapter 16.)

Localized state. A state whose spatial weight is concentrated near a defect or finite region instead of extending throughout the crystal. Localization can isolate a degree of freedom at the price of weaker controllable coupling to neighbors. (First: Chapter 5, “Crystals, bands, and localized states.”)

Logical qubit. A two-dimensional information-bearing degree of freedom inside a larger physical system. It may be a conventional-code qubit, a cluster pseudospin, or a nonlocal topological encoding; “logical” alone names no protection mechanism. (First: Chapter 3, “Qubits as controllable systems.”)

Low-energy doublet. Two eigenstates of a cluster that are selected to define an effective qubit and are separated from other states by a leakage gap.

A useful doublet additionally supports controllable projected operators, preparation, and readout; a doublet alone implies no anyon.

(First: Chapter 11, “Defect clusters as encoded qubits.”)

M–N

Majorana mode. An emergent degree of freedom represented by an operator equal to its own adjoint. In topological systems, spatially separated Majorana zero modes can encode information nonlocally. Braiding of Majorana or Ising anyons is not computationally universal by itself; Fibonacci order needs a different theory. (First: Chapter 17, “Kitaev honeycomb model.”)

Many-body gap. The energy separation between a many-body ground-state sector and the relevant excitations, defined for a specified system size and limiting procedure. This gap sets thermal and perturbative energy scales; topological order needs separate diagnostics. (First: Chapter 16, “Toric code”; feasibility conditions in Chapter 21.)

Measurement (readout). An operation that produces a classical outcome with probabilities determined by the quantum state.

A complete description accounts for readout fidelity, measurement back-action, locality, and the measured operator; a local spin signal and a topological-charge measurement are different observables.

(First: Chapter 1, “Quantum mechanics foundations”; qubit implementation in Chapter 3.)

Mixed state. A quantum state represented by a density operator with more than one nonzero eigenvalue. A mixed state can come from classical uncertainty or from tracing out part of an entangled state. (First: Chapter 2, “Composite quantum systems.”)

Non-Abelian anyon. An anyon whose exchanges act as generally noncommuting matrices on a multidimensional fusion space. “Non-Abelian” covers Ising and other theories besides Fibonacci anyons. (First: Chapter 13, “Anyons and braids.”)

Noise. Uncontrolled fluctuations or couplings that modify states, gates, measurements, or Hamiltonian parameters. An adequate noise model includes spectrum, spatial and temporal correlations, and the coupling operator, not only a single coherence time. (First: Chapter 4, “Stability, coherence, and fidelity”; defect-array model in Chapter 30.)

NV center. A nitrogen-vacancy complex in diamond, commonly considered in its neutral and negatively charged forms. Useful spin, optical, and coherence properties depend on charge state, isotopic environment, strain, temperature, and device geometry. (First: Chapter 7, “Diamond color centers.”)

O–P

Operator. A linear map defined on a space of quantum states. An operator may represent an observable, a transformation, a projector, or a Hamiltonian term, depending on its space and role. (First: Chapter 1, “Quantum mechanics foundations.”)

Partial trace. A mathematical operation applied to the density operator of a composite quantum system to eliminate an unobserved subsystem. The result is the reduced state of the subsystem that remains. Taking a partial trace shows how a subsystem of a pure entangled system can be mixed. (First: Chapter 2, “Composite quantum systems.”)

Passive protection. Error suppression produced by the system’s energy spectrum, locality properties, or Hamiltonian, without repeated syndrome measurements and recovery operations. A syndrome is measurement information identifying errors without directly measuring the encoded information. Passive protection depends on temperature, system size, noise channels, and operation timescale, and it leaves some error types unsuppressed. (First: Chapter 16, “Toric code”; limitations in Chapters 21 and 31.)

Pauli operators. The matrices X , Y , and Z , which describe observables and generate rotations for a two-level quantum system. A Pauli operator on a physical spin and one on an encoded pseudospin belong to different Hilbert spaces even when written with

the same symbols. A Hilbert space is the complex vector space of a system's allowed states. (First: Chapter 3, “Qubits as controllable systems.”)

Perturbative gadget. A construction in which auxiliary states and weak couplings are introduced so that virtual processes produce a desired effective interaction, often involving more bodies than the microscopic interactions. The intended effective term is generally smaller than the microscopic scales and comes with higher-order errors. (First: Chapter 23, “Perturbative gadgets.”)

Phonon. A quantized collective vibrational excitation of a crystal lattice. Phonons can relax or dephase defect spins, broaden optical transitions, and in some settings mediate useful interactions. (First: Chapter 5, “Crystals, bands, and localized states.”)

Physical qubit. A directly controlled two-level subsystem in a hardware platform. Multiple physical qubits may encode one logical qubit, a qubit represented within a larger state space. “Physical qubit” names the directly controlled subsystem; the physical-to-logical ratio and protection mechanism come from the chosen encoding. (First: Chapter 3, “Qubits as controllable systems.”)

Projector (P). An operator satisfying $P^2 = P$ that selects a subspace of the full state space. In cluster engineering, P selects the low-energy sector used for encoding. Projecting another operator with P sets that operator's action within the low-energy approximation. (First: Chapter 1, “Quantum mechanics foundations”; cluster use in Chapter 11.)

Pseudospin. An effective two-level degree of freedom represented using the mathematics of a spin-1/2 system. Its basis states may combine several microscopic spin or orbital states; a pseudospin is a modelling construct, not necessarily an electron's literal spin. (First: Chapter 11, “Defect clusters as encoded qubits.”)

Q–R

Quantum dimension (d_a). The asymptotic growth rate of the fusion space as many anyons of type a are added; the fusion space is the state space of their possible collective fusion outcomes.

A quantum dimension can be non-integer; for the Fibonacci anyon τ , it equals the golden ratio. It differs from the ordinary dimension of one qubit's local state space.

(First: Chapter 14, “Fusion categories without the fog.”)

Quasiparticle. A collective excitation that can be treated as a particle within an effective many-body description. An emergent anyon is a quasiparticle with topological exchange and fusion properties. Spatial localization alone does not make a bare defect spin an anyon. (First: Chapter 13, “Anyons and braids.”)

Qubit. A controllable quantum degree of freedom with a selected two-dimensional state space. A usable qubit needs the full lifecycle of initialization, coherent operations, and readout; two spectral levels alone do not suffice. (First: Chapter 3, “Qubits as controllable systems.”)

R -move. The unitary for exchanging two anyons in a specified fusion channel — one possible total topological charge from their fusion. Together with the fusion data and F -moves, which change the basis for different fusion orderings, consistent R -moves determine how braids act. (First: Chapter 14, “Fusion categories without the fog.”)

Relaxation time (T_1). The characteristic timescale for a state population to return toward equilibrium, commonly following an excitation. A long T_1 does not imply a long phase-coherence time T_2 . (First: Chapter 4, “Stability, coherence, and fidelity.”)

S

Schrieffer–Wolff transformation. A perturbative unitary transformation that block-diagonalizes a Hamiltonian to a low-energy Hamiltonian by eliminating virtual transitions between a selected low-energy sector and high-energy sectors. The approximation is controlled only when couplings are small against the energy denominators, the energy differences suppressing the virtual transitions. (First: Chapter 22, “Schrieffer–Wolff transformation.”)

Simulated or emulated anyon. A state, defect, code excitation, or gate action deliberately mapped to an anyon model using hardware that may have no intrinsic anyonic phase. Such a simulation can reproduce the model's operations correctly with active protection by intervention and correction, not passive protection. (First: Chapter 13, “Anyons and braids”; experimental treatment in Chapter 20.)

Spin. An intrinsic form of quantum angular momentum. In a solid, an effective spin label can absorb orbital character and crystal-field effects, so the label needs explicitly specified energy levels and a Hamiltonian. (First: Chapter 1, “Quantum mechanics foundations”; defect setting in Chapter 6.)

Spin–orbit coupling. An interaction between spin and orbital motion whose form and strength are strongly influenced by crystal symmetry. Spin–orbit coupling can produce optical selection rules and large splittings, and it can also open phonon-mediated relaxation pathways. (First: Chapter 6, “The defect zoo and its interactions.”)

Stabilizer. An operator for which a specified eigenvalue defines part of a code or model subspace. A mutually commuting set jointly constrains the allowed states. Measuring stabilizers applies active correction; building the same operators into a Hamiltonian applies passive energy penalties. (First: Chapter 16, “Toric code.”)

Stacking fault. An extended planar crystal defect in which the normal ordering of lattice layers is interrupted. A stacking fault can reshape local electronic structure and strain a broad region. (First: Chapter 6, “The defect zoo and its interactions.”)

Strain. A spatial deformation of a crystal. Coupling to strain shifts or mixes defect energy levels. By origin and use, strain acts as noise, spatial inhomogeneity, tuning control, or interaction channel. (First: Chapter 6, “The defect zoo and its interactions.”)

String operator. An operator formed as a product of local operators along a path. In a topologically ordered model, an open string operator can create endpoint excitations, while a closed string along a noncontractible path — one not reducible to a point in the geometry — can implement a logical operation. (First: Chapter 16, “Toric code.”)

String-net. A fluctuating network of labelled strings governed by local branching and recoupling rules. In a Levin–Wen Hamiltonian, condensation of string nets realizes doubled topological orders. A drawn network alone establishes no such phase. (First: Chapter 18, “Levin–Wen string nets.”)

Substitutional defect. A crystal defect in which a lattice site is occupied by an atomic species different from the ideal host atom. The defect's electronic behavior depends on chemistry, local symmetry, and charge compensation. (First: Chapter 6, “The defect zoo and its interactions.”)

Superexchange. An effective interaction between localized spins that is mediated by virtual processes through intermediate orbitals or lattice sites. Microscopic hopping amplitudes and energy costs set its sign and magnitude, not distance alone. (First: Chapter 10, “Defect–defect coupling”; perturbative derivation in Chapter 22.)

Superposition. A linear combination of quantum states. A superposition predicts differently from classical statistical ignorance because its complex amplitudes interfere; it is not mere uncertainty about which classical state is present. (First: Chapter 1, “Quantum mechanics foundations.”)

Symmetry-protected subspace. A subspace in which selected transitions are forbidden or suppressed because the corresponding matrix elements are constrained by a symmetry. Breaking the symmetry can remove the protection, which differs from intrinsic topological order. (First: Chapter 11, “Defect clusters as encoded qubits.”)

T

T_2 and T_2^* . T_2 is the characteristic timescale for homogeneous phase coherence under a specified refocusing convention. T_2^* commonly characterizes free-induction decay with quasi-static inhomogeneity included. Each reported value needs its pulse sequence and fitting model. (First: Chapter 4, “Stability, coherence, and fidelity.”)

Tensor product ($\otimes$). The operation combining quantum state spaces. For two qubits, the joint space is $\mathcal{H}_1 \otimes \mathcal{H}_2$, not a choice between the two. The structure permits entangled states beyond subsystem products. (First: Chapter 2, “Composite quantum systems.”)

Topological charge. A label identifying an anyon sector and specifying the excitation’s behavior under fusion and braiding. Topological charge is conserved by the fusion rules and need not equal electric charge. (First: Chapter 14, “Fusion categories without the fog.”)

Topological error protection. Operational suppression of specified logical errors by nonlocal encoding, an energy gap, braid structure, active correction, or a combination. Protection is never total; each claim states its temperature, size, noise, leakage, preparation, and readout assumptions. (First: Chapter 16, “Toric code”; limits in Chapter 31.)

Topological invariant. A quantity that remains unchanged under a specified class of continuous deformations.

The winding number is the classical example. Classifying deformation classes is separate work from establishing quantum topological order, which needs the many-body diagnostics named in its entry.

(First: Chapter 12, “Topology for non-mathematicians.”)

Topological order. Intrinsic many-body order with nonlocal structure. Standard 2D gapped examples show long-range entanglement, topology-dependent ground-state sectors, and anyonic excitations — not ordinary symmetry breaking, a locally encoded cluster, or a digitally prepared wavefunction by itself. (First: Chapter 12, “Topology for non-mathematicians”; concrete model in Chapter 16.)

Topological phase. A phase of matter whose low-energy states and excitations have a specified topological organization. Here the unqualified term means intrinsic topological order unless stated otherwise; symmetry-protected phases are named explicitly. (First: Chapter 16, “Toric code.”)

Topological quantum computation. The storage and processing of quantum information in nonlocal topological degrees of freedom, often by creating, fusing, measuring, and braiding anyons. Gate set and protection depend on both the anyon theory and its implementation. (First: Chapter 13, “Anyons and braids”; Fibonacci case in Chapter 15.)

Topology. The mathematical study of properties that remain invariant under continuous deformations performed according to specified rules. Here topology supplies concepts and tools for describing global sectors; the word alone establishes neither error protection nor a quantum phase. (First: Chapter 12, “Topology for non-mathematicians.”)

Toric code. An exactly solvable spin model whose Hamiltonian contains mutually commuting star and plaquette terms. It has topological ground-state degeneracy and Abelian e and m anyons with Abelian fusion spaces, unlike Fibonacci anyons' non-Abelian structure. The toric code benchmarks topological order; it is not a Fibonacci model. (First: Chapter 16, “Toric code.”)

U–Z

Unit cell. A repeating block of a crystal lattice with the basis atoms reconstructing the ideal periodic structure; distinct from a defect cluster selected for encoding. (First: Chapter 5, “Crystals, bands, and localized states.”)

Unitary evolution. Reversible quantum time evolution of the form $|\psi(t)\rangle = U(t)|\psi(0)\rangle$, where the unitary operator $U(t)$ preserves inner products and total probability for a closed system. An open subsystem exchanging information or energy with its environment generally needs a quantum channel, not a unitary on that subsystem alone. (First: Chapter 1, “Quantum mechanics foundations.”)

Vacancy. A missing atom at a normally occupied lattice site. A vacancy may move, carry charge, lack an optical transition, or join a useful complex; structure alone does not fix its qubit suitability or properties. (First: Chapter 6, “The defect zoo and its interactions.”)

Virtual excitation. An intermediate high-energy state that appears in perturbation theory but is not populated as a long-lived real excitation. Paths through virtual excitations can generate superexchange or gadget terms, weakened by energy denominators. (First: Chapter 22, “Schrieffer–Wolff transformation.”)

Wilson loop. A closed, nonlocal operator used to diagnose or manipulate gauge sectors and topological sectors. A locally measured stabilizer differs from a noncontractible Wilson loop. (First: Chapter 16, “Toric code”; measurement in Chapter 36.)

Winding number. An integer that counts how many times a directed loop encircles a puncture under the relevant conditions. The winding number is invariant under allowed deformations without constituting a quantum-protected qubit by itself. (First: Chapter 12, “Topology for non-mathematicians.”)

Zero-field splitting. An energy splitting between spin sublevels that exists without an externally applied magnetic field. Internal spin–spin interactions and crystal-field anisotropy commonly produce it. The numerical value is a spectral parameter, not a coherence time. (First: Chapter 7, “Diamond color centers.”)

Annotated bibliography

The references are grouped by subject. Each work has one entry and a reference label that stays the same throughout the notes.

An entry links to the publication through its digital object identifier (DOI) when the record includes one. A DOI identifies a publication even if its web address changes. Entries posted on arXiv also link to the preprint record.

Introductory quantum mechanics

[R001] S. Axler, *Linear Algebra Done Right*, 3rd ed., Springer, 2015. DOI: [10.1007/978-3-319-11080-6](https://doi.org/10.1007/978-3-319-11080-6).

[R002] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013. DOI: [10.1017/CBO9781139020411](https://doi.org/10.1017/CBO9781139020411).

[R003] B. C. Hall, *Quantum Theory for Mathematicians*, Springer, 2013. DOI: [10.1007/978-1-4614-7116-5](https://doi.org/10.1007/978-1-4614-7116-5).

[R004] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed., Cambridge University Press, 2010. DOI: [10.1017/CBO9780511976667](https://doi.org/10.1017/CBO9780511976667).

[R005] A. Peres, *Quantum Theory: Concepts and Methods*, Kluwer Academic, 1995. DOI: [10.1007/0-306-47120-5](https://doi.org/10.1007/0-306-47120-5).

[R006] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, 2018. DOI: [10.1017/9781316848142](https://doi.org/10.1017/9781316848142).

[R007] W. F. Stinespring, "Positive functions on C-algebras," *Proceedings of the American Mathematical Society** 6, 211–216 (1955). DOI: [10.1090/S0002-9939-1955-0069403-4](https://doi.org/10.1090/S0002-9939-1955-0069403-4).

[R008] M.-D. Choi, "Completely positive linear maps on complex matrices," *Linear Algebra and its Applications* 10, 285–290 (1975). DOI: [10.1016/0024-3795\(75\)90075-0](https://doi.org/10.1016/0024-3795(75)90075-0).

[R009] G. Lindblad, "On the generators of quantum dynamical semigroups," *Communications in Mathematical Physics* 48, 119–130 (1976). DOI: [10.1007/BF01608499](https://doi.org/10.1007/BF01608499).

[R010] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, "Completely positive dynamical semigroups of N-level systems," *Journal of Mathematical Physics* 17, 821–825 (1976).

DOI: [10.1063/1.522979](https://doi.org/10.1063/1.522979).

[R011] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press, 2002. DOI: [10.1093/acprof:oso/9780199213900.001.0001](https://doi.org/10.1093/acprof:oso/9780199213900.001.0001).

[R037] M. Born, "Zur Quantenmechanik der Stoßvorgänge," *Zeitschrift für Physik* 37, 863–867 (1926). DOI: [10.1007/BF01397477](https://doi.org/10.1007/BF01397477).

[R038] W. Gerlach and O. Stern, "Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld," *Zeitschrift für Physik* 9, 349–352 (1922). DOI: [10.1007/BF01326983](https://doi.org/10.1007/BF01326983).

[R039] B. Friedrich and D. Herschbach, “Stern and Gerlach: How a Bad Cigar Helped Reorient Atomic Physics,” *Physics Today* 56(12), 53–59 (2003). DOI: [10.1063/1.1650229](https://doi.org/10.1063/1.1650229).

[R040] P. A. M. Dirac, *The Principles of Quantum Mechanics*, 4th ed., Oxford University Press (1958; reissued 1981). ISBN: 978-0-19-852011-5. Stable URL: [Oxford Academic](https://www.oxfordacademic.com/doi/10.1093/oxfordacademic/9780198520115).

[R041] E. Schrödinger, “Quantisierung als Eigenwertproblem (Vierte Mitteilung),” *Annalen der Physik* 386, 109–139 (1926). DOI: [10.1002/andp.19263861802](https://doi.org/10.1002/andp.19263861802).

[R042] A. Einstein, B. Podolsky, and N. Rosen, “Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?,” *Physical Review* 47, 777–780 (1935).

DOI: [10.1103/PhysRev.47.777](https://doi.org/10.1103/PhysRev.47.777).

[R043] J. S. Bell, “On the Einstein Podolsky Rosen Paradox,” *Physics Physique Fizika* 1, 195–200 (1964). DOI: [10.1103/PhysicsPhysiqueFizika.1.195](https://doi.org/10.1103/PhysicsPhysiqueFizika.1.195).

[R044] R. F. Werner, “Quantum states with Einstein-Podolsky-Rosen correlations admitting a hidden-variable model,” *Physical Review A* 40, 4277–4281 (1989). DOI: [10.1103/PhysRevA.40.4277](https://doi.org/10.1103/PhysRevA.40.4277).

[R045] K. Kraus, “General state changes in quantum theory,” *Annals of Physics* 64, 311–335 (1971). DOI: [10.1016/0003-4916\(71\)90108-4](https://doi.org/10.1016/0003-4916(71)90108-4).

[R046] W. H. Zurek, “Decoherence, einselection, and the quantum origins of the classical,” *Reviews of Modern Physics* 75, 715–775 (2003). DOI: [10.1103/RevModPhys.75.715](https://doi.org/10.1103/RevModPhys.75.715); arXiv: [quant-ph/0105127](https://arxiv.org/abs/quant-ph/0105127).

[R047] M. Schlosshauer, “Decoherence, the measurement problem, and interpretations of quantum mechanics,” *Reviews of Modern Physics* 76, 1267–1305 (2005). DOI: [10.1103/RevModPhys.76.1267](https://doi.org/10.1103/RevModPhys.76.1267); arXiv: [quant-ph/0312059](https://arxiv.org/abs/quant-ph/0312059).

[R048] B. Hensen et al., “Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres,” *Nature* 526, 682–686 (2015). DOI: [10.1038/nature15759](https://doi.org/10.1038/nature15759); arXiv: [1508.05949](https://arxiv.org/abs/1508.05949).

[R049] F. Bloch, “Nuclear Induction,” *Physical Review* 70, 460–474 (1946). DOI: [10.1103/PhysRev.70.460](https://doi.org/10.1103/PhysRev.70.460).

[R050] D. P. DiVincenzo, “The Physical Implementation of Quantum Computation,” *Fortschritte der Physik* 48, 771–783 (2000). DOI: [10.1002/1521-3978\(200009\)48:9<113.O.CO;2-E](https://doi.org/10.1002/1521-3978(200009)48:9<113.O.CO;2-E); arXiv: [quant-ph/0002077](https://arxiv.org/abs/quant-ph/0002077).

[R051] D. Loss and D. P. DiVincenzo, “Quantum Computation with Quantum Dots,” *Physical Review A* 57, 120–126 (1998). DOI: [10.1103/PhysRevA.57.120](https://doi.org/10.1103/PhysRevA.57.120).

[R052] F. H. L. Koppens, C. Buizert, K. J. Tielrooij, I. T. Vink, K. C. Nowack, T. Meunier, L. P. Kouwenhoven, and L. M. K. Vandersypen, “Driven Coherent Oscillations of a Single Electron Spin in a Quantum Dot,” *Nature* 442, 766–771 (2006). DOI: [10.1038/nature05065](https://doi.org/10.1038/nature05065).

[R053] J. M. Elzerman, R. Hanson, L. H. Willems van Beveren, B. Witkamp, L. M. K. Vandersypen, and L. P. Kouwenhoven, “Single-Shot Read-Out of an Individual Electron Spin in a Quantum Dot,” *Nature* 430, 431–435 (2004).

DOI: [10.1038/nature02693](https://doi.org/10.1038/nature02693).

- [Ro54] D. D. Awschalom, R. Hanson, J. Wrachtrup, and B. B. Zhou, “Quantum Technologies with Optically Interfaced Solid-State Spins,” *Nature Photonics* 12, 516–527 (2018). DOI: [10.1038/s41566-018-0232-2](https://doi.org/10.1038/s41566-018-0232-2).
- [Ro55] E. Knill and R. Laflamme, “Theory of Quantum Error-Correcting Codes,” *Physical Review A* 55, 900–911 (1997). DOI: [10.1103/PhysRevA.55.900](https://doi.org/10.1103/PhysRevA.55.900); arXiv: [quant-ph/9604034](https://arxiv.org/abs/quant-ph/9604034).
- [Ro56] N. F. Ramsey, “A Molecular Beam Resonance Method with Separated Oscillating Fields,” *Physical Review* 78, 695–699 (1950). DOI: [10.1103/PhysRev.78.695](https://doi.org/10.1103/PhysRev.78.695).
- [Ro57] E. L. Hahn, “Spin Echoes,” *Physical Review* 80, 580–594 (1950). DOI: [10.1103/PhysRev.80.580](https://doi.org/10.1103/PhysRev.80.580).
- [Ro58] G. Ithier et al., “Decoherence in a superconducting quantum bit circuit,” *Physical Review B* 72, 134519 (2005). DOI: [10.1103/PhysRevB.72.134519](https://doi.org/10.1103/PhysRevB.72.134519); arXiv: [cond-mat/0508588](https://arxiv.org/abs/cond-mat/0508588).
- [Ro59] Ł. Cywiński, R. M. Lutchyn, C. P. Nave, and S. Das Sarma, “How to enhance dephasing time in superconducting qubits,” *Physical Review B* 77, 174509 (2008). DOI: [10.1103/PhysRevB.77.174509](https://doi.org/10.1103/PhysRevB.77.174509); arXiv: [0712.2225](https://arxiv.org/abs/0712.2225).
- [Ro60] R. Jozsa, “Fidelity for Mixed Quantum States,” *Journal of Modern Optics* 41, 2315–2323 (1994). DOI: [10.1080/09500349414552171](https://doi.org/10.1080/09500349414552171).
- [Ro61] M. A. Nielsen, “A simple formula for the average gate fidelity of a quantum dynamical operation,” *Physics Letters A* 303, 249–252 (2002). DOI: [10.1016/S0375-9601\(02\)01272-0](https://doi.org/10.1016/S0375-9601(02)01272-0); arXiv: [quant-ph/0205035](https://arxiv.org/abs/quant-ph/0205035).
- [Ro62] E. Magesan et al., “Efficient Measurement of Quantum Gate Error by Interleaved Randomized Benchmarking,” *Physical Review Letters* 109, 080505 (2012). DOI: [10.1103/PhysRevLett.109.080505](https://doi.org/10.1103/PhysRevLett.109.080505); arXiv: [1203.4550](https://arxiv.org/abs/1203.4550).
- [Ro63] D. Keith et al., “Benchmarking high fidelity single-shot readout of semiconductor qubits,” (2018). arXiv: [1811.03630](https://arxiv.org/abs/1811.03630).
- [Ro64] C. J. Wood and J. M. Gambetta, “Quantification and characterization of leakage errors,” *Physical Review A* 97, 032306 (2018). DOI: [10.1103/PhysRevA.97.032306](https://doi.org/10.1103/PhysRevA.97.032306); arXiv: [1704.03081](https://arxiv.org/abs/1704.03081).
- [Ro65] Google Quantum AI and R. Acharya et al., “Suppressing quantum errors by scaling a surface code logical qubit,” *Nature* 614, 676–681 (2023). DOI: [10.1038/s41586-022-05434-1](https://doi.org/10.1038/s41586-022-05434-1); arXiv: [2207.06431](https://arxiv.org/abs/2207.06431).

Defect centers

- [Ro66] F. Bloch, “Über die Quantenmechanik der Elektronen in Kristallgittern,” *Zeitschrift für Physik* 52, 555–600 (1929). DOI: [10.1007/BF01339455](https://doi.org/10.1007/BF01339455).
- [Ro67] J. C. Slater and G. F. Koster, “Simplified LCAO Method for the Periodic Potential Problem,” *Physical Review* 94, 1498–1524 (1954). DOI: [10.1103/PhysRev.94.1498](https://doi.org/10.1103/PhysRev.94.1498).
- [Ro68] W. Kohn, “Analytic Properties of Bloch Waves and Wannier Functions,” *Physical Review* 115, 809–821 (1959). DOI: [10.1103/PhysRev.115.809](https://doi.org/10.1103/PhysRev.115.809).
- [Ro69] C. G. Van de Walle and J. Neugebauer, “First-principles calculations for defects and impurities: Applications to III-nitrides,” *Journal of Applied Physics* 95, 3851–3879 (2004).

DOI: [10.1063/1.1682673](https://doi.org/10.1063/1.1682673).

[Ro70] C. Freysoldt, B. Grabowski, T. Hickel, J. Neugebauer, G. Kresse, A. Janotti, and C. G. Van de Walle, “First-principles calculations for point defects in solids,” *Reviews of Modern Physics* 86, 253–305 (2014). DOI: [10.1103/RevModPhys.86.253](https://doi.org/10.1103/RevModPhys.86.253).

This review presents a systematic treatment of first-principles calculations, meaning electronic-structure calculations based on fundamental quantum-mechanical models, for charged point defects. A point defect is a localized departure from the ideal crystal structure, and a charged point defect carries a net charge relative to the corresponding defect-free crystal. The review covers finite-size corrections, which compensate for artifacts introduced by finite computational cells; transition levels, which specify the Fermi-level conditions at which a defect changes charge state; and formation energies, which quantify the energetic cost of creating a defect under specified conditions. It provides a basis for determining both the conclusions that electronic-structure calculations can support and the limitations of such calculations when evaluating proposed qubit defects.

[Ro71] J. R. Weber, W. F. Koehl, J. B. Varley, A. Janotti, B. B. Buckley, C. G. Van de Walle, and D. D. Awschalom, “Quantum computing with defects,” *Proceedings of the National Academy of Sciences* 107, 8513–8518 (2010).

DOI: [10.1073/pnas.1003052107](https://doi.org/10.1073/pnas.1003052107).

[Ro72] A. Alkauskas, Q. Yan, and C. G. Van de Walle, “First-principles theory of nonradiative carrier capture via multiphonon emission,” *Physical Review B* 90, 075202 (2014). DOI: [10.1103/PhysRevB.90.075202](https://doi.org/10.1103/PhysRevB.90.075202).

[Ro73] G. Wolfowicz et al., “Quantum guidelines for solid-state spin defects,” *Nature Reviews Materials* 6, 906–925 (2021).

DOI: [10.1038/s41578-021-00306-y](https://doi.org/10.1038/s41578-021-00306-y). Wolfowicz and collaborators integrate the materials, optical, spin, and coherence requirements for useful solid-state defects. Here, coherence is the preservation of the phase relationships required for quantum-state control.

The review is important because neither long coherence nor bright optical emission, considered independently, is sufficient to establish that a defect system can form a scalable qubit platform.

[Ro78] A. Norambuena, E. Muñoz, H. T. Dinani, A. Jarmola, P. Maletinsky, D. Budker, and J. R. Maze, “Spin-lattice relaxation of individual solid-state spins,” *Physical Review B* 97, 094304 (2018).

DOI: [10.1103/PhysRevB.97.094304](https://doi.org/10.1103/PhysRevB.97.094304); arXiv: [1711.10280](https://arxiv.org/abs/1711.10280).

[R100] H. L. Stern, Q. Gu, J. Jarman, et al., “Room-temperature optically detected magnetic resonance of single defects in hexagonal boron nitride,” *Nature Communications* 13, 618 (2022). DOI: [10.1038/s41467-022-28169-z](https://doi.org/10.1038/s41467-022-28169-z).

[R101] M. Zhong, M. P. Hedges, R. L. Ahlefeldt, et al., “Optically addressable nuclear spins in a solid with a six-hour coherence time,” *Nature* 517, 177–180 (2015).

DOI: [10.1038/nature14025](https://doi.org/10.1038/nature14025).

[R102] J. M. Kindem, A. Ruskuc, J. G. Bartholomew, et al., “Control and single-shot readout of an ion embedded in a nanophotonic cavity,” *Nature* 580, 201–204 (2020).

DOI: [10.1038/s41586-020-2160-9](https://doi.org/10.1038/s41586-020-2160-9); arXiv: [1907.12161](https://arxiv.org/abs/1907.12161).

[R103] B. Voisin, J. Bocquel, A. Tankasala, et al., “Valley interference and spin exchange at the atomic scale in silicon,” *Nature Communications* 11, 6124 (2020).

DOI: [10.1038/s41467-020-19835-1](https://doi.org/10.1038/s41467-020-19835-1).

[R105] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor, “CODATA recommended values of the fundamental physical constants,” NIST/CODATA database (2022 adjustment, accessed 2026), stable URL: <https://physics.nist.gov/cuu/Constants/>.

[R106] F. Dolde et al., “High-fidelity spin entanglement using optimal control,” *Nature Communications* 5, 3371 (2014).

DOI: [10.1038/ncomms4371](https://doi.org/10.1038/ncomms4371); arXiv: [1309.4430](https://arxiv.org/abs/1309.4430). This experiment applied optimal control, in which control fields are designed to implement a target evolution with high accuracy, to produce high-fidelity entanglement between coupled solid-state spins. Entanglement is a quantum correlation for which the joint state cannot be expressed as a product of independent subsystem states.

The result supports the availability of coherent interactions between pairs of spins. It also demonstrates that an effective two-spin Hamiltonian, meaning a reduced energy operator that models the relevant pair interaction, can omit the substantial experimental calibration required to realize that interaction accurately.

[R108] P. W. Anderson, “Antiferromagnetism. Theory of superexchange interaction,” *Physical Review* 79, 350–356 (1950). DOI: [10.1103/PhysRev.79.350](https://doi.org/10.1103/PhysRev.79.350).

[R112] P. Zanardi and M. Rasetti, “Noiseless quantum codes,” *Physical Review Letters* 79, 3306–3309 (1997). DOI: [10.1103/PhysRevLett.79.3306](https://doi.org/10.1103/PhysRevLett.79.3306).

[R113] E. Knill, R. Laflamme, and L. Viola, “Theory of quantum error correction for general noise,” *Physical Review Letters* 84, 2525–2528 (2000).

DOI: [10.1103/PhysRevLett.84.2525](https://doi.org/10.1103/PhysRevLett.84.2525); arXiv: [quant-ph/9908066](https://arxiv.org/abs/quant-ph/9908066).

[R114] D. P. DiVincenzo, D. Bacon, J. Kempe, G. Burkard, and K. B. Whaley, “Universal quantum computation with the exchange interaction,” *Nature* 408, 339–342 (2000).

DOI: [10.1038/35042541](https://doi.org/10.1038/35042541); arXiv: [quant-ph/0005116](https://arxiv.org/abs/quant-ph/0005116).

[R115] E. A. Laird, J. M. Taylor, D. P. DiVincenzo, C. M. Marcus, M. P. Hanson, and A. C. Gossard, “Coherent spin manipulation in an exchange-only qubit,” *Physical Review B* 82, 075403 (2010). DOI: [10.1103/PhysRevB.82.075403](https://doi.org/10.1103/PhysRevB.82.075403); arXiv: [1005.0273](https://arxiv.org/abs/1005.0273).

[R116] A. Reiserer, N. Kalb, M. S. Blok, K. J. M. van Bemmelen, D. J. Twitchen, M. Markham, T. H. Taminiau, and R. Hanson, “Robust quantum-network memory using decoherence-protected subspaces of nuclear spins,” *Physical Review X* 6, 021040 (2016). DOI: [10.1103/PhysRevX.6.021040](https://doi.org/10.1103/PhysRevX.6.021040); arXiv: [1603.01602](https://arxiv.org/abs/1603.01602).

Reiserer and collaborators encoded quantum information in a decoherence-protected subspace of diamond spins. A decoherence-protected subspace is an encoded subspace in which selected environmental couplings act trivially or identically on the encoded states, thereby suppressing the corresponding noise channels. The experiment provides direct evidence that small defect clusters can suppress selected noise channels. It does not establish that these clusters generate passive topological protection, in which encoded information is protected by the nonlocal properties of a topological phase without continuous active correction.

[R117] J. Cramer, N. Kalb, M. A. Rol, B. Hensen, M. S. Blok, M. Markham, D. J. Twitchen, R. Hanson, and T. H. Taminiau, “Repeated quantum error correction on a continuously encoded qubit by real-time feedback,” *Nature Communications* 7, 11526 (2016).

DOI: [10.1038/ncomms11526](https://doi.org/10.1038/ncomms11526); arXiv: [1508.01388](https://arxiv.org/abs/1508.01388). Cramer and collaborators repeatedly detected and corrected errors in a diamond nuclear-spin register.

The result establishes the capability for local encoded control. It also distinguishes active quantum error correction, which uses repeated error detection and corrective operations, from an equilibrium topological phase, which is a stable phase of matter characterized by topological order under equilibrium conditions.

[R118] H. P. Bartling, M. H. Abobeih, B. Pingault, M. J. Degen, S. J. H. Loenen, C. E. Bradley, J. Randall, M. Markham, D. J. Twitchen, and T. H. Taminiau, “Entanglement of spin-pair qubits with intrinsic dephasing times exceeding a minute,” *Physical Review X* 12, 011048 (2022). DOI: [10.1103/PhysRevX.12.011048](https://doi.org/10.1103/PhysRevX.12.011048); arXiv: [2103.07961](https://arxiv.org/abs/2103.07961).

[R119] C. E. Bradley, J. Randall, M. H. Abobeih, R. C. Berrevoets, M. J. Degen, M. A. Bakker, M. Markham, D. J. Twitchen, and T. H. Taminiau, “A ten-qubit solid-state spin register with quantum memory up to one minute,” *Physical Review X* 9, 031045 (2019). DOI: [10.1103/PhysRevX.9.031045](https://doi.org/10.1103/PhysRevX.9.031045); arXiv: [1905.02094](https://arxiv.org/abs/1905.02094).

Diamond

[R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* 528, 1–45 (2013). DOI: [10.1016/j.physrep.2013.02.001](https://doi.org/10.1016/j.physrep.2013.02.001); arXiv: [1302.3288](https://arxiv.org/abs/1302.3288).

This comprehensive review of the nitrogen-vacancy (NV) center, a point defect consisting of a substitutional nitrogen atom adjacent to a vacant carbon site in diamond, provides the electronic structure, spin Hamiltonian, optical cycle, strain response, and relaxation channels used throughout the manuscript. These results constrain the proposed architecture to interactions and selection rules supported by the physical properties of diamond.

[R075] Á. Gali, “Ab initio theory of the nitrogen-vacancy center in diamond,” *Reviews of Modern Physics* 91, 015004 (2019). DOI: [10.1103/RevModPhys.91.015004](https://doi.org/10.1103/RevModPhys.91.015004); arXiv: [1906.00047](https://arxiv.org/abs/1906.00047).

[R076] M. W. Doherty, N. B. Manson, P. Delaney, and L. C. L. Hollenberg, “The negatively charged nitrogen-vacancy centre in diamond: the electronic solution,” *New Journal of Physics* 13, 025019 (2011). DOI: [10.1088/1367-2630/13/2/025019](https://doi.org/10.1088/1367-2630/13/2/025019); arXiv: [1008.5224](https://arxiv.org/abs/1008.5224).

[R077] P. Udvarhelyi, V. O. Shkolnikov, A. Gali, G. Burkard, and A. Pályi, “Spin-strain interaction in nitrogen-vacancy centers in diamond,” *Physical Review B* 98, 075201 (2018). DOI: [10.1103/PhysRevB.98.075201](https://doi.org/10.1103/PhysRevB.98.075201); arXiv: [1712.02684](https://arxiv.org/abs/1712.02684).

[R079] G. Balasubramanian et al., “Ultralong spin coherence time in isotopically engineered diamond,” *Nature Materials* 8, 383–387 (2009). DOI: [10.1038/nmat2420](https://doi.org/10.1038/nmat2420).

[R080] F. Dolde et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* 9, 139–143 (2013).

DOI: [10.1038/nphys2545](https://doi.org/10.1038/nphys2545); arXiv: [1212.2804](https://arxiv.org/abs/1212.2804). Dolde and collaborators demonstrated room-temperature entanglement between two individually addressable NV electronic spins.

This result establishes pairwise coherent coupling under favorable geometric conditions. It does not demonstrate a dense and uniform many-body array.

[Ro81] M. Iuliano et al., “Unconditionally teleported quantum gates between remote solid-state qubit registers,” *Nature Communications* 17 (2026). DOI: [10.1038/s41467-026-72818-6](https://doi.org/10.1038/s41467-026-72818-6).

[Ro82] C. Bradac, W. Gao, J. Forneris, M. E. Trusheim, and I. Aharonovich, “Quantum nanophotonics with group IV defects in diamond,” *Nature Communications* 10, 5625 (2019). DOI: [10.1038/s41467-019-13332-w](https://doi.org/10.1038/s41467-019-13332-w).

[Ro83] D. D. Sukachev et al., “Silicon-vacancy spin qubit in diamond: a quantum memory exceeding 10 ms with single-shot state readout,” *Physical Review Letters* 119, 223602 (2017). DOI: [10.1103/PhysRevLett.119.223602](https://doi.org/10.1103/PhysRevLett.119.223602); arXiv: [1708.08852](https://arxiv.org/abs/1708.08852).

[Ro84] K. Senkalla, G. Genov, M. H. Metsch, P. Siyushev, and F. Jelezko, “Germanium vacancy in diamond quantum memory exceeding 20 ms,” *Physical Review Letters* 132, 026901 (2024). DOI: [10.1103/PhysRevLett.132.026901](https://doi.org/10.1103/PhysRevLett.132.026901); arXiv: [2308.09666](https://arxiv.org/abs/2308.09666).

[Ro85] I. Karapatzakis et al., “Microwave control of the tin-vacancy spin qubit in diamond with a superconducting waveguide,” *Physical Review X* 14, 031036 (2024). DOI: [10.1103/PhysRevX.14.031036](https://doi.org/10.1103/PhysRevX.14.031036).

[Ro86] R. E. Evans et al., “Photon-mediated interactions between quantum emitters in a diamond nanocavity,” *Science* 362, 662–665 (2018). DOI: [10.1126/science.aau4691](https://doi.org/10.1126/science.aau4691); arXiv: [1807.04265](https://arxiv.org/abs/1807.04265).

[Ro87] X. Cheng et al., “Laser activation of single group-IV colour centres in diamond,” *Nature Communications* 16, 5124 (2025). DOI: [10.1038/s41467-025-60373-5](https://doi.org/10.1038/s41467-025-60373-5).

[R107] V. R. Kortan, C. Şahin, and M. E. Flatté, “Nanometer-scale exchange interactions between spin centers in diamond,” *Physical Review B* 93, 220402(R) (2016). DOI: [10.1103/PhysRevB.93.220402](https://doi.org/10.1103/PhysRevB.93.220402); arXiv: [1603.03485](https://arxiv.org/abs/1603.03485).

[R109] M.-A. Lemonde et al., “Phonon networks with silicon-vacancy centers in diamond waveguides,” *Physical Review Letters* 120, 213603 (2018). DOI: [10.1103/PhysRevLett.120.213603](https://doi.org/10.1103/PhysRevLett.120.213603); arXiv: [1801.01904](https://arxiv.org/abs/1801.01904).

[R110] A. Bermudez, F. Jelezko, M. B. Plenio, and A. Retzker, “Electron-mediated nuclear-spin interactions between distant nitrogen-vacancy centers,” *Physical Review Letters* 107, 150503 (2011).

DOI: [10.1103/PhysRevLett.107.150503](https://doi.org/10.1103/PhysRevLett.107.150503); arXiv: [1107.2617](https://arxiv.org/abs/1107.2617).

[R111] J. J. Nakane, K. Tahara, K. Kutsuki, and A. Yamakage, “Phonon-mediated spin-spin interaction: A general theory and application to diamond nitrogen vacancy centers,” *Physical Review B* 110, 064428 (2024).

DOI: [10.1103/PhysRevB.110.064428](https://doi.org/10.1103/PhysRevB.110.064428).

[R120] M.-R. Yun, F.-Q. Guo, L.-L. Yan, E. Liang, Y. Zhang, S.-L. Su, C. X. Shan, and Y. Jia, “Parallel-path implementation of nonadiabatic geometric quantum gates in a decoherence-free subspace with nitrogen-vacancy centers,” *Physical Review A* 105, 012611 (2022).

DOI: [10.1103/PhysRevA.105.012611](https://doi.org/10.1103/PhysRevA.105.012611).

[R191] F. Dolde, H. Fedder, M. W. Doherty, et al., “Electric-field sensing using single diamond spins,” *Nature Physics* 7, 459–463 (2011). DOI: [10.1038/nphys1969](https://doi.org/10.1038/nphys1969).

- [R193] E. R. MacQuarrie, T. A. Gosavi, A. M. Moehle, N. R. Jungwirth, S. A. Bhawe, and G. D. Fuchs, “Coherent control of a nitrogen-vacancy center spin ensemble with a diamond mechanical resonator,” *Optica* 2, 233–238 (2015). DOI: [10.1364/OPTICA.2.000233](https://doi.org/10.1364/OPTICA.2.000233).
- [R194] A. Jarmola, V. M. Acosta, K. Jensen, S. Chemerisov, and D. Budker, “Temperature- and magnetic-field-dependent longitudinal spin relaxation in nitrogen-vacancy ensembles in diamond,” *Physical Review Letters* 108, 197601 (2012). DOI: [10.1103/PhysRevLett.108.197601](https://doi.org/10.1103/PhysRevLett.108.197601).
- [R197] S. Felton et al., “Hyperfine interaction in the ground state of the negatively charged nitrogen vacancy center in diamond,” *Physical Review B* 79, 075203 (2009). DOI: [10.1103/PhysRevB.79.075203](https://doi.org/10.1103/PhysRevB.79.075203).
- [R198] L. Robledo et al., “Spin dynamics in the optical cycle of single nitrogen-vacancy centres in diamond,” *New Journal of Physics* 13, 025013 (2011). DOI: [10.1088/1367-2630/13/2/025013](https://doi.org/10.1088/1367-2630/13/2/025013); arXiv: [1010.1192](https://arxiv.org/abs/1010.1192).
- [R200] S. Meesala et al., “Strain engineering of the silicon-vacancy center in diamond,” *Physical Review B* 97, 205444 (2018). DOI: [10.1103/PhysRevB.97.205444](https://doi.org/10.1103/PhysRevB.97.205444); arXiv: [1801.09833](https://arxiv.org/abs/1801.09833).
- [R201] K. D. Jahnke et al., “Electron–phonon processes of the silicon-vacancy centre in diamond,” *New Journal of Physics* 17, 043011 (2015). DOI: [10.1088/1367-2630/17/4/043011](https://doi.org/10.1088/1367-2630/17/4/043011); arXiv: [1411.2871](https://arxiv.org/abs/1411.2871).
- [R212] S. Pezzagna and J. Meijer, “Quantum computer based on color centers in diamond,” *Applied Physics Reviews* 8, 011308 (2021). DOI: [10.1063/5.0007444](https://doi.org/10.1063/5.0007444).
- [R214] L. Childress et al., “Coherent dynamics of coupled electron and nuclear spin qubits in diamond,” *Science* 314, 281–285 (2006). DOI: [10.1126/science.1131871](https://doi.org/10.1126/science.1131871).
- [R216] T. Rosskopf et al., “Investigation of surface magnetic noise by shallow spins in diamond,” *Physical Review Letters* 112, 147602 (2014). DOI: [10.1103/PhysRevLett.112.147602](https://doi.org/10.1103/PhysRevLett.112.147602).
- [R217] P. Jamonneau et al., “Competition between electric field and magnetic field noise in the decoherence of a single spin in diamond,” *Physical Review B* 93, 024305 (2016). DOI: [10.1103/PhysRevB.93.024305](https://doi.org/10.1103/PhysRevB.93.024305).
- [R218] S. Dhomkar, H. Jayakumar, P. R. Zangara, and C. A. Meriles, “Charge dynamics in near-surface, variable-density ensembles of nitrogen-vacancy centers in diamond,” *Nano Letters* 18, 4046–4052 (2018). DOI: [10.1021/acs.nanolett.8b01739](https://doi.org/10.1021/acs.nanolett.8b01739).
- [R219] J. Teissier, A. Barfuss, P. Appel, E. Neu, and P. Maletinsky, “Strain coupling of a nitrogen-vacancy center spin to a diamond mechanical oscillator,” *Physical Review Letters* 113, 020503 (2014). DOI: [10.1103/PhysRevLett.113.020503](https://doi.org/10.1103/PhysRevLett.113.020503).
- [R234] M. H. Abobeih et al., “Fault-tolerant operation of a logical qubit in a diamond quantum processor,” *Nature* 606, 884–889 (2022). DOI: [10.1038/s41586-022-04819-6](https://doi.org/10.1038/s41586-022-04819-6); arXiv: [2108.01646](https://arxiv.org/abs/2108.01646).
- [R235] S. Pezzagna, D. Wildanger, P. Mazarov et al., “Nanoscale Engineering and Optical Addressing of Single Spins in Diamond,” *Small* 6, 2117–2121 (2010). DOI: [10.1002/sml.201000902](https://doi.org/10.1002/sml.201000902).

[R236] H. Zhang, K. Arai, C. Belthangady, J.-C. Jaskula, and R. L. Walsworth, “Selective addressing of solid-state spins at the nanoscale via magnetic resonance frequency encoding,” *npj Quantum Information* 3, 31 (2017). DOI: [10.1038/s41534-017-0033-3](https://doi.org/10.1038/s41534-017-0033-3).

[R237] B. J. Shields, Q. P. Unterreithmeier, N. P. de Leon, H. Park, and M. D. Lukin, “Efficient Readout of a Single Spin State in Diamond via Spin-to-Charge Conversion,” *Physical Review Letters* 114, 136402 (2015). DOI: [10.1103/PhysRevLett.114.136402](https://doi.org/10.1103/PhysRevLett.114.136402); arXiv: [1410.0370](https://arxiv.org/abs/1410.0370).

[R238] A. Sipahigil, R. E. Evans, D. D. Sukachev et al., “An integrated diamond nanophotonics platform for quantum-optical networks,” *Science* 354, 847–850 (2016).

DOI: [10.1126/science.aah6875](https://doi.org/10.1126/science.aah6875); arXiv: [1608.05147](https://arxiv.org/abs/1608.05147).

[R239] M. Cambria, S. Chand, C. M. Reiter, and S. Kolkowitz, “Scalable Parallel Measurement of Individual Nitrogen-Vacancy Centers,” *Physical Review X* 15, 031015 (2025).

DOI: [10.1103/jdzq-jbfz](https://doi.org/10.1103/jdzq-jbfz); arXiv: [2408.11715](https://arxiv.org/abs/2408.11715). Cambria and collaborators demonstrate scalable parallel optical measurement of individual NV centers.

This result directly addresses readout throughput. However, crosstalk, calibration, and measurement of the topological sector remain problems that must be resolved at the architectural level.

[R242] L. B. Hughes, S. A. Meynell, W. Wu, S. Parthasarathy, L. Chen, Z. Zhang, Z. Wang, E. J. Davis, K. Mukherjee, N. Y. Yao, and A. C. Bleszynski Jayich, “Strongly Interacting, Two-Dimensional, Dipolar Spin Ensembles in (111)-Oriented Diamond,” *Physical Review X* 15, 021035 (2025). DOI: [10.1103/PhysRevX.15.021035](https://doi.org/10.1103/PhysRevX.15.021035); arXiv: [2404.10075](https://arxiv.org/abs/2404.10075).

[R244] N. Y. Yao, L. Jiang, A. V. Gorshkov, P. C. Maurer, G. Giedke, J. I. Cirac, and M. D. Lukin, “Scalable architecture for a room temperature solid-state quantum information processor,” *Nature Communications* 3, 800 (2012). DOI: [10.1038/ncomms1788](https://doi.org/10.1038/ncomms1788); arXiv: [1012.2864](https://arxiv.org/abs/1012.2864).

[R246] M. Ruf, N. H. Wan, H. Choi, D. Englund, and R. Hanson, “Quantum networks based on color centers in diamond,” *Journal of Applied Physics* 130, 070901 (2021). DOI: [10.1063/5.0056534](https://doi.org/10.1063/5.0056534); arXiv: [2105.04341](https://arxiv.org/abs/2105.04341).

Silicon carbide

[R096] D. J. Christle, A. L. Falk, P. Andrich, et al., “Isolated electron spins in silicon carbide with millisecond coherence times,” *Nature Materials* 14, 160–163 (2015).

DOI: [10.1038/nmat4144](https://doi.org/10.1038/nmat4144); arXiv: [1406.7325](https://arxiv.org/abs/1406.7325). Christle and collaborators established optically addressable isolated defect spins with millisecond-scale coherence in silicon carbide (SiC).

This work provides a foundational demonstration that SiC is a viable solid-state host for spin defects, rather than only a theoretical alternative to diamond.

[R097] C. P. Anderson, E. O. Glen, C. Zeledon, et al., “Five-second coherence of a single spin with single-shot readout in silicon carbide,” *Science Advances* 8, eabm5912 (2022).

DOI: [10.1126/sciadv.abm5912](https://doi.org/10.1126/sciadv.abm5912); arXiv: [2110.01590](https://arxiv.org/abs/2110.01590).

[R099] H. Hu, Y. Zhou, A. Yi, et al., “Room-temperature waveguide integrated quantum register in a semiconductor photonic platform,” *Nature Communications* 15, 10256 (2024).

DOI: [10.1038/s41467-024-54606-2](https://doi.org/10.1038/s41467-024-54606-2). This work integrates a room-temperature multiqubit register with a SiC photonic waveguide.

The result provides important system-level evidence for integrating the host material with photonic components. It does not demonstrate autonomous topological order.

[R104] T. Nishikawa, N. Morioka, H. Abe, et al., “Coherent photoelectrical readout of single spins in silicon carbide at room temperature,” *Nature Communications* 16, 3405 (2025).

DOI: [10.1038/s41467-025-58629-1](https://doi.org/10.1038/s41467-025-58629-1).

[R205] J. Wang et al., “Efficient generation of an array of single silicon-vacancy defects in silicon carbide,” *Physical Review Applied* 7, 064021 (2017).

DOI: [10.1103/PhysRevApplied.7.064021](https://doi.org/10.1103/PhysRevApplied.7.064021). Wang and collaborators generated arrays of individual silicon-vacancy defects in SiC.

This result provides direct fabrication evidence for creating defect arrays in SiC. It does not provide evidence for coherent many-body topological interactions.

Corundum/sapphire

[R088] V. K. Sewani, R. J. Stöhr, R. Kolesov, H. H. Vallabhapurapu, T. Simmet, A. Morello, and A. Laucht, “Spin thermometry and spin relaxation of optically detected Cr³⁺ ions in Al₂O₃ (ruby),” *Physical Review B* 102, 104114 (2020). DOI: [10.1103/PhysRevB.102.104114](https://doi.org/10.1103/PhysRevB.102.104114); arXiv: [2007.07493](https://arxiv.org/abs/2007.07493).

This experiment quantifies optical spin thermometry, the inference of temperature from spin-dependent optical signals, and spin relaxation for a confocally probed chromium-ion ensemble in ruby. The result provides direct evidence for optically detected ensemble spin physics in corundum. Single-ion addressability, scalable coupling, and array fabrication remain unresolved.

[R089] Z. Velluire-Pellat, E. Maréchal, C. Feuillet-Palma, and N. Bergeal, “Spin-photon interaction between a ruby crystal and a high-critical-temperature superconducting microwave cavity,” *Communications Physics* 8, 236 (2025). DOI: [10.1038/s42005-025-02159-1](https://doi.org/10.1038/s42005-025-02159-1).

[R090] W. G. Farr, D. L. Creedon, M. Goryachev, K. Benmessai, and M. E. Tobar, “Ultrasensitive microwave spectroscopy of paramagnetic impurities of sapphire crystals at millikelvin temperatures,” *Physical Review B* 88, 224426 (2013). DOI: [10.1103/PhysRevB.88.224426](https://doi.org/10.1103/PhysRevB.88.224426); arXiv: [1311.1049](https://arxiv.org/abs/1311.1049).

[R091] P. F. Moulton, “Spectroscopic and laser characteristics of Ti:Al₂O₃,” *Journal of the Optical Society of America B* 3, 125–133 (1986). DOI: [10.1364/JOSAB.3.000125](https://doi.org/10.1364/JOSAB.3.000125).

[R092] K. H. Lee and J. H. Crawford Jr., “Luminescence of the F center in sapphire,” *Physical Review B* 19, 3217–3221 (1979). DOI: [10.1103/PhysRevB.19.3217](https://doi.org/10.1103/PhysRevB.19.3217).

[R093] C. Wilhelmer, M. E. Turiansky, D. Waldhör, L. Cvitkovich, C. G. Van de Walle, and T. Grasser, “Optical properties of vacancies in aluminum oxide (α -Al₂O₃) from first principles,” *Physical Review Materials* 9, 096202 (2025). DOI: [10.1103/5kvl-hktd](https://doi.org/10.1103/5kvl-hktd).

[R095] B. D. Evans and M. Stapelbroek, “Optical properties of the F^+ center in crystalline Al₂O₃,” *Physical Review B* 18, 7089–7098 (1978).

DOI: [10.1103/PhysRevB.18.7089](https://doi.org/10.1103/PhysRevB.18.7089).

Quantum spin systems

- [R147] G. Jackeli and G. Khaliullin, “Mott Insulators in the Strong Spin-Orbit Coupling Limit: From Heisenberg to a Quantum Compass and Kitaev Models,” *Physical Review Letters* 102, 017205 (2009). DOI: [10.1103/PhysRevLett.102.017205](https://doi.org/10.1103/PhysRevLett.102.017205); arXiv: [0809.4658](https://arxiv.org/abs/0809.4658).
- [R148] M. Hermanns, I. Kimchi, and J. Knolle, “Physics of the Kitaev Model: Fractionalization, Dynamic Correlations, and Material Connections,” *Annual Review of Condensed Matter Physics* 9, 17–33 (2018). DOI: [10.1146/annurev-conmatphys-033117-053934](https://doi.org/10.1146/annurev-conmatphys-033117-053934); arXiv: [1705.01740](https://arxiv.org/abs/1705.01740).
- [R149] H. Takagi, T. Takayama, G. Jackeli, G. Khaliullin, and S. E. Nagler, “Concept and realization of Kitaev quantum spin liquids,” *Nature Reviews Physics* 1, 264–280 (2019). DOI: [10.1038/s42254-019-0038-2](https://doi.org/10.1038/s42254-019-0038-2); arXiv: [1903.08081](https://arxiv.org/abs/1903.08081).
- [R150] Y. Matsuda, T. Shibauchi, and H.-Y. Kee, “Kitaev quantum spin liquids,” *Reviews of Modern Physics* 97, 045003 (2025). DOI: [10.1103/3m4m-3v59](https://doi.org/10.1103/3m4m-3v59).
- [R151] A. Banerjee et al., “Neutron scattering in the proximate quantum spin liquid α -RuCl₃,” *Science* 356, 1055–1059 (2017). DOI: [10.1126/science.aah6015](https://doi.org/10.1126/science.aah6015).
- [R152] Y. Kasahara et al., “Majorana quantization and half-integer thermal quantum Hall effect in a Kitaev spin liquid,” *Nature* 559, 227–231 (2018). DOI: [10.1038/s41586-018-0274-0](https://doi.org/10.1038/s41586-018-0274-0); arXiv: [1805.05022](https://arxiv.org/abs/1805.05022).
- [R153] É. Lefrançois et al., “Evidence of a Phonon Hall Effect in the Kitaev Spin Liquid Candidate α -RuCl₃,” *Physical Review X* 12, 021025 (2022). DOI: [10.1103/PhysRevX.12.021025](https://doi.org/10.1103/PhysRevX.12.021025); arXiv: [2111.05493](https://arxiv.org/abs/2111.05493).
- [R154] A. Shragai, E. Horsley, S. Kim, Y.-J. Kim, and B. J. Ramshaw, “Phonon Hall viscosity and the intrinsic thermal Hall effect of α -RuCl₃,” *Nature* 652, 1166–1172 (2026). DOI: [10.1038/s41586-026-10420-y](https://doi.org/10.1038/s41586-026-10420-y).
- [R167] M. B. Hastings and T. Koma, “Spectral gap and exponential decay of correlations,” *Communications in Mathematical Physics* 265, 781–804 (2006). DOI: [10.1007/s00220-006-0030-4](https://doi.org/10.1007/s00220-006-0030-4); arXiv: [math-ph/0507008](https://arxiv.org/abs/math-ph/0507008).
- [R168] S. Bravyi and B. Terhal, “A no-go theorem for a two-dimensional self-correcting quantum memory based on stabilizer codes,” *New Journal of Physics* 11, 043029 (2009). DOI: [10.1088/1367-2630/11/4/043029](https://doi.org/10.1088/1367-2630/11/4/043029); arXiv: [0810.1983](https://arxiv.org/abs/0810.1983).

This no-go theorem excludes self-correcting quantum memory, meaning quantum information storage whose lifetime is protected passively against thermal errors, for broad classes of two-dimensional stabilizer Hamiltonians. A stabilizer Hamiltonian is constructed from mutually commuting operators whose common

eigenspace defines the encoded quantum states. The theorem therefore imposes a central constraint: two-dimensional topological encoding does not by itself provide indefinite passive storage.

- [R169] B. J. Brown, D. Loss, J. K. Pachos, C. N. Self, and J. R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* 88, 045005 (2016). DOI: [10.1103/RevModPhys.88.045005](https://doi.org/10.1103/RevModPhys.88.045005); arXiv: [1411.6643](https://arxiv.org/abs/1411.6643).

This review surveys thermal stability and finite-temperature memory in topological codes. It is important for distinguishing four concepts: stability of a phase at zero temperature, the storage lifetime of a finite system, active decoding based on measured error information, and genuine self-correction through passive physical dynamics.

- [R170] C. Stark, L. Pollet, A. Imamoğlu, and R. Renner, “Localization of toric code defects,” *Physical Review Letters* 107, 030504 (2011). DOI: [10.1103/PhysRevLett.107.030504](https://doi.org/10.1103/PhysRevLett.107.030504); arXiv: [1101.6028](https://arxiv.org/abs/1101.6028).
- [R181] P. Neumann, R. Kolesov, B. Naydenov, et al., “Quantum register based on coupled electron spins in a room-temperature solid,” *Nature Physics* 6, 249–253 (2010). DOI: [10.1038/nphys1536](https://doi.org/10.1038/nphys1536); arXiv: [1004.5090](https://arxiv.org/abs/1004.5090).
- [R182] E. L. Rosenfeld, L. M. Pham, M. D. Lukin, and R. L. Walsworth, “Sensing coherent dynamics of electronic spin clusters in solids,” *Physical Review Letters* 120, 243604 (2018). DOI: [10.1103/PhysRevLett.120.243604](https://doi.org/10.1103/PhysRevLett.120.243604).
- [R192] A. Barfuss, J. Teissier, E. Neu, A. Nunnenkamp, and P. Maletinsky, “Strong mechanical driving of a single electron spin,” *Nature Physics* 11, 820–824 (2015). DOI: [10.1038/nphys3411](https://doi.org/10.1038/nphys3411).
- [R195] T.-Y. Hwang, J. Lee, S.-W. Jeon, et al., “Sub-10 nm precision engineering of solid-state defects via nanoscale aperture array mask,” *Nano Letters* 22, 1672–1679 (2022). DOI: [10.1021/acs.nanolett.1c04699](https://doi.org/10.1021/acs.nanolett.1c04699).
- [R196] Bureau International des Poids et Mesures, *The International System of Units (SI Brochure)*, 9th ed., version 3.01, 2024. Stable URL: <https://www.bipm.org/en/publications/si-brochure>. DOI: [10.59161/AUEZ1291](https://doi.org/10.59161/AUEZ1291).
- [R199] L. Robledo et al., “High-fidelity projective read-out of a solid-state spin quantum register,” *Nature* 477, 574–578 (2011). DOI: [10.1038/nature10401](https://doi.org/10.1038/nature10401); arXiv: [1301.0392](https://arxiv.org/abs/1301.0392).
- [R208] P. J. Mohr et al., “CODATA recommended values of the fundamental physical constants: 2022,” *Reviews of Modern Physics* 97, 025002 (2025). DOI: [10.1103/RevModPhys.97.025002](https://doi.org/10.1103/RevModPhys.97.025002).
- [R213] C. L. Degen, F. Reinhard, and P. Cappellaro, “Quantum sensing,” *Reviews of Modern Physics* 89, 035002 (2017). DOI: [10.1103/RevModPhys.89.035002](https://doi.org/10.1103/RevModPhys.89.035002).
- [R215] G. de Lange, Z. H. Wang, D. Ristè, V. V. Dobrovitski, and R. Hanson, “Universal dynamical decoupling of a single solid-state spin from a spin bath,” *Science* 330, 60–63 (2010). DOI: [10.1126/science.1192739](https://doi.org/10.1126/science.1192739).

Topological order

- [R028] Daniel Gottesman, *Stabilizer Codes and Quantum Error Correction*, PhD thesis, California Institute of Technology (1997). DOI: [10.7907/rzr7-dt72](https://doi.org/10.7907/rzr7-dt72); arXiv: [quant-ph/9705052](https://arxiv.org/abs/quant-ph/9705052).
- [R029] A. R. Calderbank, E. M. Rains, P. W. Shor, and N. J. A. Sloane, “Quantum error correction via codes over GF(4),” *IEEE Transactions on Information Theory* 44, 1369–1387 (1998). DOI: [10.1109/18.681315](https://doi.org/10.1109/18.681315); arXiv: [quant-ph/9608006](https://arxiv.org/abs/quant-ph/9608006).

- [R030] Alexei Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* 303, 2–30 (2003). DOI: [10.1016/S0003-4916\(02\)00018-0](https://doi.org/10.1016/S0003-4916(02)00018-0); arXiv: [quant-ph/9707021](https://arxiv.org/abs/quant-ph/9707021).

Kitaev’s toric-code paper established that logical information can be encoded nonlocally in a gapped two-dimensional many-body system. Nonlocal encoding distributes the logical state across extended degrees of freedom rather than associating it with a single local variable, while a spectral gap is a nonzero energy separation between the relevant low-energy states and excited states. The paper also showed how anyonic processes can manipulate the encoded information; anyons are quasiparticle excitations in two spatial dimensions whose exchanges can have statistics more general than those of bosons or fermions. This work remains the standard reference for distinguishing topological encoding from self-correction and active error correction.

- [R031] Scott Aaronson and Daniel Gottesman, “Improved simulation of stabilizer circuits,” *Physical Review A* 70, 052328 (2004). DOI: [10.1103/PhysRevA.70.052328](https://doi.org/10.1103/PhysRevA.70.052328); arXiv: [quant-ph/0406196](https://arxiv.org/abs/quant-ph/0406196).
- [R121] N. D. Mermin, “The topological theory of defects in ordered media,” *Reviews of Modern Physics* 51, 591–648 (1979). DOI: [10.1103/RevModPhys.51.591](https://doi.org/10.1103/RevModPhys.51.591).
- [R122] X.-G. Wen and Q. Niu, “Ground-state degeneracy of the fractional quantum Hall states in the presence of a random potential and on high-genus Riemann surfaces,” *Physical Review B* 41, 9377–9396 (1990). DOI: [10.1103/PhysRevB.41.9377](https://doi.org/10.1103/PhysRevB.41.9377).
- [R123] M. B. Hastings and X.-G. Wen, “Quasi-adiabatic continuation of quantum states: The stability of topological ground-state degeneracy and emergent gauge invariance,” *Physical Review B* 72, 045141 (2005). DOI: [10.1103/PhysRevB.72.045141](https://doi.org/10.1103/PhysRevB.72.045141); arXiv: [cond-mat/0503554](https://arxiv.org/abs/cond-mat/0503554).
- [R124] X.-G. Wen, “Colloquium: Zoo of quantum-topological phases of matter,” *Reviews of Modern Physics* 89, 041004 (2017). DOI: [10.1103/RevModPhys.89.041004](https://doi.org/10.1103/RevModPhys.89.041004); arXiv: [1610.03911](https://arxiv.org/abs/1610.03911).
- [R141] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, “Topological quantum memory,” *Journal of Mathematical Physics* 43, 4452–4505 (2002). DOI: [10.1063/1.1499754](https://doi.org/10.1063/1.1499754); arXiv: [quant-ph/0110143](https://arxiv.org/abs/quant-ph/0110143).

Dennis and collaborators related toric-code storage to error chains, decoding, and statistical-mechanical thresholds. Error chains are connected sequences of physical errors, decoding is the computational inference of a correction from measured error syndromes, and a threshold is a critical error rate below which increasing code size can suppress logical failure under the assumed noise and decoding models. This paper is foundational for understanding why topological encoding in realistic noisy systems still requires syndrome extraction, meaning measurement of error-diagnostic observables, followed by decoding.

- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* 51, 093512 (2010). DOI: [10.1063/1.3490195](https://doi.org/10.1063/1.3490195); arXiv: [1001.0344](https://arxiv.org/abs/1001.0344).

Bravyi, Hastings, and Michalakis proved that topological quantum order is stable under sufficiently weak local perturbations for a defined class of Hamiltonians. The theorem applies to an existing topological phase with a nonzero spectral gap. It does not guarantee that a proposed perturbative simulator reaches that phase or produces a gap large enough to be useful.

- [R143] R. Alicki, M. Fannes, and M. Horodecki, “On thermalization in Kitaev’s 2D model,” *Journal of Physics A: Mathematical and Theoretical* 42, 065303 (2009). DOI: [10.1088/1751-8113/42/6/065303](https://doi.org/10.1088/1751-8113/42/6/065303); arXiv: [0810.4584](https://arxiv.org/abs/0810.4584).
- [R144] B. M. Terhal, “Quantum error correction for quantum memories,” *Reviews of Modern Physics* 87, 307–346 (2015). DOI: [10.1103/RevModPhys.87.307](https://doi.org/10.1103/RevModPhys.87.307); arXiv: [1302.3428](https://arxiv.org/abs/1302.3428).

- [R145] S. Krinner et al., “Realizing repeated quantum error correction in a distance-three surface code,” *Nature* 605, 669–674 (2022). DOI: [10.1038/s41586-022-04566-8](https://doi.org/10.1038/s41586-022-04566-8); arXiv: [2112.03708](https://arxiv.org/abs/2112.03708).
- [R146] M. Iqbal et al., “Topological order from measurements and feed-forward on a trapped ion quantum computer,” *Communications Physics* 7, 205 (2024). DOI: [10.1038/s42005-024-01698-3](https://doi.org/10.1038/s42005-024-01698-3); arXiv: [2302.01917](https://arxiv.org/abs/2302.01917).
- [R160] M. Müger, “From subfactors to categories and topology II: The quantum double of tensor categories and subfactors,” *Journal of Pure and Applied Algebra* 180, 159–219 (2003). DOI: [10.1016/S0022-4049\(02\)00248-7](https://doi.org/10.1016/S0022-4049(02)00248-7); arXiv: [math/0111205](https://arxiv.org/abs/math/0111205).
- [R220] A. G. Fowler, “Coping with qubit leakage in topological codes,” *Physical Review A* 88, 042308 (2013). DOI: [10.1103/PhysRevA.88.042308](https://doi.org/10.1103/PhysRevA.88.042308); arXiv: [1308.6642](https://arxiv.org/abs/1308.6642).
- [R221] E. Novais, E. R. Mucciolo, and H. U. Baranger, “Resilient quantum computation in correlated environments: A quantum phase transition perspective,” *Physical Review Letters* 98, 040501 (2007). DOI: [10.1103/PhysRevLett.98.040501](https://doi.org/10.1103/PhysRevLett.98.040501); arXiv: [quant-ph/0607155](https://arxiv.org/abs/quant-ph/0607155).
- [R247] S. Jansen, M.-B. Ruskai, and R. Seiler, “Bounds for the adiabatic approximation with applications to quantum computation,” *Journal of Mathematical Physics* 48, 102111 (2007). DOI: [10.1063/1.2798382](https://doi.org/10.1063/1.2798382); arXiv: [quant-ph/0603175](https://arxiv.org/abs/quant-ph/0603175).

Jansen, Ruskai, and Seiler derive rigorous bounds for the adiabatic approximation, which describes evolution that remains close to an instantaneous eigenspace when a Hamiltonian changes sufficiently slowly. Their bounds make the dependence on the spectral gap explicit. Consequently, a small engineered many-body gap can require a prohibitively long state-preparation time even when the final Hamiltonian has the intended form.

- [R248] T. M. Stace, S. D. Barrett, and A. C. Doherty, “Thresholds for topological codes in the presence of loss,” *Physical Review Letters* 102, 200501 (2009). DOI: [10.1103/PhysRevLett.102.200501](https://doi.org/10.1103/PhysRevLett.102.200501); arXiv: [0904.3556](https://arxiv.org/abs/0904.3556).

Anyons

- [R012] E. Artin, “Theory of Braids,” *Annals of Mathematics* 48, 101–126 (1947). DOI: [10.2307/1969218](https://doi.org/10.2307/1969218).
- [R013] C. Kassel and V. Turaev, *Braid Groups*, Graduate Texts in Mathematics 247, Springer (2008). DOI: [10.1007/978-0-387-68548-9](https://doi.org/10.1007/978-0-387-68548-9).
- [R014] B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed., Springer (2015). DOI: [10.1007/978-3-319-13467-3](https://doi.org/10.1007/978-3-319-13467-3).
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian Anyons and Topological Quantum Computation,” *Reviews of Modern Physics* 80, 1083–1159 (2008). DOI: [10.1103/RevModPhys.80.1083](https://doi.org/10.1103/RevModPhys.80.1083); arXiv: [0707.1889](https://arxiv.org/abs/0707.1889).

This review provides a standard connection between non-Abelian statistics and fault-tolerant gate constructions. Non-Abelian statistics occur when exchanging quasiparticles acts through noncommuting transformations on a degenerate state space. The review also distinguishes topological protection in an ideal anyon model from the physical engineering required to realize and control that model.

- [R016] E. C. Rowell and Z. Wang, “Mathematics of Topological Quantum Computing,” *Bulletin of the American Mathematical Society* 55, 183–238 (2018). DOI: [10.1090/bull/1605](https://doi.org/10.1090/bull/1605); arXiv: [1705.06206](https://arxiv.org/abs/1705.06206).

- [R017] A. Kitaev, “Anyons in an Exactly Solved Model and Beyond,” *Annals of Physics* 321, 2–111 (2006). DOI: [10.1016/j.aop.2005.10.005](https://doi.org/10.1016/j.aop.2005.10.005); arXiv: [cond-mat/0506438](https://arxiv.org/abs/cond-mat/0506438).

Kitaev’s exact solution of the honeycomb model provides a rare microscopic spin Hamiltonian with fractionalized excitations and a non-Abelian phase. Fractionalization is the emergence of quasiparticles whose quantum numbers or statistics differ from those of the microscopic constituents. The model defines a benchmark for the properties that an engineered spin array must reproduce; it does not constitute evidence that a defect array already realizes the same phase.

- [R019] B. Bakalov and A. Kirillov Jr., *Lectures on Tensor Categories and Modular Functors*, University Lecture Series 21, American Mathematical Society (2001). DOI: [10.1090/ulect/021](https://doi.org/10.1090/ulect/021).
- [R020] P. Etingof, S. Gelaki, D. Nikshych, and V. Ostrik, *Tensor Categories*, Mathematical Surveys and Monographs 205, American Mathematical Society (2015). DOI: [10.1090/surv/205](https://doi.org/10.1090/surv/205).
- [R021] M. Barkeshli, P. Bonderson, M. Cheng, and Z. Wang, “Symmetry Fractionalization, Defects, and Gauging of Topological Phases,” *Physical Review B* 100, 115147 (2019). DOI: [10.1103/PhysRevB.100.115147](https://doi.org/10.1103/PhysRevB.100.115147); arXiv: [1410.4540](https://arxiv.org/abs/1410.4540).
- [R022] G. Moore and N. Seiberg, “Classical and Quantum Conformal Field Theory,” *Communications in Mathematical Physics* 123, 177–254 (1989). DOI: [10.1007/BF01238857](https://doi.org/10.1007/BF01238857).
- [R023] P. Bonderson, K. Shtengel, and J. K. Slingerland, “Interferometry of Non-Abelian Anyons,” *Annals of Physics* 323, 2709–2755 (2008). DOI: [10.1016/j.aop.2008.01.012](https://doi.org/10.1016/j.aop.2008.01.012); arXiv: [0707.4206](https://arxiv.org/abs/0707.4206).
- [R127] J. M. Leinaas and J. Myrheim, “On the theory of identical particles,” *Il Nuovo Cimento B* 37, 1–23 (1977). DOI: [10.1007/BF02727953](https://doi.org/10.1007/BF02727953).
- [R128] F. Wilczek, “Quantum mechanics of fractional-spin particles,” *Physical Review Letters* 49, 957–959 (1982). DOI: [10.1103/PhysRevLett.49.957](https://doi.org/10.1103/PhysRevLett.49.957).
- [R129] Y.-S. Wu, “General theory for quantum statistics in two dimensions,” *Physical Review Letters* 52, 2103–2106 (1984). DOI: [10.1103/PhysRevLett.52.2103](https://doi.org/10.1103/PhysRevLett.52.2103).
- [R130] H. Bartolomei et al., “Fractional statistics in anyon collisions,” *Science* 368, 173–177 (2020). DOI: [10.1126/science.aaz5601](https://doi.org/10.1126/science.aaz5601); arXiv: [2006.13157](https://arxiv.org/abs/2006.13157).
- [R133] A. Stern, “Anyons and the quantum Hall effect—a pedagogical review,” *Annals of Physics* 323, 204–249 (2008). DOI: [10.1016/j.aop.2007.10.008](https://doi.org/10.1016/j.aop.2007.10.008); arXiv: [0711.4697](https://arxiv.org/abs/0711.4697).
- [R134] D. A. Ivanov, “Non-Abelian statistics of half-quantum vortices in p-wave superconductors,” *Physical Review Letters* 86, 268–271 (2001). DOI: [10.1103/PhysRevLett.86.268](https://doi.org/10.1103/PhysRevLett.86.268); arXiv: [cond-mat/0005069](https://arxiv.org/abs/cond-mat/0005069).

Fibonacci anyons

- [R135] S. Trebst, M. Troyer, Z. Wang, and A. W. W. Ludwig, “A short introduction to Fibonacci anyon models,” *Progress of Theoretical Physics Supplement* 176, 384–407 (2008). DOI: [10.1143/PTPS.176.384](https://doi.org/10.1143/PTPS.176.384); arXiv: [0902.3275](https://arxiv.org/abs/0902.3275).

- [R136] M. H. Freedman, M. Larsen, and Z. Wang, “A modular functor which is universal for quantum computation,” *Communications in Mathematical Physics* 227, 605–622 (2002). DOI: [10.1007/s002200200645](https://doi.org/10.1007/s002200200645); arXiv: [quant-ph/0001108](https://arxiv.org/abs/quant-ph/0001108).
- [R137] L. Hormozi, G. Zikos, N. E. Bonesteel, and S. H. Simon, “Topological quantum compiling,” *Physical Review B* 75, 165310 (2007). DOI: [10.1103/PhysRevB.75.165310](https://doi.org/10.1103/PhysRevB.75.165310); arXiv: [quant-ph/0610111](https://arxiv.org/abs/quant-ph/0610111).
- [R139] N. Read and E. Rezayi, “Beyond paired quantum Hall states: Parafermions and incompressible states in the first excited Landau level,” *Physical Review B* 59, 8084–8092 (1999). DOI: [10.1103/PhysRevB.59.8084](https://doi.org/10.1103/PhysRevB.59.8084); arXiv: [cond-mat/9809384](https://arxiv.org/abs/cond-mat/9809384).

Read and Rezayi identified a sequence of clustered quantum Hall states whose quasiparticles include the Fibonacci case relevant to universal topological computation. Fibonacci anyons are non-Abelian anyons with Fibonacci fusion structure. The work provides a candidate physical route to these excitations, but identification of a filling fraction alone does not constitute conclusive evidence for the corresponding quasiparticles.

- [R225] E. H. Rezayi and N. Read, “Non-Abelian quantized Hall states of electrons at filling factors $12/5$ and $13/5$ in the first excited Landau level,” *Physical Review B* 79, 075306 (2009). DOI: [10.1103/PhysRevB.79.075306](https://doi.org/10.1103/PhysRevB.79.075306); arXiv: [cond-mat/0608346](https://arxiv.org/abs/cond-mat/0608346).

Rezayi and Read provided numerical evidence for non-Abelian quantum Hall states at filling factors $12/5$ and $13/5$. This result supports a candidate phase assignment, but it does not directly observe Fibonacci fusion or braiding. Fusion specifies the possible collective topological charges of multiple anyons, whereas braiding denotes their exchange along trajectories in two spatial dimensions.

- [R226] C. Zhang, C. Huan, J. S. Xia, N. S. Sullivan, W. Pan, K. W. Baldwin, K. W. West, L. N. Pfeiffer, and D. C. Tsui, “Spin polarization of the $\nu = 12/5$ fractional quantum Hall state,” *Physical Review B* 85, 241302(R) (2012). DOI: [10.1103/PhysRevB.85.241302](https://doi.org/10.1103/PhysRevB.85.241302).

This experiment measured the spin polarization of the $12/5$ fractional quantum Hall state. The measurement constrains candidate topological orders at that filling but does not uniquely establish the presence of Fibonacci quasiparticles.

- [R227] W. Zhu, S. S. Gong, F. D. M. Haldane, and D. N. Sheng, “Fractional quantum Hall states at $\nu = 13/5$ and $12/5$ and their non-Abelian nature,” *Physical Review Letters* 115, 126805 (2015). DOI: [10.1103/PhysRevLett.115.126805](https://doi.org/10.1103/PhysRevLett.115.126805); arXiv: [1505.03050](https://arxiv.org/abs/1505.03050).
- [R228] R. S. K. Mong, M. P. Zaletel, F. Pollmann, and Z. Papić, “Fibonacci anyons and charge density order in the $12/5$ and $13/5$ quantum Hall plateaus,” *Physical Review B* 95, 115136 (2017). DOI: [10.1103/PhysRevB.95.115136](https://doi.org/10.1103/PhysRevB.95.115136); arXiv: [1505.02843](https://arxiv.org/abs/1505.02843).

Mong and collaborators found numerical evidence connecting the $12/5$ and $13/5$ quantum Hall plateaus to Fibonacci topological order and charge-density structure. The conclusion depends on the model and finite system sizes used in the calculations and is therefore not a direct measurement of quasiparticle properties.

String nets

[R018] M. A. Levin and X.-G. Wen, “String-Net Condensation: A Physical Mechanism for Topological Phases,” *Physical Review B* 71, 045110 (2005).

DOI: [10.1103/PhysRevB.71.045110](https://doi.org/10.1103/PhysRevB.71.045110); arXiv: [cond-mat/0404617](https://arxiv.org/abs/cond-mat/0404617). Levin and Wen introduced string-net condensation, in which extended string-like degrees of freedom form a collective ground-state structure, as a commuting-projector mechanism for broad classes of doubled topological phases. A commuting-projector Hamiltonian is a sum of mutually commuting local projection operators. A doubled topological phase combines a topological theory with its conjugate and is therefore nonchiral.

This paper specifies the structural objective of the manuscript's proposal for engineering a doubled-Fibonacci phase.

[R155] Lukasz Fidkowski, Michael Freedman, Chetan Nayak, Kevin Walker, and Zhenghan Wang, "From string nets to nonabelions," *Communications in Mathematical Physics* 287, 805–827 (2009). DOI: [10.1007/s00220-009-0757-9](https://doi.org/10.1007/s00220-009-0757-9); arXiv: [cond-mat/0610583](https://arxiv.org/abs/cond-mat/0610583).

[R156] Alexei Kitaev and Liang Kong, "Models for gapped boundaries and domain walls," *Communications in Mathematical Physics* 313, 351–373 (2012). DOI: [10.1007/s00220-012-1500-5](https://doi.org/10.1007/s00220-012-1500-5); arXiv: [1104.5047](https://arxiv.org/abs/1104.5047).

[R157] Alexander Kirillov Jr., "String-net model of Turaev–Viro invariants," (2011). DOI: [10.48550/arXiv.1106.6033](https://doi.org/10.48550/arXiv.1106.6033); arXiv: [1106.6033](https://arxiv.org/abs/1106.6033).

[R158] Robert König, Greg Kuperberg, and Ben W. Reichardt, "Quantum computation with Turaev–Viro codes," *Annals of Physics* 325, 2707–2749 (2010). DOI: [10.1016/j.aop.2010.08.001](https://doi.org/10.1016/j.aop.2010.08.001); arXiv: [1002.2816](https://arxiv.org/abs/1002.2816).

[R159] Oliver Buerschaper, Miguel Aguado, and Guifré Vidal, "Explicit tensor network representation for the ground states of string-net models," *Physical Review B* 79, 085119 (2009). DOI: [10.1103/PhysRevB.79.085119](https://doi.org/10.1103/PhysRevB.79.085119); arXiv: [0809.2393](https://arxiv.org/abs/0809.2393).

[R161] C.-H. Lin, M. Levin, and F. J. Burnell, "Generalized string-net models: A thorough exposition," *Physical Review B* 103, 195155 (2021). DOI: [10.1103/PhysRevB.103.195155](https://doi.org/10.1103/PhysRevB.103.195155); arXiv: [2012.14424](https://arxiv.org/abs/2012.14424).

[R183] M. D. Schulz, S. Dusuel, K. P. Schmidt, and J. Vidal, "Topological phase transitions in the golden string-net model," *Physical Review Letters* 110, 147203 (2013). DOI: [10.1103/PhysRevLett.110.147203](https://doi.org/10.1103/PhysRevLett.110.147203); arXiv: [1212.4109](https://arxiv.org/abs/1212.4109).

[R184] H. Yao and S. A. Kivelson, "An exact chiral spin liquid with non-Abelian anyons," *Physical Review Letters* 99, 247203 (2007).

DOI: [10.1103/PhysRevLett.99.247203](https://doi.org/10.1103/PhysRevLett.99.247203); arXiv: [0708.0040](https://arxiv.org/abs/0708.0040).

[R185] M. Kargarian, H. Bombin, and M. A. Martin-Delgado, "Topological color codes and two-body quantum lattice Hamiltonians," *New Journal of Physics* 12, 025018 (2010). DOI: [10.1088/1367-2630/12/2/025018](https://doi.org/10.1088/1367-2630/12/2/025018); arXiv: [0906.4127](https://arxiv.org/abs/0906.4127).

[R186] G. Kells, J. Kailasvuori, J. K. Slingerland, and J. Vala, "Kaleidoscope of topological phases with multiple Majorana species," *New Journal of Physics* 13, 095014 (2011).

DOI: [10.1088/1367-2630/13/9/095014](https://doi.org/10.1088/1367-2630/13/9/095014); arXiv: [1012.5276](https://arxiv.org/abs/1012.5276).

[R187] R. Moessner and S. L. Sondhi, "Resonating valence bond phase in the triangular lattice quantum dimer model," *Physical Review Letters* 86, 1881–1884 (2001).

DOI: [10.1103/PhysRevLett.86.1881](https://doi.org/10.1103/PhysRevLett.86.1881); arXiv: [cond-mat/0007378](https://arxiv.org/abs/cond-mat/0007378).

[R188] L. Balents, M. P. A. Fisher, and S. M. Girvin, “Fractionalization in an easy-axis kagome antiferromagnet,” *Physical Review B* 65, 224412 (2002). DOI: [10.1103/PhysRevB.65.224412](https://doi.org/10.1103/PhysRevB.65.224412); arXiv: [cond-mat/0110005](https://arxiv.org/abs/cond-mat/0110005).

[R209] Roberto Oliveira and Barbara M. Terhal, “The complexity of quantum spin systems on a two-dimensional square lattice,” *Quantum Information & Computation* 8, 900–924 (2008). arXiv: [quant-ph/0504050](https://arxiv.org/abs/quant-ph/0504050).

[R211] N. Bar-Gill, L. M. Pham, A. Jarmola, D. Budker, and R. L. Walsworth, “Solid-state electronic spin coherence time approaching one second,” *Nature Communications* 4, 1743 (2013).

DOI: [10.1038/ncomms2771](https://doi.org/10.1038/ncomms2771).

Perturbative gadgets

[R024] J. R. Schrieffer and P. A. Wolff, “Relation between the Anderson and Kondo Hamiltonians,” *Physical Review* 149, 491–492 (1966). DOI: [10.1103/PhysRev.149.491](https://doi.org/10.1103/PhysRev.149.491).

[R025] Sergey Bravyi, David P. DiVincenzo, and Daniel Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” *Annals of Physics* 326, 2793–2826 (2011).

DOI: [10.1016/j.aop.2011.06.004](https://doi.org/10.1016/j.aop.2011.06.004); arXiv: [1105.0675](https://arxiv.org/abs/1105.0675). This work provides a controlled formulation of Schrieffer–Wolff perturbation theory for many-body systems, including effective Hamiltonians and error bounds. The Schrieffer–Wolff transformation perturbatively decouples low- and high-energy subspaces so that the dynamics within the low-energy subspace can be represented by an effective Hamiltonian.

This paper is the appropriate reference for claims that a low-energy interaction emerges after excited cluster states have been eliminated.

[R026] A. H. MacDonald, S. M. Girvin, and D. Yoshioka, “ t/U expansion for the Hubbard model,” *Physical Review B* 37, 9753–9756 (1988).

DOI: [10.1103/PhysRevB.37.9753](https://doi.org/10.1103/PhysRevB.37.9753).

[R027] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Springer (1976; corrected printing 1995). DOI: [10.1007/978-3-642-66282-9](https://doi.org/10.1007/978-3-642-66282-9).

[R171] H. Feshbach, “Unified theory of nuclear reactions,” *Annals of Physics* 5, 357–390 (1958). DOI: [10.1016/0003-4916\(58\)90007-1](https://doi.org/10.1016/0003-4916(58)90007-1).

[R172] J. Hubbard, “Electron correlations in narrow energy bands,” *Proceedings of the Royal Society A* 276, 238–257 (1963). DOI: [10.1098/rspa.1963.0204](https://doi.org/10.1098/rspa.1963.0204).

[R173] P. W. Anderson, “New Approach to the Theory of Superexchange Interactions,” *Physical Review* 115, 2–13 (1959). DOI: [10.1103/PhysRev.115.2](https://doi.org/10.1103/PhysRev.115.2).

[R174] S. P. Jordan and E. Farhi, “Perturbative gadgets at arbitrary orders,” *Physical Review A* 77, 062329 (2008).

DOI: [10.1103/PhysRevA.77.062329](https://doi.org/10.1103/PhysRevA.77.062329); arXiv: [0802.1874](https://arxiv.org/abs/0802.1874).

[R175] S. Bravyi, D. P. DiVincenzo, D. Loss, and B. M. Terhal, “Quantum simulation of many-body Hamiltonians using perturbation theory with bounded-strength interactions,” *Physical Review Letters* 101, 070503 (2008). DOI: [10.1103/PhysRevLett.101.070503](https://doi.org/10.1103/PhysRevLett.101.070503); arXiv: [0803.2686](https://arxiv.org/abs/0803.2686).

Bravyi and collaborators quantify both the capabilities and the costs of perturbative gadgets. A perturbative gadget is an auxiliary Hamiltonian construction in which simpler interactions reproduce a desired effective interaction within a low-energy subspace. Their analysis shows that bounded-strength two-body resources can reproduce many-body terms only under a controlled hierarchy of energy scales. The result therefore provides both a construction method and a constraint: the effective gaps can decrease, while fabrication errors can be amplified.

[R176] J. Vidal, K. P. Schmidt, and S. Dusuel, “Perturbative approach to an exactly solved problem: Kitaev honeycomb model,” *Physical Review B* 78, 245121 (2008). DOI: [10.1103/PhysRevB.78.245121](https://doi.org/10.1103/PhysRevB.78.245121); arXiv: [0809.1553](https://arxiv.org/abs/0809.1553).

[R177] C. G. Brell, S. T. Flammia, S. D. Bartlett, and A. C. Doherty, “Toric codes and quantum doubles from two-body Hamiltonians,” *New Journal of Physics* 13, 053039 (2011). DOI: [10.1088/1367-2630/13/5/053039](https://doi.org/10.1088/1367-2630/13/5/053039); arXiv: [1011.1942](https://arxiv.org/abs/1011.1942).

Brell and collaborators explicitly construct toric-code and quantum-double interactions from two-body Hamiltonians by using perturbative gadgets. This paper is directly relevant to the proposed defect architecture because it identifies the required overhead in ancillary degrees of freedom and energy scales.

[R178] R. König, “Simplifying quantum double Hamiltonians using perturbative gadgets,” *Quantum Information and Computation* 10, 292–324 (2010). DOI: [10.26421/QIC10.3-4-9](https://doi.org/10.26421/QIC10.3-4-9); arXiv: [0901.1333](https://arxiv.org/abs/0901.1333).

[R179] C. G. Brell, S. D. Bartlett, and A. C. Doherty, “Perturbative 2-body parent Hamiltonians for projected entangled pair states,” *New Journal of Physics* 16, 123056 (2014). DOI: [10.1088/1367-2630/16/12/123056](https://doi.org/10.1088/1367-2630/16/12/123056); arXiv: [1407.4829](https://arxiv.org/abs/1407.4829).

[R180] S. A. Ocko and B. Yoshida, “Nonperturbative gadget for topological quantum codes,” *Physical Review Letters* 107, 250502 (2011).

DOI: [10.1103/PhysRevLett.107.250502](https://doi.org/10.1103/PhysRevLett.107.250502); arXiv: [1107.2697](https://arxiv.org/abs/1107.2697).

[R245] T. Albash and D. A. Lidar, “Adiabatic quantum computation,” *Reviews of Modern Physics* 90, 015002 (2018).

DOI: [10.1103/RevModPhys.90.015002](https://doi.org/10.1103/RevModPhys.90.015002); arXiv: [1611.04471](https://arxiv.org/abs/1611.04471).

Quantum simulation

[R032] Frank Verstraete and J. Ignacio Cirac, “Renormalization algorithms for quantum-many body systems in two and higher dimensions,” (2004). arXiv: [cond-mat/0407066](https://arxiv.org/abs/cond-mat/0407066).

[R033] Norbert Schuch, Michael M. Wolf, Frank Verstraete, and J. Ignacio Cirac, “Computational complexity of projected entangled pair states,” *Physical Review Letters* 98, 140506 (2007). DOI: [10.1103/PhysRevLett.98.140506](https://doi.org/10.1103/PhysRevLett.98.140506); arXiv: [quant-ph/0611050](https://arxiv.org/abs/quant-ph/0611050).

[R034] Norbert Schuch, J. Ignacio Cirac, and David Pérez-García, “PEPS as ground states: Degeneracy and topology,” *Annals of Physics* 325, 2153–2192 (2010). DOI: [10.1016/j.aop.2010.05.008](https://doi.org/10.1016/j.aop.2010.05.008); arXiv: [1001.3807](https://arxiv.org/abs/1001.3807).

[R035] Ulrich Schollwöck, “The density-matrix renormalization group in the age of matrix product states,” *Annals of Physics* 326, 96–192 (2011). DOI: [10.1016/j.aop.2010.09.012](https://doi.org/10.1016/j.aop.2010.09.012); arXiv: [1008.3477](https://arxiv.org/abs/1008.3477).

- [R036] Román Orús, “A practical introduction to tensor networks: Matrix product states and projected entangled pair states,” *Annals of Physics* 349, 117–158 (2014). DOI: [10.1016/j.aop.2014.06.013](https://doi.org/10.1016/j.aop.2014.06.013); arXiv: [1306.2164](https://arxiv.org/abs/1306.2164).
- [R164] Mohsin Iqbal et al., “Non-Abelian topological order and anyons on a trapped-ion processor,” *Nature* 626, 505–511 (2024). DOI: [10.1038/s41586-023-06934-4](https://doi.org/10.1038/s41586-023-06934-4); arXiv: [2305.03766](https://arxiv.org/abs/2305.03766).
- [R166] Tomoya Hayata, Yoshimasa Hidaka & Yuta Kikuchi, “Digital quantum simulation of q -deformed SU(2) Yang–Mills theory on a trapped-ion quantum computer,” *Physical Review Research* 8, 033137 (2026). DOI: [10.1103/vlpv-n8dy](https://doi.org/10.1103/vlpv-n8dy); arXiv: [2601.13530](https://arxiv.org/abs/2601.13530).
- [R229] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson, and W. D. Oliver, “A quantum engineer’s guide to superconducting qubits,” *Applied Physics Reviews* 6, 021318 (2019). DOI: [10.1063/1.5089550](https://doi.org/10.1063/1.5089550).
- [R231] A. Browaeys and T. Lahaye, “Many-body physics with individually controlled Rydberg atoms,” *Nature Physics* 16, 132–142 (2020). DOI: [10.1038/s41567-019-0733-z](https://doi.org/10.1038/s41567-019-0733-z).
- [R232] S. J. Evered et al., “High-fidelity parallel entangling gates on a neutral-atom quantum computer,” *Nature* 622, 268–272 (2023). DOI: [10.1038/s41586-023-06481-y](https://doi.org/10.1038/s41586-023-06481-y).
- [R249] J. R. Johansson, P. D. Nation, and F. Nori, “QuTiP: An open-source Python framework for the dynamics of open quantum systems,” *Computer Physics Communications* 183, 1760–1772 (2012). DOI: [10.1016/j.cpc.2012.02.021](https://doi.org/10.1016/j.cpc.2012.02.021).
- [R250] M. Fishman, S. R. White, and E. M. Stoudenmire, “The ITensor Software Library for Tensor Network Calculations,” *SciPost Physics Codebases* 4 (2022). DOI: [10.21468/SciPostPhysCodeb.4](https://doi.org/10.21468/SciPostPhysCodeb.4); arXiv: [2007.14822](https://arxiv.org/abs/2007.14822).
- [R251] J. Hauschild and F. Pollmann, “Efficient numerical simulations with Tensor Networks: Tensor Network Python (TeNPy),” *SciPost Physics Lecture Notes* 5 (2018). DOI: [10.21468/SciPostPhysLectNotes.5](https://doi.org/10.21468/SciPostPhysLectNotes.5); arXiv: [1805.00055](https://arxiv.org/abs/1805.00055).
- [R252] J. I. Cirac, D. Pérez-García, N. Schuch, and F. Verstraete, “Matrix product states and projected entangled pair states: Concepts, symmetries, theorems,” *Reviews of Modern Physics* 93, 045003 (2021). DOI: [10.1103/RevModPhys.93.045003](https://doi.org/10.1103/RevModPhys.93.045003); arXiv: [2011.12127](https://arxiv.org/abs/2011.12127).
- [R253] J. Jordan, R. Orús, G. Vidal, F. Verstraete, and J. I. Cirac, “Classical simulation of infinite-size quantum lattice systems in two spatial dimensions,” *Physical Review Letters* 101, 250602 (2008). DOI: [10.1103/PhysRevLett.101.250602](https://doi.org/10.1103/PhysRevLett.101.250602); arXiv: [cond-mat/0703788](https://arxiv.org/abs/cond-mat/0703788).
- [R254] A. Kitaev and J. Preskill, “Topological entanglement entropy,” *Physical Review Letters* 96, 110404 (2006). DOI: [10.1103/PhysRevLett.96.110404](https://doi.org/10.1103/PhysRevLett.96.110404); arXiv: [hep-th/0510092](https://arxiv.org/abs/hep-th/0510092).
- [R255] M. Levin and X.-G. Wen, “Detecting topological order in a ground state wave function,” *Physical Review Letters* 96, 110405 (2006). DOI: [10.1103/PhysRevLett.96.110405](https://doi.org/10.1103/PhysRevLett.96.110405); arXiv: [cond-mat/0510613](https://arxiv.org/abs/cond-mat/0510613).

Fabrication

[R094] T. M. Hensen, M. J. A. de Dood, and A. Polman, “Luminescence quantum efficiency and local optical density of states in thin film ruby made by ion implantation,” *Journal of Applied Physics* 88, 5142–5147 (2000).

DOI: [10.1063/1.1314322](https://doi.org/10.1063/1.1314322).

[R098] C. Babin, R. Stöhr, N. Morioka, et al., “Fabrication and nanophotonic waveguide integration of silicon carbide colour centres with preserved spin-optical coherence,” *Nature Materials* 21, 67–73 (2022).

DOI: [10.1038/s41563-021-01148-3](https://doi.org/10.1038/s41563-021-01148-3).

[R189] D. M. Toyli, C. D. Weis, G. D. Fuchs, T. Schenkel, and D. D. Awschalom, “Chip-scale nanofabrication of single spins and spin arrays in diamond,” *Nano Letters* 10, 3168–3172 (2010).

DOI: [10.1021/nl102066q](https://doi.org/10.1021/nl102066q).

[R190] T. Schröder et al., “Scalable focused ion beam creation of nearly lifetime-limited single quantum emitters in diamond nanostructures,” *Nature Communications* 8, 15376 (2017). DOI: [10.1038/ncomms15376](https://doi.org/10.1038/ncomms15376).

[R202] P. Spinicelli et al., “Engineered arrays of NV color centers in diamond based on implantation of CN[−] molecules through nanoapertures,” *New Journal of Physics* 13, 025014 (2011). DOI: [10.1088/1367-2630/13/2/025014](https://doi.org/10.1088/1367-2630/13/2/025014); arXiv: [1008.1483](https://arxiv.org/abs/1008.1483).

[R203] Y.-C. Chen et al., “Laser writing of individual nitrogen-vacancy defects in diamond with near-unity yield,” *Optica* 6, 662–667 (2019).

DOI: [10.1364/OPTICA.6.000662](https://doi.org/10.1364/OPTICA.6.000662). Chen and collaborators demonstrated laser writing of individual nitrogen-vacancy (NV) defects with near-unity creation yield after optimization. An NV defect consists of a substitutional nitrogen atom adjacent to a vacant lattice site in diamond.

The experiment addresses fabrication yield and three-dimensional defect placement. Optical aberrations, residual material damage, and control of pair geometry at nanometer length scales remain independent constraints.

[R204] K. Ohno et al., “Engineering shallow spins in diamond with nitrogen delta-doping,” *Applied Physics Letters* 101, 082413 (2012). DOI: [10.1063/1.4748280](https://doi.org/10.1063/1.4748280); arXiv: [1207.2784](https://arxiv.org/abs/1207.2784).

[R206] P. Räcke, L. Pietzonka, J. Meijer, D. Spemann, and R. Wunderlich, “Vacancy diffusion and nitrogen-vacancy center formation near the diamond surface,” *Applied Physics Letters* 118, 204003 (2021).

DOI: [10.1063/5.0046031](https://doi.org/10.1063/5.0046031).

[R207] S. Kim et al., “Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures,” *Nature Communications* 16, 9803 (2025).

DOI: [10.1038/s41467-025-64758-4](https://doi.org/10.1038/s41467-025-64758-4); arXiv: [2502.01198](https://arxiv.org/abs/2502.01198). This work demonstrates scalable nanoscale placement of coherent color centers in prefabricated diamond nanostructures.

It is among the strongest fabrication results that combine spatial registration with coherence preservation. However, uniform short-range exchange interactions require more demanding fabrication tolerances.

[R243] M. Haruyama et al., “Triple nitrogen-vacancy centre fabrication by $C_5N_4H_n$ ion implantation,” *Nature Communications* 10, 2664 (2019).

DOI: [10.1038/s41467-019-10529-x](https://doi.org/10.1038/s41467-019-10529-x). Haruyama and collaborators used molecular implantation to fabricate correlated triples of NV centers.

This result is particularly relevant to proposals based on defect clusters. However, creating a triple does not ensure the geometry, charge state, coherence, or coupling uniformity required to realize a target Hamiltonian.

[R257] D. Scarabelli, M. Trusheim, O. Gaathon, D. Englund, and S. J. Wind, “Nanoscale engineering of closely-spaced electronic spins in diamond,” *Nano Letters* 16, 4982–4990 (2016). DOI: [10.1021/acs.nanolett.6b01692](https://doi.org/10.1021/acs.nanolett.6b01692).

[R258] K. Groot-Berning, G. Jacob, C. Osterkamp, F. Jelezko, and F. Schmidt-Kaler, “Fabrication of $^{15}NV^-$ centers in diamond using a deterministic single ion implanter,” *New Journal of Physics* 23, 063067 (2021).

DOI: [10.1088/1367-2630/ac0753](https://doi.org/10.1088/1367-2630/ac0753); arXiv: [2101.01979](https://arxiv.org/abs/2101.01979). Groot-Berning and collaborators demonstrated deterministic single-ion implantation for creating nitrogen-vacancy centers.

This work supports claims about placement yield at the single-defect level. The conversion yield from an implanted ion to the desired defect and the coherence after implantation remain separate performance metrics.

[R259] A. Persaud, J. A. Liddle, T. Schenkel, J. Bokor, Tzv. Ivanov, and I. W. Rangelow, “Ion implantation with scanning probe alignment,” *Journal of Vacuum Science & Technology B* 23, 2798–2800 (2005). DOI: [10.1116/1.2062628](https://doi.org/10.1116/1.2062628).

Persaud and collaborators demonstrated ion implantation registered by scanning-probe alignment. This method directly controls the ion’s entry coordinate. However, stopping straggle—the statistical variation in the ion’s trajectory and final depth—and post-implant defect conversion continue to determine the final three-dimensional distribution of defect sites.

[R260] S. Pezzagna, B. Naydenov, F. Jelezko, J. Wrachtrup, and J. Meijer, “Creation efficiency of nitrogen-vacancy centres in diamond,” *New Journal of Physics* 12, 065017 (2010).

DOI: [10.1088/1367-2630/12/6/065017](https://doi.org/10.1088/1367-2630/12/6/065017). Pezzagna and collaborators measured the creation efficiency of implanted nitrogen-vacancy centers.

This source is important for distinguishing the number of delivered ions from the substantially smaller probability of producing the desired optically active defect.

Experimental topological quantum computing

[R125] K. J. Satzinger et al., “Realizing topologically ordered states on a quantum processor,” *Science* 374, 1237–1241 (2021). DOI: [10.1126/science.abi8378](https://doi.org/10.1126/science.abi8378); arXiv: [2104.01180](https://arxiv.org/abs/2104.01180).

This experiment used a superconducting quantum processor to prepare and probe a toric-code-type topologically ordered state. Topological order is a form of quantum order characterized by global, nonlocal properties rather than a conventional local order parameter. The toric code is a lattice model that realizes such order and supports topological quantum error-correcting codes. Because the state was assembled through digital quantum operations

and characterized on programmable hardware, the experiment demonstrates controllable quantum simulation rather than a naturally gapped topological material, in which an intrinsic energy gap separates the topological ground-state sector from excitations.

[R126] G. Semeghini et al., “Probing topological spin liquids on a programmable quantum simulator,” *Science* 374, 1242–1247 (2021).

DOI: [10.1126/science.abi8794](https://doi.org/10.1126/science.abi8794); arXiv: [2104.04119](https://arxiv.org/abs/2104.04119). Semeghini and collaborators used a programmable array of Rydberg atoms to probe signatures associated with spin liquids and topology. A spin liquid is a quantum phase in which interacting spins remain disordered even at low temperature while retaining nontrivial quantum correlations. Rydberg atoms are atoms excited to states with large principal quantum numbers, which produce strong and controllable interactions.

This work is a major milestone in quantum simulation. However, the simulator is externally driven and measured, so its behavior should not be identified with passive material protection, in which the material’s intrinsic Hamiltonian suppresses relevant errors without continuous digital control.

[R131] J. Nakamura, S. Liang, G. C. Gardner, and M. J. Manfra, “Direct observation of anyonic braiding statistics,” *Nature Physics* 16, 931–936 (2020). DOI: [10.1038/s41567-020-1019-1](https://doi.org/10.1038/s41567-020-1019-1); arXiv: [2006.14115](https://arxiv.org/abs/2006.14115).

[R132] M. B. Andersen et al. (Google Quantum AI and Collaborators), “Non-Abelian braiding of graph vertices in a superconducting processor,” *Nature* 618, 264–269 (2023).

DOI: [10.1038/s41586-023-05954-4](https://doi.org/10.1038/s41586-023-05954-4); arXiv: [2210.10255](https://arxiv.org/abs/2210.10255). This experiment implemented non-Abelian braiding of graph defects on a superconducting processor. Non-Abelian braiding denotes exchanges whose associated transformations depend on their order and therefore do not generally commute. Here, graph defects are digitally engineered features of the implemented interaction or code graph.

The results establish controlled braiding transformations within a digitally engineered code space, defined as the subspace used to encode the relevant quantum states. They do not establish intrinsic non-Abelian quasiparticles in a defect crystal, where such excitations would arise from the material itself.

[R138] S. Xu et al., “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* 20, 1469–1475 (2024).

DOI: [10.1038/s41567-024-02529-6](https://doi.org/10.1038/s41567-024-02529-6); arXiv: [2404.00091](https://arxiv.org/abs/2404.00091). Xu and collaborators digitally prepared, fused, and braided Fibonacci anyons on a superconducting processor. Fibonacci anyons are non-Abelian anyons whose allowed fusion outcomes follow the Fibonacci fusion rules; fusion refers to combining anyonic excitations and determining their resulting topological charge.

The experiment provides pivotal evidence that the target algebra of fusion and braiding operations can be implemented. It explicitly does not establish an autonomous Fibonacci material phase, meaning a material whose intrinsic dynamics realize the corresponding topological order without digital construction.

[R140] C. F. B. Lo et al., “Universal gates from braiding and fusing anyons on quantum hardware,” *Nature* 655, 591–597 (2026).

DOI: [10.1038/s41586-026-10709-y](https://doi.org/10.1038/s41586-026-10709-y); arXiv: [2601.20956](https://arxiv.org/abs/2601.20956).

[R162] Shibo Xu et al., “Digital Simulation of Projective Non-Abelian Anyons with 68 Superconducting Qubits,” *Chinese Physics Letters* 40, 060301 (2023). DOI: [10.1088/0256-307X/40/6/060301](https://doi.org/10.1088/0256-307X/40/6/060301).

[R163] Yu-ang Fan et al., “Experimental quantum simulation of a topologically protected Hadamard gate via braiding Fibonacci anyons,” *The Innovation* 4, 100480 (2023). DOI: [10.1016/j.xinn.2023.100480](https://doi.org/10.1016/j.xinn.2023.100480); arXiv: [2210.12145](https://arxiv.org/abs/2210.12145).

[R165] Zlatko K. Mineev et al., “Realizing string-net condensation: Fibonacci anyon braiding for universal gates and sampling chromatic polynomials,” *Nature Communications* 16, 6225 (2025). DOI: [10.1038/s41467-025-61493-8](https://doi.org/10.1038/s41467-025-61493-8); arXiv: [2406.12820](https://arxiv.org/abs/2406.12820).

[R222] R. M. Lutchyn, J. D. Sau, and S. Das Sarma, “Majorana fermions and a topological phase transition in semiconductor-superconductor heterostructures,” *Physical Review Letters* 105, 077001 (2010).

DOI: [10.1103/PhysRevLett.105.077001](https://doi.org/10.1103/PhysRevLett.105.077001); arXiv: [1002.4033](https://arxiv.org/abs/1002.4033).

[R223] K. Flensberg, F. von Oppen, and A. Stern, “Engineered platforms for topological superconductivity and Majorana zero modes,” *Nature Reviews Materials* 6, 944–958 (2021). DOI: [10.1038/s41578-021-00336-6](https://doi.org/10.1038/s41578-021-00336-6); arXiv: [2103.05548](https://arxiv.org/abs/2103.05548).

[R224] Microsoft Azure Quantum, M. Aghaee, A. Alcaraz Ramirez, Z. Alam, et al., “Interferometric single-shot parity measurement in InAs–Al hybrid devices,” *Nature* 638, 651–655 (2025).

DOI: [10.1038/s41586-024-08445-2](https://doi.org/10.1038/s41586-024-08445-2); arXiv: [2401.09549](https://arxiv.org/abs/2401.09549).

[R230] Google Quantum AI and Collaborators, “Quantum error correction below the surface code threshold,” *Nature* 638, 920–926 (2025; published online 2024). DOI: [10.1038/s41586-024-08449-y](https://doi.org/10.1038/s41586-024-08449-y); arXiv: [2408.13687](https://arxiv.org/abs/2408.13687).

[R233] D. Bluvstein et al., “Logical quantum processor based on reconfigurable atom arrays,” *Nature* 626, 58–65 (2024). DOI: [10.1038/s41586-023-06927-3](https://doi.org/10.1038/s41586-023-06927-3).

[R240] A. Gruber, A. Dräbenstedt, C. Tietz, L. Fleury, J. Wrachtrup, and C. von Borczyskowski, “Scanning Confocal Optical Microscopy and Magnetic Resonance on Single Defect Centers,” *Science* 276, 2012–2014 (1997).

DOI: [10.1126/science.276.5321.2012](https://doi.org/10.1126/science.276.5321.2012).

[R241] H.-Y. Huang, R. Kueng, and J. Preskill, “Predicting many properties of a quantum system from very few measurements,” *Nature Physics* 16, 1050–1057 (2020).

DOI: [10.1038/s41567-020-0932-7](https://doi.org/10.1038/s41567-020-0932-7); arXiv: [2002.08953](https://arxiv.org/abs/2002.08953).

[R256] Y. Zhang, T. Grover, A. Turner, M. Oshikawa, and A. Vishwanath, “Quasiparticle statistics and braiding from ground-state entanglement,” *Physical Review B* 85, 235151 (2012).

DOI: [10.1103/PhysRevB.85.235151](https://doi.org/10.1103/PhysRevB.85.235151); arXiv: [1111.2342](https://arxiv.org/abs/1111.2342).